Analysis II

Several Variables — Exercise Class Notes & Transcript

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PART I

Foundations

Analysis II begins where **Analysis I** stopped, and the first thing to settle is how much of it carries over. The definitions of convergence, of continuity, and of a set being open survive the move to several variables essentially verbatim. What does not survive is every statement that quietly used the ordering of $\mathbb{R}$, and each of those has to be rebuilt on a new hypothesis. This part draws that line, fixes the notation for $\mathbb{R}^n$, and rehearses two habits used constantly later on: sketching a subset of $\mathbb{R}^n$ from the inequalities defining it, and reading a function of two variables off its level sets. It also proves Young's inequality, which looks like an isolated exercise here and turns out to be the engine behind the Hölder and Minkowski inequalities in the next part.

Prerequisites and a First Look at $\mathbb{R}^n$ (Chapter 1)

This chapter collects the **Analysis I** material the course takes for granted, together with the $\mathbb{R}^n$ warm-up that precedes any of its theory. Two things are worth settling before the theory starts. The first is which definitions survive the passage to several variables unchanged and which statements do not, since the two look alike and behave very differently. The second is fluency at turning a set-builder description into a picture and back again, a skill every later chapter assumes without ever teaching. Young's inequality (Proposition 1.b.3) is proved here as well, slightly out of sequence: it belongs to no chapter in particular, and it is wanted as soon as p-norms appear.

From Analysis I to several variables (Section 1.a)

Several of the definitions you will meet in the coming chapters are exactly the definitions you already know from **Analysis I**, only stripped of their reliance on the real line. A "*sequence*" (German: „*Folge*") $(x_k)_{k\in\mathbb{N}}$ in a set X converging to $x \in X$, a function being continuous at a point, a set being open or closed: all of these notions carry over to $\mathbb{R}^n$ and, as Section 2.a will show, to spaces far more general than $\mathbb{R}^n$.

What does **not** carry over automatically is anything that relied on the **order** of $\mathbb{R}$: there is no "$\leq$" on $\mathbb{R}^n$ for $n \geq 2$, so statements like the intermediate value theorem or "*every bounded sequence has a convergent subsequence*" need new proofs, or new hypotheses, once we leave $\mathbb{R}$.

Remark 1.a.1: The second of those statements is worth stating carefully, because it is false as written and true after one word is changed. Bounded does not imply convergent. What is true is that a bounded sequence in $\mathbb{R}^n$ always has a convergent **subsequence**, which is the Bolzano–Weierstrass theorem, revisited via Exercise 1.a.2 below. It reappears in a more general form as the Heine–Borel theorem once compactness is introduced in Section 7.a.

Exercise 1.a.2 (*1.1 — Recap Questions*):

1. Let $\{x_k\}_{k\in\mathbb{N}} \subset [1,2] \subset \mathbb{R}$ be a sequence. Is it necessarily convergent?
2. Describe all differentiable functions $f : \mathbb{R}\setminus\{0\} \to \mathbb{R}$ such that $f'(x) = 1/x$ for all $x \neq 0$.
3. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and bounded. Is it true that
$$\int_0^{\pi/2} f(\cos(x))\,dx = \int_0^1 \frac{f(t)}{\sqrt{1-t^2}}\,dt\,?$$
4. Let $f : [0,\infty) \to \mathbb{R}$ have $f''(x) > 0$ for all $x \geq 0$. Can f have a global maximum point?
5. Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map and let $Q := [0,1]\times[0,1]$ be the unit square. How does $\phi(Q)$ look like? What are the possibilities?
6. Consider the vector space V of polynomials of degree less than 4, and the function $D : V \to V$ which associates to a polynomial p its derivative p'. Is D a linear map? If so, what is the dimension of its null space?

Young's inequality (Section 1.b)

The following inequality is not part of the lecture's main line of development, but it is the cleanest route to comparing different norms on $\mathbb{R}^n$ (see Section 2.c) and to the Hölder-type estimates that appear whenever two L^p-type quantities are played off against each other. It is worth proving once and citing from then on.

Proposition 1.b.3 (*Young's inequality*): Let $a, b > 0$ and let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Remark 1.b.4: The case $p = q = 2$ recovers the familiar inequality $ab \leq \frac{a^2}{2} + \frac{b^2}{2}$, which is just a rewriting of $(a-b)^2 \geq 0$.

Proof:

Fix conjugate exponents $p, q > 1$, and define $g : (0, \infty) \to \mathbb{R}$ by $g(t) := \frac{t}{p} + \frac{1}{q} - t^{1/p}$. We first show that $g \geq 0$, with equality only at $t = 1$. Differentiating,

$$g'(t) = \frac{1}{p}\left(1 - t^{-(p-1)/p}\right).$$

Since $p > 1$, the exponent $-(p-1)/p$ is negative, so $t^{-(p-1)/p} > 1$ for $t \in (0,1)$ and $t^{-(p-1)/p} < 1$ for $t > 1$. Hence $g' < 0$ on $(0,1)$ and $g' > 0$ on $(1, \infty)$, so g attains a global minimum at $t = 1$. As $g(1) = \frac{1}{p} + \frac{1}{q} - 1 = 0$, we obtain

$$t^{1/p} \leq \frac{t}{p} + \frac{1}{q} \qquad \text{for all } t > 0. \tag{1.1}$$

Now fix $a, b > 0$ and apply (1.1) with $t := a^p / b^q$. Multiplying both sides by $b^q > 0$ gives

$$\left(\frac{a^p}{b^q}\right)^{1/p} b^q \;\leq\; \frac{a^p}{p} + \frac{b^q}{q}.$$

It remains to identify the left-hand side, and this is where the conjugacy of p and q enters. Using $1 - \frac{1}{p} = \frac{1}{q}$,

$$\left(\frac{a^p}{b^q}\right)^{1/p} b^q \;=\; a\, b^{-q/p}\, b^q \;=\; a\, b^{q\left(1-\frac{1}{p}\right)} \;=\; a\, b^{q\cdot\frac{1}{q}} \;=\; ab,$$

which is exactly the asserted inequality. **Q.E.D.**

Remark 1.b.5 (*From Young to Hölder*): The exercise sheet also asks for the vector version

$$\langle a, b\rangle \leq \frac{\|a\|_p^p}{p} + \frac{\|b\|_q^q}{q}, \qquad a, b \in \mathbb{R}^n, \tag{1.2}$$

where $\|x\|_p := \left(\sum_i |x_i|^p\right)^{1/p}$. This is **not** yet Hölder's inequality: it is the scalar case applied coordinatewise and summed (see the solution of Exercise 1.b.6 below), and the right-hand side is a sum of p-th and q-th powers rather than the product $\|a\|_p \|b\|_q$. The step from one to the other is a normalisation, and it is worth seeing once, because the same trick recurs whenever an inequality is homogeneous.

Assume $a, b \neq 0$ and apply (1.2) to the rescaled vectors $a/\|a\|_p$ and $b/\|b\|_q$, each of which has p- resp. q-norm equal to 1:

$$\frac{\langle a, b\rangle}{\|a\|_p \, \|b\|_q} \;\leq\; \frac{1}{p} + \frac{1}{q} \;=\; 1,$$

which is exactly "*Hölder's inequality*" (German: „*Höldersche Ungleichung*“) $\langle a, b\rangle \leq \|a\|_p \|b\|_q$. The conjugacy condition $\frac{1}{p} + \frac{1}{q} = 1$, which so far looked like an arbitrary bookkeeping device, is what makes the two constants add up to exactly 1 here.

This is also a first hint of why p-norms other than $p = 2$ are worth naming: Section 2.c only treats $p = q = 2$, where Hölder's inequality is Cauchy–Schwarz, but the same machinery bounds $\langle a, b\rangle$ for every conjugate pair (p, q).

Exercise 1.b.6 (*1.4 — Young's inequality*): Let $a, b > 0$, and let $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Prove that $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$.

1. Prove that for all $t > 0$, $t^{1/p} \le \frac{t}{p} + \frac{1}{q}$.
2. Multiply the inequality by b^q and choose an appropriate value of t to conclude.

A bonus: show that $\langle a, b\rangle \le \frac{\|a\|_p^p}{p} + \frac{\|b\|_q^q}{q}$ for all $a, b \in \mathbb{R}^n$.

Visualizing sets in $\mathbb{R}^2$ and $\mathbb{R}^3$ (Section 1.c)

Much of the geometric intuition for this course (open sets, level sets, submanifolds) rests on a skill that has nothing to do with any theorem: reading a set-builder description and producing an accurate picture, or going the other way. Two techniques are worth naming explicitly, both used repeatedly in the solutions below.

- **Slicing.** To sketch a set $S \subseteq \mathbb{R}^3$ defined by an equation involving z, fix $z = t$ and sketch the resulting planar slice $S \cap \{z = t\}$ as t varies. Stacking the slices gives the full picture. Two questions settle most cases: for which t is the slice non-empty, and how does its size depend on t?
- **Level sets.** If a set is defined by $f(x, y) = t$ for a fixed t, first work out the level sets $\{(x, y) : f(x, y) = c\}$ of f for varying c. Many natural functions have level sets one can simply name: those of $f(x, y) = \sqrt{x^2 + y^2}$ are circles, those of $f(x, y) = \max\{|x|, |y|\}$ are squares, and those of $f(x, y) = |x| + |y|$ are diamonds. Naming them turns a three-dimensional sketch into a stack of familiar plane figures.

Those last three functions are not a random selection. They are the Euclidean, supremum and Manhattan norms of (x, y), and their level sets at height 1 are the three unit balls drawn in Section 3.c. It is worth carrying the picture forward: a great deal of what looks like metric-space abstraction in the next two chapters is this one diagram in disguise.

Both techniques are applied to Exercises 1.c.7 and 1.c.8 below, and worked out in Section 1.d.

Exercise 1.c.7 (*1.2 — From equations to drawings*): Sketch the following subsets of $\mathbb{R}^2$, $\mathbb{R}^3$.

1. $A := \{(x, y) : x^2 + y^2 < 1,\ y \ge x^2\}$
2. $B := \{(x, y) : y^2 = x^3 - x\}$
3. $C := \{(x, y) : \frac{\pi}{6} < \arctan(y/x) < \frac{\pi}{3},\ 0 < x < y\}$
4. $D := \{(x, y, z) : \sqrt{x^2 + y^2} = 1 - z,\ 0 < z < 1\}$
5. $E := \{(x, y, z) : z = \sqrt{1 + x^2 + y^2}\}$
6. $F := \{(x, y, z) : z = \max\{|x|, |y|\}\}$

Exercise 1.c.8 (*1.3 — From drawings to equations*): Express in terms of functions and equations the following sets.

1. The coordinates of a point in the plane that goes endlessly back and forth from the origin to the point $(-1, 1)$.
2. A planar domain with two holes in it.
3. A bi-dimensional disk sitting inside the plane $\{x + y + z = 0\}$ in $\mathbb{R}^3$.
4. A Pac-Man shaped domain in the plane.
5. A pyramid, whose base is a square, sitting in $\mathbb{R}^3$.

Solutions (Section 1.d)

Solution of Exercise 1.a.2:

1. No. The sequence $x_k := \frac{3}{2} + \frac{1}{2}(-1)^k$ lies in $[1, 2]$ but oscillates between 1 and 2 without settling, so it does not converge (it has two distinct accumulation points, 1 and 2). What

is true, by the Bolzano–Weierstrass theorem, is that every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

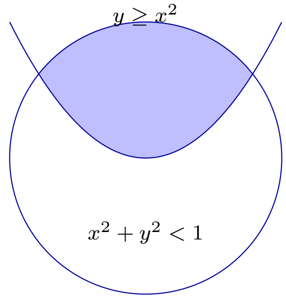
2. On each of the intervals $(-\infty, 0)$ and $(0, \infty)$ separately, $f'(x) = 1/x$ forces $f(x) = \log|x| + c$ for some constant c, since two antiderivatives on a common interval differ by a constant. The subtlety is that $\mathbb{R} \setminus \{0\}$ is **disconnected** into these two intervals, so the two constants need not agree:
$$f(x) = \begin{cases} \alpha + \log(x) & x > 0, \\ \beta + \log(-x) & x < 0, \end{cases} \qquad \alpha, \beta \in \mathbb{R} \text{ arbitrary.}$$
The answer $f(x) = \log|x| + c$ for a **single** constant c is wrong; connectedness of the domain is exactly what would be needed to force $\alpha = \beta$, and it is precisely what Section 8.a formalizes.
3. Yes. Substituting $t = \cos(x)$, $dt = -\sin(x)\,dx$, and $\sin(x) = \sqrt{1-t^2}$ for $x \in [0, \pi/2]$ turns the left integral into the right one; this is the one-variable change of variables formula, which the change-of-variables theorem (Section 20.a) generalizes to $\mathbb{R}^n$.
4. Yes: take $f(x) := e^{-x}$. Then $f''(x) = e^{-x} > 0$ for all x, yet f attains its global maximum on $[0, \infty)$ at the boundary point $x = 0$. Everywhere-positive second derivative (strict convexity) rules out an **interior** local maximum, but says nothing whatever about the boundary of the domain.
5. $\phi(Q)$ is the parallelogram spanned by $\phi(1,0)$ and $\phi(0,1)$, with vertices at 0, $\phi(1,0)$, $\phi(0,1)$, and $\phi(1,0) + \phi(0,1)$. Two degenerate cases occur: if $\phi(1,0)$ and $\phi(0,1)$ are linearly dependent (parallel or one is zero) but not both zero, the parallelogram collapses to a line segment; if $\phi \equiv 0$, it collapses to the single point $\{0\}$.
6. Yes: differentiation is linear on functions in general, hence in particular on the subspace of polynomials of degree < 4. With respect to the basis $\mathcal{B} = \{1, X, X^2, X^3\}$, the map D sends $1 \mapsto 0$, $X \mapsto 1$, $X^2 \mapsto 2X$, $X^3 \mapsto 3X^2$, giving matrix
$$[D]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$
This matrix has rank 3 (its three nonzero rows are independent), so by the rank–nullity theorem the null space has dimension $4 - 3 = 1$, and it is spanned by the constant polynomials.

Q.E.D.

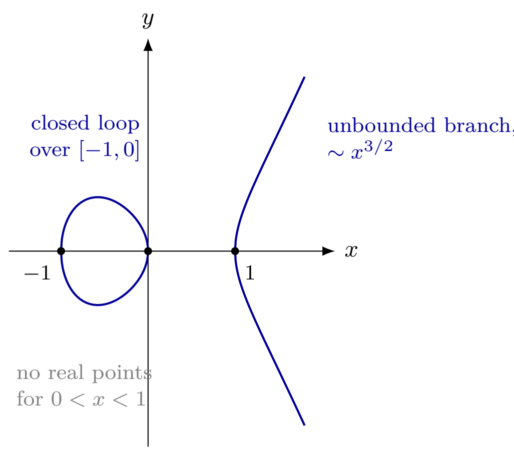
Remark 1.d.9 (*The two parts most often answered wrongly*): Linus Lüchinger's first exercise-session slides record where this sheet went wrong most often, and two of the parts above account for most of it. Part 2 was one: the single-constant answer $f(x) = \log|x| + c$ was the usual first guess, and it fails for the reason just given, namely that the domain falls apart into two intervals. Part 4 was the other: a function with everywhere positive second derivative was widely assumed to admit no global maximum at all, when in fact it may attain one at a boundary point. Both mistakes have the same shape. A hypothesis that controls the **interior** of a domain was read as though it controlled the domain.

Solution of Exercise 1.c.7:

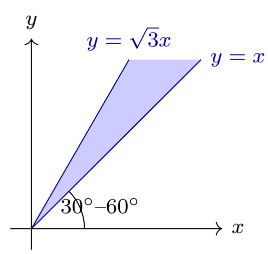
1. A is the intersection of the open unit disk with the region on or above the parabola $y = x^2$, a lens-shaped region:



2. Since $x^3 - x = x(x+1)(x-1) \geq 0$ exactly for $x \in [-1,0] \cup [1,\infty)$, the curve $y = \pm\sqrt{x^3 - x}$ is only real on these two intervals; B is the union of the graphs of $f(x) := \sqrt{x^3 - x}$ and $-f(x)$, giving a closed loop over $[-1,0]$ and an unbounded branch over $[1,\infty)$ growing like $x^{3/2}$. Note that the two branches meet exactly where $f = 0$, i.e. at the three roots $x = -1, 0, 1$, which is why the loop closes up and the right branch starts with a vertical tangent.



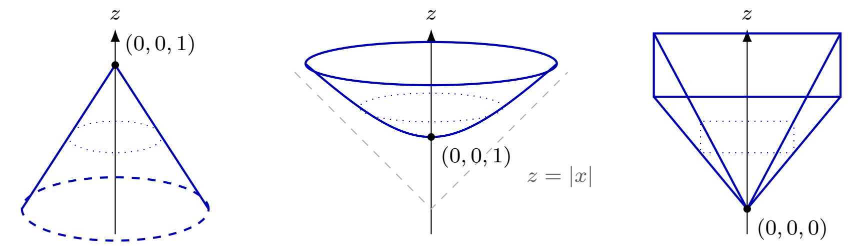
3. Since $0 < x < y$, points of C lie in the first octant with $y > x$; the condition $\pi/6 < \arctan(y/x) < \pi/3$ says the angle to the positive x-axis is between $30°$ and $60°$. So C is the open angular sector between the lines $y = x$ and $y = \sqrt{3}\,x$:



4. Slicing at height $z = t \in (0,1)$: the equation $\sqrt{x^2 + y^2} = 1 - t$ describes a circle of radius $1 - t$ centered on the z-axis. As t ranges over $(0,1)$, the radius shrinks linearly from 1 to 0, so D is the lateral surface of a right circular cone of height 1 over the unit circle, apex at $(0,0,1)$, base circle excluded (the inequalities are strict).
5. Slicing at height $z = t$: the equation $\sqrt{1 + x^2 + y^2} = t$ has solutions only for $t \geq 1$, and the slice is then a circle of radius $\sqrt{t^2 - 1}$. Note that the radius is $\sqrt{t^2 - 1}$ and not t; this is the single most common slip on this part, and it turns the answer into a cone. E is instead a bowl-shaped surface of revolution with vertex at $(0,0,1)$, obtained by rotating the curve $y = \sqrt{1 + x^2}$ about the z-axis. It approaches slope 1 for large x, y without ever attaining it, which is exactly how it differs from the cone $z = \sqrt{x^2 + y^2}$ that it is so often mistaken for.

6. The level sets $\{\max\{|x|,|y|\}=t\}$ are the boundaries of the squares $[-t,t]^2$. Stacking these square outlines at height $z=t$ as t grows from 0 upward sweeps out four flat triangular faces meeting at the apex $(0,0,0)$. So F is an upward-opening **square cone**: the exact analogue of the circular cone of part 4, with the round level circles of $\sqrt{x^2+y^2}$ replaced by the square level curves of $\max\{|x|,|y|\}$.

Parts 4, 5 and 6 are the same construction carried out three times, stacking level sets, and are best compared side by side. In each panel the dotted curve is one intermediate level set:



(4) $\sqrt{x^2+y^2}=1-z$
radius shrinks $1\to 0$

(5) $z=\sqrt{1+x^2+y^2}$
radius $\sqrt{z^2-1}$, **not** z

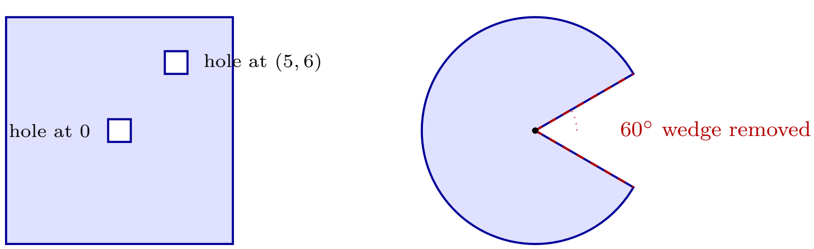
(6) $z=\max\{|x|,|y|\}$
squares instead of circles

Q.E.D.

Solution of Exercise 1.c.8:

1. A point moving back and forth along the segment from $(0,0)$ to $(-1,1)$ can be parametrized as $t\mapsto(-f(t),f(t))$ for $t\geq 0$, where $f:[0,\infty)\to[0,1]$ oscillates endlessly; $f(t):=\frac{1}{2}(1+\sin t)$ is one valid choice.
2. Remove two small disjoint squares from a large square, e.g. $\{10>\max\{|x|,|y|\}>1\}\setminus\{\max\{|x-5|,|y-6|\}\leq 1\}$.
3. Intersect the closed unit ball with the plane: $\{x^2+y^2+z^2\leq 1\}\cap\{x+y+z=0\}$.
4. Remove a $60°$ wedge from a disk: $\{x^2+y^2\leq 1\}\setminus\{|\arctan(y/x)|\leq\pi/6,\ x>0\}$.
5. Combining the square level sets from Exercise 1.c.7 part 6 with the conical slicing from part 4: $\{1-z=\max\{|x|,|y|\},\ 0\leq z\leq 1\}$ is a pyramid of height 1 over the square $[-1,1]^2$, apex at $(0,0,1)$.

The two planar answers are worth drawing, since in both cases the description *"remove a piece"* translates directly into a set difference:



2. a domain with two holes:
$\{10>\max\{|x|,|y|\}>1\}$
minus a unit square at $(5,6)$

4. Pac-Man: the closed unit disk
minus the wedge $\{|\arctan(y/x)|\leq\pi/6,\ x>0\}$

Q.E.D.

Solution of Exercise 1.b.6:

Parts 1 and 2 are exactly the proof of Proposition 1.b.3 above. For the bonus vector version, apply the scalar inequality coordinatewise, with $|a_i|$ in place of a and $|b_i|$ in place of b, and sum over i:

$$\begin{aligned} \langle a, b\rangle \;&\le\; \sum_{i=1}^{n} |a_i|\,|b_i| \;\le\; \sum_{i=1}^{n} \left(\frac{|a_i|^p}{p} + \frac{|b_i|^q}{q} \right) \\ &=\; \frac{1}{p}\sum_{i=1}^{n} |a_i|^p + \frac{1}{q}\sum_{i=1}^{n} |b_i|^q \;=\; \frac{\|a\|_p^p}{p} + \frac{\|b\|_q^q}{q}. \end{aligned}$$

Note that the two sums separate only because the right-hand side of Young's inequality is a **sum** of powers rather than a product; this is precisely the feature that Remark 1.b.5 then removes by rescaling. **Q.E.D.**

PART II

Metric Spaces and Topology

Very little of one-variable analysis actually used the arithmetic of $\mathbb{R}$. Convergence, continuity and compactness need only a notion of distance, and this part takes that observation seriously by developing all three for an arbitrary metric space. The payoff is not generality for its own sake. Function spaces such as $C^0([0,1],\mathbb{R})$ are metric spaces too, so once the theory is stated at this level, the same arguments apply to a sequence of functions and to a sequence of points alike. Two properties do most of the work: completeness, which guarantees that a limit one has constructed actually exists, and compactness, which allows an infinite problem to be replaced by a finite one. Almost every existence proof in the remaining five parts leans on one of them.

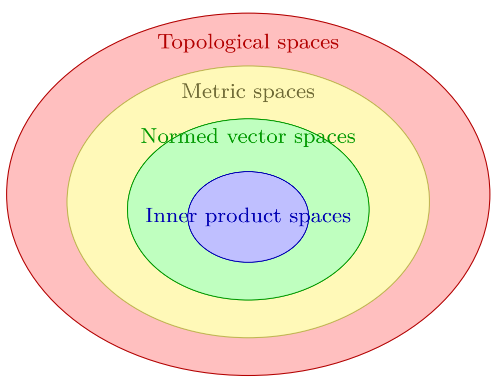
Metric, Normed and Inner-Product Spaces (Chapter 2)

How much structure does a set need before one can do analysis on it? This chapter answers that by taking $\mathbb{R}^n$ apart into layers. Distance comes first, and distance alone already carries convergence and continuity. Adding a linear structure gives a norm, which measures the length of one vector rather than the gap between two. Adding an inner product on top of that gives angles, and with them orthogonal projection. The layers nest, each one strictly inside the last, and the chapter closes with the inequality that makes the innermost layer work at all: Cauchy–Schwarz.

Structured spaces (Section 2.a)

This semester, you will be introduced to certain structures that can be given to a set, which in this context is also called a "*space*" (German: „*Raum*“) and which we will denote by X:

- **0. Topological spaces:** fourth semester.
- **1. Metric spaces** $(X,\ d : X \times X \to \mathbb{R}_{\geq 0})$
  - Defines a **distance** between points.
  - X is **not** necessarily a vector space.
  - Covered in *Analysis*.
- **2. Normed metric spaces** $(X,\ \|\cdot\| : X \to \mathbb{R}_{\geq 0})$
  - Defines the **length** of a vector.
  - Presupposes a linear structure: X must be a **vector space**.
  - Covered in *Linalg / Analysis*.
- **3. Inner product spaces** $(X,\ \langle\cdot,\cdot\rangle : X \times X \to \mathbb{R})$
  - Defines **both** lengths **and** angles!
  - Super important!
  - Covered in *Linalg* (and sometimes used in *Analysis*). X is a **vector space**.



You will see that an **inner product** $\langle\cdot,\cdot\rangle$ induces a **norm** by the formula

$$\|x\| := \sqrt{\langle x, x\rangle}, \qquad x \in X.$$

Furthermore, a **norm** induces a **metric** by

$$d(x, y) := \|x - y\|, \qquad x, y \in X.$$

Definition 2.a.1 (*Inner product / scalar product*): Let V be a **vector space** over $\mathbb{R}$. An "*inner product*" (German: „*Skalarprodukt*“) on V is a map

$$\langle\cdot,\cdot\rangle : V \times V \to \mathbb{R}$$

satisfying, for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$:

(a) Bilinearity: $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$, and equivalently in the second variable.
(b) Symmetry: $\langle u, v \rangle = \langle v, u \rangle$.
(c) Definiteness: $\langle u, u \rangle \geq 0$, and $\langle u, u \rangle = 0 \iff u = 0$.

Definition 2.a.2 (*Norm*): Let V be a **real** vector space. A "*norm*" (German: „*Norm*") on V is a map $\|\cdot\| : V \to [0, \infty)$ satisfying, for all $u, v \in V$ and $\alpha \in \mathbb{R}$:
(a) Definiteness: $\|v\| \geq 0$, and $\|v\| = 0 \iff v = 0$.
(b) Homogeneity: $\|\alpha v\| = |\alpha| \|v\|$.
(c) $\triangle$-inequality: $\|u + v\| \leq \|u\| + \|v\|$.

AI-Note 2.1: The official script (Remark 9.94, p. 31 onward) adopts the opposite notational convention from this document: from that point on **it** writes the Euclidean norm as $|x|$ and reserves $\|\cdot\|$ for non-standard norms. This document instead uses $\|\cdot\|$ (`\lVert...\rVert`) uniformly for every norm, Euclidean or not, per the house style, and keeps that convention throughout. Neither choice is more standard than the other; this note exists only to warn anyone cross-referencing the script directly that a bare $|\cdot|$ there and a bare $\|\cdot\|$ here can both denote *"the"* Euclidean norm. The warning matters most in the script's chapters 13–14, where the $|\cdot|$ convention is used densely.

Example 2.a.3 (*Hilbert–Schmidt norm on matrix spaces*): A very important example of a norm on the space of real $n \times m$ matrices $\mathbb{R}^{n \times m}$ is the "*Hilbert–Schmidt norm*" (also known as the Frobenius norm). It is defined by

$$\|M\|_2 := \sqrt{\operatorname{Tr}(M^\mathsf{T} M)},$$

where Tr denotes the trace of a matrix. Only the diagonal of $M^\mathsf{T} M$ is needed, so the off-diagonal entries are left unevaluated below. For a 2×3 matrix $M = \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right)$, we have

$$\begin{aligned} M^\mathsf{T} M &= \begin{pmatrix} a & d \\ b & e \\ c & f \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \\ &= \begin{pmatrix} a^2 + d^2 & * & * \\ * & b^2 + e^2 & * \\ * & * & c^2 + f^2 \end{pmatrix}. \end{aligned}$$

The trace is the sum of the diagonal entries, so $\|M\|_2 = \sqrt{a^2 + b^2 + c^2 + d^2 + e^2 + f^2}$. This norm is therefore nothing but the standard Euclidean norm of the matrix entries read as a single vector in $\mathbb{R}^6$, or in general in $\mathbb{R}^{nm}$. The inner product inducing it is $\langle A, B \rangle := \operatorname{Tr}(A^\mathsf{T} B)$.

Comparing the three lists of axioms is already informative. A norm has exactly the three properties a metric has, restated for a single vector instead of a pair of points, plus homogeneity. That extra axiom has no metric analogue at all, since stating it requires scalar multiplication. An inner product goes further and replaces the triangle inequality by bilinearity, which is far stronger, and then recovers the triangle inequality as a **theorem** (Proposition 2.a.6) rather than assuming it. That is the precise sense in which each layer carries more structure than the one outside it.

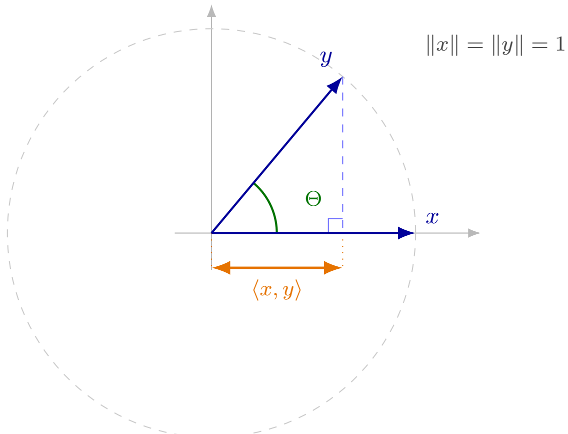
Remark 2.a.4 (*Geometric angle and projection in inner product spaces*): In an inner product space $(X, \langle \cdot, \cdot \rangle)$ over $\mathbb{R}$, the Cauchy–Schwarz inequality $|\langle x, y \rangle| \leq \|x\| \, \|y\|$ guarantees that for any non-zero vectors $x, y \in X$,

$$-1 \leq \frac{\langle x, y \rangle}{\|x\| \, \|y\|} \leq 1.$$

This allows us to formally define the "*angle*" $\Theta(x, y) \in [0, \pi]$ between x and y via

$$\Theta(x, y) := \arccos\left(\frac{\langle x, y \rangle}{\|x\| \, \|y\|}\right).$$

Geometrically, when $\|x\| = \|y\| = 1$, the inner product $\langle x, y\rangle$ is the signed length of the orthogonal projection of y onto the direction of x. In that reading Cauchy–Schwarz becomes visually obvious: a projection of a unit vector cannot be longer than the unit vector itself, so $\langle x, y\rangle \le 1$, with equality exactly when y already points along x.



Proposition 2.a.5 (*Geometric derivation of the Cauchy–Schwarz inequality*)**:** Let $(V, \langle\cdot,\cdot\rangle)$ be a real inner product space. Then for all $v, w \in V$,

$$|\langle v, w\rangle| \le \|v\|\,\|w\|.$$

Proof:

If $w = 0$, both sides vanish and the inequality holds trivially. For $w \neq 0$, consider the squared distance from v to the points of the line $L := \operatorname{span}_{\mathbb{R}}\{w\}$, parametrized by $\lambda \mapsto \lambda w$:

$$f(\lambda) := \|v - \lambda w\|^2 = \langle v - \lambda w, v - \lambda w\rangle = \|v\|^2 - 2\lambda\langle v, w\rangle + \lambda^2\|w\|^2.$$

Since $f(\lambda) \ge 0$ for all $\lambda \in \mathbb{R}$, f is a convex quadratic polynomial in λ with positive leading coefficient $\|w\|^2 > 0$. Differentiating or completing the square shows that f attains its global minimum at $\lambda_0 := \frac{\langle v,w\rangle}{\|w\|^2}$. Evaluating f at λ_0 yields

$$0 \le f(\lambda_0) = \|v\|^2 - 2\frac{\langle v, w\rangle^2}{\|w\|^2} + \frac{\langle v, w\rangle^2}{\|w\|^2} = \|v\|^2 - \frac{\langle v, w\rangle^2}{\|w\|^2}.$$

Multiplying by $\|w\|^2$ and taking square roots gives $|\langle v, w\rangle| \le \|v\|\,\|w\|$. Geometrically, $\pi_L(v) := \lambda_0 w = \frac{\langle v,w\rangle}{\|w\|^2} w$ is the unique orthogonal projection of v onto the line L, achieving the minimal distance $d(v, L) = \sqrt{f(\lambda_0)}$. **Q.E.D.**

Proposition 2.a.6 (*Inner product $\Rightarrow$ norm $\Rightarrow$ metric*)**:** Let V be a real vector space.

(a) If $\langle\cdot,\cdot\rangle$ is an inner product on V (Definition 2.a.1), then $\|v\| := \sqrt{\langle v, v\rangle}$ is a norm on V (Definition 2.a.2).

(b) If $\|\cdot\|$ is a norm on V, then $d(x, y) := \|x - y\|$ is a metric on V (Definition 2.b.8).

Proof:

(a) Definiteness of the norm is definiteness of the inner product, and homogeneity follows from bilinearity: $\|\alpha v\|^2 = \langle \alpha v, \alpha v\rangle = \alpha^2\|v\|^2$. For the triangle inequality, expand and apply Proposition 2.a.5:

$$\begin{aligned} \|u + v\|^2 &= \|u\|^2 + 2\langle u, v\rangle + \|v\|^2 \\ &\le \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2 \\ &\le \|u\|^2 + 2\|u\|\,\|v\| + \|v\|^2 \;=\; \big(\|u\| + \|v\|\big)^2, \end{aligned}$$

and taking square roots gives $\|u+v\| \le \|u\| + \|v\|$.

(b) Definiteness: $d(x,y) = 0$ means $\|x-y\| = 0$, hence $x = y$. Symmetry: homogeneity with $\alpha = -1$ gives $\|x-y\| = \|-(y-x)\| = \|y-x\|$. Triangle inequality: writing $x-z = (x-y)+(y-z)$,

$$d(x,z) = \|(x-y)+(y-z)\| \le \|x-y\| + \|y-z\| = d(x,y) + d(y,z).$$

Q.E.D.

Remark 2.a.7 (*Neither arrow reverses*): The nesting drawn above is strict at both steps.

- **Not every metric comes from a norm.** The discrete metric (Example 2.b.11**(a)**) on a vector space cannot: a norm-induced metric satisfies $d(\alpha x, 0) = |\alpha|\, d(x,0)$, which fails as soon as $\alpha \neq 0, \pm 1$.
- **Not every norm comes from an inner product.** A norm induced by an inner product obeys the "*parallelogram law*" $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$ (expand both sides by bilinearity). The supremum norm on $\mathbb{R}^2$ violates it: for $u := (1,0)$ and $v := (0,1)$ the left side is $1+1 = 2$ and the right side is $2+2 = 4$.

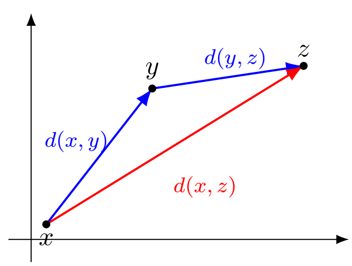
Consequently, each layer of the hierarchy really does carry strictly more structure than the one outside it. The picture at the start of this section is a genuine nesting of four distinct classes, not a chain of equivalent descriptions of the same thing.

Metric spaces (Section 2.b)

Definition 2.b.8 (*Metric space*): A "*metric space*" (German: „*metrischer Raum*") (X,d) is any set X with a "*metric*" (German: „*Metrik*") $d : X \times X \to \mathbb{R}_{\ge 0}$ such that for all $x, y, z \in X$:

- **(a) Definiteness:** $d(x,y) \ge 0$, with equality if and only if $x = y$. In words: the distance is non-negative, and a point has zero distance to itself.
- **(b) Symmetry:** $d(x,y) = d(y,x)$, that is, x has the same distance to y as y has to x.
- **(c) Triangle inequality ($\triangle$-inequality):** $d(x,z) \le d(x,y) + d(y,z)$. In words, *"detours make the way longer"*.

AI-Note 2.2: Corsin's gloss on definiteness reads *"the distance positive"*; the formula $d \ge 0$ is correct, but the gloss omits the "non-". Corrected in the prose above.



$d(x,z) \le d(x,y) + d(y,z)$

Exercise 2.b.9 (*Polygon inequality*): Let (X,d) be a metric space and $N \ge 3$. Prove, by induction on N, that

$$d(x_1, x_N) \le \sum_{i=1}^{N-1} d(x_i, x_{i+1})$$

for all $x_1, \dots, x_N \in X$.

AI-Example 2.b.10 (*Each axiom rules out something, and none is redundant*): Three functions on $\mathbb{R}$, each failing exactly one axiom of Definition 2.b.8. It is worth checking these once: a definition only becomes memorable once you know what it is keeping out.

(a) Definiteness fails: $d(x,y) := |x^2 - y^2|$. This is symmetric and satisfies the triangle inequality (it is $|\cdot|$ composed with $x \mapsto x^2$), but $d(-1,1) = 0$ while $-1 \neq 1$. Distinct points at distance zero: the space cannot tell -1 from 1, and no sequence could ever have a unique limit.

(b) Symmetry fails: $d(x,y) := \max\{\, y-x,\ 2(x-y)\,\}$, i.e. moving to the right costs 1 per unit and moving to the left costs 2. Here $d(0,1) = 1$ but $d(1,0) = 2$, so the return trip is more expensive than the outward one. Definiteness still holds, since $d(x,y) = 0$ forces both $y - x \leq 0$ and $x - y \leq 0$, hence $x = y$. So does the triangle inequality: writing $d(x,y) = \varphi(y-x)$ for $\varphi(t) := \max\{t, -2t\}$, the function φ is a maximum of two linear functions vanishing at 0, hence convex and positively homogeneous, and therefore subadditive. Such a d is called a "*quasi-metric*" and is genuinely useful elsewhere, though note that $B_r(x)$ would then depend on which argument the centre sits in.

(c) Triangle inequality fails: $d(x,y) := (x-y)^2$. Definiteness and symmetry are clear, but

$$d(0,2) = 4 \quad > \quad d(0,1) + d(1,2) = 1 + 1 = 2,$$

so the detour through 1 is **shorter** than going direct. This is exactly the *"detours make the way longer"* slogan running backwards.

Examples (Subsection 2.b.1)

Three axioms are few enough that a great many things qualify as metrics, and the point of the list below is exactly that breadth. Two of the examples are not what one would draw on a blackboard as *"distance"*: the discrete metric measures nothing but *"same or different"*, and the supremum metric measures the distance between two **functions**. Both are legitimate, and it is worth letting them stretch the intuition now. The second in particular is the setting in which uniform convergence turns out to be nothing more than an ordinary instance of Definition 4.a.1; see Exercise 4.a.14.

Example 2.b.11 (*Standard metrics on $\mathbb{R}^n$*):

(a) Let X be a set. The "*discrete metric*" is given by

$$d(x,y) := \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

Important for counterexamples!

(b) Let $X := \mathbb{R}^n$. Then we have many choices for a metric; some common ones are:

(i) Manhattan metric:

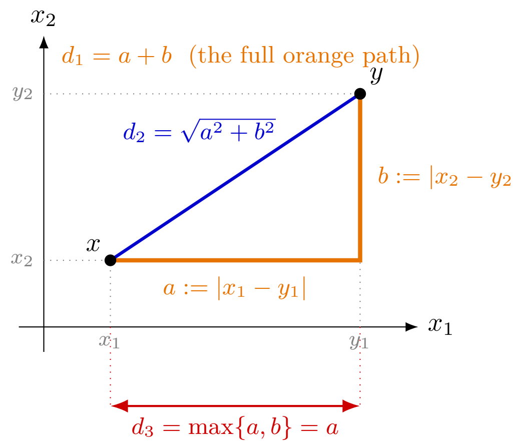
$$d_1(x,y) := |x_1 - y_1| + \cdots + |x_n - y_n| = \sum_{k=1}^{n} |x_k - y_k|$$

(ii) Standard metric:

$$d_2(x,y) := \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

(iii) Supremum metric:

$$d_3(x,y) := \sup_{k=1,\dots,n} |x_k - y_k|$$



Note that d_1 is the **sum** of the two legs and d_3 the **larger** of them. Neither is a single leg on its own. Only d_2 is the straight-line distance one would draw by reflex, and it is worth remembering that the other two are just as legitimate.

Remark 2.b.12 (*Why d_1 is the "Manhattan" metric*): The official script motivates the name explicitly. In a grid of streets running only north–south and east–west, as in much of Manhattan, a taxi cannot cut across a block diagonally. To travel from $x = (x_1, x_2)$ to $y = (y_1, y_2)$ it moves horizontally from x to (y_1, x_2) and then vertically to y (or the two legs in the other order), covering exactly $|x_1 - y_1| + |x_2 - y_2| = d_1(x, y)$ either way. This is precisely the orange path in the figure above.

In general,

$$d_m(x, y) := \sqrt[m]{\sum_{k=1}^{n} |x_k - y_k|^m}$$

is a metric on $\mathbb{R}^n$ for every $m \geq 1$, with d_1, d_2 above as the cases $m = 1, 2$.

AI-Note 2.3: Corsin writes this formula as $\sqrt[m]{\sum_k (x_k - y_k)^m}$, without the absolute values, and for any m. Two corrections have been applied above. The powers must be taken of **absolute values**, since for odd m the radicand can otherwise be negative. And the result is a metric only for $m \geq 1$: for $m \in (0, 1)$ the triangle inequality fails, as Exercise 2.c.39.**6** shows with $x = e_1$ and $y = e_2$, where $|x + y|_m = 2^{1/m} > 2 = |x|_m + |y|_m$.

Proposition 2.b.13 (*Concave reshaping preserves the triangle inequality*): Let (X, d) be a metric space and let $h : [0, \infty) \to [0, \infty)$ satisfy

(a) $h(0) = 0$ and $h(t) > 0$ for $t > 0$;

(b) h is non-decreasing;

(c) h is concave.

Then $h \circ d$ is again a metric on X.

Proof:

Definiteness is **(a)** together with definiteness of d, and symmetry is inherited from d directly. Only the triangle inequality needs an argument, and the key step is that **(a)** and **(c)** together force h to be "*subadditive*": $h(a + b) \leq h(a) + h(b)$ for all $a, b \geq 0$. Indeed, for $a, b > 0$ concavity applied to the convex combination writing a as a blend of $a + b$ and 0 gives

$$h(a) = h\Big(\tfrac{a}{a+b}(a + b) + \tfrac{b}{a+b} \cdot 0\Big) \geq \tfrac{a}{a+b} h(a + b) + \tfrac{b}{a+b} \underbrace{h(0)}_{=0} = \tfrac{a}{a+b} h(a + b),$$

and symmetrically $h(b) \geq \frac{b}{a+b} h(a+b)$. Adding the two gives $h(a) + h(b) \geq h(a+b)$. Now, for $x, y, z \in X$, monotonicity **(b)** applied to the triangle inequality for d and then subadditivity give

$$h(d(x,z)) \leq h\big(d(x,y) + d(y,z)\big) \leq h(d(x,y)) + h(d(y,z)).$$

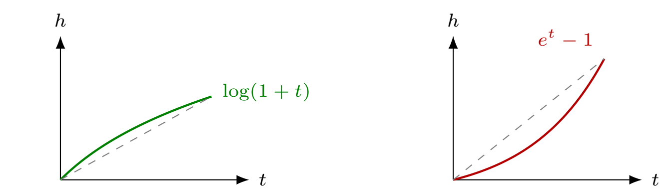
Q.E.D.

Remark 2.b.14 (*Concave is safe, convex is not*): This gives a quick reflex for the standard exercise *"is this a metric?"* when the candidate has the form $h(|x-y|)$ on $\mathbb{R}$: **sketch h and look at its curvature.** A concave h bends **towards** the axis, so distances grow ever more slowly and long jumps become relatively cheap, which is exactly the direction the triangle inequality wants. A convex h does the opposite, making one long jump more expensive than two short ones, and that is precisely what the triangle inequality forbids.

The two textbook cases sit on either side.

- $h(t) := \log(1+t)$ is increasing and concave with $h(0) = 0$, so $d(x,y) := \log(1 + |x-y|)$ **is** a metric on $\mathbb{R}$ by Proposition 2.b.13.
- $h(t) := e^t - 1$ is increasing with $h(0) = 0$ but **convex**, and $d(x,y) := e^{|x-y|} - 1$ is **not** a metric: taking $x = 0,\ y = 1,\ z = 2$,
$$d(0,2) = e^2 - 1 \approx 6.389 \quad > \quad 2(e-1) \approx 3.437 = d(0,1) + d(1,2).$$

The heuristic runs in one direction only, and is worth stating carefully. Concavity **guarantees** a metric, whereas convexity merely **suggests** failure. Convexity is not a proof of anything, and a convex h must still be refuted by an explicit triple, as above. Finally, note that the m-metrics of the preceding Example 2.b.11 split at $m = 1$ for the same reason, seen along each axis: $t \mapsto t^{1/m}$ is concave for $m \geq 1$ and convex for $m \in (0,1)$.



concave: curve **above** chord
⇒ subadditive ⇒ metric

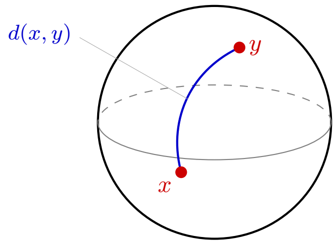
convex: curve **below** chord
⇒ triangle inequality fails

Exercise 2.b.15 (*Reshaping a metric does not change its convergent sequences*): In the setting of Proposition 2.b.13, show that a sequence $(x_n)_n$ in X converges to x with respect to d if and only if it converges to x with respect to $h \circ d$.

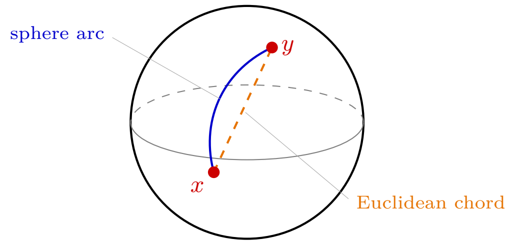
Example 2.b.16 (*continued*):

(c) Let $X := C^0([0,1], \mathbb{R})$ be the set of continuous functions $[0,1] \to \mathbb{R}$. Then we can define the "*supremum metric*" for $f, g \in X$:
$$d(f,g) := \sup_{x \in [0,1]} |f(x) - g(x)|$$
This is well defined, meaning that the supremum is a real number rather than $+\infty$, because $|f-g|$ is continuous on the compact interval $[0,1]$ and therefore bounded, by the extreme value theorem from **Analysis I** (revisited in this generality as Lemma 7.a.7 below). Compactness of the domain is doing real work here: on $C^0(\mathbb{R}, \mathbb{R})$ the same formula would produce ∞ for $f(x) := x,\ g := 0$.

(d) Let $X := S^2 := \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$. S^n is called the "*n-dimensional sphere*", or "*n-sphere*". The closest path between two points on the sphere is an **arc of a great circle** (a *geodesic*). This defines a metric, similar to how straight lines define a metric on $\mathbb{R}^n$.



Distance on sphere



Sphere arc vs. Euclidean chord

Remark 2.b.17 (*Two metrics on the sphere, made explicit*): The two pictures above compare two different metrics that S^2 inherits or acquires, and it is worth writing both down. Restricting the ambient Euclidean metric of $\mathbb{R}^3$ gives the "*chord metric*"

$$d_0(x, y) := \|x - y\|, \qquad x, y \in S^2,$$

the orange dashed segment in the right-hand panel; since $\|x\| = \|y\| = 1$, expanding $\|x - y\|^2 = \langle x - y, x - y\rangle$ gives the closed form $d_0(x, y) = \sqrt{2 - 2\langle x, y\rangle}$. The geodesic metric described above measures arc length along a great circle instead, and is the solid blue curve in both panels. It is given by

$$d_1(x, y) := \arccos\big(\langle x, y\rangle\big) \in [0, \pi].$$

Both are genuine metrics on S^2 (Example 2.b.11**(d)** already noted d_0; d_1 measures the same distance a great-circle flight would fly). They disagree as soon as $x \neq y$, since the chord always cuts inside the sphere. Therefore, $d_0(x, y) \leq d_1(x, y)$, with equality only in the limit $y \to x$. This is the sharpest instance in the chapter of a general warning. A subset inherits a metric from its ambient space, but that inherited metric need not be the one a geometer studying the subset **intrinsically** would choose.

Example 2.b.18 (*The British Rail metric*): Let $(X, \|\cdot\|)$ be a normed space and define

$$d(x, y) := \begin{cases} 0, & x = y, \\ \|x\| + \|y\|, & x \neq y. \end{cases}$$

The name records the joke that to travel between any two towns you must change trains in London: every journey is routed through the origin, and its cost is the distance in plus the distance out.

This is a metric. Definiteness holds by construction and symmetry is evident. For the triangle inequality, take $x \neq z$, so $d(x, z) = \|x\| + \|z\|$. If y equals x or z then one of the two terms on the right vanishes and the other **is** $d(x, z)$, so the inequality holds with equality; otherwise

$$d(x, y) + d(y, z) = \|x\| + 2\|y\| + \|z\| \geq \|x\| + \|z\| = d(x, z).$$

Remark 2.b.19 (*The topology it generates*): This metric is not induced by any norm, and the quickest way to see that something strange is happening is to look at small balls. For $x \neq 0$ and any radius $r \leq \|x\|$, a point $y \neq x$ has $d(y, x) = \|x\| + \|y\| \geq \|x\| \geq r$, so

$$B_r(x) = \{x\}.$$

Every non-zero point is therefore **isolated**, and $\{x\}$ is open. The origin is the sole exception: $d(y, 0) = \|y\|$ for $y \neq 0$, so $B_r(0)$ is the ordinary ball of radius r. The space is thus discrete everywhere except at one point, where it looks entirely normal. That is a useful shape to keep in mind whenever a proof seems to be using more about a metric space than the three axioms actually give.

Remark 2.b.20 (*Naming points and their components*): A practical point, which Linus reports as the single most common source of avoidable errors in submitted solutions. Once a proof involves several points, each with several components, two independent naming decisions have to be made, and mixing the two conventions is what causes the damage:

- **Subscripts for components**, different letters or primes for points: $x = (x_1, x_2)$ and $y = (y_1, y_2)$, or $x = (x_1, x_2)$ and $x' = (x'_1, x'_2)$.
- **Subscripts for points**, letters for components: (x_1, y_1) and (x_2, y_2).

Either is internally consistent, and the recommendation is the first: reserve subscripts for components, and distinguish points by x, y, z or by x, x', x''. When many points are needed at once, index them as x_k and write $x_{k,i}$ for the i-th component of the k-th point. For a single point of $\mathbb{R}^2$ or $\mathbb{R}^3$, plain (x, y) or (x, y, z) is of course fine.

The reason this matters more here than in **Analysis I** is that the product metrics of Exercise 2.b.27 and the continuity proofs that follow genuinely need both indices at once. An expression such as $d(x_1, y_1) + d(x_2, y_2)$ is therefore ambiguous until the convention is fixed, and means quite different things under the two readings.

Exercise 2.b.21 (*Supremum metric*): Show that the supremum metric from Example 2.b.11 **(c)** is a metric.

Proof of Exercise 2.b.21:

Clearly, symmetry and definiteness hold. For the triangle inequality, let $f, g, h \in C^0([0,1])$ and set $p := f - h$ and $q := h - g$. Then for all $x \in [0,1]$, we have:

$$|p(x)| \le \sup |p| \qquad \text{and} \qquad |q(x)| \le \sup |q|.$$

Adding the inequalities, we obtain for all $x \in [0,1]$:

$$\begin{aligned} |f(x) - g(x)| &= |p(x) + q(x)| \\ &\le |p(x)| + |q(x)| \\ &\le \sup |p| + \sup |q| \\ &= \sup |f - h| + \sup |h - g|. \end{aligned}$$

Q.E.D.

Exercise 2.b.22 (*L^1-metric*): Let $X := C^0([0,1], \mathbb{R})$ be the set of continuous functions $[0,1] \to \mathbb{R}$. For $f, g \in X$, define

$$d(f, g) := \int_0^1 |f(t) - g(t)|\, dt.$$

Is (X, d) a metric space? Provide a proof.

Proof of Exercise 2.b.22:

Yes, it is a metric space. We verify the three axioms of a metric:

(a) **Definiteness:** Clearly $d(f, g) \ge 0$ since the integrand $|f(t) - g(t)|$ is non-negative. If $f = g$, then $d(f, f) = \int_0^1 0\, dt = 0$. Conversely, suppose $d(f, g) = 0$. Since $|f - g|$ is a continuous, non-negative function, its integral over $[0,1]$ can only be zero if the integrand is identically zero. Thus $f(t) = g(t)$ for all $t \in [0,1]$, which means $f = g$.

(b) **Symmetry:** $d(f, g) = \int_0^1 |f(t) - g(t)|\, dt = \int_0^1 |g(t) - f(t)|\, dt = d(g, f)$.

(c) Triangle inequality: Let $f, g, h \in X$. The triangle inequality for real numbers yields $|f(t) - h(t)| \leq |f(t) - g(t)| + |g(t) - h(t)|$ for all $t \in [0,1]$. Integrating both sides, we obtain:

$$\begin{aligned} d(f,h) &= \int_0^1 |f(t) - h(t)|\, dt \\ &\leq \int_0^1 (|f(t) - g(t)| + |g(t) - h(t)|)\, dt \\ &= \int_0^1 |f(t) - g(t)|\, dt + \int_0^1 |g(t) - h(t)|\, dt \\ &= d(f,g) + d(g,h). \end{aligned}$$

Q.E.D.

AI-Note 2.4 (*Where the exercise-sheet problems live*): The official problem sheets are not reproduced as blocks anywhere in this document. Each problem is quoted verbatim (from `exercises/ExN_Analysis2_eng.pdf`; attribution: Prof. Joaquim Serra, D-MATH, ETH Zürich) and placed with the material it tests, so a problem from sheet 3 may well appear several chapters away from a problem from sheet 2. Its solution is then in the **Solutions** section of that same chapter. Corsin's priority tags (**important**, **semi-important**, **optional**) and the official **(*)** difficulty marker are carried along in each problem's title.

Notation 2.b.23: Some problems in this document have a closed-answer format, similar to what you might find on the final exam. *"Multiple Choice"* means that zero, one or more answers can be correct. Questions marked with (*) are a bit more complex, you might want to skip them at the first read.

AI-Exercise 2.b.24 (*Equivalence of Manhattan and Supremum Metrics*): Let $x := (x_1, x_2) \in \mathbb{R}^2$. Show that the Manhattan metric $d_1(x,0) := |x_1| + |x_2|$ and the supremum metric $d_3(x,0) := \max\{|x_1|, |x_2|\}$ satisfy the sharp inequalities:

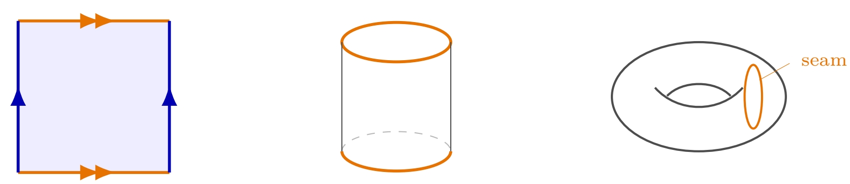
$$d_3(x,0) \leq d_1(x,0) \leq 2\, d_3(x,0).$$

Exercise 2.b.25 (*2.1 — Examples and Non-examples of Metric spaces*): Which of the following pairs are metric spaces? Prove it or provide a counterexample.

1. $(B(X), d)$, where $B(X)$ denotes the set of all bounded functions from a non-empty set X to $\mathbb{R}$ and
$$d(f,g) := \sup_{x \in X} |f(x) - g(x)|.$$
2. $(\mathbb{Q}_+, d)$, where $\mathbb{Q}_+$ are the positive rational numbers and $d(x,y) := \left|\frac{1}{x} - \frac{1}{y}\right|$.
3. $(\mathbb{R}^2, d)$, where $d(x,y) := (x_1 - y_1)^2 + |x_2 - y_2|$.
4. $(\mathbb{R}^2, d)$, where $d(x,y) := |x_1 - y_1|^{1/2} + |x_2 - y_2|$.
5. $(\mathbb{R}^{n\times n}, d)$ with $d(X,Y) := \left(\mathrm{Tr}\{(X-Y)^\mathsf{T}(X-Y)\}\right)^{1/2}$.
6. **(*)** $(\mathbb{R}^{n\times n}, d)$ with
$$d(X,Y) := \sup\{|v^\mathsf{T}(X-Y)v| : v \in \mathbb{R}^n, \|v\| = 1\},$$
and $\mathbb{R}^{n\times n}$ denoting the set of square matrices.
7. **(*)** $(\mathbb{R}^2/\mathbb{Z}^2, d)$ where the flat 2-dimensional torus $\mathbb{R}^2/\mathbb{Z}^2$ is the set of equivalence classes of pairs of real numbers under the equivalence relation
$$x, y \in \mathbb{R}^2, \quad x \sim y \iff x_1 - y_1 \in \mathbb{Z},\ x_2 - y_2 \in \mathbb{Z},$$
and $d([x],[y]) := \inf_{k,h\in\mathbb{Z}} \|x - y + (k,h)\|$, with $\|\cdot\|$ denoting the Euclidean distance and $[x] \in \mathbb{R}^2/\mathbb{Z}^2$ denoting the equivalence class of $x \in \mathbb{R}^2$.

Official hints: **2.1.3** and **2.1.4:** ignore what happens in the second variable, it is there only to distract you. **2.1.5:** rewrite, for a general matrix $A := X - Y$, what $\mathrm{Tr}\{A^\mathsf{T} A\}^{1/2}$ actually means; it should look familiar. **2.1.6:** see what happens if $(X - Y)$ is anti-symmetric. **2.1.7:** convince yourself first that the "inf" is in fact a "min".

Remark 2.b.26 (*The flat torus $\mathbb{R}^2/\mathbb{Z}^2$, folded up*): Part **2.1.7** is much easier to think about with a picture. Every equivalence class in $\mathbb{R}^2/\mathbb{Z}^2$ has a representative in the unit square $[0,1] \times [0,1]$, so the torus is that square with its opposite edges identified: $(0,t) \sim (1,t)$ glues left to right, and $(t,0) \sim (t,1)$ glues bottom to top. Performing the two gluings in turn folds the square first into a cylinder and then into a torus $\mathbb{T}^2 := \mathbb{R}^2/\mathbb{Z}^2$.



(1) the square $[0,1]^2$ opposite edges identified

(2) glue left to right: a cylinder, two orange rims

(3) glue the two rims: the torus $\mathbb{T}^2$

With the picture in hand the formula reads off directly. The distance $d([x],[y])$ is the shortest Euclidean distance between representatives, which is to say the length of the shortest path on the folded torus. The infimum over $k, h \in \mathbb{Z}$ in the problem statement is just the search over which copy of y in the tiling of the plane by unit squares lies nearest to x, and since only finitely many copies can be candidates, that infimum is attained. This is what the official hint means when it says to convince yourself the infimum is really a minimum.

Exercise 2.b.27 (*2.5 — Product of metric spaces*): Let (X, d_X) and (Y, d_Y) be a pair of metric spaces. Recall that the set of ordered pairs (x,y) with $x \in X$ and $y \in Y$ is denoted by $X \times Y$. Consider the following functions $X \times Y \to [0,\infty)$:

$$\begin{aligned} d_1((x,y),(x',y')) &:= \max\{d_X(x,x'), d_Y(y,y')\} \\ d_2((x,y),(x',y')) &:= d_X(x,x') + d_Y(y,y') \\ d_3((x,y),(x',y')) &:= \sqrt{d_X(x,x')^2 + d_Y(y,y')^2}. \end{aligned}$$

1. Show that they are all valid distance functions on $X \times Y$.
2. Show that they are all equivalent, i.e. there is a number $C > 0$ such that
$$d_1((x,y),(x',y')) \le C\, d_2((x,y),(x',y')) \le C^2 d_3((x,y),(x',y')) \le C^3 d_1((x,y),(x',y'))$$
for all $x, x' \in X$, $y, y' \in Y$.
3. Show that a sequence $(x_n, y_n) \to (x,y)$ with respect to $(X \times Y, d_3)$ if and only if $x_n \to x$ with respect to d_X and $y_n \to y$ with respect to d_Y.

Official hint: don't get distracted by the abstract set-up, since you already know all these things for $\mathbb{R} \times \mathbb{R}$; start from there, then rewrite those arguments in this general framework.

Cauchy–Schwarz (Section 2.c)

Recap: inner products and norms (Subsection 2.c.1)

Corsin opens his Friday session by recalling the two definitions, which this chapter has already stated in Section 2.a, where the hierarchy of structured spaces was drawn and where the first supplements needed them. Rather than repeat them, here is what to have in mind:

- An **inner product** (Definition 2.a.1) is **bilinear**, **symmetric** and **definite**. It measures length **and** angle.
- A **norm** (Definition 2.a.2) is **definite**, **homogeneous** and satisfies the **triangle inequality**. It measures length only.
- Each induces the next, via $\|v\| := \sqrt{\langle v, v\rangle}$ and $d(x,y) := \|x-y\|$, and neither implication reverses (Proposition 2.a.6, Remark 2.a.7).

What this section adds is the **geometry**: why $\langle x, y\rangle$ deserves to be read as an angle at all, and the resulting second proof of Cauchy–Schwarz.

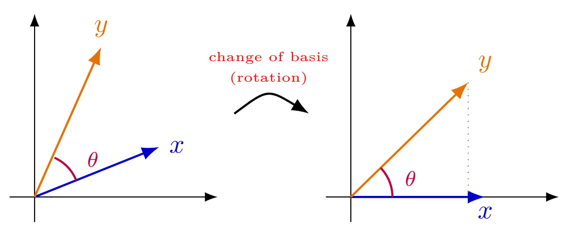
Geometric meaning of the inner product (Subsection 2.c.2)

In $\mathbb{R}^2$, the following identity holds:

$$\langle x, y\rangle := x_1y_1 + x_2y_2 = \|x\|\|y\|\cos\theta$$

where θ is the angle between $x, y \in \mathbb{R}^2$.

To see this, conveniently choose our basis vectors such that $x = (x_1, 0)$, $y = (y_1, y_2)$.



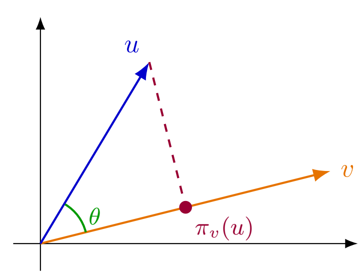
Then

$$\langle x, y\rangle = x_1y_1 = x_1\|y\|\cos\theta = \|x\|\|y\|\cos\theta.$$

For more rigour, you will show in Linalg that the rotations in $\mathbb{R}^2$ are given by matrices R satisfying $\langle Rx, Ry\rangle = \langle x, y\rangle$.

In a general inner product space V, this becomes the **definition** of the angle θ. It then makes sense to define the "*projection onto v*", π_v:

$$\pi_v(u) := \frac{\langle u, v\rangle}{\|v\|^2}\, v = \|u\|\cos\theta\, \frac{v}{\|v\|}.$$



In words, $\pi_v(u)$ is the part of u which points along v. In particular,

$$\|\pi_v(u)\| = \|u\|\,|\cos\theta| = \frac{|\langle u, v\rangle|}{\|v\|}.$$

In this sense, Cauchy–Schwarz says something visually obvious. Reading the picture above, all three of the following are the same statement:

$$\frac{|\langle u, v\rangle|}{\|v\|} \le \|u\| \iff \|\pi_v(u)\| \le \|u\| \iff \|u\|\,|\cos\theta| \le \|u\|.$$

Remark 2.c.28 (*Hint for the proof of Cauchy–Schwarz*): Use definiteness and expand the term

$$\langle u - \pi_v(u),\ u - \pi_v(u)\rangle = \dots$$

where $u - \pi_v(u)$ is the **perpendicular part** of u to v.

The Cauchy–Schwarz inequality (Subsection 2.c.3)

Theorem 2.c.29 (*Cauchy–Schwarz inequality*): Let V be an inner product space (Definition 2.a.1) with induced norm $\|\cdot\| := \sqrt{\langle\cdot,\cdot\rangle}$ (Definition 2.a.2). Then for all $u, v \in V$,

$$|\langle u, v\rangle| \le \|u\|\,\|v\|,$$

with equality if and only if u, v are linearly dependent.

Proof:

If $v = 0$, both sides vanish and the claim holds trivially, so assume $v \neq 0$. Following Remark 2.c.28, definiteness (Definition 2.a.1) gives

$$0 \ \le\ \langle u - \pi_v(u),\ u - \pi_v(u)\rangle \ =\ \langle u, u\rangle - 2\langle u, \pi_v(u)\rangle + \langle \pi_v(u), \pi_v(u)\rangle.$$

Using $\pi_v(u) = \frac{\langle u,v\rangle}{\|v\|^2} v$ and bilinearity, the two right-hand terms are

$$\langle u, \pi_v(u)\rangle = \frac{\langle u, v\rangle^2}{\|v\|^2},$$

$$\langle \pi_v(u), \pi_v(u)\rangle = \frac{\langle u, v\rangle^2}{\|v\|^4}\langle v, v\rangle = \frac{\langle u, v\rangle^2}{\|v\|^2}.$$

Note that the two are equal, so substituting them leaves a single copy behind:

$$0 \ \le\ \|u\|^2 - \frac{2\langle u, v\rangle^2}{\|v\|^2} + \frac{\langle u, v\rangle^2}{\|v\|^2} \ =\ \|u\|^2 - \frac{\langle u, v\rangle^2}{\|v\|^2}.$$

This rearranges to $\langle u, v\rangle^2 \le \|u\|^2\|v\|^2$, that is, $|\langle u, v\rangle| \le \|u\|\,\|v\|$. By definiteness, equality holds if and only if $u - \pi_v(u) = 0$, which says precisely that u is a scalar multiple of v. **Q.E.D.**

Remark 2.c.30 (*The same proof twice*): This is the second proof of Cauchy–Schwarz in these notes; Proposition 2.a.5 gave another, following Diego Torres Tejeda. The two look different at first. One minimises the quadratic $f(\lambda) := \|v - \lambda w\|^2$ over λ, while the other expands $\|u - \pi_v(u)\|^2 \ge 0$. But the minimiser found in the first is $\lambda_0 = \frac{\langle v,w\rangle}{\|w\|^2}$, which is exactly the coefficient defining $\pi_w(v)$. Consequently, the first proof **derives** the projection by optimisation, and the second **postulates** it and checks that it works. Keeping both is worthwhile precisely because the pair shows where the projection comes from. It is not a lucky guess, but the closest point of the line to v.

Exercise 2.c.31 (*3.7 — Cauchy–Schwarz*): Let V be a vector space over $\mathbb{R}$, let $\langle\cdot,\cdot\rangle$ be an inner product on V, and let $\|\cdot\| : V \to \mathbb{R}$ be given by $\|v\| = \sqrt{\langle v, v\rangle}$. Then the inequality

$$|\langle v, w\rangle| \le \|v\|\|w\| \tag{1}$$

holds for all $v, w \in V$. Furthermore, equality in (1) holds if and only if v and w are linearly dependent.

Remark 2.c.32: Corsin devotes the whole second half of Friday (Section 2.c) to this problem, including the geometric reading and an explicit proof hint.

Exercise 2.c.33 (*3.8 — Inner product and norm*): Let V be a vector space over $\mathbb{R}$, let $\langle -, -\rangle$ be an inner product on V. The map defined by

$$\|\cdot\| : V \to \mathbb{R}, \qquad \|v\| = \sqrt{\langle v, v\rangle}$$

satisfies the triangular inequality and is a norm.

Exercise 2.c.34 (*3.9 — All norms are equivalent in $\mathbb{R}^n$*): Let $|\cdot|$ denote the standard Euclidean norm in $\mathbb{R}^n$ and let $f : \mathbb{R}^n \to [0, \infty)$ be another norm (that is, a function satisfying the properties of *Definition 9.89*).

1. Expressing x in a basis and using the "abstract" properties that f must have, show that there is a constant $C_1 > 0$ such that $f(x) \le C_1|x|$ for all $x \in \mathbb{R}^n$.
2. Show that f is continuous in $\mathbb{R}^n$ (with respect to the standard distance of $\mathbb{R}^n$!).
3. Show that there is a number $c_2 > 0$ such that $f(x) \ge c_2$ for all $|x| = 1$.
4. Conclude that $f(x) \ge c_2|x|$ for all $x \in \mathbb{R}^n$.
5. Show that if $\tilde{f}$ is yet another norm, then there is $C > 0$ such that $C^{-1}f(x) \le \tilde{f}(x) \le Cf(x)$ for all $x \in \mathbb{R}^n$.

Official hints: **3.9.2:** recall that from the definition it follows that $|f(x) - f(y)| \le f(x - y)$, and that Lipschitz functions are always continuous. **3.9.3:** apply the Weierstrass theorem to f on $S := \{x \in \mathbb{R}^n : |x| = 1\}$ and use that f, being a norm, is nondegenerate. **3.9.5:** if f is equivalent to $|\cdot|$ and $\tilde{f}$ is equivalent to $|\cdot|$, it follows by transitivity that f is equivalent to $\tilde{f}$.

Remark 2.c.35: Hint 3.9.3 is exactly Corsin's reason #2 for caring about compactness (Section 7.c), namely the Weierstrass extreme value theorem applied on the compact unit sphere.

Exercise 2.c.36 (*The polynomial ring $\mathbb{R}[x]$ is infinite-dimensional*): Show that the vector space $\mathbb{R}[x]$ of polynomials with real coefficients is infinite dimensional, and exhibit a basis.

Remark 2.c.37: This is exactly the hypothesis Exercise 2.c.34 needs and cannot do without. Remark 2.d.42 already gave $C^0([0,1], \mathbb{R})$ as an infinite-dimensional space where norm equivalence fails. Note that $\mathbb{R}[x]$ is the algebraically simplest such example, with an explicit basis and no analysis required at all.

Exercise 2.c.38 (*3.10 — Hilbert–Schmidt norm of the composition*): Take two linear functions $\varphi : \mathbb{R}^d \to \mathbb{R}^n$ and $\psi : \mathbb{R}^n \to \mathbb{R}^m$, and denote with Φ, Ψ the matrices that represent them in the canonical bases. Recall that the linear map $\psi \circ \varphi : \mathbb{R}^d \to \mathbb{R}^m$ is represented in these bases by the matrix $\Psi \cdot \Phi$. Show that

$$\|\Psi \cdot \Phi\| \le \|\Psi\|\|\Phi\|,$$

where $\|\cdot\|$ is the Hilbert–Schmidt norm of a matrix (see *Definition 9.96* in the notes).

Official hint: recall that $\|M\|^2$ is the sum of the squares of the entries of M. Notice that $(\Psi \cdot \Phi)^i_j$ (i-th row, j-th column) is the scalar product of Ψ^i and Φ_j, which are vectors of $\mathbb{R}^n$. Apply Cauchy–Schwarz to each of them.

Exercise 2.c.39 (*4.3 — p-norms* (semi-important; parts 4 and 6 important)): For $p \ge 1$ and $x \in \mathbb{R}^n$ define the **p-norm** of x as

$$|x|_p := \Big(\sum_{i=1}^n |x_i|^p\Big)^{1/p}.$$

1. For $n = 2$ and $p = 1, 2, 10$ sketch the sets $\{x \in \mathbb{R}^2 : |x|_p \le 1\}$.
2. For a given $x \in \mathbb{R}^n$, compute the limit $|x|_\infty := \lim_{p\to\infty} |x|_p$.
3. Using an appropriate inequality that you have seen in class, prove that
$$\Big(\sum_{i=1}^n a_i^{p-1}b_i\Big)^2 \le \Big(\sum_{i=1}^n a_i^p\Big)\Big(\sum_{i=1}^n a_i^{p-2}b_i^2\Big),$$
whenever a_i, b_i are n-tuples of positive numbers.
4. Fix $x, y \in \mathbb{R}^n$ and consider the function $f : [0,1] \to \mathbb{R}$ defined as
$$f(t) := |tx + (1-t)y|_p = \Big(\sum_{i=1}^n |tx_i + (1-t)y_i|^p\Big)^{1/p}. \tag{1}$$

Show that f is convex. You may assume that the coordinates of x and y are all strictly positive and use the inequality of the previous point.

5. Deduce from the previous point that the triangular inequality holds, i.e. $|x+y|_p \le |x|_p + |y|_p$ for all $x, y \in \mathbb{R}^n$.
6. What happens for $p \in (0,1)$?

Official hints: **4.3.3:** use Cauchy–Schwarz and the fact that $a_i^{p-1} b_i = a_i^{p/2} \cdot a_i^{(p-2)/2} b_i$. **4.3.4:** show that $f''(t) \ge 0$; it is not the simplest derivative, but if you get it right the inequality in 4.3.3 will be just what you need. **4.3.5:** write the convexity inequality between x, y and the middle point $(x+y)/2$. To get the general case from the one with positive coordinates, observe that for $x \in \mathbb{R}^n$, denoting $\hat{x} := (|x_1|, \dots, |x_n|)$, we have $|x+y|_p \le |\hat{x}+\hat{y}|_p$ and $|\hat{x}|_p = |x|_p$ for all $x, y \in \mathbb{R}^n$.

Exercise 2.c.40 (*4.4 — p-means* (optional)): For $x \in \mathbb{R}^n$ with positive coordinates and $p \neq 0$ define the **p-mean** as

$$\mu_p(x) := \left(\frac{x_1^p + \cdots + x_n^p}{n} \right)^{1/p}.$$

1. Compute the limits $p \to \pm\infty$, $p \to 0$ and define accordingly $\mu_{-\infty}(x)$, $\mu_0(x)$, $\mu_{+\infty}(x)$.
2. For any n-tuple of numbers $a_i > 0$ show that
$$\sum_{i=1}^{n} \frac{a_i \log(a_i)}{a_1 + \cdots + a_n} \ge \log\left(\frac{a_1 \cdots + a_n}{n} \right).$$
3. For a fixed x, show that the function $f : \mathbb{R} \to \mathbb{R}$, given by $f(t) := \mu_t(x)$, is continuous and increasing.
4. Prove the Arithmetic–Geometric inequality and Arithmetic–Quadratic inequality:
$$n(x_1 x_2 \cdots x_n)^{1/n} \le x_1 + \cdots + x_n, \qquad (x_1 + \cdots + x_n)^2 \le n(x_1^2 + \cdots + x_n^2).$$
5. **(*)** Is f continuously differentiable in the whole $\mathbb{R}$?

Official hints: **4.4.2:** combine the concavity of $\log(\cdot)$ and the Cauchy–Schwarz inequality; use the generalisation of the concavity inequality: for any concave $f : \mathbb{R} \to \mathbb{R}$, $f(\lambda_1 x_1 + \cdots + \lambda_n x_n) \ge \lambda_1 f(x_1) + \cdots + \lambda_n f(x_n)$ for all $x_i \in \mathbb{R}$, $0 \le \lambda_i \le 1$ with $\lambda_1 + \cdots + \lambda_n = 1$. **4.4.3:** it might be more convenient to work with $\log f(t)$.

Solutions (Section 2.d)

Proof of Exercise 2.b.9:

For $N = 3$ this is the triangle inequality itself. For the inductive step, assume the claim holds for some $N \ge 3$ and let $x_1, \dots, x_{N+1} \in X$. Applying the triangle inequality to the triple (x_1, x_N, x_{N+1}) and then the inductive hypothesis to $x_1, \dots, x_N$,

$$d(x_1, x_{N+1}) \;\le\; d(x_1, x_N) + d(x_N, x_{N+1}) \;\le\; \sum_{i=1}^{N-1} d(x_i, x_{i+1}) + d(x_N, x_{N+1}) \;=\; \sum_{i=1}^{N} d(x_i, x_{i+1}),$$

which is the claim for $N+1$. **Q.E.D.**

Proof of Exercise 2.b.15:

Suppose $x_n \to x$ with respect to d, so $d(x_n, x) \to 0$. Concave functions on $[0, \infty)$ are continuous, so in particular h is continuous at 0; since $h(0) = 0$, this gives $h(d(x_n, x)) \to 0$, i.e. $x_n \to x$ with respect to $h \circ d$.

Conversely, suppose $x_n \to x$ with respect to $h \circ d$. Fix $\varepsilon > 0$; by hypothesis **(a)** of Proposition 2.b.13, $\delta := h(\varepsilon) > 0$. Choose N such that $h(d(x_n, x)) < \delta$ for all $n \ge N$. If $d(x_n, x) \ge \varepsilon$

held for some such n, monotonicity **(b)** would give $h(d(x_n, x)) \geq h(\varepsilon) = \delta$, a contradiction. So $d(x_n, x) < \varepsilon$ for all $n \geq N$, i.e. $x_n \to x$ with respect to d. **Q.E.D.**

Proof of AI-Exercise 2.b.24:

For any $x = (x_1, x_2) \in \mathbb{R}^2$, we have $\max\{|x_1|, |x_2|\} \leq |x_1| + |x_2|$, giving $d_3(x, 0) \leq d_1(x, 0)$. Conversely, since $|x_1| \leq \max\{|x_1|, |x_2|\}$ and $|x_2| \leq \max\{|x_1|, |x_2|\}$, summing these yields $|x_1| + |x_2| \leq 2\max\{|x_1|, |x_2|\}$, i.e., $d_1(x, 0) \leq 2\, d_3(x, 0)$. Both bounds are sharp: equality on the left holds for $x = (1, 0)$, and equality on the right holds for $x = (1, 1)$. **Q.E.D.**

Solution to Exercise 2.b.25:

Throughout, non-negativity and symmetry are immediate in every part and are not belaboured; the interest is in **definiteness** and the **triangle inequality**.

1. **Yes.** Boundedness of f and g makes the supremum finite, so d really maps into $\mathbb{R}$. Definiteness: $d(f, g) = 0$ forces $|f(x) - g(x)| = 0$ for every x, i.e. $f = g$. For the triangle inequality, estimate pointwise and only then take the supremum:
$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)| \leq d(f, g) + d(g, h) \quad \text{for all } x,$$
so the right-hand side is an upper bound for $|f - h|$ and hence dominates $d(f, h)$. This is Exercise 2.b.21 verbatim, with $[0, 1]$ replaced by an arbitrary set X and continuity replaced by boundedness. Note that it was boundedness, and not continuity, that the argument there actually used.
2. **Yes.** Let $\varphi(x) := 1/x$, an injective map $\mathbb{Q}_+ \to \mathbb{Q}_+$, and observe that $d(x, y) = |\varphi(x) - \varphi(y)|$. Symmetry and the triangle inequality are inherited from $|\cdot|$ without any work, and definiteness is exactly injectivity of φ: from $d(x, y) = 0$ we get $1/x = 1/y$, hence $x = y$. This is the same pullback construction as Exercise 6.a.10, and the same warning applies. Pulling a metric back along an injection **always** produces a metric, which is why the interesting question about such a d is never whether it is one, but whether it is complete.
3. **No:** the triangle inequality fails. Ignore the second coordinate, as the official hint suggests, and take three collinear points $x := (0, 0)$, $y := (1, 0)$ and $z := (2, 0)$:
$$d(x, z) = 2^2 = 4 \quad > \quad d(x, y) + d(y, z) = 1 + 1 = 2.$$
Squaring produces the same failure as in AI-Example 2.b.10**(c)**: a function growing faster than linearly makes one long step more expensive than two short ones.
4. **Yes.** A sum of two metrics acting on separate coordinates is a metric, so everything reduces to the first coordinate, where the claim is that $(s, t) \mapsto |s - t|^{1/2}$ satisfies the triangle inequality on $\mathbb{R}$. That follows from **subadditivity of the square root**, $\sqrt{a + b} \leq \sqrt{a} + \sqrt{b}$ for $a, b \geq 0$, which holds because squaring the right-hand side gives $a + b + 2\sqrt{ab} \geq a + b$. Hence
$$|x_1 - z_1|^{1/2} \leq \left(|x_1 - y_1| + |y_1 - z_1|\right)^{1/2} \leq |x_1 - y_1|^{1/2} + |y_1 - z_1|^{1/2}.$$
Contrast this with part 3: $\sqrt{\cdot}$ is concave and vanishes at 0, hence subadditive, while $(\cdot)^2$ is convex, hence super-additive. That single distinction settles both parts.
5. **Yes.** Write $A := X - Y$. Then
$$\operatorname{Tr}(A^\mathsf{T} A) = \sum_{i=1}^{n} (A^\mathsf{T} A)_{ii} = \sum_{i,j=1}^{n} A_{ji}^2,$$
so $d(X, Y)$ is nothing but the Euclidean distance between X and Y regarded as vectors of $\mathbb{R}^{n^2}$, and it is a metric by Proposition 2.a.6**(b)**. The associated norm is the "*Frobenius norm*" (German: „*Frobeniusnorm*“), also called the Hilbert–Schmidt norm.
6. **No**, and here it is **definiteness** that fails, not the triangle inequality. Following the hint, suppose $A := X - Y$ is **anti-symmetric**, $A^\mathsf{T} = -A$. For any v the number $v^\mathsf{T} A v$ is a 1×1

matrix, hence equal to its own transpose:

$$v^\mathsf{T} A v = (v^\mathsf{T} A v)^\mathsf{T} = v^\mathsf{T} A^\mathsf{T} v = -\,v^\mathsf{T} A v,$$

so $v^\mathsf{T} A v = 0$ for **every** v. Taking $n = 2$ with

$$X := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Y := \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

gives $X \neq Y$ but $d(X, Y) = 0$. Every other axiom does hold, so d is a "*pseudo-metric*" (German: „*Pseudometrik*“): it cannot separate two matrices differing by an anti-symmetric one. Equivalently, it restricts to a genuine metric on the symmetric matrices.

7. **Yes.** Three things need checking, and the first is the one that is easy to skip.

   Well defined. Replacing x by $x + (m, n)$ and y by $y + (m', n')$ shifts $x - y$ by the integer vector $(m - m',\, n - n')$, which merely re-indexes the infimum. So $d([x], [y])$ does not depend on the representatives chosen.

   The infimum is a minimum. Only finitely many integer vectors (k, h) satisfy $\|x - y + (k, h)\| \le \|x - y\|$, because $\mathbb{Z}^2$ is discrete; the infimum is therefore taken over a finite non-empty set of candidates and is attained. This is what the official hint asks you to settle first, and it is what makes the next step legitimate.

   Definiteness. If $d([x], [y]) = 0$ then, the minimum being attained, $x - y + (k, h) = 0$ for some integers k, h, that is $x \sim y$ and $[x] = [y]$. Note exactly where *"the infimum is a minimum"* was needed: an infimum equal to 0 but not attained would say nothing at all.

   Triangle inequality. Choose minimisers (k_1, h_1) for the pair (x, y) and (k_2, h_2) for (y, z). Their sum is a legitimate candidate for the pair (x, z), so

$$\begin{aligned} d([x], [z]) &\le \big\| x - z + (k_1{+}k_2,\ h_1{+}h_2) \big\| \\ &= \big\| \big(x - y + (k_1, h_1)\big) + \big(y - z + (k_2, h_2)\big) \big\| \\ &\le d([x], [y]) + d([y], [z]). \end{aligned}$$

Q.E.D.

AI-Note 2.5: Parts 3, 4 and 6 are the ones worth revisiting, because each breaks (or rescues) a different axiom and together they cover every way the definition can fail. Part 3 kills the triangle inequality by growing too fast; part 4 rescues it by growing slowly enough; part 6 satisfies the triangle inequality perfectly and fails on definiteness instead. Read alongside AI-Example 2.b.10, they make the case that the three axioms are genuinely independent.

Solution to Exercise 2.b.27:

Write $s := d_X(x, x')$ and $t := d_Y(y, y')$ throughout, so that the three candidates are $d_1 = \max\{s, t\}$, $d_2 = s + t$ and $d_3 = \sqrt{s^2 + t^2}$. These are precisely the supremum, Manhattan and standard metrics of Example 2.b.11, applied to the pair (s, t). Almost everything below is that one observation.

1. **All three are metrics.** Each vanishes exactly when $s = t = 0$, i.e. when $x = x'$ and $y = y'$, which is definiteness; symmetry is inherited from d_X and d_Y. For the triangle inequality, set

$$a := \big(d_X(x, x'),\, d_Y(y, y')\big), \quad b := \big(d_X(x', x''),\, d_Y(y', y'')\big), \quad c := \big(d_X(x, x''),\, d_Y(y, y'')\big),$$

   all lying in $[0, \infty)^2$. The triangle inequalities in X and in Y say precisely that $c \le a + b$ **componentwise**. Each of $\max$, the sum and $\|\cdot\|_2$ is monotone on $[0, \infty)^2$ and subadditive, so

$$N(c) \ \le\ N(a + b) \ \le\ N(a) + N(b)$$

   for $N \in \{\max,\ \sum,\ \|\cdot\|_2\}$, which is the required inequality in all three cases at once.

2. **Equivalence.** For all $s, t \ge 0$,

$$\max\{s, t\} \ \le\ \sqrt{s^2 + t^2} \ \le\ s + t \ \le\ 2\max\{s, t\},$$

i.e. $d_1 \le d_3 \le d_2 \le 2d_1$; the middle inequality holds because $(s+t)^2 = s^2+t^2+2st \ge s^2+t^2$. Hence $C := 2$ satisfies the chain in the statement: $d_1 \le 2d_2$, then $2d_2 \le 4d_3$ (since $d_2 \le \sqrt{2}\, d_3$ by Cauchy–Schwarz, and $\sqrt{2} \le 2$), and finally $4d_3 \le 8d_1$ (since $d_3 \le \sqrt{2}\, d_1$).

3. **Convergence is coordinatewise.** From part 2,

$$\max\big\{d_X(x_n,x),\ d_Y(y_n,y)\big\} \ \le\ d_3\big((x_n,y_n),(x,y)\big) \ \le\ d_X(x_n,x)+d_Y(y_n,y).$$

If the middle term tends to 0, the left inequality forces both coordinates to converge; if both coordinates converge, the right inequality forces the middle term to 0. This is Proposition 6.a.5**(a)** with $n=2$ and the two factors allowed to be arbitrary metric spaces. By part 2 the same conclusion holds for d_1 and d_2, since equivalent metrics have the same convergent sequences.

Q.E.D.

Proof of Exercise 2.c.31:

This is Theorem 2.c.29, proved in Section 2.c above by expanding $\|u-\pi_v(u)\|^2 \ge 0$, including the equality case. A second, independent proof, by minimising $\lambda \mapsto \|v-\lambda w\|^2$, is Proposition 2.a.5. Remark 2.c.30 explains why the two are the same computation approached from opposite ends.

Remark 2.d.41 (*A third route: the discriminant*): The official solution gives a third proof, worth recording because it is the one that generalises most mechanically. By definiteness, the quadratic polynomial

$$\lambda \ \longmapsto\ \langle \lambda v + w,\ \lambda v + w\rangle = \lambda^2\langle v,v\rangle + 2\lambda\langle v,w\rangle + \langle w,w\rangle$$

is non-negative for every real λ. A real quadratic with positive leading coefficient is non-negative exactly when its discriminant is ≤ 0, i.e.

$$\big(2\langle v,w\rangle\big)^2 - 4\langle v,v\rangle\langle w,w\rangle \le 0,$$

which rearranges to the inequality. Equality forces the discriminant to vanish, so the polynomial has a real root λ_0, and definiteness turns $\langle \lambda_0 v + w, \lambda_0 v + w\rangle = 0$ into $\lambda_0 v + w = 0$, which is the asserted linear dependence. Note that this is the previous proof with the geometry stripped out. Here λ_0 is again the projection coefficient, now arriving as a root rather than as a minimiser.

Q.E.D.

Proof of Exercise 2.c.33:

This is Proposition 2.a.6**(a)**, proved immediately after Proposition 2.a.5, which is exactly the input the triangle inequality needs and the reason the two problems appear adjacently on this sheet. In one line: expanding $\|u+v\|^2 = \|u\|^2 + 2\langle u,v\rangle + \|v\|^2$ and bounding the middle term by $2\|u\|\,\|v\|$ turns the right-hand side into $(\|u\|+\|v\|)^2$. **Q.E.D.**

Solution to Exercise 2.c.34:

Write $e_1,\dots,e_n$ for the standard basis and $|\cdot|$ for the Euclidean norm.

1. Expand $x = \sum_{i=1}^n x_i e_i$ and use the triangle inequality and homogeneity of f (Definition 2.a.2):

$$f(x) \ \le\ \sum_{i=1}^n f(x_i e_i) \ =\ \sum_{i=1}^n |x_i|\, f(e_i) \ \le\ \Big(\max_i f(e_i)\Big) \sum_{i=1}^n |x_i| \ \le\ \Big(\max_i f(e_i)\Big)\sqrt{n}\, |x|,$$

the last step by Cauchy–Schwarz applied to $(|x_1|,\dots,|x_n|)$ and $(1,\dots,1)$. So $C_1 := \sqrt{n}\max_i f(e_i)$ works, and $C_1 > 0$ because $e_i \ne 0$ forces $f(e_i) > 0$ by definiteness.

2. The reverse triangle inequality for f, namely $f(x) \le f(y) + f(x-y)$ together with its mirror image, gives $|f(x) - f(y)| \le f(x-y)$. Combined with part 1 this yields
$$|f(x) - f(y)| \;\le\; f(x-y) \;\le\; C_1|x-y|.$$
So f is C_1-Lipschitz with respect to the **Euclidean** distance (Definition 5.b.20), hence continuous (Remark 5.b.21). Note which distance is which here: the claim is continuity of f as a function on the metric space $(\mathbb{R}^n, |\cdot|)$, and part 1 is what makes that meaningful.
3. The unit sphere $S := \{x : |x| = 1\}$ is closed (preimage of $\{1\}$ under the continuous $|\cdot|$) and bounded, hence **compact** by Theorem 7.b.23. By part 2 the function f is continuous on S, so it attains a minimum there (Item **(b)**), say $c_2 := f(x_0)$ with $|x_0| = 1$. Since $x_0 \neq 0$, definiteness gives $c_2 = f(x_0) > 0$.
4. For $x \neq 0$ the vector $x/|x|$ lies in S, so $f(x/|x|) \ge c_2$, and homogeneity gives
$$f(x) = |x|\, f\left(\frac{x}{|x|}\right) \;\ge\; c_2\,|x|.$$
For $x = 0$ both sides vanish. Together with part 1, $c_2|x| \le f(x) \le C_1|x|$.
5. Apply parts 1 and 4 to both norms: $c_2|x| \le f(x) \le C_1|x|$ and $\tilde{c}_2|x| \le \tilde{f}(x) \le \tilde{C}_1|x|$, all four constants positive. Then
$$\tilde{f}(x) \;\le\; \tilde{C}_1|x| \;\le\; \frac{\tilde{C}_1}{c_2}\, f(x), \qquad \tilde{f}(x) \;\ge\; \tilde{c}_2|x| \;\ge\; \frac{\tilde{c}_2}{C_1}\, f(x),$$
so $C := \max\{\tilde{C}_1/c_2,\ C_1/\tilde{c}_2\}$ satisfies $C^{-1}f \le \tilde{f} \le Cf$. Equivalence of norms is thus an equivalence relation, and the Euclidean norm serves as the common reference point.

Remark 2.d.42 (*Where finite dimension is indispensable*): The result is false in infinite dimensions, and the proof shows exactly which step breaks: part 1 expands x in a **finite** basis and part 3 uses compactness of the unit sphere. In an infinite-dimensional normed space the closed unit ball is never compact, and one can write down inequivalent norms. Take for instance $\|\cdot\|_\infty$ and $\int_0^1|\cdot|$ on $C^0([0,1])$, where a tall thin spike has supremum norm 1 and integral norm arbitrarily small. Every appearance of *"all norms on $\mathbb{R}^n$ are equivalent"* later in the course silently uses $n < \infty$.

Q.E.D.

Proof of Exercise 2.c.36:

The monomials $\mathcal{B} := \{1, x, x^2, x^3, \dots\}$ form a basis: every polynomial is by definition a **finite** linear combination $\sum_{k=0}^n a_k x^k$ of them, so $\mathcal{B}$ spans $\mathbb{R}[x]$; and $\sum_{k=0}^n a_k x^k = 0$ as a polynomial forces every coefficient $a_k = 0$ (a nonzero polynomial of degree n has at most n roots, so it cannot vanish identically on all of $\mathbb{R}$), so $\mathcal{B}$ is linearly independent.

To see $\mathbb{R}[x]$ is infinite-dimensional, suppose for contradiction it had finite dimension d. Then $1, x, \dots, x^d$, a set of $d+1$ vectors of $\mathbb{R}[x]$, would have to be linearly dependent, since any $d+1$ vectors in a d-dimensional space are. But they are linearly independent by the argument above (restricted to $n \le d$): a contradiction. So no finite basis exists, and the (necessarily infinite) $\mathcal{B}$ above is one. **Q.E.D.**

Proof of Exercise 2.c.38:

Write $\Psi^i \in \mathbb{R}^n$ for the i-th **row** of Ψ and $\Phi_j \in \mathbb{R}^n$ for the j-th **column** of Φ, so that by the definition of matrix multiplication
$$(\Psi\Phi)_{ij} = \langle \Psi^i,\ \Phi_j \rangle.$$

Cauchy–Schwarz (Theorem 2.c.29) applied to this pair of vectors of $\mathbb{R}^n$ gives $\left|(\Psi\Phi)_{ij}\right|^2 \le \|\Psi^i\|^2\,\|\Phi_j\|^2$. Summing over all entries and factorising the resulting double sum,

$$\|\Psi\Phi\|^2 = \sum_{i=1}^{m}\sum_{j=1}^{d}\left|(\Psi\Phi)_{ij}\right|^2 \le \sum_{i=1}^{m}\sum_{j=1}^{d}\|\Psi^i\|^2\|\Phi_j\|^2 = \left(\sum_{i=1}^{m}\|\Psi^i\|^2\right)\left(\sum_{j=1}^{d}\|\Phi_j\|^2\right) = \|\Psi\|^2\,\|\Phi\|^2,$$

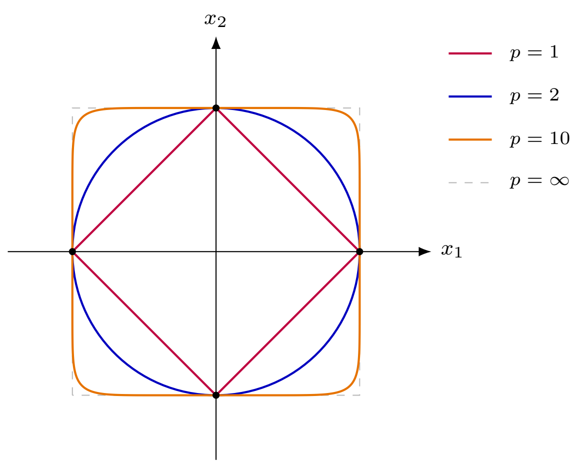
using that the Hilbert–Schmidt norm squared is the sum of the squares of all entries, read row by row for Ψ and column by column for Φ. Taking square roots gives $\|\Psi\Phi\| \le \|\Psi\|\,\|\Phi\|$.

Remark 2.d.43 (*Submultiplicativity, and what it later buys*): The Hilbert–Schmidt norm is the same object as the Frobenius norm met in Exercise 2.b.25 part 5 on the previous sheet, where it was shown to be the Euclidean norm of the matrix read as a vector of $\mathbb{R}^{n^2}$. The inequality just proved says that this norm is "*submultiplicative*" (German: „*submultiplikativ*"), which is exactly the property that makes it useful for estimating compositions of linear maps. It is also what will later let $\|Df\|$ be handled like an absolute value in the mean value inequality (Important remark 11.a.2).

Q.E.D.

Solution to Exercise 2.c.39:

1. For $n = 2$, $\{x : |x|_1 \le 1\}$ is the diamond with vertices $(\pm1,0)$, $(0,\pm1)$; $\{x : |x|_2 \le 1\}$ is the Euclidean unit disc; $\{x : |x|_{10} \le 1\}$ is a curve close to the square $[-1,1]^2$, with rounded corners that flatten as p grows. All three contain the four points $(\pm1,0)$, $(0,\pm1)$, since $|e_i|_p = 1$ for every p, and they increase with p, which is the geometric content of part 2.



2. Let $M := \max_i |x_i|$. Then

$$M \le |x|_p = M\left(\sum_{i=1}^{n}\left(\frac{|x_i|}{M}\right)^p\right)^{1/p} \le M\,n^{1/p}.$$

Since $n^{1/p} \to 1$ as $p \to \infty$, the squeeze theorem gives $|x|_\infty = \lim_{p\to\infty}|x|_p = M = \max_i |x_i|$.

3. Set $u_i := a_i^{p/2}$ and $v_i := a_i^{(p-2)/2}b_i$, so that $u_iv_i = a_i^{p-1}b_i$, $u_i^2 = a_i^p$, and $v_i^2 = a_i^{p-2}b_i^2$. Cauchy–Schwarz in $\mathbb{R}^n$ gives

$$\left(\sum_{i=1}^{n}a_i^{p-1}b_i\right)^2 = \left(\sum_{i=1}^{n}u_iv_i\right)^2 \le \left(\sum_{i=1}^{n}u_i^2\right)\left(\sum_{i=1}^{n}v_i^2\right) = \left(\sum_{i=1}^{n}a_i^p\right)\left(\sum_{i=1}^{n}a_i^{p-2}b_i^2\right).$$

4. Fix x, y with all coordinates strictly positive, let $d := x - y$, and write $z(t) := tx + (1-t)y = y + td$, so $z_i(t) > 0$ for all $t \in [0,1]$. Let $g(t) := f(t)^p = \sum_i z_i(t)^p$, so $f = g^{1/p}$. Since $z_i'(t) = d_i$,

$$g'(t) = p\sum_i z_i(t)^{p-1}d_i.$$

Differentiating $f = g^{1/p}$ twice and simplifying,

$$f''(t) = (p-1)\,g(t)^{1/p-2}\left[g(t)\sum_i z_i(t)^{p-2}d_i^2 - \Big(\sum_i z_i(t)^{p-1}d_i\Big)^2\right].$$

Applying the inequality of the previous point with $a_i = z_i(t) > 0$ and $b_i = d_i$ (any sign is allowed, since the inequality only uses b_i^2 on the right and squares the left) shows the bracket is ≥ 0. Since $g(t) > 0$ and $p \geq 1$, we get $f''(t) \geq 0$ for $p > 1$ (for $p = 1$, f is affine in t and hence convex). Thus f is convex on $[0,1]$.

5. Convexity of f at $t = \frac{1}{2}$ gives $f(\frac{1}{2}) \leq \frac{1}{2}(f(0)+f(1))$, i.e.

$$\left|\frac{x+y}{2}\right|_p \leq \frac{|x|_p + |y|_p}{2} \quad\Longrightarrow\quad |x+y|_p \leq |x|_p + |y|_p$$

for x, y with strictly positive coordinates. For general $x, y \in \mathbb{R}^n$, write $\hat{x} := (|x_1|, \ldots, |x_n|)$, $\hat{y} := (|y_1|, \ldots, |y_n|)$; these have non-negative coordinates (the strictly positive case above extends to this case by continuity), and $|x+y|_p \leq |\hat{x}+\hat{y}|_p \leq |\hat{x}|_p + |\hat{y}|_p = |x|_p + |y|_p$.

6. For $p \in (0,1)$, the triangle inequality fails, so $|\cdot|_p$ is not a norm: taking $x = e_1$, $y = e_2$ gives $|x|_p = |y|_p = 1$ but $|x+y|_p = 2^{1/p} > 2 = |x|_p + |y|_p$.

Q.E.D.

Solution to Exercise 2.c.40:

Throughout write $S(t) := \sum_{i=1}^n x_i^t$, so that $\mu_t(x) = \big(S(t)/n\big)^{1/t}$ for $t \neq 0$, and set

$$g(t) := \log \mu_t(x) = \frac{h(t)}{t}, \qquad h(t) := \log \frac{S(t)}{n}.$$

Note $h(0) = \log(n/n) = 0$; this single fact does most of the work below.

1. Let $M := \max_i x_i$ and $m := \min_i x_i$, and let k be the number of indices attaining M. Factoring out the largest term,

$$\mu_t(x) = M\left(\frac{1}{n}\sum_{i=1}^n \Big(\frac{x_i}{M}\Big)^t\right)^{1/t},$$

and as $t \to +\infty$ the bracket tends to k/n, whose t-th root tends to 1. Hence $\mu_{+\infty}(x) = M = \max_i x_i$. Replacing t by $-t$ turns the maximum into the minimum, so $\mu_{-\infty}(x) = m = \min_i x_i$.

For $t \to 0$ the exponent and the logarithm both degenerate, which is exactly the $\frac{0}{0}$ that $g = h/t$ resolves: h is differentiable with $h(0) = 0$, so

$$\lim_{t\to 0} g(t) = h'(0) = \frac{S'(0)}{S(0)} = \frac{1}{n}\sum_{i=1}^n \log x_i,$$

using $S'(t) = \sum_i x_i^t \log x_i$ and $S(0) = n$. Exponentiating,

$$\mu_0(x) = \big(x_1 x_2 \cdots x_n\big)^{1/n},$$

the "*geometric mean*" (German: „*geometrisches Mittel*").

2. The inequality is Jensen applied to $\varphi(s) := s\log s$, which is convex on $(0,\infty)$ because $\varphi''(s) = 1/s > 0$. With the uniform weights $\frac{1}{n}$,

$$\frac{1}{n}\sum_{i=1}^n a_i \log a_i \;\geq\; \varphi\left(\frac{1}{n}\sum_{i=1}^n a_i\right) = \frac{A}{n}\log\frac{A}{n}, \qquad A := a_1 + \cdots + a_n.$$

Multiplying by $n/A > 0$ gives exactly

$$\sum_{i=1}^n \frac{a_i \log a_i}{a_1 + \cdots + a_n} \;\geq\; \log\left(\frac{a_1 + \cdots + a_n}{n}\right).$$

3. *Continuity.* On $\mathbb{R} \setminus \{0\}$, $g = h/t$ is a quotient of smooth functions with non-vanishing denominator. At $t = 0$ the value $g(0) := h'(0)$ computed in part 1 is precisely the limit, so g, and hence also $f = e^g$, is continuous everywhere.
   Monotonicity. Differentiating $g = h/t$ for $t \neq 0$,
   $$t^2 g'(t) = t\,h'(t) - h(t) = \frac{t\,S'(t)}{S(t)} - \log\frac{S(t)}{n}.$$
   Now substitute $a_i := x_i^t$. Then $\log a_i = t\log x_i$, so $t\,S'(t) = \sum_i a_i \log a_i$ and $S(t) = \sum_i a_i$, and the displayed quantity becomes
   $$\sum_{i=1}^n \frac{a_i \log a_i}{a_1 + \cdots + a_n} \;-\; \log\!\left(\frac{a_1 + \cdots + a_n}{n}\right) \;\geq\; 0$$
   by part 2. Hence $g' \geq 0$ on $t \neq 0$, and by continuity g is non-decreasing on all of $\mathbb{R}$; so is $f = e^g$.
   This is why part 2 was placed where it is: it is not a separate exercise but the derivative computation of part 3 in disguise.
4. Both inequalities are instances of $\mu_s(x) \leq \mu_t(x)$ for $s \leq t$, i.e. of part 3.
   - **Arithmetic–geometric:** $\mu_0(x) \leq \mu_1(x)$ reads $(x_1 \cdots x_n)^{1/n} \leq \frac{1}{n}(x_1 + \cdots + x_n)$; multiplying by n gives $n(x_1 x_2 \cdots x_n)^{1/n} \leq x_1 + \cdots + x_n$.
   - **Arithmetic–quadratic:** $\mu_1(x) \leq \mu_2(x)$ reads $\frac{1}{n}\sum_i x_i \leq \big(\frac{1}{n}\sum_i x_i^2\big)^{1/2}$. Both sides are positive, so squaring preserves the inequality: $\frac{1}{n^2}\big(\sum_i x_i\big)^2 \leq \frac{1}{n}\sum_i x_i^2$, and multiplying by n^2 gives $(x_1 + \cdots + x_n)^2 \leq n(x_1^2 + \cdots + x_n^2)$.
5. **(*) Yes**, and considerably more than C^1. Each $x_i^t = e^{t\log x_i}$ is real-analytic in t, hence so is S, and $S > 0$ everywhere, so $h = \log(S/n)$ is real-analytic too. Because $h(0) = 0$, the quotient $h(t)/t$ has a **removable** singularity at the origin: writing $h(t) = \sum_{k\geq 1} c_k t^k$ gives $g(t) = \sum_{k\geq 1} c_k t^{k-1}$, a convergent power series near 0. So g, and therefore $f = e^g$, is real-analytic on all of $\mathbb{R}$, in particular C^∞, and so certainly C^1.
   The value at the origin is worth recording. From $g(t) = h'(0) + \frac{1}{2}h''(0)t + O(t^2)$,
   $$f'(0) = f(0)\,g'(0) = \mu_0(x)\cdot\tfrac{1}{2}h''(0) = \mu_0(x)\cdot\frac{1}{2}\left(\frac{1}{n}\sum_{i=1}^n(\log x_i)^2 - \Big(\frac{1}{n}\sum_{i=1}^n \log x_i\Big)^2\right),$$
   using $h'' = (S''S - (S')^2)/S^2$ at $t = 0$ with $S(0) = n$, $S'(0) = \sum_i \log x_i$ and $S''(0) = \sum_i(\log x_i)^2$. The bracket is the variance of the numbers $\log x_i$, so $f'(0) \geq 0$, with equality exactly when all x_i agree. In that case $\mu_t(x)$ is that common value for every t, and f is constant.

Q.E.D.

Remark 2.d.44 (*Why parts 2 and 3 are one statement*): Jensen's inequality, used above without being stated, is available as Proposition 14.a.8 (Chapter 14). The official hint for 4.4.2 routes through the concavity of log together with Cauchy–Schwarz. The solution above instead applies Jensen once, to the convex function $\varphi(s) = s\log s$. Both arrive at the same inequality, but the second makes visible why 4.4.2 and 4.4.3 are really a single statement: substituting $a_i := x_i^t$ turns the inequality of part 2 into the sign of $g'(t)$ in part 3. That is worth seeing, because monotonicity of the power means is the whole point of the problem, and the classical AM–GM inequality falls out of it as the single comparison $\mu_0 \leq \mu_1$.

Open and Closed Sets (Chapter 3)

With a distance in hand, the next question is which subsets a metric singles out. The answer is built up from a single object, the open ball, and everything in this chapter is defined in terms of it: open and closed sets first, then the interior, closure and boundary of an arbitrary set.

The chapter ends by drawing the unit ball of each of the three standard metrics on $\mathbb{R}^2$, and the result is worth anticipating. The three balls are a square, a diamond and a disk, so they are as visibly different as three convex sets can be. Yet they induce exactly the same open sets. That gap between what a metric looks like and what it decides is the first genuinely topological phenomenon in the course.

Open balls, open and closed sets (Section 3.a)

Definition 3.a.1 (*Open ball*): Let (X, d) be a metric space, let $x \in X$, and let $r \in \mathbb{R}$. Then we define the "*open ball*" of radius r as

$$B_r(x) := \{ \boxed{y \in X} : d(y, x) < r \}$$

Important remark 3.a.2: Corsin marks the boxed part in red as an *"important subtlety"*: the ambient set X in $y \in X$ is what makes openness relative to the space, and it is exactly what the common misconception on Important remark 4.a.13 turns on.

AI-Note 3.1: Definition 3.a.1 is stated for $r \in \mathbb{R}$ rather than $r > 0$. Harmless here, but loose.

AI-Example 3.a.3 (*Open Balls in the Discrete Metric*): Let (X, d_{disc}) be a discrete metric space. The open ball $B_r(x)$ centered at $x \in X$ with radius $r > 0$ is

$$B_r(x) = \begin{cases} \{x\} & \text{if } 0 < r \leq 1, \\ X & \text{if } r > 1. \end{cases}$$

Hence every singleton $\{x\}$ is open in the discrete metric, and consequently every subset of X is both open and closed ("*clopen*"). This is exactly what Exercise 3.b.16 below asks you to verify for X and $\emptyset$ in a general metric space.

Remark 3.a.4 (*Two degenerate balls worth knowing*): Taking Definition 3.a.1 literally at the edges of its range is a cheap way to test whether you have read it correctly.

- If $r \leq 0$, then $B_r(x) = \emptyset$, and **not** $\{x\}$. The condition is $d(y, x) < r$, strictly, and $d(x, x) = 0 \not< 0$. So even the centre is missing from $B_0(x)$.
- In the discrete metric $B_1(x) = \{x\}$, while $B_{1.01}(x) = X$: a ball can jump from a single point to the entire space without passing through anything in between. Radii do not have to behave like radii.

These are the two cases that make *"shrink the ball a little"* arguments go wrong if applied without thinking.

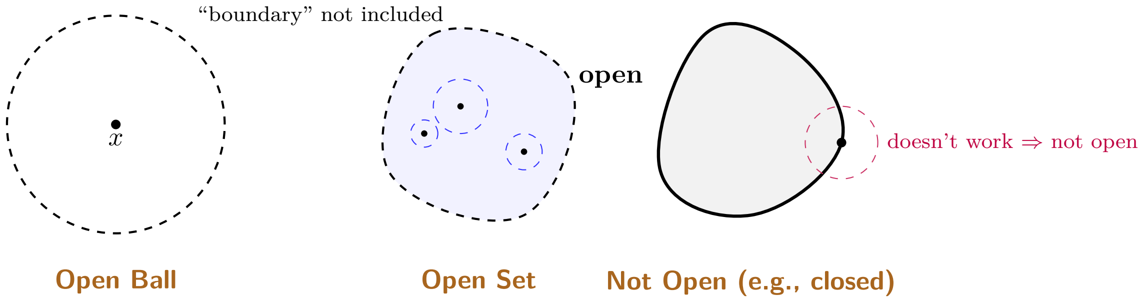
AI-Exercise 3.a.5 (*Open balls are open*): The name promises it, but Definition 3.a.1 does not say it. Show that in any metric space (X, d) the ball $B_r(x)$ is an open set in the sense of Definition 3.a.6.

Hint: given $y \in B_r(x)$, the number $\varepsilon := r - d(y, x)$ is strictly positive. Draw the picture, then let the triangle inequality do the work.

Definition 3.a.6 (*Open and closed sets*): Then, we call a subset $U \subseteq X$ "*open*" (German: „*offen*") if and only if

$$\forall y \in U \ \exists \varepsilon > 0 \text{ such that } B_\varepsilon(y) \subseteq U.$$

And we call $A \subseteq X$ "*closed*" (German: „*abgeschlossen*") if and only if $X \setminus A$ is open.



Example 3.a.7 (*Clopen sets other than X and $\emptyset$*)**:** Exercise 3.b.16 shows X and $\emptyset$ are always clopen, but nothing in Definition 3.a.6 forces these to be the **only** clopen subsets of X. Take $X := (0,1) \cup (2,3) \subset \mathbb{R}$ with the standard metric: both $(0,1)$ and $(2,3)$ are clopen **in X**. Each is open in $\mathbb{R}$, hence open in X; and each is the complement in X of the other, so each is also closed in X. Whether a space has clopen subsets besides X and $\emptyset$ turns out to be exactly the question Chapter 8 is organised around. A space with none is called "*connected*".

Proposition 3.a.8 (*How open and closed sets combine*)**:** Let (X,d) be a metric space.

(a) **Arbitrary** unions of open sets are open: if $\{U_i\}_{i\in I}$ are open, so is $\bigcup_{i\in I} U_i$.

(b) **Finite** intersections of open sets are open: if $U_1,\dots,U_k$ are open, so is $\bigcap_{i=1}^k U_i$.

(c) **Arbitrary** intersections of closed sets are closed, and **finite** unions of closed sets are closed.

Proof:

(a) Let $x \in \bigcup_i U_i$. Then $x \in U_{i_0}$ for some i_0, and since U_{i_0} is open there is $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq U_{i_0} \subseteq \bigcup_i U_i$.

(b) Let $x \in \bigcap_{i=1}^k U_i$. For each i pick $\varepsilon_i > 0$ with $B_{\varepsilon_i}(x) \subseteq U_i$, and set $\varepsilon := \min\{\varepsilon_1,\dots,\varepsilon_k\}$. Then $\varepsilon > 0$, and **this is the only place finiteness is used**: a minimum of finitely many positive numbers is positive, whereas an infimum of infinitely many can be 0. Since $B_\varepsilon(x) \subseteq U_i$ for every i, we conclude that $B_\varepsilon(x) \subseteq \bigcap_i U_i$.

(c) Take complements and apply **(a)**, **(b)** via De Morgan: $X \setminus \bigcap_i A_i = \bigcup_i (X \setminus A_i)$ and $X \setminus \bigcup_{i=1}^k A_i = \bigcap_{i=1}^k (X \setminus A_i)$. **Q.E.D.**

Important remark 3.a.9 (*The asymmetry is real, not an artefact of the proof*)**:** It is tempting to read *"finite"* in **(b)** as laziness. It is not: an infinite intersection of open sets genuinely need not be open. In $\mathbb{R}$,

$$\bigcap_{n=1}^{\infty} \left(-\tfrac{1}{n},\ \tfrac{1}{n}\right) = \{0\},$$

an intersection of open intervals which is not open. The proof shows exactly where it breaks: $\varepsilon := \min_i \varepsilon_i$ is the whole argument, and an infimum of infinitely many positive numbers can be 0. Dually, $\bigcup_{n\ge1} \left[\frac{1}{n}, 1\right] = (0,1]$ is an infinite union of closed sets that is not closed.

AI-Note 3.2: These four closure properties are exactly the axioms of a "*topology*", which is how the subject is set up when one drops the metric altogether (Section 2.a, *"fourth semester"*). That is why they are worth isolating: they are the only facts about open sets that survive into the general theory.

Example 3.a.10 (*Open and closed sets in $\mathbb{R}$*)**:** Consider the following subsets of $\mathbb{R}$ with the standard metric:

- **(a) Is $\mathbb{N}$ open, closed, or neither?** It is **closed**, since its complement $\mathbb{R} \setminus \mathbb{N} = (-\infty, 1) \cup \bigcup_{n=1}^{\infty}(n, n+1)$ is a union of open intervals, which is open by Proposition 3.a.8**(a)**. It is not open, because no open ball around any $n \in \mathbb{N}$ is contained in $\mathbb{N}$.
- **(b) Let $A := \{\frac{1}{n} \mid n = 1, 2, \dots\}$.** The set A is **neither** open nor closed. It is not open for the same reason as $\mathbb{N}$. It is not closed because 0 is an accumulation point of A, but $0 \notin A$, meaning its complement $\mathbb{R} \setminus A$ is not open (any open ball around 0 intersects A).
- **(c) Let $F := \{\frac{1}{n} \mid n = 1, 2, \dots\} \cup \{0\}$.** The set F is **closed**. By adding the limit point 0, the complement $\mathbb{R} \setminus F$ becomes a union of open intervals, which is open.

Exercise 3.a.11 (*A longer self-test*): Linus sets the following list as a self-test at the end of his first session. Each subset of $\mathbb{R}$ carries the standard metric; decide for each whether it is **open**, **closed**, **both**, or **neither**.

- **(a)** $[0, \infty)$
- **(b)** $\mathbb{Q}$
- **(c)** $\{x - \lfloor x \rfloor : x \in \mathbb{R}\}$
- **(d)** $S := \{x \in \mathbb{R} : x \cdot 2^p \in [1, 2] \text{ for some integer } p \geq 0\}$
- **(e)** $T := \{x \in \mathbb{R} : x \cdot 2^p \in (1, 2) \text{ for some integer } p \geq 0\}$
- **(f)** $\{x_n : n \in \mathbb{N}\}$, where $x_0 := 1$ and $x_n := \frac{1}{2}\left(x_{n-1} + x_{n-1}^2\right)$
- **(g)** $\{\mathrm{Re}(c) : c \in \mathbb{C},\ \zeta(c) = 0\}$, where ζ is the Riemann zeta function

Compare **(d)** with **(e)** before reading the solution: the two differ only in one bracket.

Exercise 3.a.12 (*Relative openness*): Let $X := (\{0\} \times \mathbb{R}) \cup (0, 1)^2$ be equipped with the subspace metric from $\mathbb{R}^2$. Consider the following subsets of X:

$$A := \{(x, y) \in \mathbb{R}^2 \mid x = 0, y \in (2, 3)\}$$
$$B := \left\{(x, y) \in \mathbb{R}^2 \,\middle|\, \left(x - \tfrac{1}{2}\right)^2 + \left(y - \tfrac{1}{2}\right)^2 < \tfrac{1}{16}\right\}$$

Are A and B open with respect to X? Are they open with respect to $\mathbb{R}^2$?

Proof of Exercise 3.a.12:

- A is **open with respect to X**, but **not open with respect to $\mathbb{R}^2$**. It is open in X because $A = X \cap U$ where $U = (-\varepsilon, \varepsilon) \times (2, 3)$ is an open set in $\mathbb{R}^2$. It is not open in $\mathbb{R}^2$ because any open ball around a point in A will contain points with $x \neq 0$, which are not in A.
- B is **open with respect to both X and $\mathbb{R}^2$**. B is exactly the open ball $B_{1/4}((\frac{1}{2}, \frac{1}{2}))$ in $\mathbb{R}^2$, which is open in $\mathbb{R}^2$. Since $B \subseteq (0, 1)^2 \subset X$, we have $B = X \cap B$, so B is also open relative to X.

Q.E.D.

Interior, closure and boundary (Section 3.b)

Definition 3.b.13 (*Interior, closure, and boundary*): Let (X, d) be a metric space and $A \subseteq X$. The “*interior*” (German: „*Inneres*“) of A is

$$\mathrm{int}(A) := \{x \in A : \exists\, r > 0 \text{ such that } B_r(x) \subseteq A\},$$

the largest open subset of X contained in A. The “*closure*” (German: „*Abschluss*“) of A is

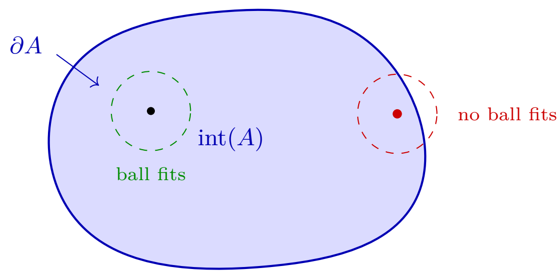
$$\overline{A} := \{x \in X : B_r(x) \cap A \neq \emptyset \text{ for all } r > 0\},$$

the smallest closed subset of X containing A. The “*boundary*” (German: „*Rand*“) of A is $\partial A := \overline{A} \setminus \mathrm{int}(A)$.

AI-Example 3.b.14 (*Interior, Closure, and Boundary of an Interval*): Applying Definition 3.b.13 to $A := (0, 1] \subset \mathbb{R}$ under the standard Euclidean metric:

- The **interior** is $\mathrm{int}(A) = (0, 1)$, the largest open set contained in A.
- The **closure** is $\overline{A} = [0, 1]$, the smallest closed set containing A.
- The **boundary** is $\partial A = \overline{A} \setminus \mathrm{int}(A) = \{0, 1\}$.

Note that $1 \in A$ while $0 \notin A$: the boundary is not required to lie inside the set, nor outside it. That is the point of defining it as a difference of two other sets rather than as *"the edge of A"*.



$\mathrm{int}(A)$: points with a ball inside A · ∂A: points every ball of which meets both A and $X \setminus A$ · $\overline{A} = \mathrm{int}(A) \cup \partial A$

Remark 3.b.15 (*Interior and closure are dual*): The two constructions are not independent: taking complements swaps them,

$$X \setminus \mathrm{int}(A) = \overline{X \setminus A}, \qquad X \setminus \overline{A} = \mathrm{int}(X \setminus A).$$

Both read off Definition 3.b.13 directly, since x fails to have a ball inside A exactly when every ball around x meets $X \setminus A$. In particular $\partial A = \partial(X \setminus A)$: a set and its complement share their boundary, which is why the boundary of the open disk and of its closed complement are the same circle.

Exercise 3.b.16 (*X and $\emptyset$ are clopen*): Let (X, d) be a metric space. Show that X and $\emptyset$ are both open and closed.

Proof of Exercise 3.b.16:

X is open, since $B_r(x) \subseteq X$ for all $r > 0$ and $x \in X$ (Definition 3.a.1). The empty set contains no points and is therefore vacuously open. Since X and $\emptyset$ are each other's complement in X, they are also closed. **Q.E.D.**

Example 3.b.17 (*Interior, closure, and boundary in $\mathbb{R}$ and $\mathbb{R}^2$*): Applying Definition 3.b.13 to various sets:

(a) Let $\Omega := \mathbb{Z} \subset \mathbb{R}$. Then $\mathrm{int}(\Omega) = \emptyset$ (no interval fits inside $\mathbb{Z}$), $\overline{\Omega} = \mathbb{Z}$ (it is already closed), and $\partial\Omega = \mathbb{Z} \setminus \emptyset = \mathbb{Z}$.

(b) Let $\Omega := B_1(0) \subset \mathbb{R}^2$. Then $\mathrm{int}(\Omega) = B_1(0)$ (it is already open), the closure is the closed disk $\overline{\Omega} = \{x \in \mathbb{R}^2 \mid d(x, 0) \leq 1\}$, and the boundary is the unit circle $\partial\Omega = \{x \in \mathbb{R}^2 \mid d(x, 0) = 1\}$.

(c) Let $\Omega := \mathbb{Q} \subset \mathbb{R}$. Then $\mathrm{int}(\Omega) = \emptyset$ (every open interval contains irrationals), $\overline{\Omega} = \mathbb{R}$ (every open interval contains rationals, so every real number is a closure point), and $\partial\Omega = \mathbb{R} \setminus \emptyset = \mathbb{R}$.

Exercise 3.b.18 (*Closure and boundary of a ball in $\mathbb{R}^n$*): For balls $B_r(x)$ in $\mathbb{R}^n$ with the Euclidean metric, prove that

$$\overline{B_r(x)} = \{y \in \mathbb{R}^n : d(x, y) \leq r\}$$

and deduce $\partial B_r(x) = \{y \in \mathbb{R}^n : d(x, y) = r\}$.

Remark 3.b.19: Item **(b)** of Example 3.b.17 above is exactly the case $r = 1$, $x = 0$, $n = 2$ of this exercise, worked here for general x, r and n.

Exercise 3.b.20 (*Interior, closure, and boundary of a mixed set*): Let $E := (\{0\} \times \mathbb{R}) \cup (0,1)^2 \subset \mathbb{R}^2$. Find the interior $\operatorname{int}(E)$, the closure $\overline{E}$, and the boundary ∂E with respect to the standard metric on $\mathbb{R}^2$.

Proof of Exercise 3.b.20:

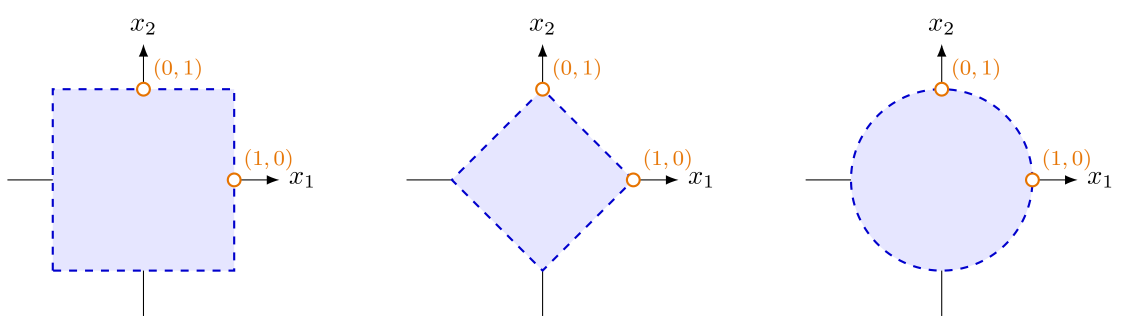
- **Interior:** $\operatorname{int}(E) = (0,1)^2$. The line $\{0\} \times \mathbb{R}$ contains no 2D open balls, so it contributes nothing to the interior. The open square $(0,1)^2$ is already open in $\mathbb{R}^2$.
- **Closure:** $\overline{E} = (\{0\} \times \mathbb{R}) \cup [0,1]^2$. The line $\{0\} \times \mathbb{R}$ is closed. The closure of the open square is the closed square $[0,1]^2$. The closure of a finite union is the union of the closures.
- **Boundary:** $\partial E = \overline{E} \setminus \operatorname{int}(E) = (\{0\} \times \mathbb{R}) \cup \big([0,1]^2 \setminus (0,1)^2\big)$. The second term is the perimeter of the square.

Q.E.D.

Unit balls of the three standard metrics (Section 3.c)

Warm up 3.c.21: What is the shape of $B_1(0) \subseteq \mathbb{R}^2$ with the supremum metric? With the Manhattan metric?

Answer 3.c.22: Supremum: the axis-aligned **open square** $(-1,1)^2$. Manhattan: the **open diamond** $\{|x_1| + |x_2| < 1\}$. In both cases the strict inequality in Definition 3.a.1 matters: the points $(\pm 1, 0)$ and $(0, \pm 1)$ sit on the boundary and are **not** in the ball. They are the corners of the diamond and the edge midpoints of the square, and are drawn hollow below.



Supremum metric d_3: open square $(-1,1)^2$ | Manhattan metric d_1: open diamond | Standard metric d_2: open disk

Remark 3.c.23 (*The unit ball remembers which metric it came from*): Three metrics, three unmistakably different unit balls: the square for d_3, the disk for d_2, the diamond for d_1. This is the quickest way to tell the three apart, and it also makes AI-Exercise 2.b.24 visible, since the inclusions

$$B_1^{d_1}(0) \;\subseteq\; B_1^{d_2}(0) \;\subseteq\; B_1^{d_3}(0)$$

are precisely the inequalities $d_3 \leq d_2 \leq d_1$ read backwards. The larger the metric, the harder it is to be within distance 1, and hence the smaller the ball. The same family, continued to all $p \geq 1$, is sketched in Exercise 2.c.39.**1**, where the square reappears as the limiting case $p \to \infty$.

None of this changes which sets are open. All three metrics have the same open sets, as AI-Exercise 2.b.24 shows. The balls look different, and the topology does not notice.

AI-Exercise 3.c.24 (*Boundary of the Open Unit Disk*)**:** Let $D := \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\}$ be the open unit disk in $\mathbb{R}^2$. Show that the boundary of D is the unit circle $\partial D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$.

Exercise 3.c.25 (*2.2 — Multiple choice (closure)*)**:** Let (X,d) be a metric space, and $Y_1, Y_2 \subset X$ subsets. Select all the statements below that are necessarily true.

(a) $\overline{Y_1 \cup Y_2} = \overline{Y_1} \cup \overline{Y_2}$
(b) $\overline{Y_1} \cap \overline{Y_2} \subset \overline{Y_1 \cap Y_2}$
(c) $\overline{Y_1 \cap Y_2} \subset \overline{Y_1} \cap \overline{Y_2}$
(d) $\overline{Y_1 \cap Y_2} = \overline{Y_1} \cap \overline{Y_2}$

Official hints: **(b):** compare with Exercise 2.b.25.3. **(d):** play with a set of three points (a triangle).

Exercise 3.c.26 (*2.3 — Multiple choice (distance from a set)*)**:** Let (X,d) be a metric space, and $A \subset X$ a non-empty subset. We define the function "distance from A" as

$$d(\cdot, A) : X \to [0,\infty), \qquad d(x,A) := \inf_{a \in A} d(x,a).$$

Select all the statements below that are necessarily true.

(a) If A is closed and $x \in A^c$, then $d(x,A) > 0$.
(b) The set $M := \{x \in X : d(x,A) \geq 1\}$ is closed in X.
(c) For $x, y \in X$, $d(x,A) \leq d(x,y) + d(y,A)$ holds.
(d) If A° is non-empty and $x \in X$, then $d(x,A) = d(x,A^\circ)$.

Official hint: first convince yourself with a drawing that the name of this function is appropriate; then use the characterization of open/closed sets with sequences.

Exercise 3.c.27 (*2.4 — Boundary, Interior etc.*)**:** Determine the interior, closure, and boundary of the following subsets Y of $\mathbb{R}$, for the standard topology on $\mathbb{R}$. No need to justify the answer.

(1)	$Y = [0,1]$	(2)	$Y = \mathbb{Q}$
(3)	$Y = \emptyset$	(4)	$Y = (0,1)$
(5)	$Y = [-1,1) \setminus \{0\}$	(6)	$Y = [0,\infty)$
(7)	$Y = \{0\}$	(8)	$Y = \{\frac{1}{n} \mid n \in \mathbb{N} \setminus \{0\}\}$

Solutions (Section 3.d)

Proof of AI-Exercise 3.a.5:

Let $y \in B_r(x)$, so $d(y,x) < r$, and set $\varepsilon := r - d(y,x) > 0$. We claim that $B_\varepsilon(y) \subseteq B_r(x)$, which is exactly what Definition 3.a.6 requires.

Let $z \in B_\varepsilon(y)$, so $d(z,y) < \varepsilon$. By the triangle inequality (Definition 2.b.8),

$$d(z,x) \leq d(z,y) + d(y,x) < \varepsilon + d(y,x) = \big(r - d(y,x)\big) + d(y,x) = r,$$

so $z \in B_r(x)$. Since $y \in B_r(x)$ was arbitrary, $B_r(x)$ is open.

Remark 3.d.28: The entire content is the choice $\varepsilon := r - d(y,x)$: the radius left over after travelling from the centre out to y. Note that it shrinks to 0 as y approaches the edge of the ball, so no **single** ε works for every y at once. That is the same *"the constant depends on the point"* phenomenon which separates continuity from uniform continuity in Important remark 5.a.5, met here for the first time.

Q.E.D.

Proof of Exercise 3.b.18:

Write $C := \{y \in \mathbb{R}^n : d(x,y) \leq r\}$ for the target set.

$\overline{B_r(x)} \subseteq C$. C is closed (it is $d(x,\cdot)^{-1}([0,r])$, the preimage of a closed set under the continuous map $d(x,\cdot)$) and contains $B_r(x)$, and the closure is the **smallest** closed set containing $B_r(x)$ (Definition 3.b.13).

$C \subseteq \overline{B_r(x)}$. Let $y \in C$, so $d(x,y) \leq r$; we must show every ball $B_\varepsilon(y)$ meets $B_r(x)$. If $d(x,y) < r$ then $y \in B_r(x)$ itself, so this is immediate. If $d(x,y) = r$, consider the points $z_t := x + t(y-x)$ for $t \in (0,1)$, which lie on the segment from x to y: since $\mathbb{R}^n$ has a vector space structure (unlike a general metric space, where this argument would not make sense), $d(x,z_t) = t\,d(x,y) = tr < r$, so $z_t \in B_r(x)$, and $d(z_t,y) = (1-t)d(x,y) = (1-t)r \to 0$ as $t \to 1$. So every ball around y eventually contains some z_t, hence meets $B_r(x)$, and $y \in \overline{B_r(x)}$.

For the boundary, $\operatorname{int}(B_r(x)) = B_r(x)$ since $B_r(x)$ is itself open (AI-Exercise 3.a.5), so

$$\partial B_r(x) = \overline{B_r(x)} \setminus \operatorname{int}(B_r(x)) = C \setminus B_r(x) = \{y \in \mathbb{R}^n : d(x,y) = r\}.$$

Q.E.D.

Proof of AI-Exercise 3.c.24:

Since D is open, its interior is $\operatorname{int}(D) = D$. The closure of D is $\overline{D} = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 1\}$. By Definition 3.b.13, the boundary is:

$$\begin{aligned} \partial D &= \overline{D} \setminus \operatorname{int}(D) \\ &= \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 1\} \setminus \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 < 1\} \\ &= \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 = 1\}. \end{aligned}$$

Q.E.D.

Solution to Exercise 3.c.25:

(a) and (c) are necessarily true; (b) and (d) are not.

(a) True. Both inclusions come from general properties of the closure. Since $Y_i \subseteq Y_1 \cup Y_2$ and the closure is monotone, $\overline{Y_i} \subseteq \overline{Y_1 \cup Y_2}$ for $i = 1,2$, which gives "$\supseteq$". Conversely $\overline{Y_1} \cup \overline{Y_2}$ is a **finite** union of closed sets, hence closed (Proposition 3.a.8**(c)**), and it contains $Y_1 \cup Y_2$; since $\overline{Y_1 \cup Y_2}$ is the smallest closed set containing $Y_1 \cup Y_2$ (Definition 3.b.13), "$\subseteq$" follows.

(c) True, and it is the easy half: $Y_1 \cap Y_2 \subseteq Y_i$ gives $\overline{Y_1 \cap Y_2} \subseteq \overline{Y_i}$ for each i by monotonicity, hence $\overline{Y_1 \cap Y_2} \subseteq \overline{Y_1} \cap \overline{Y_2}$.

(b) False, and therefore **(d) False** as well, (d) being (b) and (c) combined. In $\mathbb{R}$ take

$$Y_1 := (0,1), \qquad Y_2 := (1,2).$$

Then $\overline{Y_1} \cap \overline{Y_2} = [0,1] \cap [1,2] = \{1\}$, whereas $Y_1 \cap Y_2 = \emptyset$ and so $\overline{Y_1 \cap Y_2} = \emptyset$. The two sets meet only in the limit, and the closure of the intersection cannot see a point belonging to neither set.

Remark 3.d.29 (*Where finiteness enters, again*): Part **(a)** is false for infinite unions, for exactly the reason given in Important remark 3.a.9. Taking $Y_q := \{q\}$ for each $q \in \mathbb{Q}$,

$$\bigcup_{q\in\mathbb{Q}} \overline{Y_q} = \mathbb{Q} \qquad \text{but} \qquad \overline{\bigcup_{q\in\mathbb{Q}} Y_q} = \overline{\mathbb{Q}} = \mathbb{R}.$$

So the honest statement is: the closure commutes with **finite** unions, and only one-directionally with intersections. That is the same asymmetry that runs through the whole of Proposition 3.a.8.

Q.E.D.

Solution to Exercise 3.c.26:

(a), (b) and (c) are necessarily true; (d) is not. It is efficient to prove **(c)** first, since **(b)** then follows from it.

(c) True. Fix $x, y \in X$. For every $a \in A$ the triangle inequality gives $d(x,a) \le d(x,y) + d(y,a)$, and the left-hand side is bounded below by $d(x,A)$, so

$$d(x,A) \le d(x,y) + d(y,a) \qquad \text{for every } a \in A.$$

Taking the infimum over a on the right yields $d(x,A) \le d(x,y) + d(y,A)$. Exchanging x and y and combining,

$$\big|d(x,A) - d(y,A)\big| \le d(x,y),$$

so $d(\cdot, A)$ is 1-**Lipschitz** (Definition 5.b.20), and in particular continuous. That is what the remaining parts use.

(a) True. If A is closed then A^c is open, so for $x \in A^c$ there is $r > 0$ with $B_r(x) \subseteq A^c$. No point of A lies in that ball, i.e. $d(x,a) \ge r$ for every $a \in A$, and taking the infimum gives $d(x,A) \ge r > 0$.

(b) True. By **(c)** the function $d(\cdot, A)$ is continuous, and $M = d(\cdot, A)^{-1}\big([1,\infty)\big)$ is the preimage of a closed set under a continuous map, hence closed (Definition 5.a.1**(c)**, applied to complements).

(d) False. In $\mathbb{R}$ take $A := [0,1] \cup \{2\}$, whose interior $A^\circ = (0,1)$ is non-empty as required. At the point $x := 2$,

$$d(x,A) = 0 \quad (\text{because } 2 \in A), \qquad\qquad d(x, A^\circ) = \inf_{a \in (0,1)} |2-a| = 1.$$

The isolated point 2 of A contributes to $d(\cdot, A)$ but is invisible to A°. The hypothesis *"A° is non-empty"* is far too weak, because it constrains A nowhere near x; the statement would become true under a hypothesis such as $A = \overline{A^\circ}$. **Q.E.D.**

Solution to Exercise 3.c.27:

All with respect to the standard metric on $\mathbb{R}$, using Definition 3.b.13.

	Y	$\operatorname{int}(Y)$	$\overline{Y}$	∂Y
(1)	$[0,1]$	$(0,1)$	$[0,1]$	$\{0,1\}$
(2)	$\mathbb{Q}$	$\emptyset$	$\mathbb{R}$	$\mathbb{R}$
(3)	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
(4)	$(0,1)$	$(0,1)$	$[0,1]$	$\{0,1\}$
(5)	$[-1,1) \setminus \{0\}$	$(-1,0) \cup (0,1)$	$[-1,1]$	$\{-1,0,1\}$
(6)	$[0,\infty)$	$(0,\infty)$	$[0,\infty)$	$\{0\}$
(7)	$\{0\}$	$\emptyset$	$\{0\}$	$\{0\}$
(8)	$\left\{\frac{1}{n} : n \ge 1\right\}$	$\emptyset$	$\left\{\frac{1}{n} : n \ge 1\right\} \cup \{0\}$	$\left\{\frac{1}{n} : n \ge 1\right\} \cup \{0\}$

Three rows repay a second look.

- **(1) versus (4).** The closed interval and the open interval have the same interior, the same closure and the same boundary. These three operations simply cannot distinguish a set from any other set squeezed between its interior and its closure.
- **(5).** Removing the single point 0 from $[-1,1)$ adds 0 to the boundary without changing the closure: a removed interior point becomes a boundary point. Note also that $1 \in \partial Y$ although $1 \notin Y$, while $-1 \in \partial Y$ and $-1 \in Y$. The boundary is indifferent to membership.
- **(2) and (8).** Both have empty interior, so $\partial Y = \overline{Y}$: once a set has no interior, the boundary swallows the entire closure. That is the general reason for $\partial\mathbb{Q} = \mathbb{R}$ in Example 3.b.17**(c)**, and not an accident of $\mathbb{Q}$.

Q.E.D.

Solution to Exercise 3.a.11:

- **(a)** **Closed.** The complement is $(-\infty, 0)$, which is open, so $[0, \infty)$ is closed by Definition 3.a.6. It is not open: every ball $B_\varepsilon(0)$ contains negative numbers.
- **(b)** **Neither.** Not open, since every ball around a rational contains irrationals. Not closed, since $\sqrt{2}$ is a limit of rationals without lying in $\mathbb{Q}$; indeed $\overline{\mathbb{Q}} = \mathbb{R}$.
- **(c)** **Neither.** The map $x \mapsto x - \lfloor x \rfloor$ is the fractional part, whose range is exactly $[0, 1)$. This is not open (0 has no ball inside it) and not closed (1 is a limit point that is missing). The two defects sit at the two ends.
- **(d)** **Neither.** Multiplying by 2^{-p} turns the condition into a union of closed blocks,
$$S = \bigcup_{p \geq 0} \left[2^{-p},\, 2^{-p+1}\right] = [1, 2] \cup \left[\tfrac{1}{2}, 1\right] \cup \left[\tfrac{1}{4}, \tfrac{1}{2}\right] \cup \cdots = (0, 2],$$
since consecutive blocks share an endpoint and their left ends decrease to 0. Now $(0, 2]$ is not open (no ball around 2 fits) and not closed (0 is a missing limit point).
- **(e)** **Open**, and this is the point of putting **(d)** next to it. The same computation with open blocks gives
$$T = \bigcup_{p \geq 0} \left(2^{-p},\, 2^{-p+1}\right) = (0, 2) \setminus \left\{1, \tfrac{1}{2}, \tfrac{1}{4}, \ldots\right\},$$
because now the shared endpoints 2^{-k} are excluded by **both** neighbouring blocks and so belong to neither. An arbitrary union of open sets is open (Proposition 3.a.8), so T is open. It is not closed: 0 and each removed point 2^{-k} are limit points of T outside T.

  The lesson is that the union in **(d)** was closed-looking but glued into a half-open interval, whereas the union in **(e)** punctures itself at every join. Being a union of closed sets buys nothing; being a union of open sets buys everything.
- **(f)** **Closed.** Compute one step before doing anything else: $x_1 = \frac{1}{2}(1 + 1^2) = 1 = x_0$, so the recursion is stationary and $x_n = 1$ for every n. The set is therefore $\{1\}$, a single point, which is closed (finite sets are closed) and not open.
- **(g)** **Not known.** The trivial zeros of ζ are $-2, -4, -6, \ldots$, contributing the real parts $\{-2, -4, -6, \ldots\}$; the non-trivial zeros all lie in the critical strip $0 < \mathrm{Re}(c) < 1$. If the Riemann hypothesis holds, every non-trivial zero has $\mathrm{Re}(c) = \frac{1}{2}$, so the set is
$$\left\{\tfrac{1}{2}\right\} \cup \{-2n : n \geq 1\},$$
which is closed: its points are isolated and the only way to escape is towards $-\infty$, which is not a real accumulation point. If the Riemann hypothesis fails, the real parts of the off-line zeros are not understood well enough to say. So this entry is a joke with a sharp edge. It is a perfectly well-posed question about a perfectly explicit set, and nobody can currently answer it.

Q.E.D.

Sequences and Convergence (Chapter 4)

Open and closed sets describe a space statically. They say which points sit comfortably inside a set and which sit on its edge, but they say it all at once, with no notion of approach. Sequences make the same information dynamic, and in a metric space the two descriptions turn out to be equivalent: a set is closed exactly when it catches the limits of its own convergent sequences.

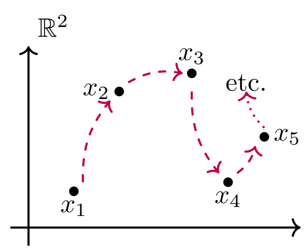
That equivalence is what makes this short chapter worth its own place. Almost every later argument that needs to produce a point produces it as a limit, and the sequential criterion is what converts that limit back into a statement about open or closed sets. The chapter also isolates a fact easy to pass over, namely that convergence depends only on the topology and not on the metric generating it, so two visibly different metrics can have exactly the same convergent sequences.

Sequences and convergence (Section 4.a)

Open and closed sets describe a space statically: which points sit comfortably inside a set and which sit on its edge. Sequences describe the same thing dynamically, by asking where a moving point ends up. The two descriptions turn out to be equivalent (Proposition 4.a.7), and having both is what makes metric spaces pleasant to work in. One description is usually easier to verify, the other easier to apply. Note that the definition below is word for word the one from **Analysis I**, with $|x_n - x|$ replaced by $d(x_n, x)$. That substitution is the pattern for the whole chapter.

Definition 4.a.1 (*Convergent sequence*): A "*sequence*" $(x_n)_{n=0}^{\infty}$ in a metric space (X, d) is a function $x : \mathbb{N} \to X$. We say that $(x_n)_{n=0}^{\infty}$ "*converges*" to $x \in X$ if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N} \text{ such that } d(x_n, x) < \varepsilon \quad \forall n \geq N.$$



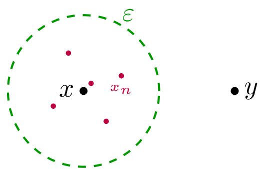
Notation 4.a.2 (*"Eventually"*): Let $(x_n)_{n\geq 0}$ be a sequence in a set X and let $\mathcal{P}$ be a property that a point of X may or may not have. One says x_n satisfies $\mathcal{P}$ "*eventually*" (German: „*schließlich*“) if there exists $N \in \mathbb{N}$ such that $\mathcal{P}(x_n)$ holds for all $n \geq N$, or equivalently if $\mathcal{P}(x_n)$ fails for at most finitely many n. In this vocabulary, Definition 4.a.1 reads: $(x_n)_n$ converges to x if, for every $\varepsilon > 0$, x_n eventually lies in $B_\varepsilon(x)$. The same compression applies throughout the rest of this chapter and the next three.

Example 4.a.3 (*A convergent sequence in $\mathbb{R}^3$*): Define a sequence $(x_n)_{n=1}^{\infty}$ in $\mathbb{R}^3$ by $x_n := \left(\frac{1}{n}, \frac{2}{n}, 1\right)$ for all $n \in \mathbb{N}$. Does a limit exist? If yes, what is it? Yes, the sequence converges. Its limit can be computed coordinate-wise:

$$\lim_{n\to\infty} x_n = \left(\lim_{n\to\infty} \frac{1}{n}, \lim_{n\to\infty} \frac{2}{n}, \lim_{n\to\infty} 1\right) = (0, 0, 1).$$

Proposition 4.a.4 (*Uniqueness of limits*): Let (X, d) be a metric space. If a sequence $(x_n)_{n=0}^{\infty}$ in X converges to both $x \in X$ and $x' \in X$, then $x = x'$.

Proof:



Proof idea: Let $\varepsilon > 0$. Since $x_n \to x$, for large n, $d(x_n, x) < \varepsilon$. For ε small enough, y will be outside the green circle ◯ so the x_n will never get closer to y.

Fix $\varepsilon > 0$. By Definition 4.a.1, there exist $N, N' \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon/2$ for all $n \geq N$ and $d(x_n, x') < \varepsilon/2$ for all $n \geq N'$. For any $n \geq \max\{N, N'\}$, the triangle inequality (Definition 2.b.8) gives

$$d(x, x') \leq d(x, x_n) + d(x_n, x') < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $d(x, x') = 0$, so $x = x'$ by definiteness (Definition 2.b.8). **Q.E.D.**

This justifies speaking of *"the"* limit of a convergent sequence, as we do implicitly throughout.

Definition 4.a.5 (*Accumulation point*): Let $(x_n)_{n=0}^{\infty}$ be a sequence in a metric space (X, d). A point $x \in X$ is an "*accumulation point*" (German: „*Häufungspunkt*") of $(x_n)_{n=0}^{\infty}$ if some subsequence $(x_{n_k})_{k=0}^{\infty}$ converges to x.

Remark 4.a.6 (*A claim in the lecture script that is not fully correct*): The official lecture script states that a sequence converges to x if and only if x is its **only** accumulation point (Definition 4.a.5). Sascha Brack's class notes flag this as false in general, with two counterexamples worth recording. First, a sequence can have a unique accumulation point without converging: on $[0,1)$, let $x_{2n+1} := \frac{1}{n}$ and $x_{2n} := 1 - \frac{1}{n}$; the only accumulation point is 0, yet (x_n) does not converge, since the terms $x_{2n} \to 1 \notin [0,1)$ leave every neighbourhood of 0 infinitely often. Second, even in $\mathbb{R}$ itself, the sequence with $x_n := \frac{1}{n}$ for odd n and $x_n := n$ for even n has 0 as its only accumulation point but is unbounded, hence divergent. What **is** true is the easier direction: if (x_n) converges to x, then x is its unique accumulation point (this follows directly from Proposition 4.a.4 together with the fact that every subsequence of a convergent sequence converges to the same limit). The converse needs an extra hypothesis, such as compactness of the ambient space (Section 7.a) together with some control ruling out the first counterexample's boundary-escaping behaviour.

Proposition 4.a.7 (*Open/closed sets via sequences*):

- **(a)** $U \subseteq X$ is **open** if and only if, for every convergent sequence (Definition 4.a.1) in X with limit in U, the sequence eventually lies in U.
- **(b)** $A \subseteq X$ is **closed** if and only if every convergent sequence in A has its limit in A.

Proof:

(a) **(⇒)** Let U be open and let $x_n \to x \in U$. Choose $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq U$ (Definition 3.a.6). By Definition 4.a.1 there is N with $d(x_n, x) < \varepsilon$ for all $n \geq N$, so $x_n \in B_\varepsilon(x) \subseteq U$ from N onwards.

(⇐) By contraposition. If U is not open, some $x \in U$ has $B_{1/n}(x) \not\subseteq U$ for every $n \geq 1$, so we may pick $x_n \in B_{1/n}(x) \setminus U$. Then $d(x_n, x) < \frac{1}{n} \to 0$, so $x_n \to x \in U$, and yet **no** term of the sequence lies in U.

(b) **(⇒)** Let A be closed, (x_n) a sequence in A with $x_n \to x$. If $x \notin A$, then x lies in the open set $X \setminus A$, so by **(a)** the sequence eventually lies in $X \setminus A$, contradicting $x_n \in A$ for all n. Hence $x \in A$.

(⇐) We show $X \setminus A$ is open. If it were not, the same construction as in **(a)** yields $x \in X \setminus A$ and points $x_n \in B_{1/n}(x) \cap A$. Then (x_n) is a sequence in A converging to x, so the hypothesis forces $x \in A$, which is the required contradiction.

Q.E.D.

Remark 4.a.8 (*The shrinking-ball construction*): Both "⇐" directions used the same device: negating *"some ball fits inside U"* gives a bad point in $B_{1/n}(x)$ for **every** n, and collecting those points produces a sequence converging to x. It is worth recognising, because it is the standard bridge from a statement about balls to a statement about sequences, and it is exactly what makes Definition 3.a.6 and Proposition 4.a.7 interchangeable in practice. In the usual division of labour one proves closedness by the sequence criterion and openness by the ball criterion. Note that it needs the radii $\frac{1}{n}$ to shrink to 0, which is where the metric enters; in a general topological space the equivalence can fail.

Lemma 4.a.9 (*Convergence is a topological notion*): Let (X,d) be a metric space. A sequence $(x_n)_n$ in X converges to x if and only if, for every open set $U \ni x$, $(x_n)_n$ eventually (Notation 4.a.2) lies in U.

Proof:

If $x_n \to x$ and $U \ni x$ is open, some $\varepsilon > 0$ has $B_\varepsilon(x) \subseteq U$ (Definition 3.a.6), and x_n eventually lies in $B_\varepsilon(x)$ (Definition 4.a.1), hence in U. Conversely, taking $U := B_\varepsilon(x)$ itself for each $\varepsilon > 0$ recovers exactly Definition 4.a.1. Q.E.D.

Corollary 4.a.10 (*Two metrics with the same topology have the same convergent sequences*): Let X be a set endowed with two metrics d_1 and d_2. Then (X,d_1) and (X,d_2) have the same convergent sequences if and only if the topologies they generate coincide.

Proof:

By Lemma 4.a.9, whether $x_n \to x$ depends only on the collection of open sets, that is, only on the topology d_1 or d_2 generates. Consequently, equal topologies give equal convergent sequences.

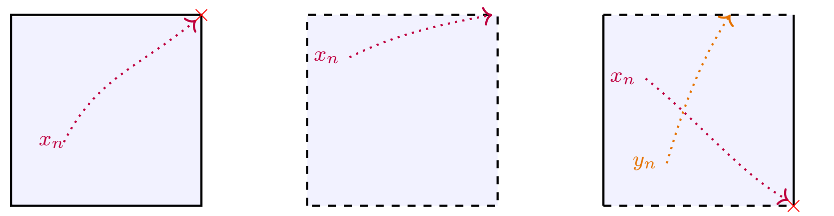
For the converse, suppose $U \subseteq X$ is open with respect to d_1 but not with respect to d_2; we exhibit a sequence distinguishing the two. Since U is not d_2-open, some $x \in U$ has, for every $k \geq 0$, a point $x_k \in B^{d_2}_{2^{-k}}(x) \setminus U$. Then $x_k \to x$ with respect to d_2 (Definition 4.a.1, since $2^{-k} \to 0$), so by hypothesis also $x_k \to x$ with respect to d_1. But U is d_1-open and contains x, so Lemma 4.a.9 forces $(x_k)_k$ to eventually lie in U, contradicting $x_k \notin U$ for every k. So every d_1-open set is d_2-open, and by symmetry the two topologies coincide. Q.E.D.

Remark 4.a.11: This is the precise sense in which *"only the topology matters, not the metric itself"* for convergence. Two visibly different metrics, even one bounded and one unbounded, generate the same convergent sequences exactly when they generate the same open sets, and one never has to compare the metrics directly. Exercise 2.b.27.**2**'s three product metrics are one instance of this; so is Proposition 2.b.13 together with Exercise 2.b.15.

Example 4.a.12 (*Open, closed, and neither in $\mathbb{R}^2$*): In the euclidean metric space $\mathbb{R}^2$, and with Proposition 4.a.7 in hand, each answer is a one-line sequence argument:

- **(a)** $[0,1]^2$ is **closed**: a sequence in $[0,1]^2$ converges coordinatewise (Proposition 6.a.5), and each coordinate limit stays in $[0,1]$ because weak inequalities survive limits.
- **(b)** $(0,1)^2$ is **open**: given $x \in (0,1)^2$, the distance from x to the nearest edge is some $\varepsilon > 0$, and $B_\varepsilon(x) \subseteq (0,1)^2$.
- **(c)** $[0,1] \times (0,1)$ is **neither open nor closed**: it is not open because no ball around $(0,\frac{1}{2})$ fits inside it, and it is not closed because $(\frac{1}{2},\frac{1}{n}) \to (\frac{1}{2},0)$ is a convergent sequence in the set whose limit is not.

The picture below shows exactly these three witnesses. In **(a)** the escaping sequence stays trapped, whereas in **(c)** one sequence escapes through the open edge while another does not.

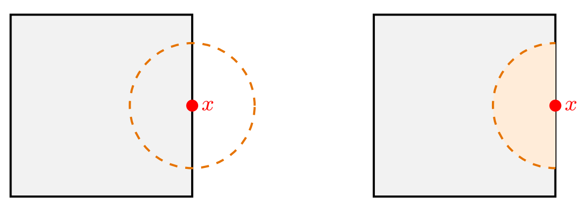


(a) $[0,1]^2$ (**closed**) (b) $(0,1)^2$ (**open**) (c) $[0,1]\times(0,1)$ (**neither**)

Important remark 4.a.13 (*Common misconception!*): In the euclidean metric space $X = [0,1]^2$, the set $[0,1]^2$ **is open**!

$[0,1]^2 \subseteq \mathbb{R}^2$
Not open, since $B_\varepsilon(x)$ is not contained in $[0,1]^2$ if x is on the boundary.

$[0,1]^2 \subseteq [0,1]^2$
Open, since the full space is always open, as shown before.



$[0,1]^2 \subseteq \mathbb{R}^2$ (**not open**) $[0,1]^2 \subseteq [0,1]^2$ (**open**)

Exercise 4.a.14 (*Uniform convergence*):

(a) Write down the definition of convergence of a sequence $(f_n)_{n=0}^\infty$ to f in the metric space X of bounded functions $[0,1]\to\mathbb{R}$ with the supremum metric.

(b) You know the resulting expression from Analysis 1. What is it called?

(c) *(Optional)* Use the fact that $C^0([0,1]) \subseteq X$ is closed and Proposition 4.a.7 on *"sequentially closed"* to conclude a big result from Analysis 1.

Solution to Exercise 4.a.14:

(a) $\forall \varepsilon > 0\ \exists N \in \mathbb{N} : \sup\limits_{x\in[0,1]} |f_n(x) - f(x)| < \varepsilon \quad \forall n \geq N$.

(b) This is *"uniform convergence"* (German: *„gleichmässige Konvergenz"*)!

(c) $C^0([0,1])$ is closed in the space of bounded functions with the supremum metric.

Proof. Let $f \in X \setminus C^0([0,1])$, i.e., $f : [0,1] \to \mathbb{R}$ is bounded and has at least one discontinuity at a point $x_0 \in [0,1]$. That is, we find $\varepsilon > 0$ such that for all $\delta > 0$ there exists $y \in (x_0-\delta, x_0+\delta)$ with $|f(x_0) - f(y)| \geq \varepsilon$. Then the $\frac{\varepsilon}{3}$-ball in X,

$$B_{\varepsilon/3}(f) := \left\{ g \in X : \sup_{x\in[0,1]} |f(x) - g(x)| < \tfrac{\varepsilon}{3} \right\},$$

is contained in $X \setminus C^0([0,1])$.

Why: let $g \in B_{\varepsilon/3}(f)$ and let $\delta > 0$ be arbitrary. Choose $y \in (x_0-\delta, x_0+\delta)$ with $|f(x_0) - f(y)| \geq \varepsilon$. Then

$$\varepsilon \leq |f(x_0) - f(y)| \leq \underbrace{|f(x_0) - g(x_0)|}_{<\ \varepsilon/3} + |g(x_0) - g(y)| + \underbrace{|g(y) - f(y)|}_{<\ \varepsilon/3},$$

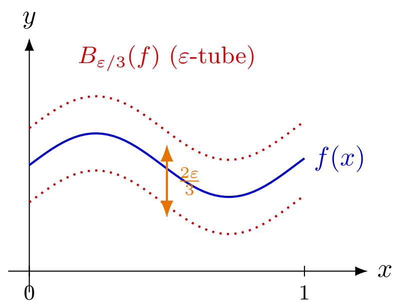
so $|g(x_0) - g(y)| > \varepsilon/3$. As δ was arbitrary, g is discontinuous at x_0 too, i.e. $g \notin C^0([0,1])$. The whole point of the $\frac{\varepsilon}{3}$ is that two of the three terms are spent on moving from f to g and back, leaving a third of the jump behind.

Given this fact, Proposition 4.a.7 ensures that the uniform limit f of $(f_n)_{n=0}^{\infty}$ in $C^0([0,1])$ is also in $C^0([0,1])$, i.e., continuous, given that $f : [0,1] \to \mathbb{R}$ is bounded. This is the case, since f_n is bounded for all $n \in \mathbb{N}$, and with $\varepsilon = 1$ we find $N \in \mathbb{N}$ such that

$$\sup_{x \in [0,1]} |f_N(x) - f(x)| < 1,$$

which implies that f is bounded.

Q.E.D.



Continuity (Chapter 5)

With distance, shape and convergence in place, we can finally ask the question the whole apparatus was built for. What does it mean for a map **between** two metric spaces to respect that structure?

Three answers are given, and they are equivalent: the $\varepsilon-\delta$ condition inherited from **Analysis I**, a sequential condition, and a topological one asking that preimages of open sets be open. Having all three is not redundance. The first is what one verifies for a concrete function, the second is what combines with the sequential arguments of the previous chapter, and the third is the only one that survives when the metric is taken away. The chapter then adds the two strengthenings needed later, uniform continuity and Lipschitz continuity, and locates them in a hierarchy of increasingly demanding conditions.

Continuity (Section 5.a)

Definition 5.a.1 (*Continuous map*): For (X, d_X) and (Y, d_Y) metric spaces, a function $f : X \to Y$ is called "*continuous*" (German: „*stetig*") if one of the following equivalent conditions holds:

- **(a) ($\varepsilon-\delta$):** $\forall x_0 \in X, \varepsilon > 0\ \exists \delta > 0$ such that for all $x \in X$:
$$d(x_0, x) < \delta \implies d(f(x_0), f(x)) < \varepsilon$$
- **(b) (Sequentially continuous):** for any sequence $(x_n)_{n=0}^{\infty}$ in X with limit $x \in X$, $\lim_{n\to\infty} f(x_n) = f(x)$ (and the limit exists).
- **(c) (Topological):** for all $U \subseteq Y$ open, $f^{-1}(U) \subseteq X$ is open (Definition 3.a.6).

AI-Note 5.1: Corsin writes $d(x_0, x) < \varepsilon$ in the $(\varepsilon-\delta)$ hypothesis above, which must read δ, since otherwise the quantified δ is never used at all. This is a slip of the pen, and it is corrected in Definition 5.a.1.

Proposition 5.a.2 (*The three definitions of continuity agree*): The conditions **(a)**, **(b)** and **(c)** of Definition 5.a.1 are equivalent, so Definition 5.a.1 is well posed.

Proof:

We prove the cycle **(a)** ⇒ **(b)** ⇒ **(c)** ⇒ **(a)**.

(a) ⇒ (b). Let $x_n \to x$ and let $\varepsilon > 0$. Pick $\delta > 0$ as in **(a)** for the point x, then N with $d_X(x_n, x) < \delta$ for $n \geq N$. For those n, $d_Y(f(x_n), f(x)) < \varepsilon$, so $f(x_n) \to f(x)$.

(b) ⇒ (c). Let $U \subseteq Y$ be open and suppose $f^{-1}(U)$ were not. By the shrinking-ball construction (Remark 4.a.8) there are $x \in f^{-1}(U)$ and points $x_n \in B_{1/n}(x)$ with $x_n \notin f^{-1}(U)$. Then $x_n \to x$, so $f(x_n) \to f(x) \in U$ by **(b)**, and since U is open, Proposition 4.a.7**(a)** forces $f(x_n) \in U$ for large n. That is, $x_n \in f^{-1}(U)$, which is a contradiction.

(c) ⇒ (a). Fix $x_0 \in X$ and $\varepsilon > 0$. The ball $B_\varepsilon(f(x_0)) \subseteq Y$ is open (AI-Exercise 3.a.5), so $f^{-1}\big(B_\varepsilon(f(x_0))\big)$ is open by **(c)** and contains x_0. Hence there is $\delta > 0$ with $B_\delta(x_0) \subseteq f^{-1}\big(B_\varepsilon(f(x_0))\big)$, which says precisely that $d_X(x_0, x) < \delta$ implies $d_Y(f(x_0), f(x)) < \varepsilon$. **Q.E.D.**

Remark 5.a.3 (*Which of the three to reach for*): Condition **(c)** is the only one that never mentions the metric. That is why it, and not the $\varepsilon-\delta$ formulation, is what survives into general topology (Section 2.a). Condition **(b)** is usually the fastest way to prove a function is **dis**continuous, since a single badly chosen sequence suffices; this is exactly how AI-Example 9.b.6 works. Condition **(a)** is the one to use when a quantitative modulus is wanted, as in Definition 5.a.4 immediately below.

Definition 5.a.4 (*Uniform continuity*): For (X, d_X) and (Y, d_Y) metric spaces, a function $f : X \to Y$ is "*uniformly continuous*" (German: „*gleichmässig stetig*") if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, x' \in X : \quad d_X(x, x') < \delta \text{ implies } d_Y(f(x), f(x')) < \varepsilon.$$

Important remark 5.a.5 (*Where the quantifiers sit*): Compare Definition 5.a.1 **(a)** with Definition 5.a.4: the **formulas** differ only in the order of two quantifiers, and that order is the whole content.

- **Continuity:** $\forall x_0\ \forall \varepsilon\ \exists \delta$. The point x_0 is chosen **before** δ, so δ may depend on it, and a different δ is allowed at every point.
- **Uniform continuity:** $\forall \varepsilon\ \exists \delta\ \forall x$. Now δ is chosen **before** seeing the point, so one and the same δ must work everywhere at once.

Uniform continuity is therefore strictly stronger. The standard witness is $f(x) := 1/x$ on $(0,1)$: it is continuous, but near 0 the graph steepens without bound, so the δ that works at $x_0 = \frac{1}{2}$ is hopeless at $x_0 = \frac{1}{1000}$, and no single δ survives. Proposition 7.a.17 below says that on a **compact** space this can never happen. The gap between the two notions closes exactly when there is no room left to run off towards a bad point.

Example 5.a.6 ($f(x) = x^2$): Let $f : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$. Let $a < b \in \mathbb{R}$. Then

- for $a < 0 < b$: $f^{-1}((a,b)) = (-\sqrt{b}, \sqrt{b})$, which is open.
- for $a < b < 0$: $f^{-1}((a,b)) = \emptyset$.
- for $0 < a < b$: $f^{-1}((a,b)) = (-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b})$.

Why is it enough to check only these cases? Hint: $f^{-1}(U_1 \cup U_2) = f^{-1}(U_1) \cup f^{-1}(U_2)$.

Example 5.a.7 (*Limits in $\mathbb{R}^2$ depend on the path*): When computing limits in $\mathbb{R}^n$, one must ensure that the limit exists and is identical along **all** possible paths approaching the point. For example, consider the function $f : \mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{R}$ given by $f(x,y) = \frac{x^2}{x^2+y^2}$. If we approach the origin along the x-axis (setting $y = 0$ first), we get

$$\lim_{x \to 0} f(x, 0) = \lim_{x \to 0} \frac{x^2}{x^2} = 1.$$

However, if we approach along the y-axis (setting $x = 0$ first), we get

$$\lim_{y \to 0} f(0, y) = \lim_{y \to 0} \frac{0}{y^2} = 0.$$

Since these limits disagree, $\lim_{(x,y) \to (0,0)} f(x,y)$ does not exist. A robust method to check limits at the origin uniformly across all directions is to switch to "*polar coordinates*" ($x = r\cos\varphi, y = r\sin\varphi$). If the limit as $r \to 0$ is independent of φ, then the function converges. (This is explored further in Exercise 5.a.15).

Question 5.a.8: Let (X, d_X) and (Y, d_Y) be metric spaces, where d_Y is the discrete metric. Is **every** function $f : (X, d_X) \to (Y, d_Y)$ continuous?

Answer 5.a.9: No. For example, take the spaces $X = Y = \mathbb{R}$, d_X the standard metric, and $f = \mathrm{id} : X \to Y, z \mapsto z$. Then $\{0\} \subseteq Y$ is open, but $\{0\} = \mathrm{id}^{-1}(\{0\})$ is not open in (X, d_X).

Question 5.a.10: And what if the roles are reversed, with d_X the discrete metric and d_Y arbitrary?

Answer 5.a.11: Any subset of X is open; therefore, $f^{-1}(U)$ is open for any open $U \subseteq Y$. So every such f is continuous.

Exercise 5.a.12 (*2.6 — Continuity of the distance*): Let (X, d) be a metric space, and endow $X \times X$ with the product distance $d_2(x, y) := d(x_1, y_1) + d(x_2, y_2)$. Show that the distance function $d : X \times X \to \mathbb{R}$ is continuous with respect to d_2 and, more precisely, that it is 1-Lipschitz. Is d also continuous with respect to the distance $d_1(x, y) := \max\{d(x_1, y_1), d(x_2, y_2)\}$? Is d also Lipschitz continuous with respect to the distance $d_3(x,y) := \sqrt{d(x_1,y_1)^2 + d(x_2,y_2)^2}$? (Notation consistent with Exercise 2.b.27.)

Exercise 5.a.13 (*2.7 — Continuity of the composition*): Let X, Y, Z be metric spaces and let $f : X \to Y$, $g : Y \to Z$ be continuous functions. Show that $g \circ f : X \to Z$ is continuous using at least two of the three equivalent definitions of continuity seen in class.

Remark 5.a.14: The three equivalent definitions of continuity referenced in Exercise 5.a.13 are exactly the ones stated in Section 5.a below (ε–δ, sequential, topological).

Exercise 5.a.15 (*3.2 — Problems at the origin*): Show that there is no continuous function $f : \mathbb{R}^2 \to \mathbb{R}$ such that $f(x,y) = xy/(x^2+y^2)$ for all $(x,y) \neq (0,0)$. On the other hand, show that there is exactly one continuous function $g : \mathbb{R}^2 \to \mathbb{R}$ such that $g(x,y) = xy/\sqrt{x^2+y^2}$ for all $(x,y) \neq (0,0)$.
Official hint: if a function is continuous at 0 and $x_k \to 0$ and $y_k \to 0$, then $f(x_k)$ and $f(y_k)$ must converge to the same value.

Exercise 5.a.16 (*Distance to a point is continuous*): Let (X,d) be a metric space and fix $a \in X$. Show that the function $f : X \to [0,\infty)$ defined by $f(x) := d(x,a)$ is continuous.

Proof of Exercise 5.a.16:

We use ε–δ continuity. Let $x \in X$ and $\varepsilon > 0$. We need to find $\delta > 0$ such that for all $x' \in X$ with $d(x,x') < \delta$, we have $|f(x) - f(x')| < \varepsilon$. Using the reverse triangle inequality, we have

$$|f(x) - f(x')| = |d(x,a) - d(x',a)| \leq d(x,x').$$

Therefore, we can simply choose $\delta := \varepsilon$. Then $d(x,x') < \delta$ directly implies $|f(x) - f(x')| < \varepsilon$, meaning f is continuous at x. Since x was arbitrary, f is continuous everywhere. **Q.E.D.**

Corollary 5.a.17 (*Closed sets are exactly the zero sets of a distance function*): Let (X,d) be a metric space and $\emptyset \neq E \subseteq X$. The function $f_E(x) := d(x,E) = \inf_{z \in E} d(x,z)$ is 1-Lipschitz, which is Exercise 3.c.26**(c)**, proved there by the same reverse-triangle-inequality argument as Exercise 5.a.16 above. Moreover,

$$E \text{ is closed} \quad \Longleftrightarrow \quad E = \{x \in X : f_E(x) = 0\}.$$

Proof:

The inclusion $E \subseteq \{f_E = 0\}$ always holds, since $z \in E$ gives $f_E(z) \leq d(z,z) = 0$. So only the reverse inclusion carries content, and it is exactly where closedness is needed.

($\Rightarrow$) Let E be closed and $f_E(x) = 0$. By definition of the infimum, choose $z_n \in E$ with $d(x,z_n) \to 0$; then $z_n \to x$, and closedness of E gives $x \in E$ (Proposition 4.a.7**(b)**).

($\Leftarrow$) Suppose $E = \{f_E = 0\}$ and let (z_n) be a sequence in E with $z_n \to x$. Then $f_E(x) \leq d(x,z_n) \to 0$, so $f_E(x) = 0$, i.e. $x \in E$ by hypothesis. Proposition 4.a.7**(b)** then gives that E is closed. **Q.E.D.**

Remark 5.a.18: This sharpens Exercise 3.c.26**(a)**, which showed only one direction (E closed $\Rightarrow$ $f_E > 0$ off E) for a general subset. It is exactly closedness, and nothing weaker, that lets f_E detect membership in E by vanishing.

Exercise 5.a.19 (*Evaluation map is continuous*): Let $X := C^0([0,1])$ equipped with the supremum metric $d(f,g) := \sup_{t\in[0,1]} |f(t) - g(t)|$. Define the function $F : X \to \mathbb{R}$ by $F(f) := f(0)$. Show that F is continuous.

Proof of Exercise 5.a.19:

Let $f \in X$ and $\varepsilon > 0$. Choose $\delta := \varepsilon$. If $g \in X$ satisfies $d(f,g) < \delta$, then by the definition of the supremum metric, we have $|f(t) - g(t)| < \delta$ for all $t \in [0,1]$. In particular, for $t = 0$, we get

$$|F(f) - F(g)| = |f(0) - g(0)| \leq \sup_{t\in[0,1]} |f(t) - g(t)| = d(f,g) < \delta = \varepsilon.$$

Thus F is continuous at f. Since f was arbitrary, F is continuous on all of X. **Q.E.D.**

Lipschitz continuity (Section 5.b)

AI-Note 5.2: The course never formally introduces Lipschitz continuity, even though a Lipschitz constant is already used informally for contraction mappings (Definition 6.c.16, below). The definition and the hierarchy below are therefore supplied here, and will not be found under this name in the lecture material.

Definition 5.b.20 (*Lipschitz continuity*): Let (X, d_X), (Y, d_Y) be metric spaces. A function $f : X \to Y$ is "*Lipschitz continuous*" (German: „*Lipschitz-stetig*") if there exists $L \geq 0$, a "*Lipschitz constant*" for f, such that

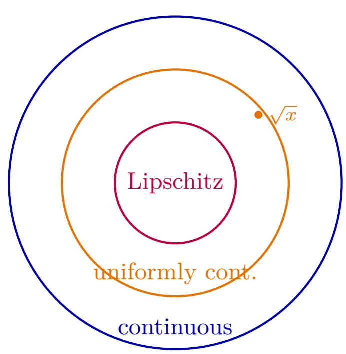
$$d_Y(f(x), f(x')) \leq L \cdot d_X(x, x') \quad \text{for all } x, x' \in X.$$

Remark 5.b.21 (*Hierarchy of continuity notions*): Lipschitz continuity is stronger than uniform continuity (Definition 5.a.4), which in turn is stronger than continuity (Definition 5.a.1). For the first implication, assume $L > 0$ (for $L = 0$ the function is constant and there is nothing to prove); then $\delta := \varepsilon/L$ satisfies the definition of uniform continuity, since

$$d_X(x, x') < \frac{\varepsilon}{L} \quad \text{gives} \quad d_Y(f(x), f(x')) \leq L\, d_X(x, x') < \varepsilon$$

for **every** pair x, x'. This is one and the same δ at every point, which is exactly what uniform continuity asks for. The second implication is immediate: a δ that works at all points in particular works at each one.

Both converses fail. Every continuously differentiable function on a compact **interval** is Lipschitz continuous, with $L := \sup|f'|$ by the mean value theorem. The word **interval** matters here, since the mean value theorem needs the segment between the two points to lie in the domain. But $f(x) := \sqrt{x}$ on $[0, \infty)$ is uniformly continuous and **not** Lipschitz. A Lipschitz constant would force δ to shrink only linearly in ε, whereas here $|\sqrt{x} - \sqrt{x'}| < \varepsilon$ near 0 already requires δ of order ε^2. The reason is visible in the graph, since the tangent at 0 is vertical and no finite slope L can dominate it.



Exercise 5.b.22 (*Classifying continuity*): For each of the following functions, determine whether it is continuous, uniformly continuous, and/or Lipschitz continuous on the given domain:

Function	Domain
$\sqrt{x}$	$[0,\infty)$
$1/x$	$(0,1)$
$\operatorname{sgn}(x)$	$(-1,1)$
$\lvert x\rvert^{3/2}$	$(-1,1)$

Exercise 5.b.23 (*3.1 — Lipschitz or not*)**:** Give an example of:

(a) A $\frac{1}{2}$-Lipschitz function $f:[0,1)\to[0,1)$ that has no fixed point.

(b) A map $f:\mathbb{R}^2\to\mathbb{R}^2$ that has no fixed point, but is an isometry, i.e. $\|f(x)-f(x')\|=\|x-x'\|$ for all $x,x'\in\mathbb{R}^2$.

Exercise 5.b.24 (*3.6 — Multiple choice (uniform continuity)*)**:** Select all the statements below that are true.

(a) The function $(x,y)\mapsto x+y$ is uniformly continuous in $\mathbb{R}^2$.

(b) The function $(x,y)\mapsto xy$ is uniformly continuous in $\mathbb{R}^2$.

(c) The function $(x,y)\mapsto x+y$ is uniformly continuous in $[0,1]^2$.

(d) The function $(x,y)\mapsto xy$ is uniformly continuous in $[0,1]^2$.

Remark 5.b.25: Corsin's third reason to care about compactness (Section 7.c) is precisely that a continuous function on a compact space is uniformly continuous, which settles Exercise 5.b.24(c) and (d) at once.

Exercise 5.b.26 (*3.11 — (*) Uniform continuity and moduli of continuity*)**:** Let (X,d_X) and (Y,d_Y) be metric spaces and let $f:X\to Y$.

A **modulus of continuity** is a function $\omega:[0,\infty)\to[0,\infty]$ such that

$$\omega(0)=0,\qquad \omega\text{ is nondecreasing},\qquad \lim_{t\downarrow 0}\omega(t)=0.$$

(Here $[0,\infty]$ means we also allow the value $+\infty$.) We say that f **has modulus of continuity** ω if

$$d_Y\big(f(x),f(y)\big)\le\omega\big(d_X(x,y)\big)\qquad\forall x,y\in X.$$

1. Prove that f is uniformly continuous on X if and only if f has a modulus of continuity.
2. *(Bonus 1)* Assume $X\subset\mathbb{R}$ is an interval with the usual distance $d_X(x,y)=|x-y|$. Show that the modulus constructed in (1) can be chosen **subadditive**, i.e. $\omega(t_1+t_2)\le\omega(t_1)+\omega(t_2)$ for all $t_1,t_2\ge 0$.
3. *(Bonus 2)* Assume $X\subset\mathbb{R}^n$ is **convex**, meaning: for all $x_1,x_2\in X$ and all $\lambda\in[0,1]$, $(1-\lambda)x_1+\lambda x_2\in X$. With $d_X(x,y)=\|x-y\|$ (Euclidean distance), show again that the modulus from (1) can be chosen subadditive.

Remark 5.b.27 (*official*)**:** In general metric spaces a modulus of continuity may have infinite value for finite $t>0$. Example: $X=\mathbb{Z}$ with the usual distance, $Y=\mathbb{R}$, $f(n)=n^2$. This f is uniformly continuous on $\mathbb{Z}$ (because points closer than 1 must be equal), but no finite $\omega(1)$ can bound $|f(n+1)-f(n)|=2n+1$ for all n. Allowing the value $+\infty$ avoids this issue, and in many familiar settings (e.g. intervals/convex sets in $\mathbb{R}^n$) the resulting $\omega_f(t)$ is finite for all t.

Official hints: **3.11.1:** define the "worst oscillation at scale t" $\omega_f(t):=\sup\{d_Y(f(x),f(y)):d_X(x,y)\le t\}$. **3.11.2 / 3.11.3:** if $d_X(x,y)\le t_1+t_2$, choose an intermediate point z on the segment from x to y so that $d_X(x,z)\le t_1$ and $d_X(z,y)\le t_2$, then use the triangle inequality in Y.

Solutions (Section 5.c)

Solution to Exercise 5.a.12:

d is 1-Lipschitz with respect to d_2. Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be points of $X \times X$. Applying the triangle inequality in X twice,

$$d(x_1, x_2) \;\le\; d(x_1, y_1) + d(y_1, y_2) + d(y_2, x_2),$$

so $d(x_1, x_2) - d(y_1, y_2) \le d(x_1, y_1) + d(x_2, y_2) = d_2(x, y)$. Exchanging the roles of x and y bounds the opposite difference in the same way, hence

$$\big|d(x_1, x_2) - d(y_1, y_2)\big| \;\le\; d_2(x, y),$$

which is 1-Lipschitz continuity (Definition 5.b.20) and therefore continuity (Remark 5.b.21).

For d_1 and d_3: yes, still Lipschitz, but the constant changes. By Exercise 2.b.27 part 2 we have $d_2 \le 2d_1$ and $d_2 \le \sqrt{2}\, d_3$, so

$$\big|d(x_1, x_2) - d(y_1, y_2)\big| \le 2\, d_1(x, y) \qquad \text{and} \qquad \big|d(x_1, x_2) - d(y_1, y_2)\big| \le \sqrt{2}\, d_3(x, y).$$

This is a general principle rather than a computation: **Lipschitz continuity survives passage to an equivalent metric**, with the constant multiplied by the comparison factor. Continuity survives too, being topological, and by Exercise 2.b.27 the three product metrics induce the same topology.

Remark 5.c.28 (*The constant 2 cannot be improved*): The constant 2 for d_1 is sharp, so d is genuinely **not** 1-Lipschitz with respect to d_1. Take $X = \mathbb{R}$ with $x := (0, 0)$ and $y := (-1, 1)$. Then $d(x_1, x_2) = 0$ and $d(y_1, y_2) = 2$, while $d_1(x, y) = \max\{1, 1\} = 1$, so the difference equals exactly $2 = 2\, d_1(x, y)$. This is worth checking, because the problem asks only **whether** d is Lipschitz for d_1 and d_3, and it is easy to answer *“yes, with the same constant”* without noticing that the constant has in fact changed.

Q.E.D.

Solution to Exercise 5.a.13:

All three arguments are short; the point of giving more than one is to see how differently the same fact reads through the three definitions of Definition 5.a.1.

Topological (Definition 5.a.1(c)). This is the shortest, because preimages compose. Let $W \subseteq Z$ be open. Then

$$(g \circ f)^{-1}(W) = f^{-1}\big(g^{-1}(W)\big),$$

and $g^{-1}(W)$ is open in Y by continuity of g, so its preimage under f is open in X by continuity of f. Hence $g \circ f$ is continuous.

Sequential (Definition 5.a.1(b)). Let $x_n \to x$ in X. Continuity of f gives $f(x_n) \to f(x)$ in Y, and continuity of g applied to **that** sequence gives $g(f(x_n)) \to g(f(x))$ in Z.

ε–δ (Definition 5.a.1(a)). Fix $x_0 \in X$ and $\varepsilon > 0$. Continuity of g at $f(x_0)$ supplies $\eta > 0$ such that $d_Y(f(x_0), y) < \eta$ implies $d_Z\big(g(f(x_0)), g(y)\big) < \varepsilon$. Continuity of f at x_0, applied to **that** η, supplies $\delta > 0$ such that $d_X(x_0, x) < \delta$ implies $d_Y(f(x_0), f(x)) < \eta$. Chaining the two, $d_X(x_0, x) < \delta$ implies $d_Z\big((g \circ f)(x_0), (g \circ f)(x)\big) < \varepsilon$.

Remark 5.c.29 (*Why the topological proof is the short one*): All three arguments have the same shape, applying the hypothesis for g first and then feeding the result to f. Only the topological version needs no quantifiers at all, because the set identity $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ does the book-keeping the other two proofs carry out by hand. This is a fair advertisement for Definition 5.a.1**(c)**, which can look like the least intuitive of the three on first meeting: it is the formulation in which composition is trivial, and that is exactly why general topology takes it as **the** definition.

Q.E.D.

Solution to Exercise 5.b.22:

- $\sqrt{x}$ on $[0,\infty)$: **uniformly continuous, not Lipschitz.** Uniform continuity follows from continuity on $[0,1]$ (compact, Item **(c)**) together with $\frac{1}{2}$-Lipschitz behaviour on $[1,\infty)$ (since $|\sqrt{x}-\sqrt{y}| \le \frac{1}{2}|x-y|$ there by the mean value theorem, as $|(\sqrt{\cdot})'| = \frac{1}{2\sqrt{x}} \le \frac{1}{2}$ for $x \ge 1$). Not Lipschitz near 0: for $x_n := 1/n^2 \to 0$, $\dfrac{|\sqrt{x_n}-\sqrt{0}|}{|x_n - 0|} = n \to \infty$, so no global L works.
- $1/x$ on $(0,1)$: **continuous, neither uniformly nor Lipschitz continuous.** For $x_n := 1/n$, $y_n := 1/(2n)$, $|x_n - y_n| \to 0$ but $|1/x_n - 1/y_n| = n \to \infty$, so no uniform $\delta(\varepsilon)$ (and hence no Lipschitz constant) can work near 0.
- $\operatorname{sgn}(x)$ on $(-1,1)$: **none of the three**, being discontinuous at 0.
- $|x|^{3/2}$ on $(-1,1)$: **Lipschitz, hence also uniformly continuous and continuous.** $|x|^{3/2}$ is C^1 on $(-1,1)\setminus\{0\}$ with $\left|\frac{d}{dx}|x|^{3/2}\right| = \frac{3}{2}|x|^{1/2} \le \frac{3}{2}$, and the bound extends across 0 by direct estimation, so $L := \frac{3}{2}$ works globally.

Q.E.D.

Solution to Exercise 5.b.23:

Both parts ask for a map with no fixed point, and each defeats one hypothesis of Theorem 6.c.17 while keeping the other, which is exactly what Remark 6.c.18 predicts must be possible.

(a) Take $f(x) := \frac{x+1}{2}$. It maps $[0,1)$ into $[\frac{1}{2},1) \subseteq [0,1)$, and $|f(x)-f(y)| = \frac{1}{2}|x-y|$, so it is $\frac{1}{2}$-Lipschitz. A fixed point would satisfy $x = \frac{x+1}{2}$, i.e. $x=1$, which is not in $[0,1)$. **What fails is completeness:** $[0,1)$ is not a complete metric space, the iterates $x_{n+1} = f(x_n)$ march towards the missing endpoint 1, and Theorem 6.c.17 does not apply.

(b) Take the translation $f(x) := x + (1,0)$. It is an isometry, since $\|f(x)-f(x')\| = \|x-x'\|$ exactly, and $f(x) = x$ would force $(1,0) = 0$. **What fails is the strict contraction:** an isometry has Lipschitz constant $L = 1$, not $L < 1$, and $\mathbb{R}^2$ is perfectly complete. Together with part **(a)** this shows neither hypothesis of the fixed point theorem can be dropped.

Q.E.D.

Solution to Exercise 5.a.15:

No continuous extension exists for $xy/(x^2+y^2)$. Suppose f were continuous on $\mathbb{R}^2$ and agreed with $xy/(x^2+y^2)$ off the origin. Approach the origin along two different lines:

$$f\left(\tfrac{1}{k},\tfrac{1}{k}\right) = \frac{1/k^2}{2/k^2} = \frac{1}{2}, \qquad f\left(\tfrac{1}{k},0\right) = 0 \quad \text{for every } k.$$

Both sequences tend to $(0,0)$, so continuity would force $f(0,0) = \frac{1}{2}$ and $f(0,0) = 0$ simultaneously. Hence no such f exists. (This is the same function, and the same argument, as AI-Example 9.b.6, where it reappears to show that the existence of both partial derivatives implies nothing.)

Exactly one continuous extension exists for $xy/\sqrt{x^2+y^2}$. Write $(x,y) = (r\cos\theta, r\sin\theta)$ with $r > 0$. Then

$$g(x,y) = \frac{r^2\cos\theta\sin\theta}{r} = r\cos\theta\sin\theta, \qquad \text{so} \qquad |g(x,y)| \le \frac{r}{2} = \frac{\|(x,y)\|}{2}.$$

The bound is uniform in θ, so $g(x,y) \to 0$ as $(x,y) \to (0,0)$, and setting $g(0,0) := 0$ produces a continuous function on all of $\mathbb{R}^2$.

Uniqueness is the part worth spelling out, and it needs no computation at all: the value at the origin is **forced**. Since $(0,0)$ lies in the closure of $\mathbb{R}^2\setminus\{(0,0)\}$, any two continuous extensions agree on a dense subset of $\mathbb{R}^2$, and two continuous functions agreeing on a dense set agree everywhere. To see the latter, take a sequence from the dense set converging to the point in question and apply

Definition 5.a.1**(b)** to both. So the extension is unique whenever it exists, which is also why the first half of the problem had only to exhibit **two** conflicting limits. **Q.E.D.**

Solution to Exercise 5.b.24:

(a), (c) and (d) are true; only (b) is false.

(a) True. Addition is Lipschitz, hence uniformly continuous (Remark 5.b.21):

$$\big|(x+y)-(x'+y')\big| \le |x-x'|+|y-y'| \le \sqrt{2}\,\big\|(x,y)-(x',y')\big\|,$$

the last step by Cauchy–Schwarz. So $\delta := \varepsilon/\sqrt{2}$ works everywhere at once.

(b) False. Multiplication is continuous but not uniformly so on the unbounded domain $\mathbb{R}^2$. Take

$$p_k := \big(k,\ \tfrac{1}{k}\big)\,, \qquad q_k := (k,\ 0)\,.$$

Then $\|p_k - q_k\| = \frac{1}{k} \to 0$, while the values are $k \cdot \frac{1}{k} = 1$ and 0, a constant gap of 1. So no δ can work for $\varepsilon < 1$. The failure is exactly the one described in Important remark 5.a.5: the function steepens without bound as one runs off to infinity, so the local δ shrinks without a positive floor.

(c) and (d) True, and both for free: $[0,1]^2$ is closed and bounded, hence **compact** by Theorem 7.b.23, and a continuous function on a compact metric space is automatically uniformly continuous by Proposition 7.a.17 (Corsin's third reason to care about compactness, Item **(c)**). No estimate is needed for either.

Remark 5.c.30: The pair **(b)** versus **(d)** is the whole lesson: the **same** function xy is uniformly continuous on $[0,1]^2$ and not on $\mathbb{R}^2$. Uniform continuity is therefore not a property of a formula but of a formula **together with its domain**, and compactness of the domain is what removes the room to run off to a bad point.

Q.E.D.

Solution to Exercise 5.b.26:

1. **($\Leftarrow$) A modulus of continuity forces uniform continuity.** Suppose $d_Y(f(x),f(y)) \le \omega(d_X(x,y))$ for all x,y, with ω as in the statement. Let $\varepsilon > 0$. Since $\omega(t) \to 0$ as $t \downarrow 0$, there is $\delta > 0$ with $\omega(\delta) < \varepsilon$. If $d_X(x,y) < \delta$ then, ω being nondecreasing,

$$d_Y\big(f(x),f(y)\big) \le \omega\big(d_X(x,y)\big) \le \omega(\delta) < \varepsilon.$$

The δ did not depend on x or y, which is precisely Definition 5.a.4.

($\Rightarrow$) Uniform continuity produces one. Following the official hint, define the "*modulus of continuity of f*" as its worst oscillation at scale t:

$$\omega_f(t) := \sup\big\{\, d_Y(f(x),f(y)) \ :\ x,y \in X,\ d_X(x,y) \le t \,\big\} \ \in\ [0,\infty].$$

It has all three required properties. $\omega_f(0) = 0$, because $d_X(x,y) \le 0$ forces $x = y$ by definiteness and then $d_Y(f(x),f(y)) = 0$. It is nondecreasing, because enlarging t enlarges the set over which the supremum is taken. And $\omega_f(t) \to 0$ as $t \downarrow 0$: given $\varepsilon > 0$, uniform continuity supplies $\delta > 0$ with $d_X(x,y) < \delta \Rightarrow d_Y(f(x),f(y)) < \varepsilon$, so every $t < \delta$ has $\omega_f(t) \le \varepsilon$. Finally f has modulus ω_f by construction, since $d_Y(f(x),f(y))$ is one of the numbers the supremum defining $\omega_f(d_X(x,y))$ ranges over.

2. **(Bonus 1) On an interval, ω_f is already subadditive.** Let $t_1,t_2 \ge 0$ and take any $x,y \in X$ with $|x-y| \le t_1+t_2$. Choose an intermediate point z on the segment from x to y as follows: if $|x-y| \le t_1$ put $z := y$; otherwise let z be the point of that segment at distance exactly t_1 from x, so that $|z-y| = |x-y| - t_1 \le t_2$. Either way $z \in X$, because X is an interval and z lies between x and y, and $|x-z| \le t_1$, $|z-y| \le t_2$. The triangle inequality in Y now gives

$$d_Y\big(f(x),f(y)\big) \ \le\ d_Y\big(f(x),f(z)\big) + d_Y\big(f(z),f(y)\big) \ \le\ \omega_f(t_1) + \omega_f(t_2).$$

Taking the supremum over all admissible x,y yields $\omega_f(t_1+t_2) \le \omega_f(t_1) + \omega_f(t_2)$.

3. **(Bonus 2) Convexity is exactly what the argument needed.** Repeat part 2 verbatim with $|x-y|$ replaced by $\|x-y\|$: the only property of an interval that was used is that the segment from x to y lies inside X, and that is the definition of convexity. The intermediate point is $z := x + \min\{t_1, \|x-y\|\} \frac{y-x}{\|y-x\|}$ for $x \neq y$, which lies on the segment and satisfies the same two estimates.

Remark 5.c.31 (*Why the value $+\infty$ has to be allowed*): The official remark accompanying this problem gives the reason, and it is worth restating: on $X = \mathbb{Z} \subseteq \mathbb{R}$ the map $f(n) := n^2$ is uniformly continuous for the trivial reason that any two distinct points are at distance at least 1. Take $\delta := \frac{1}{2}$, and the implication holds vacuously. Yet $|f(n+1) - f(n)| = 2n+1$ is unbounded, so $\omega_f(1) = +\infty$. Parts 2 and 3 rule this out by hypothesis rather than by luck: on a convex set the intermediate-point construction lets a large displacement be broken into many small ones, so $\omega_f(t) \leq \lceil t/s \rceil \, \omega_f(s)$ is finite as soon as some $\omega_f(s)$ is. A space like $\mathbb{Z}$ has no intermediate points to subdivide with.

Q.E.D.

Completeness and the Banach Fixed Point Theorem (Chapter 6)

Having examined how maps behave between metric spaces, we return to a property of a single space. Does every sequence that *"looks like"* it ought to converge (Definition 4.a.1) actually do so? A space where the answer is yes is called complete.

Completeness is the first genuinely **metric** rather than topological property met in this part, and the distinction is sharp: two metrics can generate exactly the same open sets while only one of them is complete. What completeness is usually good for is producing objects one cannot write down. The Banach fixed point theorem at the end of the chapter is the standard instrument for that, and it reappears as the engine of both the inverse function theorem and the Picard–Lindelöf existence theorem later on.

Completeness (Section 6.a)

Definition 6.a.1 (*Cauchy sequence and complete metric space*): Let (X,d) be a metric space and $(x_n)_{n=0}^{\infty}$ a sequence in X. We say that $(x_n)_{n=0}^{\infty}$ is "*Cauchy*" if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N} \text{ such that } d(x_n, x_m) < \varepsilon \quad \forall n, m \geq N.$$

(X,d) is a "*complete metric space*" (German: „*vollständiger metrischer Raum*“) if every Cauchy sequence in X converges with respect to d.

Exercise 6.a.2 (*Elementary facts about Cauchy sequences*): Let (X,d) be a metric space and $(x_n)_{n=0}^{\infty}$ a sequence in X. Prove:

- **(a)** Every Cauchy sequence is bounded (meaning $\{d(x_n,x_0) : n \geq 0\} \subset \mathbb{R}$ is a bounded set).
- **(b)** Every convergent sequence is Cauchy.
- **(c)** A Cauchy sequence converges if and only if it has a convergent subsequence.

Remark 6.a.3: Part **(c)** is the standard route to proving completeness of a space **without** guessing the limit in advance: instead of constructing a candidate limit directly, one only has to extract **some** convergent subsequence, and **(c)** upgrades that to convergence of the whole sequence. This is exactly the shape of the argument used throughout Chapter 7.

Remark 6.a.4 (*The converse construction*): Incompleteness is a defect that can always be repaired. Given **any** metric space (X,d), complete or not, one can build a complete space $(\overline{X},\overline{d})$ containing an isometric copy of X as a dense subset. This is the "*completion*" of (X,d), and $\mathbb{R}$ as the completion of $\mathbb{Q}$ is the motivating case. The construction is carried out in Section 6.b below.

That specific route to $\mathbb{R}$ itself is not repeated here, the course script marking it as extra material belonging to **Grundstrukturen**. What is worth taking from the general construction is that no metric space is irreparably incomplete, so the interesting question is never whether a completion exists, but whether the one you already have is complete.

Proposition 6.a.5 (*Convergence in $\mathbb{R}^n$ is coordinatewise; $\mathbb{R}^n$ is complete*): Equip $\mathbb{R}^n$ with the standard metric d_2, and write $x_m = (x_{m,1}, \ldots, x_{m,n})$.

- **(a)** $x_m \to x$ in $\mathbb{R}^n$ if and only if $x_{m,i} \to x_i$ in $\mathbb{R}$ for every $i = 1, \ldots, n$.
- **(b)** $(\mathbb{R}^n, d_2)$ is complete.

Proof:

Everything follows from the two-sided estimate, valid for every j:

$$|x_{m,j} - x_j| \ \leq\ \|x_m - x\| \ =\ \sqrt{\sum_{i=1}^{n} |x_{m,i} - x_i|^2} \ \leq\ \sqrt{n}\, \max_i |x_{m,i} - x_i|.$$

(a) If $\|x_m - x\| \to 0$, the left inequality forces each coordinate to converge. Conversely, if every coordinate converges then $\max_i |x_{m,i} - x_i| \to 0$ (a maximum of **finitely** many null sequences), so the right inequality forces $\|x_m - x\| \to 0$.

(b) Let (x_m) be Cauchy in $\mathbb{R}^n$. The same left inequality gives $|x_{m,j} - x_{k,j}| \le \|x_m - x_k\|$, so each coordinate sequence $(x_{m,j})_m$ is Cauchy in $\mathbb{R}$. Now $\mathbb{R}$ is complete, this being the completeness axiom from **Analysis I** and the only input the proof has, so each coordinate sequence converges, say $x_{m,j} \to x_j$. Setting $x := (x_1, \dots, x_n)$, part **(a)** gives $x_m \to x$. **Q.E.D.**

Remark 6.a.6 (*How little was actually used*): Completeness of $\mathbb{R}^n$ is completeness of $\mathbb{R}$, applied n times. Nothing else enters, which is worth noticing because it shows exactly where the argument would break in infinite dimensions. There $\max_i$ runs over infinitely many coordinates and need not tend to 0 even when every individual coordinate does. That failure is the same phenomenon behind the ε-net argument in Lemma 7.a.12.

Example 6.a.7 (*Completeness examples*):

- The Euclidean spaces $(\mathbb{R}^n, \langle\cdot,\cdot\rangle_{\text{eucl}})$ are **complete**.
- Any **non-closed** subset $U \subseteq \mathbb{R}^n$ (standard metric) and the rationals $\mathbb{Q}$ are **not complete**.
- $\big(C^0([0,1]), \sup_{x\in[0,1]} |\cdot|\big)$ is **complete** (Zorich, p. 22).
- Every set X equipped with the **discrete metric** is **complete** (e.g., the finite grid $X := \{(x,y) \in \mathbb{Z}^2 \mid |x|, |y| \le 3\}$). In such a space, every Cauchy sequence is eventually constant, hence convergent.

Remark 6.a.8 (*Completeness is not the same as closedness*): The second bullet is easy to over-read as *"complete means closed"*. It does not, and the difference is worth pinning down now (because it recurs).

Closedness is relative, completeness is intrinsic. *"Closed"* only means anything once you say **closed in what**, since it is a statement about how a set sits inside an ambient space. Completeness asks nothing about an ambient space, but only whether the Cauchy sequences of the space converge **within it**. So:

- $\mathbb{Q}$ is **closed in** $\mathbb{Q}$, every space being closed in itself (Exercise 3.b.16), and yet **not complete**: the decimal approximations $1, 1.4, 1.41, 1.414, \dots$ form a Cauchy sequence in $\mathbb{Q}$ that converges to nothing there. (Alternatively, the sequence defined recursively by $x_0 := 1$ and $x_{n+1} := \frac{x_n}{2} + \frac{1}{x_n}$ is Cauchy in $\mathbb{Q}$, but its limit $\sqrt{2}$ is not in $\mathbb{Q}$.)
- $(0,1)$ is **not closed in** $\mathbb{R}$ and also not complete, which is the second bullet's case. The two failures nevertheless have different causes, and the first does not imply the second in general.

What **is** true is a one-directional statement, and only inside a complete ambient space: if X is complete and $A \subseteq X$, then

$$A \text{ closed in } X \iff A \text{ complete.}$$

This is why the second bullet is correct **as stated for** $\mathbb{R}^n$. It silently uses completeness of the ambient $\mathbb{R}^n$. Drop that hypothesis and the equivalence dies, as $\mathbb{Q}$ inside $\mathbb{Q}$ shows.

The same relative-versus-intrinsic distinction lies behind Important remark 4.a.13 ($[0,1]^2$ is open in itself, not in $\mathbb{R}^2$), and it returns as Remark 7.b.30. Compactness, by contrast, is intrinsic in the way completeness is: a subset K is compact exactly when the metric space $(K, d|_{K\times K})$ is, with no reference to any ambient space. That is precisely the reduction used in Definition 7.a.6.

Remark 6.a.9: The second bullet of Remark 6.a.8's example, in which $(0,1) \subset \mathbb{R}$ is open and not complete, is not a quirk of that particular interval: script Exercise 9.76 asks for exactly this in general, and the fact is already in this document as Exercise 7.c.36**(b)** below. **Every** proper, non-empty open subset of $\mathbb{R}^n$ fails to be complete, and for the same underlying reason. That solution gives the topological proof via connectedness, and a remark alongside it records the script's more elementary, connectedness-free alternative.

Exercise 6.a.10 (*Completeness is not a topological property*): Corsin does not address whether completeness survives a change of metric; Diego Torres Tejeda sets this exercise, which settles it. Let $X := (0,1]$, let $d_{\mathrm{eucl}}(x,y) := |x-y|$ be the standard metric on X, and define

$$d(x,y) := \left|\frac{1}{x} - \frac{1}{y}\right|, \qquad x, y \in X.$$

(a) Show that d is a metric on X.
(b) Show that (X, d_{eucl}) is **not** complete.
(c) Show that (X, d) **is** complete.
(d) Show that d_{eucl} and d induce the **same topology** on X.

Diego's hint for **(c)**: let (x_n) be d-Cauchy. Then $(1/x_n)$ is Cauchy for the standard metric, so it has a limit $\ell \in \mathbb{R}$; show $\ell \geq 1$ and that $x_n \to 1/\ell$.

AI-Note 6.1: Diego states this exercise on $X = (0,\infty)$, where part **(c)** is false. On $(0,\infty)$ the map $x \mapsto 1/x$ is a bijection **onto** $(0,\infty)$, so (X,d) is isometric to $\big((0,\infty), |\cdot|\big)$, which is not complete: concretely, $x_n := n$ satisfies $d(x_n, x_m) = |\frac{1}{n} - \frac{1}{m}| \to 0$ yet has no limit in X. His own hint asks to show $\ell > 0$, and that is precisely the step that fails. Restricting to a bounded interval repairs it, since on $(0,1]$ the image is $[1,\infty)$, which **is** closed in $\mathbb{R}$ and hence complete. The exercise is therefore typeset with $X := (0,1]$ above.

Compactness will turn out to be a strictly stronger finiteness-type property than completeness: every compact metric space is complete (Exercise 7.a.20), but the converse fails. Note that $\mathbb{R}^n$ itself is complete and yet, being unbounded, not compact, as we will see next (Theorem 7.a.3, Section 7.b).

The completion of a metric space (Section 6.b)

Remark 6.a.8 left one question open. We know $\mathbb{Q}$ is not complete and $\mathbb{R}$ is, and that $\mathbb{R}$ is built to repair exactly that defect. Is the repair a one-off trick special to the reals, or can any incomplete space be fixed the same way?

It can, and the construction below is the general answer. Every metric space embeds isometrically, and densely, into a complete one, obtained by taking the space of its own Cauchy sequences and declaring two of them equal when they ought to have the same limit. The reals from the rationals is then just the first instance.

Definition 6.b.11 (*Completion of a metric space*): Let (X,d) be a metric space. Write $\mathcal{C}_X$ for the set of all Cauchy sequences in X, and define an equivalence relation on $\mathcal{C}_X$ by

$$(x_n)_{n=0}^{\infty} \sim (y_n)_{n=0}^{\infty} \quad :\Longleftrightarrow \quad \lim_{n\to\infty} d(x_n, y_n) = 0.$$

The quotient set $\overline{X} := \mathcal{C}_X/\sim$, equipped with the map

$$\overline{d}\big([(x_n)_{n=0}^{\infty}],\, [(y_n)_{n=0}^{\infty}]\big) := \lim_{n\to\infty} d(x_n, y_n),$$

is called the "*completion*" (German: „*Vervollständigung*") of (X,d). The injection $\iota : X \to \overline{X}$ sending x to the class of the constant sequence $(x, x, x, \dots)$ is called the "*canonical embedding*". For all $x, y \in X$,

$$d(x,y) = \overline{d}\big(\iota(x), \iota(y)\big),$$

which in particular shows ι is injective.

Exercise 6.b.12 (*Well-definedness of the completion*): Show that the objects introduced in Definition 6.b.11 are well defined. In particular, verify that $\overline{d}$ is indeed a metric on $\overline{X}$.

Exercise 6.b.13 (*The completion is complete*): As the name suggests, the completion $(\overline{X}, \overline{d})$ of a metric space is complete, meaning every Cauchy sequence in $\overline{X}$ converges. A sequence in $\overline{X}$ is essentially a sequence of sequences, i.e. $\left[(x_{m,n})_{n=0}^{\infty}\right]_{m=0}^{\infty}$. Show that $\left[(x_{n,n})_{n=0}^{\infty}\right]$ is a limit of this sequence.

Exercise 6.b.14 (*Universal property of the completion*): Let (X,d) be a metric space with completion $(\overline{X}, \overline{d})$. Let (Y, d_Y) be a complete metric space, and let $f : X \to Y$ be a function such that

$$d(x,y) = d_Y\big(f(x), f(y)\big) \qquad \text{for all } x, y \in X.$$

Show that there exists a unique function $\overline{f} : \overline{X} \to Y$ such that $f = \overline{f} \circ \iota_X$ and

$$\overline{d}(\overline{x}, \overline{y}) = d_Y\big(\overline{f}(\overline{x}), \overline{f}(\overline{y})\big) \qquad \text{for all } \overline{x}, \overline{y} \in \overline{X}.$$

This can be interpreted as: *"$\overline{X}$ is the **smallest** complete metric space containing X."*

Remark 6.b.15: Exercise 6.b.14 is what makes *"the"* completion well-posed rather than just *"a"* completion: any other complete space that X embeds into isometrically receives a unique isometric copy of $\overline{X}$ as well, via $\overline{f}$. This is the same uniqueness-up-to-isomorphism pattern behind *"the"* real numbers as *"the"* complete ordered field, and it is exactly how the script uses this construction in §9.1.4, taking $X = \mathbb{Q}$.

Contraction mappings and the Banach fixed point theorem (Section 6.c)

Completeness guarantees that certain limits exist, but on its own it does not say what they are. The following theorem is the standard way of turning that guarantee into a construction: it identifies a single point, proves it exists, proves it is unique, and supplies an algorithm that converges to it. Almost every existence proof in the second half of this course goes through it.

Definition 6.c.16 (*Contraction mapping*): Let (X,d) be a metric space. A map $f : X \to X$ is a "*contraction*" (German: „*Kontraktion*") if it is Lipschitz continuous (Definition 5.b.20) with some Lipschitz constant $L \in [0,1)$, i.e.,

$$d(f(x), f(y)) \leq L \cdot d(x,y) \quad \text{for all } x, y \in X.$$

Theorem 6.c.17 (*Banach fixed point theorem*): Let (X,d) be a nonempty complete metric space (Definition 6.a.1) and let $f : X \to X$ be a contraction (Definition 6.c.16). Then f has a unique "*fixed point*" (German: „*Fixpunkt*") $x^* \in X$, i.e., $f(x^*) = x^*$; moreover, for any $x_0 \in X$, the iterates $x_{n+1} := f(x_n)$ converge to x^*.

Proof idea:

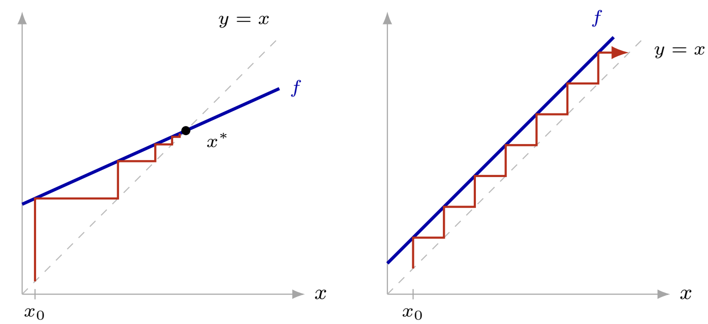
The core idea is to create a Cauchy sequence by repeatedly applying the contraction mapping f. Fix an arbitrary starting point $x_0 \in X$, and define the sequence by $x_{n+1} := f(x_n)$ for all $n \geq 0$. Comparing consecutive terms gives:

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \leq \lambda \cdot d(x_n, x_{n-1}).$$

Iterating this inequality yields $d(x_{n+1}, x_n) \leq \lambda^n d(x_1, x_0)$. Since $\lambda < 1$, one can use the triangle inequality and the geometric series to show that the distance between any two terms x_m and x_n (for $m > n$) becomes arbitrarily small as $n \to \infty$. Thus, the sequence is Cauchy. Because X is a complete metric space, this sequence converges to some limit $x^* \in X$. Since contraction mappings are continuous, one can pass the limit inside the function to conclude $f(x^*) = x^*$. **Q.E.D.**

Uniqueness of the fixed point:

Suppose $f(x) = x$ and $f(y) = y$. Then
$$d(x,y) = d\big(f(x), f(y)\big) \le L\, d(x,y),$$
so $(1-L)\, d(x,y) \le 0$. Since $L < 1$ we have $1 - L > 0$, forcing $d(x,y) \le 0$ and hence $d(x,y) = 0$, i.e. $x = y$. **Q.E.D.**



contraction, $L = 0.45$
iterates climb to the unique x^*

$L = 1$: graph parallel to $y = x$
no fixed point, iterates escape

The left panel is the proof in miniature. Each step moves vertically to the graph of f and then horizontally back to the diagonal, and because the graph is shallower than the diagonal, the steps shrink by a factor of at least L every time. That is precisely the geometric-series estimate the proof runs on. The right panel shows what a single missing hypothesis costs: with slope exactly 1 the graph never meets the diagonal, so no fixed point exists and the same iteration marches off instead of converging.

Remark 6.c.18 (*Where each hypothesis is used*): Both hypotheses of Theorem 6.c.17 are indispensable, and each fails in a different way.

- **Completeness** is what supplies the limit. On the incomplete space $X := (0,1]$, the map $f(x) := x/2$ is a contraction with $L = \frac{1}{2}$ and has no fixed point in X, since the iterates march towards 0, which is exactly the point missing.
- $L < 1$ **strictly** is what forces the iterates together. The map $f(x) := x + \frac{1}{1+e^x}$ on the complete space $\mathbb{R}$ satisfies the strict inequality $|f(x) - f(y)| < |x - y|$ for $x \neq y$, yet has no fixed point, since $f(x) > x$ everywhere. A supremum of 1 that is never attained is still not good enough.

The uniqueness proof above shows the second point cleanly: the whole argument is the division by $1 - L$, which is exactly what $L = 1$ forbids.

Exercise 6.c.19 (*Both hypotheses of Banach's theorem are sharp*): Find examples for:

(a) A Lipschitz contraction $T : X \to X$ on a non-complete metric space X that has no fixed point.

(b) A complete metric space (X, d) and an "*isometry*" (a mapping $T : X \to X$ with $d(T(x_1), T(x_2)) = d(x_1, x_2)$ for all x_1, x_2) that has no fixed point.

Remark 6.c.20: Part **(b)** sharpens Remark 6.c.18's second bullet: that example already showed a Lipschitz constant of exactly 1 can defeat the theorem, but its map T was not an isometry ($|T(x) - T(y)| < |x - y|$ strictly, just with no uniform gap). An isometry is the most rigid failure imaginable, not shrinking distances even a little, and it **still** need not have a fixed point, even on a complete space. Note that unboundedness is doing real work in the counterexample: on a **compact** space an isometry into itself is automatically surjective, and even that is not enough to force a fixed point. An irrational rotation of the circle S^1, which is compact and an isometry, has none either.

Fixed points of isometries are a genuinely separate question from Theorem 6.c.17, not one it settles as a special case.

AI-Exercise 6.c.21 (*Contraction Mapping on* $[0,1]$): Let $f\colon [0,1] \to [0,1]$ be defined by $f(x) := \frac{1}{2}\cos(x)$.

- **(a)** Show that f is a contraction mapping (Definition 6.c.16) with Lipschitz constant $L = \frac{1}{2}\sin(1) < 1$.
- **(b)** Deduce, using Theorem 6.c.17, that f has a unique fixed point $x^* \in [0,1]$.

Exercise 6.c.22 (*Does the fixed-point theorem apply?*): Remark 6.c.18 shows that each hypothesis is needed by exhibiting one failure apiece. Linus drills the same point the other way round, as a checklist: for each row below, decide whether Theorem 6.c.17 **applies** to $T : X \to X$, and separately determine the set of fixed points of T in X. The two questions have different answers more often than one expects.

$T(x)$	X	Applies? / fixed points
$\sin(x)$	$\mathbb{R}$	
$\frac{9}{10}\cos(x)$	$\mathbb{R}$	
x^2	$\mathbb{R}$	
$\frac{1}{2}x^2$	$(-1,1)$	
$\frac{1}{2}\left(x+\frac{2}{x}\right)$	$\mathbb{Q}\cap(1,2)$	

Exercise 6.c.23 (*3.5 — Multiple choice (fixed points)*): Select all the statements below that are true.

- **(a)** If you spread a map of Zurich on your desk, then one point of the desk will coincide with its representation on the map (in an ideal world).
- **(b)** If $f : [0,1] \to [0,2]$ is $\frac{1}{2}$-Lipschitz, then f has a fixed point.
- **(c)** If $f : [0,2] \to [0,1]$ is $\frac{1}{2}$-Lipschitz, then f has a fixed point.
- **(d)** If $f : [0,1]\cup[2,3] \to [0,1]\cup[2,3]$ is continuously differentiable with $|f'(x)| < 1$ for all $x \in [0,1]\cup[2,3]$, then it has a fixed point.
- **(e)** **(*)** If $f : [0,1] \to [0,1]$ is differentiable with $|f'(x)| < 1$ for all $x \in (0,1)$, then it has a fixed point.

Solutions (Section 6.d)

Proof of Exercise 6.a.2:

(a) Bounded. Take $\varepsilon = 1$: there is N with $d(x_n, x_m) < 1$ for all $n, m \geq N$; in particular $d(x_n, x_N) < 1$ for $n \geq N$. Setting $M := \max\{d(x_0, x_N), \dots, d(x_{N-1}, x_N), 1\}$, the triangle inequality gives $d(x_n, x_0) \leq d(x_n, x_N) + d(x_N, x_0) \leq 1 + M$ for every n, so $(x_n)_n \subseteq B_{1+M}(x_0)$.

(b) Convergent $\Rightarrow$ Cauchy. If $x_n \to x$, fix $\varepsilon > 0$ and choose N with $d(x_n, x) < \varepsilon/2$ for $n \geq N$. Then for $n, m \geq N$, $d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \varepsilon$.

(c) Cauchy + convergent subsequence $\Rightarrow$ convergent. Let (x_n) be Cauchy and $x_{n_k} \to x$. The *"only if"* direction is part **(b)** applied to the whole sequence. For *"if"*: fix $\varepsilon > 0$. Cauchyness gives N with $d(x_n, x_m) < \varepsilon/2$ for $n, m \geq N$; convergence of the subsequence gives K with $d(x_{n_k}, x) < \varepsilon/2$ for $k \geq K$. Since $n_k \to \infty$, pick $k \geq K$ with $n_k \geq N$. Then for every $n \geq N$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon,$$

so $x_n \to x$. **Q.E.D.**

Solution to Exercise 6.a.10:

Write $\varphi : (0,1] \to [1,\infty)$, $\varphi(x) := 1/x$. It is a bijection, and by construction $d(x,y) = |\varphi(x)-\varphi(y)|$. Consequently, d is nothing but the Euclidean metric of $[1,\infty)$ transported back to $(0,1]$ along φ. Every part follows from that.

(a) Symmetry and the triangle inequality are inherited from $|\cdot|$ directly. For definiteness, $d(x,y) = 0$ forces $\varphi(x) = \varphi(y)$, and φ is injective, so $x = y$.

(b) The sequence $x_n := \frac{1}{n}$ lies in $(0,1]$ and is Cauchy for d_{eucl}, being Cauchy in $\mathbb{R}$. Its limit in $\mathbb{R}$ is $0 \notin X$, and limits in $\mathbb{R}$ are unique (Proposition 4.a.4), so it has no limit in X.

(c) Let (x_n) be d-Cauchy. Then $\big(\varphi(x_n)\big)$ is Cauchy in $\mathbb{R}$, hence converges to some $\ell \in \mathbb{R}$. Since $\varphi(x_n) \geq 1$ for every n, also $\ell \geq 1$; in particular $\ell \neq 0$, so $x := 1/\ell$ lies in $(0,1]$. Then

$$d(x_n, x) = \big|\varphi(x_n) - \varphi(x)\big| = \big|\varphi(x_n) - \ell\big| \longrightarrow 0,$$

so $x_n \to x$ in (X,d). Note where the boundedness of X entered: it is what forces $\ell \geq 1$ rather than merely $\ell \geq 0$.

(d) φ and $\varphi^{-1}(t) = 1/t$ are continuous, so φ is a homeomorphism. A set is d-open exactly when its φ-image is open in $[1,\infty)$, exactly when it is d_{eucl}-open. Concretely, the same two-sided-ball argument as in Exercise 7.b.27**(b)** works: on $(0,1]$ one has $|x-y| \leq d(x,y) \leq \frac{1}{xy}\,|x-y|$ with $\frac{1}{xy}$ locally bounded.

Remark 6.d.24: So d_{eucl} and d have the **same open sets**, the same convergent sequences, the same continuous functions and the same compact sets, and yet one is complete and the other is not. **Completeness is a property of the metric, not of the topology it generates.**

This completes a trio worth holding together, all three proved by the same device of comparing two metrics that induce one topology:

- **Compactness is topological.** It is defined by open covers alone (Definition 7.a.1), so it cannot distinguish two metrics with the same open sets.
- **Boundedness is not:** see Remark 7.b.30.
- **Completeness is not:** this exercise.

That is exactly why *"compact if and only if closed and bounded"* needs the standard metric, and why the correct general replacement is *"complete and totally bounded"* (Remark 7.a.14). Both ingredients of the replacement are non-topological, and only their combination is.

Q.E.D.

Proof of Exercise 6.b.12:

$\sim$ is an equivalence relation on $\mathcal{C}_X$. Reflexivity is $d(x_n,x_n) = 0 \to 0$. Symmetry is symmetry of d. For transitivity, if $(x_n) \sim (y_n)$ and $(y_n) \sim (z_n)$, the triangle inequality gives $0 \leq d(x_n,z_n) \leq d(x_n,y_n) + d(y_n,z_n) \to 0 + 0 = 0$, so $(x_n) \sim (z_n)$.

$\overline{d}$ is well-defined. First, the limit defining $\overline{d}$ exists: for $(x_n),(y_n) \in \mathcal{C}_X$, the reverse triangle inequality gives

$$\big|d(x_n,y_n) - d(x_m,y_m)\big| \leq d(x_n,x_m) + d(y_n,y_m),$$

and the right-hand side $\to 0$ as $n,m \to \infty$ since (x_n) and (y_n) are each Cauchy, so $\big(d(x_n,y_n)\big)_n$ is a Cauchy sequence of **real numbers**, hence convergent. Note that this is the completeness axiom for $\mathbb{R}$ itself from **Analysis I**, and not an instance of Definition 6.a.1 being proved circularly. Second, the limit does not depend on the choice of representative: if $(x_n) \sim (x'_n)$ and $(y_n) \sim (y'_n)$, the same reverse triangle inequality gives $\big|d(x_n,y_n) - d(x'_n,y'_n)\big| \leq d(x_n,x'_n) + d(y_n,y'_n) \to 0$, so the two limits agree.

$\overline{d}$ is a metric. Fix representatives $(x_n) \in \overline{x}$, $(y_n) \in \overline{y}$, $(z_n) \in \overline{z}$.

- **Definiteness.** $\overline{d}(\overline{x},\overline{y}) = 0$ means $\lim_n d(x_n,y_n) = 0$, i.e. exactly $(x_n) \sim (y_n)$, i.e. $\overline{x} = \overline{y}$.

- **Symmetry.** Immediate from $d(x_n, y_n) = d(y_n, x_n)$ for every n.
- **Triangle inequality.** Take $n \to \infty$ in $d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n)$, using that limits preserve weak inequalities.

Finally $d(x, y) = \overline{d}(\iota(x), \iota(y))$ holds because the constant sequences $(x, x, \dots)$ and $(y, y, \dots)$ give $d(x_n, y_n) = d(x, y)$ for every n, so the limit is $d(x, y)$ itself; and this identity makes ι injective, since $d(x, y) = 0 \iff x = y$. **Q.E.D.**

Proof of Exercise 6.b.13:

Let $(\xi_m)_{m=0}^{\infty}$ be Cauchy in $\overline{X}$, with $\xi_m = [(x_{m,n})_{n=0}^{\infty}]$ for some Cauchy sequence $(x_{m,n})_n$ in X representing ξ_m. Replacing $(x_{m,n})_n$ by a tail of itself if necessary, which is harmless because a Cauchy sequence and any of its tails represent the same class, differing in only finitely many terms, we may assume each representative is already "*regularized*":

$$d(x_{m,n}, x_{m,n'}) < \tfrac{1}{m} \qquad \text{for all } n, n' \geq 0.$$

Set $y_m := x_{m,m}$, the diagonal sequence the exercise asks about. Letting $n' \to \infty$ in the regularization bound (with $n = m$) gives

$$\overline{d}\big(\iota(y_m),\, \xi_m\big) = \lim_{n\to\infty} d(x_{m,m}, x_{m,n}) \;\leq\; \tfrac{1}{m}. \tag{6.1}$$

$(y_m)_m$ is Cauchy in X. By the triangle inequality for $\overline{d}$ (already established in Exercise 6.b.12) and (6.1),

$$d(y_m, y_{m'}) = \overline{d}\big(\iota(y_m), \iota(y_{m'})\big) \;\leq\; \overline{d}(\iota(y_m), \xi_m) + \overline{d}(\xi_m, \xi_{m'}) + \overline{d}(\xi_{m'}, \iota(y_{m'})) \;\leq\; \tfrac{1}{m} + \overline{d}(\xi_m, \xi_{m'}) + \tfrac{1}{m'}.$$

Given $\varepsilon > 0$, Cauchyness of $(\xi_m)_m$ gives M_1 with $\overline{d}(\xi_m, \xi_{m'}) < \varepsilon/3$ for $m, m' \geq M_1$; taking also $M \geq \max\{M_1,\, 3/\varepsilon\}$ makes all three terms $< \varepsilon/3$ for $m, m' \geq M$, so $(y_m)_m$ is Cauchy in X and defines a class $\overline{x} := [(y_m)_m] \in \overline{X}$.

$\xi_m \to \overline{x}$. Using (6.1) again and the same triangle inequality,

$$\overline{d}(\xi_m, \overline{x}) \;\leq\; \overline{d}(\xi_m, \iota(y_m)) + \overline{d}(\iota(y_m), \overline{x}) \;\leq\; \tfrac{1}{m} + \lim_{k\to\infty} d(y_m, y_k).$$

The first term $\to 0$ as $m \to \infty$ by construction, and the second does too, being (for each fixed m) bounded by $\sup_{k \geq m} d(y_m, y_k)$, which $\to 0$ as $m \to \infty$ since $(y_m)_m$ is Cauchy. Hence $\overline{d}(\xi_m, \overline{x}) \to 0$, i.e. $\xi_m \to \overline{x}$ in $\overline{X}$, as required. **Q.E.D.**

Proof of Exercise 6.b.14:

Construction. For $\overline{x} = [(x_n)_n] \in \overline{X}$, the sequence $(f(x_n))_n$ is Cauchy in Y, since $d_Y(f(x_n), f(x_m)) = d(x_n, x_m) \to 0$ as $(x_n)_n$ is Cauchy in X. As Y is complete, $f(x_n) \to y$ for a unique $y \in Y$ (Proposition 4.a.4); define $\overline{f}(\overline{x}) := y$.

Well-defined. If $(x_n) \sim (x'_n)$, then $d_Y(f(x_n), f(x'_n)) = d(x_n, x'_n) \to 0$, so $(f(x_n))_n$ and $(f(x'_n))_n$ have the same limit in Y by uniqueness of limits; hence $\overline{f}(\overline{x})$ does not depend on the representative chosen.

$f = \overline{f} \circ \iota_X$. For $x \in X$, $\iota_X(x)$ is represented by the constant sequence $(x, x, \dots)$, so $\overline{f}(\iota_X(x)) = \lim_n f(x) = f(x)$.

$\overline{f}$ satisfies the displayed identity. For $\overline{x} = [(x_n)_n]$, $\overline{y} = [(y_n)_n]$, continuity of d_Y in both arguments (itself 1-Lipschitz, by the same reverse-triangle-inequality argument as in Exercise 6.b.12) gives

$$d_Y\big(\overline{f}(\overline{x}), \overline{f}(\overline{y})\big) = d_Y\Big(\lim_n f(x_n),\ \lim_n f(y_n)\Big) = \lim_n d_Y\big(f(x_n), f(y_n)\big) = \lim_n d(x_n, y_n) = \overline{d}(\overline{x}, \overline{y}).$$

Uniqueness. Let $g : \overline{X} \to Y$ satisfy the same two conditions. The displayed identity makes g (like $\overline{f}$) 1-Lipschitz, hence continuous. For $\overline{x} = [(x_n)_n]$, the sequence $\iota_X(x_n)$ converges to $\overline{x}$ in $\overline{X}$:

indeed $\overline{d}(\iota_X(x_n), \overline{x}) = \lim_k d(x_n, x_k) \to 0$ as $n \to \infty$, since $(x_n)_n$ is Cauchy. By continuity of g and $f = g \circ \iota_X$,

$$g(\overline{x}) = \lim_n g(\iota_X(x_n)) = \lim_n f(x_n) = \overline{f}(\overline{x}),$$

using uniqueness of limits in Y once more for the last equality. As $\overline{x}$ was arbitrary, $g = \overline{f}$.

Q.E.D.

Proof of Exercise 6.c.19:

(a) This is Remark 6.c.18's first bullet: $X := (0,1]$ is not complete, and $T(x) := x/2$ is a $\frac{1}{2}$-contraction on X with iterates marching towards the missing point 0, hence no fixed point in X.

(b) Take $X := \mathbb{R}$ (complete, Proposition 6.a.5) with the translation $T(x) := x + 1$. Then $d(T(x), T(y)) = |(x+1)-(y+1)| = |x-y| = d(x,y)$, so T is an isometry, and in particular Lipschitz with constant $L = 1$ rather than $L < 1$. Consequently, this does not contradict Theorem 6.c.17. A fixed point would need $x + 1 = x$, which is impossible, so T has none. **Q.E.D.**

Proof of AI-Exercise 6.c.21:

- **(a)** By the Mean Value Theorem, for any $x, y \in [0,1]$, there exists $\xi \in (0,1)$ such that $|f(x) - f(y)| = |f'(\xi)||x-y| = \frac{1}{2}\sin(\xi)|x-y|$. Since $\sin(x)$ is strictly increasing on $[0,1]$, $\sup_{\xi\in[0,1]} \frac{1}{2}\sin(\xi) = \frac{1}{2}\sin(1) \approx 0.4207 < 1$. Thus f is a contraction (Definition 6.c.16) with $L = \frac{1}{2}\sin(1) < 1$.
- **(b)** Since $[0,1]$ equipped with the Euclidean metric is a complete metric space (Definition 6.a.1) and $f([0,1]) \subseteq [0,1]$, Theorem 6.c.17 guarantees the existence and uniqueness of a fixed point $x^* \in [0,1]$ satisfying $f(x^*) = x^*$.

Q.E.D.

Solution to Exercise 6.c.23:

(a), (c) and (e) are true; (b) and (d) are false.

(a) True, and this is the classic illustration of Theorem 6.c.17. The map of Zurich lying on the desk defines a function from the (compact, hence complete) region of the city depicted onto a smaller region of the desk, and that function is a similarity: a rotation and translation composed with a scaling by the map's ratio $L < 1$. It is therefore a contraction of a complete metric space into itself, and so has exactly one fixed point: one spot on the map lying directly above the point of the city it represents.

(b) False. The hypothesis says nothing about f mapping its domain into itself, and the codomain $[0,2]$ is strictly larger than the domain $[0,1]$. Take

$$f(x) := \tfrac{x}{2} + \tfrac{3}{2}, \qquad f : [0,1] \to [\tfrac{3}{2}, 2] \subseteq [0,2],$$

which is $\frac{1}{2}$-Lipschitz. A fixed point would need $x = \frac{x}{2} + \frac{3}{2}$, i.e. $x = 3$, which is not in $[0,1]$.

(c) True. Now the inclusion runs the right way: f maps $[0,2]$ into $[0,1] \subseteq [0,2]$, so f is a self-map of the complete space $[0,2]$ (closed and bounded in $\mathbb{R}$, hence compact, hence complete by Exercise 7.a.20), and it is a contraction with $L = \frac{1}{2} < 1$. Theorem 6.c.17 applies directly.

(d) False, and this is the instructive one: the domain is **disconnected**, and the condition $|f'| < 1$ is purely local, so it cannot see across the gap. Define

$$f(x) := \begin{cases} \frac{5}{2} & x \in [0,1], \\ \frac{1}{2} & x \in [2,3]. \end{cases}$$

Then f is continuously differentiable with $f' \equiv 0$, it maps $[0,1] \cup [2,3]$ into itself, and it has no fixed point, because each of the two pieces is sent into the **other** one. Compare the local-to-global reading of Remark 7.c.32: a pointwise derivative bound upgrades to a global contraction only along paths that stay inside the domain, and here there are none.

(e) True, but **not** by the fixed point theorem, and that is the point of the star. The hypothesis $|f'(x)| < 1$ gives no uniform $L < 1$, since the supremum of $|f'|$ may well equal 1, so Theorem 6.c.17 is unavailable. Existence follows from the intermediate value theorem instead. Put $g(x) := f(x) - x$, which is continuous on $[0,1]$. Since f takes values in $[0,1]$,

$$g(0) = f(0) \geq 0 \qquad \text{and} \qquad g(1) = f(1) - 1 \leq 0,$$

so g vanishes somewhere in $[0,1]$ by Corollary 8.a.7, and that zero is a fixed point of f. Note that the derivative hypothesis was never used: **every** continuous self-map of $[0,1]$ has a fixed point. What $|f'| < 1$ buys is **uniqueness**, by the same two-line argument as in the proof of uniqueness for Theorem 6.c.17 combined with the mean value theorem. **Q.E.D.**

Solution to Exercise 6.c.22:

Recall that Theorem 6.c.17 needs three things at once: X complete, T mapping X into X, and a Lipschitz constant $\lambda < 1$ that is **actually attained as a bound**, not merely approached.

- **$T(x) = \sin(x)$ on $\mathbb{R}$. Does not apply; fixed points $\{0\}$.** $\mathbb{R}$ is complete and T maps it into itself, but $\lambda = \sup_x |\cos x| = 1$, attained at $x = 0$. So the theorem is unavailable. A fixed point exists all the same: $\sin x = x$ holds at $x = 0$ and nowhere else, since $|\sin x| < |x|$ for $x \neq 0$. This is the first lesson of the table: the theorem is **sufficient**, never necessary.
- **$T(x) = \frac{9}{10}\cos(x)$ on $\mathbb{R}$. Applies; one fixed point.** Here $|T'(x)| = \frac{9}{10}|\sin x| \leq \frac{9}{10} < 1$, so by the mean value theorem T is a $\frac{9}{10}$-contraction; $\mathbb{R}$ is complete and $T(\mathbb{R}) \subseteq [-\frac{9}{10}, \frac{9}{10}] \subseteq \mathbb{R}$. All three hypotheses hold, so there is exactly one fixed point. It has no closed form; iterating from any starting value gives $x^* \approx 0.693$.
- **$T(x) = x^2$ on $\mathbb{R}$. Does not apply; fixed points $\{0,1\}$.** T is not Lipschitz at all on $\mathbb{R}$, since $T'(x) = 2x$ is unbounded. That it cannot apply is visible without any computation: $x^2 = x$ has the two solutions 0 and 1, and the theorem asserts **uniqueness**, so any set of hypotheses implying it must fail here.
- **$T(x) = \frac{1}{2}x^2$ on $(-1,1)$. Does not apply; fixed points $\{0\}$.** Two hypotheses fail together. The domain $(-1,1)$ is not complete, and $\lambda = \sup_{|x|<1}|T'(x)| = \sup|x| = 1$. This supremum is not attained, but a supremum of 1 is still not < 1, which is exactly the trap of Remark 6.c.18. The map does send $(-1,1)$ into $[0, \frac{1}{2}) \subseteq (-1,1)$. Solving $\frac{1}{2}x^2 = x$ gives $x \in \{0,2\}$, of which only 0 lies in the domain, so a unique fixed point exists regardless.
- **$T(x) = \frac{1}{2}\left(x + \frac{2}{x}\right)$ on $\mathbb{Q} \cap (1,2)$. Does not apply; no fixed points.** This row is the interesting one, and it is the Babylonian iteration for $\sqrt{2}$. Everything works except completeness:
  - $T'(x) = \frac{1}{2} - \frac{1}{x^2}$, so on $(1,2)$ we have $-\frac{1}{2} \leq T'(x) \leq \frac{1}{4}$ and hence $\lambda = \frac{1}{2} < 1$: a genuine contraction.
  - T maps X into X: it sends rationals to rationals, and on $(1,2)$ its minimum is $T(\sqrt{2}) = \sqrt{2} \approx 1.414$ while $T(1) = T(2) = \frac{3}{2}$, so $T\big((1,2)\big) \subseteq [\sqrt{2}, \frac{3}{2}] \subseteq (1,2)$.
  - But $\mathbb{Q} \cap (1,2)$ is **not complete**.

  And there is indeed no fixed point: $x = \frac{1}{2}(x + \frac{2}{x})$ forces $x^2 = 2$, so $x = \sqrt{2} \notin \mathbb{Q}$. The iterates are the familiar rational approximations $\frac{3}{2}, \frac{17}{12}, \frac{577}{408}, \ldots$, which form a Cauchy sequence in X converging to a point that X does not contain.

  This is worth contrasting with Remark 6.c.18, where completeness failed on $(0,1]$ with the missing point sitting at an endpoint one is already suspicious of. Here the domain is an interval of rationals, the hole is invisible, and the contraction is a thoroughly practical algorithm. Completeness is not a technicality.

Q.E.D.

Compactness (Chapter 7)

Compactness is the least self-explanatory definition in this part, so it is worth saying in advance what it is **for**. Finite sets are easy to handle. One can take a maximum over them, choose a smallest radius, or check every case in turn. Infinite sets allow none of that. Compactness is the property that lets an infinite space be treated as though it were finite, not because it is small, but because every attempt to spread it out over infinitely many open pieces collapses back to finitely many.

The chapter defines compactness by open covers, proves it equivalent to sequential compactness, records the Heine–Borel theorem for $\mathbb{R}^n$ together with how badly it fails outside $\mathbb{R}^n$, and closes with the three payoffs that make the definition worth having.

Compactness (Section 7.a)

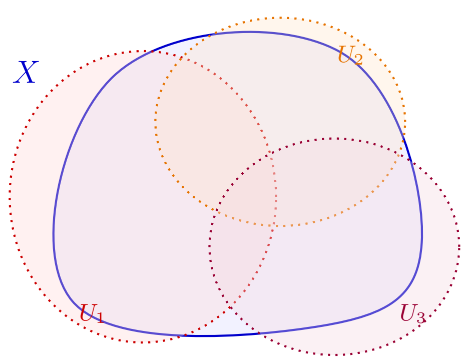
The definition itself is one line long, and the difficulty is entirely in the order of its quantifiers. It is worth reading it twice before reading anything else.

Definition 7.a.1 (*Compact metric space*): A metric space (X, d) is "*compact*" (German: „*kompakt*") if every open cover (by sets open in the sense of Definition 3.a.6) has a finite subcover.

Notation 7.a.2 (*Open covers*): An "*open cover*" (German: „*offene Überdeckung*") of X is a family $\{U_i\}_{i\in I}$ of open sets with $X = \bigcup_{i\in I} U_i$; the index set I is allowed to be arbitrary, in particular uncountable. A "*subcover*" is a subfamily $\{U_i\}_{i\in J}$, $J \subseteq I$, that still covers X, and it is **finite** when J is. The order of quantifiers is what carries the content: **every** cover must admit **some** finite subcover. Note the asymmetry this produces. Exhibiting one cover with a finite subcover proves nothing at all, since every space has the one-element cover $\{X\}$. Exhibiting one cover **without** a finite subcover, on the other hand, disproves compactness outright, and that is how Exercise 7.b.27 and the open-interval example below both work.

Theorem 7.a.3 (*Sequential compactness*): A metric space (X, d) is compact if and only if every sequence in X has a convergent (Definition 4.a.1) subsequence in X.

AI-Note 7.1: Neither Corsin nor the official lecture notes prove this here; the proof is given below as Theorem 7.a.13, once the tools it needs (total boundedness) are available.



$X = U_1 \cup U_2 \cup U_3$

Lemma 7.a.4 (*Nesting principle*): Let X be a metric space. A subset $K \subseteq X$ is topologically compact (Item **(b)**) if and only if it has the following property: for every collection $\mathcal{A} = \{A_i\}_{i\in I}$ of closed subsets of X, if every intersection of finitely many sets in $\mathcal{A}$ meets K, then $K \cap \bigcap_{i\in I} A_i$ is non-empty.

Proof:

(⇒) Let K be compact and let $\mathcal{A}$ be a collection of closed subsets of X with $K \cap \bigcap_{i\in I} A_i = \emptyset$. The complements $\mathcal{U} := \{X \setminus A_i : i \in I\}$ then form an open cover of K, so compactness gives a finite subcover $K \subseteq (X \setminus A_{i_1}) \cup \cdots \cup (X \setminus A_{i_n})$, i.e. $K \cap A_{i_1} \cap \cdots \cap A_{i_n} = \emptyset$: some finite subcollection already fails to meet K.

(⇐) Let K have the nesting property and let $\mathcal{U}$ be an open cover of K. The complements $\mathcal{A} := \{X \setminus U : U \in \mathcal{U}\}$ form a collection of closed sets with $K \cap \bigcap_{U\in\mathcal{U}}(X \setminus U) = K \setminus \bigcup_{U\in\mathcal{U}} U = \emptyset$, since $\mathcal{U}$ covers K. By the contrapositive of the nesting property, some finite subcollection $X \setminus U_1, \ldots, X \setminus U_n$ already has $K \cap (X \setminus U_1) \cap \cdots \cap (X \setminus U_n) = \emptyset$, i.e. $K \subseteq U_1 \cup \cdots \cup U_n$: a finite subcover. **Q.E.D.**

Remark 7.a.5: The nesting principle is Definition 7.a.1 read through De Morgan's laws. Open covers become closed families, unions become intersections, and *"some finite piece already covers"* becomes *"some finite piece already misses"*. It earns its own name because closed sets, not open ones, are the natural language for several later arguments (nested closed sets of shrinking size, in particular), where phrasing compactness this way saves a complementation step every time.

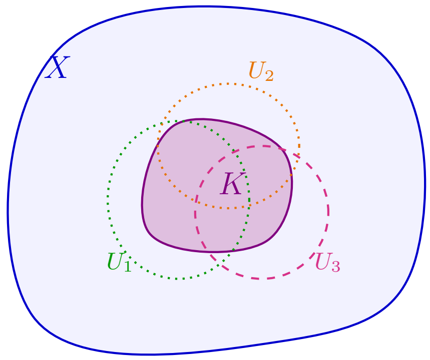
Compact subsets (Subsection 7.a.1)

A subset K of a metric space (X, d) is compact if the metric space $(K, d|_{K\times K})$ is compact (Definition 7.a.1). This is equivalent to the definition in the script:

Definition 7.a.6 (*Definition 9.62, quoted from the official lecture notes*): Let (X, d) be a metric space. A subset $K \subset X$ is called **compact** if one of the following equivalent conditions hold:

- **(a)** K is **sequentially compact**: every sequence $(x_n)_{n\in\mathbb{N}}$ in K has a subsequence that is convergent (Definition 4.a.1) in K.
- **(b)** K is **topologically compact**: every family of open sets (Definition 3.a.6) $\{U_i\}_{i\in I}$ that cover K, has a finite subcover.
- **(c)** K is **complete** (every Cauchy sequence contained in K has a limit in K) and **totally bounded** (Definition 7.a.9): for every $r > 0$ there exist finitely many $x_1, \ldots, x_n \in K$ such that the balls $B_r(x_1), \ldots, B_r(x_n)$ cover K.

AI-Note 7.2: Corsin's own citation of this definition (matching the numbering in an earlier printing of the script) reads *"9.63"* and quotes only the first two of the three equivalent conditions above; in the current script (v1, 10 July 2024) this is **Definition 9.62**, and **9.63** is instead the unnumbered remark explaining why condition **(b)** is called *"topological"*. Corrected above, with the third condition restored. Nothing downstream was ever missing mathematically, since this document already proves the three-way equivalence at Theorem 7.a.13 and has total boundedness at Definition 7.a.9. The citation was simply under-quoting its own source.



$K \subseteq U_1 \cup U_2 \cup U_3 \subseteq X$

Sequential and topological compactness agree (Subsection 7.a.2)

AI-Note 7.3: Neither this equivalence nor Theorem 7.a.3 above is proved in the transcribed notes. A complete, self-contained proof is supplied here from a second tutor's addendum. By the reduction already noted above, compactness of $K \subseteq X$ being compactness of the metric space $(K, d|_{K\times K})$, it suffices to treat a whole metric space X rather than a subset K.

Lemma 7.a.7 (*Continuous functions attain extrema on sequentially compact spaces*): Let (X,d) be sequentially compact (Item **(a)**) and $f : X \to \mathbb{R}$ continuous. Then f attains a maximum and a minimum on X.

Proof:

It suffices to find a minimum (apply the result to $-f$ for the maximum). Let $m := \inf_{x\in X} f(x) \in [-\infty,\infty)$ and choose a sequence $(x_n)_n$ with $f(x_n) \to m$. By sequential compactness, some subsequence $x_{n_k} \to x \in X$; by continuity, $f(x) = \lim_k f(x_{n_k}) = m$. In particular $m \neq -\infty$, so f attains its minimum m at x. **Q.E.D.**

Remark 7.a.8: This is exactly the extreme value theorem (Item **(b)**) proved from first principles, without presupposing Theorem 7.a.3. Avoiding that appeal is essential here, since that theorem is precisely what we are about to prove.

Definition 7.a.9 (*Totally bounded*): A metric space (X,d) is "*totally bounded*" (German: „*total beschränkt*") if for every $\varepsilon > 0$ there exist finitely many points $x_1, \dots, x_n \in X$ such that

$$X = \bigcup_{k=1}^{n} B_\varepsilon(x_k).$$

Exercise 7.a.10 (*Totally bounded is strictly stronger than bounded*): Let X be a metric space. Show that if X is totally bounded, then it is bounded, i.e. $\sup_{x,x'\in X} d(x,x') < \infty$. Find an example of a bounded metric space that is **not** totally bounded.

Proof of Exercise 7.a.10:

Totally bounded $\Rightarrow$ bounded. Apply Definition 7.a.9 with $\varepsilon := 1$: there are $x_1, \dots, x_n$ with $X = \bigcup_{k=1}^n B_1(x_k)$. Let $M := \max_{i,j} d(x_i, x_j)$ (finite, a maximum over finitely many pairs). For any $x, x' \in X$, choose i, j with $x \in B_1(x_i)$, $x' \in B_1(x_j)$; the triangle inequality gives $d(x,x') \le d(x,x_i) + d(x_i,x_j) + d(x_j,x') < 1 + M + 1$, so X is bounded.

Not conversely. Let X be any infinite set with the discrete metric. Then $\sup_{x,x'} d(x,x') = 1 < \infty$, so X is bounded. But X is not totally bounded: for $\varepsilon := \frac{1}{2}$, every ball $B_{1/2}(x) = \{x\}$ (Definition 3.a.1), so no **finite** union of such balls can cover the infinite set X. **Q.E.D.**

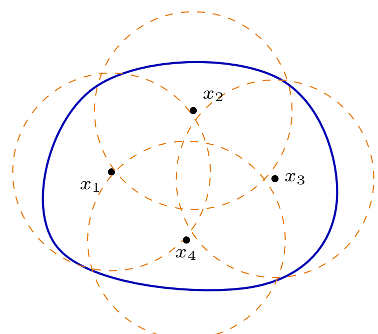
Remark 7.a.11: This is the same asymmetry Lemma 7.a.12 proves in one direction only: sequential compactness forces **both** boundedness and total boundedness, and this exercise shows the two are genuinely different hypotheses. Total boundedness is the strictly stronger of the two, and the gap opens up in infinite dimensions.

Lemma 7.a.12 (*Sequential compactness gives (total) boundedness*): Let (X,d) be sequentially compact. Then X is bounded and totally bounded (Definition 7.a.9).

Proof:

Bounded. Fix $x_0 \in X$; the function $f(y) := d(x_0, y)$ is continuous, so by Lemma 7.a.7 it attains a maximum $M \geq 0$ on X. Then $X = B_{M+1}(x_0)$.

Totally bounded. Fix $\varepsilon > 0$. Construct a sequence inductively: let $x_1 \in X$ be arbitrary, and, having chosen $x_1, \ldots, x_n$, let $U_n := \bigcup_{k=1}^{n} B_\varepsilon(x_k)$. If $U_n = X$, stop, since we then have the desired finite cover. Otherwise pick $x_{n+1} \in X \setminus U_n$, so that $d(x_{n+1}, x_k) \geq \varepsilon$ for all $k \leq n$. If this process never terminates, sequential compactness gives a convergent, hence Cauchy, subsequence $(x_{n_k})_k$; but $d(x_{n_k}, x_{n_{k+1}}) \geq \varepsilon$ for all k contradicts the Cauchy property. So, the process must terminate after finitely many steps, giving the desired finite ε-net.



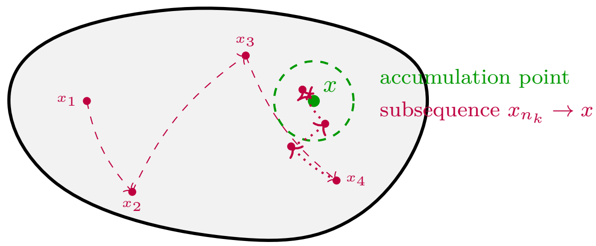
$X = B_\varepsilon(x_1) \cup \cdots \cup B_\varepsilon(x_4)$

Q.E.D.

Theorem 7.a.13 (*Sequential and topological compactness agree*): A metric space (X,d) is sequentially compact (Item **(a)**) if and only if it is topologically compact (Definition 7.a.1). In particular, Theorem 7.a.3 holds.

Proof:

(⇐) Suppose X is not sequentially compact, so some sequence $(x_n)_{n\in\mathbb{N}}$ has no convergent subsequence. Then no $x \in X$ is the limit of any subsequence, so there exist $\varepsilon_x > 0$ and $N_x \in \mathbb{N}$ such that $B_{\varepsilon_x}(x) \cap \{x_n : n \geq N_x\} = \emptyset$. The balls $\{B_{\varepsilon_x}(x)\}_{x\in X}$ cover X, so topological compactness gives finitely many $x_1, \ldots, x_m$ with $X = \bigcup_{k=1}^{m} B_{\varepsilon_{x_k}}(x_k)$. With $N := \max\{N_{x_1}, \ldots, N_{x_m}\}$, every x_n with $n \geq N$ then avoids $B_{\varepsilon_{x_1}}(x_1) \cup \cdots \cup B_{\varepsilon_{x_m}}(x_m) = X$, which is absurd. Therefore, X is sequentially compact.



(⇒) Suppose X is sequentially compact and let $\{U_i\}_{i\in I}$ be an open cover; we may assume no single U_i equals X (else $\{U_i\}$ is itself a finite subcover). Define

$$r(x) := \tfrac{1}{2}\sup\{r \geq 0 : B_r(x) \subseteq U_i \text{ for some } i \in I\}, \qquad x \in X.$$

r is finite and positive. Since $\{U_i\}$ covers X and each U_i is open, some $U_{i_0} \ni x$ contains a ball around x, so $r(x) > 0$; and since X is bounded (Lemma 7.a.12) but no U_i equals X, $r(x) < \infty$.

r is Lipschitz, hence continuous. (That is, $|r(x) - r(y)| \leq L\, d(x,y)$ for a constant L; the notion is named and placed in context as Definition 5.b.20, but only the displayed estimate is used here.) Write $R(x) := \sup\{\rho \geq 0 : B_\rho(x) \subseteq U_i \text{ for some } i\}$, so that $r = \frac{1}{2}R$; it is R that the argument bounds, and the factor $\frac{1}{2}$ must be carried through.

Let $x, y \in X$ with $d(x,y) < r(x)$. Since $r(x) < R(x)$, we have $B_{r(x)}(x) \subseteq U_i$ for some i, so $y \in U_i$, and the triangle inequality gives

$$B_{r(x)-d(x,y)}(y) \subseteq B_{r(x)}(x) \subseteq U_i \qquad \Longrightarrow \qquad R(y) \geq r(x) - d(x,y).$$

Halving, $r(y) = \frac{1}{2}R(y) \geq \frac{1}{2}\big(r(x) - d(x,y)\big)$, which also holds trivially when $d(x,y) \geq r(x)$ since $r(y) \geq 0$. By symmetry,

$$|r(x) - r(y)| \leq \tfrac{1}{2}\, d(x,y),$$

so r is $\frac{1}{2}$-Lipschitz, and in particular continuous, which is all that is needed below.

By Lemma 7.a.7, r attains a minimum $\varepsilon := \min_{x\in X} r(x) > 0$ on X. Then $B_\varepsilon(x) \subseteq U_i$ for some i, for every $x \in X$. By Lemma 7.a.12, X is totally bounded, so there are $x_1, \ldots, x_n \in X$ with $X = \bigcup_{k=1}^n B_\varepsilon(x_k)$; choosing, for each k, an index i_k with $B_\varepsilon(x_k) \subseteq U_{i_k}$, the sets $U_{i_1}, \ldots, U_{i_n}$ form a finite subcover of X. **Q.E.D.**

Remark 7.a.14 (*Total boundedness, more generally*): Diego's note records a sharper characterization as a further exercise: (X,d) is compact if and only if it is complete and totally bounded. Lemma 7.a.12 gives one direction (compact $\Longrightarrow$ complete, by Exercise 7.a.20 below, and totally bounded); for the converse, a Cauchy subsequence can be extracted from any sequence using total boundedness (cover by finitely many $\frac{1}{k}$-balls for $k = 1, 2, \ldots$ and pass to nested subsequences), and it converges by completeness.

Definition 7.a.15 (*Lebesgue number*): Let (X,d) be a metric space and $\{U_i\}_{i\in I}$ an open cover of X. A number $\varepsilon > 0$ is a "*Lebesgue number*" (German: „*Lebesgue-Zahl*") for the cover if

$$\forall x \in X\ \exists i \in I : B_\varepsilon(x) \subseteq U_i.$$

Remark 7.a.16 (*Lebesgue's covering lemma*): The proof of Theorem 7.a.13 (⇒) shows more than stated: on a compact metric space, **every** open cover has a Lebesgue number (take $\varepsilon := \min_{x\in X} r(x)$ as constructed there). This is known as "*Lebesgue's covering lemma*" (German: „*Lebesguesches Überdeckungslemma*").

Proposition 7.a.17 (*Compactness implies uniform continuity*): Let (X, d_X), (Y, d_Y) be metric spaces with X compact, and let $f : X \to Y$ be continuous. Then f is uniformly continuous (Definition 5.a.4).

Proof:

Let $\varepsilon > 0$. The sets $f^{-1}(B_{\varepsilon/2}(y))$, $y \in Y$, are open (Definition 5.a.1) and cover X, so they admit a Lebesgue number $\delta > 0$ (Definition 7.a.15). For any $x, x' \in X$ with $d_X(x, x') < \delta$, the ball $B_\delta(x)$ lies inside some $f^{-1}(B_{\varepsilon/2}(y))$, so $x' \in B_\delta(x)$ gives $f(x), f(x') \in B_{\varepsilon/2}(y)$, hence $d_Y(f(x), f(x')) < \varepsilon$ by the triangle inequality. Since δ did not depend on x, f is uniformly continuous. **Q.E.D.**

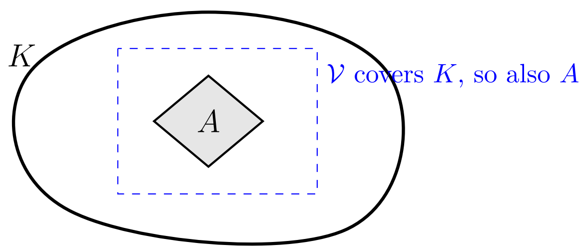
Proposition 7.a.18 (*A closed subset of a compact set is compact*): Let (X, d) be a metric space, $K \subseteq X$ compact, and $A \subseteq K$ closed in X. Then A is compact.

Proof:

Let $\mathcal{U}$ be an open cover of A. Since A is closed, $X \setminus A$ is open, so

$$\mathcal{U}' := \mathcal{U} \cup \{X \setminus A\}$$

is an open cover of K, covering A by assumption and everything outside A by the added set. Compactness of K gives a finite subcover $U_1, \ldots, U_m \in \mathcal{U}'$ of K, hence of A.



Now simply discard $X \setminus A$ from that list if it occurs: it contains no point of A, so the remaining sets still cover A. They all belong to $\mathcal{U}$, and there are finitely many. As $\mathcal{U}$ was arbitrary, A is compact. **Q.E.D.**

Remark 7.a.19 (*Enlarge the cover, then discard the extra set*): The trick just used is the standard way to transfer compactness downward, and it is worth remembering in the form **enlarge the cover by the complement of the set you care about, then throw that complement away again**. It reappears whenever one needs a compact subset of a compact set. Note that the hypothesis *"closed"* is essential, since $(0, 1) \subseteq [0, 1]$ sits inside a compact set without being compact.

Exercise 7.a.20 (*Compact sets are complete*): Let (X, d) be a metric space.

(a) Show that every finite subset $\{x_1, \ldots, x_n\} \subseteq X$ is compact.

(b) If X is compact, is it complete (Definition 6.a.1)?

Proof of Exercise 7.a.20:

(a) Let $\{U_\alpha\}_{\alpha \in I}$ be an open cover of $\{x_1, \ldots, x_n\}$. For each $i = 1, \ldots, n$, pick U_{α_i} such that $x_i \in U_{\alpha_i}$. Then $\{U_{\alpha_i}\}_{i=1,\ldots,n}$ is a finite subcover.

(b) **Yes.** Let $(x_n)_{n=0}^\infty$ be a Cauchy sequence in X. Then, because X is compact, we can extract a convergent subsequence such that $\lim_{i\to\infty} x_{n_i} = x \in X$.

Fix $\varepsilon > 0$. We find $N_1 \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon \quad \forall n, m > N_1$$

and $N_2 \in \mathbb{N}$ such that

$$d(x, x_{n_i}) < \varepsilon \quad \forall i > N_2.$$

Let $N := \max\{N_1, N_2\}$. For any $k > N$, we have:

$$d(x, x_k) \le d(x, x_{n_k}) + d(x_{n_k}, x_k) < 2\varepsilon$$

where we used the fact that $n_k \geq k > N$. Thus, any Cauchy sequence converges.

Q.E.D.

AI-Note 7.4: Corsin writes $n_k > k$ above; for a subsequence the standing fact is $n_k \geq k$. Immaterial to the argument.

Exercise 7.a.21 (*Compactness of a convergent sequence*): Let (X, d) be a metric space. Let $(x_n)_{n=0}^{\infty}$ be a sequence such that $x_n \to \bar{x}$. Prove that the set

$$E := \{x_n \mid n \in \mathbb{N}\} \cup \{\bar{x}\}$$

is compact.

Proof of Exercise 7.a.21:

Using topological compactness: Let $\{U_i\}_{i \in I}$ be an open cover of E. Since $\bar{x} \in E$, there exists some $i_* \in I$ such that $\bar{x} \in U_{i_*}$. Because U_{i_*} is open and $x_n \to \bar{x}$, there exists $N \in \mathbb{N}$ such that $x_n \in U_{i_*}$ for all $n \geq N$. For the remaining finitely many points $x_0, \dots, x_{N-1}$, we can pick sets $U_{i_0}, \dots, U_{i_{N-1}}$ from the cover containing each of them. Then $\{U_{i_*}, U_{i_0}, \dots, U_{i_{N-1}}\}$ is a finite subcover of E.

Using sequential compactness: Let $(y_k)_{k=0}^{\infty}$ be a sequence in E. If (y_k) takes values in a finite subset of E, it has a constant (hence convergent) subsequence. Otherwise, it takes infinitely many distinct values from E, which means it must contain a subsequence of (x_n) progressing towards $\bar{x}$. In either case, it contains a convergent subsequence with limit in E. Thus E is sequentially compact, hence compact.

Q.E.D.

Exercise 7.a.22 (*Compactness of a finite set grid*): Let

$$X := \{(x, y) \in \mathbb{R}^2 \mid x, y \in \mathbb{Z} \wedge |x|, |y| \leq 3\}.$$

Show that X is compact.

Proof of Exercise 7.a.22:

The set X contains exactly $7 \times 7 = 49$ points, so it is finite. By Exercise 7.a.20, every finite subset of a metric space is compact. Alternatively, using sequential compactness (as hinted in the notes), any sequence in X must take at least one value infinitely often (by the pigeonhole principle), thus it contains a constant, hence convergent, subsequence.

Q.E.D.

Heine–Borel (Section 7.b)

Theorem 7.b.23 (*Heine–Borel*): A subset $K \subseteq \mathbb{R}^n$ equipped with the **standard metric** is **compact** if and only if it is "*closed*" and "*bounded*" (German: „*beschränkt*“).

AI-Note 7.5: Neither Corsin nor this document proves Theorem 7.b.23; it is proved in the lecture, and the full argument is longer than anything else in these two chapters. It is worth knowing what goes into it, though, since every ingredient has already appeared:

- **($\Rightarrow$)** is short and general. A compact metric space is bounded and complete (Lemma 7.a.12, Exercise 7.a.20), and a complete subset of the complete space $\mathbb{R}^n$ is closed (Remark 6.a.8).
- **($\Leftarrow$)** is where the real content sits, and it is exactly **Bolzano–Weierstrass**: a bounded sequence in $\mathbb{R}^n$ has a convergent subsequence, obtained by bisecting a box n times over, or by extracting coordinatewise (Proposition 6.a.5) n times in succession. Closedness then puts the limit back inside K, giving sequential compactness, which is compactness by Theorem 7.a.13.

The reason the theorem is special to $\mathbb{R}^n$ with the standard metric is that both steps use the completeness of $\mathbb{R}$ and the finiteness of the dimension. Exercise 7.b.27 below shows how thoroughly it fails otherwise.

Proposition 7.b.24 (*Topologically compact subsets of $\mathbb{R}^n$ are bounded*): If a subset $S \subseteq \mathbb{R}^n$ is topologically compact, then S is bounded.

Proof:

Consider the collection of open balls $\mathcal{U} := \{B_k(0) \mid k \in \mathbb{N}_{\geq 1}\}$ centered at the origin. Since every point $x \in \mathbb{R}^n$ has $\|x\| < k$ for some large integer k, $\mathcal{U}$ forms an open cover of $\mathbb{R}^n$, and hence an open cover of S:

$$S \subseteq \mathbb{R}^n = \bigcup_{k=1}^{\infty} B_k(0).$$

By topological compactness of S, there exists a finite subcover $\{B_{k_1}(0), B_{k_2}(0), \ldots, B_{k_m}(0)\}$. Setting $K := \max\{k_1, k_2, \ldots, k_m\}$, we obtain

$$S \subseteq \bigcup_{j=1}^{m} B_{k_j}(0) = B_K(0).$$

Therefore, S is contained in the ball of radius K centered at the origin, which means S is bounded.

$\boxed{\text{Q.E.D.}}$

Example 7.b.25 (*Compactness via Heine-Borel and continuity*): Consider the following subsets and determine whether they are compact under the standard Euclidean metric:

- **(a)** **Is $\mathbb{N} \subset \mathbb{R}$ compact?** No. While it is closed in $\mathbb{R}$, it is not bounded. By Heine-Borel, it cannot be compact.
- **(b)** **Is $[0,1] \times [0,1] \subset \mathbb{R}^2$ complete?** Yes. The square $[0,1]^2$ is closed and bounded in $\mathbb{R}^2$, hence compact by Heine-Borel. As every compact metric space is complete (Exercise 7.a.20), it is complete.
- **(c)** Let $f : \mathbb{R} \to \mathbb{R}^3, x \mapsto (\sin x, \cos x, 1)$. **Is $f([0, \pi/2])$ compact in $\mathbb{R}^3$?** Yes. We do not need to show this manually using sequences or open covers. Since the domain $[0, \pi/2]$ is compact in $\mathbb{R}$ (closed and bounded) and the function f is continuous, its image $f([0, \pi/2])$ is compact in $\mathbb{R}^3$ (by Item **(a)**).

Exercise 7.b.26 (*Topological properties of various subsets*): Decide whether the following subsets are open, closed, compact, or complete (under the standard Euclidean metric):

- **(a)** $[0,1] \subseteq \mathbb{R}$
- **(b)** $\mathbb{Q} \subseteq \mathbb{R}$
- **(c)** $B_1(0) \subseteq \mathbb{R}^3$
- **(d)** $B_1(0) \cap B_1(1) \subseteq \mathbb{R}^2$
- **(e)** $\{0\} \cup \{1/n \mid n \in \mathbb{N}_{\geq 1}\} \subseteq \mathbb{R}$
- **(f)** $\bigcup_{n=1}^{\infty} B_{1/n}(n) \subseteq \mathbb{R}$
- **(g)** $(-\infty, 1] \subseteq \mathbb{R}$
- **(h)** $f([0,1])$ where $f : \mathbb{R} \to \mathbb{R}^n$ is continuous
- **(i)** $f^{-1}([0,1])$ where $f : \mathbb{R}^n \to \mathbb{R}$ is continuous
- **(j)** $f^{-1}((0,1))$ where $f : \mathbb{R}^n \to \mathbb{R}$ is continuous
- **(k)** $S := \operatorname{span}\{(1,0,0)^\mathsf{T}, (3,1,0)^\mathsf{T}\} \subseteq \mathbb{R}^3$
- **(l)** $\{x \in \mathbb{R}^3 \mid \exists s \in S \text{ s.t. } d(x,s) < 1\}$ for a subset $S \subseteq \mathbb{R}^3$

Proof of Exercise 7.b.26:

- **(a)** $[0,1] \subseteq \mathbb{R}$: **Closed**, **compact**, **complete**. (Not open).
- **(b)** $\mathbb{Q} \subseteq \mathbb{R}$: **None**. It is neither open nor closed, not bounded (so not compact), and not complete.
- **(c)** $B_1(0) \subseteq \mathbb{R}^3$: **Open**.
- **(d)** $B_1(0) \cap B_1(1) \subseteq \mathbb{R}^2$: **Open** (finite intersection of open sets).
- **(e)** $\{0\} \cup \{1/n \mid n \in \mathbb{N}_{\geq 1}\} \subseteq \mathbb{R}$: **Closed**, **compact**, **complete** (bounded and contains its only limit point 0).
- **(f)** $\bigcup_{n=1}^{\infty} B_{1/n}(n) \subseteq \mathbb{R}$: **Open** (arbitrary union of open sets).
- **(g)** $(-\infty, 1] \subseteq \mathbb{R}$: **Closed**, **complete**.
- **(h)** $f([0,1])$: **Closed**, **compact**, **complete** (continuous image of a compact set).
- **(i)** $f^{-1}([0,1])$: **Closed**, **complete** (continuous preimage of a closed set).
- **(j)** $f^{-1}((0,1))$: **Open** (continuous preimage of an open set).
- **(k)** $S := \operatorname{span}\{(1,0,0)^\mathsf{T}, (3,1,0)^\mathsf{T}\} \subseteq \mathbb{R}^3$: **Closed**, **complete** (it is a 2-dimensional subspace, hence closed in $\mathbb{R}^3$).
- **(l)** $\{x \in \mathbb{R}^3 \mid \exists s \in S \text{ s.t. } d(x,s) < 1\}$: **Open**. This set is equivalently written as $\{x \in \mathbb{R}^3 \mid d(x,S) < 1\}$ or $\bigcup_{s \in S} B_1(s)$, which is an arbitrary union of open balls.

Q.E.D.

This captures our intuition of a *"compact"* set. We will briefly show that Heine–Borel is **false** for $\mathbb{R}^n$ with an **arbitrary metric**.

Exercise 7.b.27 (*Heine–Borel fails for arbitrary metrics*): Let (X,d) be a metric space.

- **(a)** *(Optional)* Show that $\tilde{d}(x,y) := \dfrac{d(x,y)}{1+d(x,y)}$ is a metric on X.
- **(b)** Show that, if $U \subseteq X$ is open with respect to d, then U is open with respect to $\tilde{d}$.
- **(c)** Show that, for $X = \mathbb{R}^n$, $d(x,y) = \dfrac{|x-y|}{1+|x-y|}$, $\mathbb{N} \subseteq X$ is closed and bounded but not compact.

AI-Note 7.6: Part **(c)** above sets $X = \mathbb{R}^n$ but then works with $\mathbb{N} \subseteq X$ and $|x-y|$, which only parses for $n = 1$.

AI-Example 7.b.28 (*Open Cover of* $(0,1)$ *with No Finite Subcover*): Consider the open interval $X = (0,1)$ under the standard Euclidean metric. The family of open sets

$$U_n := \left(\frac{1}{n}, 1\right) \quad \text{for } n \in \mathbb{N}_{\geq 2}$$

forms an open cover of $(0,1)$ because for any $x \in (0,1)$, there exists $n \in \mathbb{N}$ with $1/n < x$. However, any finite subcollection $\{U_{n_1}, \ldots, U_{n_k}\}$ has a maximum index $N := \max\{n_1, \ldots, n_k\}$, so its union is $U_N = (1/N, 1)$, which fails to cover points in $(0, 1/N]$. Thus $(0,1)$ is bounded but not compact.

Example 7.b.29 (*Heine–Borel needs* $\mathbb{R}^n$ *itself, not just the standard-looking metric*): $\mathbb{Q} \cap [0,2]$, with the metric inherited from the standard metric on $\mathbb{R}$, is closed and bounded **as a subset of** $\mathbb{Q}$, and yet it is not compact. The family

$$\mathbb{Q} \cap [0,2] \;=\; \big(\mathbb{Q} \cap [0, \sqrt{2})\big) \;\cup \bigcup_{p \in \mathbb{Q},\, p > \sqrt{2}} \big(\mathbb{Q} \cap (p, 2]\big)$$

is an open cover of $\mathbb{Q} \cap [0,2]$ (each set is the trace of an open subset of $\mathbb{R}$, hence open in the subspace $\mathbb{Q} \cap [0,2]$): every rational in $[0,2]$ is either $< \sqrt{2}$ or $> \sqrt{2}$, since $\sqrt{2} \notin \mathbb{Q}$. Any finite subcover uses only finitely many of the sets $\mathbb{Q} \cap (p,2]$, hence has a largest such p, call it $p_{\max}$, and then misses every rational in $(\sqrt{2}, p_{\max}]$, of which there are infinitely many. This is sharper than Example 7.b.25**(a)**'s $\mathbb{N}$. There, compactness fails because the set is unbounded. Here it fails on a closed **and** bounded set, purely because the ambient space $\mathbb{Q}$ is not $\mathbb{R}^n$. The irrationality of $\sqrt{2}$ is doing all the work, leaving a *"hole"* exactly where the cover would need to close up.

Proof of Exercise 7.b.27:

(a) Non-negativity, definiteness and symmetry are inherited directly from the corresponding properties of d, since $t \mapsto \frac{t}{1+t}$ is non-negative on $[0,\infty)$ and vanishes only at $t=0$.

Triangle inequality. We want to show
$$\frac{d(x,y)}{1+d(x,y)} \le \frac{d(x,z)}{1+d(x,z)} + \frac{d(z,y)}{1+d(z,y)}.$$
Denote $d(x,y)=d_1$, $d(x,z)=d_2$, $d(z,y)=d_3$ and multiply out:
$$\begin{aligned}
& d_1(1+d_2)(1+d_3) \le d_2(1+d_1)(1+d_3) + d_3(1+d_1)(1+d_2) \\
\iff\ & (d_1+d_1d_2)(1+d_3) \le (d_2+d_2d_1)(1+d_3) + (d_3+d_1d_3)(1+d_2) \\
\iff\ & d_1 + d_1d_3 + d_1d_2d_3 + d_1d_2 \le d_2 + d_3d_2 + d_2d_1d_3 + d_2d_1 \\
& \qquad\qquad\qquad\qquad + d_3 + d_3d_2 + d_1d_3 + d_1d_3d_2 \\
\iff\ & d_1 \le d_2 + 2d_2d_3 + d_3 + d_1d_2d_3 \\
\Longleftarrow\ & 0 \le 2d_2d_3 + d_1d_2d_3 \quad \text{together with } d_1 \le d_2+d_3 \text{ (the triangle inequality for } d)
\end{aligned}$$
which is true by non-negativity.

(b) Let $U \subseteq X$ be open. Then, given $x \in U$, we find $r>0$ such that $B_r(x) := \{y \in X : d(x,y) < r\} \subseteq U$. Let $\varepsilon := \dfrac{r}{1+r}$. Then $\tilde{B}_\varepsilon(x) \subseteq B_r(x) \subseteq U$, where $\tilde{B}_\varepsilon(x) := \{y \in X : \tilde{d}(x,y) < \varepsilon\}$.

(c) Note that for all $x,y \in \mathbb{R}$, $\dfrac{|x-y|}{1+|x-y|} < 1$. So $\mathbb{N} \subseteq B_1(0)$, thus it is **bounded** with respect to this metric. Also, $\mathbb{N}$ is **closed**, as follows from **(b)**. But $x_n = n$ defines a sequence in $\mathbb{N}$ with no convergent subsequence, thus $\mathbb{N}$ is **not compact**. **Q.E.D.**

AI-Note 7.7: The cancellation in the derivation above originally contained a stray $d_2d_1d_2$ where the expansion gives $d_2d_1d_3$; the cancelled terms and the conclusion $0 \le 2d_2d_3 + d_1d_2d_3$ are unaffected. Re-derived cleanly above.

Remark 7.b.30 (*The moral: boundedness is not a topological property*): It is worth saying out loud what Exercise 7.b.27 has just demonstrated, because the three parts are more striking together than separately.

Part **(b)** shows that d and $\tilde{d}$ have the **same open sets**. (Only one direction is asked for, but the converse follows the same way: given $\tilde{B}_\varepsilon(x)$, the choice $r := \frac{\varepsilon}{1-\varepsilon}$ gives $B_r(x) \subseteq \tilde{B}_\varepsilon(x)$.) Two metrics with the same open sets have the same convergent sequences, the same continuous functions, and, crucially, the same **compact** sets, since compactness is defined purely by open covers (Definition 7.a.1).

Part **(c)** shows that they do **not** have the same bounded sets. Indeed $\tilde{d} < 1$ always, so under $\tilde{d}$ **every** subset of **every** metric space is bounded. The notion collapses entirely.

Putting those together: **compactness is topological, boundedness is not.** Boundedness is a property of the chosen metric, not of the open sets it generates, so no theorem of the form *"compact if and only if closed and bounded"* can hold for a general metric. That is precisely why Theorem 7.b.23 carries the hypothesis **standard metric**. That hypothesis is not a technicality to be skimmed; it is the entire content of the theorem.

Remark 7.b.31 (*What boundedness should have been*): The right general replacement is already available: compact if and only if complete **and** "*totally bounded*" (Definition 7.a.9, Remark 7.a.14). Total boundedness is the notion that boundedness was trying and failing to be, and $\mathbb{N}$ under $\tilde{d}$ separates the two cleanly. For $\varepsilon < \frac{1}{2}$ the ball $\tilde{B}_\varepsilon(n)$ contains no integer other than n itself, so no finite family of ε-balls covers $\mathbb{N}$. The set is therefore bounded but not totally bounded, and that is exactly the gap through which compactness escapes.

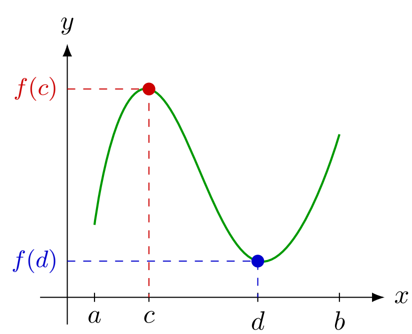
Why compactness matters (Section 7.c)

(a) **Preserved by continuous functions:** X, Y metric spaces, $K \subseteq X$ compact, $f : X \to Y$ continuous. Then $f(K) \subseteq Y$ is compact.

(b) **Weierstrass theorem / Extreme value theorem:** X metric space, $K \subseteq X$ compact, $f : K \to \mathbb{R}$ continuous. Then f attains its maximum and minimum in K. In particular, f is bounded. This is proved as Lemma 7.a.7.

AI-Note 7.8: The official script's statement of this theorem (Cor. 9.79) writes the conclusion as *"there exist $\bar{x} \in X$ such that $f(\bar{x}) = \sup_K f$"*, with $\bar{x}$ ranging over the whole ambient space X rather than over K; the proof given there repeats the slip, writing $f^{-1}(\sup f(X))$ where it must mean $f^{-1}(\sup f(K))$. As stated, the claim is false as soon as f is unbounded off K: e.g. $K := \{0\} \subset X := \mathbb{R}$, $f(x) := x$. The maximiser $\bar{x}$ must be sought in K, as in the statement above.

(c) **Uniform continuity:** X, Y metric spaces, $f : X \to Y$ continuous. If X is compact, then f is uniformly continuous (Definition 5.a.4). This is proved as Proposition 7.a.17, via the Lebesgue number of the cover $\{f^{-1}(B_{\varepsilon/2}(y))\}_y$.



(b) on the compact $K = [a, b]$, f attains its maximum at c and its minimum at d

Both extrema are attained **inside** K. That is the content of **(b)**, and it is exactly what fails on a non-compact domain: on the open interval (a, b) the same f may approach a supremum it never reaches.

Remark 7.c.32 (*What compactness and connectedness are* ***for****: local to global*)**:** The three facts above look like a list of unrelated services compactness happens to provide. Diego Torres Tejeda's notes give the idea that unifies them, and it is worth having early because it also explains why connectedness sits in the same chapter.

Local-to-global arguments are a recurring theme across mathematics. It is often easy to check a property **locally**, on some ball $B_\varepsilon(x)$ around each point, while what one actually wants is the property **globally**, on all of X. Compactness and connectedness are the two hypotheses that license that jump. Schematically, a local-to-global statement reads:

> *If for every $x \in X$ there is $\varepsilon > 0$ such that $B_\varepsilon(x)$ has property P, then X has property P.*

Each of the two hypotheses powers its own version, and the proofs are three lines each.

Connectedness turns *"locally constant"* into *"constant"*. Let (X,d) be connected and $f : X \to \mathbb{R}$ locally constant. Every level set $f^{-1}(a)$ is then open. Fix $a \in f(X)$ and split

$$X = \underbrace{f^{-1}(a)}_{=:U} \sqcup \underbrace{\bigcup_{b \neq a} f^{-1}(b)}_{=:V},$$

a disjoint union of two open sets. Since $U \neq \emptyset$, connectedness forces $V = \emptyset$, i.e. $f \equiv a$.

Compactness turns *"locally bounded"* into *"bounded"*. Let X be compact and suppose each x has an open $U_x \ni x$ on which f is bounded. The $\{U_x\}_{x\in X}$ cover X, so finitely many suffice, say $U_1, \dots, U_n$, and then

$$\sup_{x\in X} |f(x)| \;=\; \max_{1\le i\le n} \sup_{x\in U_i} |f(x)| \;<\; \infty,$$

because a **maximum of finitely many** finite numbers is finite. A continuous f is locally bounded, since $|f| < |f(x)| + 1$ near each x, so this recovers the boundedness half of Item **(b)**.

Read this way the two proofs are the same proof twice, with different notions of *"finitely many"*: connectedness says a partition into two open pieces must be trivial, compactness says a cover can be taken finite. Both convert local information into global. Every application in Items **(a)** to **(c)** is an instance. Uniform continuity is the most visible of them, since there the whole difficulty is that the δ supplied by continuity is local and one needs a single global one (Important remark 5.a.5).

The exercise below covers the same ground as Exercise 7.b.26, and is worth doing anyway. It adds the explicit **?** answers for the cases where the data is insufficient, a trap in the final item involving two **parallel** spanning vectors, and a four-column solution table which is where the lesson actually lands. Read Exercise 7.b.26 as the warm-up and this one as the real test.

Exercise 7.c.33 (*Classify these sets*): Sascha Brack sets this as a self-test at the end of his compactness lecture, and it is the fastest way to find out whether the four notions have actually separated in your head. For each subset of $\mathbb{R}^n$ (standard metric), decide whether it is **open**, **closed**, **compact**, **complete**. Throughout, f denotes a continuous function and **?** means *"not decidable from the information given"*.

1. $[0,1] \subseteq \mathbb{R}$
2. $\mathbb{Q} \subseteq \mathbb{R}$
3. $B_1(0) \subseteq \mathbb{R}^3$
4. $\left\{\frac{1}{n} : n \ge 1\right\} \subseteq \mathbb{R}$, and separately $\{0\} \cup \left\{\frac{1}{n} : n \ge 1\right\}$
5. $\bigcup_{n\ge 1} B_{1/n}(n) \subseteq \mathbb{R}$
6. $(-\infty, 1] \subseteq \mathbb{R}$
7. $f([0,1])$, $f^{-1}([0,1])$, $f^{-1}\big((0,1)\big)$
8. $S := \operatorname{span}\left\{ \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}3\\0\end{pmatrix} \right\} \subseteq \mathbb{R}^2$

Remark 7.c.34 (*What continuous maps do and do not preserve*): Diego's exercise-class notes collect Item **(a)** and the analogous fact for connectedness (Theorem 8.a.6) into one table, together with the dual statement for preimages, and a caution worth stating explicitly:

	Preserved by f^{-1}	**Preserved by f**
open	yes (Definition 5.a.1)	not in general
closed	yes	not in general
compact	not in general	yes (Item **(a)**)
connected	not in general	yes (Theorem 8.a.6)

Caution: completeness and boundedness are **not** topological properties, so in general they are preserved neither by continuous maps nor by their inverses. For instance, $\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ is a continuous bijection carrying a bounded set onto an unbounded one.

Exercise 7.c.35 (*3.3 — Open, closed, complete and compact*): For each of the following subsets of $\mathbb{R}^N$ say whether they are open / closed / none / compact (with respect to the standard Euclidean structure). Try to prove your assertions "efficiently".

1. $E_1 := \{x \in \mathbb{R}^3 : 0 < x_2 \leq 2\}$
2. $E_2 := \{x \in \mathbb{R}^n : \sin(\|x\|) \geq \frac{1}{4}\}$
3. $E_3 := \bigcup_{n\geq 1}\{x \in \mathbb{R}^3 : x_1^4 - \frac{1}{n}x_3^2 - x_2^2 > \frac{1}{n},\ x_1 + x_2 < 6\}$
4. $E_4 := \{(x, y) \in (\mathbb{R}^n \times \mathbb{R}^n) : x \cdot y > \frac{1}{2}\|x\|\|y\|\}$
5. $E_5 := \{X \in \mathbb{R}^{n\times n} : \text{the matrix } X \text{ is invertible}\}$
6. $E_6 := \{X \in \mathbb{R}^{n\times n} : \text{the matrix } X \text{ is symmetric}\}$
7. $E_7 := \{X \in \mathbb{R}^{n\times n} : \text{the entries of } X \text{ are either } 0 \text{ or } 1\}$
8. $E_8 := \{x \in \mathbb{R}^n : |x_i| \leq 6 \text{ for all } 1 \leq i \leq n\}$
9. $E_9 := E_8 \times E_3 \subset \mathbb{R}^{n+3}$
10. $E_{10} := E_2 \cap E_8 \subset \mathbb{R}^n$

Exercise 7.c.36 (*3.4 — Multiple choice (open, complete, compact)*): Select all the statements below that are true.

- **(a)** A nonempty open strict subset of $\mathbb{R}^n$ cannot be compact.
- **(b)** A nonempty open strict subset of $\mathbb{R}^n$ cannot be complete.
- **(c)** A complete subset of $\mathbb{R}^n$ contains all its accumulation points.
- **(d)** Countable union of complete subsets of $\mathbb{R}^n$ is complete.
- **(e)** A subset is closed in $\mathbb{R}^n$ if and only if it is complete as a metric space itself (with the distance inherited from $\mathbb{R}^n$).

Compactness in spaces of functions: Ascoli–Arzelà (Section 7.d)

AI-Note 7.9: Corsin's notes stop at Theorem 7.b.23, which settles compactness in $\mathbb{R}^n$ and says nothing about spaces of **functions**. Linus devotes his final session to the missing theorem (Theorem 15.47 of the lecture notes), presenting it as *"the Bolzano–Weierstrass theorem, one level up"*. It is worth having here for a concrete reason: it is the compactness input that Remark A.d.21 appeals to without naming, when it says Peano's theorem extracts a convergent subsequence from a family of approximate solutions.

The space of continuous functions on a compact set (Subsection 7.d.1)

Definition 7.d.37 (*Supremum norm on $C(K, \mathbb{R}^m)$*): Let $K \subseteq \mathbb{R}^n$ be compact. Write $C(K, \mathbb{R}^m)$ for the set of continuous maps $f : K \to \mathbb{R}^m$, a real vector space under pointwise operations, and equip it with the "*supremum norm*" (German: *„Supremumsnorm"*)

$$\|f\|_\infty := \sup_{x\in K} |f(x)|.$$

The supremum is finite because $|f|$ is continuous on a compact set, hence bounded (Section 7.c), which is the same point already made for the supremum metric in Exercise 2.b.21. Convergence in this norm is exactly **uniform** convergence.

Proposition 7.d.38 (*$C(K, \mathbb{R}^m)$ is complete*): For $K \subseteq \mathbb{R}^n$ compact, $\big(C(K, \mathbb{R}^m), \|\cdot\|_\infty\big)$ is a complete normed vector space.

Proof:

Let $(f_k)_{k\in\mathbb{N}}$ be Cauchy for $\|\cdot\|_\infty$. For each fixed $x \in K$ we have $|f_k(x) - f_\ell(x)| \le \|f_k - f_\ell\|_\infty$, so $(f_k(x))_k$ is Cauchy in $\mathbb{R}^m$ and converges, by completeness of $\mathbb{R}^m$, to a limit we call $f(x)$. This defines a function $f : K \to \mathbb{R}^m$.

The convergence is uniform. Given $\varepsilon > 0$ choose N with $\|f_k - f_\ell\|_\infty < \varepsilon$ for $k, \ell \ge N$; fixing $k \ge N$ and letting $\ell \to \infty$ in $|f_k(x) - f_\ell(x)| < \varepsilon$ gives $|f_k(x) - f(x)| \le \varepsilon$ for **every** x, i.e. $\|f_k - f\|_\infty \le \varepsilon$.

Finally f is continuous, being a uniform limit of continuous functions, by the standard $\varepsilon/3$ argument from **Analysis I**. Hence $f \in C(K, \mathbb{R}^m)$ and $f_k \to f$ in the norm.

Q.E.D.

Why Bolzano–Weierstrass fails one level up (Subsection 7.d.2)

In $\mathbb{R}^n$, bounded sequences have convergent subsequences. The naive transfer of that statement to $C(K, \mathbb{R}^m)$ is **false**, and it is worth seeing exactly how.

AI-Example 7.d.39 (*A bounded sequence of functions with no convergent subsequence*): Take $K := [0,1]$, $m := 1$ and $f_k(x) := x^k$. Every f_k is continuous with $\|f_k\|_\infty = 1$, so the sequence is bounded, and indeed lies on the unit sphere.

Yet no subsequence converges in $C([0,1])$. Pointwise,

$$f_k(x) \longrightarrow g(x) := \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1, \end{cases}$$

and the same holds for every subsequence. If some f_{k_j} converged uniformly to a limit $f \in C([0,1])$, then it would converge pointwise to f as well, forcing $f = g$; but g is not continuous. So no subsequence can converge.

Remark 7.d.40 (*The diagnosis*): Boundedness was never the issue. What goes wrong is that the f_k become arbitrarily **steep** near $x = 1$: to keep $|f_k(x) - f_k(y)|$ below a fixed ε there, one needs $|x - y|$ smaller and smaller as k grows. Each individual f_k is uniformly continuous (Definition 5.a.4), being continuous on a compact interval. What fails is that no single δ works for all of them at once. Ruling that out is precisely what the next definition does.

Equicontinuity, and the theorem (Subsection 7.d.3)

Definition 7.d.41 (*Equicontinuity*): A family $(f_k)_{k\in\mathbb{N}}$ in $C(K, \mathbb{R}^m)$ is "*equicontinuous*" (German: „*gleichgradig stetig*") if

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 \;\; \forall k \in \mathbb{N} \;\; \forall x, y \in K : \quad |x - y| < \delta \implies |f_k(x) - f_k(y)| < \varepsilon.$$

Remark 7.d.42 (*Reading the quantifiers*): Compare the three conditions, which differ only in what δ is allowed to depend on:

- **Continuity** of one f: δ may depend on ε **and on the point** x.
- **Uniform continuity** of one f: δ may depend on ε only, so there is one δ for all points.
- **Equicontinuity** of a family: δ may depend on ε only, so there is one δ for all points **and all members of the family**.

So equicontinuity is uniform continuity made uniform in one further variable, the index. Each x^k of AI-Example 7.d.39 is uniformly continuous, and the family is not equicontinuous; that is the whole gap.

Theorem 7.d.43 (*Ascoli–Arzelà*): Let $K \subseteq \mathbb{R}^n$ be compact and let $(f_k)_{k\in\mathbb{N}}$ be a sequence in $C(K, \mathbb{R}^m)$ that is

(a) **bounded**: $\sup_k \|f_k\|_\infty < \infty$; and
(b) **equicontinuous** in the sense of Definition 7.d.41.

Then (f_k) has a subsequence converging uniformly to some $f \in C(K, \mathbb{R}^m)$.

AI-Note 7.10: None of the transcribed notes prove this theorem; Linus states it and remarks that it is *"mostly just used in proofs"*. The sketch below is included because its mechanism, a diagonal argument, explains why **both** hypotheses are needed, which the statement alone does not. It is a sketch and not a proof: the details are omitted.

Remark 7.d.44 (*Proof sketch: the diagonal argument*): Fix a countable dense subset $D = \{q_1, q_2, \dots\} \subseteq K$ (the points of K with rational coordinates will do).

Step 1: converge on D, using boundedness. The values $(f_k(q_1))_k$ form a bounded sequence in $\mathbb{R}^m$, so by Bolzano–Weierstrass some subsequence converges at q_1. Pass to a further subsequence converging also at q_2, then at q_3, and so on. Each refinement is legitimate, but no single one of them handles all of D. Taking the **diagonal** sequence, whose j-th member is the j-th member of the j-th subsequence, produces one subsequence converging at every point of D simultaneously.

Step 2: upgrade to uniform convergence, using equicontinuity. Convergence on a dense set says nothing on its own about the rest of K; a sequence can converge at every rational and oscillate wildly in between. Equicontinuity forbids that: given ε, pick the δ it supplies, cover the compact K by finitely many balls of radius δ centred at points of D, and control any x by the nearby dense point. This makes the diagonal subsequence uniformly Cauchy, hence convergent by Proposition 7.d.38.

Remark 7.d.45 (*Where each hypothesis does its work*): The two hypotheses are not interchangeable, and the sketch shows which does what: **boundedness** is what lets Bolzano–Weierstrass run at each individual point, and **equicontinuity** is what converts pointwise control on a dense set into uniform control everywhere. Delete the second and AI-Example 7.d.39 is a counterexample; delete the first and the constant sequences $f_k \equiv k$ are equicontinuous (they are all constant, so any δ works) with no convergent subsequence at all.

AI-Exercise 7.d.46 (*A uniform derivative bound gives compactness*): Let $(f_k)_{k\in\mathbb{N}}$ be a sequence in $C^1([0,1], \mathbb{R})$ with

$$|f_k'(x)| \le 1 \quad \text{for all } x \in [0,1] \text{ and all } k, \qquad \text{and} \qquad f_k(0) = 0 \text{ for all } k.$$

(a) Show that the family is equicontinuous.
(b) Show that it is bounded in $\|\cdot\|_\infty$.
(c) Conclude that some subsequence converges uniformly, and explain why the argument would break down if the bound $|f_k'| \le 1$ were replaced by *"each f_k is differentiable"*.

That closes the account of compactness. What follows returns to $\mathbb{R}^n$ and to differentiation, where compactness reappears as a hypothesis rather than a subject.

Solutions (Section 7.e)

Solution to Exercise 7.c.33:

Set	open	closed	compact	complete
$[0,1]$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
$\mathbb{Q}$	$\times$	$\times$	$\times$	$\times$
$B_1(0) \subseteq \mathbb{R}^3$	$\checkmark$	$\times$	$\times$	$\times$
$\{\frac{1}{n}\}$	$\times$	$\times$	$\times$	$\times$
$\{0\} \cup \{\frac{1}{n}\}$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
$\bigcup_{n\geq 1} B_{1/n}(n)$	$\checkmark$	$\times$	$\times$	$\times$
$(-\infty, 1]$	$\times$	$\checkmark$	$\times$	$\checkmark$
$f([0,1])$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
$f^{-1}([0,1])$	?	$\checkmark$	?	$\checkmark$
$f^{-1}\big((0,1)\big)$	$\checkmark$	?	?	?
$S = \operatorname{span}\{(1,0),(3,0)\}$	$\times$	$\checkmark$	$\times$	$\checkmark$

The rows that carry the lesson:

$\{\frac{1}{n}\}$ versus $\{0\} \cup \{\frac{1}{n}\}$. Adding the single point 0 changes all four answers. Without it the set misses its own limit point, so it is not closed, and the Cauchy sequence $\frac{1}{n}$ has nowhere to converge inside the set. With it, the set is closed and bounded, hence compact by Theorem 7.b.23.

$\bigcup_{n\geq 1} B_{1/n}(n)$. Open as a union of open sets (Item **(a)**); not closed, since each endpoint $n \pm \frac{1}{n}$ is a limit point outside the set; unbounded, so certainly not compact.

S is a trap. The two spanning vectors are parallel, so S is not $\mathbb{R}^2$ but the x-axis, a line, which is closed, unbounded and not compact.

The f rows show exactly how far continuity gets you. Preimages behave: f^{-1} of a closed set is closed, of an open set is open (Definition 5.a.1). Images do not, and only compactness transfers forward (Item **(a)**), which is why $f([0,1])$ is compact but $f^{-1}([0,1])$ need not be: take $f \equiv 0$, giving $f^{-1}([0,1]) = \mathbb{R}^n$. That same f makes $f^{-1}([0,1])$ open, while $f = \mathrm{id}$ makes it $[0,1]$, which is not, hence the **?**. And $f([0,1])$ is never open: it is compact, hence closed and bounded, and a non-empty proper subset of $\mathbb{R}^m$ cannot be both open and closed, since $\mathbb{R}^m$ is connected (Definition 8.a.1).

Remark 7.e.47 (*Why the last two columns agree*): The punchline of the table is that its last two columns are **identical**. That is not a coincidence of this particular list. Inside a **complete** ambient space a subset is complete exactly when it is closed (Remark 6.a.8), and $\mathbb{R}^n$ is complete. Build the same table inside $\mathbb{Q}$ instead and the two columns come apart immediately.

Q.E.D.

AI-Note 7.11: Only the `aiexercise` problems below have solutions in this section: Corsin's own exercises in this chapter (Exercises 7.b.27 and 8.a.11) are solved inline, immediately after their statement, mirroring how he presents them in his own notes (see the *"Kept inline"* comments above each).

Solution to Exercise 7.c.35:

The efficient route is to write each set as a preimage of an open or closed set under a continuous map, and then read openness or closedness off Definition 5.a.1**(c)** rather than chasing balls. Compactness is then Heine–Borel (Theorem 7.b.23): closed **and** bounded.

	Set	open	closed	compact
1	$E_1 = \{0 < x_2 \le 2\} \subseteq \mathbb{R}^3$	×	×	×
2	$E_2 = \{\sin\lVert x\rVert \ge \frac{1}{4}\}$	×	✓	×
3	E_3 (union of open sets)	✓	×	×
4	$E_4 = \{x \cdot y > \frac{1}{2}\lVert x\rVert\lVert y\rVert\}$	✓	×	×
5	$E_5 = \{X \text{ invertible}\}$	✓	×	×
6	$E_6 = \{X \text{ symmetric}\}$	×	✓	×
7	$E_7 = \{X \text{ with entries } 0, 1\}$	×	✓	✓
8	$E_8 = [-6, 6]^n$	×	✓	✓
9	$E_9 = E_8 \times E_3$	×	×	×
10	$E_{10} = E_2 \cap E_8$	×	✓	✓

1. Half-open in the x_2 direction, so **neither**: no ball around a point with $x_2 = 2$ stays inside, and the sequence $(0, \frac{1}{k}, 0) \to 0 \notin E_1$ shows it is not closed. Unbounded, hence not compact.
2. **Closed:** $E_2 = h^{-1}\big([\frac{1}{4}, \infty)\big)$ for the continuous $h(x) := \sin(\lVert x\rVert)$. Not open, since a point with $\sin\lVert x\rVert = \frac{1}{4}$ has points of E_2^c arbitrarily close. Not compact: sin is periodic, so E_2 contains infinitely many concentric annuli and is unbounded.
3. **Open:** each set in the union is defined by two **strict** inequalities between continuous functions, hence open, and an arbitrary union of open sets is open (Item **(a)**). It is non-empty (take $x = (2, 0, 0)$ and $n = 1$) and unbounded (take $x = (-M, 0, 0)$ for large M), so it is not compact; and a non-empty proper open subset of the connected space $\mathbb{R}^3$ cannot also be closed (Definition 8.a.1).
4. **Open:** $E_4 = F^{-1}\big((0, \infty)\big)$ for the continuous $F(x, y) := x \cdot y - \frac{1}{2}\lVert x\rVert\lVert y\rVert$. Unbounded (scale any element), hence not compact, and not closed by the connectedness argument of part 3. Geometrically E_4 is the set of pairs enclosing an angle strictly less than 60° (Remark 2.a.4).
5. **Open:** $E_5 = \det^{-1}\big(\mathbb{R} \setminus \{0\}\big)$ and det is a polynomial in the entries, hence continuous. Unbounded, so not compact.
6. **Closed:** E_6 is the solution set of the linear equations $X_{ij} - X_{ji} = 0$, i.e. the preimage of the closed set $\{0\}$ under a continuous (indeed linear) map. Being a linear subspace it is unbounded, hence not compact, and for $n \ge 2$ it is a proper subspace, hence has empty interior and is not open.
7. **Closed and compact:** E_7 is **finite**, with 2^{n^2} elements. Finite sets are compact (Exercise 7.a.20**(a)**) and closed. Not open, since no ball around a matrix of 0s and 1s consists only of such matrices.
8. **Closed and compact:** $E_8 = [-6, 6]^n$ is closed (a finite intersection of the closed sets $\{|x_i| \le 6\}$) and bounded, so Theorem 7.b.23 applies. Not open: no ball around a point with some $|x_i| = 6$ fits inside.
9. **Neither, and not compact.** For non-empty A, B, the product $A \times B$ is open exactly when both factors are, closed exactly when both are, and compact exactly when both are. Here E_8 is not open and E_3 is neither closed nor compact, so E_9 fails all three.

   AI-Note 7.12: The official solution opens this part with *"Since E_8 is unbounded so is E_9"*. That is a slip, since $E_8 = [-6, 6]^n$ is bounded. It is E_3 that is unbounded, as the official solution itself established two parts earlier. The conclusion (that E_9 is not compact) is correct; only the reason is misattributed.

10. **Closed and compact:** an intersection of the closed set E_2 with the closed set E_8 is closed (Proposition 3.a.8**(c)**), and it is bounded because E_8 is. Heine–Borel again. Alternatively, quote Proposition 7.a.18: a closed subset of the compact E_8 is compact.

Remark 7.e.48 (*The efficient method, extracted*): Nine of the ten sets were settled without producing a single ε. The recipe is: **strict** inequalities between continuous functions give open sets, **weak** ones give closed sets, and mixing the two gives neither. Compactness then reduces to checking boundedness, by Heine–Borel. This is what the problem means by proving assertions *"efficiently"*, and it is the practical payoff of Definition 5.a.1**(c)** over the ε–δ formulation.

Q.E.D.

Solution to Exercise 7.c.36:

(a), (b), (c) and (e) are true; (d) is false.

The engine behind **(a)**, **(b)**, **(c)** and **(e)** is a single fact, Remark 6.a.8: inside a **complete** ambient space, a subset is complete if and only if it is closed. Here the ambient space is $\mathbb{R}^n$, which is complete by Proposition 6.a.5.

(a) True. A compact subset of $\mathbb{R}^n$ is closed and bounded (Theorem 7.b.23). If it were also open and non-empty, it would be a non-empty **clopen** proper subset of $\mathbb{R}^n$, which is impossible because $\mathbb{R}^n$ is connected. Indeed $\mathbb{R}^n$ is path-connected, any two points being joined by a segment, and hence connected by Lemma 8.b.18.

(b) True, by the same argument: complete $\Rightarrow$ closed, and non-empty, open, closed and proper is again impossible in the connected space $\mathbb{R}^n$.

(c) True. Complete $\Rightarrow$ closed, and a closed set contains all its accumulation points. Indeed, if x is an accumulation point of A then some sequence in A converges to x, and Proposition 4.a.7**(b)** puts x in A.

(d) False. Every singleton $\{q\}$ is complete (its only Cauchy sequences are eventually constant), yet

$$\mathbb{Q} = \bigcup_{q\in\mathbb{Q}} \{q\}$$

is a countable union of complete sets which is famously **not** complete (Remark 6.a.8). Equivalently, in terms of closedness: a countable union of closed sets need not be closed, exactly as in Important remark 3.a.9.

(e) True, this being Remark 6.a.8 stated for the ambient space $\mathbb{R}^n$, and it is worth stressing that the *"if"* direction is where completeness of the ambient space is used. Inside $\mathbb{Q}$ instead of $\mathbb{R}$ the statement collapses: $\mathbb{Q}$ is closed in itself but not complete. Q.E.D.

Remark 7.e.49 (*A second route to* **(b), *avoiding connectedness*):** Part **(b)** is also script Exercise 9.76, and the script proves it without invoking connectedness at all. The argument is more elementary and worth recording. Fix $x \in U$ and $y \notin U$, and let $\gamma(t) := x + t(y-x)$, $t \in [0,1]$, be the segment between them. Put $t_* := \sup\{t \in [0,1] : \gamma(t) \in U\}$. The point $\gamma(t_*)$ cannot lie in U, since openness of U would let t_* be pushed further out, and yet it is a limit of points $\gamma(t_k) \in U$ with $t_k \nearrow t_*$. That is a convergent, hence Cauchy, sequence in U whose limit escapes U.

This avoids Lemma 8.b.18 entirely, at the cost of only working in $\mathbb{R}^n$, since the segment needs the vector space structure. The connectedness proof above generalises to any connected, locally connected ambient space.

Solution to AI-Exercise 7.d.46:

(a) By the mean value theorem, for any $x, y \in [0,1]$ there is ξ between them with

$$|f_k(x) - f_k(y)| = |f_k'(\xi)|\,|x-y| \le |x-y|,$$

so every f_k is 1-Lipschitz **with the same constant**. Given $\varepsilon > 0$, the choice $\delta := \varepsilon$ then satisfies Definition 7.d.41 for every k and every pair x, y at once. A uniformly Lipschitz family is always equicontinuous, and this is by far the most common way the hypothesis is verified in practice.

(b) Apply the same bound with $y := 0$ and use $f_k(0) = 0$:

$$|f_k(x)| = |f_k(x) - f_k(0)| \le |x| \le 1 \quad \text{for all } x \in [0,1],$$

so $\|f_k\|_\infty \le 1$ for every k. Note that the normalisation $f_k(0) = 0$ is doing real work: without it the family $f_k :\equiv k$ satisfies the derivative bound and is unbounded.

(c) Both hypotheses of Theorem 7.d.43 hold on the compact set $[0,1]$, so some subsequence converges uniformly to a continuous limit.

Replacing the uniform bound by mere differentiability destroys this, and AI-Example 7.d.39 is already the counterexample: $f_k(x) := x^k$ is C^1 on $[0,1]$ with $f_k(0) = 0$, so it satisfies every hypothesis **except** the uniform one. Its derivative $f_k'(x) = kx^{k-1}$ has $\|f_k'\|_\infty = k$, which grows without bound. And we saw that no subsequence of (x^k) converges. What matters is therefore not that each f_k is well behaved, but that they are well behaved **by a common margin**.

Q.E.D.

Connectedness (Chapter 8)

Compactness told us when a space has *"enough finiteness"* to guarantee extrema and uniform behaviour. This chapter turns to a completely different, qualitative kind of structure: whether a space can be split into separate pieces at all. Connectedness and path connectedness are the two answers, they are not equivalent, and the topologist's sine curve is the standard witness.

Connectedness (Section 8.a)

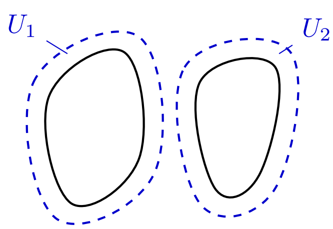
Definition 8.a.1 (*Connected and disconnected metric spaces*): A metric space (X, d) is "*disconnected*" (German: „*unzusammenhängend*“) if there exist subsets $U_1, U_2 \subseteq X$ which are

- **(a) open**,
- **(b) non-empty**, $U_1 \neq \emptyset$ and $U_2 \neq \emptyset$,
- **(c) disjoint**, $U_1 \cap U_2 = \emptyset$,

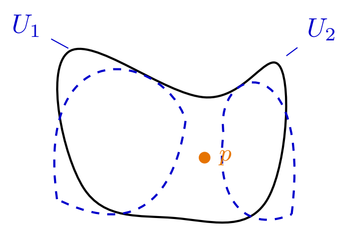
such that $X = U_1 \cup U_2$. Similarly, for a subset $A \subseteq X$, it is disconnected if it is disconnected as a metric space with the induced metric $d|_{A \times A}$.

If no such pair U_1, U_2 exists, X is "*connected*" (German: „*zusammenhängend*“). Connected is thus defined as the negation of disconnected.

Typically, proving that a set or metric space is connected is done **by contradiction**, which is why one should remember the definition of *disconnected*. Intuitively, connected means: *"the space cannot be separated by open sets."*



Disconnected
$X \subseteq U_1 \cup U_2$, and $U_1 \cap U_2 = \emptyset$



Connected
$p \in X$ lies in neither: no separation

Proposition 8.a.2 (*Intervals are connected*): Every interval $I \subseteq \mathbb{R}$, whether open, closed, half-open, bounded or unbounded, is connected (Definition 8.a.1). Conversely, a connected subset of $\mathbb{R}$ is an interval.

Proof:

Suppose $I = U_1 \sqcup U_2$ with U_1, U_2 open in I, disjoint and both non-empty. Pick $a \in U_1$ and $b \in U_2$; without loss of generality $a < b$, and $[a, b] \subseteq I$ because I is an interval. Set

$$c := \sup\{\, t \in [a, b] \;:\; [a, t] \subseteq U_1 \,\},$$

which is well defined since a belongs to the set. Now $c \in [a, b] \subseteq I$, so c lies in U_1 or in U_2, and both lead to a contradiction:

- If $c \in U_1$, openness gives $\varepsilon > 0$ with $(c - \varepsilon, c + \varepsilon) \cap I \subseteq U_1$; together with $[a, c) \subseteq U_1$ this yields $[a, t] \subseteq U_1$ for some $t > c$ unless $c = b$, contradicting either the supremum or $b \in U_2$.
- If $c \in U_2$, openness gives $\varepsilon > 0$ with $(c - \varepsilon, c] \cap I \subseteq U_2$. But $c > a$ (as $a \in U_1$ is open in I), so by definition of the supremum there is $t \in (c - \varepsilon, c]$ with $[a, t] \subseteq U_1$, contradicting disjointness.

So no such splitting exists, and I is connected. For the converse, a set $A \subseteq \mathbb{R}$ that is not an interval admits $a < c < b$ with $a, b \in A$ and $c \notin A$; then $A \cap (-\infty, c)$ and $A \cap (c, \infty)$ disconnect it. **Q.E.D.**

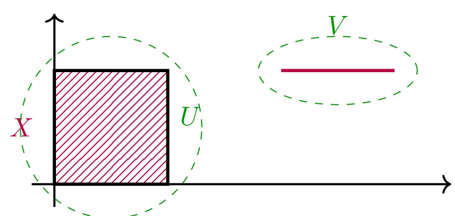
Remark 8.a.3 (*Why this is the only example one ever needs*): Every connectedness proof in this chapter reduces to Proposition 8.a.2: a path is a continuous image of $[0,1]$ (Definition 8.b.16), so path-connected sets are connected (Lemma 8.b.18) purely because the **interval** is; and the intermediate value theorem, invoked in the topologist's sine curve below, is Theorem 8.a.6 applied to an interval, since the connected subsets of $\mathbb{R}$ are exactly the intervals. In other words, all of it flows from the single fact that $\mathbb{R}$ has no gaps. That fact is the completeness axiom, and it is what powers the supremum in the proof above.

AI-Example 8.a.4 (*Disconnectedness of Rational Numbers*): The set of rational numbers $\mathbb{Q} \subset \mathbb{R}$ equipped with the subspace topology is disconnected. To see this, define

$$U := \mathbb{Q} \cap (-\infty, \sqrt{2}) \quad \text{and} \quad V := \mathbb{Q} \cap (\sqrt{2}, \infty).$$

Since $\sqrt{2} \notin \mathbb{Q}$, we have $U \cup V = \mathbb{Q}$ and $U \cap V = \emptyset$. Both U and V are non-empty and open in the subspace topology of $\mathbb{Q}$ (as intersections of $\mathbb{Q}$ with open intervals in $\mathbb{R}$). Hence U and V form a separation of $\mathbb{Q}$.

Example 8.a.5 (*A disconnected set in $\mathbb{R}^2$*): Let $X := ([0,1] \times [0,1]) \cup ([2,3] \times \{1\}) \subseteq \mathbb{R}^2$. Is X connected? No. Let $U := \{(x,y) \in X \mid x < 1.5\}$ and $V := \{(x,y) \in X \mid x > 1.5\}$. We can write these as $U = X \cap \{(x,y) \in \mathbb{R}^2 \mid x < 1.5\}$ and $V = X \cap \{(x,y) \in \mathbb{R}^2 \mid x > 1.5\}$. Both U and V are open in the subspace topology of X because they are intersections of X with open half-planes in $\mathbb{R}^2$. Furthermore, $X = U \cup V$, and U, V are disjoint and non-empty. Thus, X is disconnected.



Theorem 8.a.6 (*Continuous functions preserve connectedness*): Let X, Y be metric spaces, $f : X \to Y$ continuous. If $A \subseteq X$ is connected, then $f(A)$ is connected.

Proof:

We may assume $A = X$, replacing X by A with the induced metric and Y by $f(A)$. Suppose $f(X)$ were disconnected, say $f(X) = V_1 \sqcup V_2$ with V_1, V_2 open in $f(X)$, disjoint and non-empty. Then $U_i := f^{-1}(V_i)$ is open by Definition 5.a.1**(c)**, non-empty because V_i meets the image, and $U_1 \cap U_2 = \emptyset$ with $U_1 \cup U_2 = X$ because the V_i partition $f(X)$. So U_1, U_2 disconnect X, contrary to hypothesis. **Q.E.D.**

Corollary 8.a.7 (*Intermediate value theorem*): Let $f : [a,b] \to \mathbb{R}$ be continuous. Then f attains every value between $f(a)$ and $f(b)$.

Proof:

By Proposition 8.a.2, $[a,b]$ is connected, so $f([a,b])$ is connected by Theorem 8.a.6, hence is an interval (Proposition 8.a.2 again). An interval containing $f(a)$ and $f(b)$ contains everything between them. **Q.E.D.**

The intermediate value theorem is a result from **Analysis I**, and re-deriving it here costs one line. It is worth the line twice over. This is the first place where connectedness earns its keep, and the derivation makes visible that the theorem is really a statement about $\mathbb{R}$ having no gaps rather than a statement about f. It is used again in the topologist's sine curve (Page 97) below.

Corollary 8.a.8 (*Intermediate value theorem on a connected space*): Let X be a connected metric space and $f : X \to \mathbb{R}$ continuous. For any $a, b \in X$ and any $y \in \mathbb{R}$ between $f(a)$ and $f(b)$, there exists $x \in X$ with $f(x) = y$.

Proof:

Word for word Corollary 8.a.7's proof, with X in place of $[a, b]$: X is connected by hypothesis (rather than by Proposition 8.a.2), so $f(X)$ is connected by Theorem 8.a.6, hence an interval by Proposition 8.a.2; an interval containing $f(a)$ and $f(b)$ contains everything between them.

Q.E.D.

Remark 8.a.9: Corollary 8.a.7 is the special case $X = [a, b]$. Stating the general version separately is worth doing once: it makes clear that connectedness, not any special property of an interval, is what the intermediate value theorem actually needs. That is exactly the moral of Remark 8.a.3.

Corollary 8.a.10 (*Linear maps preserve the unit cube*): Given the n-dimensional unit cube $[0, 1]^n \subset \mathbb{R}^n$, any linear map $L : \mathbb{R}^n \to \mathbb{R}^n$ maps $[0, 1]^n$ to an n-dimensional parallelepiped. Since linear maps in Euclidean space are continuous, the image $L([0, 1]^n)$ is connected. Furthermore, since $[0, 1]^n$ is compact, $L([0, 1]^n)$ is also compact.

Exercise 8.a.11 (*Properties of connectedness*): Are the following true or false?

- **(a)** The union of connected subsets with a common point is connected.
- **(b)** The intersection of connected subsets is connected.
- **(c)** The closure (Definition 3.b.13) of a connected subset is connected.

Proof of Exercise 8.a.11:

(a) True. *Proof.* Let X be a metric space, $V_1, V_2 \subseteq X$ connected, and $x_0 \in V_1 \cap V_2$. Suppose $V_1 \cup V_2 \subseteq U_1 \cup U_2$ with $U_1, U_2 \subseteq X$ open and disjoint. Assume without loss of generality $x_0 \in U_1$. Then $U_1 \cap V_1 \neq \emptyset$ and $U_1 \cap V_2 \neq \emptyset$. So, since V_1, V_2 are connected, $V_1 \cap U_2 = V_2 \cap U_2 = \emptyset$. Therefore $(V_1 \cup V_2) \cap U_2 = \emptyset$ and $V_1 \cup V_2$ is connected.

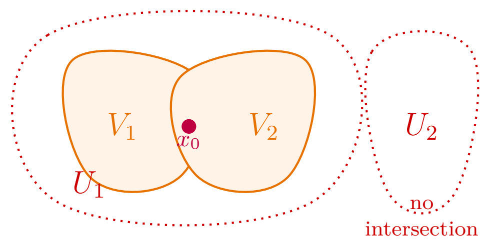
(b) False. *Counterexample:*

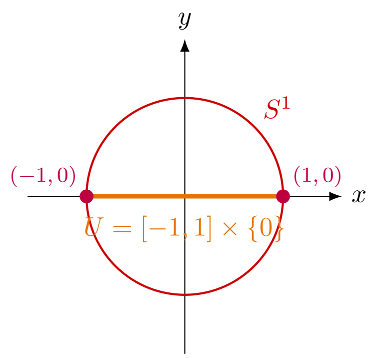
$$X = \mathbb{R}^2, \quad U = [-1, 1] \times \{0\}, \quad V = S^1 = \{x \in \mathbb{R}^2 : \|x\| = 1\},$$
$$U \cap V = \{(-1, 0), (1, 0)\}.$$

(c) True. Suppose X is a metric space, $A \subseteq X$ a subset with closure $\overline{A}$. Suppose $U_1, U_2 \subseteq X$ are disjoint open sets covering $\overline{A}$, and assume without loss of generality $\overline{A} \cap U_1 \neq \emptyset$. Then, since U_1 is open, $A \cap U_1 \neq \emptyset$. By connectedness of A, $U_2 \cap A = \emptyset$, and since U_2 is open, also $U_2 \cap \overline{A} = \emptyset$.

Q.E.D.

AI-Note 8.1: Corsin writes $\|x\| = R^2$ in the definition of S^1 above; it must be $\|x\| = 1$, as the stated intersection $\{(\pm 1, 0)\}$ confirms. Corrected here.





Corollary 8.a.12 (*Arbitrary unions of connected sets through a common point*): Let (X, d) be a metric space, $p \in X$, and $\{V_i\}_{i \in I}$ a family of connected subsets of X with $p \in V_i$ for every $i \in I$. Then $\bigcup_{i \in I} V_i$ is connected.

Proof:

Suppose $\bigcup_i V_i \subseteq U_1 \cup U_2$ with U_1, U_2 open and disjoint, and say $p \in U_1$. For each i, the argument of Exercise 8.a.11**(a)** applies to V_i alone: it meets U_1 (at p) and is connected, so it cannot also meet U_2, i.e. $V_i \subseteq U_1$. This holds for every $i \in I$, so $\bigcup_i V_i \subseteq U_1$, and U_2 cannot meet $\bigcup_i V_i$: no separation of the union exists. **Q.E.D.**

Definition 8.a.13 (*Connected component*): Let (X, d) be a metric space and $x \in X$. The "*connected component*" (German: „*Zusammenhangskomponente*“) of x is

$$C_x := \bigcup \{V \subseteq X : V \text{ connected and } x \in V\},$$

the union of all connected subsets of X containing x.

Remark 8.a.14: By Corollary 8.a.12 (with I indexing **all** connected sets containing x, and common point $p := x$), C_x is itself connected. It is therefore the **largest** connected subset of X containing x, since it contains every connected set that does. Two connected components are either equal or disjoint: if $C_x \cap C_y \neq \emptyset$, pick p in the intersection; then $C_x \cup C_y$ is connected (Corollary 8.a.12 again, common point p) and contains both x and y, so maximality of C_x and C_y forces $C_x \cup C_y \subseteq C_x$ and $C_x \cup C_y \subseteq C_y$, that is, $C_x = C_y$. Hence the connected components partition X into disjoint pieces, which is the connectedness analogue of the way an equivalence relation partitions a set into equivalence classes.

Remark 8.a.15 (*Summary of topological properties under continuous maps*): Let X, Y be metric spaces and $f : X \to Y$ a continuous function. The following table summarizes which topological properties of a subset $A \subseteq X$ or $B \subseteq Y$ are necessarily preserved under images $f(A)$ and preimages $f^{-1}(B)$.

If A or B is ...	Then $f^{-1}(B)$ is ...	Then $f(A)$ is ...
open	**open**	(not necessarily)
closed	**closed**	(not necessarily)
compact	(not necessarily)	**compact**
connected	(not necessarily)	**connected**

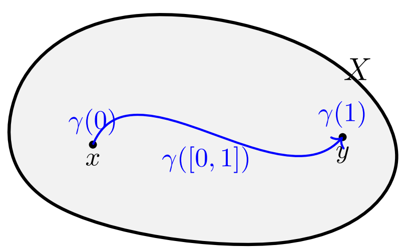
Path connectedness (Section 8.b)

Definition 8.a.1 defines connectedness negatively. A space is connected when it **cannot** be split, which is why proofs about it run by contradiction. That is mathematically convenient and intuitively unsatisfying, since it says nothing whatever about what holds a connected space together.

Path connectedness is the positive version, and it says the obvious thing: any two points can be joined by a curve running inside the space. It implies connectedness (Lemma 8.b.18), and on open subsets of $\mathbb{R}^n$ the two notions agree (Proposition 8.b.20). In general they come apart, and the topologist's sine curve at the end of this section is the standard witness.

Definition 8.b.16 (*Path*): Let (X,d) be a metric space. A continuous function $\gamma : [0,1] \to X$, where $[0,1]$ carries the standard metric, is called a "*path*" (German: „*Weg*"). More generally, we can replace $[0,1]$ by $[a,b]$ for $a < b$.

Definition 8.b.17 (*Path-connected*): A metric space X (or subset thereof) is "*path-connected*" (German: „*wegzusammenhängend*") if for all $x, y \in X$ there exists a path (Definition 8.b.16) $\gamma : [0,1] \to X$ with $\gamma(0) = x$, $\gamma(1) = y$.



Lemma 8.b.18 (*Path-connected implies connected*): Let X be a metric space. Then X path-connected (Definition 8.b.17) implies X connected (Definition 8.a.1).

Proof:

Suppose X were path-connected but disconnected, say $X = U_1 \sqcup U_2$ with U_1, U_2 open, disjoint and non-empty. Pick $x \in U_1$, $y \in U_2$ and a path $\gamma : [0,1] \to X$ with $\gamma(0) = x$, $\gamma(1) = y$. Then $\gamma^{-1}(U_1)$ and $\gamma^{-1}(U_2)$ are open in $[0,1]$ by Definition 5.a.1**(c)**, disjoint, cover $[0,1]$, and are non-empty (they contain 0 and 1 respectively). This disconnects $[0,1]$, contradicting Proposition 8.a.2. **Q.E.D.**

Exercise 8.b.19 (*The unit disk is connected*): Prove that the set $S^1 := \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ is connected.

Proof of Exercise 8.b.19:

Instead of proving connectedness directly via open sets, we can show that S^1 is path-connected, which implies connectedness (Lemma 8.b.18). Given any two points $a, b \in S^1$, we can connect them via a straight line segment. The segment can be parameterized as $\gamma(t) := (1-t)a + tb$ for $t \in [0,1]$. Since S^1 (the unit disk in $\mathbb{R}^2$) is convex, for any $t \in [0,1]$ we have $\|\gamma(t)\| \leq (1-t)\|a\| + t\|b\| \leq$

$1-t+t=1$, so the entire path lies in S^1. Because any two points can be connected by a continuous path within S^1, the set is path-connected, and therefore connected.

(Note: The standard notation for the unit disk is D^2 or $B_1(0)$, whereas S^1 usually denotes the boundary circle $\{x^2+y^2=1\}$, but we reproduce the notation S^1 here as it appears in the source notes.) **Q.E.D.**

Proposition 8.b.20 (*Connectedness and path-connectedness in open Euclidean subsets*): Let $U \subseteq \mathbb{R}^n$ be open (Definition 3.a.6). Then

$$U \text{ path-connected} \iff U \text{ connected.}$$

Proof:

($\Rightarrow$) is Lemma 8.b.18, which needs neither openness nor $\mathbb{R}^n$.

($\Leftarrow$) Assume U is connected and non-empty (the empty set is vacuously path-connected), and fix $x_0 \in U$. Let

$$A := \{x \in U \ : \text{ some path in } U \text{ joins } x_0 \text{ to } x\}.$$

We show A and $U \setminus A$ are both open; since $x_0 \in A$ (take the constant path), connectedness then forces $U \setminus A = \emptyset$, i.e. $A = U$, which is the claim.

Let $x \in U$ and choose $r > 0$ with $B_r(x) \subseteq U$, using that U is open. The ball is convex, so every $y \in B_r(x)$ is joined to x by the straight segment $t \mapsto (1-t)x + ty$, which stays inside $B_r(x) \subseteq U$. Consequently:

- If $x \in A$, concatenating a path from x_0 to x with that segment joins x_0 to any $y \in B_r(x)$, so $B_r(x) \subseteq A$ and A is open.
- If $x \notin A$, then no $y \in B_r(x)$ can lie in A, since otherwise running the segment backwards from y to x would join x_0 to x. So $B_r(x) \subseteq U \setminus A$ and $U \setminus A$ is open.

(Concatenation of two paths is again a path: traverse the first on $[0, \frac{1}{2}]$ and the second on $[\frac{1}{2}, 1]$, each at double speed; the result is continuous because the two pieces agree at $t = \frac{1}{2}$.) **Q.E.D.**

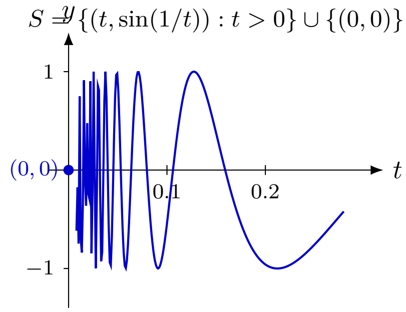
Important remark 8.b.21 (*Only for open subsets*): The implication *"connected $\Rightarrow$ path-connected"* is **false** without openness, and the proof shows exactly why: the ball $B_r(x)$ around a point of U was needed to supply the straight segment. For a set that is not open there may be points with no room around them at all, and the topologist's sine curve (Example 8.b.22) below is precisely such a case. There the origin has no neighbourhood in S that is a nice piece of curve.

Example 8.b.22 (*The topologist's sine curve*):

$$S := \left\{ \left(t, \sin\left(\tfrac{1}{t}\right)\right) : t > 0 \right\} \cup \{(0,0)\} \subseteq \mathbb{R}^2$$

S is **connected** but **not path-connected**!

AI-Note 8.2: Corsin writes the second piece as $\cup\{0\}$; since $S \subseteq \mathbb{R}^2$, the point being adjoined is the origin $(0,0) \in \mathbb{R}^2$, not the number 0. Written as $\cup\{(0,0)\}$ above, which is also what his own figure and the proof below use.



S is connected:

Write $G := S \setminus \{(0,0)\} = \{(t, \sin(1/t)) : t > 0\}$ for the graph part, and suppose for contradiction that S is disconnected, say $S = V_1 \sqcup V_2$ with V_1, V_2 non-empty, disjoint and open in S. Say $(0,0) \in V_1$.

G is the continuous image of the interval $(0, \infty)$ under $t \mapsto (t, \sin(1/t))$, hence path-connected, hence connected (Lemma 8.b.18). A connected set meeting both parts of a separation would itself be separated, so G lies entirely in V_1 or entirely in V_2.

It cannot lie in V_1: then $V_2 \subseteq S \setminus (G \cup \{(0,0)\}) = \emptyset$, contradicting $V_2 \neq \emptyset$. So $G \subseteq V_2$, and therefore $V_1 = \{(0,0)\}$.

But V_1 is open in S, so some ball $B_\varepsilon\big((0,0)\big) \cap S$ equals $\{(0,0)\}$, and that is false. For $t := \frac{1}{k\pi}$ with k large, the point $(t, \sin(1/t)) = (t, 0) \in G$ lies within ε of the origin. This is the required contradiction, so S is connected. **Q.E.D.**

S is not path-connected:

Assume by contradiction there is a path from $(0,0)$ to $(\frac{1}{\pi}, 0)$,

$$\gamma : [0,1] \to [0, \tfrac{1}{\pi}] \times [-1, 1], \qquad t \mapsto (x(t), y(t)).$$

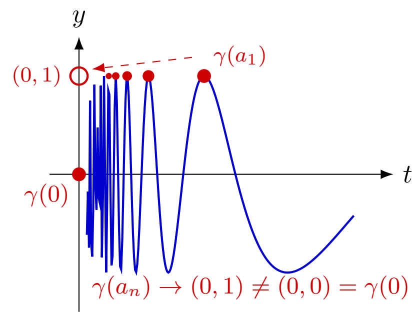
Since γ is continuous, x and y must also be continuous. By the "*intermediate value theorem*" (Corollary 8.a.7), $x : [0,1] \to [0, \frac{1}{\pi}]$ attains every value in $[0, \frac{1}{\pi}]$, since $x(0) = 0$ and $x(1) = \frac{1}{\pi}$. Therefore, we find a sequence $(a_n)_{n=1}^{\infty}$ such that

$$x(a_n) = \frac{1}{2\pi n + \frac{\pi}{2}}.$$

We may moreover choose the a_n **decreasing**: having found a_{n-1}, apply the intermediate value theorem on $[0, a_{n-1}]$ rather than on $[0,1]$, which is legitimate because $x(0) = 0$ and $x(a_{n-1}) = \frac{1}{2\pi(n-1)+\pi/2}$ already straddle the next target value. This yields $a_n \in (0, a_{n-1})$, so (a_n) is decreasing and bounded below, hence convergent, say $a_n \to \ell \in [0,1]$.

Now evaluate γ along this sequence. By construction $x(a_n) \to 0$, so continuity gives $x(\ell) = 0$; and $y(a_n) = \sin\big(2\pi n + \frac{\pi}{2}\big) = 1$ for every n, so $y(\ell) = 1$. Hence $\gamma(\ell) = (0,1)$. But $(0,1) \notin S$: the only point of S whose first coordinate vanishes is the origin. This contradicts $\gamma([0,1]) \subseteq S$, so no such path exists. **Q.E.D.**

AI-Note 8.3: Corsin's argument produces the a_n from the intermediate value theorem and then asserts $a_n \to 0$. That does not follow from the choice as stated. The intermediate value theorem returns **some** preimage, and nothing so far prevents all of them from sitting near 1. The repair is the standard one and costs one sentence: apply the theorem on the shrinking intervals $[0, a_{n-1}]$ rather than on $[0,1]$ each time, which forces the a_n down. Inserted above; the rest of his proof is unchanged.



We close the week with a change of subject: away from topology and back to the linear-algebraic structure (Section 2.a) that lets us talk about angles, not just distances.

AI-Exercise 8.b.23 (*Continuous image of a path-connected set*): Let $f\colon X \to Y$ be a continuous map between metric spaces.

(a) Prove that if X is path-connected (Definition 8.b.17), then so is $f(X)$.

(b) Conclude that the graph $\Gamma_g := \{(x, g(x)) : x \in U\}$ of a continuous $g : U \to \mathbb{R}^m$ is path-connected whenever $U \subseteq \mathbb{R}^n$ is convex. (Compare Exercise 8.b.24, which asks the same question with *"connected"* in place of *"path-connected"*.)

Exercise 8.b.24 (*4.2 — Connected graphs* (important)): Let $U \subset \mathbb{R}^n$ be open and connected and let $f \in C^1(U, \mathbb{R}^m)$. Show that its graph

$$\Gamma_f := \{(x, f(x)) : x \in U\}$$

is a connected subset of $\mathbb{R}^n \times \mathbb{R}^m$.

Solutions (Section 8.c)

Proof of AI-Exercise 8.b.23:

(a) Let $p, q \in f(X)$, say $p = f(x)$ and $q = f(y)$. Since X is path-connected there is a path $\gamma : [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$. Then $f \circ \gamma : [0,1] \to f(X)$ is continuous as a composition of continuous maps, and it satisfies $(f\circ\gamma)(0) = p$, $(f \circ \gamma)(1) = q$. So it is a path in $f(X)$ from p to q, and $f(X)$ is path-connected.

(b) A convex set U is path-connected: the straight segment $t \mapsto (1-t)x + ty$ lies in U and is continuous. The map $\varphi : U \to \mathbb{R}^n \times \mathbb{R}^m$, $\varphi(x) := (x, g(x))$, is continuous because each component is, and $\Gamma_g = \varphi(U)$. Apply **(a)**.

Remark 8.c.25: Part **(a)** is strictly easier than its connectedness counterpart (Theorem 8.a.6), because path-connectedness is a statement one verifies by **exhibiting** something, namely a path, and continuous maps push paths forward directly. Connectedness has no such witness to push forward, which is why its proof has to run by contradiction through preimages instead. This asymmetry is worth noticing: it is the reason path-connectedness is usually the easier of the two to prove, and connectedness the easier of the two to **use**.

Q.E.D.

Proof of Exercise 8.b.24:

Consider the map

$$\varphi : U \to \mathbb{R}^n \times \mathbb{R}^m, \qquad x \mapsto (x, f(x)).$$

Since $f \in C^1(U, \mathbb{R}^m)$ is in particular continuous, and the identity on U is continuous, φ is continuous. By definition $\Gamma_f = \varphi(U)$. Since U is connected and φ is continuous, Theorem 8.a.6 gives that $\varphi(U) = \Gamma_f$ is connected. **Q.E.D.**

PART III

Differential Calculus

In one variable the derivative is a number, which makes it tempting to expect the several-variable derivative to be a list of numbers. It is not. The right object is a linear map, the one that best approximates f near a point, and the partial derivatives are merely its matrix in the standard basis. Getting this order of ideas right is what turns the chain rule into a statement about composing linear maps rather than a formula to be memorised. This part sets up that differential, works out how to compute with it, and closes with Taylor's theorem, which sharpens the linear approximation to any order and supplies the quantitative estimates the optimization theory of the next part runs on.

The Differential and the Jacobi Matrix (Chapter 9)

The previous part treated continuous functions as structure-preserving maps between metric spaces. This one asks for considerably more: that a map be well approximated, near each point, by a **linear** one. That approximating linear map is the differential.

Three notions compete for the name *"derivative"* in several variables, and they are not equivalent. Partial derivatives ask only what happens along the coordinate axes, directional derivatives along an arbitrary straight line, and the differential asks for a linear map approximating f from every direction at once. Only the last deserves to be called differentiability, and the gap between them is real: a function can have every partial derivative at a point and still fail to be continuous there. This chapter sets out that hierarchy and identifies the matrix representing the differential.

The differential (Section 9.a)

For $f:\mathbb{R}\to\mathbb{R}$ differentiable at $x_0\in\mathbb{R}$:

$$f(x_0+h)=f(x_0)+f'(x_0)h+o(h)$$

$$\Updownarrow$$

$$\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}=f'(x_0)$$

$$\Updownarrow$$

$$\lim_{h\to 0}\left|\frac{f(x_0+h)-f(x_0)-f'(x_0)h}{h}\right|=0$$

For $F:\mathbb{R}^n\to\mathbb{R}^m$ differentiable at $x_0\in\mathbb{R}^n$:

$$F(x_0+h)=F(x_0)+\underbrace{DF_{x_0}(h)}_{\text{linear map}}+o(|h|)$$

$$\Updownarrow$$

$$\lim_{h\to 0}\frac{|F(x_0+h)-F(x_0)-DF_{x_0}(h)|}{|h|}=0$$

$$\Downarrow$$

$$\lim_{s\to 0}\frac{F(x_0+sv)-F(x_0)}{s}=DF_{x_0}(v)=\partial_v F(x_0)\quad\text{for each fixed } v\in\mathbb{R}^n$$

(the "*directional derivative*" of F at x_0 in direction v)

Note that the last arrow points **one way only**. The first two lines say the same thing; the third is a strictly weaker consequence, and recovering differentiability from it is exactly what fails in several variables; see AI-Example 9.b.6 below.

It is worth pausing on why the definition had to be restated rather than merely copied. The one-variable derivative is defined as a **quotient**, and that is precisely what does not survive: for $h\in\mathbb{R}^n$ with $n\geq 2$ the expression $\big(F(x_0+h)-F(x_0)\big)/h$ is meaningless, because one cannot divide by a vector. The first line above is the version that does survive, and it is the one to remember: **F is differentiable at x_0 if it is approximated near x_0 by an affine map, to an error smaller than $|h|$.** Everything that follows, the Jacobian, the chain rule and Taylor expansions alike, is bookkeeping for that single idea. The reason the derivative becomes a **linear map** rather than a number is simply that a linear map is what an approximation in n variables looks like.

AI-Note 9.1: Corsin's middle line reads $\lim_{h\to 0}\frac{F(x_0+h)-F(x_0)}{|h|}=DF_{x_0}\left(\frac{h}{|h|}\right)$, inside a chain of $\Updownarrow$. As written that limit does not exist: h is being sent to 0 on the left while still appearing on the right, and the quotient genuinely depends on the direction of approach. The intended statement is the one typeset above, in which a direction v is fixed first and only then the scalar s is sent to 0, and it is only an **implication**, not an equivalence. The distinction is not pedantic: it is the entire content of AI-Example 9.b.6 and of Important remark 10.b.7.

Definition 9.a.1 (*Differentiability*): We say that $F : \mathbb{R}^n \to \mathbb{R}^m$ is "*differentiable*" (German: „*differenzierbar*") at x_0 if a linear map DF_{x_0} as above exists, where $x_0, h \in \mathbb{R}^n$. The same definition applies verbatim to $F : U \to \mathbb{R}^m$ for any open subset $U \subseteq \mathbb{R}^n$, and we use it in that generality throughout.

Remark 9.a.2: The linear map DF_{x_0} in Definition 9.a.1, when it exists, is automatically unique: if L_1, L_2 both satisfy the defining limit, then $(L_1 - L_2)(h)/|h| \to 0$ as $h \to 0$ along every direction, which forces the linear map $L_1 - L_2$ to vanish identically. Concretely: write $L := L_1 - L_2$ and suppose $L(h) \neq 0$ for some h. Along the ray th, $t > 0$, linearity gives

$$\frac{|L(th)|}{|th|} = \frac{t\,|L(h)|}{t\,|h|} = \frac{|L(h)|}{|h|} \neq 0,$$

a nonzero **constant**. Consequently, the quotient cannot tend to 0 as $t \to 0$, and therefore $L(h) = 0$ for every h. This is the multivariable analogue of the uniqueness-of-limits argument from Proposition 4.a.4.

Proposition 9.a.3 (*Differentiability implies continuity*): If $F : U \to \mathbb{R}^m$, $U \subseteq \mathbb{R}^n$ open, is differentiable (Definition 9.a.1) at $x_0 \in U$, then F is continuous at x_0.

Proof:

By Definition 9.a.1, $F(x_0 + h) - F(x_0) = DF_{x_0}(h) + o(|h|)$ as $h \to 0$. Since DF_{x_0} is linear on the finite-dimensional space $\mathbb{R}^n$, it is bounded, i.e. there exists $C \geq 0$ with $|DF_{x_0}(h)| \leq C|h|$ for all h. Hence

$$|F(x_0 + h) - F(x_0)| \leq |DF_{x_0}(h)| + o(|h|) \leq C|h| + o(|h|) \xrightarrow{h \to 0} 0,$$

so $F(x_0 + h) \to F(x_0)$ as $h \to 0$, i.e. F is continuous at x_0. **Q.E.D.**

Remark 9.a.4: The converse fails: continuity does not imply differentiability, exactly as in one variable (e.g. $x \mapsto |x|$ at 0). AI-Example 9.b.6 below shows the gap is even wider in several variables, where **all** partial derivatives can exist at a point without F even being continuous there, let alone differentiable.

The Jacobi matrix (Section 9.b)

Since $DF_{x_0} : \mathbb{R}^n \to \mathbb{R}^m$ (Definition 9.a.1) is a linear map, there is a **1–1 correspondence with a matrix once a basis is fixed**! In the standard basis of $\mathbb{R}^n$, $(e_1, \ldots, e_n)$, this is the "*Jacobi matrix*" (German: „*Jacobi-Matrix*"):

$$DF_x \;\cong\; \mathrm{J}\,F(x) = \begin{pmatrix} \dfrac{\partial F_1}{\partial x_1} & \cdots & \dfrac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial F_m}{\partial x_1} & \cdots & \dfrac{\partial F_m}{\partial x_n} \end{pmatrix}$$

(m rows, n columns), where we wrote $F(x) = (F_1, \ldots, F_m)^\mathsf{T}$ and

$$\frac{\partial F}{\partial x_i} =: \partial_i F(x)$$

is the "*i-th partial derivative*" (German: „*partielle Ableitung*") at x:

$$\partial_i F(x) = \lim_{s\to 0} \frac{F(x+se_i) - F(x)}{s}.$$

This defines a function $\partial_i F : U \to \mathbb{R}^m$. If F is differentiable, then

$$\partial_i F(x) = DF_x(e_i).$$

Important remark 9.b.5: The partial derivatives can exist even if $F : U \to \mathbb{R}^m$ is not differentiable (recall AI-Example 9.b.6)!

AI-Example 9.b.6 (*All partial derivatives exist, yet f is not even continuous*): Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) := \begin{cases} \dfrac{xy}{x^2+y^2} & (x,y) \neq (0,0), \\ 0 & (x,y) = (0,0). \end{cases}$$

Along each axis, f vanishes identically ($f(x,0) = f(0,y) = 0$), so both partial derivatives at the origin exist and equal 0:

$$\partial_x f(0,0) = \lim_{s\to 0} \frac{f(s,0) - f(0,0)}{s} = 0, \qquad \partial_y f(0,0) = 0.$$

Yet f is not even continuous at $(0,0)$: along the diagonal $y = x$, $f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2}$ for all $x \neq 0$, so $f(x,x) \to \frac{1}{2} \neq 0 = f(0,0)$ as $x \to 0$. By Proposition 9.a.3, f is therefore not differentiable at $(0,0)$, even though both partial derivatives exist there.

Remark 9.b.7 (*The same expression, squared, behaves completely differently*): Sascha Brack pairs the function above with one that looks almost identical, as a test of whether you can tell them apart:

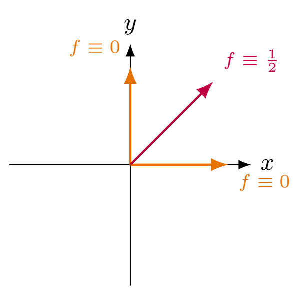
$$f_1(x,y) := \frac{xy}{x^2+y^2}, \qquad f_2(x,y) := \frac{x^2y^2}{x^2+y^2},$$

both extended by 0 at the origin. Substituting $x = r\cos\theta$, $y = r\sin\theta$ settles both at once:

$$f_1 = \frac{r^2\cos\theta\sin\theta}{r^2} = \cos\theta\sin\theta, \qquad f_2 = \frac{r^4\cos^2\theta\sin^2\theta}{r^2} = r^2\cos^2\theta\sin^2\theta.$$

In f_1 every r cancels: the function is constant along each ray and takes a different constant on different rays, so no limit exists and f_1 is discontinuous at the origin. In f_2 a factor r^2 survives, and since $|\cos^2\theta\sin^2\theta| \leq 1$ we get $|f_2| \leq r^2 \to 0$ **uniformly in θ**, so f_2 is continuous at the origin.

The pair is worth doing once because it shows what the polar substitution is actually testing. The question is never *"does the expression go to zero along each fixed ray?"*, which f_1 does not even manage. The question is **does the θ-dependence get killed by a positive power of r?** If the r's cancel completely, the direction survives into the limit and there is none. This is the same test used in Question C.a.7 on the repetition quiz, and the reason step 3 of Remark 10.c.13 insists on *"uniformly in θ"*.



$f = \frac{xy}{x^2+y^2}$ near the origin: constant along every ray,
but a **different** constant on the diagonal than on the axes

Sascha Brack's notes collect these relationships into one picture. Read it as a hierarchy: the strongest condition is at the bottom, and every arrow points at something weaker.

$$
\begin{tikzcd}
 & \begin{array}{c}\textbf{Differentiable}\\ (Df_{x_0} \text{ exists})\end{array} \arrow[dl, Rightarrow] \arrow[dr, Rightarrow] & \\
\begin{array}{c}\textbf{Partial derivatives}\\ \text{exist}\end{array} & & \begin{array}{c}\textbf{Directional derivatives}\\ \text{exist}\end{array} \\
\begin{array}{c}\textbf{Partial derivatives}\\ \text{exist}\\ \text{and are continuous}\end{array} \arrow[u, Rightarrow] \arrow[uur, Rightarrow] \arrow[rr, Leftrightarrow] & & \begin{array}{c}\textbf{Directional derivatives}\\ \text{exist}\\ \text{and are continuous}\end{array} \arrow[u, Rightarrow] \arrow[uul, Rightarrow]
\end{tikzcd}
$$

Important remark 9.b.8 (*Not one of these arrows reverses*): The diagram is easy to misread as a set of equivalences, and it is nothing of the kind: every $\Rightarrow$ in it is strict. AI-Example 9.b.6 already shows the bottom-left arrow failing backwards, since there both partial derivatives exist at the origin while f is not even continuous. The other two failures, with their counterexamples, are collected in Section 10.c, which redraws this same landscape as a single vertical chain and attaches a witness to each broken link.

The one genuine equivalence is the horizontal arrow along the bottom, and it is worth seeing why. Left to right: continuous partials make f continuously differentiable (Theorem 10.c.12), so every directional derivative exists and equals $Df_x(v)$, which depends continuously on x. Right to left is immediate: take $v = e_i$.

Examples (Subsection 9.b.1)

The three examples that follow are the three shapes the Jacobian takes, and it is worth naming them, because the layout of the matrix changes completely between them while the underlying object, the linear map DF_{x_0}, does not.

- **A curve** ($n = 1$, m arbitrary): one column, so $\mathrm{J}\,\gamma(t)$ is a **column vector**, the velocity.
- **A scalar field** (n arbitrary, $m = 1$): one row, so $\mathrm{J}\,f(x)$ is a **row vector**, namely the transposed gradient $(\nabla f)^{\mathsf{T}}$.
- **A vector field** (both > 1): the full $m \times n$ matrix, whose columns are the partial derivatives and whose rows are the gradients of the components.

The gradient being a row and the velocity a column is not a convention to memorise: it follows from *"m rows, n columns"* and nothing else. Confusing the two is the commonest source of shape errors when the chain rule is applied later in this chapter.

Example 9.b.9 (*Derivative of a curve*):

$$\gamma : \mathbb{R} \to \mathbb{R}^2, \qquad t \mapsto \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

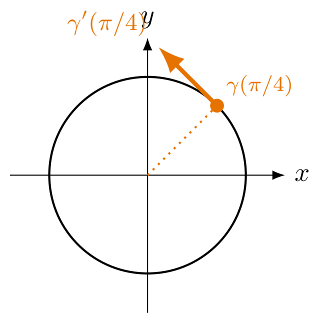
What is $\gamma'(\frac{\pi}{4}) := D\gamma_{\pi/4}$?

$$\gamma'(\tfrac{\pi}{4}) = \begin{pmatrix} \partial_t \gamma_1(\pi/4) \\ \partial_t \gamma_2(\pi/4) \end{pmatrix} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

AI-Example 9.b.10 (*Jacobian Matrix of Polar Coordinates*): Let $f\colon (0,\infty)\times(0,2\pi)\to\mathbb{R}^2$ be the polar coordinate map $f(r,\theta) := (r\cos\theta, r\sin\theta)^\mathsf{T}$. The Jacobian matrix $\mathrm{J}\,f(r,\theta)$ is:

$$\mathrm{J}\,f(r,\theta) = \begin{pmatrix} \dfrac{\partial f_1}{\partial r} & \dfrac{\partial f_1}{\partial \theta} \\ \dfrac{\partial f_2}{\partial r} & \dfrac{\partial f_2}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}.$$

Its determinant is $\det \mathrm{J}\,f(r,\theta) = r\cos^2\theta + r\sin^2\theta = r > 0$.



This vector is the **velocity of the curve** (German: „*Geschwindigkeit*“) at $t = \frac{\pi}{4}$; its length $|\gamma'(t)|$ is the **speed**. We get that

$$\gamma\left(\tfrac{\pi}{4}+h\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} + \frac{h}{\sqrt{2}}\begin{pmatrix}-1\\1\end{pmatrix} + o(|h|).$$

AI-Note 9.2: Corsin calls $\gamma'(\frac{\pi}{4})$ the *“speed”* of the curve. Strictly, $\gamma'(t)$ is the **velocity**, a vector carrying both a direction and a magnitude, whereas the speed is the scalar $|\gamma'(t)|$, here $|(-1,1)|/\sqrt{2} = 1$. Since the display below uses the vector, the distinction matters. Corrected above.

From this example, we see that the **column** vectors of the Jacobian are the partial derivatives:

$$\mathrm{J}\,F(x) = \left(\frac{\partial F}{\partial x_1} \,\middle|\, \cdots \,\middle|\, \frac{\partial F}{\partial x_n} \right)$$

and therefore, in the canonical basis, with $v := (v_1,\dots,v_n)^\mathsf{T}$:

$$\partial_v F(x) = DF_x(v) = \mathrm{J}\,F(x)\,v = \sum_{i=1}^n v_i \frac{\partial F}{\partial x_i}.$$

Example 9.b.11 (*Scalar field gradient*):

$$f:\mathbb{R}^3\to\mathbb{R}, \qquad (x,y,z)\mapsto 2x^2+y^2+3z^2-2xyz$$

What is $\mathrm{J}\,f(x,y,z)$?

$$\mathrm{J}\,f(x,y,z) = (4x-2yz,\ 2y-2xz,\ 6z-2xy) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (\nabla f)^\mathsf{T}$$

Example 9.b.12 (*Jacobian matrix of a vector field*): Consider $f:\mathbb{R}^2\to\mathbb{R}^3$ defined by

$$f(x,y) := \begin{pmatrix} x^2+y^2 \\ 2xy \\ e^x\sin(y) \end{pmatrix}.$$

The partial derivatives with respect to x and y are the column vectors:

$$\partial_1 f(x,y) = \begin{pmatrix} 2x \\ 2y \\ e^x\sin(y) \end{pmatrix}, \qquad \partial_2 f(x,y) = \begin{pmatrix} 2y \\ 2x \\ e^x\cos(y) \end{pmatrix}.$$

Because these partial derivatives exist everywhere and are continuous, f is differentiable on all of $\mathbb{R}^2$. The differential $Df_{(x_0,y_0)}$ is represented by the 3×2 Jacobian matrix:

$$\mathrm{J}\, f(x_0,y_0) = \begin{pmatrix} 2x_0 & 2y_0 \\ 2y_0 & 2x_0 \\ e^{x_0}\sin(y_0) & e^{x_0}\cos(y_0) \end{pmatrix}.$$

From this, we see that the **rows** of the Jacobian are the "*gradients*" (German: „*Gradient*“):

$$\mathrm{J}\, F(x) = \begin{pmatrix} \nabla F_1 \\ \hline \vdots \\ \hline \nabla F_m \end{pmatrix}$$

AI-Note 9.3: Corsin's sentence reads "the columns of the Jacobian are the gradients", but the displayed matrix stacks $\nabla F_1, \ldots, \nabla F_m$ as rows. The display is the correct one, and it is consistent with $(\nabla f)^\mathsf{T}$ just above. Corrected to "rows".

Example 9.b.13 (*A function that is differentiable but not C^1*): Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) := \begin{cases} (x^2+y^2)\sin\left(\dfrac{1}{\sqrt{x^2+y^2}}\right), & \text{if } (x,y)\neq(0,0), \\ 0, & \text{if } (x,y)=(0,0). \end{cases}$$

We claim that f is differentiable at $(0,0)$ with $Df_{(0,0)} = (0,0)$, but the partial derivatives are not continuous at $(0,0)$.

Step 1: Differentiability at $(0,0)$. We use the definition of the differential. Let $L = (0,0)$. We need to show that

$$\lim_{h\to(0,0)} \frac{f(h_1,h_2) - f(0,0) - L\begin{pmatrix} h_1 \\ h_2 \end{pmatrix}}{\|h\|} = 0.$$

Plugging in the definitions and changing to polar coordinates $(h_1,h_2) = (r\cos\theta, r\sin\theta)$ with $r = \|h\| = \sqrt{h_1^2+h_2^2}$:

$$\lim_{r\to 0} \frac{r^2\sin(1/r) - 0 - 0}{r} = \lim_{r\to 0} r\sin(1/r) = 0.$$

The sine term is bounded between -1 and 1, so the limit evaluates to 0. Thus, f is differentiable at $(0,0)$ and $Df_{(0,0)} = (0,0)$, which implies $\partial_1 f(0,0) = 0$ and $\partial_2 f(0,0) = 0$.

Step 2: Discontinuity of partial derivatives. Let us compute $\partial_1 f(x,y)$ for $(x,y)\neq(0,0)$. Using the product and chain rules:

$$\partial_1 f(x,y) = 2x\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) - \frac{x}{\sqrt{x^2+y^2}}\cos\left(\frac{1}{\sqrt{x^2+y^2}}\right).$$

Let us choose a sequence $(x_n,y_n) := (\frac{1}{n}, 0)$ which converges to $(0,0)$ as $n\to\infty$. Then

$$\lim_{n\to\infty} \partial_1 f\left(\frac{1}{n}, 0\right) = \lim_{n\to\infty}\left[\frac{2}{n}\sin(n) - \cos(n)\right].$$

The first term vanishes, but $\lim_{n\to\infty}\cos(n)$ does not exist. Thus, $\partial_1 f$ is not continuous at $(0,0)$, and so f is not continuously differentiable (C^1) at $(0,0)$. (By symmetry, the same holds for $\partial_2 f$).

AI-Exercise 9.b.14 (*Differential and Linearization at $(1,0)$*): Let $f(x,y) := x^2y + \sin(xy)$ defined on $\mathbb{R}^2$.

(a) Compute the gradient $\nabla f(x,y)$ and the total differential $Df(1,0)$.

(b) Find the linear approximation $L(x,y)$ of f around the point $(1,0)$.

Differentiating with respect to a matrix (Subsection 9.b.2)

Nothing in Definition 9.a.1 requires the domain to be $\mathbb{R}^n$ with n small, or even to be written as a column of numbers. A function of an $n \times m$ matrix is a function of nm real variables, and its partial derivatives $\partial f/\partial A_{ij}$ are defined exactly as before, by varying one entry and freezing the rest.

Remark 9.b.15 (*Why this is worth a subsection*): Taking derivatives with respect to a matrix argument can look like a formal exercise with no purpose. It is not, and the reason is worth recording: **it is how neural networks are trained.** The parameters of such a network are matrices. The function being minimised sends those matrices to a single number measuring how badly the network fits the data, and training consists of computing the gradient with respect to those matrices and then stepping a little way against it, because $-\nabla f$ points in the direction of steepest decrease. Every step of that loop is the material of this chapter.

The simplest instance is already worth doing by hand, and it is exactly linear regression.

AI-Example 9.b.16 (*The gradient of a least-squares fit*): Fix data matrices $X \in \mathbb{R}^{m\times k}$ (the k inputs, each a column in $\mathbb{R}^m$) and $Y \in \mathbb{R}^{n\times k}$ (the k desired outputs). The model applies an unknown matrix $A \in \mathbb{R}^{n\times m}$ and adds an unknown offset $b \in \mathbb{R}^n$ to every column, so its total error is

$$f(A,b) := \big\| \underbrace{AX + b\mathbf{1}^{\mathsf{T}} - Y}_{=:\,R} \big\|^2, \qquad \mathbf{1} := (1,\dots,1)^{\mathsf{T}} \in \mathbb{R}^k,$$

where $\|\cdot\|$ is the Hilbert–Schmidt norm of Section 2.a, so that $f(A,b) = \sum_{p,q} R_{pq}^2$ is the total squared error over all k data points. Writing out one entry of the residual,

$$R_{pq} = \sum_{s=1}^{m} A_{ps}X_{sq} + b_p - Y_{pq},$$

so that $\partial R_{pq}/\partial A_{ij} = \delta_{pi}X_{jq}$ and $\partial R_{pq}/\partial b_i = \delta_{pi}$. The chain rule then gives

$$\frac{\partial f}{\partial A_{ij}} = 2\sum_{p,q} R_{pq}\,\delta_{pi}X_{jq} = 2\sum_{q} R_{iq}X_{jq} = 2\big(RX^{\mathsf{T}}\big)_{ij},$$
$$\frac{\partial f}{\partial b_i} = 2\sum_{p,q} R_{pq}\,\delta_{pi} = 2\sum_{q} R_{iq} = 2\big(R\mathbf{1}\big)_{i}.$$

Collecting the entries back into arrays of the same shape as the variables,

$$\nabla_A f = 2\big(AX + b\mathbf{1}^{\mathsf{T}} - Y\big)X^{\mathsf{T}}, \qquad \nabla_b f = 2\big(AX + b\mathbf{1}^{\mathsf{T}} - Y\big)\mathbf{1}.$$

The gradient with respect to b is twice the vector of row sums of the residual, and the gradient with respect to A is the residual correlated against the inputs. Both are entirely explicit, and both were computed with nothing beyond partial differentiation and the chain rule.

Remark 9.b.17 (*A notational caution*): One point trips people up here. Since f above has domain $\mathbb{R}^{n\times m} \times \mathbb{R}^n$ of dimension $nm+n$, its Jacobian is formally a $1 \times (nm+n)$ row, and $\partial_v f(A,b) = \mathrm{J}\,f(A,b)\,v$ would require flattening the matrix direction v into a column of length nm. That flattening is pure bookkeeping and is almost never performed: one keeps $\nabla_A f$ shaped like A, as above. The identity $\partial_v f = \mathrm{J}\,f \cdot v$ of Section 9.b still holds, and is simply being read in a reshaped coordinate system.

Exercise 9.b.18 (*4.1 — Derivative computation* (important)): Consider the function $u : (x,y) \mapsto x^{\sin(y)}$, defined for $(x,y) \in (0,\infty) \times \mathbb{R} \subset \mathbb{R}^2$. Compute $\partial_x u$ and $\partial_y u$.

Solutions (Section 9.c)

Solution to Exercise 9.b.18:

Write $u(x,y) = x^{\sin(y)} = e^{\sin(y)\ln x}$ for $x > 0$. Differentiating with the chain rule:

$$\partial_x u = \sin(y)\, x^{\sin(y)-1}, \qquad \partial_y u = \cos(y)\ln(x)\, x^{\sin(y)} = \cos(y)\ln(x)\, e^{\sin(y)\ln x}.$$

Q.E.D.

Proof of AI-Exercise 9.b.14:

(a) Partial derivatives are $\frac{\partial f}{\partial x} = 2xy + y\cos(xy)$ and $\frac{\partial f}{\partial y} = x^2 + x\cos(xy)$. Evaluating at $(1,0)$ yields $\frac{\partial f}{\partial x}(1,0) = 0$ and $\frac{\partial f}{\partial y}(1,0) = 1 + 1 = 2$. Thus $\nabla f(1,0) = (0,2)^{\mathsf{T}}$, and the total differential $Df(1,0)\colon \mathbb{R}^2 \to \mathbb{R}$ is $Df(1,0)(h_1,h_2) = 2h_2$.

(b) Since $f(1,0) = 0$, the linearization at $(1,0)$ is $L(x,y) = f(1,0) + Df(1,0)(x-1, y-0) = 0 + 0(x-1) + 2y = 2y$.

Q.E.D.

The Chain Rule and Directional Derivatives (Chapter 10)

Having encoded the differential as a matrix, we now ask how it interacts with composition. The answer is the chain rule, and in this setting it says something cleaner than its one-variable version does: the differential of a composition is the **composition of the differentials**, which is to say the product of their matrices. Reading it that way makes it a statement of linear algebra rather than a formula to be memorised, and explains where the sums of products in the coordinate form come from.

The same machinery yields directional derivatives, and with them the standard trap of several-variable calculus. A function can possess every directional derivative at a point and still fail to be differentiable there, or even continuous. Section 10.c says what does suffice, and it is the criterion used in practice for the rest of the course.

Chain rule (Section 10.a)

Theorem 10.a.1 (*Chain rule*): The "*chain rule*" (German: „*Kettenregel*") states: for $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $g : V \subseteq \mathbb{R}^m \to \mathbb{R}^d$ differentiable, it holds:

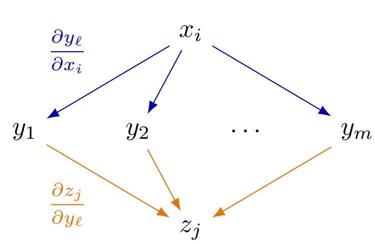
$$D(g \circ f)(x) = Dg(f(x))\, Df(x).$$

The same identity holds for the corresponding Jacobi matrices, where the composition on the right becomes an ordinary matrix product.

AI-Note 10.1: Corsin states the chain rule without proof, as does the lecture at this point. The one-line reason it is true is worth carrying, though: differentiability says f and g are each well approximated by a linear map near the relevant point, and composing two approximations composes the linear maps, with the error terms of the two compositions still $o(|h|)$. Making that precise is the whole proof, and it is the same argument as in one variable.

Remark 10.a.2 (*The chain rule as a sum over paths*): Stated as $D(g \circ f) = Dg \cdot Df$ the rule is compact but, as Sascha Brack puts it in his notes, *"pretty abstract"*. It says nothing about **why** a sum appears when the rule is written out in coordinates. His device is to draw the variables as a graph and read the formula off it.

Write $x \in \mathbb{R}^n \xmapsto{f} y \in \mathbb{R}^m \xmapsto{g} z \in \mathbb{R}^k$ and join each variable to those it feeds into. Then x_i influences z_j only by first moving some intermediate y_ℓ, and the contribution of that route is the product of the two rates along it. Summing over the routes gives the chain rule:



$$\frac{\partial z_j}{\partial x_i} = \sum_{\ell=1}^{m} \underbrace{\frac{\partial z_j}{\partial y_\ell}}_{(\mathrm{J}\,g)_{j\ell}} \underbrace{\frac{\partial y_\ell}{\partial x_i}}_{(\mathrm{J}\,f)_{\ell i}} = \big(\mathrm{J}\,g(f(x))\;\mathrm{J}\,f(x)\big)_{ji}.$$

The right-hand equality is just the definition of matrix multiplication, and that is precisely the point: **matrix multiplication is a sum over paths**, and the chain rule looks like a product of matrices precisely because composing functions composes those graphs. Sascha's summary is that computing $\partial z_j/\partial x_i$ this way is nothing more than *"computing single components of the chain rule"*.

The reading is worth having because it makes the coordinate form reconstructible instead of memorised: count the intermediate variables, one term each, each term a product of two factors, and the index that is summed over is the one belonging to the middle layer. It is also the form in which the rule is actually used in physics. Compare Section 10.d, which is this picture with a single intermediate layer $\gamma_1, \ldots, \gamma_n$ and a single source variable t.

Theorem 10.a.3 (*Euler's homogeneous function theorem*): Let $f \in C^1(\mathbb{R}^n \setminus \{0\}, \mathbb{R})$ be "*positively homogeneous of degree k*" for some $k \in \mathbb{R}$, meaning that

$$f(tx) = t^k f(x) \qquad \text{for all } x \in \mathbb{R}^n \setminus \{0\} \text{ and all } t > 0.$$

Then $\langle x, \nabla f(x)\rangle = k\, f(x)$ for all $x \in \mathbb{R}^n \setminus \{0\}$.

Proof:

Fix $x \neq 0$ and read both sides of the homogeneity identity as functions of $t > 0$ alone. The left-hand side is f composed with the path $\gamma(t) := tx$, whose velocity is $\gamma'(t) = x$, so the chain rule along a path (Section 10.d) gives

$$\frac{d}{dt} f(tx) = \big\langle \nabla f(tx),\ x \big\rangle.$$

On the right-hand side $f(x)$ is a constant in t, and therefore $\frac{d}{dt}\big(t^k f(x)\big) = kt^{k-1} f(x)$. The two derivatives agree for every $t > 0$; evaluating at $t = 1$ leaves $\langle \nabla f(x), x\rangle = kf(x)$, which is the claim by symmetry of the inner product. **Q.E.D.**

AI-Example 10.a.4 (*Chain Rule for Vector Fields*): Let $f(u, v) := u^2 + v^2$ and let $g(r, \theta) := (r\cos\theta, r\sin\theta)^\mathsf{T}$. The composite function is $(f \circ g)(r, \theta) = r^2\cos^2\theta + r^2\sin^2\theta = r^2$. By the multi-variable Chain Rule, the Jacobian matrix of $f \circ g$ satisfies:

$$D(f \circ g)(r, \theta) = Df(g(r, \theta)) \cdot Dg(r, \theta) = \begin{pmatrix} 2r\cos\theta & 2r\sin\theta \end{pmatrix} \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = \begin{pmatrix} 2r & 0 \end{pmatrix},$$

which matches the direct calculation $\frac{\partial}{\partial r}(r^2) = 2r$ and $\frac{\partial}{\partial \theta}(r^2) = 0$.

Exercise 10.a.5 (*Jacobian of a composite map*): Compute $\mathrm{J}(g \circ f)$ for

$$f : \mathbb{R}^3 \to \mathbb{R}^2,\ x \mapsto (x_1x_2,\ x_2 + x_3), \qquad g : \mathbb{R}^2 \to \mathbb{R}^2,\ x \mapsto (e^{x_1},\ x_1x_2).$$

Exercise 10.a.5:

We have

$$\mathrm{J}\, f(x) = \begin{pmatrix} x_2 & x_1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \mathrm{J}\, g(x) = \begin{pmatrix} e^{x_1} & 0 \\ x_2 & x_1 \end{pmatrix}.$$

So:

$$\begin{aligned} \mathrm{J}(g \circ f) &= \begin{pmatrix} e^{x_1x_2} & 0 \\ x_2 + x_3 & x_1x_2 \end{pmatrix} \begin{pmatrix} x_2 & x_1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} x_2e^{x_1x_2} & x_1e^{x_1x_2} & 0 \\ x_2^2 + x_2x_3 & 2x_1x_2 + x_1x_3 & x_1x_2 \end{pmatrix} \end{aligned}$$

Q.E.D.

Exercise 10.a.6 (*Computing differentials with and without the chain rule*): Let $f : \mathbb{R}^2 \to \mathbb{R}^3$ and $g : \mathbb{R}^3 \to \mathbb{R}$ be defined by

$$f(x,y) := \begin{pmatrix} x+y \\ x-y \\ x^2+y^2-1 \end{pmatrix}, \qquad g(x,y,z) := x^2+y^2+z^2.$$

Compute the differential $D(g \circ f)$ in two ways:

(a) By directly computing the composition $g \circ f$ and then differentiating.

(b) By computing Df and Dg and using the chain rule.

Exercise 10.a.6:

(a) **Directly:** First compute $g(f(x,y))$:

$$\begin{aligned}(g \circ f)(x,y) &= (x+y)^2 + (x-y)^2 + (x^2+y^2-1)^2 \\ &= (x^2+2xy+y^2) + (x^2-2xy+y^2) + (x^4+y^4+1+2x^2y^2-2x^2-2y^2) \\ &= 2x^2+2y^2+x^4+y^4+1+2x^2y^2-2x^2-2y^2 \\ &= x^4+y^4+2x^2y^2+1 \\ &= (x^2+y^2)^2+1.\end{aligned}$$

Now differentiate this scalar function with respect to x and y:

$$D(g\circ f)(x,y) = \begin{pmatrix} 2(x^2+y^2)\cdot 2x & 2(x^2+y^2)\cdot 2y \end{pmatrix} = \begin{pmatrix} 4x(x^2+y^2) & 4y(x^2+y^2) \end{pmatrix}.$$

(b) **Via chain rule:** Compute the differentials of f and g individually.

$$Df(x,y) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2x & 2y \end{pmatrix}, \qquad Dg(u,v,w) = \begin{pmatrix} 2u & 2v & 2w \end{pmatrix}.$$

Evaluate Dg at $f(x,y)$:

$$Dg(f(x,y)) = \begin{pmatrix} 2(x+y) & 2(x-y) & 2(x^2+y^2-1) \end{pmatrix}.$$

Multiply the matrices:

$$\begin{aligned}D(g\circ f)(x,y) &= Dg(f(x,y))Df(x,y) \\ &= \begin{pmatrix} 2(x+y) & 2(x-y) & 2(x^2+y^2-1) \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2x & 2y \end{pmatrix} \\ &= \begin{pmatrix} 4x+4x(x^2+y^2-1) & 4y+4y(x^2+y^2-1) \end{pmatrix} \\ &= \begin{pmatrix} 4x(x^2+y^2) & 4y(x^2+y^2) \end{pmatrix}.\end{aligned}$$

Both methods yield the same result.

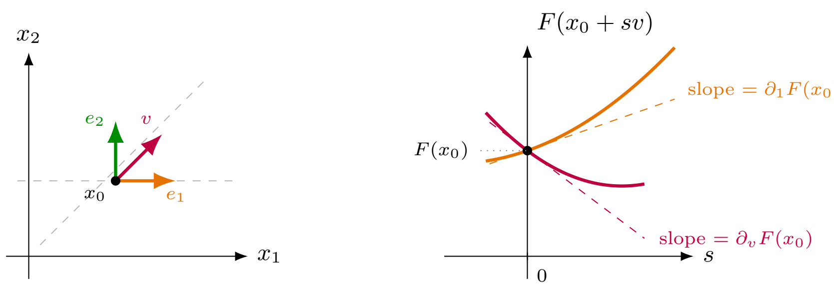
Q.E.D.

Directional derivatives (Section 10.b)

Sometimes the "*differential*" (German: „*das Differential*") is not easily expressed as a Jacobian. Then it can be convenient to use the formula for "*directional derivatives*" (German: „*Richtungsableitung*"):

$$DF_{x_0}(v) = \partial_v F(x_0) = \left.\frac{d}{ds}\right|_{s=0} F(x_0+sv) \qquad (*)$$

Formula $(*)$ describes a two-step recipe, and the picture below is worth carrying around. **Step one:** pick a direction v and walk away from x_0 along the straight ray $s \mapsto x_0 + sv$, ignoring the rest of the domain. **Step two:** record the values of F along that ray. What you get is an ordinary function of **one** variable s, and $\partial_v F(x_0)$ is simply its derivative at $s = 0$, an **Analysis I** object. The partial derivative $\partial_i F$ is the special case $v = e_i$: walk parallel to the i-th axis.



pick a direction and walk along that ray

each ray gives a one-variable function; its slope at 0 is $\partial_v F(x_0)$

The right-hand panel is also the cleanest way to see what differentiability adds. Every direction produces **some** slope; nothing so far forces those slopes to be related to one another. Being differentiable is precisely the demand that all of them are read off a **single** linear map via $\partial_v F(x_0) = DF_{x_0}(v)$, so that knowing the n partial derivatives already determines every other direction. AI-Example 10.b.8 below is the cautionary case where all the slopes exist but no such linear map governs them.

Important remark 10.b.7: The directional derivative $\partial_v F(x_0)$ can exist (depending on v and x_0) even if F is not differentiable at x_0, i.e. DF_{x_0} does not exist. However, if DF_{x_0} exists, the formula $(*)$ holds.

AI-Example 10.b.8 (*Every direction, and still not continuous*): AI-Example 9.b.6 showed that the two **partial** derivatives are too little. One might hope that asking for **all** directions is enough. It is not. Let

$$f(x,y) := \begin{cases} \dfrac{x^2 y}{x^4 + y^2} & (x,y) \neq (0,0), \\ 0 & (x,y) = (0,0). \end{cases}$$

Fix a direction $v = (a,b) \neq 0$ and compute along the ray $t \mapsto tv$:

$$\frac{f(ta,tb)}{t} = \frac{1}{t} \cdot \frac{t^2 a^2 \cdot tb}{t^4 a^4 + t^2 b^2} = \frac{a^2 b}{t^2 a^4 + b^2} \xrightarrow{t \to 0} \begin{cases} \dfrac{a^2}{b} & b \neq 0, \\ 0 & b = 0. \end{cases}$$

So $\partial_v f(0,0)$ exists for **every** direction v. Yet f is not even continuous at the origin: approaching along the **parabola** $y = x^2$ instead of along a straight line,

$$f(x,x^2) = \frac{x^2 \cdot x^2}{x^4 + x^4} = \frac{1}{2} \quad \text{for all } x \neq 0.$$

By Proposition 9.a.3, f is therefore not differentiable at the origin.

Two lessons worth keeping.

- **Straight lines cannot see everything.** Every directional derivative is computed along a line, and this f is well behaved along every line while misbehaving along a parabola. Approaching a point in $\mathbb{R}^n$ is genuinely harder than in $\mathbb{R}$, where there are only two directions.
- **Watch whether $v \mapsto \partial_v f(x_0)$ is linear.** If f were differentiable we would need $\partial_v f(0,0) = DF_0(v)$, a **linear** function of v; but a^2/b is not linear in (a,b). That failure alone rules out differentiability, without any mention of continuity, and it is precisely the test that settles Exercise 10.b.9**(b)** below.

Exercise 10.b.9 (*5.1 — Derivatives* (important)):

(a) Consider $f(x,y) := xy^2e^{-x^2-y^2}$ for $(x,y) \in \mathbb{R}^2$. Find all the critical points of f, that is all the points where the gradient of f vanishes. (The point will be given if and only if all the numerical values you find are correct... so check your computations twice.)

(b) We want to find a function $g \in C^1(\mathbb{R}^2)$ with the following directional derivative:

$$\partial_v g(x,y) = 2\cos(x^2y)xv_2 + \cos(x^2y)v_1^2 y \quad \text{for all } (x,y) \text{ and } (v_1,v_2) \in \mathbb{R}^2.$$

Give an explicit example of such a function g or prove that a function with these properties cannot exist.

Remark 10.b.10: Corsin's hint, *recall the linearity of the differential*, is the whole of Exercise 10.b.9(b): a directional derivative must be **linear** in v, and v_1^2 is not.

Exercise 10.b.11 (*5.4 — Computation of derivatives* (semi-important)): For each of the following functions compute the directional derivative in the general direction v at a general point x, that is $\partial_v f(x)$, where applicable.

1. x_1/x_2, e^{-x_1/x_2} for $x \in \mathbb{R}^2 \setminus \{x_2 = 0\}$. (Also: can they be extended continuously or differentiably across $\{x_2 = 0\}$?)
2. $e^{x_1x_2}\sin(x_1 + x_2^2)$, $\dfrac{x_1^2}{x_1^2+x_2^2}$, $\dfrac{x_1^2x_2}{x_1^2+x_2^2}$, $\dfrac{x_1^2x_2}{x_1^2+x_2^4}$ for $x \in \mathbb{R}^2 \setminus \{0\}$. (Also: can they be extended continuously or differentiably across the origin?)
3. $|x|$, $|x|^\alpha$, $x/|x|$, $g(|x|)$, $x \cdot e$, $g(x \cdot e)$, for $\alpha \in \mathbb{R}$, $e \in \mathbb{R}^n$ and $g \in C^1(\mathbb{R})$.
4. ax, x^T, $\mathrm{Tr}(x)$, x^2, x^3, $\mathrm{Tr}(x^2)$ for $x, a \in \mathbb{R}^{n\times n}$.
5. **(*)** x^{-1}, x^{-2} for $x \in \mathbb{R}^{n\times n}$ with $\det x \neq 0$.

Official hints: to prove that a function **is** differentiable at 0 you have two tools, namely the definition of differentiability and the sufficient condition of *Theorem 10.16*. To prove that a function **is not** differentiable at 0 you again have two: show that the necessary condition of *Proposition 10.14* does not hold, or show that for some C^1 curve $\gamma : (-\delta,\delta) \to \mathbb{R}^n$ the function $(f\circ\gamma)(t)$ is **not** differentiable at $t = 0$ (simpler, as you study a function of one variable). **5.4.3:** these functions are C^∞, so compute them on $x + tv$ and find an expansion in powers of t; by *Corollary 10.45* it must be the Taylor polynomial, from which the derivatives can be read off. For example $(x+tv)^2 = x^2 + (xv + vx)t + O(t^2)$.

What does imply differentiability (Section 10.c)

AI-Note 10.2: The course gives the definition of differentiability and the warnings that go with it, but never states a **sufficient** condition, so up to this point we have two counterexamples and no positive test at all. The theorem below supplies one, from a second tutor's notes. Without it the practical question *"how do I actually check differentiability?"* has no answer.

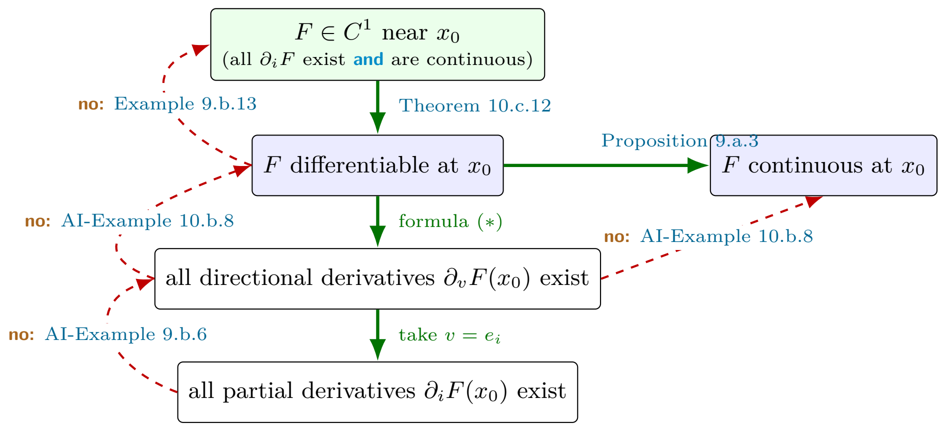
Theorem 10.c.12 (*Continuous partial derivatives suffice*): Let $U \subseteq \mathbb{R}^n$ be open and $F : U \to \mathbb{R}^m$. If all partial derivatives $\partial_i F$ exist on U **and are continuous** at every $x \in U$, then F is differentiable at every $x \in U$, and

$$DF_x = \big(\partial_1 F(x),\ \dots,\ \partial_n F(x)\big).$$

Such an F is called "*continuously differentiable*", written $F \in C^1(U)$.

This is the theorem that makes the subject workable: in practice one almost never checks the definition directly. Compute the partials, observe that they are built from continuous pieces, and differentiability follows. That is exactly why the course can write $f \in C^1$, C^2 and C^k so freely from here onward.

Together with the two counterexamples, the whole landscape is a single chain, each link one-directional:



Every downward arrow holds, and **not one of them reverses**. Each failure is witnessed by an example already in this chapter, so the diagram doubles as a map of where those counterexamples belong:

- **Differentiable $\not\Rightarrow$ C^1:** Example 9.b.13, whose partial derivatives oscillate without limit at the origin even though the differential exists there.
- **All directional derivatives $\not\Rightarrow$ differentiable:** AI-Example 10.b.8, where every ray gives a slope but $v \mapsto \partial_v f(0,0)$ is not linear. The same example defeats the branch to the right as well: it is not even **continuous** at the origin, so having every directional derivative buys strictly less than continuity does.
- **Partial derivatives $\not\Rightarrow$ all directional derivatives:** AI-Example 9.b.6 again. For $f = \frac{xy}{x^2+y^2}$ and a direction $v = (a,b)$ with $ab \neq 0$,
$$\frac{f(ta,tb)}{t} = \frac{1}{t} \cdot \frac{t^2 ab}{t^2(a^2+b^2)} = \frac{ab}{t\,(a^2+b^2)} \xrightarrow{t \to 0} \pm\infty,$$
so $\partial_v f(0,0)$ exists only along the two axes, which are precisely the two directions the partial derivatives happen to look at.

The three counterexamples are not variations on one theme; they fail at three different heights, which is precisely why the chain has three separate links.

Remark 10.c.13 (*A recipe for checking differentiability at a point*): Sascha's notes distil the above into three steps, worth following in this order.

1. **Compute the partial derivatives $\partial_i F(x_0)$.** If one of them fails to exist, F is not differentiable and you are done. If they all exist **and are continuous near** x_0, F **is** differentiable by Theorem 10.c.12, and you are done again. Most exercises end here.
2. **Otherwise, guess the candidate.** If F is differentiable at all, the differential can only be $L := \big(\partial_1 F(x_0), \dots, \partial_n F(x_0)\big)$, since the partials are forced (Section 9.b).
3. **Verify the candidate against the definition:**
$$\lim_{h \to 0} \frac{\big|F(x_0+h) - F(x_0) - L(h)\big|}{\|h\|} \stackrel{?}{=} 0.$$
In $\mathbb{R}^2$, substituting $h = (r\cos\theta,\ r\sin\theta)$ and checking that the result tends to 0 **uniformly in** θ is usually the quickest route. This is the same polar trick used in AI-Example 10.b.8 and, later, in Question C.a.7.

Step 1 is the one students skip; it settles the overwhelming majority of cases without any ε–δ work at all.

Example 10.c.14 (*Checking differentiability at the origin*): Let $f(x,y) := \frac{x^2y}{x^2+y^2}$ for $(x,y) \neq (0,0)$ and $f(0,0) := 0$. We check if f is differentiable at the origin.

1. **Partial derivatives:**
$$\partial_1 f(0,0) = \lim_{t\to 0} \frac{f(t,0)-f(0,0)}{t} = \lim_{t\to 0} \frac{0}{t} = 0, \qquad \partial_2 f(0,0) = \lim_{t\to 0} \frac{f(0,t)-f(0,0)}{t} = 0.$$
They exist, but to use Theorem 10.c.12 we would have to compute them everywhere and show they are continuous at $(0,0)$, which is tedious and (as we will see) false. We proceed to step 2.
2. **Guess the candidate:** The only possible differential is $L(h_1,h_2) = \partial_1 f(0,0)h_1 + \partial_2 f(0,0)h_2 = 0$.
3. **Verify against the definition:** We check whether the following limit is zero:
$$\lim_{h\to 0} \frac{|f(h_1,h_2)-f(0,0)-L(h)|}{\|h\|} = \lim_{h\to 0} \frac{\frac{h_1^2h_2}{h_1^2+h_2^2}}{\sqrt{h_1^2+h_2^2}} = \lim_{h\to 0} \frac{h_1^2h_2}{(h_1^2+h_2^2)^{3/2}}.$$
Passing to polar coordinates $h = (r\cos\theta, r\sin\theta)$, the expression becomes
$$\frac{r^3\cos^2\theta\sin\theta}{r^3} = \cos^2\theta\sin\theta.$$
This depends on the angle of approach θ (for instance, it is 0 along the axes but $\frac{1}{2\sqrt{2}} \neq 0$ along $h_1 = h_2$), so the limit as $r \to 0$ is not uniformly 0. Hence f is **not differentiable** at $(0,0)$.

Example 10.c.15 (*A function that is differentiable at the origin*): Let $f(x,y) := \frac{x^2y^2}{\sqrt{x^2+y^2}}$ for $(x,y) \neq (0,0)$ and $f(0,0) := 0$. We check differentiability at the origin following the recipe.

1. **Partial derivatives:** For $(x,y) \neq (0,0)$, the quotient and chain rules give
$$\partial_1 f(x,y) = xy^2 \frac{x^2+2y^2}{(x^2+y^2)^{3/2}}, \qquad \partial_2 f(x,y) = x^2y\frac{2x^2+y^2}{(x^2+y^2)^{3/2}}.$$
At the origin, computing directly from the definition yields $\partial_1 f(0,0) = 0$ and $\partial_2 f(0,0) = 0$. To see if they are continuous at $(0,0)$, we substitute $x = r\cos\theta$ and $y = r\sin\theta$:
$$\partial_1 f(r\cos\theta, r\sin\theta) = \frac{r\cos\theta\, r^2\sin^2\theta \cdot r^2(1+\sin^2\theta)}{r^3} = r^2\cos\theta\sin^2\theta(1+\sin^2\theta).$$
As $r \to 0$, this tends to 0 uniformly in θ, so $\partial_1 f$ is continuous at $(0,0)$. The same holds for $\partial_2 f$. By Theorem 10.c.12, f is differentiable at the origin.
2. **Guess the candidate:** We already know f is differentiable and $L = (0,0)$. But for practice, let's verify it with the definition directly.
3. **Verify against the definition:** We check the limit
$$\lim_{h\to 0} \frac{|f(h_1,h_2)-f(0,0)-L(h)|}{\|h\|} = \lim_{h\to 0} \frac{\frac{h_1^2h_2^2}{\sqrt{h_1^2+h_2^2}}}{\sqrt{h_1^2+h_2^2}} = \lim_{h\to 0} \frac{h_1^2h_2^2}{h_1^2+h_2^2}.$$
In polar coordinates, this is $\frac{r^4\cos^2\theta\sin^2\theta}{r^2} = r^2\cos^2\theta\sin^2\theta$. As $r \to 0$, the limit is clearly 0 uniformly in θ, confirming f is differentiable with $Df_0 = 0$.

The chain rule along a path (Section 10.d)

An important special case of the chain rule is that of a composition of a scalar field with a path. Suppose $f : \mathbb{R}^n \supseteq U \to \mathbb{R}$ and $\gamma : [0,1] \to \mathbb{R}^n$ are a C^1 "*scalar field*" (German: „*Skalarfeld*") and path, respectively. Then

$$\begin{aligned} D(f\circ\gamma)(t) &= Df(\gamma(t))\cdot D\gamma(t)\\ &= \langle \nabla f(\gamma(t)),\ \gamma'(t)\rangle\\ &= \sum_{i=1}^{n}\frac{\partial f}{\partial \gamma_i}\frac{\partial \gamma_i}{\partial t}\end{aligned}$$

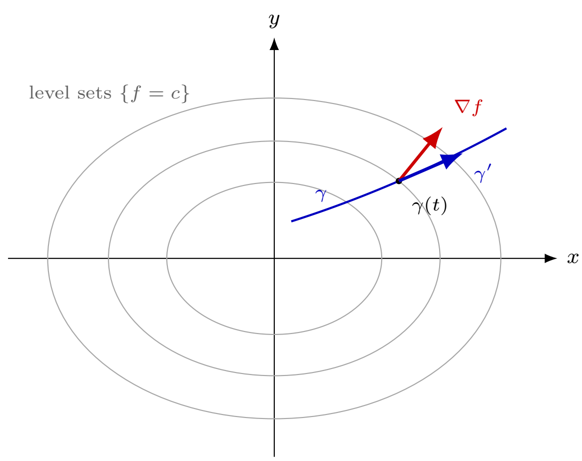
where $\gamma=(\gamma_1,\ldots,\gamma_n)$ and $\dfrac{\partial f}{\partial\gamma_i} := \partial_i f(\gamma(t))$.

This operation is so common in physics that it has its own notation: for f as above, we look at each coordinate as a function of **time** and differentiate:

$$\frac{d}{dt}f(x_1,\ldots,x_n) := Df(x_1(t),\ldots,x_n(t)) = \sum_{i=1}^{n}\frac{\partial f}{\partial x_i}\frac{\partial x_i}{\partial t}.$$

The middle line, $D(f\circ\gamma)(t) = \langle\nabla f(\gamma(t)),\gamma'(t)\rangle$, is worth dwelling on, because it is the formula that gives the gradient its meaning. It says the rate at which f changes along a path depends on the velocity only through its inner product with ∇f. Consequently, by Cauchy–Schwarz (Theorem 2.c.29), the change is fastest when γ' points along ∇f, and zero when γ' is perpendicular to it. Two standard facts fall straight out:

- **∇f points in the direction of steepest ascent**, with $|\nabla f|$ the rate of ascent in that direction.
- **∇f is perpendicular to the level sets of f.** If γ stays inside a level set $\{f=c\}$, then $f\circ\gamma$ is constant, so $\langle\nabla f(\gamma(t)),\gamma'(t)\rangle = 0$ for every t, and γ' runs through the directions tangent to the level set.



$\dfrac{d}{dt}f(\gamma(t)) = \langle\nabla f,\gamma'\rangle$: only the component of γ' along ∇f contributes, so f is constant along a level set and grows fastest along ∇f

AI-Exercise 10.d.16 (*Directional Derivative along a Unit Vector*): Let $f(x,y) := x^2y$ defined on $\mathbb{R}^2$. Compute the directional derivative $D_vf(1,2)$ of f at the point $(1,2)$ along the unit vector $v := \frac{1}{\sqrt{2}}(1,1)^{\mathsf{T}}$.

Exercise 10.d.17 (*4.6 — A directional derivative vanishes* (important)): Let $u\in C^1(\mathbb{R}^n)$ and $\nu\in\mathbb{R}^n$. Show that

1. If $\partial_1 u\equiv 0$ then "u does not depend on x_1"; more rigorously: there exists a unique function $v\in C^1(\mathbb{R}^{n-1})$ such that

$$u(x_1,\ldots,x_n) = v(x_2,\ldots,x_n)\quad\text{for all } x\in\mathbb{R}^n. \tag{2}$$

2. If $\partial_\nu u \equiv 0$ and $\nu \cdot e_1 \neq 0$ then "u is a function of $n-1$ variables"; more rigorously: there exists a unique function $w \in C^1(\mathbb{R}^{n-1})$ such that
$$u(x_1,\dots,x_n) = w\left(x_2 - \frac{x_1\nu_2}{\nu_1},\dots,x_n - \frac{x_1\nu_n}{\nu_1}\right) \quad \text{for all } x \in \mathbb{R}^n.$$
3. **(*)** What can we conclude if we assume only that $\partial_1 u = 0$ in an open connected subset $U \subset \mathbb{R}^n$?

Official hints: **4.6.2:** apply the mean value theorem to $u(x+t\nu)$. **4.6.3:** consider the domain $U := \{(x,y) : x^2 < y < x^2+1\}$ and the function
$$u(x,y) := \begin{cases} \max\{0, y-2\}^2 & \text{if } x \geq 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

Exercise 10.d.18 (*5.7 — Chain rule for a curve, or: a car on Earth* (optional — "interesting for physicists")**):** Let $R > 0$ and $\omega \in \mathbb{R}$ be constants. Let $\sigma : I \subset \mathbb{R} \to \mathbb{R}^2$, $\sigma(t) = (a(t), b(t))$, which we interpret as a curve in a "map" representing the longitude and latitude of a car at time t.

Define the "rotating" spherical coordinates $F : \mathbb{R}^2 \times \mathbb{R} \to \mathbb{R}^3$,
$$F((x,y),t) = \big(R\cos y\cos(\omega t + x),\ R\cos y\sin(\omega t + x),\ R\sin y\big),$$
representing what a point on the map looks like on Earth from a satellite (thus we consider the Earth as a rotating sphere with speed given by ω, and we forget about translation around the Sun) at time t.

Consider $r : \mathbb{R} \to \mathbb{R}^3$, $r(t) = F(\sigma(t), t)$, which is the position of the car at time t.

1. Compute the acceleration $r''(t)$ using the chain rule.
2. Assume that the acceleration is always proportional to the radial vector $r(t)$, i.e. $r''(t) = \lambda(t)\, r(t)$ for some scalar function $\lambda(t)$. Show that a and b satisfy
$$b'' + \sin b\cos b\,(\omega + a')^2 = 0, \qquad \cos b\, a'' - 2\sin b\,(\omega + a')b' = 0.$$
Equivalently, away from the poles ($\cos b \neq 0$),
$$a'' - 2\tan b\,(\omega + a')\,b' = 0.$$

Solutions (Section 10.e)

Solution to Exercise 10.d.17:

1. Define $v(x_2,\dots,x_n) := u(0, x_2,\dots,x_n)$, which is C^1 as the restriction of u to the hyperplane $x_1 = 0$. For fixed $(x_2,\dots,x_n)$, the function $t \mapsto u(t, x_2,\dots,x_n)$ has derivative $\partial_1 u \equiv 0$ everywhere, hence is constant in t; evaluating at $t = 0$ gives $u(x_1,\dots,x_n) = u(0,x_2,\dots,x_n) = v(x_2,\dots,x_n)$ for all x_1. Uniqueness: if some $\tilde v$ also satisfies $u(x) = \tilde v(x_2,\dots,x_n)$, then setting $x_1 = 0$ gives $\tilde v = v$.
2. Since $\nu \cdot e_1 = \nu_1 \neq 0$, consider the invertible linear change of coordinates $y_i := x_i - \dfrac{x_1\nu_i}{\nu_1}$ for $i = 2,\dots,n$, together with x_1 itself. Moving x_1 by t while holding $y_2,\dots,y_n$ fixed forces x_i to move by $t\nu_i/\nu_1$, i.e. moves x in the direction $\frac{1}{\nu_1}\nu$. By the hint, applying the mean value theorem to $u(x+t\nu)$ shows $t \mapsto u(x+t\nu)$ is constant since $\partial_\nu u \equiv 0$; consequently u, expressed in the coordinates $(x_1, y_2,\dots,y_n)$, has vanishing x_1-derivative and is therefore (by part 1) independent of x_1 in these coordinates. Hence, there is a (unique, by the same argument as in part 1) function $w \in C^1(\mathbb{R}^{n-1})$ with
$$u(x_1,\dots,x_n) = w\left(x_2 - \frac{x_1\nu_2}{\nu_1},\dots,x_n - \frac{x_1\nu_n}{\nu_1}\right).$$
3. In general, only **locally**: on any open subset where every fiber $\{x_1 : (x_1,x_2,\dots,x_n) \in U\}$ (for fixed $(x_2,\dots,x_n)$) is connected, the same argument as in part 1 shows u is locally of the form

$v(x_2,\dots,x_n)$. But this need not hold **globally** on all of U, even though U itself is connected: with $U := \{(x,y) : x^2 < y < x^2+1\}$ and $u(x,y) := \max\{0, y-2\}^2$ for $x \geq 0$, $u := 0$ for $x \leq 0$, one checks $\partial_1 u \equiv 0$ throughout U (near $x = 0$ both branches agree, since $y < 2$ there), yet u is not globally independent of x: for $y > 2$, the fiber of U over y splits into two disconnected intervals (one with $x > 0$, one with $x < 0$), and u takes different values on each. So, connectedness of U alone does not suffice; what is needed is connectedness of each vertical fiber of U.

Q.E.D.

Proof of AI-Exercise 10.d.16:

The gradient of $f(x,y)$ is $\nabla f(x,y) = (2xy, x^2)^\mathsf{T}$. Evaluating at $(1,2)$ gives $\nabla f(1,2) = (4,1)^\mathsf{T}$. Since f is continuously differentiable (C^∞), the directional derivative along the unit vector v is given by the inner product:

$$D_v f(1,2) = \nabla f(1,2) \cdot v = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{4+1}{\sqrt{2}} = \frac{5}{\sqrt{2}}.$$

Q.E.D.

Solution to Exercise 10.b.9:

(a) Since $e^{-x^2-y^2} > 0$ everywhere, the gradient of $f(x,y) = xy^2 e^{-x^2-y^2}$ vanishes exactly where its two factors below vanish:

$$\partial_x f = y^2(1-2x^2)e^{-x^2-y^2}, \qquad \partial_y f = 2xy(1-y^2)e^{-x^2-y^2}.$$

If $y = 0$: both partials vanish for every x (indeed $f(x,0) \equiv 0$), so the entire line $\{(x,0) : x \in \mathbb{R}\}$ consists of critical points. **If $y \neq 0$:** $\partial_x f = 0$ forces $x = \pm\frac{1}{\sqrt{2}}$, and since then $x \neq 0$, $\partial_y f = 0$ forces $y = \pm 1$. This gives the four additional critical points $\big(\pm\frac{1}{\sqrt{2}}, \pm 1\big)$.

(b) A directional derivative $\partial_v g(x,y) = \nabla g(x,y) \cdot v$ is always **linear** in v. The prescribed expression $2\cos(x^2y)xv_2 + \cos(x^2y)v_1^2 y$ contains a v_1^2 term with coefficient $y\cos(x^2y)$, which is not identically zero (e.g. at $(x,y) = (0,1)$ it equals 1). Hence the expression is not linear in v at every point, so **no such $g \in C^1(\mathbb{R}^2)$ can exist**.

Q.E.D.

Solution to Exercise 10.b.11:

Throughout, v is the direction and x the base point. Where f is differentiable the answer is just $\partial_v f(x) = \langle \nabla f(x), v\rangle$; the interest is in the points where it is not, which is what the second half of each part is asking about.

1. On $\mathbb{R}^2 \setminus \{x_2 = 0\}$ both functions are smooth, so
$$\partial_v\left(\frac{x_1}{x_2}\right) = \frac{v_1}{x_2} - \frac{x_1 v_2}{x_2^2}, \qquad \partial_v\left(e^{-x_1/x_2}\right) = -e^{-x_1/x_2}\left(\frac{v_1}{x_2} - \frac{x_1 v_2}{x_2^2}\right).$$
Neither extends continuously across $\{x_2 = 0\}$. For x_1/x_2, fix $x_1 = 1$ and let $x_2 \to 0^\pm$: the values run off to $\pm\infty$. For e^{-x_1/x_2} the failure is more interesting, since the function is **bounded** on one side: with $x_1 = 1$ fixed, $x_2 \to 0^+$ sends $-1/x_2 \to -\infty$ and the value to 0, while $x_2 \to 0^-$ sends it to $+\infty$. A limit existing from one side is not enough.
2. The first function is smooth on all of $\mathbb{R}^2$, with
$$\begin{aligned}\partial_v\big(e^{x_1x_2}\sin(x_1+x_2^2)\big) &= e^{x_1x_2}\big(x_2\sin(x_1+x_2^2) + \cos(x_1+x_2^2)\big)\,v_1 \\ &\quad + e^{x_1x_2}\big(x_1\sin(x_1+x_2^2) + 2x_2\cos(x_1+x_2^2)\big)\,v_2,\end{aligned}$$

and nothing needs extending. The other three are the point of the exercise, and they separate three different behaviours at the origin.

- $\dfrac{x_1^2}{x_1^2+x_2^2}$ is homogeneous of degree 0, with
$$\partial_v f(x) = \frac{2x_1x_2\,(x_2v_1 - x_1v_2)}{(x_1^2+x_2^2)^2}.$$
It has **no continuous extension**: it equals 1 on $\{x_2=0\}$ and 0 on $\{x_1=0\}$.
- $\dfrac{x_1^2x_2}{x_1^2+x_2^2}$ **does** extend continuously by $f(0):=0$, since $x_1^2 \le x_1^2+x_2^2$ gives $|f|\le|x_2|$. It is not differentiable there:
$$\partial_v f(0) = \lim_{t\to 0}\frac{f(tv)}{t} = \lim_{t\to 0}\frac{t^3v_1^2v_2}{t^3(v_1^2+v_2^2)} = \frac{v_1^2v_2}{v_1^2+v_2^2},$$
which exists in every direction and is **not linear** in v.
- $\dfrac{x_1^2x_2}{x_1^2+x_2^4}$ also satisfies $|f|\le|x_2|$, so it too extends continuously by 0. Here
$$\partial_v f(0) = \lim_{t\to 0}\frac{t^3v_1^2v_2}{t^3\,(v_1^2+t^2v_2^4)} = \begin{cases} v_2 & v_1\neq 0,\\ 0 & v_1 = 0,\end{cases}$$
which is not even continuous in v, let alone linear. Again continuous, not differentiable.

All three are instances of AI-Example 10.b.8: every directional derivative can exist without the map $v\mapsto\partial_v f(0)$ being linear, and linearity is exactly what differentiability demands.

3. Here $x\in\mathbb{R}^n\setminus\{0\}$ for the first four, $e\in\mathbb{R}^n$ and $g\in C^1(\mathbb{R})$:
$$\partial_v|x| = \frac{\langle x,v\rangle}{|x|},\qquad \partial_v|x|^\alpha = \alpha\,|x|^{\alpha-2}\langle x,v\rangle,\qquad \partial_v\, g(|x|) = g'(|x|)\,\frac{\langle x,v\rangle}{|x|},$$
$$\partial_v\left(\frac{x}{|x|}\right) = \frac{1}{|x|}\left(v - \frac{\langle x,v\rangle}{|x|^2}\,x\right),\qquad \partial_v\,(x\cdot e) = \langle e,v\rangle,\qquad \partial_v\, g(x\cdot e) = g'(\langle x,e\rangle)\,\langle e,v\rangle.$$
The fourth is worth reading geometrically: $x/|x|$ records only a direction, so moving x along itself changes nothing, and indeed the bracket is exactly v with its component along x removed.

4. For matrix arguments the cleanest route is the one Corsin's hint suggests: expand $f(x+tv)$ in powers of t and read off the coefficient of t. The first three maps are linear, hence their own derivatives:
$$\partial_v(ax) = av,\qquad \partial_v(x^\mathsf{T}) = v^\mathsf{T},\qquad \partial_v\,\mathrm{Tr}(x) = \mathrm{Tr}(v).$$
For the rest, $(x+tv)^2 = x^2+t(xv+vx)+t^2v^2$ and $(x+tv)^3 = x^3+t(x^2v+xvx+vx^2)+O(t^2)$, so
$$\partial_v(x^2) = xv+vx,\qquad \partial_v(x^3) = x^2v+xvx+vx^2,\qquad \partial_v\,\mathrm{Tr}(x^2) = \mathrm{Tr}(xv+vx) = 2\,\mathrm{Tr}(xv),$$
the last using cyclicity of the trace. Note that $\partial_v(x^2)\neq 2xv$: matrices do not commute, and the familiar one-variable answer is precisely what fails.

5. **(*)** Differentiate the identity $(x+tv)(x+tv)^{-1} = \mathrm{id}$ at $t=0$. The product rule gives $v\,x^{-1} + x\,\partial_v(x^{-1}) = 0$, hence
$$\partial_v(x^{-1}) = -\,x^{-1}v\,x^{-1}.$$
Writing $x^{-2} = x^{-1}x^{-1}$ and applying the product rule again,
$$\partial_v(x^{-2}) = -\,x^{-1}v\,x^{-2}\;-\;x^{-2}v\,x^{-1}.$$
Both reduce to the scalar formulas $-v/x^2$ and $-2v/x^3$ when $n=1$; for $n>1$ the two terms genuinely differ and cannot be combined.

Q.E.D.

Solution to Exercise 10.d.18:

Write $\theta(t) := \omega t + a(t)$ for the longitude actually occupied in space and $\varphi(t) := b(t)$ for the latitude, so that

$$r(t) = R\big(\cos\varphi\cos\theta,\ \cos\varphi\sin\theta,\ \sin\varphi\big).$$

1. The computation is far shorter in the moving orthonormal frame

$$\begin{aligned} e_r &:= (\cos\varphi\cos\theta,\ \cos\varphi\sin\theta,\ \sin\varphi), \\ e_\varphi &:= (-\sin\varphi\cos\theta,\ -\sin\varphi\sin\theta,\ \cos\varphi), \\ e_\theta &:= (-\sin\theta,\ \cos\theta,\ 0), \end{aligned}$$

in which $r = R\,e_r$. Differentiating the three frame vectors by the chain rule gives

$$\dot e_r = \varphi'\,e_\varphi + \theta'\cos\varphi\;e_\theta, \qquad \dot e_\varphi = -\varphi'\,e_r - \theta'\sin\varphi\;e_\theta, \qquad \dot e_\theta = \theta'\big(-\cos\varphi\,e_r + \sin\varphi\,e_\varphi\big).$$

Hence $r' = R\big(\varphi'\,e_\varphi + \theta'\cos\varphi\,e_\theta\big)$, and differentiating once more and collecting the three components,

$$\begin{aligned} r''(t) = R\Big[&-\big(\varphi'^2 + \theta'^2\cos^2\varphi\big)\,e_r \\ &+ \big(\varphi'' + \theta'^2\sin\varphi\cos\varphi\big)\,e_\varphi \\ &+ \big(\theta''\cos\varphi - 2\,\theta'\varphi'\sin\varphi\big)\,e_\theta\Big]. \end{aligned}$$

The first component is the centripetal term; the other two are tangential.

2. Substituting $\varphi = b$, $\theta' = \omega + a'$ and $\theta'' = a''$, the assumption $r''(t) = \lambda(t)\,r(t) = \lambda(t)R\,e_r$ says precisely that the e_φ and e_θ components vanish. The frame is orthonormal, so there is no cancellation between them to hope for. That is

$$b'' + \sin b\cos b\,(\omega + a')^2 = 0, \qquad \cos b\;a'' - 2\sin b\,(\omega + a')\,b' = 0,$$

which is the claim. Away from the poles $\cos b \neq 0$, so dividing the second equation by $\cos b$ gives

$$a'' - 2\tan b\,(\omega + a')\,b' = 0.$$

Remark 10.e.19 (*Reading the two equations physically*): Two features are worth noticing. First, ω enters only through the combination $\omega + a'$, which is the **absolute** angular velocity. That is what one would hope for, since the equations ought not to distinguish a car driving east on a still Earth from a parked car on a rotating one. Second, the factor 2 in the e_θ equation is the Coriolis term, and it is the same $2\theta'\varphi'$ that appears in the acceleration in polar coordinates one dimension lower.

Q.E.D.

The Mean Value Theorem and Taylor Expansions (Chapter 11)

Having linearized a map once, we ask two further questions. How good is that linear approximation, and how is it improved order by order? The answer is the several-variable analogue of the Taylor series from **Analysis I**, and it is what supplies every quantitative estimate in the optimization theory of the next part.

The mean value theorem comes first, for two reasons. It is the instrument the error terms are estimated with, and it is the first result in this course that survives the move to $\mathbb{R}^m$ only as an **inequality** rather than an equality once $m > 1$ (Exercise 11.a.7). That degradation is not a defect of the proof. There is simply no single intermediate point at which all m components can be caught at once.

The mean value theorem (Section 11.a)

AI-Note 11.1: The mean value theorem is used twice in this chapter, in the official hint to Exercise 10.d.17 and again in its solution, but the course never states it. It is therefore stated here, together with the warning that follows, which is what Exercise 10.d.17 actually turns on.

Theorem 11.a.1 (*Mean value theorem*): Let $f : U \to \mathbb{R}$ be differentiable, $x \in U$ and $h \in \mathbb{R}^n$ with $x + th \in U$ for all $t \in [0,1]$. Then there is $t_0 \in (0,1)$ such that, writing $\xi := x + t_0 h$,

$$f(x+h) - f(x) = \partial_h f(\xi).$$

Proof:

Apply the one-variable mean value theorem to $\varphi(t) := f(x+th)$ on $[0,1]$, which is differentiable with $\varphi'(t) = \partial_h f(x+th)$ by the chain rule along a path (Section 10.d). It gives $t_0 \in (0,1)$ with $\varphi(1) - \varphi(0) = \varphi'(t_0)$, which is the claim. **Q.E.D.**

Important remark 11.a.2 (*It fails for vector-valued f*): The hypothesis *"$f : U \to \mathbb{R}$"* is essential. **The theorem is false the moment the target is $\mathbb{R}^m$ with $m \geq 2$**, and this is exactly what Exercise 11.a.7 below probes.

Applying the scalar result to each component $f_1, \dots, f_m$ separately does produce points $\xi_1, \dots, \xi_m$ with

$$f(x+h) - f(x) = \begin{pmatrix} \partial_h f_1(\xi_1) \\ \vdots \\ \partial_h f_m(\xi_m) \end{pmatrix},$$

but the ξ_i are **different points** in general, and no single ξ need work for all components at once. The standard witness is the **circle** $\gamma(t) := (\cos t, \sin t)$ on $[0, 2\pi]$: here $\gamma(2\pi) - \gamma(0) = 0$, while $\gamma'(t) = (-\sin t, \cos t)$ has $|\gamma'(t)| = 1$ and so never vanishes. There is no such ξ at all. Geometrically this is entirely reasonable: a walker who returns to the starting point must have had zero **average** velocity, but need never have stood still. In one dimension *"returning"* forces a turning point; in two it does not.

What survives for vector-valued f is the "*mean value inequality*" (German: „*Mittelwertsatz-Ungleichung*"), which replaces exact equality by a sharp upper bound controlling the displacement in terms of the maximum derivative along the line segment.

Theorem 11.a.3 (*Mean value inequality*): Let $U \subseteq \mathbb{R}^n$ be open, $x \in U$, and $h \in \mathbb{R}^n$ such that the line segment $[x, x+h] := \{x + th \mid t \in [0,1]\}$ lies entirely in U. If $F : U \to \mathbb{R}^m$ is differentiable at every point of $[x, x+h]$, then

$$\|F(x+h) - F(x)\| \le \left(\sup_{t\in[0,1]} \|DF_{x+th}\|_{\mathrm{op}} \right) \|h\|,$$

where $\|DF_z\|_{\mathrm{op}} := \sup_{\|v\|\le 1} \|DF_z(v)\|$ is the operator norm of the linear map DF_z.

Proof:

Set $y := F(x+h) - F(x) \in \mathbb{R}^m$. If $y = 0$, the inequality holds trivially. Assuming $y \ne 0$, let $u := \frac{y}{\|y\|}$, so that $\|u\| = 1$ and $\langle u, y\rangle = \|y\|$.

Define the auxiliary scalar function $\varphi : [0,1] \to \mathbb{R}$ by

$$\varphi(t) := \langle u, F(x+th)\rangle.$$

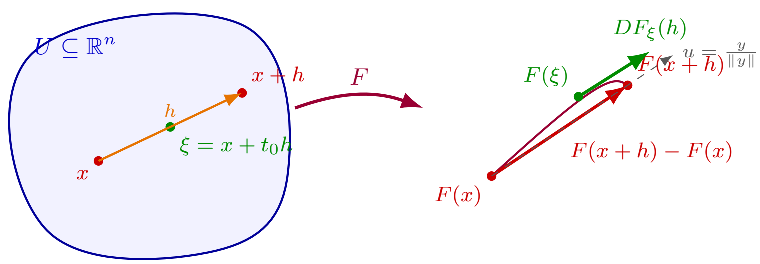
By the chain rule along a path (Section 10.d), φ is continuous on $[0,1]$ and differentiable on $(0,1)$ with derivative $\varphi'(t) = \langle u, DF_{x+th}(h)\rangle$. Applying the scalar Mean Value Theorem (Theorem 11.a.1) to φ on $[0,1]$, there exists some $t_0 \in (0,1)$ such that $\varphi(1) - \varphi(0) = \varphi'(t_0)$, which expands to

$$\|F(x+h) - F(x)\| = \langle u, F(x+h) - F(x)\rangle = \langle u, DF_{x+t_0h}(h)\rangle.$$

Applying the Cauchy–Schwarz inequality and the definition of the operator norm, we obtain

$$\|F(x+h) - F(x)\| \le \|u\|\,\|DF_{x+t_0h}(h)\| \le 1 \cdot \|DF_{x+t_0h}\|_{\mathrm{op}}\,\|h\| \le \left(\sup_{t\in[0,1]} \|DF_{x+th}\|_{\mathrm{op}} \right) \|h\|,$$

Q.E.D.



Remark 11.a.4 (*Global Lipschitz bound on convex domains*): As Diego Torres Tejeda emphasizes in his notes, Theorem 11.a.3 is the principal multivariable tool for establishing Lipschitz continuity. If $U \subseteq \mathbb{R}^n$ is **convex** and $F : U \to \mathbb{R}^m$ is differentiable with $\|DF_z\|_{\mathrm{op}} \le M$ for all $z \in U$, then for any pair $x, y \in U$, the line segment $[x, y]$ lies in U, so

$$\|F(y) - F(x)\| \le M\|y - x\|.$$

Thus F is M-Lipschitz continuous on U.

Example 11.a.5 (*Computing a Lipschitz constant via the Mean Value Inequality*): Consider the vector field $F : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $F(x,y) := (\sin(x+y), \cos(x-y))^{\mathsf{T}}$. We use the Mean Value Inequality to find an explicit Lipschitz constant for F on $\mathbb{R}^2$.

The Jacobi matrix of F at (x,y) is

$$\mathrm{J}\,F(x,y) = \begin{pmatrix} \cos(x+y) & \cos(x+y) \\ -\sin(x-y) & \sin(x-y) \end{pmatrix}.$$

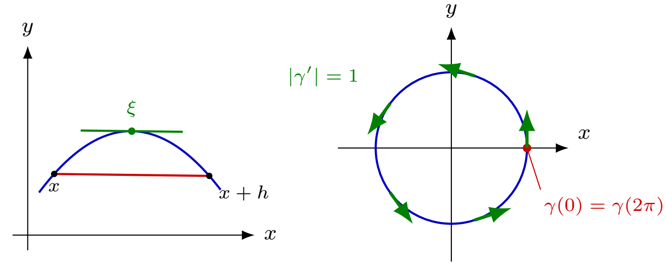
For any vector $v = (v_1, v_2)^{\mathsf{T}} \in \mathbb{R}^2$, the squared norm of $\mathrm{J}\,F(x,y)v$ is

$$\|\mathrm{J}\,F(x,y)v\|^2 = \cos^2(x+y)(v_1+v_2)^2 + \sin^2(x-y)(v_1-v_2)^2.$$

Since $\cos^2(x+y) \le 1$ and $\sin^2(x-y) \le 1$, we estimate:

$$\|\mathrm{J}\,F(x,y)v\|^2 \le (v_1+v_2)^2 + (v_1-v_2)^2 = (v_1^2 + 2v_1v_2 + v_2^2) + (v_1^2 - 2v_1v_2 + v_2^2) = 2(v_1^2+v_2^2) = 2\|v\|^2.$$

Taking square roots gives $\| \mathrm{J}\,F(x,y)v\| \leq \sqrt{2}\|v\|$, so $\|DF_{(x,y)}\|_{\mathrm{op}} \leq \sqrt{2}$ for all $(x,y) \in \mathbb{R}^2$. Since $\mathbb{R}^2$ is convex, Remark 11.a.4 implies that F is $\sqrt{2}$-Lipschitz continuous on $\mathbb{R}^2$.



scalar f: some ξ has tangent parallel to the chord **vector-valued** γ: $\gamma(2\pi) - \gamma(0) = 0$, yet γ' never vanishes, so no ξ exists

Exercise 11.a.6 (*Differential of $A \mapsto A^{\mathsf{T}}A$*): We identify the inner product space of real-valued matrices $\mathbb{R}^{n\times n}$ with inner product

$$\langle A, B\rangle = \sum_{i,j=1}^{n} A_{ij}B_{ij} = \mathrm{Tr}(A^{\mathsf{T}}B)$$

for $A, B \in \mathbb{R}^{n\times n}$, with the euclidean space $\mathbb{R}^{n^2} \cong \mathbb{R}^{n\times n}$, with the standard inner product. Then, consider the function

$$F : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}, \qquad A \mapsto A^{\mathsf{T}}A.$$

Compute the differential at the identity matrix $\mathrm{id} \in \mathbb{R}^{n\times n}$ applied to $X \in \mathbb{R}^{n\times n}$:

$$DF_{\mathrm{id}}(X) = \partial_X F(\mathrm{id}).$$

Exercise 11.a.6:

We use $(*)$:

$$\begin{aligned}
DF_{\mathrm{id}}(X) &= \frac{d}{ds}\bigg|_{s=0} F(\mathrm{id} + sX) \\
&= \frac{d}{ds}\,(\mathrm{id}+sX)^{\mathsf{T}}(\mathrm{id}+sX)\bigg|_{s=0} \\
&= \frac{d}{ds}\left[\mathrm{id} + s(X^{\mathsf{T}} + X) + s^2 X^{\mathsf{T}}X\right]\bigg|_{s=0} \\
&= X^{\mathsf{T}} + X
\end{aligned}$$

Q.E.D.

Exercise 11.a.7 (*4.5 — Mean value for vector-valued functions* (optional)): Let $f \in C^1(\mathbb{R}, \mathbb{R}^m)$ for $m > 1$. Is it true that there is $t \in [0,1]$ such that

$$f(1) - f(0) = Df_t(1) = \begin{pmatrix} f_1'(t) \\ \vdots \\ f_m'(t) \end{pmatrix}?$$

Prove it or provide a counterexample.

Official hint: try $f(t) = (\sin(2\pi t), \cos(2\pi t))$.

Taylor expansions (Section 11.b)

The general formula (Subsection 11.b.1)

In the lecture, you have seen the following formula. For $f \in C^k(U \subseteq \mathbb{R}^n, \mathbb{R})$, $\bar{x} \in U$:

$$f(\bar{x}+h) = \sum_{|\alpha|=0}^{k} \partial^\alpha f(\bar{x}) \frac{h^\alpha}{\alpha!} + o(|h|^k)$$

where α is a "*multi-index*" (German: „*Multiindex*") in $\mathbb{N}^n$.

In practice, we never use this formula directly, but it is worth unpacking once (e.g., as in Exercise 11.b.11).

Before unpacking the formula, one hypothesis hidden in the notation deserves to be made explicit. Writing ∂^α for a multi-index α presumes that it does not matter **in which order** the derivatives are taken, since otherwise $\partial^{(1,1)}$ would be ambiguous between $\partial_1\partial_2$ and $\partial_2\partial_1$. That the presumption is legitimate is itself a theorem, and it is stated next.

Theorem 11.b.8 (*Schwarz / Clairaut — mixed partial derivatives commute*): Let $U \subseteq \mathbb{R}^n$ be open and $f \in C^2(U, \mathbb{R})$. Then for all $i, j \in \{1, \dots, n\}$ and all $x \in U$,

$$\partial_i \partial_j f(x) = \partial_j \partial_i f(x).$$

More generally, for $f \in C^k(U, \mathbb{R})$ any k partial derivatives may be permuted freely, so $\partial^\alpha f$ depends only on the multi-index α and not on the order of differentiation. In particular the Hessian $\operatorname{Hess} f(x)$ of Remark 11.b.21 is a **symmetric** matrix.

Remark 11.b.9 (*The hypothesis is not decoration*): Continuity of the second derivatives is genuinely needed. The standard counterexample is

$$f(x,y) := \begin{cases} xy\,\dfrac{x^2-y^2}{x^2+y^2} & (x,y) \neq (0,0), \\ 0 & (x,y) = (0,0), \end{cases}$$

for which a direct computation from the definition gives $\partial_1\partial_2 f(0,0) = 1$ but $\partial_2\partial_1 f(0,0) = -1$. This f has both mixed partials everywhere, but they are not continuous at the origin, so Theorem 11.b.8 does not apply, and indeed fails. Since everything in Section 11.b assumes $f \in C^k$, the issue never arises here. It is worth knowing that it **could**.

Exercise 11.b.10 (*Applying Schwarz's theorem*): Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) := x^2 e^{y^2} + e^{-x^2} \sin\left(e^{x^2}\left(x^5 + 3x^2 + 100x + \frac{1}{2+\sin(x)}\right)\right).$$

(a) Show that $f \in C^2(\mathbb{R}^2, \mathbb{R})$.

(b) Compute $\partial_2\partial_1 f(x,y)$. (Make it as easy as possible for yourself.)

Exercise 11.b.10:

(a) Every term in f is a composition of C^2 functions (polynomials, exponentials, sine), and the denominator $2+\sin(x)$ is bounded away from zero. Thus f is C^2.

(b) Since $f \in C^2$, by Schwarz's theorem (Theorem 11.b.8) we have $\partial_2\partial_1 f = \partial_1\partial_2 f$. Computing ∂_1 of the second term would be tedious. However, notice that the second term depends only on x, so its derivative with respect to y is 0. Thus, we first compute $\partial_2 f$:

$$\partial_2 f(x,y) = x^2(2ye^{y^2}) + 0 = 2x^2 y e^{y^2}.$$

Then we compute ∂_1 of this result:

$$\partial_1\partial_2 f(x,y) = \partial_1(2x^2 y e^{y^2}) = 4xye^{y^2}.$$

By Schwarz's theorem, $\partial_2\partial_1 f(x,y) = 4xye^{y^2}$. **Q.E.D.**

Exercise 11.b.11 (*Maximal Taylor expansion in $\mathbb{R}^2$*): Let $f \in C^3(\mathbb{R}^2, \mathbb{R})$. Write out the maximal Taylor approximation.

Exercise 11.b.11:

The multi-indices are:

- $|\alpha| = 0$: $\alpha = (0,0)$
- $|\beta| = 1$: $\beta_1 = (1,0)$, $\beta_2 = (0,1)$
- $|\gamma| = 2$: $\gamma_1 = (1,1)$, $\gamma_2 = (2,0)$, $\gamma_3 = (0,2)$
- $|\delta| = 3$: $\delta_1 = (3,0)$, $\delta_2 = (2,1)$, $\delta_3 = (1,2)$, $\delta_4 = (0,3)$

Recall that

$$\partial^\alpha = \partial_1^{\alpha_1}\partial_2^{\alpha_2}\cdots\partial_n^{\alpha_n}, \qquad \text{e.g. } \partial^{\delta_2} = \partial^{(2,1)} = \partial_1^2\partial_2,$$

and

$$\alpha! = \alpha_1!\cdots\alpha_n!, \qquad \text{e.g. } \gamma_2! = (2,0)! = 2\cdot 1 = 2.$$

So:

$$\begin{aligned} f(\bar{x}+h) = f(\bar{x}) & & (\alpha) \\ &+ \partial_1 f(\bar{x})h_1 + \partial_2 f(\bar{x})h_2 & (\beta_1,\beta_2) \\ &+ \partial_1\partial_2 f(\bar{x})h_1h_2 + \partial_1^2 f(\bar{x})\frac{h_1^2}{2} + \partial_2^2 f(\bar{x})\frac{h_2^2}{2} & (\gamma_1,\gamma_2,\gamma_3) \\ &+ \partial_1^3 f(\bar{x})\frac{h_1^3}{6} + \partial_1^2\partial_2 f(\bar{x})\frac{h_1^2h_2}{2} & (\delta_1,\delta_2) \\ &+ \partial_1\partial_2^2 f(\bar{x})\frac{h_1h_2^2}{2} + \partial_2^3 f(\bar{x})\frac{h_2^3}{6} & (\delta_3,\delta_4) \\ &+ o(|h|^3) \end{aligned}$$

Q.E.D.

AI-Note 11.2: Corsin writes the recall line as $\partial^\alpha = \partial_{\alpha_1}\partial_{\alpha_2}\cdots\partial_{\alpha_n}$ with the example $\partial^{(2,1)} = \partial_2\partial_1$. Read literally that is the wrong object, since a multi-index α records *how many times* each ∂_i is applied, so $\partial^{(2,1)} = \partial_1^2\partial_2$. His own expansion above uses $\partial_1^2\partial_2 f(\bar{x})\,\frac{h_1^2h_2}{2}$ for δ_2, confirming the intended meaning. Corrected above.

The working formula (Subsection 11.b.2)

We can rewrite this as

$$f(\bar{x}+h) = \sum_{\ell=0}^{k} \frac{1}{\ell!}\big(h_1\partial_1 + h_2\partial_2\big)^\ell f(\bar{x}).$$

Theorem 11.b.12 (*Taylor working formula*): For $f \in C^k(U \subseteq \mathbb{R}^n, \mathbb{R})$ with $\bar{x} + th \in U$ for all $t \in [0,1]$:

$$f(\bar{x}+h) = \sum_{\ell=0}^{k} \frac{1}{\ell!}\left(\sum_{m=1}^{n} h_m\partial_m\right)^\ell f(\bar{x}) + o(|h|^k)$$

AI-Note 11.3: Both sums in Theorem 11.b.12 are written with lower limit $\ell = 1$ (resp. starting index) in the source. The $\ell = 0$ term is $f(\bar{x})$ itself, which the explicit expansion above does include, so the lower limit must be 0. Corrected above; compare the $|\alpha| = 0$ line of Exercise 11.b.11, which is the same term written in multi-index form.

Theorem 11.b.13 (*Taylor's theorem with Lagrange remainder*): Let $U \subseteq \mathbb{R}^n$ be open, $f \in C^{k+1}(U, \mathbb{R})$, $\bar{x} \in U$, and $h \in \mathbb{R}^n$ with $\bar{x} + th \in U$ for all $t \in [0,1]$. Then there exists $\theta \in (0,1)$

such that

$$f(\bar{x}+h) = \sum_{\ell=0}^{k} \frac{1}{\ell!}\Big(\sum_{m=1}^{n} h_m\partial_m\Big)^{\ell} f(\bar{x}) \;+\; \frac{1}{(k+1)!}\Big(\sum_{m=1}^{n} h_m\partial_m\Big)^{k+1} f(\bar{x}+\theta h).$$

Equivalently, in multi-index notation,

$$f(\bar{x}+h) = \sum_{|\alpha|\le k} \partial^\alpha f(\bar{x})\,\frac{h^\alpha}{\alpha!} \;+\; \sum_{|\alpha|=k+1} \partial^\alpha f(\bar{x}+\theta h)\,\frac{h^\alpha}{\alpha!}.$$

Proof:

Let $\varphi(t) := f(\bar{x}+th)$ for $t\in[0,1]$, well defined and C^{k+1} since $f\in C^{k+1}$ and the segment $\bar{x}+th$ stays in U. By induction on ℓ via the chain rule, which is exactly the computation underlying Theorem 11.b.12 itself,

$$\varphi^{(\ell)}(t) = \Big(\sum_{m=1}^{n} h_m\partial_m\Big)^{\ell} f(\bar{x}+th) \qquad \text{for } 0\le \ell\le k+1.$$

Apply the one-variable Taylor theorem with Lagrange remainder (**Analysis I**) to φ on $[0,1]$: there is $\theta\in(0,1)$ with

$$\varphi(1) = \sum_{\ell=0}^{k} \frac{\varphi^{(\ell)}(0)}{\ell!} + \frac{\varphi^{(k+1)}(\theta)}{(k+1)!}.$$

Substituting $\varphi(1)=f(\bar{x}+h)$, $\varphi^{(\ell)}(0) = \big(\sum_m h_m\partial_m\big)^{\ell} f(\bar{x})$, and $\varphi^{(k+1)}(\theta) = \big(\sum_m h_m\partial_m\big)^{k+1} f(\bar{x}+\theta h)$ gives the working-formula version. The multi-index version follows by expanding $\big(\sum_m h_m\partial_m\big)^{\ell}$ via the multinomial theorem, exactly as in Exercise 11.b.11. **Q.E.D.**

Remark 11.b.14: Unlike Theorem 11.b.12's $o(|h|^k)$ remainder, this is an **exact** equality for the specific h in question, not merely a statement about what happens as $h\to 0$. That is genuinely different information, and it is what is needed whenever a hard, uniform error bound is wanted rather than an asymptotic one (e.g. bounding $|f(\bar{x}+h)-T_k(h)|$ over an entire region at once, given a bound on the $(k{+}1)$-th derivatives there). The price is that θ depends on $\bar{x}$, h **and** k together, and is almost never computable in practice. That is exactly why the rest of this chapter uses the asymptotic form for actually computing expansions, and reserves this one for error estimates.

Corollary 11.b.15 (*A polynomial that fits to order k **is the Taylor polynomial*):** Let $U\subseteq\mathbb{R}^n$ be open, $x_0\in U$, $f\in C^{k+1}(U)$, and let P be a polynomial of degree at most k with

$$|f(x_0+h)-P(h)| = o\big(|h|^k\big) \qquad \text{as } h\to 0.$$

Then $\partial^\alpha f(x_0) = \partial^\alpha P(0)$ for every multi-index with $|\alpha|\le k$. Equivalently, P is **the** Taylor polynomial of f of degree k at x_0.

Proof:

By Theorem 11.b.12, the Taylor polynomial T of degree k also satisfies $|f(x_0+h)-T(h)| = o(|h|^k)$. Subtracting, the polynomial $Q := P-T$ has degree at most k and satisfies $|Q(h)| = o(|h|^k)$. Suppose $Q\not\equiv 0$ and let $d\le k$ be the degree of its lowest non-vanishing homogeneous part Q_d. Choosing v with $Q_d(v)\ne 0$ and restricting to the ray $h:=tv$,

$$Q(tv) = t^d Q_d(v) + O(t^{d+1}), \qquad \text{so} \qquad \frac{|Q(tv)|}{t^k} \sim |Q_d(v)|\,t^{d-k} \not\to 0$$

as $t\downarrow 0$, since $d\le k$. That contradicts $|Q(h)| = o(|h|^k)$, so $Q\equiv 0$ and $P=T$. **Q.E.D.**

Remark 11.b.16 (*Reading derivatives off an asymptotic expansion*): This is more useful than it looks, because it runs **backwards**. Normally one computes derivatives in order to produce an expansion;

the corollary says that an expansion, however it was obtained, hands back the derivatives. Given

$$f(x_0+h) = c + a^\mathsf{T} h + \tfrac{1}{2} h^\mathsf{T} B h + o(|h|^2)$$

one may read off $f(x_0) = c$, $\nabla f(x_0) = a$ and $\operatorname{Hess} f(x_0) = B$ without differentiating anything. This is the same move by which Definition 9.a.1 recognises a differential from an $o(|h|)$ estimate, now carried to higher order. It is the standard route to a Hessian for functions built out of known series.

Two cautions on the bookkeeping, both of which the exercise below turns on. The estimate must be $o(|h|^k)$, not $O(|h|^k)$: an $O(|h|^k)$ error can contain a genuine degree-k term and so corrupts the order-k coefficients, leaving only the derivatives of order $< k$ determined. And the expansion determines derivatives only up to order k, and nothing beyond it.

Exercise 11.b.17 (*What does an expansion actually tell you?*): For each of the following, state precisely which values of f and of its derivatives at the indicated point are determined by the given information, and which are not.

(a) $f : \mathbb{R}^3 \to \mathbb{R}$ with $f(x_1,x_2,x_3) - x_1x_2 \in O\big(|x|\big)$, at $x = 0$.
(b) $f : \mathbb{R}^2 \to \mathbb{R}$ with $f(1+x_1,\, -1+x_2) - 2 - x_1 \in o\big(|x|\big)$, at $(1,-1)$.
(c) $f : \mathbb{R}^2 \to \mathbb{R}$ with $f(-1+x_1,\, x_2) + 2 - 2x_1x_2 + 2x_2^2 - 3x_2^3 \in o\big(|x|^3\big)$, at $(-1,0)$.

Assume in each case that f is as smooth as the relevant statement requires.

Remark 11.b.18 (*Two ways to weight the same expansion*): Exercise 11.b.11 above and Theorem 11.b.12 appear to disagree on how heavily mixed partial derivatives are weighted. Take the coefficient of $h_1^2 h_2$, so $\alpha = (2,1)$ and $\ell = |\alpha| = 3$. The explicit multi-index expansion gives $\frac{1}{2}$ directly. The working formula instead offers $\frac{1}{\ell!}(h_1\partial_1 + h_2\partial_2)^\ell$, and expanding that by the multinomial theorem produces

$$\frac{\ell!}{\alpha!}\cdot\frac{1}{\ell!} = \frac{1}{\alpha!} = \frac{1}{2!\,1!} = \frac{1}{2}$$

for the term $h^\alpha \partial^\alpha f$. The two agree, but they arrive by different routes, and seeing why is worth the paragraph.

The working formula's $\frac{1}{\ell!}$ weights **all** ℓ-fold products of partials globally, including repeated ones such as $\partial_1\partial_2$, which appears in more than one order in the multinomial expansion of $(h_1\partial_1 + h_2\partial_2)^\ell$. By Schwarz's lemma (equality of mixed partial derivatives, valid here because $f \in C^k$) all those repeats collapse onto the same ∂^α, and there are exactly $\ell!/\alpha!$ of them. That collapsing is what turns the global $1/\ell!$ into the per-term $1/\alpha!$ seen in Exercise 11.b.11.

In short, $1/\ell!$ weights **paths** through the mixed partial derivatives, whereas $1/\alpha!$ weights the **distinct** derivatives themselves, once those paths have been identified with one another.

A worked example (Subsection 11.b.3)

Exercise 11.b.19 (*Taylor expansion of $(x+y^2)e^{-x^2-y^2}$*): Compute the Taylor expansion of

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad (x,y) \mapsto (x+y^2)e^{-x^2-y^2}$$

around $(0,0)$ up to third order.

Exercise 11.b.19, cumbersome solution:

Calculate all partial derivatives up to third order and use the definition. (Please never do this.)

Q.E.D.

Exercise 11.b.19, the easy solution:

The substitution and multiplication rules we used in Analysis 1 still apply:

$$\begin{aligned}(x+y^2)e^{-x^2-y^2} &= (x+y^2)\left(1-x^2-y^2+o(|x^2+y^2|)\right) \\ &= x+y^2-x^3-xy^2+o\big(|(x,y)|^3\big)\end{aligned}$$

Q.E.D.

Example 11.b.20 (*More Taylor expansions via Analysis 1*): The substitution and multiplication rules from Analysis 1 often make expansions much faster than using the formula directly, provided the expansion is around $(0,0)$.

(a) Expand $f(x,y) := \cos(x)e^y$ up to second order around $(0,0)$:

$$f(x,y) = \left(1-\frac{x^2}{2}+o(|(x,y)|^2)\right)\left(1+y+\frac{y^2}{2}+o(|(x,y)|^2)\right) = 1+y-\frac{1}{2}x^2+\frac{1}{2}y^2+o(|(x,y)|^2).$$

(b) Expand $f(x,y) := \cos(x+y^2)$ up to second order around $(0,0)$. Setting $z := x+y^2$:

$$\cos(z) = 1-\frac{z^2}{2}+o(|z|^2) \implies \cos(x+y^2) = 1-\frac{(x+y^2)^2}{2}+o(|(x,y)|^2) = 1-\frac{1}{2}x^2+o(|(x,y)|^2),$$

where terms like xy^2 and y^4 are absorbed into $o(|(x,y)|^2)$ since their total degree is strictly greater than 2.

(c) Expand $f(x,y) := \sqrt{y}e^{\sqrt{x}}$ to first order around $(1,1)$. Here the formula is better since the expansion point is not the origin. We have $f(1,1) = e$. $\partial_1 f(x,y) = \sqrt{y}e^{\sqrt{x}}\frac{1}{2\sqrt{x}} \implies \partial_1 f(1,1) = \frac{e}{2}$. $\partial_2 f(x,y) = \frac{1}{2\sqrt{y}}e^{\sqrt{x}} \implies \partial_2 f(1,1) = \frac{e}{2}$. Writing $h = (h_1,h_2) = (x-1, y-1)$:

$$f(1+h_1, 1+h_2) = f(1,1)+\partial_1 f(1,1)h_1+\partial_2 f(1,1)h_2+o(|h|) = e+\frac{e}{2}h_1+\frac{e}{2}h_2+o(|h|).$$

Second-order form and the Hessian (Subsection 11.b.4)

Remark 11.b.21 (*Second-order Taylor formula with Hessian*): The second-order Taylor approximation of $f \in C^2(\mathbb{R}^n,\mathbb{R})$ around $x_0 \in \mathbb{R}^n$ can be expressed compactly as

$$f(x_0+x) = f(x_0)+\nabla f(x_0)\cdot x+\tfrac{1}{2}x^\mathsf{T}\cdot \operatorname{Hess} f(x_0)\cdot x+o(|x|^2)$$

where

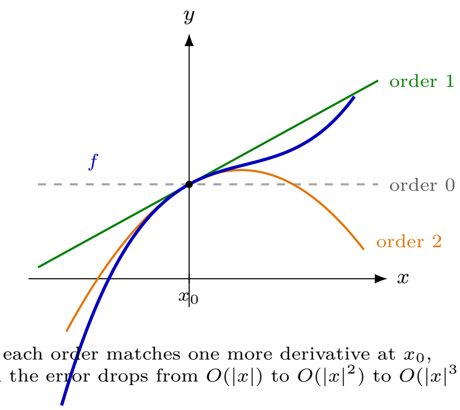
$$\operatorname{Hess} f(x_0) = \big(\partial_i\partial_j f(x_0)\big)_{i,j=1,\dots,n}$$

is the "*Hessian matrix*" (German: „*Hesse-Matrix*") of f.

This is the form the rest of the course actually uses, and it is worth reading term by term as a sequence of ever-better approximations to f near x_0:

- the **constant** term $f(x_0)$, the crudest guess, exact only at x_0;
- the **linear** term $\nabla f(x_0)\cdot x$, which is the tangent plane and hence exactly the differential of Definition 9.a.1;
- the **quadratic** term $\frac{1}{2}x^\mathsf{T}\operatorname{Hess} f(x_0)\,x$, the first term that sees curvature, and the one that decides in Section 12.b whether a critical point is a minimum, a maximum, or a saddle.

By Theorem 11.b.8 the Hessian is symmetric, which is what makes that last decision a question about the **signs of its eigenvalues** rather than anything harder.



each order matches one more derivative at x_0,
and the error drops from $O(|x|)$ to $O(|x|^2)$ to $O(|x|^3)$

In several variables the same picture holds with *"tangent line"* replaced by **tangent plane** and *"parabola"* by a paraboloid, bowl-shaped or saddle-shaped according to the signs of the eigenvalues of Hess $f(x_0)$. That is exactly the classification carried out in Section 12.b.

Question 11.b.22: For $f \in C^k(U,\mathbb{R})$, do the Taylor expansions of $t \mapsto f(x_0+th)$ in t and of $(th) \mapsto f(x_0+th)$ in (th) around 0 coincide?

Answer 11.b.23: **Yes.** This is, in fact, how we prove the multi-dimensional Taylor approximation, and why we need the assumption $x_0+th \in U$ for all $t \in [0,1]$.

Exercise 11.b.24 (*Proof of second-order Taylor formula*): Prove the formula from above:

$$f(x_0+h) = f(x_0) + \nabla f(x_0)h + \tfrac{1}{2}h^\mathsf{T}\,\mathrm{Hess}\, f(x_0)h + \dots$$

for $f \in C^k(U,\mathbb{R})$ with $x_0+th \in U$ for all $t \in [0,1]$.

Proof of Exercise 11.b.24:

We expand $t \mapsto f(x_0+th)$ in t:

$$\begin{aligned}
f(x_0+th) &= f(x_0) + t\,\frac{d}{dt}\Big|_{t=0} f(x_0+th) + \frac{t^2}{2}\,\frac{d^2}{dt^2}\Big|_{t=0} f(x_0+th) + o(|th|^2)\\
&= f(x_0) + t\sum_{i=1}^n \partial_i f(x_0)h_i + \frac{t^2}{2}\frac{d}{dt}\sum_{i=1}^n \partial_i f(x_0+th)h_i\Big|_{t=0} + o(|h|^2)\\
&= f(x_0) + t\nabla f(x_0)h + \frac{t^2}{2}\sum_{i,j=1}^n \partial_i\partial_j f(x_0)h_ih_j + o(|th|^2)\\
&= f(x_0) + t\nabla f(x_0)h + \frac{t^2}{2}h^\mathsf{T}\,\mathrm{Hess}\, f(x_0)h + o(|th|^2)
\end{aligned}$$

Setting $t=1$ gives the result. **Q.E.D.**

Exercise 11.b.25 (*5.3 — Multiple choice (Landau notation)* (important)): Mark all and only the statements which are true.

- **(a)** As $|x| \to 0$, if $f(x) = x_2x_1^2 + O(x_1^2)$, then $f(x) = O(|x|^2)$.
- **(b)** As $x \to 0$, if $f(x) = x_2x_1^2 + O(x_1^3)$, then $f(x) = O(x_1^3)$.
- **(c)** As $|x| \to 0$, if $f(x) = x_2x_1 + O(|x|^2)$, then f is differentiable at $x=0$.
- **(d)** As $x \to 0$, if $f(x) = x_2x_1 + O(|x|^2)$, then f is twice differentiable around $x=0$.
- **(e)** As $|x| \to 0$, if $f(x) = x_2x_1 + O(|x|^3)$, then f is twice differentiable around $x=0$ and $\partial_{11}f(0) = 0$.

Official hint: revise *Corollary 10.45*.

Exercise 11.b.26 (*5.5 — Multiple choice (cost of a Taylor expansion)* (optional)): Assume $g \in C^\infty(\mathbb{R})$ and $f \in C^\infty(\mathbb{R}^n)$ is such that

$$f(x) = x_1 + x_2 x_1 + O(|x|^4) \quad \text{as } x \to 0.$$

We want to compute the Taylor expansion of $(g \circ f)$ at $x = 0$ of the highest possible order with the information we have. We can ask for the value $g^{(k)}(t)$ for arbitrary $k \in \mathbb{N}$, $t \in \mathbb{R}$, but we have to pay 5 CHF each time.

For a general g, which is the degree of the best (i.e. highest-order) Taylor polynomial we can compute? How expensive will it be to compute it?

(a) degree 3 and 20 CHF
(b) degree 2 and 15 CHF
(c) degree 2 and 10 CHF
(d) degree 3 and 15 CHF

Exercise 11.b.27 (*5.6 — Polynomials* (optional — "interesting, but irrelevant for the course")): Let $P, Q \in \mathbb{R}[x_1, \ldots, x_n]$ be polynomials of n variables; assume that there are positive numbers M, σ such that

$$|P(x) - Q(x)| \le M|x|^\sigma \quad \text{for all } |x| \le 1.$$

Show that P and Q have the same coefficients of order smaller than σ. That is, if we write $P(X) = \sum_{\alpha \in \mathbb{N}^n} a_\alpha X^\alpha$, $Q(X) = \sum_{\alpha \in \mathbb{N}^n} b_\alpha X^\alpha$, then $a_\alpha = b_\alpha$ for all $|\alpha| < \sigma$.

Exercise 11.b.28 (*Evaluating partial derivatives using multi-index notation*): Let $f(x, y, z) := x^3 + 2y + e^{2x}\cos(z)$. Calculate the partial derivative $\partial^\alpha f$ for the multi-index $\alpha = (2, 0, 1)$.

Exercise 11.b.28:

The multi-index $\alpha = (2, 0, 1)$ corresponds to differentiating twice with respect to x, zero times with respect to y, and once with respect to z. Since f is smooth, the order of differentiation does not matter.

$$\begin{aligned}\partial_x f &= 3x^2 + 2e^{2x}\cos(z)\\ \partial_{xx} f &= 6x + 4e^{2x}\cos(z)\\ \partial_{xxz} f &= -4e^{2x}\sin(z).\end{aligned}$$

Hence $\partial^{(2,0,1)} f(x, y, z) = -4e^{2x}\sin(z)$. **Q.E.D.**

Exercise 11.b.29 (*Taylor expansion of* $\sin(xy)$): Compute the Taylor expansion of $f(x, y) := \sin(xy)$ to order 3 around $(0, 0)$.

Exercise 11.b.29:

We can solve this in two ways:

(a) **Using the general formula:** We list the multi-indices α with $|\alpha| \le 3$ and compute the partial derivatives at $(0, 0)$:
- $|\alpha| = 0$: $f(0, 0) = \sin(0) = 0$.
- $|\alpha| = 1$: $\partial_x f = y\cos(xy) \implies 0$. $\partial_y f = x\cos(xy) \implies 0$.
- $|\alpha| = 2$: $\partial_{xx} f = -y^2\sin(xy) \implies 0$. $\partial_{yy} f = -x^2\sin(xy) \implies 0$. $\partial_{xy} f = \cos(xy) - xy\sin(xy) \implies 1$.
- $|\alpha| = 3$: $\partial_{xxx} f = -y^3\cos(xy) \implies 0$. $\partial_{yyy} f = -x^3\cos(xy) \implies 0$. $\partial_{xxy} f = -2y\sin(xy) - xy^2\cos(xy) \implies 0$. $\partial_{xyy} f = -2x\sin(xy) - x^2y\cos(xy) \implies 0$.

The only non-zero term is $\partial^{(1,1)} f(0, 0) = 1$. The corresponding factor in the formula is $\frac{1}{\alpha!}h^\alpha = \frac{1}{(1!)(1!)}xy = xy$. Thus,

$$f(x, y) = xy + o(|(x, y)|^3).$$

(b) Using 1D expansions (the clever way): We know that $\sin(t) = t - \frac{t^3}{6} + O(t^5)$ as $t \to 0$. Substituting $t = xy$ yields

$$\sin(xy) = xy - \frac{(xy)^3}{6} + O((xy)^5) = xy + o(|(x,y)|^3).$$

Q.E.D.

Exercise 11.b.30 (*Taylor expansion of* $\sqrt{1+x+y^2}$)**:** Compute the Taylor expansion of $f(x,y) := \sqrt{1+x+y^2}$ to order 2 around $(0,0)$.

Exercise 11.b.30:

Using the 1D expansion $\sqrt{1+t} = 1 + \frac{1}{2}t - \frac{1}{8}t^2 + O(t^3)$ with $t = x + y^2$, we obtain:

$$\begin{aligned}\sqrt{1+x+y^2} &= 1 + \frac{1}{2}(x+y^2) - \frac{1}{8}(x+y^2)^2 + O(|(x,y)|^3)\\ &= 1 + \frac{1}{2}x + \frac{1}{2}y^2 - \frac{1}{8}(x^2 + 2xy^2 + y^4) + o(|(x,y)|^2)\\ &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{2}y^2 + o(|(x,y)|^2).\end{aligned}$$

Q.E.D.

Exercise 11.b.31 (*Taylor expansion of* $\exp(\arctan(x-y))$)**:** Compute the Taylor expansion of $f(x,y) := \exp(\arctan(x-y))$ to order 2 around $(0,0)$.

Exercise 11.b.31:

Write $s := x - y$, itself of order $|h|$. Since arctan is odd, $\arctan(t) = t + O(t^3)$, so

$$\arctan(s) = s + O(|h|^3) = s + o(|h|^2).$$

Since exp is smooth and bounded near 0, an $o(|h|^2)$ correction inside exp contributes only an $o(|h|^2)$ correction to the output. Consequently, to order 2, $\exp(\arctan(s))$ and $\exp(s)$ agree:

$$\exp(\arctan(x-y)) = \exp(x-y) + o(|(x,y)|^2).$$

Now $\exp(s) = 1 + s + \frac{1}{2}s^2 + O(s^3)$ with $s^2 = x^2 - 2xy + y^2$, so

$$\exp(\arctan(x-y)) = 1 + x - y + \tfrac{1}{2}x^2 - xy + \tfrac{1}{2}y^2 + o(|(x,y)|^2).$$

Q.E.D.

Exercise 11.b.32 (*Taylor expansion of* $1/(1-x^2-y^2)$)**:** Compute the Taylor expansion of $f(x,y) := \dfrac{1}{1-x^2-y^2}$ to order 2 around $(0,0)$.

Exercise 11.b.32:

Using $\frac{1}{1-t} = 1 + t + O(t^2)$ with $t := x^2 + y^2$, itself already $O(|h|^2)$, so $t^2 = O(|h|^4) = o(|h|^2)$:

$$\frac{1}{1-x^2-y^2} = 1 + (x^2+y^2) + o(|(x,y)|^2).$$

Q.E.D.

Remark 11.b.33: Together with Exercises 11.b.29 and 11.b.30 above, these two exercises are exactly script Exercise 10.36: *"compute the Taylor polynomials up to the quadratic order of* $\sin(xy)$, $\sqrt{1+x+y^2}$, $\exp(\arctan(x-y))$, $1/(1-x^2-y^2)$ *— don't use the general formula!"* All four are worked by the same *"1D substitution"* technique this section uses throughout.

Exercise 11.b.34 (*Taylor expansion of $\frac{\cos x}{1-y^2}$*): Compute the Taylor expansion of $f(x,y) := \frac{\cos x}{1-y^2}$ to order 4 around $(0,0)$.

Exercise 11.b.34:

We multiply the 1D expansions for $\cos x$ and $\frac{1}{1-y^2}$. As $x \to 0$, $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)$. As $y \to 0$, $\frac{1}{1-y^2} = 1 + y^2 + y^4 + O(y^6)$. Multiplying these together and keeping terms up to order 4:

$$\begin{aligned}\frac{\cos x}{1-y^2} &= \left(1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)\right)\left(1 + y^2 + y^4 + O(y^6)\right)\\ &= 1 + y^2 + y^4 - \frac{x^2}{2} - \frac{x^2y^2}{2} + \frac{x^4}{24} + o(|(x,y)|^4)\\ &= 1 - \frac{1}{2}x^2 + y^2 + \frac{1}{24}x^4 - \frac{1}{2}x^2y^2 + y^4 + o(|(x,y)|^4).\end{aligned}$$

Q.E.D.

Analytic functions (Section 11.c)

Just as in one variable, we can ask when a smooth function is exactly equal to its Taylor series. This requires a bound on the growth of its derivatives.

Definition 11.c.35 (*Analytic estimates*): Let $U \subseteq \mathbb{R}^n$ be open. A smooth function $f \in C^\infty(U,\mathbb{R})$ satisfies "*analytic estimates*" if for every $x_0 \in U$, there exist $\rho > 0$ and $C > 0$ such that for all multi-indices $\alpha \in \mathbb{N}^n$,

$$\max_{x \in B_\rho(x_0)} |\partial^\alpha f(x)| \le C|\alpha|!\,(n\rho)^{-|\alpha|}.$$

Such functions are called "*real analytic*".

Theorem 11.c.36 (*Convergence of the Taylor series*): If $f \in C^\infty(U,\mathbb{R})$ satisfies analytic estimates, then its Taylor series converges to $f(x)$ near x_0. That is, for all $h \in B_\rho(0)$,

$$f(x_0 + h) = \sum_{k=0}^{\infty} \sum_{|\alpha|=k} \frac{\partial^\alpha f(x_0)}{\alpha!} h^\alpha.$$

Example 11.c.37 (*An analytic function*): The function $f(x,y) = \sin(xy)$ satisfies the analytic estimates everywhere, and is therefore equal to its Taylor series.

Corollary 11.c.38 (*Unique continuation principle*): Let $U \subseteq \mathbb{R}^n$ be connected and open, and let f, g be real analytic on U (Definition 11.c.35). If $\partial^\alpha f(x_0) = \partial^\alpha g(x_0)$ for every multi-index α at some single point $x_0 \in U$, then $f = g$ on all of U. In particular, if f and g agree on a non-empty open subset $V \subseteq U$, they must agree on all of U.

Proof:

Let $U' := \{x \in U : \partial^\alpha f(x) = \partial^\alpha g(x) \text{ for all } \alpha \in \mathbb{N}^n\}$; by hypothesis $x_0 \in U'$, so U' is non-empty. Each $\partial^\alpha(f-g)$ is continuous, so

$$U \setminus U' = \bigcup_{\alpha \in \mathbb{N}^n} \big(\partial^\alpha(f-g)\big)^{-1}\big((-\infty,0) \cup (0,\infty)\big)$$

is a union of open sets, hence open. To see U' is also open, let $x \in U'$; since f and g are analytic at x, Theorem 11.c.36 gives $\rho > 0$ (the smaller of the two radii for f and g at x) such that both are equal to their Taylor series on $B_\rho(x)$; but those Taylor series agree term by term, since $x \in U'$, so $f = g$ on all of $B_\rho(x)$, giving $\partial^\alpha f = \partial^\alpha g$ throughout $B_\rho(x)$ (differentiate the identity $f \equiv g$ there) and hence $B_\rho(x) \subseteq U'$.

So, U' and $U \setminus U'$ are both open, U' is non-empty, and U is connected. Connectedness therefore forces $U \setminus U' = \emptyset$, that is, $U' = U$. In particular $f = g$ on all of U (the case $\alpha = 0$).

For the last statement, if $f = g$ on a non-empty open $V \subseteq U$, pick any $x_0 \in V$: then $\partial^\alpha f(x_0) = \partial^\alpha g(x_0)$ for every α (differentiating the identity $f \equiv g$ near x_0), so the first statement applies.

Q.E.D.

Remark 11.c.39: This is the payoff that makes real-analyticity worth having beyond C^∞: an analytic function is determined, on its **entire** connected domain, by its behaviour arbitrarily close to a single point. Knowing every derivative at one x_0, or merely agreeing with another analytic function on a tiny open patch, pins the function down everywhere. This is dramatically false for merely C^∞ functions: the standard bump function $\varphi(x) := e^{-1/x^2}$ for $x > 0$, $\varphi(x) := 0$ for $x \le 0$, has $\partial^\alpha \varphi(0) = 0$ for every α, agreeing to all orders with the zero function at 0, yet $\varphi \not\equiv 0$ on any neighbourhood of 0. That is exactly the failure Corollary 11.c.38 rules out for analytic functions.

Solutions (Section 11.d)

Solution to Exercise 11.b.17:

Throughout, the tool is Corollary 11.b.15: an $o(|h|^k)$ fit by a polynomial of degree $\le k$ pins down every derivative of order $\le k$, and nothing beyond.

(a) **Only $f(0) = 0$.** The hypothesis is an O, not an o, so the corollary does not apply at any order, and that is the entire point of the item. Setting $x = 0$ in $|f(x) - x_1x_2| \le C|x|$ gives $|f(0)| \le 0$, so $f(0) = 0$. Nothing else is determined: both

$$f(x) := x_1x_2 \qquad \text{and} \qquad g(x) := x_1x_2 + 5x_1$$

satisfy the hypothesis, yet $\nabla f(0) = (0, 0)$ while $\nabla g(0) = (5, 0)$. An $O(|x|)$ remainder is free to hide a whole linear term, so not even the first derivatives survive.

Note also that the visible x_1x_2 is a red herring: it is $O(|x|^2)$, hence already absorbed by the stated error, and contributes nothing.

(b) **Value and first derivatives; nothing further.** Put $x_0 := (1, -1)$ and $h := (x_1, x_2)$. The hypothesis reads $f(x_0 + h) = 2 + h_1 + o(|h|)$, and $P(h) := 2 + h_1$ has degree 1, so by Corollary 11.b.15 with $k = 1$ it **is** the first-order Taylor polynomial:

$$f(1, -1) = 2, \qquad \partial_1 f(1, -1) = 1, \qquad \partial_2 f(1, -1) = 0.$$

Equivalently $\nabla f(1, -1) = (1, 0)^\mathsf{T}$. Second derivatives are not determined: adding any quadratic to f preserves the hypothesis.

(c) **Everything up to third order.** Put $x_0 := (-1, 0)$. Rearranging,

$$f(x_0 + h) = \underbrace{-2 + 2h_1h_2 - 2h_2^2 + 3h_2^3}_{=:\,P(h)} + o\big(|h|^3\big),$$

with $\deg P = 3$, so P is the third-order Taylor polynomial and every derivative of order ≤ 3 can be read off.

Order 0 and 1. $f(-1, 0) = -2$, and P has no linear part, so $\nabla f(-1, 0) = (0, 0)^\mathsf{T}$, so the point is critical.

Order 2. The quadratic part of a Taylor expansion is $\frac{1}{2} h^\mathsf{T} \operatorname{Hess} f(x_0) h$, so $\frac{1}{2} h^\mathsf{T} \operatorname{Hess} f(-1, 0)\, h = 2h_1h_2 - 2h_2^2$. Comparing coefficients in $h^\mathsf{T} H h = H_{11}h_1^2 + 2H_{12}h_1h_2 + H_{22}h_2^2 = 4h_1h_2 - 4h_2^2$ gives

$$\operatorname{Hess} f(-1, 0) = \begin{pmatrix} 0 & 2 \\ 2 & -4 \end{pmatrix}.$$

Its determinant is $-4 < 0$, so the Hessian is indefinite and, by Theorem 12.b.10, the critical point $(-1, 0)$ is a **saddle**. Note that this was settled without ever differentiating f.

Order 3. A third-order term is $\partial^\alpha f(x_0)\, h^\alpha/\alpha!$. The only cubic monomial present is h_2^3, with $\alpha = (0,3)$ and $\alpha! = 6$, so $\partial_2^3 f(-1,0)/6 = 3$, giving $\partial_2^3 f(-1,0) = 18$; every other third derivative vanishes. Fourth-order derivatives are not determined.

Q.E.D.

Solution to Exercise 11.a.7:

No, and the official hint's function is the cleanest witness. Take

$$f(t) := \big(\sin(2\pi t),\ \cos(2\pi t)\big), \qquad f \in C^1(\mathbb{R}, \mathbb{R}^2).$$

Then $f(1) = (0,1) = f(0)$, so the left-hand side is

$$f(1) - f(0) = (0,0).$$

On the other hand $f'(t) = 2\pi\big(\cos(2\pi t),\ -\sin(2\pi t)\big)$, whose norm is

$$\|f'(t)\| = 2\pi\sqrt{\cos^2(2\pi t) + \sin^2(2\pi t)} = 2\pi \neq 0 \qquad \text{for every } t.$$

So $f'(t)$ never equals $(0,0)$, and no $t \in [0,1]$ can satisfy the claimed identity.

The reason is the one given in Important remark 11.a.2: applying the scalar mean value theorem componentwise produces one intermediate point **per component**,

$$f_1(1) - f_1(0) = f_1'(\xi_1), \qquad f_2(1) - f_2(0) = f_2'(\xi_2),$$

and nothing forces $\xi_1 = \xi_2$. Here $\xi_1 = \frac{1}{4}$ and $\xi_2 = \frac{1}{2}$ work componentwise. Both components do vanish, just not at the same time.

AI-Note 11.4: Note what does **not** fail. The **mean value inequality** $\|f(1) - f(0)\| \leq \sup_{t\in[0,1]}\|f'(t)\|$ holds here as $0 \leq 2\pi$, comfortably. That inequality is what survives into $\mathbb{R}^m$ and what every later estimate in this document actually uses; the **equality** is a one-dimensional accident. Geometrically the counterexample is a walker going once round a circle: they return to the start, so the total displacement is zero, but they never once stand still.

Q.E.D.

Solution to Exercise 11.b.25:

Throughout, recall the hint: *as* $x \to 0$ and *as* $|x| \to 0$ describe the same limit, so both phrasings are treated identically below. The recurring subtlety is instead between a statement holding **at** the single point $x = 0$ (which a remainder estimate as $x \to 0$ can establish) versus **around**/in a whole neighbourhood of $x = 0$ (which it cannot).

- **(a) True.** Since $|x_2 x_1^2| \leq |x| \cdot |x_1|^2 \leq |x|^3 = O(|x|^2)$ (as $|x|^3/|x|^2 = |x| \to 0$) and $O(x_1^2) = O(|x|^2)$ (since $x_1^2 \leq |x|^2$), the sum $f(x) = x_2 x_1^2 + O(x_1^2)$ is a sum of two $O(|x|^2)$ terms, hence $f(x) = O(|x|^2)$.
- **(b) False.** It suffices to show $x_2 x_1^2$ itself is not $O(x_1^3)$. Along the path $x_1 = t^2$, $x_2 = t$ (so $x \to 0$ as $t \to 0$), $\dfrac{|x_2 x_1^2|}{|x_1|^3} = \dfrac{t \cdot t^4}{t^6} = \dfrac{1}{t} \to \infty$, so no bound $|x_2 x_1^2| \leq C|x_1|^3$ can hold near 0. Adding an $O(x_1^3)$ term cannot repair this, so $f(x) \neq O(x_1^3)$ in general.
- **(c) True.** Since $x_2 x_1 = O(|x|^2)$ and the hypothesis adds a further $O(|x|^2)$ term, $f(x) = O(|x|^2) = o(|x|)$ as $|x| \to 0$. Hence $f(x) = 0 \cdot x + o(|x|)$, i.e. f is differentiable **at** $x = 0$ with $Df(0) = 0$.
- **(d) False.** The hypothesis only constrains the behaviour of f as $x \to 0$, i.e. at the single point 0; it says nothing about f at nearby points $x_0 \neq 0$, so twice differentiability **around** (in a neighbourhood of) $x = 0$ does not follow. Concretely, take $f(x) := \max\{0, x_1\}^2$: this satisfies $f(x) = x_2 x_1 + O(|x|^2)$ as $x \to 0$ (indeed even $f(x) = O(|x|^2)$, since $x_2 x_1$ itself is $O(|x|^2)$ and so contributes nothing extra to the hypothesis), yet along x_1 alone, $t \mapsto \max\{0, t\}^2$ has derivative $2\max\{0, t\}$, which is not differentiable at $t = 0$ (its one-sided difference quotients

at 0 are 2 from the right and 0 from the left). So f is not twice differentiable at $x=0$, let alone around it.

(e) **False**, for the same reason as **(d)**: even the sharper estimate $f(x)=x_2x_1+O(|x|^3)$ only pins down the second-order Taylor data of f **at** the point $x=0$ (by Corollary 10.45, this does give f twice differentiable **at** 0 with $\partial_{11}f(0)=0$), but it does not force twice differentiability in a whole neighbourhood **around** $x=0$. Without an a priori assumption such as $f\in C^2$ near 0, one can cook up genuinely pathological examples, e.g. $f(x):=x_1x_2+|x|^4\sin(|x|^{-100})$, for which $\partial_1 f(x)\to\infty$ as $x\to 0$, so $\partial_1 f$ is not even continuous at 0.

Q.E.D.

Solution to Exercise 11.b.26:

The answer is (a): degree 3, at a cost of 20 CHF.

The hypothesis $f(x)=x_1+x_2x_1+O(|x|^4)$ says two things at once: $f(0)=0$, and f is pinned down exactly up to order 3 (its degree-3 part happens to be zero). Expanding g about $f(0)=0$,

$$(g\circ f)(x)=\sum_{k\geq 0}\frac{g^{(k)}(0)}{k!}\,f(x)^k.$$

Since $f=O(|x|)$, we have $f^k=O(|x|^k)$, so every term with $k\geq 4$ is already $O(|x|^4)$ and invisible at order 3. The unknown $O(|x|^4)$ inside f likewise only perturbs the answer at order 4. So the degree-3 Taylor polynomial of $g\circ f$ is determined, and no more than that is: nothing in the data constrains the degree-4 coefficients of f.

Concretely, truncating $f=x_1+x_1x_2$ at order 3 gives $f^2=x_1^2+2x_1^2x_2+O(|x|^4)$ and $f^3=x_1^3+O(|x|^4)$, so

$$(g\circ f)(x)=g(0)+g'(0)\big(x_1+x_1x_2\big)+\frac{g''(0)}{2}\big(x_1^2+2x_1^2x_2\big)+\frac{g'''(0)}{6}\,x_1^3+O(|x|^4).$$

That uses the four values $g(0),g'(0),g''(0),g'''(0)$, so four queries at 5 CHF each, giving **20 CHF**. Options (b) and (c) under-pay and cannot reach degree 3; (d) claims degree 3 for three queries, which is one derivative short. **Q.E.D.**

Solution to Exercise 11.b.27:

Put $R:=P-Q=\sum_\alpha c_\alpha X^\alpha$ with $c_\alpha:=a_\alpha-b_\alpha$, so the hypothesis reads $|R(x)|\leq M|x|^\sigma$ for $|x|\leq 1$ and the claim is that $c_\alpha=0$ whenever $|\alpha|<\sigma$.

Suppose not, and let

$$d:=\min\{\,|\alpha|\ :\ c_\alpha\neq 0\,\}<\sigma.$$

Split $R=R_d+\widetilde{R}$, where $R_d:=\sum_{|\alpha|=d}c_\alpha X^\alpha$ is the homogeneous part of degree d, non-zero by the choice of d, and $\widetilde{R}$ collects the terms of degree $>d$, so that $\widetilde{R}(tu)=O(t^{d+1})$ uniformly over unit vectors u.

Fix a unit vector u and evaluate at $x:=tu$ for $t\in(0,1]$. Homogeneity gives $R_d(tu)=t^dR_d(u)$, so

$$\big|t^dR_d(u)+O(t^{d+1})\big|\ =\ |R(tu)|\ \leq\ M\,t^\sigma,$$

and dividing by $t^d>0$,

$$\big|R_d(u)+O(t)\big|\ \leq\ M\,t^{\sigma-d}.$$

Here is where the hypothesis $d<\sigma$ is spent: the exponent $\sigma-d$ is **strictly positive**, so the right-hand side tends to 0 as $t\to 0^+$, and the left-hand side tends to $|R_d(u)|$. Hence $R_d(u)=0$ for every unit vector u, and by homogeneity $R_d\equiv 0$ on all of $\mathbb{R}^n$.

A polynomial vanishing identically on $\mathbb{R}^n$ has all coefficients zero, so $c_\alpha=0$ for every $|\alpha|=d$ — contradicting the choice of d. Therefore no such d exists, i.e. $a_\alpha=b_\alpha$ for all $|\alpha|<\sigma$.

AI-Note 11.5: The statement is sharp in the obvious way: coefficients of order exactly σ need not agree. With $n = 1$, $\sigma = 2$, $P(x) := x^2$ and $Q := 0$ we get $|P - Q| = |x|^2 \leq 1 \cdot |x|^2$, and the order-$2$ coefficients differ. The argument above breaks at precisely the right place: for $d = \sigma$ the bound becomes $|R_d(u) + O(t)| \leq M$, which says nothing.

Q.E.D.

PART IV

Optimization and Convexity

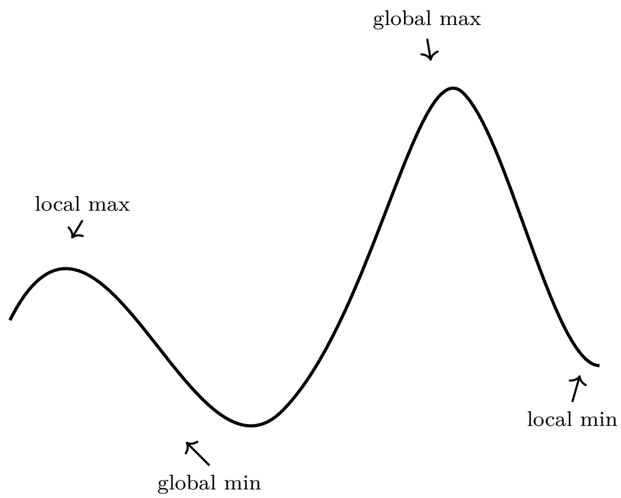
Differentiability turns the search for extrema into algebra. An interior extremum forces the gradient to vanish, so the candidates can be found by solving equations, and the sign behaviour of the Hessian then decides which kind of point each candidate is. Two complications occupy the rest of the part. When the variables are constrained to a level set, the gradient need not vanish at an extremum at all, and the method of Lagrange multipliers says what condition replaces it. When the Hessian test comes out inconclusive, or when the question asked is global rather than local, convexity is the hypothesis that settles it: for a convex function a critical point is not a candidate but an answer.

Extrema and the Hessian Test (Chapter 12)

The Taylor expansion of the previous chapter stops being a formal exercise the moment one asks what it is **for**. Its first-order term locates the candidates for a maximum or a minimum, namely the points where the gradient vanishes. Its second-order term decides which of them are which. Carrying out the second half is the business of the Hessian test, and most of the work turns out to be reading the sign of a quadratic form off a matrix.

Optimization (Section 12.a)

Four kinds of point on one graph, and telling them apart is the whole vocabulary of this section. Two of them are extreme only in comparison with their immediate neighbours, and two are extreme against the entire domain.



Definition 12.a.1 (*Local extrema and critical points*):

- "*Local minimum*" (German: „*lokales Minimum*") of $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$: f has a local minimum at $x_0 \in U$ if
$$\exists\, \varepsilon > 0 \text{ such that } \forall x \in B_\varepsilon(x_0) \cap U : f(x) \geq f(x_0).$$
- **Local maxima** are defined analogously with $f(x) \leq f(x_0)$.
- The extremum is **strict** if the inequality is strict for $x \neq x_0$: $f(x) > f(x_0)$ for a strict minimum, $f(x) < f(x_0)$ for a strict maximum.
- A "*critical point*" (German: „*kritischer Punkt*") of $f \in C^1(U, \mathbb{R})$, $U \subseteq \mathbb{R}^n$ open, is a point $x_0 \in U$ where $\nabla f(x_0) = 0$.

Proposition 12.a.2 (*Local extremum* $\implies$ *critical point*): If $f \in C^1(U, \mathbb{R})$ and $x_0 \in \operatorname{int}(U)$ is a local extremum, then $\nabla f(x_0) = 0$.

Proof:

Fix a direction $v \in \mathbb{R}^n$ and consider the one-variable function $\varphi(t) := f(x_0 + tv)$, defined for $|t|$ small enough that $x_0 + tv$ stays in U. This is where $x_0 \in \operatorname{int}(U)$ is used, and it is the only place. If f has a local extremum at x_0, then φ has one at the **interior** point $t = 0$, so by the one-variable result from Analysis I, $\varphi'(0) = 0$. By the chain rule along a path (Section 10.d),
$$0 = \varphi'(0) = \langle \nabla f(x_0), v \rangle.$$

Since v was arbitrary, $\nabla f(x_0)$ is orthogonal to every vector of $\mathbb{R}^n$, including itself: $\langle \nabla f(x_0), \nabla f(x_0)\rangle = |\nabla f(x_0)|^2 = 0$, hence $\nabla f(x_0) = 0$. **Q.E.D.**

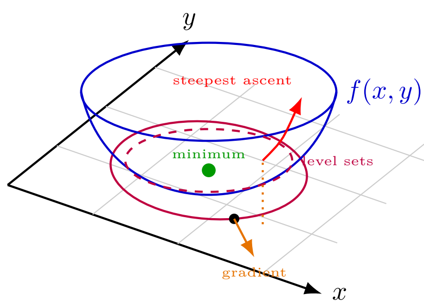
Remark 12.a.3 (*Intuition, and two standing warnings*): The proof says: at an interior extremum every direction is a *"flat"* direction, and the only vector orthogonal to everything is the zero vector,

$$\langle \nabla f(x_0), v\rangle = 0 \text{ for all } v \in \mathbb{R}^n \iff \nabla f(x_0) = 0.$$

Two things this proposition does **not** say, both of which the rest of the chapter is built around.

- **The converse is false.** A critical point need not be an extremum: $f(x,y) := x^2 - y^2$ has $\nabla f(0,0) = 0$ but a saddle there, and even in one variable $f(x) := x^3$ has $f'(0) = 0$ with no extremum. Critical points are **candidates**; separating them is exactly what Theorem 12.b.10 is for.
- **Interiority is essential.** On $U := [0,1]$ the function $f(x) := x$ attains both its minimum and its maximum, at $x = 0$ and $x = 1$, and $f' \equiv 1$ never vanishes. The hypothesis $x_0 \in \operatorname{int}(U)$ is what fails. This is why every optimization over a closed region in this chapter is done in two steps. Interior critical points come first, and the boundary is then treated separately, by the method of Lagrange multipliers.

Geometrically, the proposition says that at an interior extremum the direction of steepest ascent ceases to exist. Anywhere else the gradient is a genuine non-zero vector, perpendicular to the level set of f through the point and pointing the way f grows fastest. That perpendicularity is worth remembering past this chapter, since it is the whole geometric content of the Lagrange condition in Chapter 13.



So, the gradient produces a list of candidates and then falls silent. Which of them are minima, which are maxima, and which are neither? The first derivative cannot say, since it vanishes at all of them alike, so the question has to be put to the second.

The Hessian test (Section 12.b)

The **Hessian** (Remark 11.b.21) of $f \in C^2(U,\mathbb{R})$, $U \subseteq \mathbb{R}^n$, is given by

$$\operatorname{Hess} f(x) = (\partial_i \partial_j f)_{i,j=1,\ldots,n}$$

with respect to the standard basis of $\mathbb{R}^n$. By "*Schwarz's lemma*" (German: „*Satz von Schwarz*"), stated as Theorem 11.b.8, $\operatorname{Hess} f(x)$ is symmetric.

Remark 12.b.4 (*Why symmetry is proved here*): Everything in this section rests on symmetry of the Hessian: the spectral theorem in Remark 12.b.7, the eigenvalue criteria, the Hurwitz minors. Yet Theorem 11.b.8 was quoted without proof where it first appeared, so the proof is supplied now. It comes with the attempt that fails, which is the more informative half.

Proof of Theorem 11.b.8:

It suffices to treat $m = 1$ (apply the result to each component), $i < j$ (there is nothing to prove when $i = j$), and $n = 2$ with $i = 1$, $j = 2$ (freeze all the other variables). So, let $f \in C^2$ near (x, y) and write ∂_1, ∂_2 for the two partial derivatives.

The attempt that fails. Unwinding both mixed derivatives as iterated limits gives

$$\partial_2\partial_1 f(x,y) = \lim_{h_2\to 0}\lim_{h_1\to 0}\frac{f(x+h_1,y+h_2)-f(x+h_1,y)-f(x,y+h_2)+f(x,y)}{h_1h_2},$$

and one would like to apply the mean value theorem in each variable and be done. The obstruction is that the two intermediate points the mean value theorem returns depend on h_1 and h_2 **separately** and need not agree, so forcing them together leaves a stray factor h_2/h_1, which blows up when $h_1 \to 0$ at fixed h_2. The symmetric-looking expression above is not enough on its own; one has to keep the two increments coupled.

The proof. Couple them by fixing a single $h \neq 0$ and setting

$$F(h) := f(x+h,y+h) - f(x+h,y) - \big(f(x,y+h) - f(x,y)\big),$$

which is the second difference of f over a square of side h, and $\varphi(t) := f(x+th, y+h) - f(x+th, y)$, so that $F(h) = \varphi(1) - \varphi(0)$. The mean value theorem gives $\xi_1 \in (0,1)$ with

$$F(h) = \varphi'(\xi_1) = h\,\partial_1 f(x+\xi_1 h,\ y+h) - h\,\partial_1 f(x+\xi_1 h,\ y).$$

Now apply the mean value theorem a second time, to $\psi(t) := \partial_1 f(x+\xi_1 h,\ y+th)$. This is legitimate precisely because $f \in C^2$, so $\partial_1 f$ is itself continuously differentiable in the second variable. Since $F(h) = h\big(\psi(1) - \psi(0)\big)$, there is $\xi_2 \in (0,1)$ with

$$F(h) = h\,\psi'(\xi_2) = h^2\,\partial_2\partial_1 f(x+\xi_1 h,\ y+\xi_2 h).$$

Grouping the four terms of $F(h)$ the other way (first and third together, second and fourth together) runs the identical argument with the roles of the variables exchanged and produces $\tilde\xi_1, \tilde\xi_2 \in (0,1)$ with $F(h) = h^2\,\partial_1\partial_2 f(x+\tilde\xi_1 h,\ y+\tilde\xi_2 h)$. Dividing both expressions by $h^2 \neq 0$,

$$\partial_2\partial_1 f(x+\xi_1 h,\ y+\xi_2 h) \;=\; \partial_1\partial_2 f(x+\tilde\xi_1 h,\ y+\tilde\xi_2 h).$$

All four numbers $\xi_1, \xi_2, \tilde\xi_1, \tilde\xi_2$ lie in $(0,1)$. They depend on h, but they stay bounded, so both arguments converge to (x,y) as $h \to 0$. Continuity of the mixed second partials (again $f \in C^2$) lets us pass to the limit on both sides, giving $\partial_2\partial_1 f(x,y) = \partial_1\partial_2 f(x,y)$. **Q.E.D.**

Remark 12.b.5 (*What the failed attempt shows*): The two attempts differ in exactly one respect: the failed one lets h_1 and h_2 go to zero **independently**, the successful one ties them together as a single h and only takes a limit at the very end. That is why the second difference $F(h)$ over a **square** is the right object. It is symmetric in the two variables by construction, so the two groupings of its four terms compute the two mixed partials, and the symmetry of the answer is inherited from the symmetry of the square rather than proved by force.

It also locates precisely where continuity is used: twice, once to apply the mean value theorem to $\partial_1 f$ and once to pass to the limit at the end. Drop it and the conclusion genuinely fails. Remark 11.b.9 gives the standard counterexample.

Definition 12.b.6 (*Sign of a symmetric matrix*): A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is:

- "*positive definite*" (German: „*positiv definit*“) if $v^\mathsf{T} A v > 0$ for all $v \in \mathbb{R}^n \setminus \{0\}$;
- "*negative definite*" (German: „*negativ definit*“) if $v^\mathsf{T} A v < 0$ for all $v \in \mathbb{R}^n \setminus \{0\}$;
- **indefinite** if there exist $v, u \in \mathbb{R}^n \setminus \{0\}$ such that $v^\mathsf{T} A v > 0$ and $u^\mathsf{T} A u < 0$;
- **degenerate** (German: „*entartet*“) if there exists $v \in \mathbb{R}^n \setminus \{0\}$ such that $Av = 0$; equivalently, if $\det A = 0$, or if 0 is an eigenvalue of A.

AI-Note 12.1: Corsin defines degenerate as *"there exists $v \neq 0$ with $v^\mathsf{T} A v = 0$"*. That condition is much weaker than intended: if A is indefinite, with $v^\mathsf{T} A v > 0$ and $u^\mathsf{T} A u < 0$, then $t \mapsto$

$(tv+(1-t)u)^{\mathsf{T}}A(tv+(1-t)u)$ is continuous and changes sign, so it vanishes at some $w \neq 0$. Every indefinite matrix would then be degenerate, and the third bullet of Theorem 12.b.10 vacuous. The intended notion is a nontrivial kernel ($\det A = 0$), which is also what Exercise 12.b.20 states. Corrected above.

Remark 12.b.7 (*Linear algebra interlude: diagonalizing the Hessian*): Since the Hessian $\operatorname{Hess} f(x_0)$ is a symmetric matrix, it is diagonalizable over $\mathbb{R}$ by the spectral theorem. There exist n real eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ and a corresponding orthonormal basis of eigenvectors $v_1, \dots, v_n \in \mathbb{R}^n$ such that $\operatorname{Hess} f(x_0)v_i = \lambda_i v_i$ for all i.

Every vector $x \in \mathbb{R}^n$ can be written in this basis as $x = \sum_{i=1}^n a_i v_i$ with $a_i \in \mathbb{R}$. Applying the Hessian yields:

$$\operatorname{Hess} f(x_0)x = \operatorname{Hess} f(x_0)\sum_{i=1}^n a_i v_i = \sum_{i=1}^n a_i \operatorname{Hess} f(x_0)v_i = \sum_{i=1}^n \lambda_i a_i v_i.$$

In this eigenbasis, the Hessian matrix becomes diagonal. Consequently, the quadratic form evaluates to $x^{\mathsf{T}} \operatorname{Hess} f(x_0)x = \sum_{i=1}^n \lambda_i a_i^2$, which immediately links the definiteness of the Hessian to the signs of its eigenvalues.

Exercise 12.b.8 (*Polarisation formula*): Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth. Prove that the map

$$\mathbb{R}^n \ni e \;\longmapsto\; \left.\frac{d^2}{dt^2}\right|_{t=0} f(te) = e^{\mathsf{T}} \operatorname{Hess} f(0)\, e$$

determines all the second partial derivatives $\partial_i\partial_j f(0)$.

Proof of Exercise 12.b.8:

By the chain rule, $\frac{d}{dt} f(te) = \sum_i \partial_i f(te)e_i$, and differentiating again,

$$\left.\frac{d^2}{dt^2}\right|_{t=0} f(te) = \sum_{i,j} \partial_i\partial_j f(0)\, e_i e_j = e^{\mathsf{T}} \operatorname{Hess} f(0)\, e =: Q(e),$$

which is the displayed identity, with Q a quadratic form in e.

Taking $e := e_i$, the i-th standard basis vector, gives $Q(e_i) = \partial_i^2 f(0)$: the diagonal entries directly. For the off-diagonal entries, expand

$$Q(e_i + e_j) = (e_i + e_j)^{\mathsf{T}} \operatorname{Hess} f(0)(e_i + e_j) = Q(e_i) + 2\,\partial_i\partial_j f(0) + Q(e_j),$$

using symmetry of $\operatorname{Hess} f(0)$ (Theorem 11.b.8) to combine the two mixed terms into one. Solving,

$$\partial_i\partial_j f(0) = \tfrac{1}{2}\big(Q(e_i + e_j) - Q(e_i) - Q(e_j)\big),$$

so every entry of $\operatorname{Hess} f(0)$ is recovered from Q alone. **Q.E.D.**

Remark 12.b.9: This is the same phenomenon as recovering an inner product from its induced norm via the parallelogram-style polarisation identity: a symmetric bilinear form is determined by its values on the diagonal. Here it says that a single-variable computation, differentiating $t \mapsto f(te)$ twice at $t = 0$ for every direction e, already contains everything the full Hessian matrix does.

Theorem 12.b.10 (*The Hessian test*): If $f \in C^2(U, \mathbb{R})$ has a critical point $x_0 \in \operatorname{int}(U)$, then:

- $\operatorname{Hess} f(x_0)$ **positive definite** $\Rightarrow$ local **min** at x_0;
- $\operatorname{Hess} f(x_0)$ **negative definite** $\Rightarrow$ local **max** at x_0;
- $\operatorname{Hess} f(x_0)$ **indefinite** $\Rightarrow$ "*saddle point*" (German: „*Sattelpunkt*“) at x_0.

If $\operatorname{Hess} f(x_0)$ is degenerate but not indefinite (i.e. positive or negative semidefinite with 0 an eigenvalue), the test gives **no information**, and all three behaviours are possible.

AI-Note 12.2: Corsin's third bullet reads *"indefinite and not degenerate"*. With degeneracy corrected to $\det A = 0$ the extra hypothesis is unnecessary: indefiniteness alone already forces a saddle, since directions v with $v^\mathsf{T} \operatorname{Hess} f(x_0)v > 0$ and u with $u^\mathsf{T} \operatorname{Hess} f(x_0)u < 0$ make $t \mapsto f(x_0 + tv)$ increase and $t \mapsto f(x_0 + tu)$ decrease away from x_0. Dropped above. The genuinely inconclusive case, semidefinite **and** degenerate, is stated explicitly instead, since that is the one students actually run into (Exercise 12.b.22 is exactly this trap). The official script carries the identical redundant hypothesis (its Prop. 11.15(3): *"indefinite and non-degenerate"*), so this is not a slip specific to Corsin's notes. Both sources state the theorem with one clause more than it needs.

Remark 12.b.11 (*Why positive definite means minimum*): The statement is easy to memorise and easy to find unmotivated: it is not obvious what a condition on $v^\mathsf{T} \operatorname{Hess} f(x_0)v$ has to do with f going up. The following two-step reading, which recovers the one-dimensional test as the case $n = 1$, makes the connection direct.

Step one: the Hessian moves the gradient. The entries of $\operatorname{Hess} f$ are derivatives of the entries of ∇f, so the Hessian is to ∇f what ∇f is to f. Applying the linear approximation to ∇f at the critical point, where $\nabla f(x_0) = 0$,

$$\nabla f(x_0 + v) \approx \underbrace{\nabla f(x_0)}_{=0} + \operatorname{Hess} f(x_0)\, v = \operatorname{Hess} f(x_0)\, v.$$

So, once we step off x_0 in the direction v, the gradient we find there is approximately $\operatorname{Hess} f(x_0)v$.

Step two: the gradient moves f. The gradient measures the rate of change of f, so travelling the displacement v against a gradient of $\operatorname{Hess} f(x_0)v$ changes f by roughly the inner product of the two,

$$f(x_0 + v) - f(x_0) \;\approx\; \tfrac{1}{2}\, v^\mathsf{T} \operatorname{Hess} f(x_0)\, v,$$

the factor $\frac{1}{2}$ appearing because the gradient grows linearly from 0 along the way, so its average over the step is half its final value. This is of course the second-order Taylor expansion of Remark 11.b.21 with the first-order term deleted by criticality; the point of the derivation is that it explains **why** that expansion has the form it does.

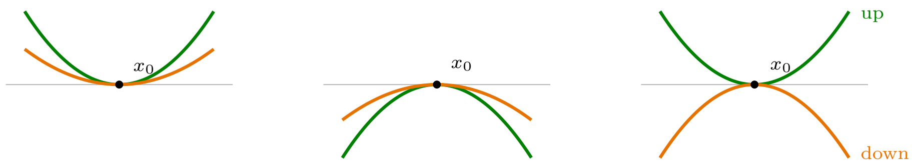
Now the theorem reads itself. If $\operatorname{Hess} f(x_0)$ is positive definite, then $v^\mathsf{T} \operatorname{Hess} f(x_0)v > 0$ for every $v \neq 0$, so f increases in **every** direction away from x_0, and that is a strict local minimum. Negative definite reverses every inequality. Indefinite means some directions give a positive value and others a negative one, so f rises along some and falls along others: a saddle. The degenerate-but-not-indefinite case is now inconclusive for a visible reason. Along an eigenvector with eigenvalue 0 the quadratic term vanishes entirely, so the approximation says nothing and the behaviour is decided by terms this expansion has discarded.

AI-Example 12.b.12 (*Classifying saddle points via the Hessian*): Consider $f(x, y) := x^2 - y^2$ on $\mathbb{R}^2$. The gradient $\nabla f(x, y) = (2x,\, -2y)^\mathsf{T}$ vanishes at $(0, 0)$, and the Hessian is constant:

$$\operatorname{Hess} f(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}.$$

Its eigenvalues are $\lambda_1 = 2 > 0$ and $\lambda_2 = -2 < 0$, so $\operatorname{Hess} f(0, 0)$ is indefinite and $(0, 0)$ is a saddle point by Theorem 12.b.10. This is the model case the word *"saddle"* is named after, since f rises along the x-axis and falls along the y-axis.

Remark 12.b.13 (*Mnemonic*): positive $= \smile \dots$ min; negative $= \frown \dots$ max; indefinite = saddle.



positive definite	**negative definite**	**indefinite**
$\lambda_1, \lambda_2 > 0$	$\lambda_1, \lambda_2 < 0$	opposite signs
local **min**	local **max**	**saddle**

The picture is the whole theorem. By Remark 12.b.7 the second-order Taylor term is $\frac{1}{2}\sum_i \lambda_i a_i^2$ in the eigenbasis, so along the i-th eigendirection f behaves like the parabola $\frac{1}{2}\lambda_i t^2$: it curves **up** when $\lambda_i > 0$ and **down** when $\lambda_i < 0$. All up gives a minimum, all down a maximum. One of each gives a saddle, since there is then a direction of ascent and a direction of descent through the same point. And when some $\lambda_i = 0$ that direction is **flat** to second order, the quadratic term says nothing about it, and the test must fall silent. That is exactly the degenerate case, illustrated by $x^4 + y^4$ against $x^4 - y^4$ below.

AI-Note 12.3: The regularity in Theorem 12.b.10 is stated by Corsin as C^3; the test needs only C^2 (which is what the Hessian's definition assumes). Relaxed to C^2 above. This is not just Corsin's own habit. The official script's Prop. 11.15 independently requires $f \in C^3$ too, and its proof indeed only uses the second-order Taylor expansion (with an $o(|h|^2)$ remainder), which needs nothing beyond C^2. Ours is both the weaker hypothesis and the standard statement of this theorem.

AI-Example 12.b.14 (*Inconclusive Hessian Test at a Degenerate Critical Point*): Consider $f(x,y) := x^4 + y^4$ and $g(x,y) := x^4 - y^4$ at the critical point $(0,0)$.

- For both functions, all second-order partial derivatives at $(0,0)$ vanish, so $\operatorname{Hess} f(0,0) = \operatorname{Hess} g(0,0) = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.
- The Hessian test is **inconclusive** since $\det \operatorname{Hess} = 0$.
- However, direct inspection reveals that $f(x,y) \geq 0 = f(0,0)$ for all (x,y), so $(0,0)$ is a strict local minimum for f, whereas $g(x,0) > 0$ and $g(0,y) < 0$ for non-zero x, y, so $(0,0)$ is a saddle point for g.

Remark 12.b.15: Script Exercise 11.16 makes the same point with a two-parameter family: $f_{a,b}(x,y) := ax^4 + by^4$ has $\operatorname{Hess} f_{a,b}(0,0)$ equal to the zero matrix **regardless of** a, b, yet the behaviour at 0 ranges over every possible case as a, b vary. Taking $a, b > 0$ recovers the strict minimum of f above ($a = b = 1$), $a, b < 0$ gives a strict maximum, and opposite signs recover the saddle of g above ($a = 1$, $b = -1$). It is the sharpest available statement of "*degenerate* $\Rightarrow$ *no information*": the identical (zero) Hessian is compatible with every possible outcome, depending only on quartic terms the second-order test cannot see.

How do I determine positive/negative definiteness? (Subsection 12.b.1)

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$. Note: $\det(A) = \lambda_1 \cdots \lambda_n$ and $\operatorname{Tr}(A) = \lambda_1 + \cdots + \lambda_n$.

For $n = 2$:

- $\det(A) = \lambda_1\lambda_2 < 0 \implies$ **indefinite**
- $\det(A) > 0,\ \operatorname{Tr}(A) > 0 \implies$ **positive definite**
- $\det(A) > 0,\ \operatorname{Tr}(A) < 0 \implies$ **negative definite**

For $n \geq 3$:

- Solve the characteristic equation $\det(A - \lambda\,\mathrm{id}) = 0$ for λ and check all eigenvalues.

- "*Hurwitz criterion*" (German: „*Hurwitz-Kriterium*"): for

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \\ a_{n1} & \cdots & & a_{nn} \end{pmatrix}$$

define the leading principal minors

$$A_1 = \det(a_{11}), \quad A_2 = \det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad \ldots, \quad A_n = \det(A).$$

Then:
  - $A_1 > 0,\ A_2 > 0,\ A_3 > 0, \ldots$ if and only if the matrix is **positive definite**
  - $A_1 < 0,\ A_2 > 0,\ A_3 < 0, \ldots$ if and only if the matrix is **negative definite**
  - **indefinite** if neither $A_1 \geq 0, A_2 \geq 0, \ldots$ nor $A_1 \leq 0, A_2 \geq 0, A_3 \leq 0, \ldots$

AI-Note 12.4: Corsin writes the characteristic equation as "$A - \lambda\,\mathrm{id} = 0$", omitting the determinant. Corrected above.

Example 12.b.16 (*Definiteness calculations*):

Matrix	Computation	Conclusion
$\begin{pmatrix} 2 & 2 \\ 2 & 5 \end{pmatrix}$	$\det = 6 > 0$, $\mathrm{Tr} = 7 > 0$	positive definite
$\begin{pmatrix} -4 & 2 \\ 2 & -4 \end{pmatrix}$	$\det = 12 > 0$, $\mathrm{Tr} = -8 < 0$	negative definite
$\begin{pmatrix} 4 & 2 \\ 2 & -4 \end{pmatrix}$	$\det = -20 < 0$	indefinite
$\begin{pmatrix} -5 & 0 & 0 \\ 0 & -4 & -2 \\ 0 & -2 & -4 \end{pmatrix}$	$A_1 = -5 < 0$, $A_2 = 20 > 0$, $A_3 = -60 < 0$	negative definite
$\begin{pmatrix} -1 & 0 & -4 \\ 0 & 2 & -2 \\ -4 & -2 & 22 \end{pmatrix}$	$A_1 = -1 < 0$, $A_2 = -2 < 0$, $A_3 = -72 < 0$	indefinite

Example 12.b.17 (*A two-parameter family running through all four Hessian cases*): Let $a, b \in \mathbb{R}$ and $f(x,y) := x\sin(y) + ax^2 + by^2$. Then $Df(0,0) = (0,0)$, and

$$\operatorname{Hess} f(0,0) = \begin{pmatrix} 2a & 1 \\ 1 & 2b \end{pmatrix}, \qquad \det \operatorname{Hess} f(0,0) = 4ab - 1.$$

Using the determinant/trace rule above:

- If $a > 0$ and $4ab - 1 > 0$: positive definite, so $(0,0)$ is a local minimum.
- If $a < 0$ and $4ab - 1 > 0$: negative definite, so $(0,0)$ is a local maximum.
- If $4ab - 1 < 0$: indefinite, so $(0,0)$ is a saddle point.
- If $4ab - 1 = 0$: degenerate, and the test says nothing.

Unlike the fixed examples above, a single formula with two free parameters realises every outcome of Theorem 12.b.10 at once, simply by moving (a,b) across the curve $4ab = 1$ in parameter space.

Exercise 12.b.18 (*Critical values of $x^3 - y^3 + 3\alpha xy$*): Find the critical values of

$$f_\alpha : \mathbb{R}^2 \to \mathbb{R}, \qquad (x,y) \mapsto x^3 - y^3 + 3\alpha xy$$

for $\alpha \in \mathbb{R} \setminus \{0\}$ and determine min/max/saddle points.

Solution to Exercise 12.b.18:

$$\frac{\partial f_\alpha}{\partial x} = 3x^2 + 3\alpha y \overset{!}{=} 0 \qquad \Longrightarrow y = -\frac{x^2}{\alpha}$$
$$\frac{\partial f_\alpha}{\partial y} = -3y^2 + 3\alpha x \overset{!}{=} 0 \qquad \Longrightarrow -\frac{x^4}{\alpha^2} + \alpha x = 0$$

1. $x = y = 0$

2. $x \neq 0 \implies x^3 = \alpha^3 \implies x = \alpha,\ y = -\alpha$

So, we have critical points $(0,0)$ and $(\alpha, -\alpha)$.

Compute the Hessian:

$$\begin{pmatrix} \partial_x^2 f & \partial_x \partial_y f \\ \partial_x \partial_y f & \partial_y^2 f \end{pmatrix} = \begin{pmatrix} 6x & 3\alpha \\ 3\alpha & -6y \end{pmatrix} = \operatorname{Hess} f(x,y)$$

$$\operatorname{Hess} f(0,0) = \begin{pmatrix} 0 & 3\alpha \\ 3\alpha & 0 \end{pmatrix}, \qquad \det = -9\alpha^2 < 0 \implies \text{indefinite}$$

$$\operatorname{Hess} f(\alpha, -\alpha) = \begin{pmatrix} 6\alpha & 3\alpha \\ 3\alpha & 6\alpha \end{pmatrix} = 3\alpha \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

$$\det = 27\alpha^2 > 0, \qquad \operatorname{Tr} = 12\alpha.$$

For $\alpha > 0$ this is positive definite; for $\alpha < 0$ it is negative definite. So:

- $(0,0) \longrightarrow$ **saddle point**
- $(\alpha, -\alpha) \longrightarrow$ **local minimum** for $\alpha > 0$; **local maximum** for $\alpha < 0$

Q.E.D.

AI-Exercise 12.b.19 (*Global Extrema on a Compact Triangle*): Find the global maximum and minimum values of $f(x,y) := xy$ on the compact triangular region $T := \{(x,y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0, x + y \leq 2\}$.

Exercise 12.b.20 (*6.2 — The signature of a 2×2 matrix* (important)): Despite the definition, it is not necessary to compute the eigenvalues of a matrix to find its signature. Prove that for a symmetric 2×2 matrix M we have the following simple rule to determine the signature in terms of $\det M$ and $\operatorname{Tr} M$:

- If $\det M > 0$, $\operatorname{Tr} M > 0$ then M is positive definite,
- If $\det M > 0$, $\operatorname{Tr} M < 0$ then M is negative definite,
- If $\det M < 0$ then M is indefinite,
- If $\det M = 0$, then M is degenerate.

Exercise 12.b.21 (*Classifying critical points with the Hessian*): Find and classify the critical points of $f(x,y) := xy - x^2 - y^2 - x$ on $\mathbb{R}^2$. (cf. Michaels Bsp 7.1.3)

Exercise 12.b.21:

First, we compute the gradient to find the critical points:

$$\nabla f(x,y) = \begin{pmatrix} y - 2x - 1 \\ x - 2y \end{pmatrix}.$$

Setting $\nabla f(x,y) = (0,0)$ gives the system:

$$\begin{aligned} -2x + y &= 1 \\ x - 2y &= 0 \implies x = 2y. \end{aligned}$$

Substituting into the first equation: $-2(2y) + y = 1 \implies -3y = 1 \implies y = -1/3$. Then $x = -2/3$. So, the only critical point is $(-2/3, -1/3)$.

To classify it, we compute the Hessian matrix:

$$Hf(x,y) = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}.$$

The Hessian is constant. We apply the second derivative test by examining its eigenvalues (or determinant and trace). The determinant is $\det Hf = (-2)(-2) - 1(1) = 3 > 0$. The trace is $\operatorname{Tr} Hf = -2 + (-2) = -4 < 0$. Since $\det > 0$ and $\operatorname{Tr} < 0$, the Hessian is negative definite at the critical point. Therefore, $(-2/3, -1/3)$ is a strict local maximum. **Q.E.D.**

Exercise 12.b.22 (*6.8 — Multiple choice* (important)): The Hessian matrix of $f \in C^2(\mathbb{R}^n)$ is positive semidefinite at a critical point x_0 of f, i.e.,

$$\langle v, \operatorname{Hess} f(x_0) v\rangle \geq 0 \quad \text{for all } v \in \mathbb{R}^n.$$

Which of the following statements necessarily hold? (There may be more than one).

(a) x_0 is a strict local minimum of f.

(b) x_0 is a local minimum of f.

(c) x_0 is not a local maximum of f.

(d) None of the above statements.

Exercise 12.b.23 (*6.9 — Minimization* (important)): The function $f : \mathbb{R}^2 \to \mathbb{R}$ is given by $f(x, y) = 2x^2 + y^2 - x$. Determine the extrema of f on...

(a) ...the unit circle $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$;

(b) ...the closed unit disk $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$.

Exercise 12.b.24 (*6.12 — Critical points* (important)): Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function $f(x, y) = (ax^2 + by^2)e^{-x^2-y^2}$ with real parameters $a, b \in \mathbb{R}$. Find all critical points and determine their nature with the Hessian test, depending on a, b.

Solutions (Section 12.c)

Proof of AI-Exercise 12.b.19:

Since T is compact and f is continuous, global extrema exist by the extreme value theorem (Item **(b)**).

- **Interior of T:** Critical points satisfy $\nabla f(x, y) = (y, x)^\mathsf{T} = (0, 0)^\mathsf{T}$, and therefore $(x, y) = (0, 0)$, which lies on the boundary. Thus, there are no critical points in $\operatorname{int}(T)$.
- **Boundary of T:**
  - Along the leg $x = 0$ $(0 \leq y \leq 2)$, $f(0, y) = 0$.
  - Along the leg $y = 0$ $(0 \leq x \leq 2)$, $f(x, 0) = 0$.
  - Along the hypotenuse $x + y = 2$ $(0 \leq x \leq 2)$, substitute $y = 2 - x$:
  $$h(x) := x(2 - x) = 2x - x^2, \qquad h'(x) = 2 - 2x,$$
  which vanishes at $x = 1$. This gives the candidate point $(1, 1)$ with $f(1, 1) = 1$. The endpoints $(2, 0)$ and $(0, 2)$ yield $f = 0$.

Comparing values, the **global maximum** is 1 at $(1, 1)$, and the **global minimum** is 0, attained everywhere on the coordinate axes legs of T. **Q.E.D.**

Proof of Exercise 12.b.20:

By the spectral theorem, since M is symmetric there exists an orthogonal matrix O (so $OO^\mathsf{T} = \mathrm{id}$, hence $\det O = \pm 1$) and $\lambda_1, \lambda_2 \in \mathbb{R}$ such that

$$OMO^\mathsf{T} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

Using $\det(AB) = \det(A)\det(B)$ and $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$,

$$\det M = \det(OMO^\mathsf{T}) = \lambda_1\lambda_2, \qquad \operatorname{Tr} M = \operatorname{Tr}(MO^\mathsf{T}O) = \operatorname{Tr}(OMO^\mathsf{T}) = \lambda_1 + \lambda_2.$$

So, $\det M$ and $\operatorname{Tr} M$ are exactly the product and sum of the eigenvalues, and the four cases follow immediately:

- $\det M > 0$: λ_1, λ_2 have the same sign. If additionally $\operatorname{Tr} M > 0$, both are positive, so M is **positive definite**; if $\operatorname{Tr} M < 0$, both are negative, so M is **negative definite**.
- $\det M < 0$: λ_1, λ_2 have opposite signs, so M is **indefinite**.
- $\det M = 0$: one of λ_1, λ_2 is zero, so M is **degenerate**.

Q.E.D.

Solution to Exercise 12.b.22:

None of **(a)**–**(c)** hold in general:

- **(a) False:** $f \equiv 0$ has $\operatorname{Hess} f(x_0) = 0$ positive semidefinite at every x_0, but x_0 is not a **strict** local minimum (every nearby value equals $f(x_0)$).
- **(b) False:** $f(x) = x_1^3$ at $x_0 = 0$ has $\operatorname{Hess} f(0) = 0$ (positive semidefinite), but f takes negative values arbitrarily close to 0, so x_0 is not a local minimum at all.
- **(c) False:** $f(x) = -x_1^4$ at $x_0 = 0$ has $\operatorname{Hess} f(0) = 0$ (positive semidefinite, vacuously, since it is the zero matrix), yet x_0 **is** a (strict) local maximum of f.

Since **(a)**–**(c)** all fail, **(d) is the correct answer**: none of the above statements necessarily hold.

Q.E.D.

Solution to Exercise 12.b.23:

(a) With $g(x,y) := x^2 + y^2 - 1$, the Lagrangian is $L(x,y,\lambda) := f(x,y) - \lambda g(x,y)$, giving the system

$$0 = \partial_x L = 2x(2-\lambda) - 1, \qquad 0 = \partial_y L = 2y(2-2\lambda), \qquad 0 = \partial_\lambda L = -(x^2+y^2-1).$$

From the middle equation, either $y = 0$ or $\lambda = 1$.

If $y = 0$: then $x^2 = 1$, so $x = \pm 1$, giving the points $(1,0)$ and $(-1,0)$ with $f(1,0) = 1$ and $f(-1,0) = 3$.

If $\lambda = 1$: the first equation gives $2x(1) = 1$, i.e. $x = \frac{1}{2}$, and the constraint gives $y = \pm\frac{\sqrt{3}}{2}$, with $f\left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right) = \frac{1}{2} + \frac{3}{4} - \frac{1}{2} = \frac{3}{4}$.

Comparing all four values, f attains a **global maximum** at $(-1,0)$ (value 3) and **global minima** at $\left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right)$ (value $\frac{3}{4}$); $(1,0)$ is a local (non-global) extremum with value 1.

Q.E.D.

AI-Note 12.5: The official solution lists $(1,0)$ alongside $(-1,0)$ as *"local extrema"* from the Lagrange system, then only mentions it again implicitly by omission from the final global-extrema summary. It is worth flagging explicitly that **Lagrange's condition finds all critical points of $f|_{S^1}$, not automatically the global extrema**. Here $(1,0)$ is a genuine local maximum along the circle (value 1) that is simply not the largest one; only comparing all candidate values at the end distinguishes it from the global maximum at $(-1,0)$ (value 3).

Solution to Exercise 12.b.23, continued:

(b) Having handled the boundary $S^1 = \partial D$ in **(a)**, we look for interior critical points: $Df(x,y) = (4x-1, 2y) = 0$ gives $(x,y) = \left(\frac{1}{4}, 0\right) \in \operatorname{int}(D)$. The Hessian $\operatorname{Hess} f = \left(\begin{smallmatrix} 4 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ is positive definite, so this is a local minimum, with value $f\left(\frac{1}{4}, 0\right) = \frac{1}{8} - \frac{1}{4} = -\frac{1}{8}$. Since $-\frac{1}{8}$ is smaller than every boundary value found in **(a)**, f on D attains a **global minimum** at $\left(\frac{1}{4}, 0\right)$ and a **global maximum** at $(-1,0)$.

Q.E.D.

Solution to Exercise 12.b.24:

Writing $r^2 := x^2 + y^2$, the gradient is

$$\nabla f(x,y) = \big(2x(a - ax^2 - by^2),\ 2y(b - ax^2 - by^2)\big)e^{-r^2}.$$

Case $a = b = 0$: $f \equiv 0$, so every point of $\mathbb{R}^2$ is critical, and (being the constant function) each is simultaneously a global maximum and minimum.

Case $a = 0, b \neq 0$ (and symmetrically $a \neq 0, b = 0$): the gradient equations reduce to $-2xby^2 = 0$ and $2by(1 - y^2) = 0$, whose common solutions are the entire line $\{(x,0) : x \in \mathbb{R}\}$ together with the two points $(0, \pm 1)$. On the critical line the Hessian is singular (degenerate, as expected since a whole line is critical); at $(0, \pm 1)$, $\operatorname{Hess} f(0, \pm 1) = \operatorname{diag}(-2b, -4b)\, e^{-1}$, giving **minima** if $b < 0$ and **maxima** if $b > 0$. The case $a \neq 0, b = 0$ is symmetric, with critical line $\{(0,y)\}$ and extra points $(\pm 1, 0)$: minima if $a < 0$, maxima if $a > 0$.

Case $a, b \neq 0$, $a \neq b$: setting $x = 0$ or $y = 0$ in the gradient equations gives the five critical points $(0,0)$, $(0, \pm 1)$, $(\pm 1, 0)$ (checking $x, y \neq 0$ simultaneously forces $a = b$, excluded here). The Hessians are

$$\operatorname{Hess} f(0,0) = \begin{pmatrix} 2a & 0 \\ 0 & 2b \end{pmatrix},$$

$$\operatorname{Hess} f(0,\pm 1) = e^{-1} \begin{pmatrix} 2(a-b) & 0 \\ 0 & -4b \end{pmatrix},$$

$$\operatorname{Hess} f(\pm 1, 0) = e^{-1} \begin{pmatrix} -4a & 0 \\ 0 & 2(b-a) \end{pmatrix}.$$

Reading off signs for each ordering of a, b against 0:

- $a > b > 0$: min at $(0,0)$, saddle at $(0, \pm 1)$, max at $(\pm 1, 0)$.
- $b > a > 0$: min at $(0,0)$, max at $(0, \pm 1)$, saddle at $(\pm 1, 0)$.
- $a > 0 > b$: saddle at $(0,0)$, min at $(0, \pm 1)$, max at $(\pm 1, 0)$.
- $b > 0 > a$: saddle at $(0,0)$, max at $(0, \pm 1)$, min at $(\pm 1, 0)$.
- $0 > a > b$: max at $(0,0)$, min at $(0, \pm 1)$, saddle at $(\pm 1, 0)$.
- $0 > b > a$: max at $(0,0)$, saddle at $(0, \pm 1)$, min at $(\pm 1, 0)$.

Case $a = b \neq 0$: beyond $(0,0)$, $(0, \pm 1)$, $(\pm 1, 0)$, setting $x, y \neq 0$ now gives $a - ax^2 - ay^2 = 0$, that is, $x^2 + y^2 = 1$, so the entire unit circle is critical. On that circle,

$$\operatorname{Hess} f(x,y) = -4a\, e^{-1} \begin{pmatrix} x^2 & xy \\ xy & y^2 \end{pmatrix},$$

whose determinant is $16a^2 e^{-2}(x^2y^2 - x^2y^2) = 0$. The Hessian is therefore degenerate (Definition 12.b.6) at every such point, and the test is inconclusive, as it must be, since a whole curve of critical points cannot consist of isolated extrema.

Q.E.D.

Lagrange Multipliers (Chapter 13)

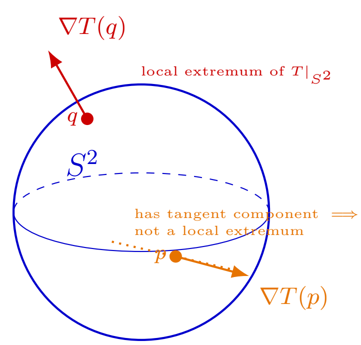
The previous chapter looked for extrema of f on an open set, where the condition $\nabla f(x_0) = 0$ is available. Restrict f to a lower-dimensional constraint set and that condition is generally too strong to hold anywhere. What replaces it is the requirement that ∇f be a linear combination of the gradients of the constraints, which says geometrically that at a constrained extremum no direction **along** the constraint set is left in which f still increases. The method carries one hypothesis on those gradients, and dropping it does not merely weaken the conclusion. It lets the method miss an extremum outright.

Lagrange multipliers (Section 13.a)

Suppose we want to find the hottest place on the surface of the earth. In theory, temperature is defined everywhere, giving some function

$$T \in C^\infty(\mathbb{R}^3, (0,\infty)) \qquad \text{(absolute zero is not attainable)}$$

If we model the earth as $S^2 := \{x \in \mathbb{R}^3 : |x| = 1\}$, then we want to find the local maxima of $T|_{S^2}$, which **are not necessarily critical points of T**!



Read the picture as follows. At p the gradient $\nabla T(p)$ has a non-zero component **along** the surface, so walking on S^2 in that direction increases T: the point p cannot be a local extremum of $T|_{S^2}$. So, if $T|_{S^2}$ **does** have a local extremum at a point, then ∇T there must have no tangential component left. It must be **perpendicular to the surface**, as at q.

Note the direction of that statement carefully: perpendicularity is **necessary, not sufficient**. Plenty of points have a perpendicular gradient without being extrema at all. On the earth, the coldest place has a perpendicular gradient just as the hottest does, and so does many a saddle. What the condition delivers is a shortlist of **candidates**. Deciding which of them are maxima, which minima, and which neither is a separate step, and in practice it is done by evaluating T at each and comparing.

Note also that S^2 is the "*level set*" (German: „*Niveaumenge*") $g(x) = 0$ for $g(x) := |x|^2 - 1$ (i.e., $S^2 = g^{-1}\{0\}$).

Remark 13.a.1 (*Gradient perpendicular to level sets*): For $g(x) := |x|^2 - 1$ we have $\nabla g(x) = 2x$, which at a point $x \in S^2$ is the outward radial direction, and therefore perpendicular to S^2. In general, the gradient is perpendicular to the level sets.

AI-Note 13.1: Corsin writes $\nabla g(x) = \frac{x}{|x|}$ here. That is the gradient of $x \mapsto |x|$, not of the $g(x) = |x|^2 - 1$ just fixed above, whose gradient is $2x$. The two differ only by the positive factor $2|x|$, so they point the same way and the geometric conclusion is unaffected. The constraint actually differentiated in the Lagrangian below is nonetheless the squared one. See Exercise 13.a.8 for the same $|x|$ versus $|x|^2$ slip in the exercise.

This intuition is made rigorous in the following.

Proposition 13.a.2 (*Constrained optimization problems*): Suppose $f : U \to \mathbb{R}$, $g : U \to \mathbb{R}^k$ are C^1 and $U \subseteq \mathbb{R}^n$ is open, and write $g := (g_1, \dots, g_k)$. If $f|_{g^{-1}\{0\}}$ has a local extremum at $x_0 \in g^{-1}\{0\}$, then

$$\{\nabla f(x_0), \nabla g_1(x_0), \dots, \nabla g_k(x_0)\} \text{ are linearly dependent.}$$

Remark 13.a.3: Once $(\nabla g_1, \dots, \nabla g_k)$ are linearly independent everywhere on $g^{-1}\{0\}$ (the standing assumption below), the constraint set $g^{-1}\{0\}$ is in fact a submanifold of $\mathbb{R}^n$. This is made precise by Theorem 17.a.3, which also explains Proposition 13.a.2 geometrically: $\nabla f(x_0)$ must be normal to the tangent space of the constraint set at a constrained extremum.

For $k = 1$, this corresponds to our intuition above of $\nabla f(x_0)$ **parallel to** $\nabla g(x_0)$.

In practice, we assume $(\nabla g_1, \dots, \nabla g_k)$ nowhere linearly dependent, since otherwise $\lambda_0 = 0$ is possible (which does not give us anything).

AI-Example 13.a.4 (*What goes wrong when ∇g vanishes*): The standing assumption is not bookkeeping. Drop it and the method can miss an extremum outright. Let

$$f(x, y) := x, \qquad M := \{(x, y) \in \mathbb{R}^2 : y^2 = x^3\},$$

the "*cuspidal cubic*" (German: „*Neilsche Parabel*“). Since $y^2 \geq 0$ forces $x^3 \geq 0$, every point of M has $x \geq 0$, and $(0, 0) \in M$. So, f attains a clear **global minimum on M at the origin**.

Now run the method with $g(x, y) := y^2 - x^3$. The Lagrange condition $\nabla f = \lambda \nabla g$ reads

$$(1, 0) = \lambda\,(-3x^2,\ 2y),$$

whose first component demands $1 = -3\lambda x^2$. At the origin this says $1 = 0$: no multiplier λ exists. The method returns **no candidates at all**, while the minimum sits there in plain sight.

The culprit is visible in $\nabla g(0, 0) = (0, 0)$: the standing assumption fails at exactly the point that matters. Geometrically, M has a **cusp** at the origin. It is not a smooth curve there, so it has no tangent line, and "*∇f perpendicular to the constraint*" has nothing to be perpendicular to.

The practical moral: before trusting a Lagrange computation, check that $\nabla g \neq 0$ (or, for several constraints, that the ∇g_j are independent) at every point of the constraint set. Where it fails, add those points to your candidate list by hand.

Remark 13.a.5 (*Where the coefficient λ_0 went*): Proposition 13.a.2 asserts linear **dependence** of $\{\nabla f(x_0), \nabla g_1(x_0), \dots, \nabla g_k(x_0)\}$, i.e. scalars $\lambda_0, \lambda_1, \dots, \lambda_k$, not all zero, with $\lambda_0 \nabla f(x_0) + \lambda_1 \nabla g_1(x_0) + \cdots + \lambda_k \nabla g_k(x_0) = 0$. The standing assumption above, that $(\nabla g_1, \dots, \nabla g_k)$ are linearly independent, is exactly what rules out $\lambda_0 = 0$ (else the ∇g_j's alone would already be dependent), so $\lambda_0 \neq 0$ and we may divide by it, silently renaming λ_j / λ_0 as λ_j. This is why the Lagrangian below carries no separate λ_0. Adrien Martelli's notes keep λ_0 explicit instead of dividing it away, pairing the dependence relation with a normalization constraint $\lambda_0^2 + \cdots + \lambda_k^2 = 1$ (any nonzero vector of multipliers can be rescaled to satisfy this). The two formulations are equivalent whenever $\lambda_0 \neq 0$; keeping λ_0 explicit only matters in degenerate settings where the independence assumption on $(\nabla g_1, \dots, \nabla g_k)$ fails and $\lambda_0 = 0$ is genuinely forced, e.g. if x_0 is itself a critical point of the constraint map g.

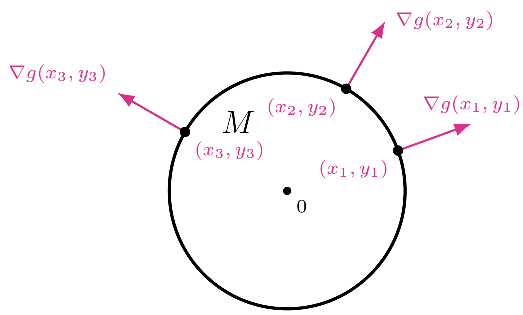
Remark 13.a.6 (*What the multiplier itself measures*): So far λ has been bookkeeping, a number produced by the method and then discarded once the candidate points are found. It carries information of its own. Consider the constrained problem with the constraint level left free,

$$m(c) := \min\{f(x) : g(x) = c\},$$

and suppose the minimiser $x^*(c)$ depends differentiably on c. Differentiating $g(x^*(c)) = c$ gives $\langle \nabla g(x^*(c)), (x^*)'(c)\rangle = 1$, while the Lagrange condition $\nabla f = \lambda \nabla g$ at $x^*(c)$ turns the chain rule for m into

$$m'(c) = \big\langle \nabla f(x^*(c)),\ (x^*)'(c)\big\rangle = \lambda\big\langle \nabla g(x^*(c)),\ (x^*)'(c)\big\rangle = \lambda.$$

So, λ is the rate at which the optimal value responds to relaxing the constraint by one unit. In economics this reading is standard enough to have its own name, the "*shadow price*" (German: „*Schattenpreis*") of the constraint. A large $|\lambda|$ says the constraint is expensive, and $\lambda = 0$ says it is not binding at all.



As the figure suggests, $\nabla g(x)$ is perpendicular to the constraint set $M := g^{-1}\{0\}$. To find extrema along M, we basically only care about the tangential part of ∇f. We can then find a multiplier λ to *"erase"* the perpendicular component of $\nabla f(x_0)$. We do this by forming the "*Lagrangian*" (German: „*Lagrange-Funktion*"):

$$\begin{aligned} L : U \times \mathbb{R}^k &\to \mathbb{R} \\ (x, \lambda) &\mapsto f(x) - \lambda \cdot g(x) = f(x) - \lambda_1 g_1(x) - \cdots - \lambda_k g_k(x) \end{aligned}$$

We then find local extrema by forcing $\nabla L(x, \lambda) = 0$, i.e.

$$\underbrace{\frac{\partial L}{\partial x_i} = 0 \quad \forall i = 1, \dots, n}_{\nabla f(x) - \lambda \cdot \nabla g(x) = 0} \qquad \text{and} \qquad \underbrace{\frac{\partial L}{\partial \lambda_j} = 0 \quad \forall j = 1, \dots, k}_{g_j(x) = 0}$$

Remark 13.a.7 (*Why the Lagrangian is built this way*): The construction can look like a trick pulled out of nowhere. It is not: L is assembled precisely so that setting **one** gradient to zero encodes **both** halves of the problem at once.

Read the two underbraces. Differentiating in the x_i reproduces the geometric condition, namely ∇f parallel to the ∇g_j, with no tangential component left. Differentiating in λ_j returns $g_j(x) = 0$, the constraint itself, because λ_j appears in L only in the term $-\lambda_j g_j(x)$, linearly. So, the multipliers are not auxiliary clutter: they are exactly the bookkeeping that lets a **constrained** problem in n variables be written as an **unconstrained** critical-point problem in $n+k$ variables, which is the one kind of problem we already know how to solve (Proposition 12.a.2).

That also explains the minus sign, which is arbitrary: writing $f + \lambda \cdot g$ merely flips the sign of every λ_j and changes nothing. And it explains why λ is discarded at the end. It is scaffolding, not part of the answer. That does not make it meaningless, and Remark 13.a.6 above says what it measures.

Exercise 13.a.8 (*Extrema of xyz on the unit ball*): Find the local extrema of

$$f : K \to \mathbb{R}, \quad (x, y, z) \mapsto xyz, \qquad K = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}.$$

Proof of Exercise 13.a.8:

We first find the **interior critical points**:

$$\nabla f(x, y, z) = (yz,\ xz,\ xy) \overset{!}{=} 0 \implies x = y = 0 \ \text{ or } \ x = z = 0 \ \text{ or } \ y = z = 0.$$

Then we look for local extrema **on the boundary**. We need to find the local extrema of $f|_{g^{-1}\{0\}}$, where $g : \mathbb{R}^3 \to \mathbb{R}$ is given by $g(x) = |x|^2 - 1$.

The Lagrangian is

$$L(x, y, z, \lambda) := f(x, y, z) - \lambda g(x, y, z) = xyz - \lambda(x^2 + y^2 + z^2 - 1),$$

giving the system of equations

$$\frac{\partial L}{\partial x} = yz - 2\lambda x = 0, \quad \frac{\partial L}{\partial \lambda} = 1 - (x^2 + y^2 + z^2) = 0.$$
$$\frac{\partial L}{\partial y} = xz - 2\lambda y = 0,$$
$$\frac{\partial L}{\partial z} = xy - 2\lambda z = 0,$$

We then have

$$2\lambda = \frac{yz}{x} = \frac{xz}{y} = \frac{xy}{z} \tag{$*$}$$

and multiplying these in pairs gives

$$\underbrace{y^2z^2 = x^2z^2}_{y^2=x^2} = \underbrace{x^2y^2}_{y^2=z^2} \qquad \text{(if any coordinate is 0, so are the others)},$$

and therefore $x^2 = y^2 = z^2$. Then $x^2 + y^2 + z^2 = 1 \implies 3x^2 = 1 \implies x = \pm\frac{1}{\sqrt{3}}$, giving **8 solutions**:

$$(x, y, z) \in \left\{ \tfrac{1}{\sqrt{3}}(\alpha, \beta, \gamma) : \alpha, \beta, \gamma \in \{\pm 1\} \right\}$$

plus the solutions $\{(x_1, x_2, x_3) \in K : x_i = x_j = 0,\ i \neq j\}$.

To justify division by z, x or y in $(*)$: if $z = 0$, then $-\lambda x = -\lambda y = xy = 0$, giving $x = 0$ or $y = 0$, and $\lambda = 0$, since $x = y = z = 0$ does not satisfy $x^2 + y^2 + z^2 = 1$. Then, since $\lambda = 0$, $\nabla f(x, y, z) = 0$ and we have already found that point in the first step. Similarly for $y = 0$ or $x = 0$.

Classifying the candidates. The work so far produced candidates, not answers (Proposition 12.a.2 and the warning after it). Since K is compact and f is continuous, f does attain a global maximum and minimum on K, so it is enough to evaluate.

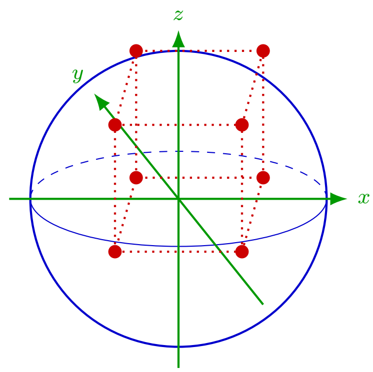
At a boundary candidate $\frac{1}{\sqrt{3}}(\alpha, \beta, \gamma)$ with $\alpha, \beta, \gamma \in \{\pm 1\}$,

$$f\Big(\tfrac{1}{\sqrt{3}}(\alpha, \beta, \gamma)\Big) = \frac{\alpha\beta\gamma}{3\sqrt{3}},$$

so the value depends only on the sign $\alpha\beta\gamma$. Four of the eight points have $\alpha\beta\gamma = +1$ and give the maximal value $\frac{1}{3\sqrt{3}}$; the other four have $\alpha\beta\gamma = -1$ and give $-\frac{1}{3\sqrt{3}}$. On the coordinate axes found in the first step, $f \equiv 0$.

Hence f attains its **global maximum** $\frac{1}{3\sqrt{3}}$ at the four points with an even number of minus signs, and its **global minimum** $-\frac{1}{3\sqrt{3}}$ at the four with an odd number. The interior candidates are **not extrema at all**: at any point of a coordinate axis $f = 0$, yet every neighbourhood contains points where $xyz > 0$ and points where $xyz < 0$, since flipping the sign of one coordinate flips the sign of the product. They are the three-variable analogue of a saddle. Q.E.D.

AI-Note 13.2: Corsin writes $g(x) = |x| - 1$ for the constraint above, but the Lagrangian uses $x^2 + y^2 + z^2 - 1$, i.e. $|x|^2 - 1$. Both cut out the same sphere; the squared version is the one actually differentiated, so it is used throughout.



8 critical points $\frac{1}{\sqrt{3}}(\pm 1, \pm 1, \pm 1)$ on S^2

Exercise 13.a.9 (*Extrema on a circle*): Find the maximum and minimum of $f(x,y) = xy$ under the condition that (x,y) lies on the circle with radius $\sqrt{2}$. (cf. Michaels Bsp 7.2.3)

Exercise 13.a.9:

We want to optimize $f(x,y) = xy$ subject to the constraint $g(x,y) = x^2 + y^2 - 2 = 0$. The Lagrangian is

$$L(x,y,\lambda) := f(x,y) - \lambda g(x,y) = xy - \lambda(x^2 + y^2 - 2).$$

Setting the gradient of L to zero yields the system:

$$\begin{aligned}
\partial_x L &= y - 2\lambda x = 0 \\
\partial_y L &= x - 2\lambda y = 0 \\
\partial_\lambda L &= 2 - (x^2 + y^2) = 0.
\end{aligned}$$

From the first two equations, $y = 2\lambda x$ and $x = 2\lambda y$. Substituting the second into the first gives $y = 4\lambda^2 y$, or $y(1 - 4\lambda^2) = 0$. If $y = 0$, then $x = 0$, which contradicts $x^2 + y^2 = 2$. Hence $y \neq 0$ and $4\lambda^2 = 1 \implies \lambda = \pm 1/2$. If $\lambda = 1/2$, then $y = x$. The constraint gives $2x^2 = 2 \implies x = \pm 1$, giving points $(1,1)$ and $(-1,-1)$. At these points, $f(\pm 1, \pm 1) = 1$. If $\lambda = -1/2$, then $y = -x$. The constraint gives $2x^2 = 2 \implies x = \pm 1$, giving points $(1,-1)$ and $(-1,1)$. At these points, $f(\pm 1, \mp 1) = -1$. Since the circle is compact, these must be the global extrema. The maximum is 1 and the minimum is -1. **Q.E.D.**

Exercise 13.a.10 (*Largest rectangle in an ellipse*): Given an ellipse $E = \{(x,y) \in \mathbb{R}^2 \mid x^2/A^2 + y^2/B^2 = 1\}$, find the rectangle (with sides parallel to the axes) whose vertices lie on E and has the largest area. (cf. Michaels Bsp 7.2.7)

Exercise 13.a.10:

Let the vertices of the rectangle in the four quadrants be $(\pm x, \pm y)$ with $x, y \geq 0$. The area of the rectangle is $f(x,y) = 4xy$. We maximize $f(x,y)$ subject to the constraint $g(x,y) = \frac{x^2}{A^2} + \frac{y^2}{B^2} - 1 = 0$ in the region $x > 0, y > 0$. The Lagrangian is

$$L(x,y,\lambda) := 4xy - \lambda\left(\frac{x^2}{A^2} + \frac{y^2}{B^2} - 1\right).$$

The critical point equations are:

$$\partial_x L = 4y - \frac{2\lambda x}{A^2} = 0 \implies \lambda = \frac{2yA^2}{x}$$
$$\partial_y L = 4x - \frac{2\lambda y}{B^2} = 0 \implies \lambda = \frac{2xB^2}{y}$$
$$\partial_\lambda L = 1 - \frac{x^2}{A^2} - \frac{y^2}{B^2} = 0.$$

Equating the expressions for λ yields $\frac{2yA^2}{x} = \frac{2xB^2}{y} \implies y^2A^2 = x^2B^2$, or $\frac{x^2}{A^2} = \frac{y^2}{B^2}$. Substituting this into the constraint gives $2\frac{x^2}{A^2} = 1 \implies x = \frac{A}{\sqrt{2}}$. Similarly, $y = \frac{B}{\sqrt{2}}$. The maximal area is $f\left(\frac{A}{\sqrt{2}}, \frac{B}{\sqrt{2}}\right) = 4 \cdot \frac{A}{\sqrt{2}} \cdot \frac{B}{\sqrt{2}} = 2AB$. **Q.E.D.**

Second-order conditions on the constraint set (Subsection 13.a.1)

Every example so far has closed the same gap in the same way. The Lagrange condition returns a list of candidates, and compactness plus a comparison of values decides among them. That works only when the constraint set is compact, and it decides only the global extrema. Is there a second-order test, of the kind Theorem 12.b.10 provides in the unconstrained case? There is, and it is that same test, applied to the Lagrangian rather than to f, and read only along directions that stay on the constraint set.

Theorem 13.a.11 (*Second-order sufficient conditions for constrained extrema*): Let $U \subseteq \mathbb{R}^n$ be open, $f \in C^2(U, \mathbb{R})$, $g = (g_1, \ldots, g_k) \in C^2(U, \mathbb{R}^k)$, and suppose $x_0 \in g^{-1}\{0\}$ satisfies:

- **(a)** $\nabla g_1(x_0), \ldots, \nabla g_k(x_0)$ are linearly independent, and
- **(b)** $\nabla f(x_0) = \sum_{j=1}^k \lambda_j \nabla g_j(x_0)$ for some $\lambda \in \mathbb{R}^k$ (a Lagrange critical point, as furnished by Proposition 13.a.2).

Write $L(x, \lambda) := f(x) - \sum_{j=1}^k \lambda_j g_j(x)$ for the Lagrangian, and let

$$T_{x_0} := \{v \in \mathbb{R}^n : Dg(x_0)v = 0\}$$

be the tangent space of the constraint set $g^{-1}\{0\}$ at x_0 (a $(n-k)$-dimensional subspace, by **(a)**). Then:

- **(1)** If $v^\mathsf{T} \operatorname{Hess}_x L(x_0, \lambda) v > 0$ for all $v \in T_{x_0} \setminus \{0\}$, then x_0 is a **strict local minimum** of $f|_{g^{-1}\{0\}}$.
- **(2)** If $v^\mathsf{T} \operatorname{Hess}_x L(x_0, \lambda) v < 0$ for all $v \in T_{x_0} \setminus \{0\}$, then x_0 is a **strict local maximum** of $f|_{g^{-1}\{0\}}$.
- **(3)** If $v^\mathsf{T} \operatorname{Hess}_x L(x_0, \lambda) v$ takes both signs on $T_{x_0} \setminus \{0\}$, then x_0 is **not** a local extremum of $f|_{g^{-1}\{0\}}$.

Here $\operatorname{Hess}_x L(x_0, \lambda)$ denotes the Hessian of $L(\cdot, \lambda)$ in the x-variable alone, evaluated at x_0.

Proof:

By **(a)** and the implicit function theorem (Theorem 16.a.1), after relabelling coordinates so that the $k \times k$ block $\partial g / \partial(x_{n-k+1}, \ldots, x_n)$ is invertible at x_0, there is an open $V \subseteq \mathbb{R}^{n-k}$ and a C^2 map $\varphi : V \to g^{-1}\{0\}$ with $\varphi(0) = x_0$, solving the constraint locally, such that $D\varphi(0) : \mathbb{R}^{n-k} \to T_{x_0}$ is an isomorphism (this is the standard local parametrization of the submanifold $g^{-1}\{0\}$; see also Theorem 17.a.3 for why it is one).

Since $g(\varphi(t)) \equiv 0$ on V, the terms $-\lambda_j g_j(\varphi(t))$ vanish identically, so

$$h(t) := f(\varphi(t)) = L(\varphi(t), \lambda) \qquad \text{for all } t \in V.$$

By the chain rule, $Dh(0) = D_x L(x_0, \lambda)\, D\varphi(0) = 0$, since $D_x L(x_0, \lambda) = \nabla f(x_0)^\mathsf{T} - \sum_j \lambda_j \nabla g_j(x_0)^\mathsf{T} = 0$ by hypothesis **(b)**: so h has a critical point at 0, i.e. $f|_{g^{-1}\{0\}}$ has a critical point at x_0, consistently with **(b)**.

Differentiating a second time, for $w \in \mathbb{R}^{n-k}$,

$$\operatorname{Hess} h(0)[w,w] = \big(D\varphi(0)w\big)^{\mathsf{T}} \operatorname{Hess}_x L(x_0,\lambda)\big(D\varphi(0)w\big) \; + \; D_x L(x_0,\lambda) \cdot D^2\varphi(0)[w,w],$$

and the second term vanishes because $D_x L(x_0,\lambda) = 0$, regardless of what $D^2\varphi(0)$ is. This is the same cancellation that already made $Dh(0) = 0$. So, writing $v := D\varphi(0)w$, which ranges over all of T_{x_0} as w ranges over $\mathbb{R}^{n-k}$ (since $D\varphi(0)$ is an isomorphism onto T_{x_0}),

$$\operatorname{Hess} h(0)[w,w] = v^{\mathsf{T}} \operatorname{Hess}_x L(x_0,\lambda)v.$$

So, $\operatorname{Hess}_x L(x_0,\lambda)$ being positive definite on T_{x_0} is exactly $\operatorname{Hess} h(0)$ being positive definite, and the unconstrained Hessian test (Theorem 12.b.10, applied to h on the open set $V \subseteq \mathbb{R}^{n-k}$) gives a strict local minimum of h at 0. Since φ is a local homeomorphism from V onto a neighbourhood of x_0 in $g^{-1}\{0\}$, that is exactly a strict local minimum of $f|_{g^{-1}\{0\}}$ at x_0. This proves **(1)**; **(2)** is identical with every inequality reversed. For **(3)**, if the quadratic form takes both signs on T_{x_0}, then $\operatorname{Hess} h(0)$ is indefinite, so h increases along one direction through $0 \in V$ and decreases along another (Theorem 12.b.10 again), and transporting those two directions through φ shows $f|_{g^{-1}\{0\}}$ does the same near x_0. **Q.E.D.**

Remark 13.a.12: This is exactly Theorem 12.b.10 run on the Lagrangian instead of on f itself, with the Hessian test's ordinary $\mathbb{R}^n$ replaced by the tangent space T_{x_0}. That is also why L, not f, is the right function to differentiate: only L has a genuine critical point at x_0 (in the full, unrestricted sense), since f itself typically does not. In practice **(1)–(3)** are checked by picking any basis $v_1,\dots,v_{n-k}$ of T_{x_0} (e.g. a basis of $\ker Dg(x_0)$) and testing the definiteness of the $(n-k)\times(n-k)$ matrix $\big(v_i^{\mathsf{T}} \operatorname{Hess}_x L(x_0,\lambda)v_j\big)_{i,j}$ with Theorem 12.b.10, or with the determinant/trace recipe from Section 12.b. There is no need to construct φ explicitly, which was only a proof device.

Example 13.a.13 (*The circle example, revisited with the second-order test*): Apply Theorem 13.a.11 to Exercise 13.a.9 ($f(x,y) = xy$ on $g(x,y) := x^2+y^2-2 = 0$) to see **why** $(1,1)$ is a maximum and $(1,-1)$ a minimum, without appealing to compactness. Here

$$\operatorname{Hess}_x L(x,y,\lambda) = \operatorname{Hess} f - \lambda \operatorname{Hess} g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} -2\lambda & 1 \\ 1 & -2\lambda \end{pmatrix},$$

which is indefinite (eigenvalues $-2\lambda \pm 1$, of opposite sign whenever $|\lambda| < \frac{1}{2}$) or degenerate (at $\lambda = \pm\frac{1}{2}$, exactly the values that occur here) on all of $\mathbb{R}^2$, so the unconstrained Hessian test would say nothing. Restricted to the 1-dimensional tangent space it is a different story.

At $(1,1)$, $\lambda = \frac{1}{2}$: $\operatorname{Hess}_x L = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, and $T_{(1,1)} = \ker \begin{pmatrix} 2 & 2 \end{pmatrix} = \operatorname{span}\{(1,-1)\}$. For $v = (1,-1)$,

$$v^{\mathsf{T}} \operatorname{Hess}_x L\, v = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 2 \end{pmatrix} = -4 < 0,$$

negative definite on $T_{(1,1)}$: a strict local maximum, matching $f(1,1) = 1$.

At $(1,-1)$, $\lambda = -\frac{1}{2}$: $\operatorname{Hess}_x L = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and $T_{(1,-1)} = \ker \begin{pmatrix} 2 & -2 \end{pmatrix} = \operatorname{span}\{(1,1)\}$. For $v = (1,1)$,

$$v^{\mathsf{T}} \operatorname{Hess}_x L\, v = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 4 > 0,$$

positive definite on $T_{(1,-1)}$: a strict local minimum, matching $f(1,-1) = -1$. The same computation at $(-1,-1)$ and $(-1,1)$ (by symmetry $f(-x,-y) = f(x,y)$) reproduces the same two conclusions. This is the sharpest illustration available of the theorem's point: the **unrestricted** Hessian of L is degenerate at both critical points, yet restricting to the 1-dimensional tangent space resolves it completely in both cases.

An application: the spectral theorem (Subsection 13.a.2)

The spectral theorem has been used freely since Remark 12.b.7, on the strength of linear algebra alone. It can also be **proved** here, and the proof is a fair test of what Proposition 13.a.2 is worth. Maximise the quadratic form $x \mapsto \langle x, Ax\rangle$ over the unit sphere, and the multiplier that falls out is an eigenvalue. Nothing is circular in this, since Proposition 13.a.2 nowhere uses diagonalisation.

Theorem 13.a.14 (*Spectral theorem, via constrained optimization*): Every symmetric $A \in \mathbb{R}^{n\times n}$ admits an orthonormal basis $\{v_1, \dots, v_n\}$ of $\mathbb{R}^n$ consisting of eigenvectors of A, with real eigenvalues.

Proof:

Let $f(x) := \langle x, Ax\rangle$, a polynomial and hence continuous, and let $g(x) := |x|^2 - 1$, so that the unit sphere is $S^{n-1} = g^{-1}\{0\}$. Since A is symmetric, $\nabla f(x) = 2Ax$; also $\nabla g(x) = 2x$.

We construct the v_j one at a time. Suppose $v_1, \dots, v_k$ are already orthonormal eigenvectors of A, with eigenvalues $\lambda_1, \dots, \lambda_k$ (for $k = 0$ there is nothing to suppose). Put

$$K_k := S^{n-1} \cap \{x : \langle x, v_j\rangle = 0 \text{ for } j = 1, \dots, k\},$$

the unit sphere of the orthogonal complement of $\operatorname{span}\{v_1, \dots, v_k\}$. This is closed and bounded, hence compact by Theorem 7.b.23, and non-empty as long as $k < n$, so f attains a maximum on it at some $v_{k+1} \in K_k$.

The constraints active at v_{k+1} are g together with the k linear functions $h_j(x) := \langle x, v_j\rangle$, whose gradients are $\nabla g(v_{k+1}) = 2v_{k+1}$ and $\nabla h_j(v_{k+1}) = v_j$. These $k+1$ vectors are orthogonal and non-zero, hence linearly independent, so Proposition 13.a.2 applies and yields multipliers with

$$2Av_{k+1} = 2\lambda_{k+1}v_{k+1} + \sum_{j=1}^{k} \mu_j v_j.$$

Pair both sides with v_i for a fixed $i \le k$. On the right, $\langle v_{k+1}, v_i\rangle = 0$ and $\langle v_j, v_i\rangle = \delta_{ji}$, leaving μ_i. On the left, symmetry of A moves it across the inner product,

$$\langle Av_{k+1}, v_i\rangle = \langle v_{k+1}, Av_i\rangle = \lambda_i\langle v_{k+1}, v_i\rangle = 0,$$

using that v_i is already an eigenvector. Hence $\mu_i = 0$ for every $i \le k$, and the displayed identity collapses to $Av_{k+1} = \lambda_{k+1}v_{k+1}$. So, v_{k+1} is a unit eigenvector orthogonal to all its predecessors, and after n steps the v_j form the required basis. Each eigenvalue is real because $\lambda_j = \langle v_j, Av_j\rangle$ is a real number by construction. **Q.E.D.**

Exercise 13.a.15 (*Antipodal points maximise distance on the sphere*): Let $n \ge 2$. Prove that two points $x, y \in S^{n-1}$ have maximal distance $\|x - y\|$ if and only if $x = -y$. Consider the function $(x, y) \mapsto \|x - y\|^2$ on $S^{n-1} \times S^{n-1} \subset \mathbb{R}^{2n}$.

Proof of Exercise 13.a.15:

Since $\|x\| = \|y\| = 1$ on S^{n-1},

$$f(x, y) := \|x - y\|^2 = \|x\|^2 - 2\langle x, y\rangle + \|y\|^2 = 2 - 2\langle x, y\rangle,$$

so maximising $\|x-y\|$ is the same as maximising f, i.e. minimising $\langle x, y\rangle$. The domain $S^{n-1}\times S^{n-1}$ is compact (a product of compact sets), so f attains a maximum somewhere by the extreme value theorem; we locate it with Lagrange multipliers under the two constraints $g_1(x, y) := \|x\|^2 - 1 = 0$ and $g_2(x, y) := \|y\|^2 - 1 = 0$.

Computing gradients in $\mathbb{R}^{2n} = \mathbb{R}^n_x \times \mathbb{R}^n_y$:

$$\nabla f = (-2y,\ -2x), \qquad \nabla g_1 = (2x,\ 0), \qquad \nabla g_2 = (0,\ 2y).$$

At a critical point, $\nabla g_1, \nabla g_2$ are linearly independent (they are non-zero and supported on disjoint blocks of coordinates), so Proposition 13.a.2 gives λ_1, λ_2 with $\nabla f = \lambda_1\nabla g_1 + \lambda_2\nabla g_2$, i.e.

$$-2y = 2\lambda_1 x, \qquad -2x = 2\lambda_2 y \qquad \Longrightarrow \qquad y = -\lambda_1 x, \quad x = -\lambda_2 y.$$

So, x and y are parallel. Combined with $\|x\| = \|y\| = 1$, this forces $y = x$ or $y = -x$. These are the only two candidates, and $f(x, x) = 0$ while $f(x, -x) = \|2x\|^2 = 4$. The maximum of f over the compact domain is therefore 4, attained exactly when $y = -x$ (i.e. exactly at antipodal pairs, where $\|x - y\| = 2$). **Q.E.D.**

Corollary 13.a.16 (*Extreme eigenvalues as Rayleigh quotients*): For symmetric $A \in \mathbb{R}^{n\times n}$,

$$\lambda_{\min}(A) = \min_{x\in S^{n-1}} \langle x, Ax\rangle, \qquad \lambda_{\max}(A) = \max_{x\in S^{n-1}} \langle x, Ax\rangle.$$

Proof:

Write $x \in S^{n-1}$ in the orthonormal eigenbasis of Theorem 13.a.14 as $x = \sum_j c_j v_j$ with $\sum_j c_j^2 = 1$. Then $\langle x, Ax\rangle = \sum_j \lambda_j c_j^2$ is a convex combination of the eigenvalues, so it lies between $\lambda_{\min}$ and $\lambda_{\max}$; both bounds are attained by taking x to be the corresponding eigenvector. **Q.E.D.**

AI-Exercise 13.a.17 (*Extrema on a Circle via Lagrange Multipliers*): Find the maximum and minimum values of $f(x, y) := x + 2y$ subject to the constraint $g(x, y) := x^2 + y^2 - 5 = 0$.

Exercise 13.a.18 (*5.2 — Lagrange multipliers* (important)): Consider the ball $U := \{(x, y, z) : x^2 + y^2 + z^2 < 1\}$ and the function

$$f(x, y, z) := 1 - x^2 - y^2 + 4x.$$

1. Motivate rigorously why f attains its absolute maximum and minimum in $\overline{U}$.
2. Find all the critical points of f which lie in the interior of U. Say whether $\sup_U f$ (or $\inf_U f$) is attained at some point in U.
3. Say whether $\max_{\partial U} f$ and $\min_{\partial U}$ exist.
4. With the method of Lagrange multipliers, find the possible critical points for the constrained problem $\max\{f(x, y, z) : (x, y, z) \in \partial U\}$, and same for min.
5. Among all the points you have found, clarify at which points the maximum/minimum of f is attained.

Exercise 13.a.19 (*6.10 — Lagrange multipliers* (important)): Consider the function $f(x, y, z) = 3x - y + 2z$ and the set

$$M = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1,\ x + y = 0\}.$$

Determine the extrema of f on M and their nature.

Exercise 13.a.20 (*6.11 — Closest point on a hyperbolic paraboloid* (important)): Find the points on the submanifold

$$M = \{(x, y, z) \in \mathbb{R}^3 : z = x^2 - y^2\}$$

that are closest to the point $(0, 0, 1)$.

Solutions (Section 13.b)

Solution to Exercise 13.a.18:

1. $\overline{U}$ is closed and bounded in $\mathbb{R}^3$, hence compact by Theorem 7.b.23, and f is a polynomial, hence continuous. By the extreme value theorem (Item **(b)**), f attains its absolute maximum and minimum on $\overline{U}$.
2. $\nabla f(x, y, z) = (-2x + 4,\ -2y,\ 0)$ vanishes only at $x = 2$, $y = 0$ (any z), and no such point lies in U (since $x = 2$ already violates $x^2 + y^2 + z^2 < 1$). So, f has **no critical point in the interior of U**; consequently, by Proposition 12.a.2, $\sup_U f$ and $\inf_U f$ are **not attained** at any interior point.

3. ∂U is compact (a sphere) and f is continuous, so $\max_{\partial U} f$ and $\min_{\partial U} f$ **both exist**, again by the extreme value theorem (Item **(b)**).
4. With $g(x,y,z) := x^2+y^2+z^2-1$, the Lagrange condition $\nabla f = \lambda \nabla g$ reads $(-2x+4,-2y,0) = \lambda(2x,2y,2z)$. The last coordinate gives $\lambda z = 0$. **If $z \neq 0$:** then $\lambda = 0$, so $-2x+4=0 \Rightarrow x = 2$, impossible on ∂U. So, $z=0$. Then $x^2+y^2=1$, and the remaining equations are $4-2x = 2\lambda x$ and $-2y = 2\lambda y$, i.e. $-2y(1+\lambda)=0$. **If $\lambda = -1$:** the first equation becomes $4-2x=-2x$, i.e. $4=0$, impossible. So $y=0$, and $x^2=1$ gives $x=\pm 1$. The Lagrange candidates are $(1,0,0)$ (with $\lambda=1$) and $(-1,0,0)$ (with $\lambda=-3$).
5. Evaluating, $f(1,0,0) = 1-1+4 = 4$ and $f(-1,0,0) = 1-1-4 = -4$. Since there are no interior critical points, these are the global extrema on $\overline{U}$: the **maximum** 4 is attained at $(1,0,0)$, and the **minimum** -4 at $(-1,0,0)$.

Q.E.D.

Proof of AI-Exercise 13.a.17:

The constraint set $S := \{(x,y)\in\mathbb{R}^2 \mid x^2+y^2=5\}$ is compact and $\nabla g(x,y) = (2x,2y)^\mathsf{T} \neq (0,0)^\mathsf{T}$ on S. By Lagrange Multipliers, candidates satisfy $\nabla f = \lambda \nabla g$, i.e.:

$$1 = 2\lambda x \quad \text{and} \quad 2 = 2\lambda y \implies y = 2x.$$

Substituting $y=2x$ into $x^2+y^2=5$ gives $5x^2=5 \implies x=\pm 1$. The candidate points are $(1,2)$ and $(-1,-2)$. Evaluating f:

$$f(1,2) = 1+4 = 5 \quad \text{(maximum)}, \qquad f(-1,-2) = -1-4 = -5 \quad \text{(minimum)}.$$

Q.E.D.

Solution to Exercise 13.a.19:

With $g_1(x,y,z) := x^2+y^2+z^2-1$ and $g_2(x,y,z) := x+y$, the Lagrangian $L := f - \lambda_1 g_1 - \lambda_2 g_2$ gives the system

$$3-2\lambda_1 x - \lambda_2 = 0, \qquad -1-2\lambda_1 y - \lambda_2 = 0, \qquad 2-2\lambda_1 z = 0, \qquad x+y=0, \qquad x^2+y^2+z^2=1.$$

From $x+y=0$, $y=-x$. Subtracting the second equation from the first eliminates λ_2:

$$4-2\lambda_1 x - 2\lambda_1(-x)\cdot(-1) = 4-2\lambda_1 x - 2\lambda_1 x = 4-4\lambda_1 x = 0 \implies \lambda_1 x = 1.$$

From $2-2\lambda_1 z = 0$, $\lambda_1 = 1/z$ (so $z\neq 0$), hence $x=z$ (from $\lambda_1 x = 1 = \lambda_1 z$). With $y=-x=-z$, the constraint becomes $3x^2=1$, i.e. $x = \pm\frac{1}{\sqrt{3}}$, giving the two points $\frac{1}{\sqrt{3}}(1,-1,1)$ and $-\frac{1}{\sqrt{3}}(1,-1,1)$. Evaluating,

$$f\left(\tfrac{1}{\sqrt{3}}(1,-1,1)\right) = \frac{3+1+2}{\sqrt{3}} = 2\sqrt{3}, \qquad f\left(-\tfrac{1}{\sqrt{3}}(1,-1,1)\right) = -2\sqrt{3}.$$

Since M is compact (closed and bounded in $\mathbb{R}^3$) and f is continuous, these are the only Lagrange candidates, so f attains its **maximum** $2\sqrt{3}$ at $\frac{1}{\sqrt{3}}(1,-1,1)$ and its **minimum** $-2\sqrt{3}$ at $-\frac{1}{\sqrt{3}}(1,-1,1)$.

Q.E.D.

Solution to Exercise 13.a.20:

Minimizing the distance to $(0,0,1)$ is the same as minimizing $f(x,y,z) := x^2+y^2+(z-1)^2$ subject to $g(x,y,z) := z-x^2+y^2 = 0$. Since f is a distance squared to a fixed point, its minimum over the (closed) constraint set is attained. Lagrange's condition $\nabla f = \lambda\nabla g$ reads

$$2x = -2\lambda x, \qquad 2y = 2\lambda y, \qquad 2(z-1) = \lambda,$$

i.e. $x(1+\lambda) = 0$ and $y(1-\lambda)=0$.

Case $x=y=0$: the constraint gives $z=0$, so $(0,0,0)$ with $d^2=1$.

Case $x \neq 0$ (so $\lambda = -1$): then $2(z-1) = -1 \implies z = \frac{1}{2}$, and $y(1-\lambda) = 2y = 0 \implies y = 0$. The constraint $z = x^2 - y^2$ gives $x^2 = \frac{1}{2}$, i.e. $x = \pm\frac{1}{\sqrt{2}}$, giving $\left(\pm\frac{1}{\sqrt{2}}, 0, \frac{1}{2}\right)$ with $d^2 = \frac{1}{2} + 0 + \frac{1}{4} = \frac{3}{4}$.

Case $y \neq 0$ (so $\lambda = 1$): then $2(z-1) = 1 \implies z = \frac{3}{2}$, and $x(1+\lambda) = 2x = 0 \implies x = 0$; the constraint becomes $-y^2 = \frac{3}{2}$, which has no real solution.

Comparing $d^2 = 1$ against $d^2 = \frac{3}{4}$, the **closest points** are $\left(\pm\frac{1}{\sqrt{2}}, 0, \frac{1}{2}\right)$, at minimal distance $\frac{\sqrt{3}}{2}$. **Q.E.D.**

Convexity (Chapter 14)

Convexity is the hypothesis under which the local information of the previous two chapters becomes global. For a convex function a critical point is not merely a local minimum but the minimum, and the Hessian test stops being a test at all, since positive semidefiniteness at every point is one of the ways of saying that a function is convex. What has to be earned is that the three natural ways of saying it (by chords, by tangent planes, by the Hessian) really are the same condition.

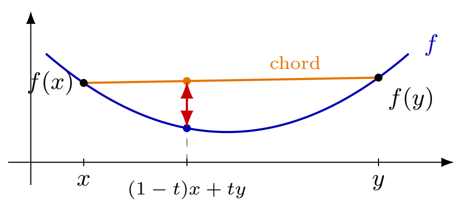
Convexity (Section 14.a)

Definitions and characterizations (Subsection 14.a.1)

Definition 14.a.1 (*Convex set and convex function*): A subset $A \subseteq \mathbb{R}^n$ is "*convex*" (German: „*konvex*“) if for all $x, y \in A$ and every $t \in [0,1]$, the point $p := (1-t)x + ty$ is an element of A.

A function $f : A \to \mathbb{R}$ is convex if A is convex and, for all $x, y \in A$ and $t \in [0,1]$,

$$f\big((1-t)x + ty\big) \le (1-t)f(x) + tf(y).$$



Definition 14.a.2 (*Semidefinite matrices*): Let $A \in \mathbb{R}^{n\times n}$ be symmetric. Then A is

- "*positive semidefinite*" (German: „*positiv semidefinit*“), also called **nonnegative definite**, if $v^\mathsf{T} A v \ge 0$ for all $v \in \mathbb{R}^n$;
- "*negative semidefinite*" (German: „*negativ semidefinit*“), or **nonpositive definite**, if $v^\mathsf{T} A v \le 0$ for all $v \in \mathbb{R}^n$.

These are the boundary cases of Definition 12.b.6: positive definite implies positive semidefinite, and the two differ exactly on the degenerate matrices, where $v^\mathsf{T} A v = 0$ is allowed for some $v \ne 0$.

Proposition 14.a.3 (*Characterizations of convexity*): Let $U \subseteq \mathbb{R}^n$ be open and convex, and let $f \in C^2(U)$. Then the following are equivalent:

(a) f is convex;

(b) $\operatorname{Hess} f(x)$ is positive semidefinite for all $x \in U$;

(c) $f(y) - f(x) \ge Df_x(y-x)$ for all $x, y \in U$.

Example 14.a.4 (*The parabola*): The parabola $p : \mathbb{R}^n \to \mathbb{R},\ x \mapsto |x|^2 = \sum_{i=1}^n x_i^2$, is convex:

$$\partial_i \partial_j p(x) = 2\delta_{ij} \implies \operatorname{Hess} p(x) = 2\,\mathrm{id}_n,$$

and every eigenvalue is $2 > 0$, so $\operatorname{Hess} p(x)$ is positive definite (hence positive semidefinite) for every x, and p is convex by Proposition 14.a.3.

Exercise 14.a.5 (*Convex or not?*): Decide which of the following are convex, and prove it. Linus's advice is worth following: attempt each one **without** plotting it, since that is the situation in an exam. To show a function **is** convex, computing the Hessian and checking Proposition 14.a.3**(b)** is usually quickest; to show one is **not**, it is usually fastest to exhibit a single triple x, y, t violating the defining inequality.

(a) $f : [0,\infty)^2 \to \mathbb{R},\ f(x,y) := xy$.
(b) $f : \mathbb{R}^2 \to \mathbb{R},\ f(x,y) := x^2 + xy + y^2$.
(c) $f : [0,\pi]^2 \to \mathbb{R},\ f(x,y) := \cos(x)\cos(y)$.
(d) If $g, h : \mathbb{R} \to \mathbb{R}$ are convex, is $(x,y) \mapsto g(x)h(y)$ necessarily convex?

Proof of the characterizations (Subsection 14.a.2)

Exercise 14.a.6 (*Equivalent characterizations of convexity*): Let $U \subseteq \mathbb{R}^n$ be open and convex.

(a) Show that $f : U \to \mathbb{R}$ is convex if and only if the functions

$$f_{x,y} : [0,1] \to \mathbb{R}, \qquad s \mapsto f(x + s(y-x))$$

are convex for all $x, y \in U$.

(b) Show that

$$f \text{ convex} \iff f(y) - f(x) \geq Df_x(y-x) \quad \text{for all } x, y \in U.$$

This also holds for $f \in C^1(U)$, $U \subseteq \mathbb{R}^n$ open and convex.

(c) Conclude that if $f \in C^1(U)$ is convex and has a critical point at $x_0 \in U$, then f has a minimum at x_0.

Solution to Exercise 14.a.6:

(a) By definition, $f_{x,y}$ is convex if and only if

$$f_{x,y}\big((1-t)s_0 + ts_1\big) \leq (1-t)f_{x,y}(s_0) + tf_{x,y}(s_1) \quad \text{for all } s_0, s_1, t \in [0,1],$$

which, unwinding the definition of $f_{x,y}$, reads

$$f\big(x + [(1-t)s_0 + ts_1](y-x)\big) \leq (1-t)f(x + s_0(y-x)) + tf(x + s_1(y-x)),$$

that is,

$$f\big((1-t)(x + s_0(y-x)) + t(x + s_1(y-x))\big) \leq (1-t)f(x + s_0(y-x)) + tf(x + s_1(y-x)).$$

Setting $s_0 = 0$, $s_1 = 1$ recovers exactly the convexity inequality for f at the points x and y, so if this holds for all $x, y \in U$, then f is convex. Conversely, if f is convex, then convexity of $f_{x,y}$ for each fixed $x, y \in U$ follows by applying convexity of f to the two points $x + s_0(y-x)$ and $x + s_1(y-x)$, which both lie in U since U is convex.

(b) Recall that $g \in C^1(I)$, for $I \subseteq \mathbb{R}$ an interval, is convex if and only if g' is increasing. Combined with **(a)**, this gives

$$f \text{ convex} \iff f'_{x,y} \text{ is increasing, for all } x, y \in U.$$

By the chain rule, for $t \in [0,1]$,

$$f'_{x,y}(t) = Df_{x+t(y-x)}(y-x),$$

and by the mean value theorem there exists $s \in (0,1)$ such that

$$f(y) - f(x) = f_{x,y}(1) - f_{x,y}(0) = f'_{x,y}(s) = Df_{x+s(y-x)}(y-x).$$

($\Rightarrow$) If f is convex, then $f_{x,y}$ is convex for all $x, y \in U$ by **(a)**, so $f'_{x,y}$ is increasing, and in particular $f'_{x,y}(s) \geq f'_{x,y}(0)$. Combined with the mean value theorem identity above, this yields

$$f(y) - f(x) \geq Df_x(y-x).$$

($\Leftarrow$) Suppose $f(y) - f(x) \geq Df_x(y-x)$ for all $x, y \in U$. We must show that $0 \leq s < t \leq 1 \implies f'_{x,y}(s) \leq f'_{x,y}(t)$. Set $u := x + s(y-x)$ and $v := x + t(y-x)$, so that $u - v = (s-t)(y-x)$. Applying the hypothesis to the pairs (u,v) and (v,u) gives

$$Df_u(v-u) \leq f(v) - f(u), \qquad Df_v(u-v) \leq f(u) - f(v).$$

Therefore,

$$(t-s)f'_{x,y}(s) = Df_u(v-u) \leq f(v) - f(u) \leq -Df_v(u-v) = Df_v(v-u) = (t-s)f'_{x,y}(t),$$

and since $t - s > 0$, this gives $f'_{x,y}(s) \le f'_{x,y}(t)$, i.e., $f'_{x,y}$ is increasing, hence $f_{x,y}$ is convex for all $x, y \in U$, and so f is convex by **(a)**.

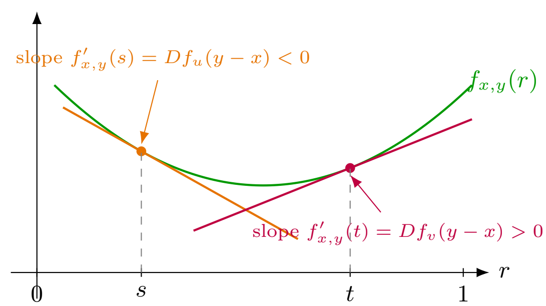
Remark 14.a.7: The argument above only ever used differentiability of f, never twice-differentiability, so the equivalence in **(b)** already holds for $f \in C^1(U)$ with $U \subseteq \mathbb{R}^n$ open and convex, one order of regularity weaker than Proposition 14.a.3, which is stated for $f \in C^2(U)$. This is exactly the fact used in Exercise 14.a.11**(b)**.

(c) Since f is convex and $x_0 \in U$ is a critical point, $Df_{x_0} = 0$, so **(b)** gives

$$0 = Df_{x_0}(y - x_0) \le f(y) - f(x_0) \quad \text{for all } y \in U,$$

i.e., $f(y) \ge f(x_0)$ for all $y \in U$. Hence f has a minimum at x_0, and in fact a global minimum on U. In the convex case this sharpens Proposition 12.a.2, which on its own only guarantees that critical points are the sole **candidates** for extrema.

Q.E.D.



The picture records both halves of **(b)** at once. Each tangent line lies weakly **below** the graph, which is the inequality $f(y) - f(x) \ge Df_x(y - x)$ read along the segment, and the slope grows from s to t, which is $f'_{x,y}$ increasing.

Proposition 14.a.8 (*Jensen's inequality*): Let $U \subseteq \mathbb{R}^n$ be open and convex and $f : U \to \mathbb{R}$ convex. Then for every finite collection of points $x_1, \dots, x_N \in U$ and weights $w_1, \dots, w_N \in (0,1)$ with $\sum_{i=1}^N w_i = 1$,

$$f\Big(\sum_{i=1}^N w_i x_i\Big) \;\le\; \sum_{i=1}^N w_i f(x_i).$$

Proof:

Let $z := \sum_i w_i x_i \in U$ (U is convex, so this weighted average lies in U). By Exercise 14.a.6**(b)** (valid already for $f \in C^1$), applied to the pair (z, x_i) for each i,

$$f(x_i) - f(z) \ge Df_z(x_i - z), \qquad i = 1, \dots, N.$$

Multiply the i-th inequality by $w_i > 0$ and sum over i:

$$\sum_i w_i f(x_i) - f(z) = \sum_i w_i\big(f(x_i) - f(z)\big) \ge Df_z\Big(\sum_i w_i(x_i - z)\Big) = Df_z\Big(\sum_i w_i x_i - z\Big) = Df_z(0) = 0,$$

using $\sum_i w_i = 1$ to pull $f(z)$ out on the left and $\sum_i w_i x_i = z$ in the last step. Rearranging gives $f(z) \le \sum_i w_i f(x_i)$, which is the claim. (For f merely convex, without C^1 regularity, the same computation goes through with Exercise 14.a.6**(b)** replaced by the general subgradient inequality, which holds for any convex function on an open set.) **Q.E.D.**

Remark 14.a.9: The case $N = 2$ is exactly Definition 14.a.1 itself: Jensen's inequality is the statement that convexity, defined only for a two-point average, automatically extends to any finite weighted average. It is used silently in the solution to Exercise 2.c.40 (Chapter 2), applied to $\varphi(s) := s \log s$, and is stated explicitly here for the first time.

Example 14.a.10 (*Convex, bounded below, and yet no minimum*): Exercise 14.a.6**(c)** guarantees that **if** a convex function has a critical point, that point is a global minimum. It does not guarantee that one exists. Take

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x_1, x_2) := e^{x_1+x_2}.$$

Its Hessian is

$$\operatorname{Hess} f(x_1, x_2) = e^{x_1+x_2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

so for every $v \in \mathbb{R}^2$ we get $v^{\mathsf{T}} \operatorname{Hess} f(x)\, v = e^{x_1+x_2}(v_1 + v_2)^2 \geq 0$. The Hessian is positive semidefinite everywhere, so f is convex by Proposition 14.a.3. But $f > 0$ everywhere, while along the ray $(-t, -t)$ we have $f(-t, -t) = e^{-2t} \to 0$ as $t \to \infty$. Hence $\inf_{\mathbb{R}^2} f = 0$ and the infimum is attained nowhere.

What fails is not convexity but compactness: the sublevel sets $\{f \leq c\}$ are unbounded, so the extreme value theorem (Lemma 7.a.7) has nothing to act on. Convexity organises the minima a function has. It does not manufacture one.

Convexity is worth the trouble because of what it removes. Under it there is no saddle to rule out, no degenerate case in which the second-order test falls silent, and no gap left between a local and a global minimum. What it does not remove is the need for a minimum to exist at all, which is the one thing Example 14.a.10 was there to show.

Exercise 14.a.11 (*6.7 — Multiple choice* (important)): Among the following statements about convex functions mark those (and only those) which are always true.

(a) If $f \in C^1(U)$ is convex in some open convex set $U \subset \mathbb{R}^n$ and f has a local maximum at $z \in U$, then $\nabla f \equiv 0$ in U.

(b) If $f \in C^1(U)$ is convex in some open convex set $U \subset \mathbb{R}^n$ and f has a global maximum at $z \in U$, then $\nabla f \equiv 0$ in U.

(c) Assume $f_n \in C^2(\mathbb{R})$ is a sequence of convex functions that converge pointwise to some $f : \mathbb{R} \to \mathbb{R}$. Is f necessarily convex?

(d) There exists a convex function $f \in C^\infty(\mathbb{R}^2)$ such that

$$f(x) = 1 - 2x_1 + x_2^3 + O(|x|^4) \quad \text{as } |x| \to 0.$$

(e) There exists a convex function $f \in C^\infty(\mathbb{R}^2)$ such that

$$f(x) = 1 - 2x_1 + x_2^4 + O(|x|^4) \quad \text{as } |x| \to 0.$$

(f) A convex set is not necessarily connected.

Solutions (Section 14.b)

Solution to Exercise 14.a.11:

(a) **False.** Let $f(x) := \max\{x_1, \frac{1}{2}\}^4$ on $U := B_1 \subset \mathbb{R}^n$. The inner function $x \mapsto \max\{x_1, \frac{1}{2}\}$ is convex and takes values in $[\frac{1}{2}, \infty)$, and $t \mapsto t^4$ is convex and increasing there. A convex increasing function composed with a convex function is convex, so f is convex. For all $x \in U$ with $x_1 \leq \frac{1}{2}$, $f(x) = (\frac{1}{2})^4$ identically, so f has a (non-strict) local maximum at $x = 0$, yet ∇f does not vanish once $x_1 > \frac{1}{2}$.

Remark 14.b.12: Worth noting: $x = 0$ here is simultaneously a (non-strict) local **minimum** too, since f is locally constant on the whole region $\{x_1 \leq \frac{1}{2}\} \cap U$. This is exactly why the counterexample doesn't contradict **(b)**: $x = 0$ is not a **global** maximum of f on U (as f grows without bound as $x_1 \to 1$), so **(b)**'s stronger hypothesis simply isn't triggered.

(b) **True.** Since z is a local maximum, $\nabla f(z) = 0$ (Proposition 12.a.2). Convexity gives $f(x) \geq f(z) + \nabla f(z) \cdot (x - z) = f(z)$ for all $x \in U$; but z being a global maximum also gives $f(x) \leq f(z)$ for all $x \in U$. Hence $f \equiv f(z)$ on U, so $\nabla f \equiv 0$ in U.

(c) **True.** Fix $x, y \in U$ and $t \in [0,1]$. Convexity of each f_n gives $f_n(tx + (1-t)y) \leq t f_n(x) + (1-t) f_n(y)$ for every n; letting $n \to \infty$ (with x, y, t fixed) and using pointwise convergence yields the same inequality for f. Since x, y, t were arbitrary, f is convex.

(d) **False.** A convex function lies above its tangent plane, so from the Taylor data at 0, any such f would satisfy $f(x) \geq 1 - 2x_1$ for all $x \in \mathbb{R}^n$. Combined with $f(x) = 1 - 2x_1 + x_2^3 + O(|x|^4)$, this forces $x_2^3 \geq -M|x|^4$ near 0 for some $M > 0$, which fails along $x = (0, -r, 0, \ldots, 0)$ as $r \downarrow 0$ (left side $\sim -r^3$, right side $\sim -Mr^4$, and $-r^3 < -Mr^4$ for small r). Contradiction, so no such convex f exists.

(e) **True.** Simply take $f(x) := 1 - 2x_1 + x_2^4$ itself (with the $O(|x|^4)$ remainder identically zero): it is convex, being a sum of the affine (hence convex) term $1 - 2x_1$ and the convex term x_2^4.

(f) **False.** A convex set is always path-connected, since for any two points the straight segment joining them lies inside the set by definition of convexity; path-connected sets are connected.

Q.E.D.

Solution to Exercise 14.a.5:

(a) **Not convex.** The Hessian is constant,

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

with eigenvalues ± 1, hence indefinite, so Proposition 14.a.3**(b)** fails. Explicitly, take $u := (1,0)$, $v := (0,1)$ and $t := \frac{1}{2}$: the midpoint is $(\frac{1}{2}, \frac{1}{2})$ with $f = \frac{1}{4}$, while $\frac{1}{2} f(u) + \frac{1}{2} f(v) = 0$. Since $\frac{1}{4} > 0$, the defining inequality is violated.

(b) **Convex.** Again the Hessian is constant,

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

with eigenvalues 2 ± 1, i.e. 3 and 1, both strictly positive. So, the Hessian is positive definite everywhere and f is convex, in fact strictly so.

Without eigenvalues: $x^2 + xy + y^2 = \left(x + \frac{y}{2}\right)^2 + \frac{3}{4} y^2$ is a sum of squares of affine functions, and such a sum is always convex.

(c) **Not convex.** Differentiating,

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} -\cos x \cos y & \sin x \sin y \\ \sin x \sin y & -\cos x \cos y \end{pmatrix}, \qquad \text{so} \qquad \operatorname{Hess} f\left(\tfrac{\pi}{2}, \tfrac{\pi}{2}\right) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which is indefinite. For an explicit violation, run along the anti-diagonal: with $u := (0, \pi)$, $v := (\pi, 0)$ and $t := \frac{1}{2}$ the midpoint is $(\frac{\pi}{2}, \frac{\pi}{2})$, where $f = 0$, whereas $\frac{1}{2} f(u) + \frac{1}{2} f(v) = \frac{1}{2}(-1) + \frac{1}{2}(-1) = -1$. Since $0 > -1$, convexity fails.

Note that the **main** diagonal would not have exposed this: from $(0,0)$ to (π, π) the midpoint value 0 sits below the average 1, consistent with convexity. Choosing the direction well is most of the work in these problems, and the indefinite Hessian tells you which direction to choose, namely an eigenvector for the negative eigenvalue, here $(1, -1)$.

(d) **No.** Part **(a)** is already a counterexample: $g(x) := x$ and $h(y) := y$ are affine, hence convex, yet $g(x)h(y) = xy$ is not. Convexity is preserved by sums and by non-negative scalar multiples, but not by products. That is why Proposition 14.a.3 is stated for a single function rather than as a calculus of convex building blocks.

Q.E.D.

PART V

Inverse and Implicit Functions; Submanifolds

Two questions run through this part. When can an equation $f(x, y) = 0$ be solved for y as a function of x, and when is a map invertible near a point? Neither has a workable global answer, but both have clean local ones, and both reduce to the same linear-algebra hypothesis: invertibility of a Jacobian. That is the content of the implicit and inverse function theorems, which turn out to be equivalent to each other and are proved here from the contraction principle of Part II. Their deeper use is definitional. A level set on which the differential is surjective is locally a graph, and being locally a graph is precisely what it means to be a submanifold, a subset of $\mathbb{R}^n$ that looks like $\mathbb{R}^m$ from close up. The closing chapters develop that notion, since submanifolds are the objects the last two parts integrate over.

The Inverse Function Theorem (Chapter 15)

So far the differential has been used to **describe** a map near a point. The inverse function theorem turns that around. If the differential at a point is invertible, then so is the map itself near that point, and the inverse is exactly as differentiable as the map was. Everything in the rest of this part rests on that one statement.

The inverse function theorem (Section 15.a)

Theorem 15.a.1 (*Inverse function theorem*): Let $U \subseteq \mathbb{R}^n$ be open, let $f \in C^k(U, \mathbb{R}^n)$, and let $x_0 \in U$ be such that Df_{x_0} is invertible. Then there exists an open neighbourhood $U_0 \subseteq U$ of x_0 such that

$$f|_{U_0} : U_0 \to f(U_0)$$

is a C^k-"*diffeomorphism*" (German: „*Diffeomorphismus*") and

$$D\big(f|_{U_0}^{-1}\big)_{f(x)} = D\big(f|_{U_0}\big)_x^{-1} \quad \text{for all } x \in U_0.$$

Remark 15.a.2 (*Why invertibility of Df_{x_0} is the right hypothesis*): In one variable, consider $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$. The inverse $f^{-1}(y) = y^{1/3}$ exists everywhere, yet it fails to be differentiable at $y = 0$. Geometrically, the graph of f^{-1} is obtained by reflecting the graph of f across the line $y = x$. The horizontal tangent line of f at $x = 0$ (where $f'(0) = 0$) flips into a vertical tangent line for f^{-1} at $y = 0$, causing the derivative of the inverse to blow up.

In multivariable calculus, for a map $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$, the tangent space to the graph of F at $(x_0, F(x_0))$ is the affine subspace

$$\big\{(x_0 + v,\ F(x_0) + DF_{x_0}(v)) \mid v \in \mathbb{R}^n\big\} \subseteq \mathbb{R}^n \times \mathbb{R}^n.$$

A *"horizontal"* direction (a non-zero input displacement producing zero output change) corresponds to a non-zero vector $v \in \mathbb{R}^n$ such that $DF_{x_0}(v) = 0$. To eliminate vertical tangents after inversion, we must require DF_{x_0} to be injective. Since F maps between spaces of the same dimension n, injectivity of DF_{x_0} is equivalent to its invertibility.

Lemma 15.a.3 (*Lipschitz perturbations of the identity*): Let $U \subseteq \mathbb{R}^n$ be open, and let $F : U \to \mathbb{R}^n$ be a map of the form

$$F(x) := x + \phi(x),$$

where $\phi : U \to \mathbb{R}^n$ is a λ-Lipschitz function with constant $\lambda \in (0, 1)$. Then:

(a) For every open ball $B_r(x) \subseteq U$, the image $F(B_r(x))$ satisfies the inclusions

$$B_{(1-\lambda)r}(F(x)) \ \subseteq\ F(B_r(x)) \ \subseteq\ B_{(1+\lambda)r}(F(x)).$$

In particular, $F(U) \subseteq \mathbb{R}^n$ is an open set.

(b) $F : U \to F(U)$ is injective, and its inverse $F^{-1} : F(U) \to U$ is $\frac{1}{1-\lambda}$-Lipschitz continuous.

Proof:

(a) Fix $y_0 \in B_{(1-\lambda)r}(F(x))$. We seek a point $x_0 \in B_r(x)$ satisfying $F(x_0) = y_0$, which is equivalent to the fixed-point equation $x_0 = y_0 - \phi(x_0)$. Define $g : \overline{B_r(x)} \to \mathbb{R}^n$ by $g(z) := y_0 - \phi(z)$. For any $z_1, z_2 \in \overline{B_r(x)}$,

$$\|g(z_1) - g(z_2)\| = \|\phi(z_1) - \phi(z_2)\| \leq \lambda \|z_1 - z_2\|,$$

so g is a λ-contraction. Next, we verify that g maps $\overline{B_r(x)}$ into itself:

$$\|g(z) - x\| \leq \|g(z) - g(x)\| + \|g(x) - x\| \leq \lambda \|z - x\| + \|y_0 - F(x)\| < \lambda r + (1 - \lambda) r = r.$$

Thus $g(\overline{B_r(x)}) \subseteq B_r(x) \subseteq \overline{B_r(x)}$. Since $\overline{B_r(x)}$ is complete, the Banach fixed-point theorem (Theorem 6.c.17) yields a unique point $x_0 \in B_r(x)$ with $g(x_0) = x_0$, i.e. $F(x_0) = y_0$. This proves $B_{(1-\lambda)r}(F(x)) \subseteq F(B_r(x))$. The outer inclusion follows directly from $\|F(y) - F(x)\| \leq \|y - x\| + \|\phi(y) - \phi(x)\| < (1+\lambda)r$.

(b) For any $x_1, x_2 \in U$, applying the reverse triangle inequality gives

$$\|F(x_1) - F(x_2)\| = \|(x_1 - x_2) + (\phi(x_1) - \phi(x_2))\| \geq \|x_1 - x_2\| - \|\phi(x_1) - \phi(x_2)\| \geq (1-\lambda)\|x_1 - x_2\|.$$

Hence $F(x_1) = F(x_2) \implies x_1 = x_2$, so F is injective. Setting $y_1 := F(x_1)$ and $y_2 := F(x_2)$, this bound rewrites as $\|F^{-1}(y_1) - F^{-1}(y_2)\| \leq \frac{1}{1-\lambda}\|y_1 - y_2\|$, establishing that F^{-1} is $\frac{1}{1-\lambda}$-Lipschitz continuous.

Q.E.D.

AI-Note 15.1 (*Where this lemma is used*): Theorem 15.a.1 is stated above without proof, and the lemma just proved is the engine of the standard proof of it. Without that connection the lemma arrives, is proved, and is never used again, so it is worth seeing how the two fit together.

Assume $x_0 = 0$ and $f(x_0) = 0$, which costs nothing, and replace f by $Df_0^{-1} \circ f$, which is legitimate because Df_0 is invertible by hypothesis and which makes $Df_0 = \mathrm{id}$. Then $\phi(x) := f(x) - x$ satisfies $D\phi_0 = 0$. Since $D\phi$ is continuous, there is a ball around 0 on which $\|D\phi\| \leq \lambda$ for any $\lambda \in (0,1)$ we like, and on a ball (a convex set) the mean value inequality upgrades that to ϕ being λ-Lipschitz. Now $f = \mathrm{id} + \phi$ is exactly the shape Lemma 15.a.3 handles, and the lemma returns, in one step, that f is open and injective there with a Lipschitz inverse. Only differentiability of f^{-1} is then left, and it follows from the definition of the differential together with the Lipschitz bound.

Remark 15.a.4 (*Checking for a global diffeomorphism*): To verify that a map $f \in C^k(U, V)$ between open sets is a global C^k-diffeomorphism, it is best to check the following two conditions:

(a) **Global bijectivity:** Check that f is bijective. This assures that a global inverse $f^{-1} : V \to U$ exists, and any local inverse will coincide with it.

(b) **Local diffeomorphism everywhere:** Check that the differential Df_x (or $\mathrm{J}\, f(x)$) is invertible at every point $x \in U$. By the inverse function theorem, this ensures f is a local C^k-diffeomorphism everywhere.

If both conditions hold, the global inverse f^{-1} inherits the C^k regularity of the local inverses, making f a global C^k-diffeomorphism.

Example 15.a.5 (*Invertible everywhere, injective nowhere near globally*): The second condition of Remark 15.a.4 does not imply the first. Let

$$f : \mathbb{R}^2 \to \mathbb{R}^2, \qquad f(x,y) := \begin{pmatrix} e^x \cos y \\ e^x \sin y \end{pmatrix},$$

which is the complex exponential written in real coordinates. Its Jacobian is

$$\mathrm{J}\, f(x,y) = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}, \qquad \det \mathrm{J}\, f(x,y) = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x} > 0,$$

so $\mathrm{J}\, f$ is invertible at **every** point and f is a local C^∞-diffeomorphism everywhere by Theorem 15.a.1. Yet $f(x, y+2\pi) = f(x,y)$ for all (x,y), so f is not injective. Indeed, every value in its image is attained infinitely often, and f is a global diffeomorphism on no horizontal strip of height 2π or more.

Restricting the codomain does not help either: f misses the origin and nothing else, so $f : \mathbb{R}^2 \to \mathbb{R}^2 \setminus \{0\}$ is surjective but still not injective. What **does** work is restricting the domain, for instance to $\mathbb{R} \times (-\pi, \pi)$, on which f is the polar-coordinate diffeomorphism of Example 15.a.6 in disguise.

Example 15.a.6 (*Polar and spherical coordinates*): The most important examples of diffeomorphisms are transformations between coordinate systems. For instance, the map assigning polar coordinates to Cartesian coordinates on $\mathbb{R}^2 \setminus ((-\infty, 0] \times \{0\})$, and similarly for spherical coordinates on $\mathbb{R}^3$, are global C^∞-diffeomorphisms onto their images.

Question 15.a.7:

Q1. If $f : \mathbb{R}^n \to \mathbb{R}^n$ is C^k and a "*homeomorphism*" (German: „*Homöomorphismus*"), that is, bijective with continuous inverse, is it necessarily a C^k-diffeomorphism?

Q2. If $f \in C^\infty(U, \mathbb{R}^n)$ has an everywhere invertible differential, i.e., Df_x is invertible for all $x \in U$ with $U \subseteq \mathbb{R}^n$ open, then f is a C^∞-diffeomorphism onto its image. True or false?

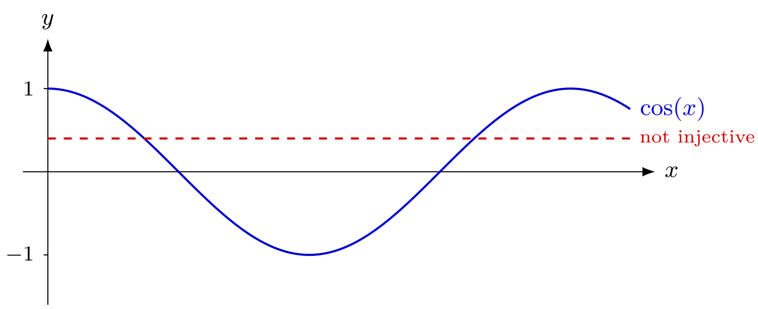
Answer 15.a.8:

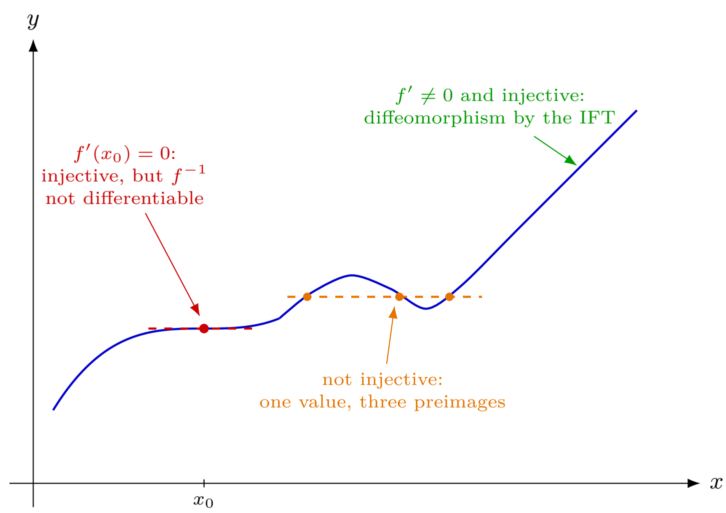
A1. **False.** Consider $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$. Then $f^{-1} : \mathbb{R} \to \mathbb{R}$, $x \mapsto \sqrt[3]{x}$, is not continuously differentiable at $0 \in \mathbb{R}$.

A2. **False.** Consider

$$\cos|_{\mathbb{R}\setminus\pi\mathbb{Z}} : \mathbb{R} \setminus \pi\mathbb{Z} \to (-1, 1), \qquad x \mapsto \cos(x).$$

For any $x \in \mathbb{R} \setminus \pi\mathbb{Z}$, $\cos'(x) = -\sin(x) \neq 0$, so the differential is everywhere invertible, but the cosine is not injective on this domain, so it fails to be a diffeomorphism onto its image.





Reading the three regions left to right makes the two answers above concrete.

- At x_0 the curve is still **strictly increasing**, so f is injective there and f^{-1} exists. But the tangent is horizontal, so the tangent to f^{-1} at $f(x_0)$ is **vertical**, and f^{-1} fails to be differentiable. Invertible, yet not a diffeomorphism: this is exactly $x \mapsto x^3$ from **A1**, and it is why Theorem 15.a.1 assumes Df_{x_0} invertible rather than merely assuming f bijective.
- On the hump, $f' \neq 0$ at every individual point, so f **is** a local diffeomorphism near each of them. Yet one value has three preimages, so f is not globally injective. This is **A2**: an everywhere-invertible differential buys locality and nothing more.

- On the final stretch both hypotheses hold, and the inverse function theorem delivers a genuine C^∞-diffeomorphism onto the image.

Exercise 15.a.9 (*Open mapping*): Let $f: U \to \mathbb{R}$ be a continuously differentiable function, where $U \subseteq \mathbb{R}^n$ is open and connected. Assume that $\nabla f(x) \neq 0$ for all $x \in U$. Prove that $f(U)$ is open in $\mathbb{R}$.

Hint: Define an auxiliary function $F: \mathbb{R}^n \to \mathbb{R}^n$ and use the Inverse Function Theorem.

Exercise 15.a.9:

Let $x_0 \in U$. Since $\nabla f(x_0) \neq 0$, there exists an index $i \in \{1, \dots, n\}$ such that $\partial_{x_i} f(x_0) \neq 0$. Without loss of generality, assume $i = 1$. Define the auxiliary function $F: U \to \mathbb{R}^n$ by

$$F(x_1, x_2, \dots, x_n) = (f(x_1, \dots, x_n), x_2, \dots, x_n).$$

The Jacobian matrix of F at x is

$$DF(x) = \begin{pmatrix} \partial_{x_1} f(x) & \partial_{x_2} f(x) & \dots & \partial_{x_n} f(x) \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

Its determinant is $\det DF(x) = \partial_{x_1} f(x)$. At $x = x_0$, we have $\det DF(x_0) = \partial_{x_1} f(x_0) \neq 0$. Thus, DF_{x_0} is invertible. By the Inverse Function Theorem (Theorem 15.a.1), there exist open neighbourhoods $V_0 \subseteq U$ of x_0 and $W_0 \subseteq \mathbb{R}^n$ of $F(x_0)$ such that $F|_{V_0}: V_0 \to W_0$ is a diffeomorphism, and in particular, $F(V_0) = W_0$ is open in $\mathbb{R}^n$.

Let $\pi_1: \mathbb{R}^n \to \mathbb{R}$, $(y_1, \dots, y_n) \mapsto y_1$ be the projection onto the first coordinate. Projections are open mappings. Notice that $f = \pi_1 \circ F$. Therefore, $f(V_0) = \pi_1(F(V_0)) = \pi_1(W_0)$ is open in $\mathbb{R}$. Thus, for every $x_0 \in U$, there is a neighbourhood $V_0 \subseteq U$ such that $f(V_0)$ is an open neighbourhood of $f(x_0)$ contained in $f(U)$. This proves that $f(U)$ is open in $\mathbb{R}$. **Q.E.D.**

Exercise 15.a.10 (*7.1 — Diffeomorphism* (important)):

(a) Give a diffeomorphism between $\mathbb{R}^2$ and $(0,1) \times (0,1)$.

(b) Is $f(x) = x^5$, $x \in \mathbb{R}$ a diffeomorphism of $\mathbb{R}$ to itself? Motivate rigorously your answer.

Solutions (Section 15.b)

Solution to Exercise 15.a.10:

(a) First build a diffeomorphism $\mathbb{R} \to (0,1)$: the function

$$\psi: \mathbb{R} \to (0,1), \qquad \psi(t) := \frac{1}{2} + \frac{1}{\pi}\arctan(t)$$

is C^∞, strictly increasing (since $\arctan' > 0$), and bijective onto $(0,1)$ (as arctan is bijective onto $(-\frac{\pi}{2}, \frac{\pi}{2})$), with C^∞ inverse $\psi^{-1}(s) = \tan\big(\pi(s - \frac{1}{2})\big)$. Applying ψ in each coordinate gives the diffeomorphism

$$\phi: \mathbb{R}^2 \to (0,1) \times (0,1), \qquad \phi(x,y) := \Big(\frac{1}{2} + \frac{1}{\pi}\arctan(x),\ \frac{1}{2} + \frac{1}{\pi}\arctan(y)\Big).$$

(b) **No.** f is a C^∞ bijection of $\mathbb{R}$ with $f^{-1}(x) = \sqrt[5]{x}$, but $f'(0) = 5 \cdot 0^4 = 0$. If f were a diffeomorphism, then by the chain rule applied to $f^{-1} \circ f = \mathrm{id}$,

$$1 = (f^{-1})'(f(x)) \cdot f'(x) \quad \text{for all } x,$$

which is impossible at $x = 0$ since $f'(0) = 0$. Equivalently, $(f^{-1})'(x) = \frac{1}{5}x^{-4/5} \to \infty$ as $x \to 0$, so f^{-1} is not even differentiable at 0, let alone continuously so.

Q.E.D.

The Implicit Function Theorem (Chapter 16)

The inverse function theorem asks when a map can be undone. The implicit function theorem generalizes the question: when can an equation $f(x,y)=0$ be solved for **some** of the variables in terms of the others, without f itself being invertible? The answer is the same condition applied to a submatrix of the Jacobian, and it is what turns level sets into graphs.

The implicit function theorem (Section 16.a)

Theorem 16.a.1 (*Implicit function theorem*): Let $1 \le n-d < n$, let $f \in C^k(\mathbb{R}^d \times \mathbb{R}^{n-d}, \mathbb{R}^{n-d})$ (or an open subset of $\mathbb{R}^d \times \mathbb{R}^{n-d}$), and suppose $f(x_0,y_0)=0$ for some $(x_0,y_0) \in \mathbb{R}^d \times \mathbb{R}^{n-d}$ such that the differential $D_y f(x_0,y_0)$ of $y \mapsto f(x_0,y)$ at y_0 is bijective (so $\mathrm{J}_y\, f(x_0,y_0)$ is invertible). Then there exist balls $B_r(x_0) \subseteq \mathbb{R}^d$ and $B_s(y_0) \subseteq \mathbb{R}^{n-d}$ and $g \in C^k(B_r(x_0), B_s(y_0))$ such that, for all $(x,y) \in B_r(x_0) \times B_s(y_0)$,

$$f(x,y)=0 \iff y=g(x).$$

Furthermore,

$$\mathrm{J}\, g(x) = -\big(\,\mathrm{J}_y\, f(x,g(x))\big)^{-1}\,\mathrm{J}_x\, f(x,g(x)).$$

In words: the equation $f(x,y)=0$ ties x and y together without telling us what y is. The theorem says that near a point where the tie is **non-degenerate in y**, the equation can be untangled. There is then an honest function g producing the right y for each nearby x, it is as smooth as f was, and its derivative is computable without ever writing g down.

Three things are worth reading off the statement before the pictures.

- **Which variables get solved for is a choice.** Splitting the n coordinates into d *"free"* ones and $n-d$ *"solved"* ones is ours to make; the theorem only demands that the block $\mathrm{J}_y\, f$ belonging to the solved variables be invertible. Choose a different split and a different block has to be invertible.
- **The conclusion is an equivalence, not just an inclusion.** Inside the box, $f(x,y)=0 \iff y=g(x)$. So g does not merely produce **some** solutions: within the box there are no others.
- **Nothing is claimed globally.** The level set $\{f(x,y)=0\}$ is a graph over x only near the chosen point. Section 17.a will call such a set a submanifold, and this local-graph property is exactly what that definition abstracts.

Example 16.a.2 (*A concrete example in $\mathbb{R}^2$*): Consider $f:\mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) := x^2+y^2-1$. Here $n=2$ and $d=1$, so $x \in \mathbb{R}$ and $y \in \mathbb{R}$. We want to see whether we can write y locally as a function of x, i.e., find $g(x)$ such that $f(x,g(x))=0$.

First, let us find a point (x_0,y_0) where $f(x_0,y_0)=0$. We guess $f(0,1)=0^2+1^2-1=0$, so $(0,1)$ works. Next, we check whether the y-derivative $\mathrm{J}_y\, f(0,1)$ is invertible. Since y is 1-dimensional, this is just the partial derivative $\partial_y f(0,1)$:

$$\partial_y f(0,1) = \partial_y(x^2+y^2-1)\big|_{(0,1)} = 2y\big|_{y=1} = 2 \neq 0.$$

Since it is non-zero, it is invertible as a 1×1 matrix. The theorem then guarantees that for small $r,s>0$, there exists $g: B_r(0) \to B_s(1)$ such that for all $(x,y) \in B_r(0) \times B_s(1)$,

$$f(x,y)=0 \iff y=g(x).$$

Since $f \in C^\infty(\mathbb{R}^2)$, we also have $g \in C^\infty(B_r(0))$. Moreover, we can compute $g'(x)=\mathrm{J}\, g(x)$ using the formula:

$$g'(x) = -\big(\partial_y f(x,g(x))\big)^{-1}\partial_x f(x,g(x)) = -(2g(x))^{-1}(2x) = -\frac{x}{g(x)}.$$

(Indeed, since $y=g(x)=\sqrt{1-x^2}$ near $(0,1)$, its derivative is exactly $-x/\sqrt{1-x^2}$.)

Remark 16.a.3 (*Why* $\mathrm{J}\,g$ *has this form*): The formula is just implicit differentiation, done in blocks. By construction g satisfies $f(x, g(x)) \equiv 0$ for every $x \in B_r(x_0)$: the left-hand side is identically zero as a function of x, so its derivative is too. Differentiating with the chain rule (Theorem 10.a.1), and remembering that x enters both directly and through g,

$$0 = \underbrace{D_x f(x, g(x))}_{\text{direct}} + \underbrace{D_y f(x, g(x)) \cdot Dg(x)}_{\text{through } g},$$

and since $D_y f(x, g(x))$ is invertible near (x_0, y_0) we may move it to the other side:

$$Dg(x) = -\big(D_y f(x, g(x))\big)^{-1} D_x f(x, g(x)),$$

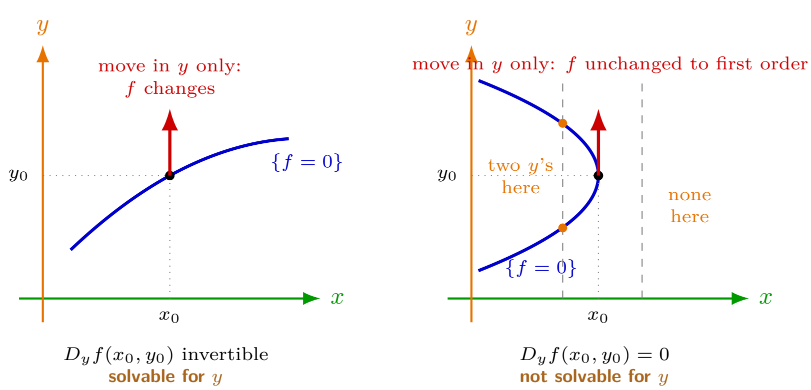
which is the formula in Theorem 16.a.1.

Read it as an exchange rate. Nudging x changes f by $D_x f$; to keep the equation satisfied, y must move by whatever compensates, and $\big(D_y f\big)^{-1}$ converts *"how much f moved"* back into *"how far y must travel"*. The minus sign is the compensation. This is also why invertibility of $D_y f$ is the hypothesis: if $D_y f$ were singular, some change in f could not be absorbed by moving y at all, and there would be nothing for g to do.

AI-Note 16.1: Corsin's own notes stop after the algebra, with the aside "the geometrical meaning of this one eludes me, sorry". The exchange-rate reading above and the two figures below are supplied here to fill that in. The chain-rule derivation itself is his.

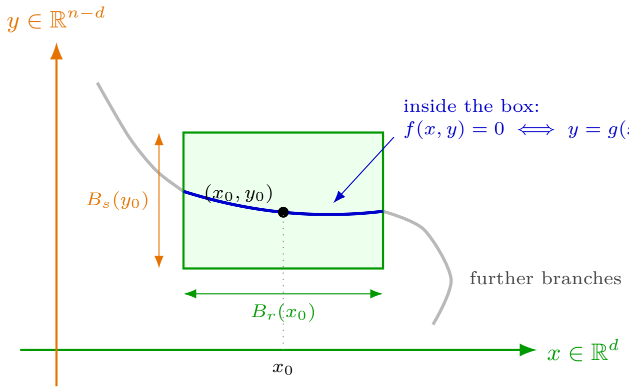
What the hypothesis means

The condition on $D_y f(x_0, y_0)$ is a statement about the **direction** in which the level set $\{f = 0\}$ runs at the point. Moving off (x_0, y_0) straight up, in the y-direction alone, changes f to first order by $D_y f(x_0, y_0)$. If that map is invertible, no vertical motion leaves f unchanged, so the level set cannot run vertically at (x_0, y_0), and a curve that is not vertical is locally a graph over x. If instead $D_y f(x_0, y_0) = 0$, the level set is free to turn vertical, and above a single x there may sit two values of y, or none.



What the conclusion gives

The theorem does not say the whole level set is a graph, only that it is one **inside a small enough box**. On $B_r(x_0) \times B_s(y_0)$ the two descriptions coincide exactly: the zero set of f and the graph of g are the same set. Outside the box the level set may do anything at all, including doubling back or carrying further branches, and that is precisely why the statement is local.



Example 16.a.4 (*The circle S^1*): The circle $S^1 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 - 1 = 0\}$ is the level set at 0 of $F(x,y) := x^2 + y^2 - 1$, i.e., $S^1 = F^{-1}\{0\}$. Locally, there are explicit functions

$$x \mapsto \pm\sqrt{1-x^2} = y(x),$$

defined on open intervals around x_0 whenever $y(x_0) \neq 0$. And indeed,

$$D_y F(x,y) = \frac{\partial F}{\partial y} = 2y$$

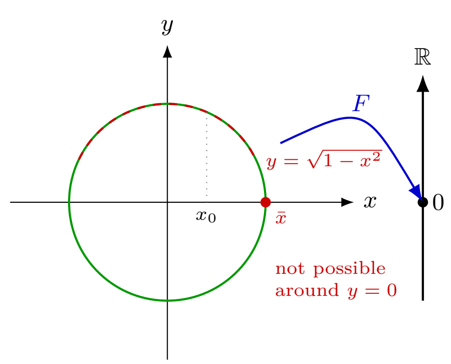
is bijective whenever $y \neq 0$, so the implicit function theorem holds there.

Remark 16.a.5 (*The circle realizes both panels*): The circle exhibits both cases of the figure above, and it is worth checking them against each other.

At a point with $y_0 \neq 0$, say $(0,1)$, we have $D_y F = 2y_0 \neq 0$, the circle meets that point at a non-vertical angle, and near it the circle really is the graph $y = \sqrt{1-x^2}$. That is the left panel.

At $\bar{x} = 1$, where $y = 0$, we get $D_y F(1,0) = 0$. The circle is **vertical** there, and no graph $y(x)$ can exist around it: for x slightly less than 1 there are two points of the circle above x, namely $\pm\sqrt{1-x^2}$, and for x slightly greater than 1 there are none. That is the right panel exactly, with two y's on one side and none on the other.

Note what has **not** gone wrong at $\bar{x} = 1$: the circle is a perfectly smooth curve there, with no corner and no break. What fails is only the attempt to describe it as a graph **over** x. Swap the roles of the two variables and everything works again, since $D_x F(1,0) = 2 \neq 0$: near $(1,0)$ the circle is the graph $x = \sqrt{1-y^2}$ of a function of y. This is the *"which variables get solved for is a choice"* point above, in its smallest possible instance.



Example 16.a.6 (*An implicit quintic*): Let $G(x,y) := y^5 + y^3 + y + x$, and consider its level set at 0, $G^{-1}\{0\} = \{(x,y) \in \mathbb{R}^2 : G(x,y) = 0\}$. Clearly $(0,0) \in G^{-1}\{0\}$, since $G(0,0) = 0^5 + 0^3 + 0 + 0 = 0$. Now both

$$\partial_x G(0,0) = 1 \neq 0, \qquad \partial_y G(0,0) = \left[5y^4 + 3y^2 + 1\right]_{y=0} = 1 \neq 0,$$

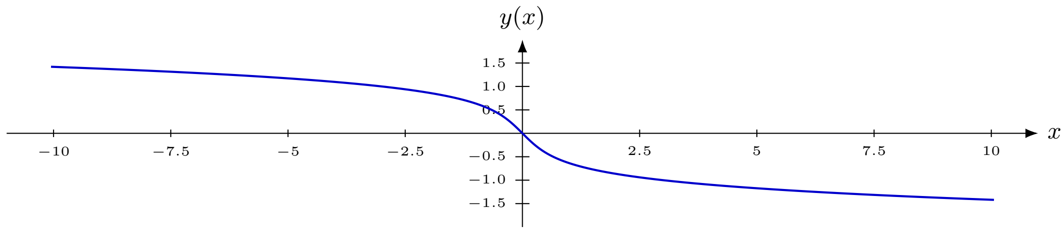
so by the implicit function theorem there are open intervals (balls) $(-r,r)$, $(-s,s)$ and functions

$$(-r,r) \to (-s,s), \quad y \mapsto x(y) = -y^5 - y^3 - y,$$
$$(-s,s) \to (-r,r), \quad x \mapsto y(x),$$

satisfying

$$y^5 + y^3 + y + x(y) = 0, \qquad y(x)^5 + y(x)^3 + y(x) + x = 0.$$

Remark 16.a.7: As Galois and Abel showed, no closed-form expression for $y(x)$ in radicals exists in general, since $y^5 + y^3 + y + x = 0$ is a genuine quintic in y. Nevertheless, the implicit function theorem guarantees that $y(x)$ exists (and is C^∞, since G is a polynomial), and it can be computed and plotted numerically.



Exercise 16.a.8 (*Implicitly defined functions from a system*): Show that one can solve the system of equations

$$\begin{cases} -2x^2 + y^2 + z^2 = 0 \\ x^2 + e^{y-1} - 2y = 0 \end{cases}$$

for y, z as functions of x (i.e. $y(x), z(x)$) close enough to the point $(1,1,1)$. Calculate $\begin{pmatrix} y'(1) \\ z'(1) \end{pmatrix}$.

(cf. Michaels Bsp 8.3.5)

Exercise 16.a.8:

Let $F(x,y,z) = \begin{pmatrix} -2x^2 + y^2 + z^2 \\ x^2 + e^{y-1} - 2y \end{pmatrix}$. We have $F(1,1,1) = \begin{pmatrix} -2+1+1 \\ 1+1-2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The Jacobian of F with respect to (y,z) is

$$D_{y,z}F(x,y,z) = \begin{pmatrix} \partial_y F_1 & \partial_z F_1 \\ \partial_y F_2 & \partial_z F_2 \end{pmatrix} = \begin{pmatrix} 2y & 2z \\ e^{y-1} - 2 & 0 \end{pmatrix}.$$

At $(x,y,z) = (1,1,1)$, we have

$$D_{y,z}F(1,1,1) = \begin{pmatrix} 2 & 2 \\ e^0 - 2 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ -1 & 0 \end{pmatrix}.$$

The determinant is $\det(D_{y,z}F(1,1,1)) = 2(0) - 2(-1) = 2 \neq 0$. Thus the matrix is invertible, and by the Implicit Function Theorem, there exist functions $y(x), z(x)$ solving the system in a neighbourhood of $x = 1$ such that $y(1) = 1$ and $z(1) = 1$.

To find the derivatives at $x = 1$, we use the formula

$$\begin{pmatrix} y'(1) \\ z'(1) \end{pmatrix} = -(D_{y,z}F(1,1,1))^{-1} D_x F(1,1,1).$$

We calculate $D_x F(x,y,z) = \begin{pmatrix} -4x \\ 2x \end{pmatrix}$, so $D_x F(1,1,1) = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$. The inverse of $D_{y,z}F(1,1,1)$ is

$$\begin{pmatrix} 2 & 2 \\ -1 & 0 \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} 0 & -2 \\ 1 & 2 \end{pmatrix}.$$

Then
$$\begin{pmatrix} y'(1) \\ z'(1) \end{pmatrix} = -\frac{1}{2}\begin{pmatrix} 0 & -2 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} -4 \\ 2 \end{pmatrix} = -\frac{1}{2}\begin{pmatrix} -4 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$
Therefore, $y'(1) = 2$ and $z'(1) = 0$.

Q.E.D.

Exercise 16.a.9 (*Implicit solving in different variables*): Consider the equation $xyz - yz^4 + x^2y = 2$ near the point $(0, -2, 1)$. Can one solve this equation for:

(a) x?
(b) y?
(c) z?

Calculate the derivative of the solution at the given point, if possible. (cf. Michaels Bsp 8.3.3)

Exercise 16.a.9:

Let $F(x, y, z) = xyz - yz^4 + x^2y - 2$. At the point $P = (0, -2, 1)$, we have $F(0, -2, 1) = 0 - (-2)(1) + 0 - 2 = 0$. We compute the partial derivatives of F:
$$\begin{aligned} \partial_x F(x, y, z) &= yz + 2xy \\ \partial_y F(x, y, z) &= xz - z^4 + x^2 \\ \partial_z F(x, y, z) &= xy - 4yz^3 \end{aligned}$$
Evaluating at $P = (0, -2, 1)$:
$$\begin{aligned} \partial_x F(0, -2, 1) &= (-2)(1) + 0 = -2 \neq 0 \\ \partial_y F(0, -2, 1) &= 0 - 1 + 0 = -1 \neq 0 \\ \partial_z F(0, -2, 1) &= 0 - 4(-2)(1) = 8 \neq 0 \end{aligned}$$
(a) Since $\partial_x F(P) = -2 \neq 0$, the Implicit Function Theorem guarantees we can solve for $x = x(y, z)$ locally. The derivative (gradient) is
$$\nabla x(y, z)\Big|_{(-2,1)} = \begin{pmatrix} \partial_y x \\ \partial_z x \end{pmatrix} = -\frac{1}{\partial_x F(P)}\begin{pmatrix} \partial_y F(P) \\ \partial_z F(P) \end{pmatrix} = -\frac{1}{-2}\begin{pmatrix} -1 \\ 8 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 4 \end{pmatrix}.$$
(b) Since $\partial_y F(P) = -1 \neq 0$, we can solve for $y = y(x, z)$ locally. The derivative is
$$\nabla y(x, z)\Big|_{(0,1)} = \begin{pmatrix} \partial_x y \\ \partial_z y \end{pmatrix} = -\frac{1}{\partial_y F(P)}\begin{pmatrix} \partial_x F(P) \\ \partial_z F(P) \end{pmatrix} = -\frac{1}{-1}\begin{pmatrix} -2 \\ 8 \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \end{pmatrix}.$$
(c) Since $\partial_z F(P) = 8 \neq 0$, we can solve for $z = z(x, y)$ locally. The derivative is
$$\nabla z(x, y)\Big|_{(0,-2)} = \begin{pmatrix} \partial_x z \\ \partial_y z \end{pmatrix} = -\frac{1}{\partial_z F(P)}\begin{pmatrix} \partial_x F(P) \\ \partial_y F(P) \end{pmatrix} = -\frac{1}{8}\begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1/4 \\ 1/8 \end{pmatrix}.$$

Q.E.D.

Exercise 16.a.10 (*7.5 — Implicit function II* (important)): Show that the system of equations
$$\begin{cases} xy^5 + yu^5 + zv^5 = 1, \\ x^5y + y^5u + z^5v = 1, \end{cases}$$
is solvable for the variables u and v in a neighborhood of the point $(x_0, y_0, z_0, u_0, v_0) = (0, 1, 1, 1, 0)$ and determine the derivatives $D_{(0,1,1)}u$ and $D_{(0,1,1)}v$ of the implicitly defined functions $u = u(x, y, z)$ and $v = v(x, y, z)$.

Solutions (Section 16.b)

Solution to Exercise 16.a.10:

Write the system as $F(x,y,z,u,v)=0$ for $F:\mathbb{R}^5\to\mathbb{R}^2$,

$$F(x,y,z,u,v) := \begin{pmatrix} xy^5+yu^5+zv^5-1 \\ x^5y+y^5u+z^5v-1 \end{pmatrix}.$$

The base point does lie on the level set, which the theorem requires before anything else:

$$F(0,1,1,1,0) = \begin{pmatrix} 0+1+0-1 \\ 0+1+0-1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The differential is

$$\mathrm{J}\,F(x,y,z,u,v) = \begin{pmatrix} y^5 & 5xy^4+u^5 & v^5 & 5yu^4 & 5zv^4 \\ 5x^4y & x^5+5y^4u & 5z^4v & y^5 & z^5 \end{pmatrix},$$

so at the base point,

$$\mathrm{J}\,F(0,1,1,1,0) = \begin{pmatrix} 1 & 1 & 0 & 5 & 0 \\ 0 & 5 & 0 & 1 & 1 \end{pmatrix}.$$

By the implicit function theorem applied with (x,y,z) as the free variables and (u,v) as the variables to solve for, we need the submatrix of partial derivatives with respect to u,v to be invertible; this is

$$D_{(u,v)}F(0,1,1,1,0) = \begin{pmatrix} 5 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \det = 5 \neq 0,$$

so the implicit function theorem applies, and the system is indeed solvable for u,v as C^∞ functions of (x,y,z) near $(0,1,1)$. Writing $G(x,y,z) := (u(x,y,z), v(x,y,z))$, the theorem gives

$$\begin{aligned} \mathrm{J}\,G(0,1,1) &= -\begin{pmatrix} 5 & 0 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 5 & 0 \end{pmatrix} \\ &= -\frac{1}{5}\begin{pmatrix} 1 & 0 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 5 & 0 \end{pmatrix} \\ &= -\frac{1}{5}\begin{pmatrix} 1 & 1 & 0 \\ -1 & 24 & 0 \end{pmatrix}. \end{aligned}$$

So, $D_{(0,1,1)}u = -\frac{1}{5}\begin{pmatrix} 1 & 1 & 0 \end{pmatrix}$ and $D_{(0,1,1)}v = -\frac{1}{5}\begin{pmatrix} -1 & 24 & 0 \end{pmatrix}$. **Q.E.D.**

Submanifolds, Tangent and Normal Spaces (Chapter 17)

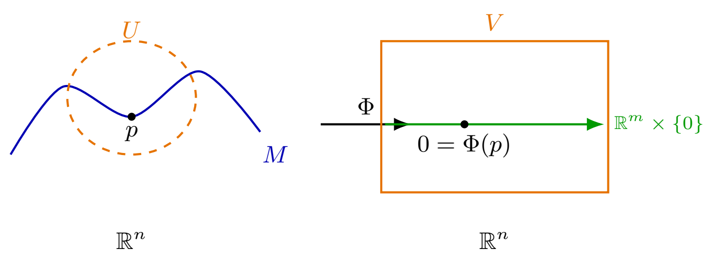
The implicit function theorem said that a level set is locally a graph. This chapter takes that as a definition. A submanifold of $\mathbb{R}^n$ is a set that looks, near each of its points, like a piece of $\mathbb{R}^m$. Three descriptions of such a set turn out to be equivalent (a level set of a submersion, a local graph, a local parametrization), and picking the convenient one is most of the work in practice. Each point then carries its own linear algebra: the tangent space of directions along the submanifold, and the normal space orthogonal to it.

Submanifolds (Section 17.a)

Recall that a "*C^k-diffeomorphism*" (Theorem 15.a.1) is a C^k bijective function whose inverse is also C^k.

Definition 17.a.1 (*Submanifold*): $M^m \subseteq \mathbb{R}^n$ is a C^k, m-dimensional "*submanifold*" (German: „*Untermannigfaltigkeit*“) of $\mathbb{R}^n$ if for any point $p \in M$ there are an open neighbourhood $U \subseteq \mathbb{R}^n$ of p, an open neighbourhood $V \subseteq \mathbb{R}^n$ of 0, and a C^k-diffeomorphism $\Phi : U \to V$ with $\Phi(p) = 0$ (a "*submanifold chart*") such that

$$\Phi(U \cap M) = (\mathbb{R}^m \times \{0\}) \cap V = \{x \in V : x_{m+1} = \cdots = x_n = 0\}.$$



Remark 17.a.2 (*Why the definition is almost never used directly*): Producing such a Φ by hand is difficult even for simple M, so the definition is rarely what one works with. Two equivalent descriptions do the job instead, the **implicit** one (the regular value theorem below) and the **explicit or graphical** one (the local parametrization theorem below). Remark 17.a.10 compares all three once both are available.

The regular value theorem (Subsection 17.a.1)

Theorem 17.a.3 (*Regular value theorem*): Suppose $U \subseteq \mathbb{R}^n$ is open, $F \in C^k(U, \mathbb{R}^\ell)$, and $y \in F(U)$. If

$$DF_x : \mathbb{R}^n \to \mathbb{R}^\ell$$

is surjective (i.e., $\mathrm{J}\,F(x)$ has full rank) for all $x \in F^{-1}\{y\} = \{\bar{x} \in U : F(\bar{x}) = y\}$, then $F^{-1}\{y\}$ is a C^k submanifold of dimension $n - \ell$.

Important remark 17.a.4 (*Full rank means two opposite things*): *"Full rank"* is decided by the **shape** of the matrix, and the two representations of a submanifold sit on opposite sides of it:

- An **implicit** description $F : \mathbb{R}^n \to \mathbb{R}^\ell$ cuts M out by ℓ equations, so $\mathrm{J}\,F$ is **wide** ($\ell \times n$ with $\ell \leq n$). Full rank $= \ell$ means DF_x is **surjective**, so the ℓ constraints are independent, each one genuinely cutting a dimension. Here $\dim M = n - \ell$.

- A **parametrization** $f : \mathbb{R}^m \to \mathbb{R}^n$ sweeps M out by m parameters, so $\mathrm{J}\,f$ is **tall** ($n \times m$ with $m \le n$). Full rank $= m$ means Df_x is **injective**, so the m parameters move in independent directions, none of them wasted. Here $\dim M = m$.

Same phrase, opposite condition. A wide matrix is never injective and a tall one is never surjective, so there is no ambiguity in practice. But it is worth checking which side you are on before writing down what *"full rank"* is supposed to give you.

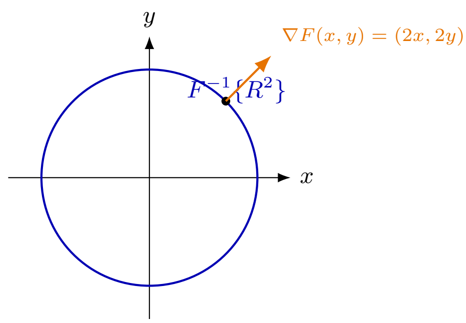
Example 17.a.5 (*Circle of radius R*): Let $F : \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x^2 + y^2$. Then

$$\mathrm{J}\,F(x, y) = (2x,\ 2y)$$

has full rank at every $(x, y) \in \mathbb{R}^2 \setminus \{0\}$. Here $\ell = 1$, so by Important remark 17.a.4 full rank means that $DF_{(x,y)} : \mathbb{R}^2 \to \mathbb{R}$ is **surjective**, which for a single row is simply the requirement that $(2x, 2y)$ be nonzero. For $R > 0$, $(0, 0) \notin F^{-1}\{R^2\}$, so

$$F^{-1}\{R^2\} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\}$$

is a submanifold of dimension $2 - 1 = 1$ by Theorem 17.a.3. Similarly, $\{x \in \mathbb{R}^n : |x|^2 = R^2\}$ is a submanifold of dimension $n - 1$ for $n \ge 1$ (the "*n-sphere*" of radius R).



AI-Example 17.a.6 (*A set that is **not a submanifold, and how to prove it*):** The regular value theorem is a one-way street: if DF_x is surjective everywhere on the level set, we get a submanifold. When it is not, we get **no information**. The level set may still be a submanifold, or it may not. It is worth seeing a case where it is not.

Let $F(x, y) := xy$ and consider $C := F^{-1}\{0\}$, the union of the two coordinate axes. Here $\mathrm{J}\,F(x, y) = (y,\ x)$, which is nonzero, and hence surjective, at every point of C **except** the origin. So, Theorem 17.a.3 does apply on $C \setminus \{(0, 0)\}$, telling us that the four open half-axes are 1-dimensional submanifolds, and says nothing at the origin.

And at the origin C genuinely fails to be a submanifold. The argument is a connectedness trick worth remembering, since it is the standard way to prove a negative here.

Suppose C were a 1-dimensional submanifold. By Definition 17.a.1 there would be a neighbourhood U of the origin and a diffeomorphism $\Phi : U \to V$ carrying $C \cap U$ onto a piece of a **line**, $(\mathbb{R} \times \{0\}) \cap V$. Now delete the centre point from each side and count connected components (Definition 8.a.1):

- Removing a point from a small piece of a line leaves **two** pieces.
- Removing the origin from $C \cap U$ leaves **four** pieces, one along each half-axis.

But Φ is a homeomorphism, and homeomorphisms preserve the number of connected components (Theorem 8.a.6, applied to Φ and to Φ^{-1}). Four cannot equal two, so no such Φ exists and C is not a submanifold.

Remark 17.a.7: The moral is that submanifolds are forbidden from having **crossings, corners or cusps**: near each of its points a submanifold must look like a flat piece of $\mathbb{R}^m$, and a crossing does not. Compare AI-Example 13.a.4, where the constraint set $\{y^2 = x^3\}$ had a cusp at exactly the point where ∇g vanished. That is the same phenomenon, seen through the Lagrange-multiplier lens back in Chapter 13.

Connection to Lagrange multipliers (Subsubsection 17.a.1.1)

Remark 17.a.8 (*Constraint sets are submanifolds*): Recall the Lagrangian from Proposition 13.a.2: when optimizing $f|_M$ for $f \in C^k(\mathbb{R}^n, \mathbb{R})$, where $g_1, \dots, g_k : U \to \mathbb{R}$,

$$M := \{x \in \mathbb{R}^n : g_1(x) = \cdots = g_k(x) = 0\}, \qquad (\nabla g_1, \dots, \nabla g_k) \text{ linearly independent for all } x \in M$$

(hence $k \le n$). This means precisely that, for $g : \mathbb{R}^n \to \mathbb{R}^k$, $x \mapsto (g_1(x), \dots, g_k(x))$, the Jacobian

$$\mathrm{J}\, g(x) = \begin{pmatrix} \nabla g_1(x) \longrightarrow \\ \vdots \\ \nabla g_k(x) \longrightarrow \end{pmatrix}$$

has full rank. Therefore, $M = g^{-1}\{0\}$ is an $(n-k)$-dimensional submanifold by the regular value theorem: the constraint sets of the Lagrange-multiplier problems in Chapter 13 were submanifolds all along.

The local parametrization theorem (Subsection 17.a.2)

Theorem 17.a.9 (*Local parametrization*): Let $M \subseteq \mathbb{R}^n$. Then M is a C^k submanifold with $\dim M = m$ if and only if for every $p \in M$ there exist an open neighbourhood V of p, an open set $U \subseteq \mathbb{R}^m$, and a C^k map $f : U \to V \cap M$ such that

- $Df_x : \mathbb{R}^m \to \mathbb{R}^n$ is injective for all $x \in U$;
- f is a homeomorphism (bijective, continuous, with continuous inverse f^{-1}).

Remark 17.a.10 (*Three descriptions, and which one to reach for*): There are now three ways to certify that M is an m-dimensional submanifold, and they are equivalent. They are not equally convenient, though, and choosing badly is what makes these exercises feel hard.

- **Chart** (Definition 17.a.1) is the definition itself. Straighten M out with a diffeomorphism, so that the first m coordinates of the chart carry the submanifold and the rest vanish. It is almost never used directly on a concrete M, since producing Φ by hand is painful. Its role is theoretical, and it is what the other two are proved from.
- **Implicit, via the regular value theorem** (Theorem 17.a.3) is the one to reach for when M arrives as **equations**, such as $\{x^2 + y^2 = R^2\}$ or $\{g_1 = \cdots = g_k = 0\}$. It rests on the same idea as the chart definition, with the diffeomorphism produced implicitly rather than by hand. Check a rank condition on $\mathrm{J}\, F$ and you are done, with no parametrization ever written. This is normally the cheapest route.
- **Parametrization** (Theorem 17.a.9) is the one for an M that arrives as a **picture you can sweep out**, such as a graph, a surface of revolution, or a curve given by a formula in a parameter. It is often the most intuitive, since it builds the submanifold out of parameters explicitly. It is also the one to use when the tangent space is wanted next, because $\partial_1 f, \dots, \partial_m f$ hand you a basis directly (Definition 17.c.21).

The rough rule: **equations $\to$ regular value theorem; a formula with parameters $\to$ local parametrization.** A graph $\{(x, f(x))\}$ satisfies both, which is why graphs are the easiest case of all.

Two cautions carried over from Important remark 17.a.4. *"Full rank"* means **surjective** in the first case and **injective** in the second, and the shape of the matrix decides which. And neither theorem yields a negative. If the rank condition fails you have learned nothing, and must argue separately (see AI-Example 17.a.6).

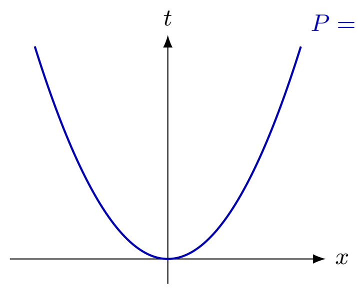
Example 17.a.11 (*The n-parabola*): Let $P := \{(x,t) \in \mathbb{R}^n \times \mathbb{R} : t = |x|^2\}$. One could use the regular value theorem, but instead take

$$f : \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}, \qquad x \mapsto (x, |x|^2).$$

1. The Jacobian matrix has full rank n at every x:
$$\mathrm{J}\, f(x) = \begin{pmatrix} \mathrm{id}_{n\times n} \\ 2x^{\mathsf{T}} \end{pmatrix}.$$
This one is tall, so by Important remark 17.a.4 full rank means that Df_x is **injective**, which is clear from the $\mathrm{id}_{n\times n}$ block alone.
2. f is bijective onto P and continuous, with inverse $\pi : f(U) \to \mathbb{R}^n$, $(x,t) \mapsto x$ (the canonical projection), which is also continuous.
3. $f(\mathbb{R}^n) = P$.

So, P is an n-dimensional submanifold by Theorem 17.a.9.

Remark 17.a.12: The graphical representation of a submanifold (as the graph of a function, here $x \mapsto |x|^2$) is therefore a common, and equivalent, special case of a local parametrization: it is always injective on its domain by construction, and its inverse is simply the projection forgetting the last coordinate.



An alternative optimization technique (Subsubsection 17.a.2.1)

Remark 17.a.13 (*Parametrized optimization, revisiting Exercise 12.b.23*): Local parametrizations give a second route to constrained-extremum problems, alongside Lagrange multipliers: parametrize the constraint set and optimize the resulting unconstrained function of fewer variables. Corsin illustrates this by revisiting $f(x,y) := 2x^2 + y^2 - x$ on $S^1 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, which is exactly Exercise 12.b.23**(a)**, solved there via Lagrange multipliers.

Parametrize S^1 by $(x(\theta), y(\theta)) := (\cos\theta, \sin\theta)$. Then
$$f(\theta) := f(x(\theta), y(\theta)) = 2\cos^2\theta + \sin^2\theta - \cos\theta = \cos^2\theta - \cos\theta + 1,$$
using $\sin^2\theta = 1 - \cos^2\theta$. Differentiating,
$$f'(\theta) = -2\cos\theta\sin\theta + \sin\theta \stackrel{!}{=} 0.$$

1. $\sin\theta = 0 \implies \theta_1 \in \{0, \pi\}$.
2. $\sin\theta \neq 0 \implies -2\cos\theta + 1 = 0 \implies \cos\theta = \frac{1}{2} \implies \theta_2 \in \{\frac{\pi}{3}, \frac{5\pi}{3}\}$.

So, the extrema are at $(\cos\theta_1, \sin\theta_1) = (\pm 1, 0)$ and $(\cos\theta_2, \sin\theta_2) = \left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right)$. These are exactly the four points found via Lagrange multipliers in the solution to Exercise 12.b.23**(a)**, so the two methods agree.

Exercise 17.a.14 (*Dimension of a submanifold*): Show that $M := \{(x,y,z) \in \mathbb{R}^3 \mid 2x + z = 1, x^2 + y^2 - z^2 = 1\}$ is a submanifold of $\mathbb{R}^3$. What is its dimension? (cf. Michaels Bsp 9.1.2 d)

Exercise 17.a.14:

Let $F(x,y,z) = \begin{pmatrix} 2x+z-1 \\ x^2+y^2-z^2-1 \end{pmatrix}$. Then $M = F^{-1}\{0\}$. We compute the Jacobian matrix of F:

$$\mathrm{J}\,F(x,y,z) = \begin{pmatrix} 2 & 0 & 1 \\ 2x & 2y & -2z \end{pmatrix}.$$

We need to check that $\operatorname{rank}(\mathrm{J}\,F(x,y,z)) = 2$ for all $(x,y,z) \in M$. The rank is less than 2 if and only if all 2×2 subdeterminants are zero:

$$\det\begin{pmatrix} 2 & 0 \\ 2x & 2y \end{pmatrix} = 4y = 0 \implies y = 0$$

$$\det\begin{pmatrix} 2 & 1 \\ 2x & -2z \end{pmatrix} = -4z-2x = 0 \implies x = -2z$$

$$\det\begin{pmatrix} 0 & 1 \\ 2y & -2z \end{pmatrix} = -2y = 0 \implies y = 0.$$

So, the rank is not maximal only when $y=0$ and $x=-2z$. Let's check if such a point can belong to M. Substituting $x=-2z$ into the first equation of M gives $2(-2z)+z = -3z = 1 \implies z = -1/3$. Then $x = 2/3$. Thus the point is $(2/3, 0, -1/3)$. Now check the second equation of M: $(2/3)^2 + 0^2 - (-1/3)^2 = 4/9 - 1/9 = 3/9 = 1/3 \neq 1$. Therefore, this point is not in M. Since $\mathrm{J}\,F(x,y,z)$ has maximal rank 2 for all $(x,y,z) \in M$, M is a submanifold of $\mathbb{R}^3$. Its dimension is $n-k = 3-2 = 1$. **Q.E.D.**

Exercise 17.a.15 (*Submanifold of matrices*): Show that $M := \{A \in \mathrm{Mat}_{2\times 2}(\mathbb{R}) \mid \operatorname{rank}(A) = 1\}$ is a submanifold of $\mathrm{Mat}_{2\times 2}(\mathbb{R})$. What is the dimension of M?

Exercise 17.a.15:

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The condition $\operatorname{rank}(A) \leq 1$ is equivalent to $\det(A) = ad-bc = 0$. The condition $\operatorname{rank}(A) = 1$ means $\det(A) = 0$ but $A \neq 0$. So, M is the zero set of $F : \mathbb{R}^4 \setminus \{0\} \to \mathbb{R}$, $F(a,b,c,d) = ad-bc$. The space $\mathrm{Mat}_{2\times 2}(\mathbb{R})$ is isomorphic to $\mathbb{R}^4$. The Jacobian of F is

$$\mathrm{J}\,F(a,b,c,d) = \begin{pmatrix} \partial_a F & \partial_b F & \partial_c F & \partial_d F \end{pmatrix} = \begin{pmatrix} d & -c & -a & b \end{pmatrix}.$$

For $\mathrm{J}\,F$ to not have maximal rank (which is 1), we must have $d=0$, $c=0$, $a=0$, and $b=0$, which means $A=0$. But the zero matrix is excluded from M because its rank is 0, not 1. Thus $\mathrm{J}\,F(A) \neq 0$ for all $A \in M$. By the regular value theorem, M is a submanifold. The ambient dimension is 4 and the number of independent constraints is 1. So, the dimension of M is $4-1=3$. **Q.E.D.**

Three equivalent descriptions are on the table, each characterising the same class of sets. What they do not yet come with is practice, both in choosing which description to write down for a surface handed to you and, harder, in arguing that a set is **not** a submanifold at all. None of the three theorems helps with the second, since each of them only ever yields a positive.

Recognising a submanifold in practice (Section 17.b)

One surface, three descriptions (Subsection 17.b.1)

The three characterisations of a submanifold are easiest to keep apart when they are applied to the same object at the same point. Take the open upper half-sphere

$$\mathbb{S}^2_+ := \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 = 1,\ z > 0\},$$

a 2-dimensional submanifold of $\mathbb{R}^3$, and throughout fix the point

$$p := \Big(0,\ \tfrac{1}{\sqrt{2}},\ \tfrac{1}{\sqrt{2}}\Big) \in \mathbb{S}^2_+.$$

Example 17.b.16 (*The upper half-sphere, three ways*):

(a) Implicit. Put $F(x,y,z) := x^2+y^2+z^2-1$ on the open set $U := \{z > 0\}$. Then $\mathbb{S}^2_+ = F^{-1}\{0\}$ and

$$\nabla F(x,y,z) = (2x,\ 2y,\ 2z), \qquad \nabla F(p) = (0,\ \sqrt{2},\ \sqrt{2}) \neq 0,$$

so $\mathrm{J}\,F$ has full rank $\ell = 1$ everywhere on U (indeed $2z > 0$ already forces the row to be nonzero). Theorem 17.a.3 gives a submanifold of dimension $3-1=2$.

(b) Graph. Over the open unit disc $V := \{(x,y) : x^2+y^2 < 1\}$ put $f(x,y) := \sqrt{1-x^2-y^2}$. Then $\mathbb{S}^2_+ = \{(x,y,f(x,y)) : (x,y) \in V\}$ is the graph of f, and $f(0,1/\sqrt{2}) = \sqrt{1-1/2} = 1/\sqrt{2}$ recovers the third coordinate of p. Here the half-sphere is globally a graph, which is exactly why the **upper half** is a friendlier example than the whole sphere.

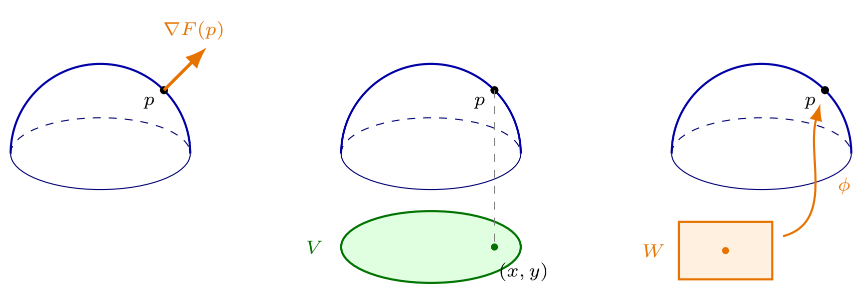
(c) Parametrization. With $W := (0,\pi/2) \times (0,2\pi)$ and

$$\phi(\theta,\psi) := (\sin\theta\cos\psi,\ \sin\theta\sin\psi,\ \cos\theta),$$

we get $p = \phi(\pi/4,\ \pi/2)$. The Jacobian

$$\mathrm{J}\,\phi(\theta,\psi) = \begin{pmatrix} \cos\theta\cos\psi & -\sin\theta\sin\psi \\ \cos\theta\sin\psi & \sin\theta\cos\psi \\ -\sin\theta & 0 \end{pmatrix}$$

is tall (3×2), and full rank $=2$ means $D\phi$ is **injective** (Important remark 17.a.4): the bottom row forces the first column to be nonzero whenever $\sin\theta \neq 0$, and the second column is then independent of it because $\sin\theta \neq 0$ makes $(-\sin\theta\sin\psi,\ \sin\theta\cos\psi)$ nonzero. On W we have $\theta \in (0,\pi/2)$, so $\sin\theta > 0$ throughout.



(a) implicit $\mathbb{S}^2_+ = F^{-1}\{0\}$ **(b)** graph $z = \sqrt{1-x^2-y^2}$ **(c)** parametrization $p = \phi(\frac{\pi}{4}, \frac{\pi}{2})$

Remark 17.b.17 (*Why all three theorems are stated **locally*):** The half-sphere is unusually obliging: one graph and one parametrization cover all of it. That is not the general situation, and each description fails to globalise for its own reason.

- **A parametrization need not be global.** The circle $\mathbb{S}^1$ is a 1-dimensional submanifold, yet no single parametrization covers it. Suppose $\gamma : I \to \mathbb{S}^1$ were a homeomorphism from an open interval, as Theorem 17.a.9 demands. Delete one point from each side: $I \setminus \{t\}$ falls into **two** pieces, while $\mathbb{S}^1 \setminus \{\gamma(t)\}$ is still an arc, so **one** piece. A homeomorphism preserves the number of connected components (Theorem 8.a.6), so no such γ exists. That is the counting argument of AI-Example 17.a.6, run in the opposite direction. Two overlapping arcs do cover the circle, and two is the minimum.

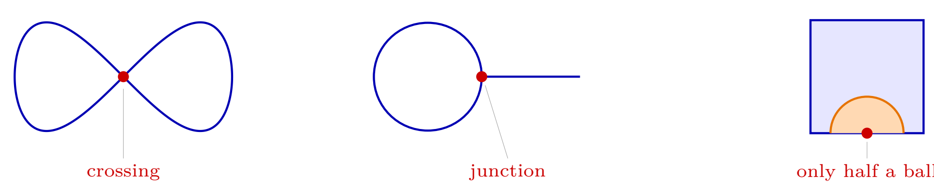
- **An implicit description need not be global either.** The Möbius strip is a 2-dimensional submanifold of $\mathbb{R}^3$, but it is not $g^{-1}\{0\}$ for any single g with $\nabla g \neq 0$: such a g would supply a globally consistent normal direction $\nabla g/|\nabla g|$, and the Möbius strip has none (Section 25.c).

Requiring the descriptions only near each point costs nothing, since being a submanifold is itself a local condition, and it buys all the examples that matter.

A gallery of non-examples (Subsection 17.b.2)

Linus prefaces the formal definitions with a list of sets that are **not** submanifolds, which is worth reproducing because each entry fails for a different reason. Being a d-dimensional submanifold means looking, near every one of its points, like a flat piece of $\mathbb{R}^d$, and there are several distinct ways to violate that.

- **The figure eight** $\{(x,y) : y^2 = x^2 - x^4\}$ has a **crossing**. Away from the origin it is a perfectly good curve. At the origin four branches meet. This is the same failure as AI-Example 17.a.6, and the same connectedness count settles it.
- **The lollipop** is a circle with a segment attached, or any *"line sticking out of a sphere"*. Here the trouble is not a crossing but a **change of dimension** at the junction: near the attachment point the set is neither 1- nor 2-dimensional.
- **The closed square** $[0,1]^2 \subseteq \mathbb{R}^2$ has a **boundary**. Its interior is a fine 2-dimensional submanifold, but no neighbourhood of a corner or edge point looks like an open piece of $\mathbb{R}^2$, because points of $[0,1]^2$ near an edge fill only a half-plane. (Sets like this are the subject of Section 25.d; a submanifold **with boundary** is a genuinely weaker notion, not a submanifold that happens to have edges.)
- $\mathbb{Q}^n \subseteq \mathbb{R}^n$ has **no dimension at all**. It is not 0-dimensional, since a 0-dimensional submanifold is a set of isolated points and no rational point is isolated. And it is not n-dimensional, since it contains no open ball.



(a) figure eight **(b)** lollipop **(c)** closed square

Proving a negative: the tangent field test (Subsection 17.b.3)

AI-Example 17.a.6 disproved a submanifold claim by counting connected components after deleting a point. That argument is sharp but narrow: it needs the bad point to disconnect the set. Linus gives a second, complementary criterion, which catches cases the counting argument misses.

Proposition 17.b.18 (*Continuous tangent field along a curve*): Let $M \subseteq \mathbb{R}^n$ be a 1-dimensional C^1 submanifold. Then every point of M has a neighbourhood on which there is a continuous map T assigning to each q a nonzero vector $T(q) \in T_qM$.

Proof:

Fix $p \in M$. By Theorem 17.a.9 there are an open interval $I \subseteq \mathbb{R}$, a neighbourhood V of p in $\mathbb{R}^n$, and a C^1 homeomorphism $\gamma : I \to V \cap M$ with $D\gamma_t$ injective for every t. Injectivity of a linear map $\mathbb{R} \to \mathbb{R}^n$ says exactly that $\gamma'(t) \neq 0$. Set

$$T(q) := \gamma'\big(\gamma^{-1}(q)\big), \qquad q \in V \cap M.$$

This is nonzero by the above, lies in T_qM by Proposition 17.c.24, and is continuous as a composition of the continuous maps γ^{-1} and γ'. **Q.E.D.**

The test is used in contrapositive form: **if no continuous nonvanishing tangent field exists near a point, the set is not a 1-dimensional submanifold there.**

AI-Example 17.b.19 (*The lollipop, disproved by tangent fields*): Let L be the lollipop of the gallery above and let q be its junction point. Deleting q leaves **three** pieces where a curve would leave two, so the counting argument of AI-Example 17.a.6 already applies. But it is instructive to see the tangent-field test reach the same verdict, since it localises the contradiction differently.

Suppose L were a 1-dimensional submanifold near q, and let T be the tangent field supplied by Proposition 17.b.18. Approach q along the stick: the tangent direction there is constant, so T tends to a vector along the stick. Approach q along the circle instead: the circle meets the stick at a right angle at q, so T tends to a vector perpendicular to the stick. A continuous map cannot have two different limits at the same point, so no such T exists.

Remark 17.b.20 (*When to reach for which*): Both tests disprove the same statement, and neither subsumes the other. Counting components is the tool when the offending point **separates** the set, and it needs no differentiability. The tangent-field test is the tool when the set stays connected but the direction of travel jumps. A cusp such as $\{y^2 = x^3\}$ at the origin is the standard case, where removing the origin leaves the expected two pieces and only the tangent direction gives the failure away.

Every argument in this section leaned on knowing which vectors point **along** M at a given point. The next section makes that space of directions an object in its own right.

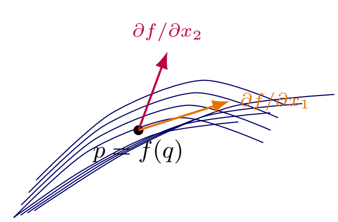
Tangent and normal spaces (Section 17.c)

Tangent space (Subsection 17.c.1)

Suppose $M^m \subseteq \mathbb{R}^n$ is a submanifold and $f : U \to f(U) \subseteq M$ is a local parametrization around some point $p \in f(U)$ with $f(q) = p$. The $n \times m$ matrix

$$\mathrm{J}\, f(q) = \begin{pmatrix} \dfrac{\partial f}{\partial q_1} & \cdots & \dfrac{\partial f}{\partial q_m} \end{pmatrix}$$

has full rank, that is, the column vectors $\partial_i f(q)$ are linearly independent. They form a basis of T_pM:



Definition 17.c.21 (*Tangent space*): For M, f as above, the "*tangent space*" (German: „*Tangentialraum*") of M at $p \in M$ is

$$T_pM := \operatorname{span}\{\partial_1 f(q), \ldots, \partial_m f(q)\} = Df_q(\mathbb{R}^m);$$

it is independent of the choice of f.

Example 17.c.22 (*Tangent space to the upper hemisphere*)**:** Parametrize the upper hemisphere $S^2_+ := \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 = 1,\ z > 0\}$ graphically, i.e.,

$$f(x,y) = \begin{pmatrix} x \\ y \\ \sqrt{1-x^2-y^2} \end{pmatrix}.$$

We determine $T_{e_3}S^2_+$, where $e_3 = (0,0,1)$:

$$\left.\frac{\partial f}{\partial x}\right|_{(x,y)=(0,0)} = \left.\begin{pmatrix} 1 \\ 0 \\ \dfrac{-x}{\sqrt{1-x^2-y^2}} \end{pmatrix}\right|_{(x,y)=(0,0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = e_1,$$

$$\left.\frac{\partial f}{\partial y}\right|_{(x,y)=(0,0)} = \left.\begin{pmatrix} 0 \\ 1 \\ \dfrac{-y}{\sqrt{1-x^2-y^2}} \end{pmatrix}\right|_{(x,y)=(0,0)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = e_2.$$

So, $T_{e_3}S^2_+ = T_{e_3}S^2 = \operatorname{span}\{e_1, e_2\} = \mathbb{R}^2 \times \{0\}$.

Example 17.c.23 (*Tangent space to the 2-Torus*)**:** Consider the 2-torus $\mathbb{T}^2 \subset \mathbb{R}^3$ parametrized by $\phi(t,\theta) : \mathbb{R}^2 \to \mathbb{R}^3$,

$$\phi(t,\theta) := \begin{pmatrix} (R_1 + R_2\cos\theta)\cos t \\ (R_1 + R_2\cos\theta)\sin t \\ R_2\sin\theta \end{pmatrix},$$

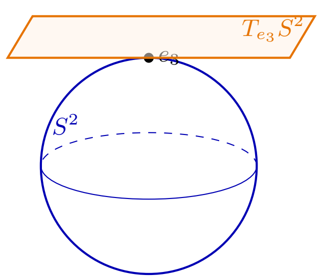
where $R_1 > R_2 > 0$ are the major and minor radii. The partial derivatives of ϕ with respect to the angles t and θ yield two linearly independent column vectors:

$$\frac{\partial\phi}{\partial t} = \begin{pmatrix} -(R_1 + R_2\cos\theta)\sin t \\ (R_1 + R_2\cos\theta)\cos t \\ 0 \end{pmatrix}, \qquad \frac{\partial\phi}{\partial\theta} = \begin{pmatrix} -R_2\sin\theta\cos t \\ -R_2\sin\theta\sin t \\ R_2\cos\theta \end{pmatrix}.$$

By Definition 17.c.21, at any point $p = \phi(t,\theta) \in \mathbb{T}^2$, the tangent space is the two-dimensional vector subspace

$$T_p\mathbb{T}^2 = \operatorname{span}\left\{ \frac{\partial\phi}{\partial t}(t,\theta),\ \frac{\partial\phi}{\partial\theta}(t,\theta) \right\}.$$

In particular, at the outer point $p_0 := (R_1 + R_2, 0, 0)^\mathsf{T}$ (where $t = 0, \theta = 0$), we obtain $\frac{\partial\phi}{\partial t}(0,0) = (0, R_1 + R_2, 0)^\mathsf{T}$ and $\frac{\partial\phi}{\partial\theta}(0,0) = (0,0,R_2)^\mathsf{T}$, so $T_{p_0}\mathbb{T}^2 = \{0\} \times \mathbb{R}^2$.



Definition 17.c.21 builds T_pM out of a parametrization, so the space is pinned down by a formula rather than by what its elements are. There is an equivalent description that says what they are. A vector is tangent to M at p exactly when some curve running inside M passes through p with that velocity, which is both how tangent vectors are usually pictured and the quicker test in practice, since no parametrization need be written down.

Proposition 17.c.24 (*Tangent space via curves*): Let $M^m \subseteq \mathbb{R}^n$ be a C^k submanifold ($k \geq 1$) and $p \in M$. Then

$$T_pM = \big\{\gamma'(0) : \gamma \in C^1\big((-\varepsilon,\varepsilon),\mathbb{R}^n\big),\ \varepsilon > 0,\ \gamma(t) \in M \text{ for all } t,\ \gamma(0) = p\big\}.$$

Proof:

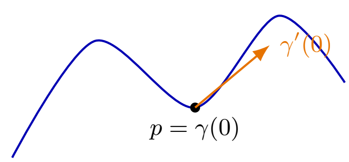
Let $\Phi : U \to V$ be a submanifold chart at p (Definition 17.a.1), so $\Phi(U \cap M) = (\mathbb{R}^m \times \{0\}) \cap V$ and $\Phi(p) = 0$; equivalently, let $f := \Phi^{-1}|_{(\mathbb{R}^m\times\{0\})\cap V}$ be the corresponding local parametrization (Theorem 17.a.9), $f(q) = p$.

($\subseteq$) Let $v \in T_pM = Df_q(\mathbb{R}^m)$, say $v = Df_q(w)$. The curve $\gamma(t) := f(q + tw)$, defined for t small enough that $q + tw$ stays in f's domain, satisfies $\gamma(t) \in M$ for all such t, $\gamma(0) = p$, and by the chain rule $\gamma'(0) = Df_q(w) = v$.

($\supseteq$) Let γ be a curve as in the statement. Since Φ is a diffeomorphism with $\Phi(\gamma(t)) \in \mathbb{R}^m \times \{0\}$ for all t near 0, write $\Phi \circ \gamma =: (\eta, 0)$ with $\eta \in C^1\big((-\varepsilon,\varepsilon),\mathbb{R}^m\big)$. Then $\gamma = \Phi^{-1}(\eta,0) = f(\eta(\cdot))$ near 0 (shrinking U if necessary so that f is defined on all of η's image), so by the chain rule,

$$\gamma'(0) = Df_{\eta(0)}\big(\eta'(0)\big) \in Df_q(\mathbb{R}^m) = T_pM$$

(using $\eta(0) = q$, since $\Phi(p) = 0 = \Phi(f(q))$ and Φ is injective). **Q.E.D.**



Remark 17.c.25 (*Four representations of the tangent space*): Depending on how a m-dimensional submanifold $M \subseteq \mathbb{R}^n$ is locally described near p, its tangent space T_pM can be computed algebraically in four equivalent ways:

- **Diffeomorphism** (from the definition, $\Psi(p) = 0$): $T_pM = (D\Psi_p)^{-1}(\mathbb{R}^m \times \{0\})$.
- **Implicit representation** ($M = F^{-1}\{0\}$): $T_pM = \ker(DF_p)$.
- **Parametric representation** (local parametrization g with $g(q) = p$): $T_pM = Dg_q(\mathbb{R}^m)$.
- **Graph representation** (graph of h, $p = (z, h(z))$): $T_pM = \{(w, Dh_z(w)) \mid w \in \mathbb{R}^m\}$.

These algebraic descriptions are all equivalent to the geometric description via velocities of curves (Proposition 17.c.24).

Normal space (Subsection 17.c.2)

Suppose $F : U \to \mathbb{R}^{n-m}$ is an implicit representation of $M^m = F^{-1}\{0\}$ (via the regular value theorem) and $f : V^m \to M$ is a local parametrization, M a submanifold. Then, by definition,

1. The Jacobian matrix:
$$\mathrm{J}\,F(x) = \begin{pmatrix} \nabla F_1(x) \longrightarrow \\ \vdots \\ \nabla F_{n-m}(x) \longrightarrow \end{pmatrix}$$
has full rank, i.e., $\{\nabla F_i(x)\}_{i=1,\dots,n-m}$ are linearly independent;
2. $T_{f(q)}M = \operatorname{span}\{\partial_i f(q)\}$, as above;
3. $F(f(x)) = 0$ for all $x \in V$, since $f(x) \in M = F^{-1}\{0\}$.

From **3**, it follows that

$$\begin{aligned} \mathrm{J}(F\circ f)(q) &= \mathrm{J}\,F(f(q))\cdot \mathrm{J}\,f(q) \\ &= \begin{pmatrix} \nabla F_1(f(q)) \longrightarrow \\ \vdots \\ \nabla F_{n-m}(f(q)) \longrightarrow \end{pmatrix} \begin{pmatrix} \frac{\partial f}{\partial q_1} & \cdots & \frac{\partial f}{\partial q_m} \end{pmatrix} = 0. \end{aligned}$$

Now, since $\big(\mathrm{J}\,F(f(q))\cdot \mathrm{J}\,f(q)\big)_{ij} = \langle \nabla F_i(f(q)), \partial_j f(q)\rangle = 0$ for all $i=1,\dots,n-m$, $j=1,\dots,m$, the vectors $\nabla F_i(f(q))$ are perpendicular, that is, "*normal*", to $T_{f(q)}M$. Since by **1** we have $n-m$ linearly independent vectors normal to the m-dimensional plane $T_{f(q)}M$, we can define:

Definition 17.c.26 (*Normal space*): Let $M^m \subseteq \mathbb{R}^n$ be a submanifold and $F: U^n \to \mathbb{R}^{n-m}$ an implicit representation, $M = F^{-1}\{0\}$. Then the "*normal space*" (German: „*Normalraum*“) of M at $p\in M$ is

$$N_pM := \operatorname{span}\{\nabla F_1(p),\dots,\nabla F_{n-m}(p)\}.$$

Example 17.c.27 (*Normal space to the sphere*): Represent S^2 implicitly: $S^2 = F^{-1}\{1\} = \{(x,y,z)\in\mathbb{R}^3 : F(x,y,z)=1\}$ where $F(x,y,z) = x^2+y^2+z^2$. We determine $N_{e_3}S^2$:

$$\nabla F(0,0,1) = (2x,2y,2z)\big|_{(x,y,z)=(0,0,1)} = 2e_3.$$

So, $N_{e_3}S^2 = \operatorname{span}\{e_3\} = \{0\}\times\mathbb{R}$.

Proposition 17.c.28 (*The gradient is normal to the level set*): Let $U\subseteq\mathbb{R}^n$ be open, $g\in C^1(U,\mathbb{R})$, and let $S := \{x\in U : g(x) = c\}$ for some $c\in\mathbb{R}$. If $\nabla g(x_0)\neq 0$ at a point $x_0\in S$, then near x_0 the set S is an $(n-1)$-dimensional submanifold and

$$N_{x_0}S = \operatorname{span}\{\nabla g(x_0)\}.$$

In particular the normal space is a **line**, and the affine **tangent** hyperplane to S at x_0 is $\{x\in\mathbb{R}^n : \langle x-x_0,\ \nabla g(x_0)\rangle = 0\}$.

Proof:

That S is a submanifold of dimension $n-1$ near x_0 is Theorem 17.a.3, since $\nabla g(x_0)\neq 0$ says exactly that $Dg_{x_0}:\mathbb{R}^n\to\mathbb{R}$ is surjective. For the normal space, let $v\in T_{x_0}S$ and pick a C^1 curve $\gamma:(-\varepsilon,\varepsilon)\to S$ with $\gamma(0)=x_0$ and $\gamma'(0)=v$. Then $g\circ\gamma\equiv c$, so differentiating at $t=0$ with the chain rule along a path (Section 10.d) gives

$$0 = \frac{d}{dt}\Big|_{t=0} g(\gamma(t)) = \big\langle \nabla g(x_0),\ v\big\rangle.$$

Hence $\nabla g(x_0)\perp T_{x_0}S$, that is, $\operatorname{span}\{\nabla g(x_0)\}\subseteq N_{x_0}S$. Both are subspaces of dimension $n-\dim T_{x_0}S = n-(n-1) = 1$, so they are equal. **Q.E.D.**

Example 17.c.29 (*Tangent and normal vectors via parametrization*): For a parametrized surface $f: V\to\mathbb{R}^3$, the tangent space at $p=f(x)$ is $T_pM = \operatorname{span}\{\partial_1 f(x), \partial_2 f(x)\}$. A normal vector can be found via the cross product: $\vec{n} = \partial_1 f(x)\times\partial_2 f(x)$.

Let us apply this to the sphere S^2 using spherical coordinates:

$$\Phi(\theta,\varphi) = \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}, \qquad (\theta,\varphi)\in(0,\pi)\times(-\pi,\pi).$$

(Note that the image of Φ is the unit sphere minus a *"thin slice"*, namely the meridian where $\varphi=\pm\pi$ together with the two poles where $\theta\in\{0,\pi\}$.) The tangent space $T_{\Phi(\theta,\varphi)}S^2$ is spanned by the partial derivatives:

$$\partial_\theta \Phi(\theta,\varphi) = \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix}, \qquad \partial_\varphi \Phi(\theta,\varphi) = \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix}.$$

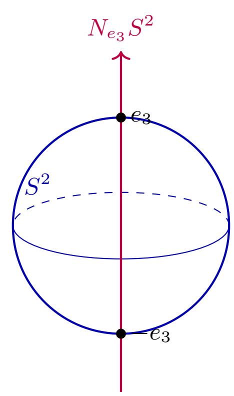
A normal vector is given by their cross product:

$$\begin{aligned} \vec{n} = \partial_\theta\Phi \times \partial_\varphi\Phi &= \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix} \times \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \sin^2\theta\cos\varphi \\ \sin^2\theta\sin\varphi \\ \sin\theta\cos\theta\cos^2\varphi + \sin\theta\cos\theta\sin^2\varphi \end{pmatrix} \\ &= \sin\theta \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}. \end{aligned}$$

Its length is $|\vec{n}| = \sin\theta$ (since $\sin\theta > 0$ for $\theta \in (0,\pi)$). The unit normal vector is therefore

$$\nu(\theta,\varphi) := \frac{\vec{n}}{|\vec{n}|} = \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix} = \Phi(\theta,\varphi).$$

This confirms that the outward normal vector at a point on the unit sphere is precisely the position vector of that point.



Exercise 17.c.30 (*7.6 — Spherical coordinates* (important)): The mapping $\Phi : (0,\infty)\times(0,\pi)\times(-\pi,\pi) \to \mathbb{R}^3$ defined by

$$\Phi(r,\theta,\varphi) = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix}$$

is called *spherical coordinates*.

1. Sketch the images of $r \mapsto \Phi(r,\theta_0,\varphi_0)$, $\theta \mapsto \Phi(r_0,\theta,\varphi_0)$ and $\varphi \mapsto \Phi(r_0,\theta_0,\varphi)$ for some fixed $r_0 \in (0,\infty)$, $\theta_0 \in (0,\pi)$, $\varphi_0 \in (-\pi,\pi)$.
2. What is the image of Φ?
3. Show that $\det(D_{(r,\theta,\varphi)}\Phi) = r^2\sin\theta$ holds.
4. Conclude that the mapping Φ is a diffeomorphism onto its image.

Solutions (Section 17.d)

AI-Note 17.1: Corsin's notes for this material cover only the theory (submanifolds, the regular value and local parametrization theorems, tangent and normal spaces), with no page working out any of the official-sheet exercises, so all solutions below are original. Each has been checked against `exercises/Sol7_Analysis2_eng.pdf`; no divergence from the official solutions was found.

Solution to Exercise 17.c.30:

1. Fixing θ_0, φ_0, the image of $r \mapsto \Phi(r, \theta_0, \varphi_0)$ is the radial half-line through the point $p_0 := (\sin\theta_0 \cos\varphi_0, \sin\theta_0 \sin\varphi_0, \cos\theta_0)$ on the unit sphere, i.e., $\{rp_0 : r \in (0, \infty)\}$. Fixing r_0, φ_0, the image of $\theta \mapsto \Phi(r_0, \theta, \varphi_0)$ is a semicircle of radius r_0 from the north pole to the south pole (excluding the poles), at longitude φ_0. Fixing r_0, θ_0, the image of $\varphi \mapsto \Phi(r_0, \theta_0, \varphi)$ is a circle of latitude of radius $r_0 \sin\theta_0$ at height $r_0 \cos\theta_0$ (excluding the point at $\varphi = \pm\pi$).
2. For each fixed r, the image sweeps out the sphere of radius r minus the half-plane at $\varphi = \pm\pi$ (the **anti**-meridian, since the prime meridian is $\varphi = 0$). Overall,
$$\operatorname{im}(\Phi) = \mathbb{R}^3 \setminus \big((-\infty, 0] \times \{0\} \times \mathbb{R}\big).$$
3. Computing,
$$D_{(r,\theta,\varphi)}\Phi = \begin{pmatrix} \sin\theta\cos\varphi & r\cos\theta\cos\varphi & -r\sin\theta\sin\varphi \\ \sin\theta\sin\varphi & r\cos\theta\sin\varphi & r\sin\theta\cos\varphi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix},$$
hence, expanding along the last row,
$$\begin{aligned}\det(D_{(r,\theta,\varphi)}\Phi) &= r^2\sin\theta\cos^2\theta\cos^2\varphi + r^2\sin^3\theta\sin^2\varphi + r^2\sin\theta\cos^2\theta\sin^2\varphi + r^2\sin^3\theta\cos^2\varphi \\ &= r^2\big(\sin\theta\cos^2\theta + \sin^3\theta\big) = r^2\sin\theta.\end{aligned}$$
4. Since $r \in (0, \infty)$ and $\theta \in (0, \pi)$, $r^2 \sin\theta$ is never 0, so $\mathrm{J}\,\Phi(r, \theta, \varphi)$ is everywhere invertible and Φ is a local diffeomorphism by Theorem 15.a.1. To see it is a **global** diffeomorphism onto its image, it remains to check Φ is injective: if $\Phi(r_1, \theta_1, \varphi_1) = \Phi(r_2, \theta_2, \varphi_2)$, then $r_1 = r_2$ follows by taking the norm of both sides; since $\cos : (0, \pi) \to (-1, 1)$ is injective, the third coordinate gives $\theta_1 = \theta_2$; finally $(\cos\varphi, \sin\varphi)$ with $\varphi \in (-\pi, \pi)$ determines a unique point on the unit circle, so $\varphi_1 = \varphi_2$. Hence Φ is injective, and (being a local diffeomorphism everywhere on its domain) a diffeomorphism onto its image.

Q.E.D.

Geodesics (Chapter 18)

With submanifolds and their tangent spaces in place, a natural question is what a straight line becomes once space is curved. This short chapter answers it. A geodesic is a curve of locally minimal length inside a submanifold, and characterising one takes the first genuine calculus-of-variations argument in the course. Differentiate the length functional along a perturbation, and require the derivative to vanish for every perturbation.

A short introduction to geodesics (Section 18.a)

AI-Note 18.1: Corsin marks this section *"not exam relevant"*. It is included anyway, since the calculation is short and the underlying idea, treating length as a functional and applying optimization to it, is the same one used later in the course for genuinely examinable variational arguments.

We will try to find the shortest path (i.e., a geodesic) from $0 \in \mathbb{R}^n$ to $x \in \mathbb{R}^n$. This is, of course, the straight line, but the essentially same calculation can be used to find the shortest path between points on a sphere or any other submanifold.

Idea. Apply *"optimization"* to the map

$$L : \mathcal{A} \to [0,\infty), \qquad c \mapsto L(c) = \int_0^1 |c'(t)|\,dt, \qquad \mathcal{A} := \{\gamma \in C^1([0,1],\mathbb{R}^n) : \gamma(0) = 0,\ \gamma(1) = x\}.$$

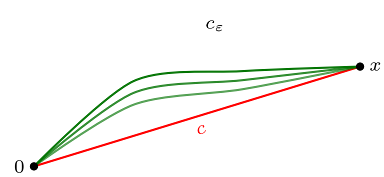
Lemma 18.a.1 (*A few differentiation identities*)**:** Let $\alpha, \beta : (0,1) \to \mathbb{R}^n$ be C^1. Then

$$\frac{d}{dt}\langle \alpha(t), \beta(t)\rangle = \langle \alpha'(t), \beta(t)\rangle + \langle \alpha(t), \beta'(t)\rangle, \qquad \frac{d}{dt}|\alpha(t)| = \frac{1}{|\alpha(t)|}\langle \alpha(t), \alpha'(t)\rangle.$$

Proposition 18.a.2 (*The shortest path is a straight line*)**:** Assume that $c \in C^\infty([0,1],\mathbb{R}^n)$ with $c(0) = 0$, $c(1) = x$ and $|c'(t)| \equiv \lambda$ is the optimal path we are looking for. Then $c(t) = xt$ and $\lambda = |x|$.

Proof:

Let $g \in C^\infty([0,1],\mathbb{R}^n)$ with $g(0) = g(1) = 0$, and define, for $\varepsilon \in \mathbb{R}$, the family of paths $c_\varepsilon(t) := c(t) + \varepsilon g(t)$, so that $c_0 = c$. The boundary conditions on g are what keep the whole family admissible: $c_\varepsilon(0) = c(0) = 0$ and $c_\varepsilon(1) = c(1) = x$ for every ε, so each c_ε is a competitor in $\mathcal{A}$ and not merely a nearby curve.



By assumption, c is a minimizer of L, so it is a critical point of the family, i.e.

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L(c_\varepsilon) = 0.$$

(This equality itself requires a proof, but the idea is exactly the same as that of optimization in $\mathbb{R}^n$.) First,

$$\left.\frac{\partial}{\partial\varepsilon}\right|_{\varepsilon=0} c_\varepsilon(t) = \left.\frac{\partial}{\partial\varepsilon}\right|_{\varepsilon=0} \big(c(t) + \varepsilon g(t)\big) = g(t).$$

Recalling $|c'| \equiv \lambda$ and using Lemma 18.a.1,

$$\begin{aligned}
\frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(c_\varepsilon) &= \frac{d}{d\varepsilon}\int_0^1 |c_\varepsilon'(t)|\,dt\bigg|_{\varepsilon=0} = \int_0^1 \frac{\partial}{\partial\varepsilon}|c_\varepsilon'(t)|\,dt\bigg|_{\varepsilon=0} \\
&= \int_0^1 \frac{\langle c_\varepsilon'(t),\ \partial_\varepsilon c_\varepsilon'(t)\rangle}{|c_\varepsilon'(t)|}\,dt\bigg|_{\varepsilon=0} = \frac{1}{\lambda}\int_0^1 \Big\langle c'(t),\ \partial_\varepsilon\big|_{\varepsilon=0} c_\varepsilon'(t)\Big\rangle\,dt \\
&= \frac{1}{\lambda}\int_0^1 \frac{d}{dt}\Big\langle c'(t), g(t)\Big\rangle - \big\langle c''(t), g(t)\big\rangle\,dt = \frac{1}{\lambda}\left[\Big\langle c'(t), g(t)\Big\rangle\Big|_0^1 - \int_0^1 \big\langle c''(t), g(t)\big\rangle\,dt\right] \\
&= -\frac{1}{\lambda}\int_0^1 \big\langle c''(t), g(t)\big\rangle\,dt \overset{!}{=} 0,
\end{aligned}$$

where the boundary term vanished since $g(0) = g(1) = 0$. Since g is arbitrary, it follows that $c''(t) \equiv 0$. This last step is the "*fundamental lemma of the calculus of variations*": a continuous function integrating to zero against **every** smooth g vanishing at the endpoints must itself vanish, since otherwise one could choose g to be a bump concentrated where c'' has a fixed sign and make the integral non-zero. Together with $c(0) = 0$ and $c(1) = x$ this gives $c(t) = xt$ and $\lambda = |x|$.

Q.E.D.

Remark 18.a.3 (*The two steps that hide real work*): The proof moves quickly over $\frac{d}{d\varepsilon}\big|_{\varepsilon=0} L(c_\varepsilon) = 0$ and over the final $\overset{!}{=} 0$, and those are precisely the two places where something is being assumed.

The first is the claim that a minimizer of L over the infinite-dimensional set $\mathcal{A}$ is a critical point of every one-parameter family through it. That is the same fact used for optimization on $\mathbb{R}^n$ in Proposition 12.a.2, and it wants $\mathcal{A}$ to be a vector space. It is not one, since $\mathcal{A}$ is an affine subspace. What rescues the argument is that the **differences** $c_\varepsilon - c$ range over $\{g \in C^\infty([0,1], \mathbb{R}^n) : g(0) = g(1) = 0\}$, which is a genuine vector space, so the perturbation directions form one even though the curves themselves do not.

The second is the fundamental lemma named in the proof, which upgrades "*the integral vanishes for every g*" to "*c'' vanishes at every t*", rather than merely on average. Corsin gives neither its name nor its proof.

Remark 18.a.4 (*Why constant speed is assumed, and why it costs nothing*): Proposition 18.a.2 assumes $|c'(t)| \equiv \lambda$, which looks like a restriction on the competitor curves and is not one. Length is invariant under reparametrization, since substituting $t = \varphi(s)$ for an increasing C^1 bijection φ leaves $\int |c'|\,dt$ unchanged. Every curve with nowhere-vanishing speed therefore has a constant-speed representative tracing the same image, obtained by running along it proportionally to arc length. The hypothesis chooses that representative.

It is worth choosing, because the functional L cannot see the difference and the equation can. Without it the computation still goes through, but $1/|c'(t)|$ stays inside the integral and the conclusion is that $\big(c'/|c'|\big)' \equiv 0$, which says the **direction** of travel is constant while the speed is free. Constant speed converts that into the cleaner $c'' \equiv 0$.

Geodesics on a submanifold (Section 18.b)

What survives of "*shortest means straight*" when the curve is not free to leave a surface? On $\mathbb{R}^n$ the answer was $c'' \equiv 0$, and on a sphere that is hopeless: no curve with zero acceleration stays on a sphere at all. Something weaker has to replace it, and the computation itself says what.

The only place the ambient space entered the proof of Proposition 18.a.2 was in the choice of perturbations g. There, g ranged over **all** smooth fields vanishing at the endpoints, and that is exactly what forced $c'' \equiv 0$. On a submanifold the competitors must stay on M, so g may only push along M, and the conclusion weakens in step: c'' has to be orthogonal to the directions g is allowed to take, and to nothing else.

Proposition 18.b.5 (*The geodesic equation*): Let $M \subseteq \mathbb{R}^n$ be a C^2 submanifold and let $c \in C^2([0,1],\mathbb{R}^n)$ with $c(t) \in M$ for all t and constant speed $|c'(t)| \equiv \lambda > 0$. If c minimises length among C^1 curves in M with the same endpoints, then

$$c''(t) \;\perp\; T_{c(t)}M \qquad \text{for every } t \in [0,1],$$

that is, $c''(t) \in N_{c(t)}M$ in the sense of Definition 17.c.26.

Proof sketch:

Let $g \in C^1$ vanish at the endpoints and be "*tangential*" along c, meaning $g(t) \in T_{c(t)}M$ for every t. Take any family c_ε of C^1 curves **in** M with $c_0 = c$, with the endpoints held fixed, and with $\partial_\varepsilon\big|_{\varepsilon=0} c_\varepsilon = g$. Such a family exists: read M near $c(t)$ through a local parametrization (Theorem 17.a.9), push g back to the parameter domain, move there, and push forward again. This is the one step taken on trust here, and it is the only place the submanifold structure is used.

From there the computation of Proposition 18.a.2 runs verbatim, since it never asked where g came from, and yields

$$0 = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(c_\varepsilon) = -\frac{1}{\lambda}\int_0^1 \big\langle c''(t),\, g(t)\big\rangle\, dt.$$

The fundamental lemma of the calculus of variations now applies to tangential g only, so it gives that the tangential **part** of $c''(t)$ vanishes for every t, rather than c'' itself. That is precisely $c''(t) \perp T_{c(t)}M$. **Q.E.D.**

Remark 18.b.6 (*Reading the equation*): The statement has a mechanical reading worth carrying: a geodesic is a curve that accelerates **only as much as the surface forces it to**. Any acceleration along M would be wasted motion that a shorter competitor avoids, so all that is left points out of M, and the surface supplies exactly that much to keep the curve on itself. Setting $M = \mathbb{R}^n$ recovers Proposition 18.a.2, since then $T_{c(t)}M = \mathbb{R}^n$ and orthogonality to everything means $c'' \equiv 0$.

The converse fails, and for the same reason it failed in Proposition 12.a.2: the equation is a first-order criticality condition, so it detects candidates. On the sphere the long way round a great circle satisfies it and is not shortest.

AI-Example 18.b.7 (*Great circles, and why a circle of latitude is not a geodesic*): Take $M := S^2 = \{x \in \mathbb{R}^3 : |x|^2 = 1\}$. By Proposition 17.c.28 applied to $g(x) := |x|^2$, the normal space at $p \in S^2$ is $N_pS^2 = \operatorname{span}\{p\}$, so Proposition 18.b.5 reads: **c is a geodesic candidate exactly when $c''(t)$ is a multiple of $c(t)$.**

A great circle satisfies it. Fix orthonormal $p, v \in \mathbb{R}^3$ and put

$$c(t) := \cos(\lambda t)\, p + \sin(\lambda t)\, v.$$

Then $|c(t)| = 1$, so c runs in S^2, and $|c'(t)| \equiv \lambda$, so the constant-speed hypothesis holds. Differentiating twice,

$$c''(t) = -\lambda^2\big(\cos(\lambda t)\, p + \sin(\lambda t)\, v\big) = -\lambda^2\, c(t),$$

which is a multiple of $c(t)$. The condition holds at every t.

A circle of latitude does not. Fix a height h with $0 < |h| < 1$, put $r := \sqrt{1-h^2}$, and run around the circle at that height at constant speed:

$$c(t) := \big(r\cos(\omega t),\; r\sin(\omega t),\; h\big), \qquad \omega \neq 0.$$

Again $|c(t)|^2 = r^2 + h^2 = 1$ and $|c'(t)| \equiv r|\omega|$ is constant. But

$$c''(t) = -\omega^2\big(r\cos(\omega t),\; r\sin(\omega t),\; 0\big),$$

whose third coordinate is 0 while the third coordinate of $c(t)$ is $h \neq 0$, and whose first two coordinates are not both zero. So, $c''(t)$ is not a multiple of $c(t)$, and no circle of latitude other than the equator ($h = 0$, where the two agree and the curve is a great circle) is a geodesic.

This is the flight-path fact in the form the equation gives it: flying due west along a parallel is not the shortest route between two points on it, which is why long-haul routes bend towards the pole.

AI-Exercise 18.b.8 (*Geodesics on the cylinder*): Let $Z := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$ be the unit cylinder, and for $a, b \in \mathbb{R}$ not both zero consider the helix

$$c(t) := \big(\cos(at),\ \sin(at),\ bt\big).$$

(a) Show that c runs in Z at constant speed, and compute that speed.

(b) Determine $N_{c(t)}Z$, and verify that c satisfies the geodesic equation of Proposition 18.b.5 for every choice of a and b.

(c) Which curves do the cases $a = 0$ and $b = 0$ describe?

Solutions (Section 18.c)

Solution to AI-Exercise 18.b.8:

(a) The first two coordinates satisfy $\cos^2(at) + \sin^2(at) = 1$ for every t, so $c(t) \in Z$. Differentiating,

$$c'(t) = \big(-a\sin(at),\ a\cos(at),\ b\big), \qquad |c'(t)| = \sqrt{a^2\sin^2(at) + a^2\cos^2(at) + b^2} = \sqrt{a^2 + b^2},$$

which is constant and non-zero, since a and b are not both zero.

(b) Write $Z = F^{-1}\{0\}$ for $F(x, y, z) := x^2 + y^2 - 1$. Then $\nabla F(x, y, z) = (2x,\ 2y,\ 0)$, which is non-zero on Z, so Theorem 17.a.3 applies and Definition 17.c.26 gives

$$N_pZ = \operatorname{span}\{(p_1,\ p_2,\ 0)\}, \qquad p \in Z.$$

The cylinder's normal at a point is horizontal and radial: it forgets the height entirely. Differentiating c a second time,

$$c''(t) = \big(-a^2\cos(at),\ -a^2\sin(at),\ 0\big) = -a^2\big(c_1(t),\ c_2(t),\ 0\big),$$

which is exactly a multiple of the spanning vector of $N_{c(t)}Z$. So $c''(t) \in N_{c(t)}Z$, i.e. $c''(t) \perp T_{c(t)}Z$, and the geodesic equation of Proposition 18.b.5 holds for every a and b.

Note that the z-component of c'' vanishes identically, which is what makes this work. The vertical direction is tangential to Z everywhere, so any vertical acceleration would have a tangential part and would violate the equation.

(c) With $a = 0$ the curve is $t \mapsto (1, 0, bt)$, a vertical straight line ruling the cylinder. With $b = 0$ it is $t \mapsto (\cos(at), \sin(at), 0)$, the horizontal unit circle. Both are limiting cases of the helix, and both are geodesics by **(b)**.

Remark 18.c.9 (*Why the cylinder has **only these geodesics*):** The three families found here are in fact all of them, and the quickest reason is that the cylinder is "*locally isometric*" to the plane: rolling a sheet of paper into a cylinder bends it without stretching, so it preserves the length of every curve drawn on it. Lengths are all the geodesic equation ever sees, so geodesics must correspond to geodesics. Unrolling sends helices to straight lines, vertical rulings to vertical lines, and horizontal circles to horizontal lines, which is every straight line in the plane and nothing else.

The sphere admits no such trick, which is the same obstruction that makes flat world maps distort areas or angles, and it is why AI-Example 18.b.7 had to be checked by computation instead.

Q.E.D.

PART VI

Integration

Integration in several variables raises a question that one variable never had to face. Before asking what $\int_A f$ is, one has to say which sets A possess a volume at all, and Jordan measure is the answer this course uses. With that settled, Fubini's theorem reduces a multiple integral to iterated one-dimensional ones, and substitution generalizes to the change of variables formula, in which the factor that appears is $|\det \mathrm{J}\,\phi|$, the local volume distortion of the substitution. The final chapter leaves flat regions behind altogether. The Gram determinant measures how much a parametrization stretches length, area or volume, and with it one can integrate over a curve or a surface sitting inside $\mathbb{R}^n$ rather than over a piece of $\mathbb{R}^n$ itself.

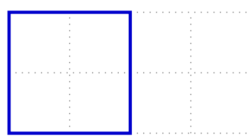
Jordan Measure and the Riemann Integral (Chapter 19)

Integration in several variables needs a notion of the volume of a domain before it can define an integral over one. This chapter sets up Jordan measure and the multidimensional Riemann integral built on it, and then spends most of its effort on the sets that are **not** Jordan measurable. That is where every counterexample on the problem sheet lives, and the reason they exist is a single one: a Jordan cover may use only finitely many cubes, all of one size.

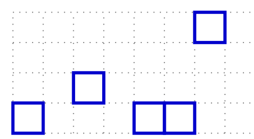
Multivariate integration (Section 19.a)

Jordan measure and the Riemann integral (Subsection 19.a.1)

The rigorous definition of the multidimensional integral relies on the "*Jordan measure*" (German: „*Jordan-Maß*"). The intuition, as presented in Sascha Brack's notes, involves "*dyadic cubes*" (pixels) that approximate a region from the inside and outside.



dyadic cube, side length $1 = 2^{-0}$



dyadic set
(union of same size cubes)

Definition 19.a.1 (*Dyadic cube and dyadic set*): For $p \in \mathbb{N}$ and $a \in \mathbb{Z}^n$, the "*dyadic cube*" (German: „*dyadischer Würfel*") of generation p at a is

$$Q_p(a) := 2^{-p}\big(a + [0,1)^n\big),$$

a half-open cube of side length 2^{-p}. A "*dyadic set*" (German: „*dyadische Menge*") is a finite union $E = \bigcup_{\ell=1}^{N} Q_p(a(\ell))$ of distinct dyadic cubes **of one common generation** p, and its volume is

$$\mu(E) := 2^{-pn} N.$$

Exercise 19.a.2 (*Inclusion–exclusion for dyadic sets*): Show that, for two dyadic sets E_1, E_2,

$$\mu(E_1 \cup E_2) = \mu(E_1) + \mu(E_2) - \mu(E_1 \cap E_2).$$

Show more generally that, for dyadic sets $E_1, \dots, E_N$, the measure of their union is given by the principle of inclusion–exclusion,

$$\mu\Big(\bigcup_{i=1}^{N} E_i\Big) = \sum_{k=1}^{N} (-1)^{k+1} \Big(\sum_{1 \le i_1 < i_2 < \cdots < i_k \le N} \mu(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_k}) \Big).$$

Proof of Exercise 19.a.2:

Any finite collection of dyadic sets can be refined to a common generation p without changing any of their measures (splitting a cube of generation p into 2^n cubes of generation $p+1$ replaces one term of $\mu(E) := 2^{-pn}N$ by 2^n terms of size $2^{-(p+1)n}$, which sum to the same value). So, we may assume $E_1, \dots, E_N$ all consist of cubes of one common generation p. Each such cube Q then either lies in none of the E_i, or in exactly the E_i indexed by some non-empty $J \subseteq \{1, \dots, N\}$, and Q contributes 2^{-pn} to $\mu(E_{i_1} \cap \cdots \cap E_{i_k})$ exactly when $\{i_1, \dots, i_k\} \subseteq J$.

Two sets. If Q lies in neither E_1 nor E_2, it contributes 0 to both sides. If it lies in exactly one, it contributes 2^{-pn} to $\mu(E_1) + \mu(E_2)$ and to $\mu(E_1 \cup E_2)$, and 0 to $\mu(E_1 \cap E_2)$: both sides agree. If it lies in both, it contributes 2^{-pn} to $\mu(E_1 \cup E_2)$, and on the right $2 \cdot 2^{-pn} - 2^{-pn} = 2^{-pn}$: both sides agree again.

N **sets.** It suffices to check the identity cube by cube. A cube Q lying in exactly the sets indexed by $J \neq \emptyset$, $|J| =: m \geq 1$, contributes 2^{-pn} to the left side (it lies in the union). On the right, for each k it contributes 2^{-pn} once for every k-element subset of J, of which there are $\binom{m}{k}$, so the total contribution is

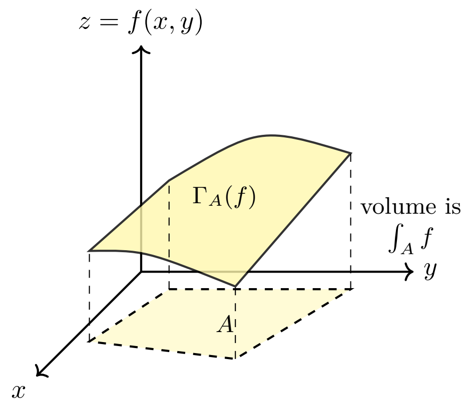
$$2^{-pn}\sum_{k=1}^{m}(-1)^{k+1}\binom{m}{k} = 2^{-pn}\Big(1-\sum_{k=0}^{m}(-1)^k\binom{m}{k}\Big) = 2^{-pn}\big(1-(1-1)^m\big) = 2^{-pn},$$

using the binomial theorem and $m \geq 1$. Both sides agree cube by cube, hence in total. **Q.E.D.**

The requirement that all cubes share a generation is not cosmetic: it is what separates the Jordan theory from the Lebesgue theory, as Remark 19.a.10 makes precise.

Similar to the 1D Riemann integral, we approximate the volume of a set $E \subseteq \mathbb{R}^n$ using an outer sequence of dyadic sets $G_i \supseteq E$ and an inner sequence $F_i \subseteq E$. If the limits of their volumes coincide as the pixel size $2^{-p} \to 0$, E is "*Jordan measurable*".

Furthermore, the multidimensional integral of a non-negative function $f : A \to \mathbb{R}$ is exactly the $(n+1)$-dimensional Jordan measure (volume) of its hypograph $\Gamma_A(f)$:

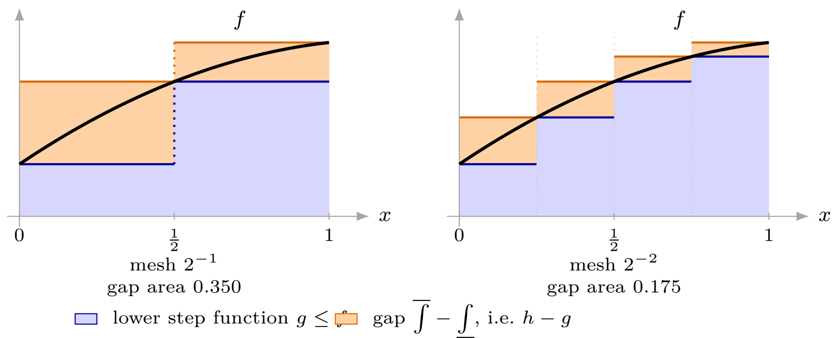


Definition 19.a.3 (*Riemann integral in $\mathbb{R}^n$*): Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}$. Analogous to Analysis I, the integral is defined via dyadic step functions (German: „*dyadische Treppenfunktionen*“). The lower and upper integrals are defined as

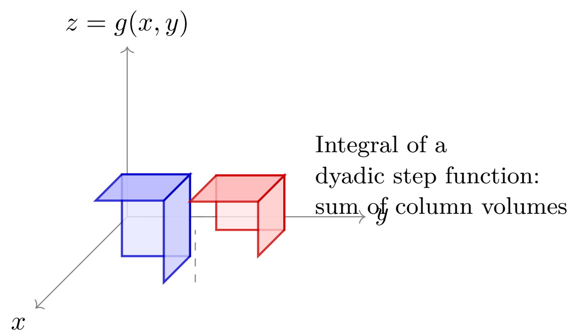
$$\underline{\int}_A f := \sup\left\{\int_A g \;\middle|\; g : \mathbb{R}^n \to \mathbb{R} \text{ dyadic step function with } g \leq f\right\},$$
$$\overline{\int}_A f := \inf\left\{\int_A h \;\middle|\; h : \mathbb{R}^n \to \mathbb{R} \text{ dyadic step function with } h \geq f\right\}.$$

The function f is "*Riemann integrable*" if $\underline{\int}_A f = \overline{\int}_A f$. In this case, their common value is denoted by $\int_A f$.

In one variable the construction can simply be drawn, and it is worth seeing once. Fix a dyadic mesh. On each dyadic interval let g be the largest constant still lying below f, and h the smallest constant still lying above it. Refining the mesh can only raise g and lower h, so the orange gap between them shrinks, and f is integrable exactly when that gap can be driven to 0.



Halving the mesh here halves the gap, from 0.350 to 0.175. That is not a coincidence of this example: for a monotone f on an interval the upper and lower steps differ, on each piece, by exactly the rise of f across it, and those rises telescope. In $\mathbb{R}^n$ the same construction stacks columns over dyadic cubes instead of intervals:



This integral satisfies all the *"good"* properties from the one-dimensional case, such as linearity. Moreover, if $E \subseteq \mathbb{R}^n$ is Jordan measurable and $f : \overline{E} \to [0, \infty)$ is continuous, then f is Riemann integrable on E.

Theorem 19.a.4 (*Jordan measurability criterion via boundary*): A bounded subset $E \subseteq \mathbb{R}^n$ is Jordan measurable if and only if its boundary ∂E is a Jordan null set, i.e. $\mu_{\text{out}}(\partial E) = 0$.

Lemma 19.a.5 (*Compact sets: Jordan null versus Lebesgue null*): Let $K \subseteq \mathbb{R}^n$ be compact. Then K is a Jordan null set if and only if K is a Lebesgue null set.

Proof:

If K is Jordan null, then for every $\varepsilon > 0$, K is covered by finitely many dyadic cubes $Q_1, \ldots, Q_m$ with total volume $\sum_{i=1}^{m} \mu(Q_i) < \varepsilon$, so K is a Lebesgue null set. Conversely, if K is Lebesgue null, there is a countable cover by dyadic cubes $\{Q_j\}_{j \geq 1}$ with $\sum_{j=1}^{\infty} \mu(Q_j) < \varepsilon$. Taking the interiors of slightly enlarged cubes yields an open cover of the compact set K, which admits a finite subcover of total volume $< 2\varepsilon$, proving K is Jordan null. **Q.E.D.**

Example 19.a.6 (*The Cantor set is a Jordan null set*): Consider the middle-thirds "*Cantor set*" $C \subseteq [0, 1]$, defined inductively by $C_0 := [0, 1]$ and

$$C_{n+1} := \tfrac{1}{3} C_n \cup \left(\tfrac{2}{3} + \tfrac{1}{3} C_n\right), \qquad C := \bigcap_{n=0}^{\infty} C_n.$$

Each C_n consists of 2^n closed intervals of length 3^{-n}, so $\mu(C_n) = (2/3)^n$. Since $C \subseteq C_n$ for all n, we have

$$\mu_{\text{out}}(C) \leq \mu(C_n) = \left(\tfrac{2}{3}\right)^n \xrightarrow{n \to \infty} 0,$$

proving that the Cantor set C is a Jordan null set.

Moreover, as Diego Torres Tejeda highlights in his notes, C consists precisely of those points $x \in [0,1]$ possessing a ternary (base-3) expansion $x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}$ composed entirely of digits $a_n \in \{0,2\}$ without the digit 1.

Example 19.a.7 (*An open set that is not Jordan measurable*): Jordan measurability can fail even for open sets. Enumerate the rational numbers in $[0,1]$ as $\mathbb{Q} \cap [0,1] = \{q_1, q_2, \dots\}$, and define the open set

$$U := \bigcup_{k=1}^{\infty} B_{2^{-(k+2)}}(q_k) \ \cap \ (0,1).$$

Since $\mathbb{Q} \cap [0,1]$ is dense in $[0,1]$, the closure of U is $\overline{U} = [0,1]$. If U were Jordan measurable, its measure would satisfy $\mu(U) = \mu(\overline{U}) = 1$. However, any finite union of intervals covering a dyadic inner approximation of U has measure bounded by $\sum_{k=1}^{\infty} 2 \cdot 2^{-(k+2)} = \frac{1}{2}$. Thus $\mu_{\text{in}}(U) \le \frac{1}{2} < 1 = \mu_{\text{out}}(U)$, showing that U is **not** Jordan measurable. Its boundary $\partial U = [0,1] \setminus U$ is a *"fat"* boundary of positive outer measure.

Proposition 19.a.8 (*Algebra of Jordan measurable sets*): If $E, F \subset \mathbb{R}^n$ are Jordan measurable, then their union $E \cup F$, intersection $E \cap F$, and differences $E \setminus F$, $F \setminus E$ are also Jordan measurable. Furthermore, if $E \cap F = \emptyset$ (or if they only intersect in a Jordan null set), then

$$\mu(E \cup F) = \mu(E) + \mu(F).$$

Remark 19.a.9 (*Lebesgue null vs. Jordan null*): A set $E \subset \mathbb{R}^n$ is "*Lebesgue null*" if for every $\varepsilon > 0$, there exist **countably** many dyadic cubes Q_i covering E with $\sum \mu(Q_i) < \varepsilon$. For a "*Jordan null*" set, the cover must be formed by **finitely** many cubes (which implicitly requires E to be bounded). Because a countable cover is allowed, the set $\mathbb{Q} \cap [0,1]$ is Lebesgue null. However, it is not Jordan null (in fact, it is not even Jordan measurable, as seen in Example 19.a.7), since any finite union of intervals covering it must also cover its closure $[0,1]$, thus having measure at least 1.

Remark 19.a.10 (*The relaxation is really about **sizes, not just counts*):** Stating the difference as *"finitely many"* against *"countably many"* is correct but hides where the extra power actually comes from. A Jordan cover is a **dyadic set**, and every cube in a dyadic set of generation p has the same side length 2^{-p} (Definition 19.a.1). So, the two conditions really read:

- **Jordan null:** for each $\varepsilon > 0$, cover E by cubes that all have **one common side length**, of total volume $< \varepsilon$. Finiteness is then automatic: only finitely many cubes of a fixed size fit into a bounded region.
- **Lebesgue null:** for each $\varepsilon > 0$, cover E by cubes of **whatever side lengths you like**, of total volume $< \varepsilon$. Allowing the sizes to vary is exactly what lets the cover be infinite while the total volume stays finite.

The two relaxations are therefore one relaxation. It is permission to shrink the cubes as you go that buys the countable cover, and it is the countable cover that makes the notions differ.

$\mathbb{N} \subseteq \mathbb{R}$ shows this in its purest form. Cover the integer k by an interval of length $\varepsilon\, 2^{-k-1}$; the total length is ε, so $\mathbb{N}$ is Lebesgue null. No Jordan cover can do this: cubes of one fixed length 2^{-p} need at least one cube per integer, infinitely many of them, of infinite total length. (Indeed $\mathbb{N}$ is unbounded, so it is not Jordan measurable at all.) The point is that the geometric series is available only when the sizes may vary.

Theorem 19.a.11 (*Linear transformations of Jordan measurable sets*): Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear, invertible map. If $E \subset \mathbb{R}^n$ is Jordan measurable, then the image $L(E)$ is also Jordan measurable, and its measure scales by the absolute value of the determinant of L:

$$\mu(L(E)) = |\det L|\, \mu(E).$$

Exercise 19.a.12 (*8.2 — True or False* (important)):

1. A bounded countable set is always Jordan-null.
2. A countable set is always Lebesgue-null.
3. Let $D \subset [0,1]$ be a dense set (i.e. $\overline{D} = [0,1]$). Then $\mu_{out}(D) = 1$. (μ_{out} was defined in Definition 13.10.)
4. Let $X, Y \subset [0,1]$ Jordan measurable sets such that $\mu(X) > 1/2$ and $\mu(Y) > 1/2$. Then $X \cap Y \neq \emptyset$.
5. Let $X, Y \subset [0,1]$ such that $\mu_{out}(X) > 1/2$ and $\mu_{out}(Y) > 1/2$. Then $X \cap Y \neq \emptyset$.

Official hint: **8.2.5:** μ_{out} can be positive and large on very sparse sets...

Sascha Brack's hints: a toolkit of sets worth testing every part against, namely $\mathbb{Q}$, the irrationals $\mathbb{R} \setminus \mathbb{Q}$, $\mathbb{Q} \cap [0,1]$, and $(\mathbb{R} \setminus \mathbb{Q}) \cap [0,1]$. For **8.2.3** specifically: monotonicity already gives $\mu_{out}(D) \leq \mu_{out}([0,1]) = 1$, so the work is showing $\mu_{out}(D) \geq 1$, essentially that the *"smallest"* dyadic set containing D also contains $[0,1]$.

Exercise 19.a.13 (*8.4 — Multiple Choice* (important)): Let $U \subset \mathbb{R}^n$ be a nonempty, open subset, $f : U \to \mathbb{R}^m$ a function, and $N \subset U$ a Jordan null set. In which of the following cases is the image $f(N) \subset \mathbb{R}^m$ necessarily a null set? *Attention: Only one answer is correct!*

1. If f is uniformly continuous.
2. If f is uniformly continuous and $m \geq n$.
3. If f is locally Lipschitz continuous.
4. If f is locally Lipschitz continuous and $m \geq n$.

Solutions (Section 19.b)

Solution to Exercise 19.a.12:

The whole problem turns on one distinction: **Jordan** outer measure is an infimum over **finite** covers, **Lebesgue** outer measure over **countable** ones. Parts 1–3 are that distinction three times; parts 4 and 5 are the difference between a measure and a merely sub-additive outer measure.

1. **False.** Take $X := \mathbb{Q} \cap [0,1]$: bounded and countable, yet $\mu_{out}(X) = 1$ by part 3, since X is dense in $[0,1]$. Countability buys nothing here, since only finitely many intervals are allowed at a time, and finitely many intervals covering a dense set must cover the whole of $[0,1]$.
2. **True**, and this is exactly where countable covers do buy something. Enumerate $X = \{x_1, x_2, \dots\}$ and, given $\varepsilon > 0$, cover x_k by an interval of length $\varepsilon 2^{-k}$. The total length is $\sum_{k \geq 1} \varepsilon 2^{-k} = \varepsilon$, and ε was arbitrary, so the Lebesgue outer measure is 0.
3. **True.** Monotonicity gives $\mu_{out}(D) \leq \mu_{out}([0,1]) = 1$, so only the lower bound needs an argument. Let $I_1, \dots, I_N$ be finitely many intervals covering D. Enlarging each to its closure changes no length, so we may assume they are closed; then $\bigcup_{k=1}^N I_k$ is **closed**, which is where finiteness enters, and contains D, hence contains $\overline{D} = [0,1]$. Therefore $\sum_{k=1}^N |I_k| \geq \mu([0,1]) = 1$. Taking the infimum over such covers, $\mu_{out}(D) \geq 1$.
4. **True.** Suppose $X \cap Y = \emptyset$. Jordan measure is additive on disjoint Jordan measurable sets, so
$$\mu(X \cup Y) = \mu(X) + \mu(Y) > \tfrac{1}{2} + \tfrac{1}{2} = 1.$$
But $X \cup Y \subseteq [0,1]$, so monotonicity forces $\mu(X \cup Y) \leq 1$, a contradiction. Hence $X \cap Y \neq \emptyset$.
5. **False**, and this is the official hint's point. Take
$$X := \mathbb{Q} \cap [0,1], \qquad Y := (\mathbb{R} \setminus \mathbb{Q}) \cap [0,1].$$
Both are dense in $[0,1]$, so part 3 gives $\mu_{out}(X) = \mu_{out}(Y) = 1 > \frac{1}{2}$; yet $X \cap Y = \emptyset$ by construction. The argument of part 4 breaks at its first step: μ_{out} is only **sub**additive,

so $\mu_{out}(X \cup Y) \leq \mu_{out}(X) + \mu_{out}(Y)$, an inequality pointing the wrong way to produce a contradiction. Neither set is Jordan measurable, which is precisely why part 4 does not apply.

AI-Note 19.1: Parts 1 and 5 are the same counterexample used twice, and part 3 is the lemma both rest on. It is worth extracting what makes it work: a **finite** union of closed sets is closed, so a finite cover of D automatically covers $\overline{D}$. Every difference between Jordan and Lebesgue theory visible on this sheet traces back to that one sentence.

Q.E.D.

Solution to Exercise 19.a.13:

The correct answer is 4: f locally Lipschitz continuous and $m \geq n$. Both hypotheses are needed, and dropping either one admits a counterexample.

Why 1 and 2 fail: uniform continuity is not enough. Let $c : [0,1] \to [0,1]$ be the Cantor function, which is continuous on a compact interval and hence uniformly continuous, and let $N \subset [0,1]$ be the Cantor set. Then N is Jordan-null (it is covered by 2^k intervals of total length $(2/3)^k \to 0$), while $c(N) = [0,1]$ is not null. Here $n = m = 1$, so this single example kills option 1 **and** option 2. The extra hypothesis $m \geq n$ does nothing to rescue mere continuity. Intuitively, a uniformly continuous map has no bound on how much it may stretch, only on how abruptly.

Why 3 fails: $m \geq n$ is not decoration. Let $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) := y$, which is Lipschitz with constant 1, and let $N := \{0\} \times [0,1]$, a segment and so Jordan-null in $\mathbb{R}^2$. Then $f(N) = [0,1]$, which is not null in $\mathbb{R}^1$. Here $m = 1 < 2 = n$: a projection loses a dimension, and what was negligible in the source stops being negligible in the smaller target.

Why 4 holds. Let N be Jordan-null. Being Jordan-null it is bounded, and we may work on a compact neighbourhood $K \subset U$ of $\overline{N}$ on which f is Lipschitz with some constant L (local Lipschitz continuity plus compactness gives a single constant). Fix $\varepsilon > 0$ and cover N by finitely many cubes $Q_1, \ldots, Q_M \subset K$ of sides $s_i \leq 1$ with $\sum_i s_i^n < \varepsilon$. Each Q_i has diameter $s_i\sqrt{n}$, so $f(Q_i \cap N)$ has diameter at most $Ls_i\sqrt{n}$ and therefore fits inside a cube of side $Ls_i\sqrt{n}$ in $\mathbb{R}^m$, of volume $(L\sqrt{n})^m s_i^m$. Since $s_i \leq 1$ and $m \geq n$ we have $s_i^m \leq s_i^n$, so the images are covered with total volume at most

$$(L\sqrt{n})^m \sum_{i=1}^{M} s_i^n \;<\; (L\sqrt{n})^m\, \varepsilon.$$

As ε was arbitrary, $f(N)$ is null.

AI-Note 19.2: The proof shows exactly where each hypothesis is spent. **Lipschitz** is what converts a bound on the side of a cube into a bound on the diameter of its image. Continuity alone gives no such conversion. $m \geq n$ is what makes $s_i^m \leq s_i^n$, i.e. what stops the volume from being measured with too few powers of s_i. The two counterexamples attack precisely these two steps.

Q.E.D.

Change of Variables, Fubini and Feynman's Trick (Chapter 20)

With the integral defined, the problem becomes computing one, and three tools do nearly all of that work. Change of variables trades an awkward domain for a convenient one and pays a Jacobian factor for the trade. Fubini reduces an n-dimensional integral to n one-dimensional ones. Differentiating under the integral sign reaches integrals with no elementary antiderivative at all, by turning the integral into an ODE in a parameter it never had.

Change of variables (Section 20.a)

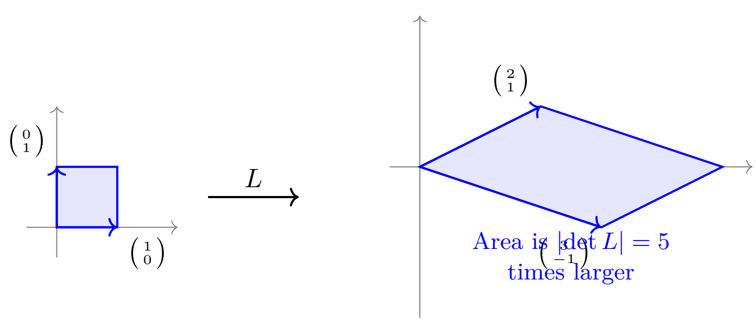
Theorem 20.a.1 (*Change of variables*): Suppose $U, V \subseteq \mathbb{R}^n$ are open sets and $\Phi : U \to V$ is a C^1-diffeomorphism. If $A \subseteq U$ is Jordan measurable with $\overline{A} \subseteq U$, and $f : A \to \mathbb{R}$ is continuous, then

$$\int_A f(x)\,dx = \int_{\Phi(A)} \frac{f \circ \Phi^{-1}(y)}{\left|\det \mathrm{J}\,\Phi\big(\Phi^{-1}(y)\big)\right|}\,dy.$$

AI-Note 20.1: Corsin uses "*Jordan measurable*" (German: „*Jordan-messbar*") here without defining it; the definition is in the lecture (ch. 13). For this chapter it is enough to know that boxes, and the bounded regions cut out by finitely many smooth curves that appear in the examples, are all Jordan measurable.

On the shape of the formula: it is often quoted the other way round, as $\int_{\Phi(A)} f = \int_A (f \circ \Phi)\,|\det \mathrm{J}\,\Phi|$. The version above is the same statement with Φ pushed to the other side, which is why the determinant appears **divided** rather than multiplied. Which form is convenient depends on whether it is Φ or Φ^{-1} that you can write down. In Example 20.b.19 below it is Φ, so the dividing form is the one that saves work.

Remark 20.a.2 (*Intuition for the Jacobian determinant*): From linear algebra, we know that for a linear map given by a matrix L, the absolute value of its determinant $|\det L|$ describes how much the map scales volumes. For example, the map $L = \left(\begin{smallmatrix} 2 & 3 \\ 1 & -1 \end{smallmatrix}\right)$ expands areas by a factor of $|\det L| = |-2-3| = 5$.



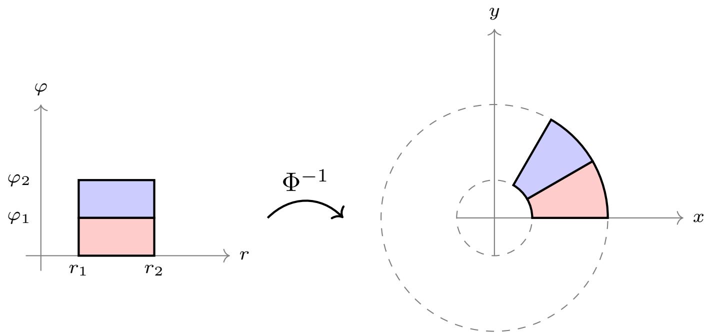
The change of variables formula generalizes this to non-linear maps: the Jacobian determinant acts as the local volume scaling factor.

Example 20.a.3 (*Gaussian integral via polar coordinates*): A classic application of the change of variables formula is computing the Gaussian integral $I := \int_{-\infty}^{\infty} e^{-x^2}\,dx$. We consider its square:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\,dy\right) = \int_{\mathbb{R}^2} e^{-(x^2+y^2)}\,\mathrm{dvol}(x,y).$$

We use polar coordinates, but strictly following Theorem 20.a.1, we define the forward map $\Phi : \mathbb{R}^2 \setminus \{(0,0)\} \to (0,\infty) \times [-\pi,\pi)$ by

$$\Phi(x,y) := \left(\sqrt{x^2+y^2}, \operatorname{atan2}(y,x)\right) =: (r,\varphi).$$



not equal in size ⇒ we must compensate
with the (Jacobian) determinant

The Jacobian matrix is

$$\mathrm{J}\,\Phi(x,y) = \begin{pmatrix} \frac{x}{\sqrt{x^2+y^2}} & \frac{y}{\sqrt{x^2+y^2}} \\ \frac{-y}{x^2+y^2} & \frac{x}{x^2+y^2} \end{pmatrix}.$$

Its determinant is

$$\det \mathrm{J}\,\Phi(x,y) = \frac{x^2}{(x^2+y^2)^{3/2}} + \frac{y^2}{(x^2+y^2)^{3/2}} = \frac{1}{\sqrt{x^2+y^2}} = \frac{1}{r}.$$

Applying the theorem (ignoring the null set $\{(0,0)\}$ where Φ is undefined), we get

$$I^2 = \int_{\Phi(\mathbb{R}^2\setminus\{(0,0)\})} \frac{e^{-r^2}}{|1/r|}\,dr\,d\varphi = \int_{-\pi}^{\pi} d\varphi \int_0^\infty re^{-r^2}\,dr = 2\pi \cdot \left[-\frac{1}{2}e^{-r^2}\right]_0^\infty = 2\pi\cdot\frac{1}{2} = \pi.$$

Since $I>0$, we conclude that $I=\sqrt{\pi}$.

Example 20.a.4 (*Change of variables on a diamond*): Compute $I := \int_E e^{x+y}(x-y)\,\mathrm{dvol}(x,y)$, where E is the region bounded by the lines $y=x$, $y=-x+2$, $y=x-2$, and $y=-x$.

Writing out the inequalities, $E=\{y\le x\}\cap\{y\le -x+2\}\cap\{y\ge -x\}\cap\{y\ge x-2\}$. Rearranging these gives $0\le x-y\le 2$ and $0\le x+y\le 2$. This suggests the substitution $u:=x+y$ and $v:=x-y$. The new domain is the square $V=[0,2]\times[0,2]$. The inverse map is $\Phi^{-1}(u,v)=(x,y)=\left(\frac{u+v}{2},\frac{u-v}{2}\right)$. Its Jacobian matrix is $\mathrm{J}\,\Phi^{-1}(u,v) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$, so $\left|\det \mathrm{J}\,\Phi^{-1}\right| = \left|-\frac{1}{4}-\frac{1}{4}\right| = \frac{1}{2}$. By the change of variables formula:

$$I = \int_0^2\int_0^2 e^u v\left|\det \mathrm{J}\,\Phi^{-1}\right|\,du\,dv = \frac{1}{2}\int_0^2\int_0^2 e^u v\,du\,dv = \frac{1}{2}\left[e^u\right]_0^2\left[\frac{v^2}{2}\right]_0^2 = \frac{1}{2}(e^2-1)\cdot 2 = e^2-1.$$

Example 20.a.5 (*Improper integral via polar coordinates*): Compute $\int_{\mathbb{R}^2}\frac{1}{(1+x^2+y^2)^2}\,\mathrm{dvol}(x,y)$. This integral is improper because the domain is unbounded. Since the integrand is non-negative and continuous, its Riemann improper integral is well-defined, and we can compute it using polar coordinates $x=r\cos\theta, y=r\sin\theta$. The Jacobian determinant is r. The domain $\mathbb{R}^2$ corresponds to $r\in[0,\infty)$ and $\theta\in[0,2\pi)$.

$$\int_{\mathbb{R}^2}\frac{1}{(1+x^2+y^2)^2}\,dx\,dy = \int_0^{2\pi}\int_0^\infty \frac{r}{(1+r^2)^2}\,dr\,d\theta.$$

Substituting $u=r^2$, $du=2r\,dr$, the inner integral is $\frac{1}{2}\int_0^\infty \frac{1}{(1+u)^2}\,du = \frac{1}{2}\left[\frac{-1}{1+u}\right]_0^\infty = \frac{1}{2}$. The outer integral then gives $\frac{1}{2}\int_0^{2\pi} d\theta = \pi$.

Example 20.a.6 (*Volume of a sphere*): Let $A := \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 \le 1\}$ be the unit ball. Given that A is Jordan measurable, its volume is $\mu(A) = \int_A 1 \, d\mathrm{vol}(x,y,z)$. We can compute this using spherical coordinates:

$$\Phi(r,\theta,\varphi) = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix}.$$

Up to a Jordan null set (which does not change the volume), the domain A corresponds to the box $V = (0,1) \times (0,\pi) \times (-\pi,\pi)$. The Jacobian determinant is $\det \mathrm{J}\,\Phi(r,\theta,\varphi) = r^2 \sin\theta$, which is positive on this domain. Applying the change of variables formula and Fubini's theorem (which allows slicing up the integral):

$$\begin{aligned} \mu(A) &= \int_{\Phi^{-1}(A)} \left|\det \mathrm{J}\,\Phi(r,\theta,\varphi)\right| dr\, d\theta\, d\varphi \\ &= \int_{-\pi}^{\pi}\int_0^{\pi}\int_0^1 r^2 \sin\theta \, dr\, d\theta\, d\varphi \\ &= \left(\int_{-\pi}^{\pi} d\varphi\right)\left(\int_0^{\pi} \sin\theta\, d\theta\right)\left(\int_0^1 r^2\, dr\right) \\ &= (2\pi)\cdot(2)\cdot\left(\frac{1}{3}\right) = \frac{4\pi}{3}. \end{aligned}$$

Remark 20.a.7 (*Polar coordinates as slicing into shells*): The factor r^2 in the computation above, and the r in the two-dimensional case, are usually produced by grinding out a Jacobian determinant. There is a reading that gets them without any computation, and that generalises to $\mathbb{R}^n$ at no extra cost.

Slice $\mathbb{R}^n$ into spheres centred at the origin. The sphere of radius r is an $(n-1)$-dimensional submanifold, and it is the unit sphere scaled by r; since scaling by r multiplies $(n-1)$-dimensional volume by r^{n-1}, its volume is

$$\mathrm{vol}_{n-1}\left(r\,\mathbb{S}^{n-1}\right) = C_n\, r^{n-1}, \qquad C_n := \mathrm{vol}_{n-1}\left(\mathbb{S}^{n-1}\right).$$

A shell between radii r and $r+dr$ is that sphere thickened by dr, so it contributes $C_n r^{n-1}\, dr$ of n-dimensional volume. Adding the shells,

$$\int_{B_R(0)} f(|x|)\, dx = C_n \int_0^R f(r)\, r^{n-1}\, dr \qquad \text{for radial } f.$$

The powers now need no memorising: $n=2$ gives r^1 and $n=3$ gives r^2, matching the two Jacobians above, and $C_2 = 2\pi$, $C_3 = 4\pi$ are the circumference of the unit circle and the area of the unit sphere. Taking $f \equiv 1$, $n=3$, $R=1$ reproduces $4\pi \int_0^1 r^2\, dr = 4\pi/3$ in one line.

This is a genuine shortcut only for **radial** integrands, where the angular integral is a constant that can be pulled out; when f depends on the angles as well, the full change of variables is unavoidable. But it explains where the radial power comes from, which the determinant computation does not.

Remark 20.a.8 (*Reference: the standard coordinate Jacobians*): The polar, cylindrical and spherical substitutions recur throughout this chapter and the next several; collected here in one place, with $r \ge 0$ throughout and each parametrizing its target domain up to a Jordan null set:

Coordinates	Transformation	$\lvert\det \mathrm{J}\,\Phi\rvert$
Polar ($\mathbb{R}^2$)	$\Phi(r,\varphi) = (r\cos\varphi,\ r\sin\varphi)$	r
Cylindrical ($\mathbb{R}^3$)	$\Phi(r,\varphi,z) = (r\cos\varphi,\ r\sin\varphi,\ z)$	r
Spherical ($\mathbb{R}^3$)	$\Phi(r,\theta,\varphi) = (r\sin\theta\cos\varphi,\ r\sin\theta\sin\varphi,\ r\cos\theta)$	$r^2\sin\theta$

The angular ranges are $\varphi \in (-\pi, \pi)$ throughout, and $\theta \in (0, \pi)$ ("*colatitude*", measured from the positive z-axis) for the spherical case. Cylindrical coordinates are just polar coordinates in the first two variables, carried along unchanged in z, which is exactly why the Jacobian is the same r as in the planar case: the extra dz contributes a factor of 1. Spherical coordinates are used already in Example 20.a.6 above; cylindrical coordinates are the natural choice whenever a region has rotational symmetry about an axis without being a ball, e.g. a solid cylinder or cone.

Example 20.a.9 (*Volume of an ellipse*): Compute the area (2-dimensional volume) of the ellipse $E := \{(x, y) \in \mathbb{R}^2 \mid (x/a)^2 + (y/b)^2 < 1\}$ for $a, b > 0$. We use the transformation $\Phi(r, \varphi) = (ar\cos\varphi, br\sin\varphi)$. This maps the rectangle $V = (0, 1) \times (0, 2\pi)$ bijectively onto E minus a null set (the positive x-axis). The Jacobian matrix is

$$\mathrm{J}\,\Phi(r, \varphi) = \begin{pmatrix} a\cos\varphi & -ar\sin\varphi \\ b\sin\varphi & br\cos\varphi \end{pmatrix},$$

and its determinant is $\det \mathrm{J}\,\Phi(r, \varphi) = abr(\cos^2\varphi + \sin^2\varphi) = abr$. Thus, by the change of variables formula,

$$\int_E 1\,dx\,dy = \int_0^{2\pi}\int_0^1 abr\,dr\,d\varphi = \left(\int_0^{2\pi} ab\,d\varphi\right)\left(\int_0^1 r\,dr\right) = 2\pi ab \cdot \frac{1}{2} = \pi ab.$$

Example 20.a.10 (*Area of a cardioid*): Compute the area of the region Ω given in polar coordinates by $\Omega := \{(r, \varphi) \mid \varphi \in [0, 2\pi),\ 0 \le r \le 1+\cos\varphi\}$. Using polar coordinates $\Phi(r, \varphi) = (r\cos\varphi, r\sin\varphi)$, the area is

$$\int_\Omega 1\,dx\,dy = \int_0^{2\pi}\int_0^{1+\cos\varphi} r\,dr\,d\varphi = \int_0^{2\pi}\left[\frac{r^2}{2}\right]_{r=0}^{r=1+\cos\varphi} d\varphi = \frac{1}{2}\int_0^{2\pi}(1+\cos\varphi)^2\,d\varphi.$$

Expanding the integrand: $(1+\cos\varphi)^2 = 1 + 2\cos\varphi + \cos^2\varphi$. We know $\int_0^{2\pi}\cos\varphi\,d\varphi = 0$ and $\int_0^{2\pi}\cos^2\varphi\,d\varphi = \pi$. Thus,

$$\int_\Omega 1\,dx\,dy = \frac{1}{2}(2\pi + 0 + \pi) = \frac{3\pi}{2}.$$

Example 20.a.11 (*Combining substitution and Fubini*): Compute $\int_\Omega (x^3y + xy^3)\,dx\,dy$ where $\Omega := \{(x, y) \in \mathbb{R}^2 \mid x, y > 0,\ 1/x \le y \le 3/x,\ 1 \le x^2 - y^2 \le 4\}$. The integrand factors as $xy(x^2+y^2)$. The boundary inequalities are equivalent to $1 \le xy \le 3$ and $1 \le x^2 - y^2 \le 4$, which strongly suggests defining $u := xy$ and $v := x^2 - y^2$. The domain in the (u, v)-plane is the rectangle $R = [1, 3] \times [1, 4]$. Let $\Psi(x, y) = (xy, x^2 - y^2) = (u, v)$. The Jacobian of this inverse transformation is

$$\mathrm{J}\,\Psi(x, y) = \begin{pmatrix} y & x \\ 2x & -2y \end{pmatrix}, \qquad \det \mathrm{J}\,\Psi(x, y) = -2y^2 - 2x^2 = -2(x^2+y^2).$$

For the forward transformation $\Phi = \Psi^{-1}$, we have $\left|\det \mathrm{J}\,\Phi(u, v)\right| = \frac{1}{|\det \mathrm{J}\,\Psi(x,y)|} = \frac{1}{2(x^2+y^2)}$. Applying the change of variables formula:

$$\int_\Omega xy(x^2+y^2)\,dx\,dy = \int_R u(x^2+y^2)\frac{1}{2(x^2+y^2)}\,du\,dv = \frac{1}{2}\int_1^4\int_1^3 u\,du\,dv = \frac{1}{2}\cdot 3\cdot\left[\frac{u^2}{2}\right]_1^3 = \frac{3}{2}\cdot 4 = 6.$$

Exercise 20.a.12 (*8.5 — Change of variables and Jacobians* (important)): For each of the following domains and change of variables find the jacobian and the appropriate transformed domain. There is no need to actually compute the integrals!

1. $A := \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0,\ x_2 < x_1,\ 1 < x_1^2 + x_2^2 < 4\}$ and $x_1 = r\cos\theta$, $x_2 = r\sin\theta$. Using the change of variables formula, complete the dots in the following formula

$$\int_A x_1^2 \sin(x_2)\,dx_1 dx_2 = \int_{\dots} \dots\,dr\,d\theta$$

2. $B := \{(x,y) \in \mathbb{R}^2 \mid 1 < xy < 2,\ x^2 < y < 2x^2\}$ and $u := xy,\ v := x^2$. Using the change of variables formula, complete the dots in the following formula
$$\int_B y^2 e^{-xy}\,dx\,dy = \int_{\dots} \dots\,du\,dv$$
3. $C := \{(x,y,z) \in \mathbb{R}^3 \mid 1 < z-2y < 2,\ 0 < z < 1,\ -2 < 3x+y+z < 0\}$ and $u := z$, $v := z-2y$, $w := 3x+y+z$. Using the change of variables formula, complete the dots in the following formula
$$\int_C xyz\,dx\,dy = \int_{\dots} \dots\,du\,dv\,dw.$$

Sascha Brack's hint on **8.5.2:** first pin down the signs. The constraints force $x \neq 0$ and $y \neq 0$; then $y > 0$ forces $x > 0$, so $u > 0$ and $v > 0$ throughout B. That is what lets you divide by x and y and rearrange the inequalities without their directions flipping.

AI-Note 20.2: The integral in **8.5.3** is written $\int_C xyz\,dx\,dy$ on the official sheet, but C is a subset of $\mathbb{R}^3$ and the right-hand side carries three differentials, so this must read $dx\,dy\,dz$. Quoted verbatim above, as the sheet is a primary source.

Fubini's theorem (Section 20.b)

Theorem 20.b.13 (*Fubini*): Let $X \subseteq \mathbb{R}^m$ and $Y \subseteq \mathbb{R}^n$ be intervals (*"boxes"*), and let $f : X \times Y \to \mathbb{R}$, $(x,y) \mapsto f(x,y)$, be continuous. Then
$$\int_{X\times Y} f(x,y)\,dx\,dy = \int_X \left(\int_Y f(x,y)\,dy\right) dx = \int_Y \left(\int_X f(x,y)\,dx\right) dy,$$
where we look at the expression in brackets as a function, e.g. $x \mapsto \int_Y f(x,y)\,dy$.

Proposition 20.b.14 (*Cavalieri's principle*): Let $A, B \subseteq \mathbb{R}^{n+1}$ be Jordan measurable sets. For each height $t \in \mathbb{R}$, denote by $A_t := \{x \in \mathbb{R}^n \mid (x,t) \in A\}$ and $B_t := \{x \in \mathbb{R}^n \mid (x,t) \in B\}$ the n-dimensional cross-sections at t. If the cross-sectional volumes agree for all $t \in \mathbb{R}$, i.e.
$$\mu_n(A_t) = \mu_n(B_t) \quad \forall t \in \mathbb{R},$$
then the total $(n+1)$-dimensional volumes are equal: $\mu_{n+1}(A) = \mu_{n+1}(B)$.

Proof:

By Fubini's Theorem (Theorem 20.b.13), integrating the indicator function over slices gives:
$$\mu_{n+1}(A) = \int_{\mathbb{R}^{n+1}} \chi_A(x,t)\,dx\,dt = \int_{\mathbb{R}} \left(\int_{\mathbb{R}^n} \chi_{A_t}(x)\,dx\right) dt = \int_{\mathbb{R}} \mu_n(A_t)\,dt.$$
Replacing $\mu_n(A_t)$ with $\mu_n(B_t)$ yields $\mu_{n+1}(B)$. **Q.E.D.**

Theorem 20.b.15 (*Volume of a cone over a measurable base*): Let $\Omega \subseteq \mathbb{R}^n$ be a bounded, Jordan measurable domain. The "*cone over* Ω" (German: „*Kegel über* Ω") with vertex at $(0,1) \in \mathbb{R}^n \times \mathbb{R}$ and base $\Omega \times \{0\}$ is defined by
$$C\Omega := \big\{(x,t) \in \mathbb{R}^n \times [0,1] \mid x \in (1-t)\Omega\big\}.$$
Then $C\Omega$ is Jordan measurable in $\mathbb{R}^{n+1}$, and its volume is given by
$$\mu_{n+1}(C\Omega) = \frac{\mu_n(\Omega)}{n+1}.$$

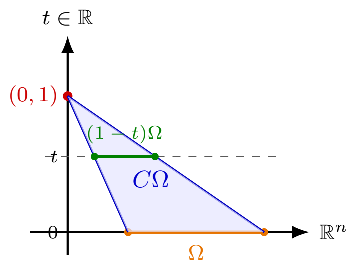
Proof:

For any $t \in [0,1]$, the cross-section of $C\Omega$ at height t is $(C\Omega)_t = (1-t)\Omega \subseteq \mathbb{R}^n$. By the scaling property of n-dimensional Jordan measure, $\mu_n\big((1-t)\Omega\big) = (1-t)^n \mu_n(\Omega)$. Applying Cavalieri's

Principle (Proposition 20.b.14):

$$\mu_{n+1}(C\Omega) = \int_0^1 \mu_n\big((1-t)\Omega\big)\,dt = \mu_n(\Omega)\int_0^1 (1-t)^n\,dt = \mu_n(\Omega)\left[-\frac{(1-t)^{n+1}}{n+1}\right]_0^1 = \frac{\mu_n(\Omega)}{n+1}.$$

Q.E.D.



Example 20.b.16 (*Fubini on the unit square*):

$$\int_{[0,1]^2} x^{y+1}\,dx\,dy = \int_0^1 dy \int_0^1 dx\; x^{y+1} = \int_0^1 dy\; \left[\frac{x^{y+2}}{y+2}\right]_0^1$$
$$= \int_0^1 dy\;\frac{1}{y+2} = \log|y+2|\,\Big|_0^1 = \log\!\left(\tfrac{3}{2}\right).$$

AI-Note 20.3: Corsin writes the first line as $\int_0^1 dx \int_0^1 dy$, but the antiderivative taken on the next line is in x, and it produces $x^{y+2}/(y+2)$, evaluated from 0 to 1. The differentials are therefore swapped relative to the computation actually performed. The order is corrected above and the result is unaffected.

The example is well chosen: integrating x^{y+1} in y first would mean integrating an exponential in the exponent, which is far worse. Fubini's real use is that you may pick the easier order, and here only one of the two is comfortable.

Example 20.b.17 (*Changing the order of integration*): Consider the iterated integral $\int_0^2 \int_{y^2}^{2\sqrt{2y}} f(x,y)\,dx\,dy$. Let us rewrite it with the integration order reversed (first y, then x).

The region is defined by $0 \le y \le 2$ and $y^2 \le x \le 2\sqrt{2y}$. Notice that for $y \in [0,2]$, the lower bound is $x \ge 0$ and the upper bound is $x \le 2\sqrt{4} = 4$, so x ranges from 0 to 4. We now solve the inequalities for y: $x \le 2\sqrt{2y} \implies x^2 \le 8y \implies y \ge \frac{1}{8}x^2$. $x \ge y^2 \implies y \le \sqrt{x}$ (since $x, y \ge 0$). Thus, the new bounds are $0 \le x \le 4$ and $\frac{1}{8}x^2 \le y \le \sqrt{x}$. The reversed integral is:

$$\int_0^4 \int_{\frac{1}{8}x^2}^{\sqrt{x}} f(x,y)\,dy\,dx.$$

Example 20.b.18 (*Surface parametrization and Fubini*): Let $f(u,v) := (u\cos v, u\sin v, u^2)^\mathsf{T} \in \mathbb{R}^3$ for $(u,v) \in \mathbb{R}^2$. Write the third coordinate $z(u,v)$ as a function of the first two, $\tilde{z}(x,y)$, and compute $\int_0^y \int_0^x \tilde{z}(x',y')\,dx'\,dy'$.

We have $x(u,v) = u\cos v$ and $y(u,v) = u\sin v$. Then $x^2 + y^2 = u^2\cos^2 v + u^2\sin^2 v = u^2 = z(u,v)$.

So, we define $\tilde{z}(x,y) := x^2+y^2$. The integral is:

$$\begin{aligned}\int_0^y\int_0^x (x'^2+y'^2)\,dx'\,dy' &= \int_0^y \left[\frac{x'^3}{3}+x'y'^2\right]_{x'=0}^{x'=x} dy' \\ &= \int_0^y \left(\frac{x^3}{3}+xy'^2\right) dy' \\ &= \left[\frac{x^3y'}{3}+x\frac{y'^3}{3}\right]_{y'=0}^{y'=y} \\ &= \frac{xy}{3}(x^2+y^2). \qquad (20.1)\end{aligned}$$

Example 20.b.19 (*A domain that Fubini alone cannot handle*): Consider

$$\int_A \frac{y}{\sqrt{x}}\,dy\,dx, \qquad A := \left\{(x,y)\in\mathbb{R}^2 \;:\; x,y>0,\;\; 1\le\frac{y}{\sqrt{x}}\le 2,\;\; 1\le xy\le 2\right\}.$$

This is not easily integrated with Fubini alone. But we can apply a transformation of variables to simplify the integration domain A: take

$$\Phi: U\to V, \qquad \begin{pmatrix}x\\y\end{pmatrix}\mapsto\begin{pmatrix}u(x,y)\\v(x,y)\end{pmatrix}=\begin{pmatrix}\frac{y}{\sqrt{x}}\\xy\end{pmatrix},$$

with U and V still to be determined, subject only to their being open.

Check injectivity. Suppose, for $x,y,x',y'>0$,

$$\textbf{(1)}\;\; \frac{y}{\sqrt{x}}=\frac{y'}{\sqrt{x'}}, \qquad\qquad \textbf{(2)}\;\; xy=x'y'.$$

From **(2)**, $\dfrac{x}{x'}=\dfrac{y'}{y}$. Inserting into **(1)**,

$$\frac{1}{\sqrt{x}}=\frac{y'}{y}\cdot\frac{1}{\sqrt{x'}}=\frac{x}{x'}\cdot\frac{1}{\sqrt{x'}} \quad\Longrightarrow\quad x^{3/2}=(x')^{3/2} \quad\Longrightarrow\quad x=x',$$

and consequently $\dfrac{y'}{y}=\dfrac{x}{x'}=1$. So, for $U:=(0,\infty)^2$ and $V:=\Phi(U)$, the map Φ is a bijection between open sets.

Functional determinant. We calculate $|\det \mathrm{J}\,\Phi|$; if it is strictly positive we may proceed, since Φ is then a local diffeomorphism as well as a bijection, hence a diffeomorphism. Now

$$\mathrm{J}\,\Phi(x,y)=\begin{pmatrix}-\dfrac{y}{2\sqrt{x}^3} & \dfrac{1}{\sqrt{x}}\\ y & x\end{pmatrix},$$

so that

$$\left|\det \mathrm{J}\,\Phi(x,y)\right| = \left|-\frac{3}{2}\frac{y}{\sqrt{x}}\right| = \frac{3}{2}\frac{y}{\sqrt{x}} = \frac{3}{2}u(x,y).$$

So, Φ is a diffeomorphism and $du\,dv=\frac{3}{2}u\,dx\,dy$. Furthermore, by design,

$$\Phi(A)=(1,2)^2.$$

So, with the change of variables (Theorem 20.a.1), followed by Fubini (Theorem 20.b.13),

$$\int_A\frac{y}{\sqrt{x}}\,dy\,dx=\int_{(1,2)^2} u\,\frac{1}{\frac{3}{2}u}\,du\,dv=\frac{2}{3}\int_1^2 dv\int_1^2 du=\frac{2}{3}.$$

AI-Note 20.4: The design principle is worth stating, because it is the reverse of how substitution is usually taught. Here Φ was not chosen to simplify the **integrand** but to straighten the **domain**: the two constraints defining A are literally $1\le u\le 2$ and $1\le v\le 2$, so Φ turns an awkward curvilinear region into a square, and only then does Fubini become available. That the integrand $y/\sqrt{x}$ is itself u, and cancels against most of the Jacobian factor, is a bonus of the same choice.

We now want to look at a powerful new integration technique using ODEs.

Feynman's trick (Section 20.c)

Theorem 20.c.20 (*Differentiation under the integral sign*): Let $U \subseteq \mathbb{R}^n \times \mathbb{R}$ be open, $f \in C^1(U)$, and let $K \subseteq \mathbb{R}^n$ be compact such that $K \times [a,b] \subseteq U$. Then the function

$$(a,b) \to \mathbb{R}, \qquad y \mapsto \int_K f(x,y)\,dx$$

is continuously differentiable, and

$$\frac{d}{dy}\int_K f(x,y)\,dx = \int_K \frac{\partial}{\partial y} f(x,y)\,dx \qquad \text{for all } y \in (a,b).$$

Example 20.c.21 ($\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$ *by Feynman's trick*): We want to compute the integral

$$I(1) = \int_0^1 \frac{\log(1+x)}{1+x^2}\,dx,$$

which has no convenient antiderivative. By introducing a parameter $\alpha \in (-1,1)$:

$$I(\alpha) := \int_0^1 \frac{\log(1+\alpha x)}{1+x^2}\,dx.$$

Note that the integrand is a C^1 function on $U := \left(-\frac{1}{1+\varepsilon},\ \frac{1}{\varepsilon}\right) \times (-\varepsilon,\ 1+\varepsilon)$ in the variables (x,α), with $K := [0,1]$ and $[a,b] := [0,1]$ satisfying $K \times [a,b] \subseteq U$ for $0 < \varepsilon < 1$ (pay attention to the logarithm, which is what forces $1+\alpha x > 0$). So, we may differentiate under the integral!

We obtain an ordinary differential equation. By Theorem 20.c.20,

$$I'(\alpha) = \frac{d}{d\alpha}\int_0^1 \frac{\log(1+\alpha x)}{1+x^2}\,dx = \int_0^1 \frac{x}{(1+\alpha x)(1+x^2)}\,dx.$$

Partial fractions: writing

$$\frac{x}{(1+\alpha x)(1+x^2)} = \frac{A}{1+\alpha x} + \frac{Bx+C}{1+x^2}$$

and clearing denominators gives $A + Ax^2 + Bx + B\alpha x^2 + C + C\alpha x = x$, so comparing coefficients,

$$-A = C, \qquad -\tfrac{A}{\alpha} = B, \qquad B + C\alpha = 1 \implies -\tfrac{A}{\alpha} - A\alpha = 1,$$

whence

$$A = -\frac{\alpha}{1+\alpha^2}, \qquad B = \frac{1}{1+\alpha^2}, \qquad C = \frac{\alpha}{1+\alpha^2}.$$

Therefore

$$\begin{aligned} I'(\alpha) &= \left(\int_0^1 \frac{-\alpha}{1+\alpha x} + \frac{x+\alpha}{1+x^2}\,dx\right)\frac{1}{1+\alpha^2} \\ &= \frac{1}{1+\alpha^2}\Big[-\log(1+\alpha x) + \tfrac{1}{2}\log(1+x^2) + \alpha\arctan(x)\Big]_0^1 \\ &= \frac{1}{1+\alpha^2}\Big[\log\sqrt{2} + \tfrac{\alpha\pi}{4} - \log(1+\alpha)\Big]. \end{aligned}$$

With initial value $I(0) = \int_0^1 \frac{\log(1)}{1+x^2}\,dx = 0$, we obtain from the fundamental theorem of calculus

$$\begin{aligned} \int_0^1 \frac{\log(1+x)}{1+x^2}\,dx &= I(1) = I(1) - I(0) \\ &= \int_0^1 \left(\frac{\log\sqrt{2}}{1+\alpha^2} + \frac{\pi}{4}\frac{\alpha}{1+\alpha^2} - \underbrace{\frac{\log(1+\alpha)}{1+\alpha^2}}_{\to\, I(1)}\right)d\alpha. \end{aligned}$$

The last term is $I(1)$ again, the integral we started from with α merely renamed, so bringing it to the left-hand side,

$$2\,I(1) = \Big[\log\sqrt{2}\,\arctan(\alpha) + \tfrac{\pi}{8}\log(1+\alpha^2)\Big]_0^1 = \frac{\pi\log\sqrt{2}}{4} + \frac{\pi\log\sqrt{2}}{4},$$

$$\implies \quad I(1) = \frac{\pi\log\sqrt{2}}{4} = \boxed{\frac{\pi\log 2}{8}}.$$

Example 20.c.22 (*Dirichlet integral via Feynman's trick*): Consider the integral $\int_0^\infty \frac{\sin x}{x}\,dx$. We can evaluate it by introducing a damping factor e^{-tx} and defining

$$I(t) := \int_0^\infty e^{-tx}\frac{\sin x}{x}\,dx \qquad \text{for } t > 0.$$

Differentiating under the integral sign with respect to t (which is justified since the integrand and its derivative decay rapidly for $t > 0$):

$$I'(t) = \int_0^\infty \frac{\partial}{\partial t}\left(e^{-tx}\frac{\sin x}{x}\right)dx = \int_0^\infty -xe^{-tx}\frac{\sin x}{x}\,dx = -\int_0^\infty e^{-tx}\sin x\,dx.$$

This standard integral can be evaluated using integration by parts twice, or by using Euler's formula $e^{ix} = \cos x + i\sin x$ and noting that $e^{-tx}\sin x = \operatorname{Im}(e^{(-t+i)x})$:

$$\int_0^\infty e^{(-t+i)x}\,dx = \left[\frac{e^{(-t+i)x}}{-t+i}\right]_0^\infty = 0 - \frac{1}{-t+i} = \frac{t+i}{t^2+1}.$$

Taking the imaginary part gives $\int_0^\infty e^{-tx}\sin x\,dx = \frac{1}{t^2+1}$. Thus,

$$I'(t) = -\frac{1}{t^2+1} \implies I(t) = -\arctan(t) + C.$$

Since $\lim_{t\to\infty} I(t) = 0$, we must have $C = \lim_{t\to\infty}\arctan(t) = \frac{\pi}{2}$. Therefore, $I(t) = \frac{\pi}{2} - \arctan(t)$. The original integral corresponds to the limit as $t \to 0^+$ (which requires an extra justification step usually provided by Abel's theorem, but we formally take the limit):

$$\int_0^\infty \frac{\sin x}{x}\,dx = \lim_{t\to 0^+}\left(\frac{\pi}{2} - \arctan(t)\right) = \frac{\pi}{2}.$$

AI-Note 20.5: Corsin's final integration writes the α-term as $\frac{\pi}{2}\cdot\frac{\alpha}{1+\alpha^2}$, integrating to $\frac{\pi}{4}\log(1+\alpha^2)$. Both are a factor 2 away from what $I'(\alpha)$ gives: his own line above has $\frac{\alpha\pi}{4}$, so the term is $\frac{\pi}{4}\cdot\frac{\alpha}{1+\alpha^2}$ and its antiderivative is $\frac{\pi}{8}\log(1+\alpha^2)$. Corrected above, and the correction is self-checking, since it is exactly what makes his two bracket terms come out equal, as his own next line asserts: $\frac{\pi}{8}\log 2 = \frac{\pi}{8}\cdot 2\log\sqrt{2} = \frac{\pi\log\sqrt{2}}{4}$. His final answer $\frac{\pi\log 2}{8}$ is correct and matches the known value.

AI-Note 20.6 (*Why this is called a trick, and when to reach for it*): The move that makes this work is not the differentiation. It is noticing that the **antiderivative** of the resulting integrand is elementary while the original one is not. The pattern is worth naming:

1. Insert a parameter α so that the target is $I(\alpha_1)$ for some α_1, and so that $I(\alpha_0)$ is trivial for some other α_0. Here $I(0) = 0$, because $\log 1 = 0$.
2. Differentiate under the integral. The log disappears, leaving a rational function that partial fractions can handle.
3. Integrate I' back from α_0 to α_1.

The self-reference in step 3, where the answer reappears on the right-hand side so that one solves for $I(1)$ rather than computing it, is specific to this integral and is the reason it is a classic. Do not expect it in general. Usually step 3 is an honest integration.

Solutions (Section 20.d)

Solution to Exercise 20.a.12:

In each part the work is the same two steps: find $|\det \mathrm{J}|$, then translate every constraint. As the exercise says, no integral is actually evaluated.

1. **Polar coordinates.** Here $\Phi(r,\theta) := (r\cos\theta, r\sin\theta)$ has $\det \mathrm{J}\,\Phi = r > 0$ (Example 15.a.6), so $dx_1 dx_2 = r\,dr\,d\theta$. The three constraints translate one at a time:
   - $1 < x_1^2 + x_2^2 < 4$ becomes $1 < r < 2$;
   - $x_1 > 0$ becomes $\cos\theta > 0$, i.e. $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$;
   - $x_2 < x_1$ becomes $r\sin\theta < r\cos\theta$; since $r > 0$ and $\cos\theta > 0$ this is $\tan\theta < 1$, i.e. $\theta < \frac{\pi}{4}$.

   So the transformed domain is the rectangle $(r,\theta) \in (1,2) \times \big(-\frac{\pi}{2}, \frac{\pi}{4}\big)$ and
   $$\int_A x_1^2 \sin(x_2)\,dx_1 dx_2 = \int_1^2 \int_{-\pi/2}^{\pi/4} r^3 \cos^2\theta\, \sin(r\sin\theta)\,d\theta\,dr,$$
   the extra power of r coming from $x_1^2 = r^2\cos^2\theta$ times the Jacobian factor r.

2. **$u := xy$, $v := x^2$.** Follow Sascha Brack's hint and fix the signs first. On B the constraint $x^2 < y$ forces $y > 0$, and then $xy > 1 > 0$ forces $x > 0$; so $u > 0$ and $v > 0$ throughout, and $x = \sqrt{v}$, $y = u/\sqrt{v}$.

   Differentiating in the convenient direction,
   $$\frac{\partial(u,v)}{\partial(x,y)} = \det\begin{pmatrix} y & x \\ 2x & 0 \end{pmatrix} = -2x^2 = -2v, \qquad \text{so} \qquad dx\,dy = \frac{du\,dv}{2v}.$$
   Now the constraints. $1 < xy < 2$ becomes $1 < u < 2$ at once. The second is the interesting one: $x^2 < y < 2x^2$ reads $v < u/\sqrt{v} < 2v$, i.e.
   $$v^{3/2} < u < 2\,v^{3/2}, \qquad \text{equivalently} \qquad \Big(\frac{u}{2}\Big)^{2/3} < v < u^{2/3}.$$
   The integrand becomes $y^2 e^{-xy} = (u^2/v)\,e^{-u}$, so
   $$\int_B y^2 e^{-xy}\,dx\,dy = \int_1^2 \int_{(u/2)^{2/3}}^{u^{2/3}} \frac{u^2 e^{-u}}{2v^2}\,dv\,du.$$
   Note what did **not** happen: straightening $1 < xy < 2$ into a pair of constant bounds did nothing for the other constraint, which stays curved. Straightening one constraint does not oblige the rest to cooperate.

3. **A linear change of variables.** With $u := z$, $v := z - 2y$, $w := 3x + y + z$,
   $$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \det\begin{pmatrix} 0 & 0 & 1 \\ 0 & -2 & 1 \\ 3 & 1 & 1 \end{pmatrix} = 6, \qquad \text{so} \qquad dx\,dy\,dz = \frac{du\,dv\,dw}{6},$$
   a constant, since the map is linear. Each constraint already **is** one of the new coordinates, so the transformed domain is literally the box
   $$(u,v,w) \in (0,1) \times (1,2) \times (-2,0).$$
   Inverting the substitution gives $z = u$, $y = \frac{u-v}{2}$ and $x = \frac{1}{6}\big(2w - 3u + v\big)$, so $xyz = \frac{1}{12}\,u\,(u-v)\,(2w-3u+v)$ and
   $$\int_C xyz\,dx\,dy\,dz = \int_0^1 \int_1^2 \int_{-2}^0 \frac{u\,(u-v)\,(2w-3u+v)}{72}\,dw\,dv\,du.$$

AI-Note 20.7: The three parts are graded deliberately, and comparing them is the lesson. In part 3 the change of variables was **chosen** to be the constraints, so the domain is a box and the Jacobian a constant, and nothing is left to do. In part 1 the domain is a box in the new coordinates but the Jacobian is not constant. In part 2 neither holds: the Jacobian varies and one of the two constraints stays curved. Reading the substitution off the constraints, as in part 3, is the move worth remembering, and it is available far more often than it looks.

Q.E.D.

Gram Determinants, Area and Arc Length (Chapter 21)

The previous chapters integrated over subsets of $\mathbb{R}^n$, where the volume element is just dx. This one asks what to integrate over a d-dimensional surface sitting inside $\mathbb{R}^n$, where $d < n$ and there is no such element to start from. The answer runs from the area of a parallelogram, through the Gram determinant of m vectors in $\mathbb{R}^n$ for $m \neq n$, to the d-volume of a parametrized submanifold, with the length of a curve falling out as the case $d = 1$.

Length, area, and volume (Section 21.a)

The area of a parallelogram in $\mathbb{R}^2$ (Subsection 21.a.1)

The relationship between the determinant and volume is a key concept in this part of the lecture. We illustrate the special case of $\mathbb{R}^2$.

Proposition 21.a.1 (*Area of a parallelogram*): Let $x, y \in \mathbb{R}^2$. The area of the parallelogram $(0, x, x+y, y)$ is

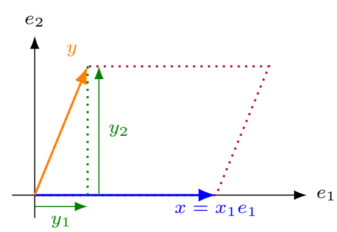
$$A = \big|\det(x \mid y)\big|.$$

Proof:

Assume w.l.o.g. that $x = x_1 e_1$, a reduction justified below. Then, writing $y = (y_1, y_2)^\mathsf{T}$,

$$A = |x_1 y_2| = \left|\det\begin{pmatrix} x_1 & y_1 \\ 0 & y_2 \end{pmatrix}\right|,$$

directly from the base-times-height formula for a parallelogram with one side on the e_1-axis.



For the reduction, let $R \in \mathrm{O}(2)$ be any rotation carrying x, y to a pair Rx, Ry with Rx along e_1; rotations preserve area, so

$$A = \big|\det(Rx \mid Ry)\big| = \big|\det(R)\det(x \mid y)\big| = \big|\det(x \mid y)\big|,$$

using $|\det R| = 1$ and multiplicativity of $\det$. So, the *"w.l.o.g."* above was justified: the formula $A = |\det(x \mid y)|$ holds for the rotated pair, and equals $|\det(x \mid y)|$ for the original one. **Q.E.D.**

The Gram determinant (Subsection 21.a.2)

Lemma 21.a.2 (*Gram determinant of a square matrix*): For $M \in \mathrm{Mat}_{n\times n}(\mathbb{R})$,

$$|\det(M)| = \sqrt{\det(M)\det(M^\mathsf{T})} = \sqrt{\det(M^\mathsf{T})\det(M)} = \sqrt{\det(M^\mathsf{T}M)}.$$

Proof:

The first equality is $|\det M| = \sqrt{(\det M)^2}$ together with $\det(M^\mathsf{T}) = \det(M)$; the last is multiplicativity of the determinant, $\det(M^\mathsf{T})\det(M) = \det(M^\mathsf{T}M)$.

Q.E.D.

The quantity $\det(M^\mathsf{T} M)$ is called the "*Gram determinant*" (German: „*Gram-Determinante*") of M. So, by Proposition 21.a.1, the volume of an n-parallelotope in $\mathbb{R}^n$ is given by the Gram determinant, in analogy to the determinant itself.

AI-Note 21.1: Terminology diverges from the official script here, worth flagging since it is exactly the kind of thing that costs a mark in an exam. The script's Definition 13.55 calls $\sqrt{\det(L^\mathsf{T} L)}$, the square root and so the **volume** itself, the *"Gram determinant"*. This document instead follows the more standard usage and calls the bare $\det(M^\mathsf{T} M)$ the Gram determinant, with the square root then giving the volume (Definition 21.a.3 below). Every downstream formula agrees regardless of which name is attached to which quantity, so this is terminology only, not a mathematical disagreement.

The script's own statement of Definition 13.55 is, independently, internally inconsistent: it introduces a linear map $L : \mathbb{R}^m \to \mathbb{R}^n$ with $m < n$, but then writes the formula using an undefined d (*"the $d \times d$ matrix $L^\mathsf{T} L$"*) where the surrounding text means m; and the prose describes *"the determinant of the ... matrix LL^T"* while the displayed formula correctly uses $L^\mathsf{T} L$. Only $L^\mathsf{T} L$ (size $m \times m$) is invertible for $m < n$, since LL^T is $n \times n$ and singular. The displayed formula is therefore the reliable half of the definition, and the surrounding prose is not.

AI-Note 21.2: The point of restating $|\det M|$ this way is that the middle expression, $\det(M^\mathsf{T} M)$, still makes sense once $M \in \mathrm{Mat}_{n\times m}(\mathbb{R})$ is rectangular: $M^\mathsf{T} M \in \mathrm{Mat}_{m\times m}(\mathbb{R})$ is square even when M is not, so its determinant is always defined, whereas $\det(M)$ itself is not. This is what lets the notion of *"volume spanned by vectors"* survive dropping the requirement $m = n$. A set of fewer than n vectors in $\mathbb{R}^n$ still spans a lower-dimensional parallelotope, and that parallelotope still has an m-dimensional volume, even though no square matrix and no ordinary determinant is available to compute it with.

Definition 21.a.3 (*Volume of a parallelotope via the Gram determinant*): For m vectors $v_1, \dots, v_m \in \mathbb{R}^n$, the "*area*"/"*length*"/ "*volume*" of the parallelotope spanned by them is

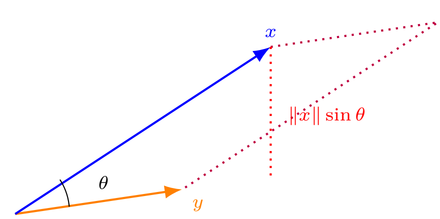
$$\sqrt{\det\left(\langle v_i, v_j\rangle\right)} = \sqrt{\det(V^\mathsf{T} V)}, \qquad V := (v_1 \mid v_2 \mid \cdots \mid v_m) \in \mathrm{Mat}_{n\times m}(\mathbb{R}).$$

Example 21.a.4: For $m = 1$, $\sqrt{\det(v_1^\mathsf{T} v_1)} = \sqrt{\langle v_1, v_1\rangle} = \|v_1\|$, which is the length of v_1.

Example 21.a.5 ($m = 2,\ n = 3$: *area of a parallelogram in* $\mathbb{R}^3$): For $m = 2$, $n = 3$, we hope that the Gram determinant of

$$M = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \end{pmatrix}$$

gives the area of the parallelogram spanned by x and y in $\mathbb{R}^3$. If θ is the angle between x and y, this should be $A = \|x\|\|y\||\sin\theta|$.



And indeed:

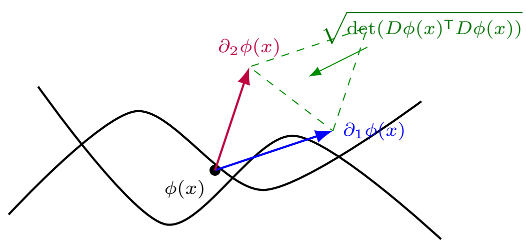
$$\begin{aligned}\sqrt{\det(M^\mathsf{T} M)} &= \sqrt{\det\begin{pmatrix}\langle x,x\rangle & \langle x,y\rangle\\ \langle x,y\rangle & \langle y,y\rangle\end{pmatrix}}\\ &= \sqrt{\langle x,x\rangle\langle y,y\rangle - \langle x,y\rangle^2}\\ &= \sqrt{\|x\|^2\|y\|^2 - \|x\|^2\|y\|^2\cos^2\theta}\\ &= \|x\|\|y\||\sin\theta| \qquad (= \ |x\times y|).\end{aligned}$$

Volume of embedded surfaces (Section 21.b)

Definition 21.b.6 (*d-volume on a parametrized d-submanifold*): Let $V \subseteq \mathbb{R}^d$ be open, and let $\phi : V \to \mathbb{R}^n$ be a parametrized submanifold. Given a Jordan measurable set $E \subseteq V$ with $\overline{E} \subseteq V$, we define the d-volume of $\phi(E)$ by

$$\mathrm{vol}_d(\phi(E)) := \int_E \sqrt{\det\left(D\phi(x)^\mathsf{T} D\phi(x)\right)}\, dx.$$

Here the Gram determinant in the integrand is the volume of the parallelotope spanned by the vectors $\partial_1\phi(x), \partial_2\phi(x), \ldots, \partial_d\phi(x)$, which by assumption (see *"parametrized submanifold"*) are linearly independent, i.e. $D\phi(x)$ has full rank, so the volume is strictly positive.



This factor gives a measure for the stretching of E as it is *"glued"* to the surface. If

$$\sqrt{\det\left(D\phi(x)^\mathsf{T} D\phi(x)\right)} = 1 \quad \text{for all } x,$$

then ϕ is called an "*isometry*" (German: „*Isometrie*"). A special case is the length of a curve, treated in Section 21.c below.

Remark 21.b.7 (*Why the transpose, and why the square root*): The shape of the formula looks arbitrary until one asks what could possibly stand in its place. The honest question is: **by what factor does ϕ stretch d-dimensional volume near x?** Near x the map ϕ is its differential, so the question is really about the linear map $D\phi(x)$, and the answer runs in three steps.

(a) **The obvious candidate does not exist.** For a map $\mathbb{R}^n \to \mathbb{R}^n$ the stretching factor is $|\det|$. Here $D\phi(x)$ is $n \times d$ with $d < n$, a **rectangular** matrix, and a rectangular matrix has no determinant. So, the one-line answer is unavailable, and something has to replace it.

(b) **A factor nevertheless exists.** A linear map still scales all d-volumes in its domain by one and the same constant: it sends the unit cube of $\mathbb{R}^d$ to a fixed parallelotope, and every other region is built from scaled copies of that cube. Call this constant $c \geq 0$. It is what we want; we only need a formula for it.

(c) **The Gram matrix supplies it.** The product $D\phi(x)^\mathsf{T} D\phi(x)$ is $d \times d$, square at last, so it **does** have a determinant, and that determinant turns out to be exactly c^2. Hence $c = \sqrt{\det(D\phi^\mathsf{T} D\phi)}$, and the square root is there for no deeper reason than to undo that squaring.

The two familiar formulas are the small cases. For $d = 1$ the matrix $D\phi(t)$ is the single column $\phi'(t)$, so $D\phi^\mathsf{T} D\phi = |\phi'(t)|^2$ and $c = |\phi'(t)|$, which is the arc-length integrand of Section 21.c. For $d = 2$, $n = 3$ one computes $c = |\partial_1\phi \times \partial_2\phi|$, the area of the parallelogram the two tangent vectors span, which is the surface-area formula used throughout this chapter.

AI-Note 21.3 (*Two objects, three notations*): From this chapter onwards the lecture notes write $D\phi(x)$ for the Jacobian **matrix** of ϕ at x, the object earlier chapters of this document call $\mathrm{J}\,\phi(x)$. That collides with the notation $D\phi_x$ used since Chapter 9 for the **differential**, the linear map itself. The two are not the same kind of object, even though they carry the same information:

$$\underbrace{D\phi_x}_{\text{linear map}}(v) \;=\; \underbrace{\mathrm{J}\,\phi(x)}_{\text{matrix}}\,v \;=\; \underbrace{D\phi(x)}_{\text{same matrix}}\,v.$$

The subscript is the whole distinction: $D\phi_x$ has x downstairs, $D\phi(x)$ has it in brackets. This document keeps each source's own convention rather than renotating transcribed material, so both spellings of the matrix appear; where it matters, read $D\phi(x)$ and $\mathrm{J}\,\phi(x)$ as synonyms.

Example 21.b.8 (*Integral on the graph of a function*): Given an open set $U \subseteq \mathbb{R}^2$ and a C^1 function $f : U \to \mathbb{R}$, we can define the parametrized surface $\phi : U \to \mathbb{R}^3$ by $\phi(x, y) = (x, y, f(x, y))^\mathsf{T}$. Assuming $M := \operatorname{Im}\phi$ is a C^1 submanifold, its area is given by $\int_M 1\,d\mathrm{vol}_2$.

We first find the Jacobian matrix of the parametrization:

$$\mathrm{J}\,\phi(x,y) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \partial_1 f(x,y) & \partial_2 f(x,y) \end{pmatrix}.$$

Then we compute the Gram determinant:

$$\begin{aligned}\sqrt{|\det(\mathrm{J}\,\phi^\mathsf{T}\,\mathrm{J}\,\phi)|} &= \sqrt{\det\begin{pmatrix} 1 + (\partial_1 f)^2 & \partial_1 f \partial_2 f \\ \partial_2 f \partial_1 f & 1 + (\partial_2 f)^2 \end{pmatrix}} \\ &= \sqrt{\big(1 + (\partial_1 f)^2\big)\big(1 + (\partial_2 f)^2\big) - (\partial_1 f)^2(\partial_2 f)^2} \\ &= \sqrt{1 + (\partial_1 f)^2 + (\partial_2 f)^2} = \sqrt{1 + |\nabla f(x,y)|^2}.\end{aligned}$$

By definition, we get the surface area formula for a graph:

$$\int_M 1\,d\underset{2}{\mathrm{vol}} = \int_U \sqrt{1 + |\nabla f(x,y)|^2}\,dx\,dy.$$

Example: area of a paraboloid patch (Subsection 21.b.1)

Example 21.b.9 (*Area of a paraboloid patch*): Compute the area of the surface

$$F = \left\{(x,y,z) \in \mathbb{R}^3 \;:\; x^2 + 4y^2 < 8,\;\; z = \tfrac{1}{2}(x^2 + 2y^2)\right\}.$$

Since $z = \frac{1}{2}(x^2 + 2y^2)$ means F is a graph, we can conveniently write F as the image of the parametrization

$$\phi : A \to F, \qquad (x,y) \mapsto \left(x,\, y,\, \tfrac{1}{2}(x^2 + 2y^2)\right), \qquad A := \{(x,y) \in \mathbb{R}^2 : x^2 + 4y^2 < 8\}.$$

Gram determinant. We compute

$$D\phi(x,y) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ x & 2y \end{pmatrix} \quad\Longrightarrow\quad D\phi(x,y)^\mathsf{T} D\phi(x,y) = \begin{pmatrix} 1 + x^2 & 2xy \\ 2xy & 1 + 4y^2 \end{pmatrix},$$

so that

$$\sqrt{\det D\phi(x,y)^\mathsf{T} D\phi(x,y)} = \sqrt{(1 + x^2)(1 + 4y^2) - 4x^2y^2} = \sqrt{1 + x^2 + 4y^2}.$$

The integral. By Definition 21.b.6,

$$\operatorname{vol}_2(F) = \int_A \sqrt{1+x^2+4y^2}\,dx\,dy, \qquad A = \{(x,y) \in \mathbb{R}^2 : x^2+4y^2 < 8\}.$$

We should choose more convenient coordinates. Polar coordinates come to mind, but then we still have problems, since A is not radially symmetric. We want to write $A = \Psi\big((0,R)\times(0,2\pi)\big)$ (up to a set of measure zero). Note that for $y=0$, $|x| < \sqrt{8} = 2\sqrt{2}$, and for $x=0$, $|y| < \sqrt{2}$. Choosing $R := \sqrt{2}$ and weighing x, y accordingly,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2r\cos\theta \\ r\sin\theta \end{pmatrix} =: \Psi(r,\theta).$$

Change of variables.

$$dx\,dy = |\det D\Psi(r,\theta)|\,dr\,d\theta = \left\| \begin{matrix} 2\cos\theta & -2r\sin\theta \\ \sin\theta & r\cos\theta \end{matrix} \right\| dr\,d\theta = 2r\,dr\,d\theta,$$

giving

$$\begin{aligned} \operatorname{vol}_2(F) &= \int_A \sqrt{1+x^2+4y^2}\,dx\,dy = \int_0^{\sqrt{2}} dr \int_0^{2\pi} d\theta\; 2r\sqrt{1+4r^2} \\ &= 4\pi \int_0^{\sqrt{2}} dr\; r(1+4r^2)^{1/2} = \frac{\pi}{2}\int_0^{\sqrt{2}} dr\; 8r(1+4r^2)^{1/2} \\ &= \frac{\pi}{2}\left[(1+4r^2)^{3/2}\cdot\frac{2}{3}\right]_0^{\sqrt{2}} = \frac{\pi}{3}(27-1) = \frac{26\pi}{3}. \end{aligned}$$

AI-Note 21.4: Worth flagging what makes A awkward: it is an ellipse, not a disc, so naive polar coordinates $(x,y) = (\rho\cos\theta, \rho\sin\theta)$ would force ρ to depend on θ along the boundary. The fix is to build the eccentricity into the change of variables itself. Scaling x by a factor of 2 relative to y turns the ellipse into the disc $r < \sqrt{2}$ before polar coordinates are even applied. This is the same idea as Example 20.b.19: choose Ψ to straighten the domain, not to simplify the integrand.

Example 21.b.10 (*Surface area of a sphere*): Given the sphere $M := \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = r^2\}$, we can use the parametrization $\phi : (0,\pi)\times(-\pi,\pi) \to \mathbb{R}^3$,

$$\phi(\theta,\varphi) = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix}.$$

The Jacobian matrix is

$$\mathrm{J}\,\phi(\theta,\varphi) = \begin{pmatrix} r\cos\theta\cos\varphi & -r\sin\theta\sin\varphi \\ r\cos\theta\sin\varphi & r\sin\theta\cos\varphi \\ -r\sin\theta & 0 \end{pmatrix},$$

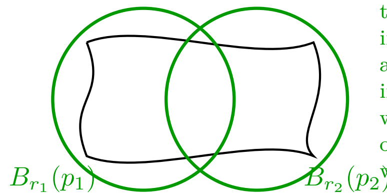
which has rank 2 for all (θ,φ) in the domain. Then we compute the Gram determinant:

$$\mathrm{J}\,\phi^{\mathsf{T}}\,\mathrm{J}\,\phi = \begin{pmatrix} r^2 & 0 \\ 0 & r^2\sin^2\theta \end{pmatrix} \quad\Longrightarrow\quad \sqrt{\det(\mathrm{J}\,\phi^{\mathsf{T}}\,\mathrm{J}\,\phi)} = r^2\sin\theta.$$

The surface area is given by the integral:

$$\begin{aligned} \int_M 1\,d\underset{2}{\operatorname{vol}} &= \int_{-\pi}^{\pi}\int_0^{\pi} \sqrt{\det(\mathrm{J}\,\phi^{\mathsf{T}}\,\mathrm{J}\,\phi)}\,d\theta\,d\varphi \\ &= \int_{-\pi}^{\pi}\int_0^{\pi} r^2\sin\theta\,d\theta\,d\varphi \\ &= r^2\left(\int_{-\pi}^{\pi} d\varphi\right)\left(\int_0^{\pi}\sin\theta\,d\theta\right) \\ &= r^2\cdot 2\pi\cdot 2 = 4\pi r^2. \end{aligned}$$

Note that here, our partition of unity simply was the identity (since the parametrization covers the sphere up to a null set).



and $\eta_\ell(x)$ is defined such that when we integrate in each ball separately and add all the integrals together, we exactly get a notion of the integral on the whole submanifold.

Exercise 21.b.11 (*9.1 — Volume* (important)): Calculate the volume of the region $B \subset \mathbb{R}^3$ enclosed by the surfaces $x^2 + y^2 + z^2 = 8$ and $2z = x^2 + y^2$. *Official hint:* use cylindrical coordinates.

Corsin's hint:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix} \qquad \text{for } (r,\theta,\varphi) \in (0,\infty) \times (0,\pi) \times (0,2\pi).$$

AI-Note 21.5 (*Corsin's hint is spherical, not cylindrical*): Corsin labels the formula above "*cylindrical coordinates*", but $(r\sin\theta\cos\varphi,\ r\sin\theta\sin\varphi,\ r\cos\theta)$ is the **spherical** parametrization, and his range $\theta \in (-\pi/2, \pi/2)$ has been corrected to $(0,\pi)$, the range on which $\theta \mapsto \cos\theta$ actually covers $[-1,1]$ bijectively (already logged in Exercise 17.c.30). The mislabelling is doubly worth noting here, since the official sheet's own hint recommends **cylindrical** coordinates (r,θ,z), and that is in fact the more natural choice for this problem, as the solution on Page 222 does: both surfaces are surfaces of revolution about the z-axis, which cylindrical coordinates respect directly, whereas spherical coordinates mix r and θ into both defining equations.

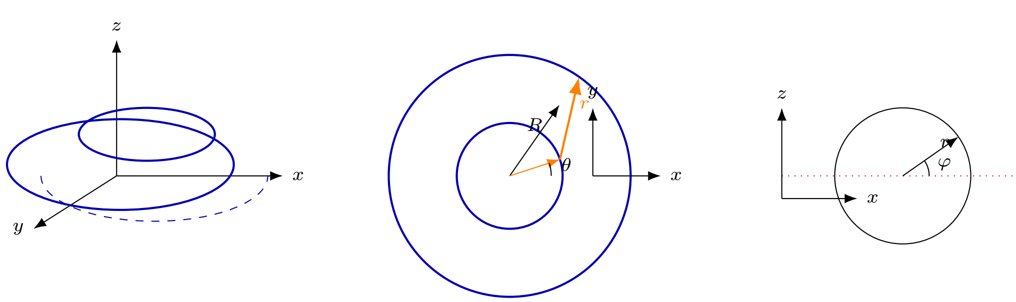
Exercise 21.b.12 (*10.1 — Spherical quadrilateral* (important)): Calculate the area of the spherical quadrilateral $S \subset \mathbb{S}^2$ bounded by the meridians $-\pi < \varphi_1 < \varphi_2 < \pi$ and parallels $-\pi/2 < \theta_1 < \theta_2 < \pi/2$.

Exercise 21.b.13 (*10.2 — The surface area of a torus* (important)): Let $S := \{(x,y,0) \in \mathbb{R}^3 \mid x^2 + y^2 = 4\}$ and $T := \{x \in \mathbb{R}^3 \mid d_S(x) \leq 1\}$, where $d_S(x) = \inf\{|x-y| \mid y \in S\}$ denotes the minimal distance from S to $x \in \mathbb{R}^3$. Parametrize ∂T and then calculate the area.

Corsin's hint, general parametrization of a torus surface:

$$x = (R + r\cos\phi)\cos\theta, \qquad y = (R + r\cos\phi)\sin\theta, \qquad z = r\sin\phi,$$

for the torus swept by a tube of radius r at distance R from the axis.



Length of a curve (Section 21.c)

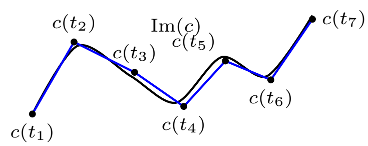
Definition 21.c.14 (*Length of a C^1-curve*): The "*length*" (German: „*Länge*") of a curve or path $\gamma \in C^1([a,b],\mathbb{R}^n)$ is given by

$$L(\gamma) := \int_a^b |\gamma'(t)|\,dt.$$

Definition 21.c.15 (*Length of a C^0-curve*): For a C^0-curve $c \in C^0([a,b],\mathbb{R}^n)$, the length is defined as

$$L(c) := \sup\left\{\sum_{i=1}^{n-1} |c(t_{i+1}) - c(t_i)| \;:\; a = t_1 < \cdots < t_n = b\right\},$$

the supremum over all lengths of inscribed polygons.



Remark 21.c.16: For a C^1-curve, Definitions 21.c.14 and 21.c.15 agree. Notice that

$$|\gamma'(t)| = \sqrt{D\gamma(t)^\mathsf{T} \cdot D\gamma(t)}$$

is exactly the Gram determinant of $D\gamma(t)$ in the sense of Definition 21.a.3, the $m = 1$ case of Example 21.a.4, now applied pointwise along the curve. This is why the length of a curve is the special case $d = 1$ of Definition 21.b.6: a curve is a 1-dimensional parametrized submanifold, and $L(\gamma) = \mathrm{vol}_1(\gamma([a,b]))$.

Example 21.c.17 (*Line integral*): Let $\phi : [-\pi/2, \pi/2] \to \mathbb{R}^2$ be given by $\phi(\theta) = \begin{pmatrix} 3\sin\theta \\ 4\sin\theta \end{pmatrix}$. Compute the length of the curve $M := \mathrm{Im}\,\phi$, i.e. $\int_M 1\,d\mathrm{vol}_1$.

First, we compute the Jacobian (which here is just the derivative vector):

$$\mathrm{J}\,\phi(\theta) = \begin{pmatrix} 3\cos\theta \\ 4\cos\theta \end{pmatrix}.$$

Then we compute the Gram determinant (which is the length of the tangent vector):

$$\sqrt{\det(\mathrm{J}\,\phi(\theta)^\mathsf{T}\,\mathrm{J}\,\phi(\theta))} = \sqrt{9\cos^2\theta + 16\cos^2\theta} = \sqrt{25\cos^2\theta} = 5|\cos\theta|.$$

Since $\cos\theta \geq 0$ for $\theta \in [-\pi/2, \pi/2]$, this simplifies to $5\cos\theta$. The length is:

$$\int_{-\pi/2}^{\pi/2} 5\cos\theta\,d\theta = 5\big[\sin\theta\big]_{-\pi/2}^{\pi/2} = 5(1 - (-1)) = 10.$$

Proposition 21.c.18 (*Length of a graph*): Let $f \in C^1([a,b],\mathbb{R})$. The graph $\Gamma_f := \{(x, f(x)) : x \in [a,b]\} \subseteq \mathbb{R}^2$ has length

$$\mathrm{vol}_1(\Gamma_f) = \int_a^b \sqrt{1 + f'(x)^2}\,dx.$$

Proof:

Parametrize by $\phi(t) := (t, f(t))^\mathsf{T}$ on $[a,b]$, which is injective and C^1 with $\phi'(t) = (1, f'(t))^\mathsf{T} \neq 0$. The Gram determinant of the single tangent vector is

$$\sqrt{\det\big(\mathrm{J}\,\phi(t)^\mathsf{T}\,\mathrm{J}\,\phi(t)\big)} = \|\phi'(t)\| = \sqrt{1 + f'(t)^2},$$

and integrating over $[a,b]$ is the definition of vol_1. **Q.E.D.**

Example 21.c.19 (*A quarter of an astroid*): Let $\gamma(t) := (\cos^3 t,\ \sin^3 t)^\mathsf{T}$ for $t \in [0, \frac{\pi}{2}]$. Then

$$\gamma'(t) = \begin{pmatrix} -3\cos^2 t\,\sin t \\ 3\sin^2 t\,\cos t \end{pmatrix},$$

so, using $\cos t \geq 0$ and $\sin t \geq 0$ on this interval,

$$\|\gamma'(t)\| = 3\cos t\,\sin t\sqrt{\cos^2 t + \sin^2 t} = 3\cos t\,\sin t = \tfrac{3}{2}\sin(2t).$$

The length is therefore

$$\int_0^{\pi/2} \tfrac{3}{2}\sin(2t)\,dt = \Big[-\tfrac{3}{4}\cos(2t)\Big]_0^{\pi/2} = -\tfrac{3}{4}\big((-1)-1\big) = \tfrac{3}{2}.$$

Note that $\gamma'(0) = \gamma'(\frac{\pi}{2}) = 0$: the astroid has cusps at its endpoints, so this is a curve whose parametrization fails the immersion condition exactly at the two boundary points. The length integral is unaffected, since two points contribute nothing to it.

Example 21.c.20 (*Spiral length*): Consider the spiral parametrized by $\phi : (0, 2\pi) \to \mathbb{R}^3$,

$$\phi(t) = \begin{pmatrix} \cos(\omega t) \\ \sin(\omega t) \\ vt \end{pmatrix},$$

where $v \in \mathbb{R}\setminus\{0\}$ and $\omega \in (0,\infty)$. Compute the length of the spiral $\operatorname{Im}\phi$.

We compute the Jacobian:

$$\mathrm{J}\,\phi(t) = \begin{pmatrix} -\omega\sin(\omega t) \\ \omega\cos(\omega t) \\ v \end{pmatrix}.$$

The Gram determinant is:

$$\sqrt{\mathrm{J}\,\phi(t)^\mathsf{T}\,\mathrm{J}\,\phi(t)} = \sqrt{\omega^2\sin^2(\omega t) + \omega^2\cos^2(\omega t) + v^2} = \sqrt{\omega^2 + v^2}.$$

The length of the spiral is the integral over the parameter domain:

$$\int_0^{2\pi} \sqrt{\omega^2 + v^2}\,dt = 2\pi\sqrt{\omega^2 + v^2}.$$

Example 21.c.21 (*Line integral of a function on a circle*): Given a function $f : \mathbb{R}^3 \to \mathbb{R}$, $f(x,y,z) = x^2 + z^2$. We want to integrate this function on the circle $M := \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 = 1,\ z = 2\}$. We parametrize the circle by $\phi : (0, 2\pi) \to \mathbb{R}^3$,

$$\phi(\varphi) = \begin{pmatrix} \cos\varphi \\ \sin\varphi \\ 2 \end{pmatrix}.$$

The Jacobian is $\mathrm{J}\,\phi(\varphi) = \phi'(\varphi) = \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix}$. The Gram determinant is $\sqrt{\mathrm{J}\,\phi^\mathsf{T}\,\mathrm{J}\,\phi} = |\phi'(\varphi)| = \sqrt{\sin^2\varphi + \cos^2\varphi} = 1$. The function evaluated on the curve is $f(\phi(\varphi)) = \cos^2\varphi + 4$. The line integral is thus:

$$\begin{aligned}\int_M f \, d\underset{1}{\mathrm{vol}} &= \int_0^{2\pi} f(\phi(\varphi))|\phi'(\varphi)| \, d\varphi \\ &= \int_0^{2\pi} (\cos^2 \varphi + 4) \cdot 1 \, d\varphi \\ &= \int_0^{2\pi} \cos^2 \varphi \, d\varphi + \int_0^{2\pi} 4 \, d\varphi \\ &= \pi + 8\pi = 9\pi.\end{aligned}$$

Exercise 21.c.22 (*9.2 — Multiple integrals*):

1. Let $D := [0,2] \times [0,1]$. Calculate $\iint_D (x^3 + 3x^2 y + y^3) \, dx \, dy$.
2. Let $D \subset \mathbb{R}^2$ be the interior of the triangle with vertices $(0,0)$, $(0,\pi)$, and (π,π). Calculate $\iint_D x \cos(x+y) \, dx \, dy$.
3. Let $D := \{(x,y) \in \mathbb{R}^2 \mid x > 1,\ y > 1,\ x + y < 3\}$. Calculate $\iint_D \frac{1}{(x+y)^3} \, dx \, dy$.

Corsin's hint (part 2): you can parametrize the triangle as

$$D = \{(x,y) \in \mathbb{R}^2 : 0 \le x \le y \le \pi\}.$$

Exercise 21.c.23 (*10.3 — Solids of revolution* (important)): Given $f \in C^1((a,b)) \cap C([a,b])$ such that $f \ge 0$, define the rotational body around the x-axis as

$$\Omega := \left\{ (x,y,z) \in \mathbb{R}^3 \mid a < x < b,\ \sqrt{y^2 + z^2} < f(x) \right\},$$
$$\partial_{\mathrm{side}}\Omega := \left\{ (x,y,z) \in \mathbb{R}^3 \mid a < x < b,\ \sqrt{y^2 + z^2} = f(x) \right\}.$$

1. Say whether Ω, $\partial_{\mathrm{side}}\Omega$ are open/closed/connected subsets of $\mathbb{R}^3$.
2. Show that $\mathrm{vol}_3(\Omega) = \pi \int_a^b f(x)^2 \, dx$.
3. Show that $\mathrm{vol}_2(\partial_{\mathrm{side}}\Omega) = 2\pi \int_a^b \sqrt{1 + f'(x)^2} \, f(x) \, dx$. *Hint:* parametrize $(x,\theta) \mapsto (x, f(x)\cos\theta, f(x)\sin\theta)$.
4. Calculate the volume and surface area of the 'improper rotational body' that arises when $f(x) = 1/x$, $a = 1$, $b = \infty$.
5. (*) Show that Ω is a bounded C^1 domain (as in Definition 22.a.1) if and only if

$$f(x) > 0 \text{ in } (a,b), \qquad f'(a^+) = +\infty, \qquad f'(b^-) = -\infty. \tag{$*$}$$

Integration over submanifolds (Section 21.d)

While Section 21.b explains how to compute the d-volume of a parametrized submanifold, it requires the entire relevant portion of the surface to be covered by a single parametrization ϕ. For more complicated submanifolds $M \subseteq \mathbb{R}^n$, we need to integrate scalar functions $f : M \to \mathbb{R}$ globally.

To achieve this, we cover the m-dimensional submanifold M with the images of local parametrizations $\Phi_l : V_l \to \mathbb{R}^n$, called an "*atlas*". By employing a smooth "*partition of unity*" $\eta_l \in C_c^\infty(\mathbb{R}^n, [0,1])$ (see AI-Note 25.4 and Proposition 25.b.8 for a rigorous construction) subordinate to this cover, we ensure that:

1. $\sum_{l=1}^N \eta_l \equiv 1$ on M,
2. $\mathrm{supp}(\eta_l) \subset \mathrm{Im}(\Phi_l)$.

Definition 21.d.24 (*Integral of a scalar function over a submanifold*): Given an m-dimensional submanifold $M \subseteq \mathbb{R}^n$, an atlas of local parametrizations $\Phi_l : V_l \to \mathbb{R}^n$, and a subordinate partition of unity $\{\eta_l\}$, the integral of a function $f: M \to \mathbb{R}$ over M is defined as:

$$\int_M f \underset{m}{\mathrm{dvol}} := \sum_{l=1}^{N} \int_{V_l} f(\Phi_l(x))\eta_l(\Phi_l(x))\sqrt{\det(\mathrm{J}\,\Phi_l^{\mathsf{T}}(x)\,\mathrm{J}\,\Phi_l(x))}\,dx.$$

This definition pieces together the local integrals (weighted by η_l) to compute the global integral over the entire submanifold, generalizing the volume integral from Definition 21.b.6.

Solutions (Section 21.e)

Solution to Exercise 21.b.11:

Following the official hint rather than Corsin's (AI-Note 21.5), we use cylindrical coordinates $(x, y, z) = (r\cos\theta, r\sin\theta, z)$, under which $x^2 + y^2 = r^2$. The two bounding surfaces become $r^2 + z^2 = 8$ and $z = r^2/2$; intersecting them,

$$z^2 + 2z - 8 = 0 \quad\Longrightarrow\quad z = 2 \text{ or } z = -4,$$

and since $z = r^2/2 \geq 0$ on the paraboloid, the surfaces meet at $z = 2$, $r = 2$. For $r \in (0, 2)$ the paraboloid $z = r^2/2$ lies below the upper half of the sphere $z = \sqrt{8 - r^2}$, so

$$B = \left\{(r, \theta, z) : 0 < r < 2,\ 0 < \theta < 2\pi,\ \tfrac{1}{2}r^2 < z < \sqrt{8 - r^2}\right\}.$$

Hence, with $dx\,dy\,dz = r\,dr\,d\theta\,dz$,

$$\mathrm{vol}(B) = \int_0^{2\pi}\int_0^2 r\left(\sqrt{8 - r^2} - \frac{r^2}{2}\right)dr\,d\theta = 2\pi\int_0^2\left(r\sqrt{8 - r^2} - \frac{r^3}{2}\right)dr.$$

For the first term, substitute $u = 8 - r^2$, $du = -2r\,dr$:

$$\int_0^2 r\sqrt{8 - r^2}\,dr = \left[-\frac{u^{3/2}}{3}\right]_{u=8}^{u=4} = \frac{8^{3/2} - 4^{3/2}}{3} = \frac{16\sqrt{2} - 8}{3}.$$

For the second, $\int_0^2 \frac{1}{2}r^3\,dr = \frac{1}{2}\cdot\frac{16}{4} = 2$. So

$$\mathrm{vol}(B) = 2\pi\left(\frac{16\sqrt{2} - 8}{3} - 2\right) = 2\pi\cdot\frac{16\sqrt{2} - 14}{3} = \boxed{\frac{4\pi}{3}(8\sqrt{2} - 7)}.$$

Q.E.D.

Solution to Exercise 21.c.22:

Part 1. By Fubini (Theorem 20.b.13),

$$\iint_D (x^3 + 3x^2y + y^3)\,dx\,dy = \int_0^2\int_0^1 (x^3 + 3x^2y + y^3)\,dy\,dx = \int_0^2\left(x^3 + \frac{3}{2}x^2 + \frac{1}{4}\right)dx.$$

The last integral is $\left[\frac{x^4}{4} + \frac{x^3}{2} + \frac{x}{4}\right]_0^2 = 4 + 4 + \frac{1}{2} = \boxed{\frac{17}{2}}$.

Part 2. Using Corsin's parametrization $D = \{0 \leq x \leq y \leq \pi\}$,

$$\iint_D x\cos(x + y)\,dx\,dy = \int_0^\pi\int_x^\pi x\cos(x + y)\,dy\,dx = \int_0^\pi x\big[\sin(x + y)\big]_{y=x}^{y=\pi}\,dx.$$

Since $\sin(x + \pi) = -\sin x$, the bracket equals $-\sin x - \sin(2x)$, so

$$\iint_D x\cos(x + y)\,dx\,dy = -\int_0^\pi x\sin x\,dx - \int_0^\pi x\sin(2x)\,dx.$$

Integrating by parts, $\int_0^\pi x\sin x\,dx = [-x\cos x + \sin x]_0^\pi = \pi$, and $\int_0^\pi x\sin(2x)\,dx = \left[-\frac{x}{2}\cos(2x) + \frac{1}{4}\sin(2x)\right]_0^\pi = -\frac{\pi}{2}$. Hence

$$\iint_D x\cos(x + y)\,dx\,dy = -\pi - \left(-\frac{\pi}{2}\right) = \boxed{-\frac{\pi}{2}}.$$

Part 3. Here $D = \{1 < x < 2,\ 1 < y < 3 - x\}$, so

$$\iint_D \frac{1}{(x+y)^3}\,dx\,dy = \int_1^2 \int_1^{3-x} (x+y)^{-3}\,dy\,dx = \int_1^2 \left[-\frac{1}{2}(x+y)^{-2}\right]_{y=1}^{y=3-x} dx.$$

Since $x + (3-x) = 3$, the bracket is $-\frac{1}{2}\left(3^{-2} - (x+1)^{-2}\right) = \frac{1}{2}(x+1)^{-2} - \frac{1}{18}$, and

$$\int_1^2 \left(\frac{1}{2(x+1)^2} - \frac{1}{18}\right) dx = \left[-\frac{1}{2(x+1)}\right]_1^2 - \frac{1}{18} = \left(-\frac{1}{6} + \frac{1}{4}\right) - \frac{1}{18} = \frac{1}{12} - \frac{1}{18} = \boxed{\frac{1}{36}}.$$

Q.E.D.

Solution to Exercise 21.b.12:

We parametrize $\mathbb{S}^2$ (up to a null set) using latitude/longitude spherical coordinates

$$\Phi : (-\pi, \pi) \times \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right) \to \mathbb{S}^2, \qquad (\varphi, \theta) \mapsto \begin{pmatrix} \cos\varphi\cos\theta \\ \sin\varphi\cos\theta \\ \sin\theta \end{pmatrix}.$$

Note that this is **not** the $(r \equiv 1)$ parametrization used elsewhere in this chapter, whose polar angle runs over $(0, \pi)$. Here θ is measured from the equator, matching the exercise's range $\theta \in (-\pi/2, \pi/2)$. Computing

$$\partial_\varphi\Phi \times \partial_\theta\Phi = \begin{pmatrix} -\sin\varphi\cos\theta \\ \cos\varphi\cos\theta \\ 0 \end{pmatrix} \times \begin{pmatrix} -\cos\varphi\sin\theta \\ -\sin\varphi\sin\theta \\ \cos\theta \end{pmatrix} = \begin{pmatrix} \cos\varphi\cos^2\theta \\ \sin\varphi\cos^2\theta \\ \sin\theta\cos\theta \end{pmatrix},$$

its length is $|\partial_\varphi\Phi \times \partial_\theta\Phi| = \cos\theta$ (for $\theta \in (-\pi/2, \pi/2)$, so $\cos\theta > 0$). By Proposition 23.b.2(2), this is the Gram determinant, so $dA = \cos\theta\,d\theta\,d\varphi$, and

$$\int_S dA = \int_{\varphi_1}^{\varphi_2} \int_{\theta_1}^{\theta_2} \cos\theta\,d\theta\,d\varphi = \boxed{(\varphi_2 - \varphi_1)\left(\sin\theta_2 - \sin\theta_1\right)}.$$

Q.E.D.

Solution to Exercise 21.b.13:

The circle S has radius 2, and T is the solid torus swept by a tube of radius 1 around it, so ∂T is Corsin's general torus surface with $R = 2$, $r = 1$ fixed. Using his parametrization,

$$\phi(\theta, \varphi) = \begin{pmatrix} (2+\cos\varphi)\cos\theta \\ (2+\cos\varphi)\sin\theta \\ \sin\varphi \end{pmatrix}, \qquad \theta, \varphi \in (0, 2\pi).$$

Then $\partial_\theta\phi = \big(-(2+\cos\varphi)\sin\theta,\ (2+\cos\varphi)\cos\theta,\ 0\big)$ has length $2 + \cos\varphi$, and $\partial_\varphi\phi = (-\sin\varphi\cos\theta,\ -\sin\varphi\sin\theta,\ \cos\varphi)$ has length 1; the two are orthogonal (their dot product is $\sin\varphi\cos\varphi\sin\theta\cos\theta - \sin\varphi\cos\varphi\sin\theta\cos\theta = 0$), so

$$\sqrt{\det D\phi^{\mathsf{T}} D\phi} = |\partial_\theta\phi|\,|\partial_\varphi\phi| = 2 + \cos\varphi.$$

Hence

$$\operatorname{vol}_2(\partial T) = \int_0^{2\pi}\int_0^{2\pi} (2+\cos\varphi)\,d\theta\,d\varphi = 2\pi\left(2\cdot 2\pi + \underbrace{\int_0^{2\pi}\cos\varphi\,d\varphi}_{=0}\right) = \boxed{8\pi^2}.$$

Q.E.D.

Solution to Exercise 21.c.23:

Part 1. $\partial_{\text{side}}\Omega$ is not closed (it omits, e.g., the limit point $(a, f(a), 0)$ as $x \to a^+$) and not open (a 2-dimensional surface has empty interior in $\mathbb{R}^3$), but it **is** path-connected: from any point, first move along the graph of f in the x-direction, then move around the circle in the (y,z)-variables. Ω is open (as the strict sublevel set of a continuous function), not closed, and its connectedness depends on f: if f vanishes somewhere in (a,b), the cross-section pinches to a point there and Ω is disconnected into the pieces on either side; if $f > 0$ throughout (a,b), Ω is connected.

Part 2. In cylindrical coordinates (x, ρ, θ) with $y = \rho\cos\theta$, $z = \rho\sin\theta$, $dx\,dy\,dz = \rho\,d\rho\,d\theta\,dx$, the domain becomes the box-like region $\{a < x < b,\ 0 < \rho < f(x),\ \theta \in [0, 2\pi)\}$, so

$$\mathrm{vol}_3(\Omega) = \int_a^b \int_0^{2\pi} \int_0^{f(x)} \rho\,d\rho\,d\theta\,dx = \int_a^b \int_0^{2\pi} \frac{1}{2} f(x)^2\,d\theta\,dx = \boxed{\pi \int_a^b f(x)^2\,dx}.$$

Part 3. Using the suggested parametrization $\Phi(x,\theta) = (x, f(x)\cos\theta, f(x)\sin\theta)$, write the orthonormal frame $\xi_1 := (1,0,0)$, $\xi_2 := (0, \cos\theta, \sin\theta)$, $\xi_3 := (0, -\sin\theta, \cos\theta)$, so that

$$\partial_x\Phi = \xi_1 + f'(x)\xi_2, \qquad \partial_\theta\Phi = f(x)\xi_3.$$

Since ξ_1, ξ_2, ξ_3 are orthonormal and right-handed, $\xi_1 \times \xi_3 = -\xi_2$ and $\xi_2 \times \xi_3 = \xi_1$, so

$$\partial_x\Phi \times \partial_\theta\Phi = f(x)\big(\xi_1 \times \xi_3 + f'(x)\,\xi_2 \times \xi_3\big) = f(x)\big(-\xi_2 + f'(x)\xi_1\big), \qquad |\partial_x\Phi \times \partial_\theta\Phi| = f(x)\sqrt{1 + f'(x)^2}.$$

By Proposition 23.b.2(2), this is the Gram determinant, so

$$\mathrm{vol}_2(\partial_{\text{side}}\Omega) = \int_a^b \int_0^{2\pi} f(x)\sqrt{1 + f'(x)^2}\,d\theta\,dx = \boxed{2\pi \int_a^b f(x)\sqrt{1 + f'(x)^2}\,dx}.$$

Part 4. With $f(x) = 1/x$ on $[1, \infty)$, both integrands are non-negative, so the improper integrals are unambiguous (possibly $+\infty$). The volume is finite:

$$\pi \int_1^\infty \frac{1}{x^2}\,dx = \pi\Big[-\frac{1}{x}\Big]_1^\infty = \boxed{\pi}.$$

For the surface area, $f'(x) = -1/x^2$, so the integrand is $\sqrt{1 + x^{-4}}/x \geq 1/x$ pointwise, and since $\int_1^\infty \frac{1}{x}\,dx$ diverges, so does

$$2\pi \int_1^\infty \frac{\sqrt{1 + x^{-4}}}{x}\,dx = \boxed{+\infty}.$$

This is the classical "*Gabriel's horn*" (German: „*Torricellis Trompete*"): finite volume, infinite surface area.

Part 5. ($\Leftarrow$) Suppose $(*)$ holds. Since $f'(a^+) = +\infty$, the inverse function theorem applies to $f|_{(a, a+\varepsilon)}$ for ε small, giving a C^1, increasing inverse there; extend it by

$$g(y) := \begin{cases} f^{-1}(y) & f(a) < y < f(a) + \varepsilon, \\ a & y \leq f(a). \end{cases}$$

Since $f'(a^+) = +\infty$, $\lim_{y \to f(a)^+} g'(y) = \lim_{x \to a^+} 1/f'(x) = 0 = \lim_{y \to f(a)^-} g'(y)$, so $g \in C^1$ across $y = f(a)$. Around a boundary point $P := (a, f(a)\cos\phi, f(a)\sin\phi)$, the map $(t,\theta) \mapsto \big(g(t), f(g(t))\cos\theta, f(g(t))\sin\theta\big)$ is then a regular C^1 parametrization of $\partial\Omega$ (using $f(a) > 0$ to keep the angular direction non-degenerate). The symmetric argument at b, using $f'(b^-) = -\infty$, handles the other end, so $\partial\Omega$ is a C^1 submanifold and Ω is a bounded C^1 domain.

($\Rightarrow$) Suppose Ω is a bounded C^1 domain. If f vanished at some interior point, Ω would be disconnected by Part 1, which Definition 22.a.1 excludes, since it requires Ω itself to make sense as a single bounded open set with $(n-1)$-manifold boundary (implicit in the exercise's setup). Therefore $f > 0$ on (a,b). For the endpoint condition at a: the point $P := (a, f(a), 0) \in \partial\Omega$ is a smooth boundary point, so $T_P(\partial\Omega)$ is 2-dimensional and contains

$$\frac{d}{dt}\Big|_{t=0^+} \big(a, f(a) - t, 0\big) = -e_2, \qquad \frac{d}{dt}\Big|_{t=0^-} \big(a, f(a)\cos t, f(a)\sin t\big) = f(a)e_3,$$

so $T_P(\partial\Omega) = \operatorname{span}\{e_2, e_3\}$. But $T_P(\partial\Omega)$ must also contain the limit of the (rescaled, if necessary) tangent vectors

$$v_\varepsilon := \frac{d}{dt}\Big|_{t=0^+} \big(a+\varepsilon+t,\ f(a+\varepsilon+t),\ 0\big) = \big(1,\ f'(a+\varepsilon),\ 0\big)$$

as $\varepsilon \to 0^+$ (these are tangent to $\partial\Omega$ at nearby points converging to P). If $f'(a+\varepsilon)$ stayed bounded along some sequence $\varepsilon_k \to 0^+$, the direction of v_{ε_k} would converge to some $\alpha e_1 + \beta e_2 \notin \operatorname{span}\{e_2, e_3\}$ (its e_1-component is the nonzero constant 1, up to normalization), contradicting $T_P(\partial\Omega) = \operatorname{span}\{e_2, e_3\}$. So $f'(a^+) = +\infty$; the symmetric argument at b gives $f'(b^-) = -\infty$.

Q.E.D.

AI-Note 21.6: All parts of Exercises 21.b.12, 21.b.13 and 21.c.23 were worked out independently before consulting `Sol10_Analysis2_eng.pdf`. The three main answers ($(\varphi_2-\varphi_1)(\sin\theta_2 - \sin\theta_1)$, $8\pi^2$, $\pi\int_a^b f(x)^2dx$ / $2\pi\int_a^b f(x)\sqrt{1+f'(x)^2}dx$ / volume π, area $+\infty$) match exactly. Part 5 was reconstructed independently along the same two-directional strategy (build an explicit C^1 parametrization near each endpoint using the inverse function theorem; for the converse, derive a contradiction from the tangent space being forced to both contain $\operatorname{span}\{e_2, e_3\}$ and the limiting direction of v_ε) and agrees with the official solution in substance, though the write-up above is condensed relative to the official one.

PART VII

Vector Calculus and Differential Forms

The classical integral theorems of vector calculus look at first like three unrelated results. Green's theorem converts a line integral into a double integral, the divergence theorem converts a flux through a closed surface into a triple integral, and the theorem of Stokes converts a circulation into a flux. This part proves all three, and then explains why they are one theorem. The apparatus that achieves this consists of differential forms and the exterior derivative d, a single operator that absorbs grad, curl and div at once. In that language each of the statements above reads $\int_M d\omega = \int_{\partial M} \omega$, and the classical versions are what it becomes when M is a plane region, a solid, or a surface. Everything built earlier is needed here: submanifolds to integrate over, orientation to fix the sign, and compactness to keep the integrals finite.

Vector Calculus Operators (Chapter 22)

The integral theorems of this part relate what happens inside a domain to what happens on its boundary, so two things have to come first. One is a class of domains whose boundary is well behaved enough to integrate over. The other is the differential operators (gradient, divergence, curl) whose interplay the theorems express. This chapter supplies both, and says why the definition of a bounded C^1 domain is stated in the awkward way it is.

Bounded C^1 domains and why the definition in the script is so terrible (Section 22.a)

Definition 22.a.1 (*Bounded C^k domain*): Given $k \geq 1$, a bounded open set $\Omega \subset \mathbb{R}^n$ is called a "*bounded C^k domain*" if $M := \partial\Omega$ is an $(n-1)$-dimensional submanifold of class C^k.

Citing the 2024 lecture notes, Corsin builds up the machinery needed to actually **use** this definition. To define "*flux*" we need the concept of interior and exterior normal vectors. For Ω as above, a normal vector $v \in N_p(\partial\Omega)$ is "*interior*" if

$$p + hv \in \Omega \quad \text{for all } h > 0 \text{ small,}$$

and "*exterior*" if

$$p + hv \in \mathbb{R}^n \setminus \Omega \quad \text{for all } h > 0 \text{ small.}$$

A "*Gauss map*" is a continuous map of unit normal vectors on $\partial\Omega$. There exists a unique interior and exterior Gauss map; write

$$\nu \in C^0(\partial\Omega, \mathbb{S}^{n-1}) \text{ such that } \nu(p) \text{ is exterior normal}$$

for the exterior unit normal map.

Theorem 22.a.2 (*Jordan–Brouwer separation theorem*): If $M \subseteq \mathbb{R}^{n+1}$ is an n-dimensional compact and connected submanifold, then $\mathbb{R}^{n+1} \setminus M$ has exactly two connected components, an "*inside*" and an "*outside*".

Vector calculus operators (Section 22.b)

Definition 22.b.3 (*Vector field*): Let $U \subseteq \mathbb{R}^n$ be open. A "*C^k vector field*" (German: „*Vektorfeld*“) is a function $F : U \to \mathbb{R}^n$ of class $C^k(U, \mathbb{R}^n)$. The scalar functions $F_1, \ldots, F_n$ in $F(x) = (F_1(x), \ldots, F_n(x))^\mathsf{T}$ are called the "*components*".

Definition 22.b.4 (*Gradient*): Let $f : U \to \mathbb{R}$ be a function of class C^k on an open set $U \subseteq \mathbb{R}^n$. The "*gradient*" of f is the C^{k-1} vector field

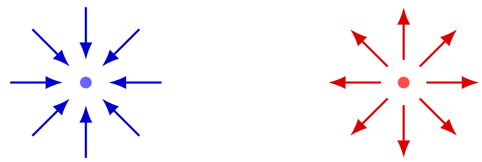
$$\operatorname{grad} f(x) = \nabla f(x) := \begin{pmatrix} \partial_1 f(x) \\ \vdots \\ \partial_n f(x) \end{pmatrix}.$$

Definition 22.b.5 (*Divergence*): For a C^1 vector field $F : U \to \mathbb{R}^n$, the "*divergence*" is the scalar function

$$\operatorname{div} F(x) := \sum_{i=1}^{n} \partial_i F_i(x).$$

In 3D, physicists write this as $\vec{\nabla} \cdot \vec{F}(x) = \partial_1 F_1(x) + \partial_2 F_2(x) + \partial_3 F_3(x)$.

AI-Note 22.1: Intuitively, $\operatorname{div} F(x)$ tells us how much flow of F is generated at x. If $\operatorname{div} F(x) = 0$, then the total generated flow is zero. If $\operatorname{div} F(x) > 0$, then x is a "*source*" (German: „*Quelle*"), meaning more flow is generated than removed. If $\operatorname{div} F(x) < 0$, then x is a "*sink*" (German: „*Senke*"), meaning more flow is removed than generated.



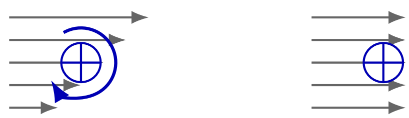
sink: $\operatorname{div} F < 0$ source: $\operatorname{div} F > 0$

Definition 22.b.6 (*Rotation/Curl*): Let F be a C^k vector field on an open set $U \subseteq \mathbb{R}^3$. The "*rotation*" or "*curl*" of F is the C^{k-1} vector field

$$\operatorname{curl} F(x) := \begin{pmatrix} \partial_2 F_3(x) - \partial_3 F_2(x) \\ \partial_3 F_1(x) - \partial_1 F_3(x) \\ \partial_1 F_2(x) - \partial_2 F_1(x) \end{pmatrix}.$$

Physicists write $\operatorname{curl} F(x) = \vec{\nabla} \times \vec{F}(x)$. In $\mathbb{R}^2$, the rotation of a vector field $F = (F_1, F_2)^\mathsf{T}$ is the scalar function $\operatorname{curl} F(x) := \partial_1 F_2(x) - \partial_2 F_1(x)$.

AI-Note 22.2: Intuitively, $\operatorname{curl} F(x)$ tells us how much the field tends to rotate around x. If $\operatorname{curl} F(x) \neq 0$, a small wheel centered at x would rotate.



rotates does not rotate

Example 22.b.7 (*Computing vector calculus operators*): Given $\phi : \mathbb{R}^3 \to \mathbb{R}$, $\phi(x, y, z) := x^2 + y^2 + z^2 e^{-xy}$. Let us compute the gradient, the divergence of the gradient, and the rotation of the gradient.

- **Gradient:**
$$\nabla \phi(x, y, z) = \begin{pmatrix} 2x - yz^2 e^{-xy} \\ 2y - xz^2 e^{-xy} \\ 2z e^{-xy} \end{pmatrix}.$$
- **Divergence of the gradient** (denoted $\operatorname{div}(\nabla \phi)$ or $\Delta \phi$):
$$\begin{aligned} \operatorname{div}(\nabla \phi) &= \partial_x (2x - yz^2 e^{-xy}) + \partial_y (2y - xz^2 e^{-xy}) + \partial_z (2z e^{-xy}) \\ &= (2 + y^2 z^2 e^{-xy}) + (2 + x^2 z^2 e^{-xy}) + (2 e^{-xy}) \\ &= 4 + (x^2 z^2 + y^2 z^2 + 2) e^{-xy}. \end{aligned}$$
- **Rotation of the gradient** ($\operatorname{curl}(\nabla \phi)$):
$$\begin{aligned} \operatorname{curl}(\nabla \phi) &= \begin{pmatrix} \partial_y (2z e^{-xy}) - \partial_z (2y - xz^2 e^{-xy}) \\ \partial_z (2x - yz^2 e^{-xy}) - \partial_x (2z e^{-xy}) \\ \partial_x (2y - xz^2 e^{-xy}) - \partial_y (2x - yz^2 e^{-xy}) \end{pmatrix} \\ &= \begin{pmatrix} -2xz e^{-xy} - (-2xz e^{-xy}) \\ -2yz e^{-xy} - (-2yz e^{-xy}) \\ -z^2 e^{-xy} - (-z^2 e^{-xy}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

  This is not a coincidence: for any C^2 function f, $\operatorname{curl}(\nabla f) = 0$ by Schwarz's theorem.

Exercise 22.b.8 (curl ∘ grad = 0 *and* div ∘ curl = 0): Let f be a twice continuously differentiable function and F a twice continuously differentiable vector field on an open set $U \subseteq \mathbb{R}^3$. Show that

$$\operatorname{curl}(\operatorname{grad} f) = 0 \qquad \text{and} \qquad \operatorname{div}(\operatorname{curl} F) = 0.$$

Proof of Exercise 22.b.8:

Both identities follow from Schwarz's theorem (Theorem 11.b.8), which applies since $f, F \in C^2$.

$\operatorname{curl}(\operatorname{grad} f) = 0$. The first component of $\operatorname{curl}(\operatorname{grad} f)$ is $\partial_2(\partial_3 f) - \partial_3(\partial_2 f) = 0$ by Schwarz, and the other two components vanish the same way, cyclically permuting the indices. This is exactly Example 22.b.7's observation above, now stated and proved in general.

$\operatorname{div}(\operatorname{curl} F) = 0$.

$$\operatorname{div}(\operatorname{curl} F) = \partial_1(\partial_2 F_3 - \partial_3 F_2) + \partial_2(\partial_3 F_1 - \partial_1 F_3) + \partial_3(\partial_1 F_2 - \partial_2 F_1),$$

and every one of the six terms cancels against another by Schwarz: $\partial_1\partial_2 F_3$ cancels $-\partial_2\partial_1 F_3$, $-\partial_1\partial_3 F_2$ cancels $\partial_3\partial_1 F_2$, and $\partial_2\partial_3 F_1$ cancels $-\partial_3\partial_2 F_1$. **Q.E.D.**

Remark 22.b.9: This is the classical-vector-calculus shadow of *"$d^2 = 0$"* (Theorem 24.b.12**(2)**, Chapter 24): grad, curl and div are exactly the 0-, 1- and 2-form incarnations of the exterior derivative in $\mathbb{R}^3$ (Example 24.b.13), and both identities above are the single fact $d \circ d = 0$ read off once in each degree.

Proposition 22.b.10 (*Product rule for the curl*): Let $U \subseteq \mathbb{R}^3$ be open, $F : U \to \mathbb{R}^3$ a C^1 vector field, and $\varphi : U \to \mathbb{R}$ a C^1 function. Then

$$\operatorname{curl}(\varphi F) = \varphi \operatorname{curl}(F) + (\operatorname{grad} \varphi) \times F.$$

Proof:

By the product rule applied componentwise, e.g. for the first component,

$$\partial_2(\varphi F_3) - \partial_3(\varphi F_2) = \varphi(\partial_2 F_3 - \partial_3 F_2) + (\partial_2\varphi)F_3 - (\partial_3\varphi)F_2,$$

and the first term is $\varphi \operatorname{curl}(F)$'s first component, the last two are $((\operatorname{grad}\varphi) \times F)$'s first component. The other two components follow identically, cyclically permuting the indices. **Q.E.D.**

AI-Note 22.3: The official script sets this identity as an exercise (Ex. 14.61) but states it without the first term, as $\operatorname{curl}(\varphi F) = (\operatorname{grad}\varphi) \times F$, which is false in general. Taking $\varphi \equiv 1$ would force $\operatorname{curl} F \equiv 0$ for every F. The missing term is exactly $\varphi \operatorname{curl}(F)$, restored above. (The script's exercise also names the vector field f in its statement and F in its conclusion, as though the two were interchangeable.)

Exercise 22.b.11 (*11.1 — Flow of vector field* (important)): Compute the flow of the vector field

$$V : (x, y, z) \mapsto (x, y, z - x^2 - y^2)$$

through the upper hemisphere $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1,\ z \geq 0\}$ from *"inside"* to *"outside"*. That is the number

$$\int_S V(x) \cdot \nu(x)\, d\operatorname{vol}_2.$$

Solutions (Section 22.c)

Solution of Exercise 22.b.11:

The vector field $V(x, y, z) = (x, y, z - x^2 - y^2)$ has divergence

$$\operatorname{div} V = \partial_x(x) + \partial_y(y) + \partial_z(z - x^2 - y^2) = 1 + 1 + 1 = 3.$$

S is not itself the boundary of a bounded C^1 domain, so close it up with the equatorial disk $D := \{(x,y,0) : x^2+y^2 \leq 1\}$: then $\Omega := \{(x,y,z) \in B_1(0) : z \geq 0\}$, the upper half ball, is a bounded C^1 domain with $\partial\Omega = S \cup D$, and S's *"inside to outside"* normal agrees with the outward normal of Ω on S. By Theorem 23.b.4,

$$\int_S V \cdot \nu \, d\,\mathrm{vol}_2 + \int_D V \cdot \nu_D \, d\,\mathrm{vol}_2 = \int_\Omega \operatorname{div} V \, dx = 3\,\mathrm{vol}_3(\Omega) = 3 \cdot \frac{2\pi}{3} = 2\pi,$$

using $\mathrm{vol}_3(\Omega) = \frac{1}{2} \cdot \frac{4}{3}\pi = \frac{2\pi}{3}$. On D the outward normal is $\nu_D = -e_3$, and $V(x,y,0) = (x,y,-x^2-y^2)$, so $V \cdot \nu_D = x^2+y^2$; in polar coordinates,

$$\int_D V \cdot \nu_D \, d\,\mathrm{vol}_2 = \int_0^{2\pi} \int_0^1 r^2 \cdot r \, dr \, d\varphi = 2\pi \left[\frac{r^4}{4}\right]_0^1 = \frac{\pi}{2}.$$

Hence

$$\int_S V \cdot \nu \, d\,\mathrm{vol}_2 = 2\pi - \frac{\pi}{2} = \boxed{\frac{3\pi}{2}}.$$

Q.E.D.

Flux and the Divergence Theorem (Chapter 23)

The divergence theorem is the first of the integral theorems: the flux of a vector field out through the boundary of a domain equals the integral of its divergence inside. The chapter opens with the case $n = 1$, where the statement degenerates into the fundamental theorem of calculus and the mechanism is visible without any geometry, and then works up to the general theorem and its physical reading as a conservation law.

Gauss' theorem revisited: the case $n = 1$, and a direct proof for a square (Section 23.a)

Gauss' theorem (Theorem 23.b.4), proved in Section 23.b below, is, in a precise sense, a higher-dimensional fundamental theorem of calculus: it trades a derivative integrated over a region for the original quantity evaluated on the region's boundary. Setting $n = 1$ makes the analogy literal. Let $f \in C^1([a,b])$, viewed as a (one-dimensional) vector field. Since $\operatorname{div} f = f'$ in dimension 1, Gauss' formula becomes

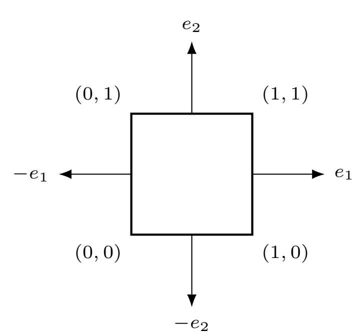
$$\int_a^b \frac{\partial f}{\partial x}\,dx = \int_{[a,b]} \operatorname{div} f \, d\operatorname{vol}_1 = \int_{\partial[a,b]} \langle f, \nu\rangle \, d\operatorname{vol}_0 = f(b) - f(a) \qquad \text{(FTC, almost).}$$

The last step identifies $\int_{\partial[a,b]}\langle f,\nu\rangle\,d\operatorname{vol}_0$ with $f(b) - f(a)$ by treating $\partial[a,b] = \{a,b\}$ as a 0-dimensional submanifold with outward unit normals $\nu(a) = -1$, $\nu(b) = +1$, and vol_0 as counting measure.

AI-Note 23.1: Corsin flags this himself: *"there are several problems with this argument, for example what a 0-dim Jordan measure or 0-dim submanifold is"*, and notes that *"the definitions from the lecture can be extended to include this case"* without carrying out the extension. The word *"almost"* above is his. Not pursued further here; the point of the calculation is the analogy with the FTC, not a rigorous $n = 1$ instance of Theorem 23.b.4.

Proof of Theorem 23.b.4 for $n = 2$, $B = (0,1)^2$:

Conversely, the fundamental theorem of calculus can be used to **prove** Gauss' theorem; we carry this out for the unit square. Let $F : [0,1]^2 \to \mathbb{R}^2$ be a C^1 vector field and $B := (0,1)^2$. The boundary ∂B splits into four edges, with outward unit normals $-e_1$, e_1, $-e_2$, e_2 on the left, right, bottom, and top edges respectively.



Parametrizing each edge by $t \mapsto (0,t)$, $(1,t)$, $(t,0)$, $(t,1)$ for $t \in [0,1]$,

$$\int_{\partial B}\langle F,\nu\rangle\,d\operatorname{vol}_1 = \int_0^1 \langle F(0,t), -e_1\rangle\,dt + \int_0^1 \langle F(1,t), e_1\rangle\,dt + \int_0^1 \langle F(t,0), -e_2\rangle\,dt + \int_0^1 \langle F(t,1), e_2\rangle\,dt.$$

Grouping the first pair and the second pair,

$$\int_{\partial B} \langle F, \nu \rangle \, d\,\mathrm{vol}_1 = \int_0^1 \big[F_1(1,t) - F_1(0,t)\big]\, dt + \int_0^1 \big[F_2(t,1) - F_2(t,0)\big]\, dt.$$

By the fundamental theorem of calculus applied to $t \mapsto F_1(s,t)$ along s and to $s \mapsto F_2(t,s)$ along the second variable,

$$\int_{\partial B} \langle F, \nu \rangle \, d\,\mathrm{vol}_1 = \int_0^1 \left(\int_0^1 \partial_1 F_1(s,t)\, ds\right) dt + \int_0^1 \left(\int_0^1 \partial_2 F_2(t,s)\, ds\right) dt,$$

and by Fubini,

$$\int_{\partial B} \langle F, \nu \rangle \, d\,\mathrm{vol}_1 = \int_B \partial_1 F_1 \, d\,\mathrm{vol}_2 + \int_B \partial_2 F_2 \, d\,\mathrm{vol}_2 = \int_B \operatorname{div} F \, d\,\mathrm{vol}_2 \,.$$

Q.E.D.

Flux and the divergence theorem (Section 23.b)

Definition 23.b.1 (*Flux*): Suppose $F : U \to \mathbb{R}^n$ is a C^1 vector field, $U \subseteq \mathbb{R}^n$ open, and $\Omega \subseteq U$ is a C^1 bounded domain. The "*flux*" (German: „*Fluss*“) of F through $\partial\Omega$ is defined as

$$\int_{\partial\Omega} \langle F, \nu \rangle \, d\,\mathrm{vol}_{n-1} \qquad \text{(analysis notation)} \qquad = \qquad \int_{\partial\Omega} \vec{F}(x) \cdot d\vec{A} \qquad \text{(physics notation)}$$

$$= \int_B \big\langle F \circ \phi(x),\ \nu \circ \phi(x) \big\rangle \sqrt{\det D\phi(x)^{\mathsf{T}} D\phi(x)}\, dx \qquad \text{(explicit)},$$

where $\nu : \partial\Omega \to \mathbb{S}^{n-1}$ is the exterior unit normal, i.e. $\nu(y)$ is the outward-facing unit normal of $\partial\Omega$ at $y \in \partial\Omega$, and $\phi : B \to \partial\Omega$ is a (local) parametrization of $\partial\Omega$. In general, multiple integrals in the explicit formulation may be necessary.

Proposition 23.b.2 (*Flux in* $\mathbb{R}^3$): Let $\Omega \subseteq \mathbb{R}^3$ be a bounded C^1 domain and $F \in C^1(\overline{\Omega}, \mathbb{R}^3)$ a vector field, and let $\phi : B \to \partial\Omega \setminus N$ be a parametrization of $\partial\Omega \setminus N$, where $N \subseteq \partial\Omega$ is a null set, $B \subseteq \mathbb{R}^2$. Then:

1. $\nu(x) := \dfrac{\partial_1\phi(x) \times \partial_2\phi(x)}{|\partial_1\phi(x) \times \partial_2\phi(x)|}$ is an (interior or exterior) Gauss map (more precisely, $\nu \circ \phi^{-1}$ is a Gauss map, i.e. $\nu(x) \in N_{\phi(x)}(\partial\Omega \setminus N)$).
2. The Gram determinant is given by $\sqrt{\det D\phi(x)^{\mathsf{T}} D\phi(x)} = |\partial_1\phi(x) \times \partial_2\phi(x)|$.
3. Therefore,
$$\int_{\partial\Omega} \vec{F} \cdot d\vec{A} = \pm \int_B \big\langle F \circ \phi(x),\ \partial_1\phi(x) \times \partial_2\phi(x) \big\rangle\, dx,$$
where the sign depends on whether $\nu(x)$ is interior $(-)$ or exterior $(+)$.

Example 23.b.3 (*Flux through the unit sphere*): Compute the flux of $F(x,y,z) := (xz,\ z,\ y)$ through the surface of $\mathbb{S}^2$ with interior $B_1(0)$.

We use spherical coordinates with $r \equiv 1$:

$$\phi : (0, 2\pi) \times (0, \pi) \to \mathbb{R}^3, \qquad (\varphi, \theta) \mapsto \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}.$$

Normal vector.

$$\partial_\varphi\phi = \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix}, \qquad \partial_\theta\phi = \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix} \qquad \Longrightarrow \qquad \partial_\varphi\phi \times \partial_\theta\phi = \begin{pmatrix} -\sin^2\theta\cos\varphi \\ -\sin^2\theta\sin\varphi \\ -\sin\theta\cos\theta \end{pmatrix}.$$

Sanity check: does this make sense, and is it exterior or interior? We have

$$\partial_\varphi\phi \times \partial_\theta\phi = -\sin\theta\ \phi(\varphi,\theta) = -\sin\theta\,(x,y,z),$$

and the exterior normal of $\mathbb{S}^2$ is $+(x,y,z)$, so $\partial_\varphi\phi\times\partial_\theta\phi$ is an **interior** normal (for $\theta\in(0,\pi)$, $\sin\theta>0$). So we multiply by -1 to switch the order in the cross product, matching the exterior sign in Proposition 23.b.2.

The integral.

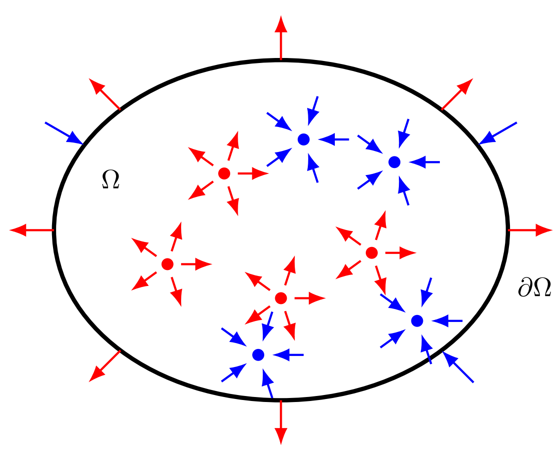
$$\begin{aligned}\int_{\mathbb{S}^2}\vec F\cdot d\vec A &= \int_0^{2\pi}\int_0^\pi \begin{pmatrix} xz\\ z\\ y\end{pmatrix}\cdot\begin{pmatrix}\sin\theta\cos\varphi\\ \sin\theta\sin\varphi\\ \cos\theta\end{pmatrix}\sin\theta\,d\theta\,d\varphi\\ &= \int_0^{2\pi}\int_0^\pi\left[\sin^3\theta\cos\theta\cos^2\varphi+\sin^2\theta\cos\theta\sin\varphi+\sin^2\theta\cos\theta\sin\varphi\right]d\theta\,d\varphi\\ &= \pi\int_0^\pi\sin^3\theta\cos\theta\,d\theta = \pi\left[\frac{\sin^4\theta}{4}\right]_0^\pi = 0,\end{aligned}$$

using $\int_0^{2\pi}\cos^2\varphi\,d\varphi=\pi$ and $\int_0^{2\pi}\sin\varphi\,d\varphi=0$ for the middle two terms.

Theorem 23.b.4 (*Gauss*): Let $\Omega\subseteq\mathbb{R}^n$ be a bounded C^1 domain and $F:\overline{\Omega}\to\mathbb{R}^n$ a C^1 vector field. Then

$$\int_\Omega \operatorname{div}F\,d\operatorname{vol}_n = \int_{\partial\Omega}\langle F,\nu\rangle\,d\operatorname{vol}_{n-1},\qquad\text{or, in physics notation,}\qquad \int_\Omega\vec\nabla\cdot\vec F\,dx=\int_{\partial\Omega}\vec F\cdot d\vec A.$$

AI-Note 23.2: Corsin dates the theorem to Lagrange (discovered 1762) and Ostrogradsky (proven 1828), and adds, next to the physics form, an unusually enthusiastic *“how cool is this!”* The remark is earned: a volume integral of a derivative is reduced to a boundary integral of the original field, which is the shape of the fundamental theorem of calculus.



Example 23.b.5 (*Recomputing Example 23.b.3 via the divergence theorem*): With $\Omega=B_1(0)$ and $F(x,y,z)=(xz,z,y)$ as before, $\operatorname{div}F=\partial_x(xz)+\partial_y(z)+\partial_z(y)=z+0+0=z$. So, in spherical coordinates ($z=r\cos\theta$),

$$\int_{\partial\Omega}\vec F\cdot d\vec A=\int_\Omega\operatorname{div}F\,dx=\int_\Omega z\,dx=\int_0^1\int_0^{2\pi}\int_0^\pi\underbrace{\cos\theta\sin\theta}_{=\frac12\sin(2\theta),\ \pi\text{-periodic}}r^3\,d\theta\,d\varphi\,dr=0,$$

since $\int_0^\pi\sin(2\theta)\,d\theta=0$. This confirms Example 23.b.3 with none of its cross-product bookkeeping.

AI-Note 23.3: Corsin writes the divergence as $\operatorname{div}F=z$ directly without the intermediate $\partial_x(xz)+\partial_z(z)+\partial_y(y)$ step; spelled out above for clarity. Worth noting how much the two computations of the same flux differ in character: the direct definition needs an explicit Gauss map, an orientation sanity check, and three cross terms in the integrand, while the divergence theorem needs only $\operatorname{div} F$ and a coordinate change already used in Example 21.b.9. This is the theorem's main practical use, and it is not conceptual elegance but the plain fact that $\operatorname{div} F$ is often far simpler than F restricted to a curved boundary.

Example 23.b.6 (*Computing flux using the divergence theorem*)**:** Let $\Omega := [-1,1]^3 \subset \mathbb{R}^3$ and $F(x,y,z) := (x^2, y^2, z^2)^\mathsf{T}$. We compute the flux of F out of Ω using the divergence theorem. First, we find the divergence of F:

$$\operatorname{div} F(x,y,z) = \partial_x(x^2) + \partial_y(y^2) + \partial_z(z^2) = 2x + 2y + 2z.$$

Then, integrating over Ω:

$$\begin{aligned}\int_{\partial\Omega} \langle F(p),\ \nu(p)\rangle \, d\underset{2}{\mathrm{vol}}(p) &= \int_\Omega \operatorname{div} F(x,y,z)\, d\underset{3}{\mathrm{vol}}(x,y,z) \\ &= \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (2x+2y+2z)\, dx\, dy\, dz.\end{aligned}$$

By symmetry, the integral of $2x$ over $[-1,1]$ is 0, and similarly for y and z. Therefore, the total flux is 0.

Example 23.b.7 (*Flux out of a cylinder*)**:** Let $\Omega := \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 \le R^2,\ 0 \le z \le H\}$ be a cylinder of radius R and height H. Consider the vector field $F(x,y,z) := (x^3 + \sin(z), y^3 + xz, z^3 + xy)^\mathsf{T}$. We compute the outward flux of F through $\partial\Omega$ using the divergence theorem. First, we compute the divergence of F:

$$\operatorname{div} F(x,y,z) = 3x^2 + 3y^2 + 3z^2 = 3(x^2+y^2+z^2).$$

By the divergence theorem, the flux is the integral of the divergence over Ω. We compute this using cylindrical coordinates (r,θ,z):

$$\begin{aligned}\int_{\partial\Omega} \langle F(p),\ \nu(p)\rangle \, d\underset{2}{\mathrm{vol}}(p) &= \int_\Omega 3(x^2+y^2+z^2)\, d\underset{3}{\mathrm{vol}}(x,y,z) \\ &= \int_0^H \int_0^{2\pi} \int_0^R 3(r^2+z^2) r\, dr\, d\theta\, dz \\ &= 2\pi \int_0^H \left[\frac{3}{4} r^4 + \frac{3}{2} r^2 z^2\right]_0^R dz \\ &= 2\pi \int_0^H \left(\frac{3}{4} R^4 + \frac{3}{2} R^2 z^2\right) dz \\ &= 2\pi \left[\frac{3}{4} R^4 z + \frac{1}{2} R^2 z^3\right]_0^H \\ &= 2\pi \left(\frac{3}{4} R^4 H + \frac{1}{2} R^2 H^3\right) = \frac{\pi}{2} R^2 H (3R^2 + 2H^2).\end{aligned}$$

Example 23.b.8 (*Archimedes' principle*)**:** A solid $\Omega \subset \mathbb{R}^3$ (a bounded C^1 domain) is fully submerged in water, with the coordinates chosen so that $x_3 = 0$ is the water's surface. At depth $-x_3$ the water pressure is $P(x) := -\rho g x_3$, where ρ is the density of water and g the gravitational acceleration. The force the water exerts on a surface element near $q \in \partial\Omega$ is proportional to the pressure and points inward, along $-\nu(q)$, so the total buoyant force is

$$\vec{F} := -\int_{\partial\Omega} P\, \nu\, dS.$$

Substituting the pressure and applying the divergence theorem to each component,

$$F_i = \int_{\partial\Omega} \rho g x_3 \nu_i \, dS = \rho g \int_{\partial\Omega} \langle x_3 e_i, \nu\rangle \, dS = \rho g \int_\Omega \operatorname{div}(x_3 e_i)\, dV.$$

Since $\operatorname{div}(x_3 e_i) = \partial_i x_3$ equals 1 if $i = 3$ and 0 otherwise, this gives $F_1 = F_2 = 0$ and

$$F_3 = \rho g \int_\Omega 1 \, dV = \rho g \ \mathrm{vol}_3(\Omega).$$

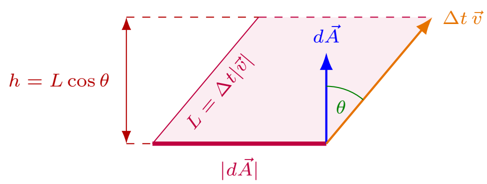
The buoyant force is vertical, with magnitude $\rho g \, \mathrm{vol}_3(\Omega)$, which is exactly the weight of the water displaced by Ω, confirming Archimedes' principle. The divergence theorem converts a surface integral of an unknown, position-dependent pressure into a volume integral that only sees the shape of Ω through its volume.

Gauss' theorem is also a powerful tool in physics. Consider a fluid of density ϱ moving with velocity $\vec{v}$, and a surface element $d\vec{A}$ whose normal makes an angle θ with $\vec{v}$. The fluid crossing that element in time Δt is exactly the fluid filling the column of slant length $L = \Delta t \, |\vec{v}|$ erected on it along $\vec{v}$, and that column stands at perpendicular height $L \cos\theta$. Its volume is therefore

$$V = |d\vec{A}| \, L \cos\theta = \Delta t \, \vec{v} \cdot d\vec{A},$$

and the mass of fluid carried through the element is ϱV.

AI-Note 23.4: Corsin writes this as $V = \Delta t \, \vec{v} \cdot d\vec{A} \cdot \varrho$ and calls the result a volume. The right-hand side is a mass: $\Delta t \, \vec{v} \cdot d\vec{A}$ is already the volume swept, and the density only converts it. The two have been separated above, since the distinction is needed immediately. The integral below is of $\varrho\vec{v}$ and therefore measures mass per unit time, which is what the continuity equation balances against $\partial\varrho/\partial t$.



Therefore, the fluid flowing out of any bounded C^1 domain Ω per unit time is

$$\int_{\partial\Omega} \varrho\vec{v} \cdot d\vec{A} = \int_\Omega \vec{\nabla} \cdot (\varrho\vec{v}) \, dx.$$

Of course, since the total amount of fluid is conserved, this is also the change in the amount of fluid in Ω over time:

$$\int_\Omega \vec{\nabla} \cdot (\varrho\vec{v}) \, dx = -\frac{d}{dt} \int_\Omega \varrho \, dx = -\int_\Omega \frac{\partial\varrho}{\partial t} \, dx.$$

Since this holds for any Ω, the "*continuity equation*" (German: „*Kontinuitätsgleichung*") follows:

$$\vec{\nabla} \cdot (\varrho\vec{v}) + \frac{\partial\varrho}{\partial t} = 0.$$

Exercise 23.b.9 (*11.5 — Radial vector fields* (important)): For $\alpha \in \mathbb{R}$ consider the vector field

$$V_\alpha(x) = |x|^\alpha x, \qquad x \in \mathbb{R}^n \setminus \{0\},$$

where $|x|$ is the usual Euclidean norm.

1. Compute $\operatorname{div} V_\alpha(x)$ for all $x \neq 0$, and show that it vanishes identically for $\alpha = -n$.
2. Using the Divergence theorem on V_0, show the following relation between the n-volume of a sphere and the $n-1$ volume of its boundary:

$$r \ \mathrm{vol}_{n-1}(\partial B_r) = n \ \mathrm{vol}_n(B_r), \qquad \forall r > 0.$$

3. Show that

$$\int_{\partial B_1} V_{-n} \cdot \nu \, d\,\mathrm{vol}_{n-1} > 0, \qquad \int_{B_1} \operatorname{div} V_{-n}(x) \, dx = 0.$$

Why doesn't this contradict the Divergence Theorem?

Solutions (Section 23.c)

Solution of Exercise 23.b.9:

(1) Writing $V_\alpha^i(x) = |x|^\alpha x_i$,

$$\partial_i\big(|x|^\alpha x_i\big) = \alpha|x|^{\alpha-2}x_i^2 + |x|^\alpha \qquad \text{(no sum)},$$

so, summing over $i = 1, \ldots, n$,

$$\operatorname{div} V_\alpha(x) = \sum_{i=1}^{n}\Big(\alpha|x|^{\alpha-2}x_i^2 + |x|^\alpha\Big) = \alpha|x|^{\alpha-2}|x|^2 + n|x|^\alpha = (\alpha+n)|x|^\alpha.$$

Since $|x|^\alpha \neq 0$ for $x \neq 0$, this vanishes identically exactly when $\alpha = -n$.

(2) For $\alpha = 0$, $V_0(x) = x$ and, by part (1), $\operatorname{div} V_0 \equiv n$. Applying Theorem 23.b.4 on B_r,

$$\int_{B_r} \operatorname{div} V_0 \, dx = n \operatorname{vol}_n(B_r).$$

On ∂B_r the outward unit normal is $\nu(x) = x/r$, so $V_0(x)\cdot\nu(x) = x \cdot x/r = |x|^2/r = r$ (using $|x| = r$ on ∂B_r), giving

$$\int_{\partial B_r} V_0 \cdot \nu \, d\operatorname{vol}_{n-1} = r \operatorname{vol}_{n-1}(\partial B_r).$$

Equating the two expressions of $\int_{B_r} \operatorname{div} V_0\, dx = \int_{\partial B_r} V_0 \cdot \nu \, d\operatorname{vol}_{n-1}$ gives $r \operatorname{vol}_{n-1}(\partial B_r) = n \operatorname{vol}_n(B_r)$.

(3) By part (1), $\operatorname{div} V_{-n}(x) = 0$ for every $x \neq 0$, so $\int_{B_1} \operatorname{div} V_{-n}(x)\, dx = 0$ trivially (the integrand vanishes on $B_1 \setminus \{0\}$, a set of full measure in B_1). On ∂B_1, $\nu(x) = x$ and $|x| = 1$, so $V_{-n}(x) = |x|^{-n}x = x$, giving

$$\int_{\partial B_1} V_{-n}\cdot\nu\, d\operatorname{vol}_{n-1} = \int_{\partial B_1} x \cdot x \, d\operatorname{vol}_{n-1} = \int_{\partial B_1} 1 \, d\operatorname{vol}_{n-1} = \operatorname{vol}_{n-1}(\partial B_1) > 0.$$

This does not contradict Theorem 23.b.4: the theorem requires F to be C^1 on all of $\overline{B_1}$, but $V_{-n}(x) = |x|^{-n}x$ is not even defined, let alone continuously differentiable, at $x = 0 \in \overline{B_1}$. Its hypotheses simply do not hold on this domain. **Q.E.D.**

Alternating Forms, the Wedge Product and the Exterior Derivative

(Chapter 24)

Gradient, divergence and curl look like three unrelated operators that happen to satisfy two unrelated identities ($\operatorname{curl}\operatorname{grad} = 0$ and $\operatorname{div}\operatorname{curl} = 0$). This chapter builds the machinery that makes them one operator satisfying one identity. Antisymmetric k-linear forms and the wedge product supply the objects; the exterior derivative d supplies the operator; and $d \circ d = 0$ supplies the identity.

Antisymmetric k-linear forms and the wedge product (Section 24.a)

AI-Note 24.1 (*Motivation*): Suppose one studies a submanifold $\mathcal{M} \subseteq \mathbb{R}^n$ via a parametrization $\phi : D \to \mathcal{M}$, and integrates some quantity over $\mathcal{M}$ (for instance the Gauss curvature, as in the Gauss–Bonnet theorem). The natural question is whether the resulting number is a property of ϕ, or a property of $\mathcal{M}$ itself: if it is invariant under reparametrization, it must be the latter. "*Differential forms*" are built to satisfy exactly this property. For a diffeomorphism $\Psi : V \to D$ with $\det D\Psi > 0$ and a form ω,

$$\begin{aligned}\int_{\mathcal{M}} \omega &= \int_D \omega_{\phi(x)}(D\phi_x)\,dx \\ &\overset{\text{change of var.}}{=} \int_V \omega_{\phi(\Psi(y))}(D\phi_{\Psi(y)})|\det D\Psi_y|\,dy \\ &= \int_V \omega_{(\phi\circ\Psi)(y)}(D(\phi\circ\Psi)_y)\,dy = \int_{\mathcal{M}} \omega,\end{aligned}$$

where the middle equality uses the chain rule $D(\phi\circ\Psi)_y = D\phi_{\Psi(y)}\,D\Psi_y$ together with multilinearity and $\det D\Psi_y = |\det D\Psi_y|$. This calculation is precisely why ω must be built from multilinear, antisymmetric maps: multilinearity lets the Jacobian $D\Psi_y$ factor out as $\det D\Psi_y$, matching the $|\det D\Psi_y|$ produced by the change-of-variables formula, exactly when $\det D\Psi_y > 0$.

Definition 24.a.1 (*Antisymmetric k-linear form*): Let V be an n-dimensional real vector space and let $L : V^k \to \mathbb{R}$. Then L is an "*antisymmetric k-linear form*" (German: „*antisymmetrische k-lineare Form*") if

(a) (k-linear): L is linear in every entry:

$$L(v_1, \ldots, \alpha v_i + \beta w_i, \ldots, v_k) = \alpha L(v_1, \ldots, v_i, \ldots, v_k) + \beta L(v_1, \ldots, w_i, \ldots, v_k);$$

(b) (antisymmetric): swapping any two entries negates the value:

$$L(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = -L(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k).$$

Remark 24.a.2: For $k = n$ and $V = \mathbb{R}^n$, the normalization $L(e_1, \ldots, e_n) = 1$ pins down a **unique** such L, namely $L = \det$.

Notation 24.a.3: For $v \in V$ and a fixed basis $(e_1, \ldots, e_n)$ of V, superscripts index the coordinates of a single vector, $v = \sum_{i=1}^n v^i e_i = (v^1, \ldots, v^n)$, while subscripts index a list of vectors $v_1, \ldots, v_k \in V$. This distinguishes *"the i-th coordinate of v"* from *"the i-th vector in the list"*. From here on, $V = \mathbb{R}^n$ with the standard basis.

AI-Example 24.a.4 (*Two families of examples*):

1. Fix $A \in \mathrm{Mat}_{k\times n}(\mathbb{R})$. For $v_1, \ldots, v_k \in \mathbb{R}^n$, write $V := (v_1 \mid \cdots \mid v_k) \in \mathrm{Mat}_{n\times k}(\mathbb{R})$ for the matrix with these columns; then $AV \in \mathrm{Mat}_{k\times k}(\mathbb{R})$, and $L(v_1, \ldots, v_k) := \det(AV)$ is an antisymmetric k-linear form, since $\det$ is multilinear and antisymmetric in the columns of its argument, and $v \mapsto Av$ is linear.

2. For $k = 1$, $L : \mathbb{R}^n \to \mathbb{R}$ is a "*covector*", $L \in (\mathbb{R}^n)^*$. The standard basis $(e_1^*, \dots, e_n^*)$ of $(\mathbb{R}^n)^*$, satisfying $e_i^*(e_j) = \delta_{ij}$, consists of 1-forms; in particular $e_i^*(v) = v^i$.

Remark 24.a.5 (*An equivalent characterization*): In the lecture, antisymmetry was replaced by the equivalent condition

$$L(VA) = L(V)\det(A), \qquad V := (v_1, \dots, v_k) \in \mathrm{Mat}_{n\times k}(\mathbb{R}), \quad A \in \mathrm{Mat}_{k\times k}(\mathbb{R}),$$

writing $L(V)$ for $L(v_1, \dots, v_k)$. We verify the equivalence in the special case $n = 3$, $k = 2$: let $V = (v \mid w) \in \mathrm{Mat}_{3\times 2}(\mathbb{R})$ and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, so that $VA = (av + cw \mid bv + dw)$. Rewriting $L(VA)$ in bilinear form and using $L(v,v) = -L(v,v) = 0$ (from antisymmetry) together with $L(w,v) = -L(v,w)$,

$$\begin{aligned} L(VA) &= L(av + cw,\, bv + dw) \\ &= ab\,L(v,v) + ad\,L(v,w) + cb\,L(w,v) + cd\,L(w,w) \\ &= (ad - bc)\,L(v,w) = L(V)\det(A). \end{aligned}$$

Definition 24.a.6 (*Wedge product*): For a k-form α and an m-form β on $\mathbb{R}^n$, the "*wedge product*" (German: „*Dachprodukt*") $\alpha \wedge \beta$ is the $(k+m)$-form

$$(\alpha \wedge \beta)(v_1, \dots, v_k, v_{k+1}, \dots, v_{k+m}) = \sum_{\sigma \in S_{k,m}} \mathrm{sgn}(\sigma)\, \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)})\, \beta(v_{\sigma(k+1)}, \dots, v_{\sigma(k+m)}),$$

where $S_{k,m}$ is the set of permutations σ of $\{1, \dots, k+m\}$ satisfying $\sigma(1) < \cdots < \sigma(k)$ and $\sigma(k+1) < \cdots < \sigma(k+m)$ (the "*(k,m)-shuffles*").

AI-Note 24.2: Corsin calls this formula *"not terribly important for our purposes"* and instead has us work directly with the five properties below.

The following properties of $\wedge$ are the ones actually used:

(1) $\wedge$ is bilinear.

(2) For α a k-form and β an m-form, $\alpha \wedge \beta$ is a $(k+m)$-form.

(3) $(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma) = \alpha \wedge \beta \wedge \gamma$ (associativity).

(4) For 1-forms $L_1, \dots, L_k$ and vectors $v_1, \dots, v_k \in \mathbb{R}^n$,

$$L_1 \wedge L_2 \wedge \cdots \wedge L_k(v_1, \dots, v_k) = \det\big(L_i(v_j)\big)_{1 \le i,j \le k}.$$

(5) For $\alpha(T) := \det(AT)$, $\beta(S) := \det(BS)$ as in Definition 24.a.1 example 1,

$$\alpha \wedge \beta(X) = \det\left(\begin{bmatrix} A \\ B \end{bmatrix} X\right).$$

Example 24.a.7: By property (4), for $1 \le i_1 < \cdots < i_k \le n$,

$$e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*(v_1, \dots, v_k) = \det\left(\begin{pmatrix} v_1^{i_1} & \cdots & v_k^{i_1} \\ \vdots & \ddots & \vdots \\ v_1^{i_k} & \cdots & v_k^{i_k} \end{pmatrix}\right) = \det\left(\begin{bmatrix} e_{i_1}^\top \\ \vdots \\ e_{i_k}^\top \end{bmatrix} \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}\right).$$

Proposition 24.a.8 (*Every k-linear form is a combination of wedge products of covectors*): Every antisymmetric k-linear form L on $\mathbb{R}^n$ can be written as

$$L = \sum_{1 \le i_1 < \cdots < i_k \le n} \ell_{i_1 \cdots i_k}\, e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$$

for uniquely determined scalars $\ell_{i_1 \cdots i_k} \in \mathbb{R}$.

Proof:

Expand each argument $v_i = \sum_{m=1}^n v_i^m e_m$ and use k-linearity:

$$L(v_1, \dots, v_k) = L\left(\sum_{i_1=1}^n v_1^{i_1} e_{i_1}, \dots, \sum_{i_k=1}^n v_k^{i_k} e_{i_k}\right) = \sum_{1 \le i_1, \dots, i_k \le n} L(e_{i_1}, \dots, e_{i_k})\, v_1^{i_1} \cdots v_k^{i_k}.$$

By antisymmetry, $L(e_{i_1}, \dots, e_{i_k}) = 0$ whenever two indices coincide, so the sum may be restricted to $i_1, \dots, i_k$ pairwise distinct; regrouping each such tuple by the permutation σ that sorts it into increasing order $i_{\sigma(1)} < \cdots < i_{\sigma(k)}$,

$$\begin{aligned} L(v_1, \dots, v_k) &= \sum_{1 \le i_1 < \cdots < i_k \le n} L(e_{i_1}, \dots, e_{i_k}) \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\, v_1^{i_{\sigma(1)}} \cdots v_k^{i_{\sigma(k)}} \\ &= \sum_{1 \le i_1 < \cdots < i_k \le n} \ell_{i_1 \cdots i_k}\, e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*(v_1, \dots, v_k), \end{aligned}$$

with $\ell_{i_1 \cdots i_k} := L(e_{i_1}, \dots, e_{i_k})$, using that the inner sum over σ is exactly the determinant expansion of $e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*(v_1, \dots, v_k)$ from property (4) above. Q.E.D.

Differential forms and the exterior derivative (Section 24.b)

Definition 24.b.9 (*Differential form*): A "*differential p-form*" ω (German: „*Differentialform*") is defined on a domain $D \subseteq \mathbb{R}^n$ if, for every $x \in D$, $\omega(x) = \omega_x : (\mathbb{R}^n)^p \to \mathbb{R}$ is an antisymmetric p-linear map. Since $(e_1(x), \dots, e_n(x))$ is a basis of $\mathbb{R}^n$ for every $x \in D$, Proposition 24.a.8 lets us write

$$\omega(x) = \sum_{1 \le i_1 < \cdots < i_p \le n} \omega_{i_1 \cdots i_p}(x)\, e_{i_1}^*(x) \wedge \cdots \wedge e_{i_p}^*(x)$$

for functions $\omega_{i_1 \cdots i_p} : \mathbb{R}^n \to \mathbb{R}$.

Notation 24.b.10: We abbreviate $e_{i_1}^* \wedge \cdots \wedge e_{i_p}^* =: e_I^*$ for $I = (i_1, \dots, i_p)$, and write $\Omega^p(U)$ for the set of p-forms on $U \subseteq \mathbb{R}^n$. By convention, $\Omega^0(U) = C^\infty(U)$.

Example 24.b.11: Let $U \subseteq \mathbb{R}^n$.

1. Any $f \in C^\infty(U)$ is a 0-form.
2. A vector field $F : U \to \mathbb{R}^n$ has an associated 1-form $\omega_F(x) : v \mapsto \langle F(x), v \rangle$, that is,
$$\omega_F(x)(v) = \sum_{i=1}^n F^i(x)\, v^i = \sum_{i=1}^n F^i(x)\, e_i^*(x)(v) \qquad \Longrightarrow \qquad \omega_F = \sum_{i=1}^n F^i\, e_i^*.$$
3. $F : U \to \mathbb{R}^n$ also naturally induces an $(n-1)$-form
$$\omega_F^{n-1}(x)(v_1, \dots, v_{n-1}) := \det\big(F(x) \mid v_1 \mid \cdots \mid v_{n-1}\big).$$
Developing this determinant along the first column,
$$\begin{aligned} \omega_F^{n-1}(x)(v_1, \dots, v_{n-1}) &= \sum_{i=1}^n (-1)^{i-1} F^i(x) \det \begin{pmatrix} v_1^1 & \cdots & v_{n-1}^1 \\ \vdots & & \vdots \\ v_1^{i-1} & \cdots & v_{n-1}^{i-1} \\ v_1^{i+1} & \cdots & v_{n-1}^{i+1} \\ \vdots & & \vdots \\ v_1^n & \cdots & v_{n-1}^n \end{pmatrix} \\ &= \sum_{i=1}^n (-1)^{i-1} F^i(x)\, e_1^* \wedge \cdots \wedge \widehat{e_i^*} \wedge \cdots \wedge e_n^*(v_1, \dots, v_{n-1}), \end{aligned}$$
where $\widehat{e_i^*}$ denotes an omitted factor, so that $\omega_F^{n-1} = \sum_{i=1}^n (-1)^{i-1} F^i\, e_1^* \wedge \cdots \wedge \widehat{e_i^*} \wedge \cdots \wedge e_n^*$.

4. In particular, for $F : \mathbb{R}^3 \to \mathbb{R}^3$,
$$\omega_F^2(x)(v_1, v_2) = \langle F(x), v_1 \times v_2 \rangle = \det\big(F(x) \mid v_1 \mid v_2\big).$$

AI-Note 24.3: Item 4 identifies ω_F^2 with the flux integrand from Section 23.b: the flux $\int_S \langle F, \nu \rangle \, d\,\mathrm{vol}_2$ over a parametrized surface is exactly the integral, in the sense of the motivating change-of-variables calculation above, of the 2-form ω_F^2.

Theorem 24.b.12 (*Exterior derivative*): There exists a unique linear operator $d : \Omega^k(U) \to \Omega^{k+1}(U)$ satisfying

(1) for $f \in C^\infty(U) = \Omega^0(U)$: $df = Df$, the ordinary differential;
(2) for $\omega \in \Omega^k(U)$, $\theta \in \Omega^s(U)$: $d(\omega \wedge \theta) = d\omega \wedge \theta + (-1)^k \, \omega \wedge d\theta$;
(3) $d \circ d\,\omega = d^2 \omega = 0$.

AI-Note 24.4: Corsin states this as a fact seen in the lecture, without proof, and moves directly to its consequences. The proof can be found in the official script (Analysis_II_Script_v1.pdf, §14.3, pp. 136–138), where the operator is constructed explicitly in coordinates and properties (1)–(3) are verified.

Denote by dx^i the differential of $x \mapsto x^i$ in a moving frame $(e_1(x), \dots, e_n(x))$, that is, $dx^i = e_i^*(x)$. Every p-form ω on $U \subseteq \mathbb{R}^n$ then takes the shape
$$\omega = \sum_{1 \le i_1 < \cdots < i_p \le n} \omega_{i_1 \cdots i_p} \, dx^{i_1} \wedge \cdots \wedge dx^{i_p} = \sum_I \omega_I \, dx^I, \qquad \omega_I \in C^\infty(U),$$
with exterior derivative
$$d\omega = \sum_I d\omega_I \wedge dx^I.$$

Example 24.b.13 (*The exterior derivative recovers the gradient and the curl*):

1. For $f \in C^\infty(\mathbb{R}^n) = \Omega^0(\mathbb{R}^n)$,
$$df_x(v) = \left(\frac{\partial f}{\partial x^1} \;\middle|\; \cdots \;\middle|\; \frac{\partial f}{\partial x^n} \right) \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix} = \frac{\partial f}{\partial x^1} \, dx^1(v) + \cdots + \frac{\partial f}{\partial x^n} \, dx^n(v),$$
so $df = \frac{\partial f}{\partial x^1} \, dx^1 + \cdots + \frac{\partial f}{\partial x^n} \, dx^n = \langle \nabla f, \cdot \rangle$: the exterior derivative of a 0-form is exactly the 1-form associated (as in the example above) to the gradient vector field ∇f. This is the shape that makes Stokes' theorem specialize to the FTC, as in Section 23.a.

2. Define a 1-form on $U \subseteq \mathbb{R}^3$ by $\omega(x, y, z) = \langle F(x, y, z), \cdot \rangle = F^1 \, dx + F^2 \, dy + F^3 \, dz$ for a vector field F. By property (2) of Theorem 24.b.12 and $d(dx^i) = 0$ (since $dx^i = d(x^i)$ and $d^2 = 0$),
$$d\omega = dF^1 \wedge dx + dF^2 \wedge dy + dF^3 \wedge dz,$$
and expanding each $dF^i = \partial_x F^i \, dx + \partial_y F^i \, dy + \partial_z F^i \, dz$, only the cross terms survive the wedge (since $dx^i \wedge dx^i = 0$ by antisymmetry):
$$d\omega = \Big(\frac{\partial F^3}{\partial y} - \frac{\partial F^2}{\partial z} \Big) dy \wedge dz + \Big(\frac{\partial F^1}{\partial z} - \frac{\partial F^3}{\partial x} \Big) dz \wedge dx + \Big(\frac{\partial F^2}{\partial x} - \frac{\partial F^1}{\partial y} \Big) dx \wedge dy.$$
Comparing with example item 4 in Section 24.b, this is exactly $\omega^2_{\nabla \times F} = \langle \nabla \times F, \cdot \times \cdot \rangle = \det(\nabla \times F \mid \cdot \mid \cdot)$: the exterior derivative of the 1-form associated to F is the 2-form associated to $\operatorname{curl} F$.

3. Recall the $(n-1)$-form $\omega_F^{n-1} = \sum_{i=1}^n (-1)^{i-1} F^i \, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$ from Section 24.b (example item 3), with $\widehat{dx^i}$ denoting an omitted factor. In $d(F^i \, dx^1 \wedge \cdots \widehat{dx^i} \cdots \wedge dx^n) = dF^i \wedge dx^1 \wedge \cdots \widehat{dx^i} \cdots \wedge dx^n$, only the $\partial_i F^i \, dx^i$ summand of $dF^i = \sum_j \partial_j F^i \, dx^j$ survives,

since every $j \neq i$ repeats a factor already present and so kills the wedge by antisymmetry. Therefore

$$d\omega_F^{n-1} = \sum_{i=1}^{n} (-1)^{i-1} \partial_i F^i \; dx^i \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n.$$

Moving dx^i past the $i-1$ factors $dx^1, \ldots, dx^{i-1}$ ahead of it, into its natural position, costs another sign of $(-1)^{i-1}$, cancelling the one already there:

$$dx^i \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n = (-1)^{i-1}\, dx^1 \wedge \cdots \wedge dx^n \quad \Longrightarrow \quad d\omega_F^{n-1} = \sum_{i=1}^{n} \partial_i F^i \; dx^1 \wedge \cdots \wedge dx^n = (\operatorname{div} F)\, dx^1 \wedge \cdots \wedge dx$$

So, the exterior derivative of ω_F^{n-1} is $\operatorname{div} F$ times the volume form. That completes the pattern of items 1 and 2 above: grad, curl and div are exactly the 0-, 1- and $(n-1)$-form faces of the single operator d.

4. Let $\omega = x_3\, dx_3 + x_1 x_2\, dx_2$ be a 1-form on $\mathbb{R}^3$. Let us compute its exterior derivative using the rules:

$$\begin{aligned} d\omega &= d(x_3) \wedge dx_3 + d(x_1 x_2) \wedge dx_2 \\ &= dx_3 \wedge dx_3 + (x_2\, dx_1 + x_1\, dx_2) \wedge dx_2 \\ &= 0 + x_2\, dx_1 \wedge dx_2 + x_1 \underbrace{dx_2 \wedge dx_2}_{=0} \\ &= x_2\, dx_1 \wedge dx_2. \end{aligned}$$

AI-Note 24.5: Corsin writes the middle expansion of $d\omega$ out in full, all nine terms, one for each pair (dF^i, dx^j), before collecting the six that survive into the three coefficient pairs above. It is condensed here to the surviving terms directly. Nothing is lost, since $dx \wedge dx = dy \wedge dy = dz \wedge dz = 0$ kills exactly the three diagonal terms, and antisymmetry ($dy \wedge dx = -dx \wedge dy$, and so on) pairs up the rest.

Pullbacks, Integration of Forms and Orientability (Chapter 25)

To integrate a differential form over a submanifold one has to pull it back to a parameter domain, and then check that the answer does not depend on which parametrization was chosen. It does not, provided the parametrizations agree on a direction. That proviso is orientability, and it is what this chapter is really about. The last section fixes the induced orientation on a boundary, which is the convention that makes the sign in Stokes' theorem come out right.

Pullback of forms (Section 25.a)

Definition 25.a.1 (*Pullback of a form*): Let $U \subseteq \mathbb{R}^m$, $V \subseteq \mathbb{R}^n$ be open, $F : U \to V$ a smooth map, and $\omega \in \Omega^p(V)$ a differential p-form on V. The "*pullback*" (German: „*Zurückziehung*") of ω under F is the differential p-form on U defined by

$$(F^*\omega)_x(v_1, \dots, v_p) := \omega_{F(x)}\big(DF_x(v_1), \dots, DF_x(v_p)\big).$$

AI-Note 25.1: Corsin also draws a commutative diagram $TU \xrightarrow{dF} TV \xrightarrow{\omega} \mathbb{R}$ over $U \xrightarrow{F} V$, with tangent-bundle projections π_U, π_V, and labels it *"not exam relevant"*. Omitted here for that reason; the formula above is the operative content.

The pullback satisfies two properties that make it the right tool for transporting forms along smooth maps:

(1) $F^*(\omega \wedge \theta) = F^*\omega \wedge F^*\theta$;
(2) $F^*(d\omega) = d(F^*\omega)$ (the pullback commutes with the exterior derivative).

AI-Note 25.2: This is precisely the operation used, informally, in the reparametrization-invariance motivation for Definition 24.a.1: writing $\phi^*\omega$ and $(\phi \circ \Psi)^*\omega$ for the two sides of that calculation makes it read as $\int_D \phi^*\omega = \int_V (\phi \circ \Psi)^*\omega = \int_V \Psi^*(\phi^*\omega)$, i.e. property (1) applied to $\phi^*\omega$ and the fact that pullback along a composition is the composition of pullbacks.

Example 25.a.2 (*Computing a pullback*): Let $\omega = x_3\, dx_3 + x_1 x_2\, dx_2$ be a 1-form on $\mathbb{R}^3$. Consider the map $F : \mathbb{R}^2 \to \mathbb{R}^3$ given by

$$F(u, v) = \begin{pmatrix} u + v \\ e^u \\ u^2 - v \end{pmatrix}.$$

We want to compute the pullback $F^*\omega$, which will be a 1-form on $\mathbb{R}^2$. Using the properties of the pullback, we have:

$$\begin{aligned} F^*\omega &= F^*(x_3\, dx_3 + x_1 x_2\, dx_2) \\ &= F^*(x_3) F^*(dx_3) + F^*(x_1 x_2) F^*(dx_2) \\ &= (x_3 \circ F) d(F^*(x_3)) + (x_1 x_2 \circ F) d(F^*(x_2)) \\ &= (u^2 - v) d(u^2 - v) + (u + v) e^u d(e^u) \\ &= (u^2 - v)(2u\, du - dv) + (u + v) e^u (e^u\, du) \\ &= \big(2u(u^2 - v) + (u + v) e^{2u}\big)\, du - (u^2 - v)\, dv. \end{aligned}$$

Integration of differential forms on submanifolds (Section 25.b)

If $\mathcal{M}$ is an m-dimensional submanifold of $\mathbb{R}^m$ (i.e. simply an open set) and $\omega \in \Omega^m(\mathcal{M})$ is an m-form, then

$$\int_{\mathcal{M}} \omega := \int_{\mathcal{M}} \omega_x(e_1, \ldots, e_m)\, dx;$$

in particular, for $f \in C^\infty(\mathcal{M})$,

$$\int_{\mathcal{M}} f\, dx^1 \wedge \cdots \wedge dx^m = \int_{\mathcal{M}} f(x)\, dx,$$

recovering the ordinary Lebesgue integral of f.

Definition 25.b.3 (*Integral of a form over a parametrized submanifold*): If $L^k \subseteq \mathbb{R}^n$ is a k-dimensional submanifold, $n > k$, and $\phi : U \to L$ is a parametrization with $U \subseteq \mathbb{R}^k$ open, then the integral of $\omega \in \Omega^k(L)$ is defined as

$$\int_L \omega := \int_U \phi^* \omega = \int_U \omega_{\phi(x)}\left(\frac{\partial \phi}{\partial x^1}, \ldots, \frac{\partial \phi}{\partial x^k}\right) dx.$$

In particular, for $f \in C^\infty(L)$ and $1 \le i_1 < \cdots < i_k \le n$,

$$\int_L f\, dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \int_U f(x) \det\left(\frac{\partial \phi^{i_s}}{\partial x^t}\right)_{1 \le s,t \le k} dx,$$

because, by property (4) of Definition 24.a.6,

$$dx^{i_1} \wedge \cdots \wedge dx^{i_k}(\partial_1 \phi, \ldots, \partial_k \phi) = \det\begin{pmatrix} dx^{i_1}(\partial_1\phi) & \cdots & dx^{i_1}(\partial_k\phi) \\ \vdots & \ddots & \vdots \\ dx^{i_k}(\partial_1\phi) & \cdots & dx^{i_k}(\partial_k\phi) \end{pmatrix} = \det\left(\frac{\partial \phi^{i_s}}{\partial x^t}\right)_{1 \le s,t \le k}.$$

AI-Example 25.b.4 (*Evaluating a 2-form on a concrete pair of vectors*): Everything above is stated symbolically, and the mechanics are easy to lose track of. Here is the whole computation on numbers. It is worth doing once, because every integral in this chapter is an integral of exactly this evaluation.

The recipe implicit in the displays above is: **a wedge $dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ evaluated on k vectors selects rows $i_1, \ldots, i_k$ from the matrix whose columns are those vectors, and takes the determinant.** The coefficient function is then evaluated at the point and multiplied on.

Take $\omega := 3\cos(x^1)\, dx^1 \wedge dx^3 \in \Omega^2(\mathbb{R}^3)$, the point $p := (0, 1, 1)$, and the two vectors

$$v_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad v_2 := \begin{pmatrix} 1 \\ 1 \\ \frac{1}{2} \end{pmatrix}, \qquad \text{so} \qquad \begin{pmatrix} v_1 & v_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

The index set is $I = (1, 3)$, so we keep rows 1 and 3 and discard row 2:

$$dx^1 \wedge dx^3(v_1, v_2) = \det\begin{pmatrix} dx^1(v_1) & dx^1(v_2) \\ dx^3(v_1) & dx^3(v_2) \end{pmatrix} = \det\begin{pmatrix} 1 & 1 \\ 0 & \frac{1}{2} \end{pmatrix} = 1 \cdot \tfrac{1}{2} - 1 \cdot 0 = \tfrac{1}{2}.$$

The coefficient at p is $3\cos(0) = 3$, and therefore

$$\omega_p(v_1, v_2) = 3 \cdot \tfrac{1}{2} = \tfrac{3}{2}.$$

Two things are worth noticing. The second components of v_1 and v_2 never entered the computation, since a form built from dx^1 and dx^3 is blind to motion in the x^2 direction, which is what makes it the natural thing to integrate over a surface in the x^1x^3-plane. And the answer is **signed**: swapping v_1 and v_2 swaps the determinant's columns and returns $-\frac{3}{2}$. That sign is exactly the orientation data of Section 25.c, and it is why $\int_L \omega$ needs an oriented L to be well defined.

AI-Note 25.3: Applying Definition 25.b.3 to ω_F^{n-1} or ω_F^2 from Chapter 24 recovers exactly the flux integrals of Chapter 23. Those were, all along, integrals of forms in this sense. And Definition 25.a.1 is what makes the motivating reparametrization-invariance calculation precise: $\int_L \omega$ does not depend on the parametrization ϕ chosen, only on L and its orientation.

AI-Note 25.4 (*Integrating over non-graphical submanifolds via partitions of unity*): Lukas Krause notes that when a submanifold $\mathcal{M} \subseteq \mathbb{R}^n$ cannot be covered by a single local parametrization chart (for instance, a sphere S^2 or a compact manifold with boundary), one cannot directly evaluate $\int_{\mathcal{M}} \omega$ using a single coordinate domain U.

The solution is to decompose any continuous function f on $\mathcal{M}$ into a sum of functions $f = \sum_{l=1}^N \eta_l f$, each supported inside a single coordinate chart domain U_l. Integrating each term $\eta_l f$ separately and summing the results yields the total integral over $\mathcal{M}$. The key technical tool for this decomposition is a "*smooth partition of unity*" (German: „*glatte Zerlegung der Eins*").

Definition 25.b.5 (*Support of a function*): Let $U \subseteq \mathbb{R}^n$ be open and $f : U \to \mathbb{R}$ a continuous function. The "*support*" (German: „*Träger*") of f is the closure of the set where f is non-zero:

$$\operatorname{supp}(f) := \overline{\{x \in U \mid f(x) \neq 0\}}.$$

If $\operatorname{supp}(f)$ is a bounded set (and $\operatorname{supp}(f) \subset U$), we say that f is "*compactly supported*" (German: „*kompakt getragen*").

Example 25.b.6 (*Supports of common functions*):

(a) For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sin(x)$, the function is non-zero on $\mathbb{R} \setminus \pi\mathbb{Z}$. The closure of this set is all of $\mathbb{R}$, so $\operatorname{supp}(f) = \mathbb{R}$. This function is not compactly supported.

(b) Define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) = 0$ for $x \le 0$ and $g(x) = e^{-1/x}$ for $x > 0$. The set where $g(x) \neq 0$ is $(0, \infty)$. Its closure is $[0, \infty)$. Thus $\operatorname{supp}(g) = [0, \infty)$. This is also not compactly supported (since it is not bounded).

Definition 25.b.7 (*Standard smooth bump function*): The canonical C^∞ "*bump function*" (German: „*Höckerfunktion*") $\psi : \mathbb{R}^n \to \mathbb{R}$ supported on the unit ball $B_1(0)$ is defined by

$$\psi(x) := \begin{cases} \exp\left(-\dfrac{1}{1 - \|x\|^2}\right) & \text{if } \|x\| < 1, \\ 0 & \text{if } \|x\| \ge 1. \end{cases}$$

Since all higher derivatives of $t \mapsto \exp(-1/(1-t))$ approach 0 faster than any polynomial as $t \to 1^-$, $\psi \in C^\infty(\mathbb{R}^n)$ with $\operatorname{supp}(\psi) = \overline{B_1(0)}$.

Proposition 25.b.8 (*Existence of a smooth partition of unity subordinate to a finite cover*): Let $K \subseteq \mathbb{R}^n$ be compact, and let $\{B_{r_l}(p_l)\}_{l=1}^N$ be a finite open cover of K. Then there exist an open set $U \supset K$ and $N+1$ non-negative smooth functions $\eta_1, \dots, \eta_N, \tilde{\eta} \in C^\infty(\mathbb{R}^n, [0,1])$ such that:

(a) $\tilde{\eta}(x) + \sum_{l=1}^N \eta_l(x) = 1$ for all $x \in \mathbb{R}^n$;

(b) $\operatorname{supp}(\eta_l) \subseteq B_{r_l}(p_l)$ for each $l \in \{1, \dots, N\}$;

(c) $\operatorname{supp}(\tilde{\eta}) \subseteq \mathbb{R}^n \setminus U$.

In particular, on the neighborhood U of K, $\tilde{\eta} \equiv 0$ so that $\sum_{l=1}^N \eta_l(x) = 1$.

Exercise 25.b.9 (*12.9 — Change of variables with differential forms* (important)): Let $U, V \subseteq \mathbb{R}^n$ be open sets, and let $\psi = (\psi^1, \dots, \psi^n) : U \to V$ be a diffeomorphism with $\det\left(\frac{\partial \psi^j}{\partial x^i}\right) > 0$. Define an n-differential form ω on U by $\omega = d\psi^1 \wedge d\psi^2 \wedge \cdots \wedge d\psi^n$.

(a) Show that $\omega = \det\left(\frac{\partial \psi^j}{\partial x^i}\right) dx^1 \wedge \cdots \wedge dx^n$.

(b) From the relation in (a), rewrite the change of variables formula using differential forms.

Orientable manifolds (Section 25.c)

AI-Note 25.5: The two definitions below are reproduced by Corsin directly from the lecture script (numbered there as Definitions 14.68 and 14.65), rather than written out in his own hand; he adds the margin annotations quoted and figured below.

Definition 25.c.10 (*Support of a k-form*): Let $U \subseteq \mathbb{R}^n$ be open and $\alpha \in \Omega^k(U)$. The "*support*" (German: „*Träger*“) of α is the closed set

$$\operatorname{supp} \alpha := \overline{\{x \in U : \alpha_x \neq 0\}} \subseteq \overline{U}.$$

We say α has "*compact support*" in U if $\operatorname{supp} \alpha$ is bounded and contained in U.

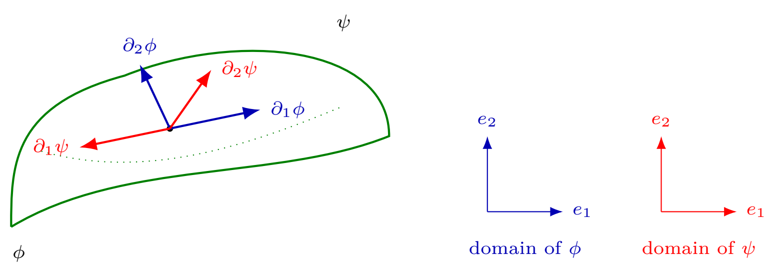
Definition 25.c.11 (*Orientable manifold*): Let $M \subseteq \mathbb{R}^n$ be a k-dimensional C^∞ submanifold. We say M is "*orientable*" (German: „*orientierbar*“) if there exists a family of local parametrizations $\phi_i : V_i \to M \cap U_i$, $i \in I$, whose images cover M, and such that for every $i, j \in I$ with $M \cap U_i \cap U_j \neq \emptyset$, the transition map $\widetilde{\phi}_i^{-1} \circ \widetilde{\phi}_j : \widetilde{V}_j \to \widetilde{V}_i$ has positive Jacobian determinant:

$$\det D(\widetilde{\phi}_i^{-1} \circ \widetilde{\phi}_j) > 0 \qquad \text{on } \widetilde{V}_j.$$

Such a family $\{\phi_i\}$ is called an "*atlas*" of M; if it satisfies the determinant condition above it is called an "*orientation*" of M. Once such a family is chosen, M is said to be "*oriented*".

AI-Note 25.6: Corsin marks the family $\{\phi_i\}$ itself with the margin note *"is called an Atlas"* next to the covering condition, and the positivity condition with *"Atlas"* again next to the determinant inequality, flagging that *"atlas"* names the family regardless of whether the determinant condition holds, while *"orientation"* is the stronger notion once it does. This matches the definition's own wording above.

Corsin illustrates the failure of the orientation condition with a hand-drawn counterexample: two overlapping local parametrizations ϕ, ψ of the same patch of a surface, whose coordinate frames $(\partial_1\phi, \partial_2\phi)$ and $(\partial_1\psi, \partial_2\psi)$ at a shared point disagree in handedness.



AI-Note 25.7: At the shared point, $(\partial_1\phi, \partial_2\phi)$ turns counterclockwise while $(\partial_1\psi, \partial_2\psi)$ turns clockwise, so the two frames are mirror images of each other. Corsin's caption: *"Here,* $\det(D(\phi^{-1} \circ \psi)) < 0$*!"* This is exactly the atlas $\{\phi, \psi\}$ failing Definition 25.c.11: it covers the patch, but its transition map has **negative** Jacobian determinant there, so $\{\phi, \psi\}$ is an atlas but not an orientation.

AI-Note 25.8: Exercise 26.e.31 and Exercise 12.6 (the latter not reproduced below, see the priority table) invoke "*Stokes' theorem*" in its classical $\mathbb{R}^3$ form, which none of Corsin's transcribed pages state explicitly. Recorded here for self-containedness, since the exercise-sheet solutions below rely on it.

Theorem 25.c.12 (*Stokes' theorem, classical form in $\mathbb{R}^3$*): Let $S \subseteq \mathbb{R}^3$ be a compact, oriented C^1 surface with boundary ∂S carrying the induced orientation, and let F be a C^1 vector field on a neighborhood of S. Then

$$\int_S \operatorname{curl} F \cdot d\vec{n} = \int_{\partial S} F \cdot d\vec{s}.$$

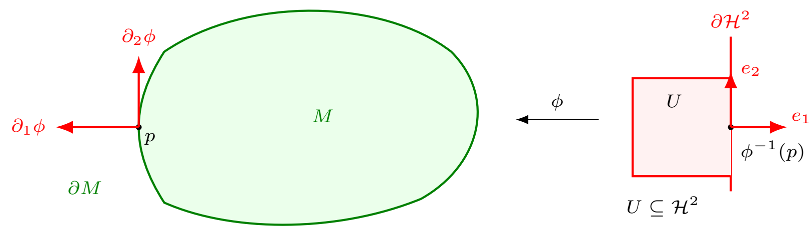
Submanifolds with boundary and induced orientation (Section 25.d)

AI-Note 25.9: Corsin recaps Theorem 17.a.9 verbatim (for every $p \in M$, an open neighbourhood V of p, an open set $U \subseteq \mathbb{R}^k$, and $\phi : U \to V \cap M$ that is smooth, has $D\phi_x$ injective for all $x \in U$, and is a homeomorphism onto $V \cap M$) before extending it to submanifolds with boundary below. It is not repeated here; see Section 17.a.

Definition 25.d.13 (*Submanifold with boundary*): For a k-dimensional submanifold **with boundary**, we require similarly that for a local parametrization $\phi : U \to V \cap M$ as in Theorem 17.a.9, either $U \subseteq \mathbb{R}^k$ is open, or

$$U = U' \cap \mathcal{H}^k, \qquad U' \subseteq \mathbb{R}^k \text{ open}, \qquad \mathcal{H}^k := \{(x_1, \dots, x_k) \in \mathbb{R}^k : x_1 \le 0\}$$

the (closed) "*left half-space*". If $p \in \partial M$ and $\phi : U \subseteq \mathcal{H}^k \to V \cap M$ is a local parametrization of M around p, then $\phi|_{\partial \mathcal{H}^k}$ (identifying $\partial \mathcal{H}^k = \{0\} \times \mathbb{R}^{k-1} \cong \mathbb{R}^{k-1}$) is a local parametrization of ∂M around p.



Definition 25.d.14 (*Induced orientation*): If $\{\partial_1\phi, \dots, \partial_k\phi\}$ is the oriented basis of T_pM given by a local parametrization ϕ around $p \in \partial M$, then $\{\partial_2\phi, \dots, \partial_k\phi\}$ is the "*induced*" oriented basis of $T_p(\partial M)$ around p.

Example: cylinder (Subsection 25.d.1)

Consider the cylindrical surface

$$C = \{(x, y, z) \in \mathbb{R}^3 : y^2 + z^2 = r^2,\ 0 \le x \le L\},$$

a 2-dimensional submanifold with boundary, and the two parametrizations $\phi, \psi : [-L, 0] \times [0, 2\pi] \to C$ (both domains $\subseteq \mathcal{H}^2$),

$$\phi(x, \theta) = \begin{pmatrix} -x \\ r\cos\theta \\ -r\sin\theta \end{pmatrix}, \qquad \psi(x, \theta) = \begin{pmatrix} x + L \\ r\cos\theta \\ r\sin\theta \end{pmatrix}.$$

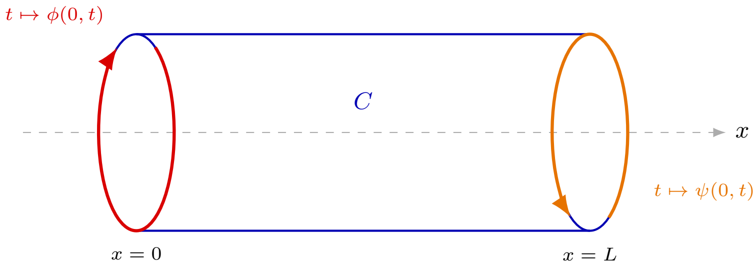
Both cover all of C: as the local x-coordinate ranges over $[-L, 0]$, $-x$ and $x + L$ both range over $[0, L]$.

The two are positively oriented with respect to each other: writing out the composition,

$$\phi^{-1} \circ \psi(x, \theta) = \phi^{-1}(x + L, r\cos\theta, r\sin\theta) = (-x - L, -\theta) = -(x + L, \theta),$$

so $D(\phi^{-1} \circ \psi) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\det D(\phi^{-1} \circ \psi) = 1 > 0$: $\{\phi, \psi\}$ is a valid orientation of C by Definition 25.c.11.

But restricted to the line $\partial\mathcal{H}^2 = \{(x, \theta) : x = 0\}$, ϕ and ψ parametrize **opposite** boundary components of the cylinder. Here $\phi(0, \cdot)$ traces the circle at $x = 0$ and $\psi(0, \cdot)$ the circle at $x = L$, and Definition 25.d.14 gives them opposite rotational senses when both are drawn in the same (y, z)-view.



one cylinder, both rims seen from the same side:
the two induced senses turn opposite ways

Remark 25.d.15 (*Why the opposite senses are not a contradiction*): Nothing has gone wrong. The pair $\{\phi, \psi\}$ is a perfectly good orientation of C as a whole, since the overlap check above is exactly what Definition 25.c.11 demands and it passes. What the picture shows is that a **consistently** oriented cylinder induces boundary orientations on its two ends that turn opposite ways when both circles are viewed from the same side.

That is the familiar fact from the divergence theorem, in another costume: the outward normal at the two ends of a solid cylinder points in opposite x-directions. Induced orientation is doing its job here, not failing at it.

Solutions (Section 25.e)

Solution of Exercise 25.b.9:

(a) Writing each $d\psi^j$ in the coordinate basis, $d\psi^j = \sum_{i=1}^n \frac{\partial \psi^j}{\partial x^i}\, dx^i$, so

$$\omega = \left(\sum_{i_1=1}^n \frac{\partial \psi^1}{\partial x^{i_1}} dx^{i_1}\right) \wedge \cdots \wedge \left(\sum_{i_n=1}^n \frac{\partial \psi^n}{\partial x^{i_n}} dx^{i_n}\right) = \sum_{i_1,\ldots,i_n} \frac{\partial \psi^1}{\partial x^{i_1}} \cdots \frac{\partial \psi^n}{\partial x^{i_n}}\, dx^{i_1} \wedge \cdots \wedge dx^{i_n}$$

by bilinearity of $\wedge$ (property (1) of Definition 24.a.6). Any term with a repeated index vanishes, since $dx^i \wedge dx^i = 0$ by antisymmetry, so only tuples $(i_1, \ldots, i_n) = (\sigma(1), \ldots, \sigma(n))$ for a permutation $\sigma \in S_n$ contribute; using $dx^{\sigma(1)} \wedge \cdots \wedge dx^{\sigma(n)} = \operatorname{sgn}(\sigma)\, dx^1 \wedge \cdots \wedge dx^n$,

$$\omega = \left(\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \frac{\partial \psi^1}{\partial x^{\sigma(1)}} \cdots \frac{\partial \psi^n}{\partial x^{\sigma(n)}}\right) dx^1 \wedge \cdots \wedge dx^n.$$

The coefficient is exactly the Leibniz expansion of $\det(D\psi)$, so

$$\boxed{d\psi^1 \wedge \cdots \wedge d\psi^n = \det(D\psi)\, dx^1 \wedge \cdots \wedge dx^n.}$$

(b) Let $\eta = f(y)\, dy^1 \wedge \cdots \wedge dy^n$ be an n-form on V. Its pullback under ψ is $\psi^*\eta = f(\psi(x))\, d\psi^1 \wedge \cdots \wedge d\psi^n$, which by part (a) equals $f(\psi(x)) \det(D\psi(x))\, dx^1 \wedge \cdots \wedge dx^n$. By Definition 25.b.3 (applied with U itself as the n-dimensional submanifold),

$$\int_U \psi^*\eta = \int_U f(\psi(x)) \det(D\psi(x))\, dx.$$

Since $\det(D\psi) = |\det(D\psi)| > 0$ by hypothesis, the right-hand side is exactly the ordinary change-of-variables formula, so

$$\int_U \psi^*\eta = \int_U f(\psi(x))\, |\det(D\psi(x))|\, dx = \int_V f(y)\, dy = \int_V \eta.$$

Thus $\boxed{\int_U \psi^*\eta = \int_V \eta}$: the change-of-variables formula is exactly the statement that integration of forms is invariant under pullback by an orientation-preserving diffeomorphism. **Q.E.D.**

Stokes' and Green's Theorems (Chapter 26)

Everything in this part converges here. Stokes' theorem says that integrating $d\omega$ over a submanifold is the same as integrating ω over its boundary. That one statement contains the fundamental theorem of calculus, Green's theorem, the divergence theorem and the classical Stokes theorem as special cases. The chapter follows Corsin's own order, taking the special cases first, where the content is visible, then the general theorem, then what it buys. Among other things it buys a formula computing an area from a boundary integral alone.

Sneak peek at Stokes' theorem (Section 26.a)

AI-Note 26.1: Before the formal introduction of Stokes' theorem (Section 26.d), Sascha presented introductory examples to build intuition. The classical version of Stokes' theorem states that for a surface $A \subset \mathbb{R}^3$ and a vector field F,

$$\int_A (\operatorname{curl} F) \cdot \nu \, d\underset{2}{\mathrm{vol}} = \int_{\partial A} F \cdot \tau \, dL,$$

where ν is the unit normal to A and τ is the unit tangent vector to the boundary curve ∂A.

Example 26.a.1 (*Stokes' theorem on a hemisphere*): Let $A := \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 1,\ z \geq 0\}$ be the upper hemisphere. Given the vector field $F(x,y,z) := (z^2, x^2, y^2)^\mathsf{T}$, we can evaluate both sides of Stokes' theorem.

1. **The rotation of F:** $\operatorname{curl} F = \nabla \times F = (2y, 2z, 2x)^\mathsf{T}$.
2. **The surface normal:** On the unit sphere, the outward normal is simply $\nu(x,y,z) = (x,y,z)^\mathsf{T}$.
3. **The boundary ∂A:** The boundary of the upper hemisphere is the unit circle in the xy-plane, $x^2+y^2 = 1$ with $z = 0$. This can be parametrized by the closed path $\gamma : [0, 2\pi] \to \mathbb{R}^3$,

$$\gamma(t) = \begin{pmatrix} \cos t \\ \sin t \\ 0 \end{pmatrix}.$$

Evaluating the path integral (the right-hand side of Stokes' theorem):

$$\int_{\partial A} F \cdot \tau \, dL = \int_0^{2\pi} F(\gamma(t)) \cdot \gamma'(t) \, dt.$$

Since $z = 0$ on γ, we have $F(\gamma(t)) = (0, \cos^2 t, \sin^2 t)^\mathsf{T}$. The derivative is $\gamma'(t) = (-\sin t, \cos t, 0)^\mathsf{T}$.

$$\int_0^{2\pi} \begin{pmatrix} 0 \\ \cos^2 t \\ \sin^2 t \end{pmatrix} \cdot \begin{pmatrix} -\sin t \\ \cos t \\ 0 \end{pmatrix} dt = \int_0^{2\pi} \cos^3 t \, dt = 0.$$

Example 26.a.2 (*Path integral using Stokes' theorem*): Compute $\oint_\Gamma F \cdot \tau \, dL$ where $F(x,y,z) := (x,y,z)^\mathsf{T}$ and Γ is the unit circle in the xy-plane (oriented counter-clockwise).

Instead of computing the line integral directly, we can use Stokes' theorem. The curve Γ bounds the unit disk $A = \{(x,y,0) \mid x^2+y^2 \leq 1\}$. First, we compute the rotation of F:

$$\operatorname{curl} F = \nabla \times F = \begin{pmatrix} \partial_y(z) - \partial_z(y) \\ \partial_z(x) - \partial_x(z) \\ \partial_x(y) - \partial_y(x) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

By Stokes' theorem, the line integral along Γ bounds a surface A on the sphere (for example, the upper hemisphere). The surface integral is:

$$\oint_\Gamma F \cdot \tau \, dL = \int_A (\operatorname{curl} F) \cdot \nu \, d\underset{2}{\mathrm{vol}} = 0.$$

Green's theorem and conservative fields (Section 26.b)

AI-Note 26.2: Green's theorem, conservative vector fields, and path integrals were also discussed in the Monday session.

Line integrals (Wegintegrale) (Subsection 26.b.1)

Definition 26.b.3 (*Work / Line integral*): Let $U \subseteq \mathbb{R}^n$ be an open set, $F : U \to \mathbb{R}^n$ a continuous vector field, and $\gamma : [a,b] \to U$ a piecewise C^1 path. We define the "*work*" (or "*line integral*") of F along γ as

$$\int_\gamma F \cdot d\gamma := \int_a^b F(\gamma(t)) \cdot \gamma'(t)\, dt.$$

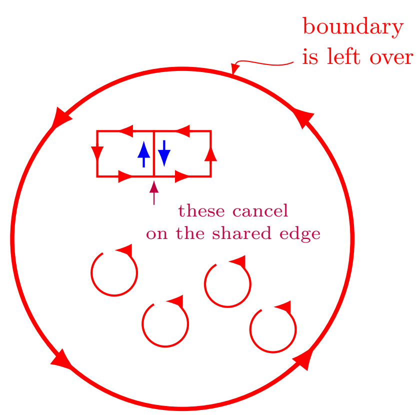
The work measures how much the vector field *"pushes"* along the path.

Green's theorem (Subsection 26.b.2)

Theorem 26.b.4 (*Green's Theorem*): Let $F : U \to \mathbb{R}^2$ be a continuously differentiable vector field on an open set $U \subseteq \mathbb{R}^2$. Then, for any compact C^1 domain $\Omega \subseteq U$,

$$\int_\Omega \operatorname{curl} F \, d\underset{2}{\mathrm{vol}} = \int_{\partial\Omega} F \cdot \tau \, dL,$$

where τ is the unit tangent vector to the boundary $\partial\Omega$ oriented counter-clockwise (i.e. $\tau = (-\nu_2, \nu_1)^\mathsf{T}$, with ν being the outward unit normal).



AI-Note 26.3:

We integrate the rotation inside the domain. Basically all the *"intermediate values"* cancel out because for two points close to each other, the rotation points in opposite directions. Only the boundary is left over in the end!

Example 26.b.5 (*Area of a region via Green's theorem*): Consider a region $\Omega \subseteq \mathbb{R}^2$ defined by the region inside the curve $x^{2/3} + y^{2/3} = 1$ in the first quadrant, bounded by the axes. The curve can be parametrized as $\varphi(t) = (\cos^3 t, \sin^3 t)^\mathsf{T}$ for $t \in [0, \pi/2]$.

The area is $\mathrm{vol}_2(\Omega) = \int_\Omega 1 \, d\mathrm{vol}_2$. We look for a vector field F such that $\operatorname{curl} F = \partial_x F_2 - \partial_y F_1 \equiv 1$ on Ω. A simple choice is $F(x,y) := (0, x)^\mathsf{T}$. By Green's theorem, we have:

$$\mathrm{vol}_2(\Omega) = \int_\Omega \operatorname{curl} F \, d\underset{2}{\mathrm{vol}} = \int_{\partial\Omega} F \cdot \tau \, dL = \int_0^{\pi/2} F(\varphi(t)) \cdot \varphi'(t)\, dt.$$

Since $\tau(\varphi(t))\, dL = \varphi'(t)\, dt$ along the parametrization, we compute the dot product:

$$F(\varphi(t)) \cdot \varphi'(t) = \begin{pmatrix} 0 \\ \varphi_1(t) \end{pmatrix} \cdot \begin{pmatrix} \varphi_1'(t) \\ \varphi_2'(t) \end{pmatrix} = \varphi_1(t)\varphi_2'(t).$$

This turns the area computation into a single 1D integral in t.

Example 26.b.6 (*Line integral via Green's theorem*): We want to compute the work of $F(x,y) := (0, xy^2)^\mathsf{T}$ along the ellipse γ given by $x^2+(y/2)^2 = 1$, oriented counter-clockwise. (This is Michael's Bsp. 18.1.1(b)). Let $\Omega = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2/4 \le 1\}$. By Green's theorem,

$$\int_\gamma F \cdot d\gamma = \int_{\partial\Omega} F \cdot \tau \, dL = \int_\Omega \operatorname{curl} F \, d\operatorname{vol}_2 .$$

We have $\operatorname{curl} F = \partial_x(xy^2) - \partial_y(0) = y^2$. We compute the double integral using modified polar coordinates $x = r\cos\theta$, $y = 2r\sin\theta$ with Jacobian $2r$:

$$\int_\Omega y^2 \, d\operatorname{vol}_2 = \int_0^{2\pi}\int_0^1 (2r\sin\theta)^2(2r)\, dr\, d\theta = 8\int_0^{2\pi} \sin^2\theta \, d\theta \int_0^1 r^3\, dr = 8(\pi)\left(\frac{1}{4}\right) = 2\pi.$$

Example 26.b.7 (*Line integral of a curl-free field*): Let Ω be the unit disk $B_1(0) \subseteq \mathbb{R}^2$ and $F(x,y) := (xy^2, x^2y)^\mathsf{T}$. We compute $\int_{\partial\Omega} F\cdot\tau\, dL$ (Michael's Bsp. 20.2.1). By Green's theorem, we need to integrate the curl of F over Ω:

$$\operatorname{curl} F = \partial_x(x^2y) - \partial_y(xy^2) = 2xy - 2xy = 0.$$

Thus, the integral is simply $\int_\Omega 0 \, d\operatorname{vol}_2 = 0$.

Example 26.b.8 (*2D Divergence theorem*): Green's theorem can also be viewed as a 2D version of the divergence theorem. Consider the vector field $F(x,y) := (x-y^2, y-x^2)^\mathsf{T}$. We want to compute the outward flux of F through the curve γ given by the graph of $y = \cos x$ for $x \in [-\pi/2, \pi/2]$ (Michael's Bsp. 19.6.2).

We can close the curve by adding the segment σ on the x-axis from $\pi/2$ to $-\pi/2$, forming a closed region Ω (the area under the cosine bump). By the 2D divergence theorem:

$$\int_\gamma F\cdot\nu\, dL + \int_\sigma F\cdot\nu\, dL = \int_\Omega \operatorname{div} F\, d\operatorname{vol}_2 .$$

The divergence is $\operatorname{div} F = \partial_x(x-y^2) + \partial_y(y-x^2) = 1+1 = 2$. The integral of the divergence is $2\operatorname{vol}_2(\Omega) = 2\int_{-\pi/2}^{\pi/2}\cos x\, dx = 2(2) = 4$. For the segment $\sigma(x) = (x,0)$ from $\pi/2$ to $-\pi/2$, the outward normal from Ω is $\nu = (0,-1)^\mathsf{T}$. The flux through σ is:

$$\int_\sigma F\cdot\nu\, dL = \int_{-\pi/2}^{\pi/2} \begin{pmatrix} x \\ -x^2\end{pmatrix}\cdot\begin{pmatrix}0\\-1\end{pmatrix} dx = \int_{-\pi/2}^{\pi/2} x^2\, dx = \left[\frac{x^3}{3}\right]_{-\pi/2}^{\pi/2} = \frac{\pi^3}{12}.$$

Therefore, the flux through γ is $4 - \frac{\pi^3}{12}$.

Example 26.b.9 (*The work of a field can depend on the path*): Let $F(x,y) := (-y,x)^\mathsf{T}$, and let $\gamma_0,\gamma_1,\gamma_2 : [0,1] \to \mathbb{R}^2$ be the three paths from $(0,0)$ to $(1,1)$ given by $\gamma_0(t) := (t,t)$ (the diagonal), and

$$\gamma_1(t) := \begin{cases}(2t,0) & t\in[0,\frac12]\\ (1,2t{-}1) & t\in[\frac12,1]\end{cases} \qquad \gamma_2(t) := \begin{cases}(0,2t) & t\in[0,\frac12]\\ (2t{-}1,1) & t\in[\frac12,1]\end{cases}$$

(right-then-up, and up-then-right, along the sides of the unit square). Direct computation gives three different answers:

$$\int_{\gamma_0} F\cdot d\gamma_0 = \int_0^1 \langle(-t,t),(1,1)\rangle\, dt = 0, \qquad \int_{\gamma_1} F\cdot d\gamma_1 = 1, \qquad \int_{\gamma_2} F\cdot d\gamma_2 = -1.$$

Moving **perpendicular** to F costs no work, moving **with** F does positive work, and moving **against** F does negative work. Since the three paths share endpoints but disagree on the work done, F cannot be conservative (Definition 26.b.10 below): Proposition 26.b.13 would otherwise force all three integrals to agree.

Conservative vector fields (Subsection 26.b.3)

Definition 26.b.10 (*Conservative vector field*): Given a C^k vector field F on an open set $U \subseteq \mathbb{R}^n$, we call it "*conservative*" if there exists a scalar field $f : U \to \mathbb{R}$ of class C^{k+1} such that $F \equiv \nabla f$. The function f is called a "*potential*" for F.

Lemma 26.b.11 (*Integrability conditions*): Let $U \subseteq \mathbb{R}^n$ be open, and let F be a C^1 conservative vector field on U with components $F_1, \ldots, F_n$. Then we necessarily have the integrability conditions:

$$\partial_j F_k = \partial_k F_j \qquad \text{for all } j, k \in \{1, \ldots, n\}.$$

Proof:

Let f be a potential such that $F = \nabla f = (\partial_1 f, \ldots, \partial_n f)^\mathsf{T}$. Then by Schwarz's theorem (symmetry of second derivatives),

$$\partial_j F_k = \partial_j(\partial_k f) = \partial_j \partial_k f = \partial_k \partial_j f = \partial_k(\partial_j f) = \partial_k F_j.$$

Q.E.D.

Example 26.b.12 (*A non-conservative field*): Not every vector field is conservative. For example, a frictional force is not conservative. Mathematically, consider $F : \mathbb{R}^3 \to \mathbb{R}^3$ given by $F(x, y, z) := (xz - y, x^2 + z^3, 3xz^3 - xy)^\mathsf{T}$. If F were conservative, it would have to satisfy the integrability conditions (Lemma 26.b.11). But

$$\partial_x F_2 = 2x \quad \text{and} \quad \partial_y F_1 = -1.$$

Since they are not equal, F cannot be conservative.

Proposition 26.b.13 (*Work along a path as difference of potential*): Let $U \subseteq \mathbb{R}^n$ be open, and let $F : U \to \mathbb{R}^n$ be a continuous conservative vector field with potential f. Then for any C^1 path $\gamma : [0, 1] \to U$, the line integral of F along γ depends only on the endpoints:

$$\int_\gamma F \cdot d\gamma = f(\gamma(1)) - f(\gamma(0)).$$

Proof:

By definition of the path integral and the chain rule,

$$\begin{aligned}\int_\gamma F \cdot d\gamma &= \int_0^1 F(\gamma(t)) \cdot \gamma'(t)\, dt = \int_0^1 \nabla f(\gamma(t)) \cdot \gamma'(t)\, dt \\ &= \int_0^1 Jf(\gamma(t)) J\gamma(t)\, dt = \int_0^1 J(f \circ \gamma)(t)\, dt = \int_0^1 (f \circ \gamma)'(t)\, dt.\end{aligned}$$

By the fundamental theorem of calculus, this equals $(f \circ \gamma)(1) - (f \circ \gamma)(0) = f(\gamma(1)) - f(\gamma(0))$.

Q.E.D.

Example 26.b.14 (*Line integral of a conservative field*): Consider the vector field $F : [0, 1] \times [0, 2] \to \mathbb{R}^2$ given by $F(x, y) := (y^2, 2xy)^\mathsf{T}$. First, we show that F is conservative by finding a potential f such that $\nabla f = F$. We need $\partial_x f = y^2$ and $\partial_y f = 2xy$. Integrating the first equation with respect to x gives $f(x, y) = xy^2 + c(y)$. Differentiating with respect to y gives $\partial_y f = 2xy + c'(y)$, which must equal $2xy$. Thus $c'(y) = 0$, and we can choose $c(y) = 0$. The potential is $f(x, y) = xy^2$.

Now, let Γ be a closed path on the boundary ∂U of the domain, starting and ending at $\gamma(0) = \gamma(1) = (0, 0)^\mathsf{T}$. By Proposition 26.b.13, the line integral around this closed loop is simply the difference in potential at the endpoints:

$$\oint_\Gamma F \cdot d\gamma = f(\gamma(1)) - f(\gamma(0)) = f(0, 0) - f(0, 0) = 0.$$

Example 26.b.15 (*Line integral of a conservative field via potential*): Consider the vector field $F(x,y) := (10x^4 - 2xy^3, -3x^2y^2)^\mathsf{T}$ and a path γ connecting $(0,0)$ to $(2,1)$, for instance, the curve defined by $x^4 - 6xy^3 = 4y^2$ (Michael's Bsp. 18.2.7).

Since F is defined on all of $\mathbb{R}^2$ and satisfies $\partial_y F_1 = -6xy^2 = \partial_x F_2$, it is conservative. We find a potential $f(x,y)$ by integration:

$$\partial_x f = 10x^4 - 2xy^3 \implies f(x,y) = 2x^5 - x^2y^3 + c(y),$$
$$\partial_y f = -3x^2y^2 + c'(y).$$

Equating this with $F_2 = -3x^2y^2$, we get $c'(y) = 0$, so we can choose $c(y) = 0$. The potential is $f(x,y) = 2x^5 - x^2y^3$. By Proposition 26.b.13, the work along γ is path-independent and given by:

$$\int_\gamma F \cdot d\gamma = f(2,1) - f(0,0) = (2(32) - 4(1)) - 0 = 60.$$

Notice that the equation of the curve $x^4 - 6xy^3 = 4y^2$ was irrelevant, as long as it connects the two endpoints!

Exercise 26.b.16 (*Which α makes this field conservative?*): For which $\alpha \in \mathbb{R}$ is the vector field $F(x,y) := (\alpha x e^y,\ (y + 1 + x^2)e^y)^\mathsf{T}$ conservative? Find a potential for that value of α.

Proof of Exercise 26.b.16:

By Lemma 26.b.11, a necessary condition is $\partial_y F_1 = \partial_x F_2$, i.e. $\alpha x e^y = 2xe^y$ for all (x,y), forcing $\alpha = 2$. (This is also sufficient, since F is C^1 on all of $\mathbb{R}^2$, which is simply connected, so a potential exists.) With $\alpha = 2$, $F(x,y) = (2xe^y,\ (y+1+x^2)e^y)^\mathsf{T}$; integrating $\partial_x f = 2xe^y$ gives $f(x,y) = x^2e^y + c(y)$, and matching $\partial_y f = x^2e^y + c'(y)$ against $F_2 = (x^2 + y + 1)e^y$ gives $c'(y) = (y+1)e^y$, so $c(y) = ye^y$ (check: $\frac{d}{dy}(ye^y) = e^y + ye^y = (y+1)e^y$). A potential is $f(x,y) = x^2e^y + ye^y$. **Q.E.D.**

Exercise 26.b.17 (*Choosing parameters for an irrotational field*): Determine real parameters α, β, γ such that the vector field

$$F(x,y,z) := (x + 2y + \alpha z,\ \beta x - 3y - z,\ 4x + \gamma y + 2z)$$

on $\mathbb{R}^3$ becomes irrotational, i.e. $\operatorname{curl}(F) = 0$. Determine, after choosing these parameters, a potential for F.

Proof of Exercise 26.b.17:

Setting $\operatorname{curl} F = 0$ componentwise (Definition 22.b.6):

$$\partial_y F_3 - \partial_z F_2 = \gamma - (-1) \overset{!}{=} 0, \qquad \partial_z F_1 - \partial_x F_3 = \alpha - 4 \overset{!}{=} 0, \qquad \partial_x F_2 - \partial_y F_1 = \beta - 2 \overset{!}{=} 0,$$

giving $\gamma = -1$, $\alpha = 4$, $\beta = 2$, so $F(x,y,z) = (x + 2y + 4z,\ 2x - 3y - z,\ 4x - y + 2z)$. This F is C^1 on the simply connected $\mathbb{R}^3$ and curl-free, hence conservative (the 3-dimensional case of Corollary 26.c.19). Integrating $\partial_x f = x + 2y + 4z$ gives $f = \frac{1}{2}x^2 + 2xy + 4xz + c(y,z)$; matching $\partial_y f = 2x + c_y$ against $F_2 = 2x - 3y - z$ gives $c_y = -3y - z$, so $c = -\frac{3}{2}y^2 - yz + d(z)$; matching $\partial_z f = 4x - y + d'(z)$ against $F_3 = 4x - y + 2z$ gives $d'(z) = 2z$, so $d(z) = z^2$. A potential is

$$f(x,y,z) = \tfrac{1}{2}x^2 + 2xy + 4xz - \tfrac{3}{2}y^2 - yz + z^2.$$

Q.E.D.

The Poincaré lemma and simply connected domains (Section 26.c)

Theorem 26.c.18 (*Poincaré Lemma*): Let $U \subseteq \mathbb{R}^n$ be open, and let $\alpha \in \Omega^1(U)$ be a C^1 closed 1-form, that is, $d\alpha = 0$. Let $\gamma_0, \gamma_1 : [0,1] \to U$ be piecewise C^1 paths with the same initial point

x_0 and the same endpoint x_1. Suppose that γ_0 and γ_1 are "*homotopic*" in U with fixed endpoints. For simplicity, assume that the homotopy $H : [0,1]^2 \to U$ is smooth. Then

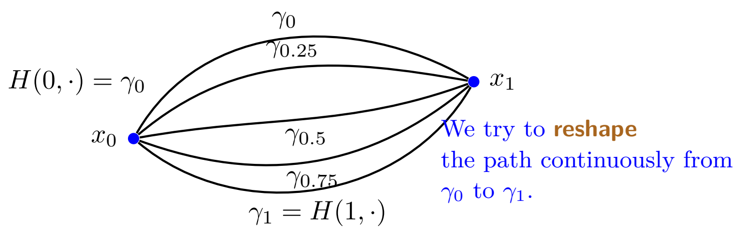
$$\int_{\gamma_0} \alpha = \int_{\gamma_1} \alpha.$$

Corollary 26.c.19 (*Existence of potentials in simply connected domains*): Let $U \subseteq \mathbb{R}^n$ be an open "*simply connected*" set. Then every closed C^1 1-form on U is exact. That is, if $\alpha \in \Omega^1(U)$ satisfies $d\alpha = 0$, then there exists a function $f \in C^2(U)$ such that $\alpha = df$. Equivalently, if $F \in C^1(U, \mathbb{R}^n)$ satisfies the integrability conditions

$$\partial_j F_k = \partial_k F_j \qquad \text{for all } j,k \in \{1,\dots,n\},$$

then F is conservative: there exists a potential $f \in C^2(U)$ such that $\nabla f = F$.

AI-Note 26.4 (*Intuition on simply connected spaces*): A space U is simply connected if any closed loop in U can be continuously contracted to a point within U (i.e. any two paths with the same endpoints are homotopic).



Examples of simply connected spaces:

- $\mathbb{R}^n$
- $\mathbb{R}^3 \setminus \{(0,0,0)\}$

Examples of spaces that are **not** simply connected:

- $S^1 \subseteq \mathbb{R}^2$ (the unit circle)
- $\mathbb{R}^3 \setminus \operatorname{span}\{(0,0,1)^\mathsf{T}\}$ (the z-axis removed)
- The torus

This topological property is what allows the integrability conditions to globally imply the existence of a potential.

Example 26.c.20 (*Integrability without a potential: why simple connectedness is needed*): Let $U := \mathbb{R}^2 \setminus \{0\}$ (which is **not** simply connected, unlike its 3-dimensional analogue $\mathbb{R}^3 \setminus \{0\}$ listed above) and

$$F(x,y) := \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right), \qquad (x,y) \in U.$$

A direct computation gives

$$\partial_1 F_2 = \partial_x \left(\frac{x}{x^2+y^2} \right) = \frac{-x^2+y^2}{(x^2+y^2)^2} = \partial_y \left(\frac{-y}{x^2+y^2} \right) = \partial_2 F_1,$$

so F satisfies the integrability conditions (Lemma 26.b.11) **everywhere** on U. Yet F is not conservative: for the loop $\gamma(t) := (\cos(2\pi t), \sin(2\pi t))$, $t \in [0,1]$, tracing the unit circle once counterclockwise,

$$\oint_\gamma F \cdot d\gamma = 2\pi \int_0^1 \langle (-\sin(2\pi t), \cos(2\pi t)), (-\sin(2\pi t), \cos(2\pi t)) \rangle \, dt = 2\pi \neq 0,$$

which by Proposition 26.b.13 is impossible for a conservative field (a closed loop would have to return $f(\gamma(1)) - f(\gamma(0)) = 0$). This is exactly the witness that Corollary 26.c.19 needs the hypothesis it has: the integrability conditions are necessary but, on a domain that is not simply connected, not sufficient.

Stokes' theorem (Section 26.d)

Theorem 26.d.21 (*Stokes' theorem*): Let $M \subseteq \mathbb{R}^n$ be an orientable m-manifold with (possibly empty) boundary, and let $\omega \in \Omega^{m-1}(\mathbb{R}^n)$ have compact support in $\mathbb{R}^n$. Then

$$\int_{\partial M} \omega = \int_M d\omega.$$

AI-Note 26.5: Theorem 25.c.12 was stated ahead of Corsin's own notes, as background needed to solve Exercise 26.e.31, in its classical $\mathbb{R}^3$ curl-theorem form. It is exactly the case $m = 2$, $n = 3$, $\omega = \omega_F$ of the general theorem above, as Section 26.d.3 below confirms directly from Corsin's own derivation.

Example: Green's theorem (Subsection 26.d.1)

Example 26.d.22 (*The exterior derivative of a 1-form on $\mathbb{R}^2$*): Let $\omega(x,y) = P(x,y)\,dx + Q(x,y)\,dy$ be a 1-form on $\mathbb{R}^2$. Then

$$d\omega = dP \wedge dx + dQ \wedge dy = \left(\frac{\partial P}{\partial x}dx + \frac{\partial P}{\partial y}dy\right) \wedge dx + \left(\frac{\partial Q}{\partial x}dx + \frac{\partial Q}{\partial y}dy\right) \wedge dy = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy,$$

using $dx \wedge dx = dy \wedge dy = 0$ and $dy \wedge dx = -dx \wedge dy$.

Example: proof of Green's theorem (Subsection 26.d.2)

AI-Note 26.6: The theorem below is reproduced by Corsin directly from the lecture script (numbered there as Theorem 14.24), as with the two definitions quoted in Section 25.c.

Theorem 26.d.23 (*Green's theorem in the plane*): Let $F : U \to \mathbb{R}^2$ be a continuously differentiable vector field on an open set $U \subseteq \mathbb{R}^2$. Then, for any compact C^1 domain $\Omega \subseteq U$,

$$\int_\Omega \operatorname{curl} F\, dx = \int_{\partial\Omega} F \cdot \tau\, dL,$$

with $\tau = \nu^\perp = (-\nu_2, \nu_1)$, where ν is the exterior unit normal.

Writing $F(x,y) = (P(x,y), Q(x,y))$, so that $\operatorname{curl} F = \partial_x Q - \partial_y P$ (the scalar 2-dimensional curl), Theorem 26.d.23 follows from Theorem 26.d.21 and Example 26.d.22 applied to $\omega = P\,dx + Q\,dy$:

$$\int_\Omega \operatorname{curl} F\, dx\, dy = \int_\Omega \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy = \int_\Omega d\omega \overset{\text{Stokes}}{=} \int_{\partial\Omega} P(x,y)\,dx + Q(x,y)\,dy = \int_{\partial\Omega} F \cdot \tau\, dL.$$

Justification of the last step:

Suppose $\gamma : [0, L] \to \partial\Omega$ is a C^1 curve of unit speed and positive orientation, i.e. $\gamma'(t) = \tau(\gamma(t))$. Then

$$\langle F, \tau \rangle = \langle F, \gamma'(t) \rangle = (P\,dx + Q\,dy)(\gamma'(t)),$$

since $\tau = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \nu$ is the tangent vector consistent with the induced orientation, so integrating $(P\,dx + Q\,dy)(\gamma'(t))$ over $t \in [0, L]$ is exactly $\int_{\partial\Omega} P\,dx + Q\,dy$ by Definition 25.b.3. **Q.E.D.**

Example: Stokes' theorem in $\mathbb{R}^3$ (Subsection 26.d.3)

Recall from Example 24.b.13 that for $\omega_F = \langle F, \cdot \rangle$, $d\omega_F = \langle \operatorname{curl} F, \cdot \times \cdot \rangle$. So, if $\Omega \subseteq \mathbb{R}^3$ is a 2-dimensional submanifold with boundary, parametrized by ϕ with tangent vectors $\tau_i := \partial_i \phi$ and exterior unit normal ν, then

$$\int_\Omega d\omega_F = \int_\Omega \langle \operatorname{curl} F, \tau_1 \times \tau_2 \rangle\, dx = \int_\Omega \langle \operatorname{curl} F, \nu \rangle\, d\operatorname{vol}_2 = \int_\Omega \vec{\nabla} \times \vec{F} \cdot d\vec{A},$$

and
$$\int_{\partial\Omega} \omega_F = \int_{\partial\Omega} \langle F, \tau \rangle \, d\,\mathrm{vol}_1 = \int_{\partial\Omega} \vec{F} \cdot d\vec{s}.$$
By Theorem 26.d.21, these are equal, which recovers Theorem 25.c.12.

Theorem 26.d.24 (*Stokes' theorem in 3D*): For an orientable, parametrized 2-dimensional submanifold $M \subseteq \mathbb{R}^3$ with boundary ∂M, and a C^1 vector field F:
$$\int_M \mathrm{curl}(F) \cdot \nu \, d\,\mathrm{vol}_2 = \int_{\partial M} F \cdot \tau \, d\,\mathrm{vol}_1 .$$
The orientation of ∂M is chosen such that when traversing ∂M in the direction of τ, the surface M (given by the normal ν) is to the left (the "right-hand rule").

Example 26.d.25 (*Stokes' theorem on a hemisphere*): Let $F(x, y, z) := (z, x, y)^\mathsf{T}$. We want to compute the flux of the curl of F through the upper hemisphere $H := \{(x, y, z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 1,\ z \geq 0\}$ (Michael's Bsp. 20.1.2).

First, the curl of F is:
$$\mathrm{curl}(F) = \begin{pmatrix} \partial_y F_3 - \partial_z F_2 \\ \partial_z F_1 - \partial_x F_3 \\ \partial_x F_2 - \partial_y F_1 \end{pmatrix} = \begin{pmatrix} 1-0 \\ 1-0 \\ 1-0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$
By Theorem 26.d.24, the flux of $\mathrm{curl}(F)$ through H equals the line integral of F along the boundary ∂H. The boundary is the unit circle in the xy-plane, which we parametrize by $\gamma(t) = (\cos t, \sin t, 0)^\mathsf{T}$ for $t \in [0, 2\pi]$. The tangent vector is $\gamma'(t) = (-\sin t, \cos t, 0)^\mathsf{T}$. The line integral is:
$$\begin{aligned} \int_{\partial H} F \cdot d\gamma &= \int_0^{2\pi} F(\gamma(t)) \cdot \gamma'(t) \, dt \\ &= \int_0^{2\pi} \begin{pmatrix} 0 \\ \cos t \\ \sin t \end{pmatrix} \cdot \begin{pmatrix} -\sin t \\ \cos t \\ 0 \end{pmatrix} dt \\ &= \int_0^{2\pi} \cos^2 t \, dt = \pi. \end{aligned}$$
Thus, $\int_H \mathrm{curl}(F) \cdot \nu \, d\,\mathrm{vol}_2 = \pi$.

Exercise 26.d.26 (*Flux through a cylinder, direct and via Stokes*): Let $F(x, y, z) := (yz, x^2, 1)^\mathsf{T}$ on $\mathbb{R}^3$, and let M be the lateral surface of the cylinder $M := \{(x, y, z) : x^2 + y^2 = 1,\ 0 < z < 1\}$, with the outward-pointing normal $\nu(x, y, z) = (x, y, 0)$. Compute $\int_M \mathrm{curl}(F) \cdot \nu \, d\,\mathrm{vol}_2$ directly, and again using Stokes' theorem.

Proof of Exercise 26.d.26:

First, $\mathrm{curl}(F) = (\partial_y F_3 - \partial_z F_2,\ \partial_z F_1 - \partial_x F_3,\ \partial_x F_2 - \partial_y F_1) = (0,\ y,\ 2x - z)$.

Directly. Parametrize M by $\phi(\theta, z) := (\cos\theta, \sin\theta, z)$, $(\theta, z) \in [0, 2\pi) \times (0, 1)$; then $\partial_\theta \phi \times \partial_z \phi = (\cos\theta, \sin\theta, 0) = \nu(\phi(\theta, z))$, already a unit vector, so $d\,\mathrm{vol}_2 = d\theta \, dz$. Then $\mathrm{curl}(F)(\phi(\theta, z)) = (0, \sin\theta, 2\cos\theta - z)$, and
$$\int_M \mathrm{curl}(F) \cdot \nu \, d\,\mathrm{vol}_2 = \int_0^1 \int_0^{2\pi} \big\langle (0, \sin\theta, 2\cos\theta - z), (\cos\theta, \sin\theta, 0) \big\rangle \, d\theta \, dz = \int_0^1 \int_0^{2\pi} \sin^2\theta \, d\theta \, dz = \pi.$$
Via Stokes. The boundary ∂M consists of the two circles $z = 0$ and $z = 1$; the orientation induced by the outward normal ν (right-hand rule) traverses the **bottom** circle ($z = 0$) with θ increasing and the **top** circle ($z = 1$) with θ **decreasing**. The parameter rectangle $[0, 2\pi] \times [0, 1]$ has $z = 0$ as part of its positively-oriented boundary traversed left to right and $z = 1$ traversed

right to left, and the two vertical edges $\theta = 0, 2\pi$ cancel since they are the same curve in M. With $\gamma_0(\theta) := (\cos\theta, \sin\theta, 0)$ and $\gamma_1(\theta) := (\cos\theta, \sin\theta, 1)$, both for $\theta \in [0, 2\pi]$,

$$F(\gamma_0(\theta)) = (0, \cos^2\theta, 1), \qquad F(\gamma_1(\theta)) = (\sin\theta, \cos^2\theta, 1),$$

and $\gamma_0'(\theta) = \gamma_1'(\theta) = (-\sin\theta, \cos\theta, 0)$, so

$$\int_0^{2\pi} F(\gamma_0(\theta)) \cdot \gamma_0'(\theta)\, d\theta = \int_0^{2\pi} \cos^3\theta\, d\theta = 0,$$

$$\int_0^{2\pi} F(\gamma_1(\theta)) \cdot \gamma_1'(\theta)\, d\theta = \int_0^{2\pi} \left(-\sin^2\theta + \cos^3\theta\right) d\theta = -\pi.$$

The bottom circle is traversed with θ increasing (contributing $+0$) and the top circle with θ **decreasing** (contributing $-(-\pi) = \pi$), so $\int_{\partial M} F \cdot \tau \, d\,\mathrm{vol}_1 = 0 + \pi = \pi$, agreeing with the direct computation. **Q.E.D.**

Exercise 26.d.27 (*Flux through a spherical cap, via Stokes and via Gauss*): Let $0 < h < 1$ and let $S := \{(x, y, z) \in S^2 : z > -h\}$ be the part of the unit sphere above height $-h$. Consider $F(x, y, z) := (-y, x, 0)^\mathsf{T}$. Calculate the flux $\int_S \operatorname{curl}(F) \cdot \nu \, d\,\mathrm{vol}_2$ (outward normal) directly, once using Stokes' theorem, and once using the divergence theorem.

Proof of Exercise 26.d.27:

Here $\operatorname{curl}(F) = (0, 0, 2)$, a constant field, and on S^2 the outward normal is $\nu(x, y, z) = (x, y, z)$ itself, so $\langle \operatorname{curl}(F), \nu \rangle = 2z$.

Directly. Using colatitude $\varphi \in [0, \arccos(-h)]$ (so $z = \cos\varphi > -h$) and $d\,\mathrm{vol}_2 = \sin\varphi\, d\varphi\, d\theta$,

$$\int_S 2z \, d\,\mathrm{vol}_2 = \int_0^{2\pi} \int_0^{\arccos(-h)} 2\cos\varphi \sin\varphi \, d\varphi\, d\theta = 2\pi \left[-\tfrac{1}{2}\cos(2\varphi)\right]_0^{\arccos(-h)} = 2\pi(1 - h^2),$$

using $\cos(2\arccos(-h)) = 2h^2 - 1$.

Via Stokes. ∂S is the circle $x^2 + y^2 = 1 - h^2$, $z = -h$, of radius $r := \sqrt{1 - h^2}$, oriented (right-hand rule, outward normal) by $\gamma(\theta) := (r\cos\theta, r\sin\theta, -h)$, $\theta \in [0, 2\pi]$. Then $\gamma'(\theta) = (-r\sin\theta, r\cos\theta, 0)$ and $F(\gamma(\theta)) = (-r\sin\theta, r\cos\theta, 0)$, so

$$\int_{\partial S} F \cdot d\gamma = \int_0^{2\pi} \left(r^2 \sin^2\theta + r^2 \cos^2\theta\right) d\theta = 2\pi r^2 = 2\pi(1 - h^2),$$

agreeing with the direct computation.

Via Gauss. Since $\operatorname{div}(\operatorname{curl} F) \equiv 0$ (Exercise 22.b.8), the flux of $\operatorname{curl}(F)$ through any surface with boundary ∂S is the same. Replace S by the flat disk $D := \{x^2 + y^2 \le 1 - h^2,\ z = -h\}$, with upward normal $(0, 0, 1)$: $S \cup (-D)$ (the cap together with the disk, reversed) bounds the solid cap Ω, and the divergence theorem gives $\int_S \operatorname{curl}(F) \cdot \nu \, d\,\mathrm{vol}_2 - \int_D \operatorname{curl}(F) \cdot (0, 0, 1) \, d\,\mathrm{vol}_2 = \int_\Omega \operatorname{div}(\operatorname{curl} F) \, d\,\mathrm{vol}_3 = 0$. Since $\langle \operatorname{curl} F, (0, 0, 1) \rangle \equiv 2$,

$$\int_S \operatorname{curl}(F) \cdot \nu \, d\,\mathrm{vol}_2 = \int_D 2 \, d\,\mathrm{vol}_2 = 2 \cdot \mathrm{vol}_2(D) = 2\pi(1 - h^2),$$

the area of a disk of radius $\sqrt{1 - h^2}$ times 2, again the same answer. **Q.E.D.**

Remark 26.d.28 (*One theorem, four costumes*): Every integral theorem proved in this part of the course is the single identity $\int_M d\omega = \int_{\partial M} \omega$ of Theorem 26.d.21, for a different choice of the dimension m of M and the degree of ω. Collected in one place:

Theorem	M	m	ω	$d\omega$
Fundamental theorem of calculus	$[a,b] \subset \mathbb{R}$	1	f (a 0-form)	$f'\,dx$
Green's theorem (Theorem 26.d.23)	$\Omega \subseteq \mathbb{R}^2$	2	$P\,dx + Q\,dy$	$(\partial_x Q - \partial_y P)\,dx \wedge dy$
Classical Stokes (Theorem 25.c.12)	surface $S \subset \mathbb{R}^3$	2	ω_F	$\omega^2_{\operatorname{curl} F}$
Divergence theorem (Theorem 23.b.4)	$\Omega \subseteq \mathbb{R}^n$	n	ω_F^{n-1}	$(\operatorname{div} F)\,dx^1 \wedge \cdots \wedge dx^n$

Each row's $\int_M d\omega = \int_{\partial M} \omega$ reads off as exactly that row's theorem. The top row is Section 23.a's $n = 1$ computation seen from the other side: there, f played the role of a 1-dimensional vector field and the identity was reached via the divergence theorem; here f is instead a 0-form and the same identity is the $m = 1$ case of Theorem 26.d.21 directly. These are two routes to the same fact, matching Example 24.b.13's remark that $df = \langle \nabla f, \cdot \rangle$ is exactly the shape that makes this specialization work. The bottom row uses $d\omega_F^{n-1} = (\operatorname{div} F)\,dx^1 \wedge \cdots \wedge dx^n$ (Example 24.b.13, item 3) together with the identification of $\int_{\partial\Omega} \omega_F^{n-1}$ with the flux $\int_{\partial\Omega} \langle F, \nu \rangle \, d\operatorname{vol}_{n-1}$ already noted in Chapter 25.

The pattern is worth stating in words, because it is easy to lose under the notation: a k-dimensional derivative measured **inside** a region always equals the corresponding $(k-1)$-dimensional quantity measured on its **boundary**. Differentiation and taking a boundary are, in this precise sense, the same operation viewed from two sides. That is also exactly the content of $d^2 = 0$ (Theorem 24.b.12**(3)**): a boundary has no boundary, just as $d\omega$ has no further exterior derivative to take that isn't already zero.

Area via Green's formula (Section 26.e)

Exercise 26.e.29 (*Area as a boundary functional*): Suppose $\gamma : [0, L) \to \mathbb{R}^2$ traces the boundary of a connected C^∞-domain $U \subseteq \mathbb{R}^2$ in the positive direction. Express the area of U as a functional of γ.

Solution:

By definition, $A(U) = \int_U dx\,dy = \int_U 1\,dx \wedge dy$. By Green's formula (Theorem 26.d.23, in its 1-form guise $\int_\Omega d\omega = \int_{\partial\Omega} \omega$),

$$\int_U 1\,dx \wedge dy = \int_0^L \langle F \circ \gamma(t), \gamma'(t) \rangle \, dt$$

for any $F : \overline{U} \to \mathbb{R}^2$ satisfying $\partial_x F^2 - \partial_y F^1 = 1$. This is satisfied by $F(x, y) = \frac{1}{2}(-y, x)$, giving the "*shoelace formula*" (German: „*Gaußsche Trapezformel*")

$$A(\gamma) = \frac{1}{2} \int_0^L \left[\gamma_1(t)\gamma_2'(t) - \gamma_2(t)\gamma_1'(t) \right] dt.$$

Q.E.D.

Example 26.e.30 (*Ellipse*): The path $\gamma : [0, 2\pi) \to \mathbb{R}^2$, $t \mapsto (a \cos t, b \sin t)$, traces the boundary of an ellipse. By Exercise 26.e.29,

$$A(\gamma) = \frac{1}{2} \int_0^{2\pi} \left[a \cos t \cdot b \cos t - b \sin t \cdot (-a \sin t) \right] dt = \frac{ab}{2} \int_0^{2\pi} dt = \pi ab.$$

For the circle $a = b =: r$, this gives the familiar $A = \pi r^2$.

AI-Note 26.7: Corsin adds, in the same breath as the ellipse's area: *"Famously, there exists no closed form solution for the length of the boundary of an ellipse"*. The ellipse's perimeter is a complete elliptic integral of the second kind, not an elementary function, unlike its area.

Exercise 26.e.31 (*12.3 — Work along paths and Stokes* (important)): Let $F : \mathbb{R}^3\backslash(\{(0,0)\}\times\mathbb{R}) \to \mathbb{R}^3$ be the vector field defined by

$$F(x,y,z) = \left(\frac{2(xz+y)}{x^2+y^2},\ \frac{2(yz-x)}{x^2+y^2},\ \log(x^2+y^2)\right)$$

and let $\gamma_1, \gamma_2, \gamma_3 : [0,2\pi] \to \mathbb{R}^3$ be the paths

$$\gamma_1(t) = (\cos t, \sin t, 0), \qquad \gamma_2(t) = (2\cos t, 2\sin t, 2), \qquad \gamma_3(t) = (3\cos t + 6, 3\sin t, 3).$$

1. Describe the domain of definition of F. Is it connected? Is it simply connected?
2. Compute the rotation $\mathrm{curl}(F)$ of F and the path integral $\int_{\gamma_1} F \cdot d\gamma_1$ of F along γ_1. Is F conservative?
3. Explain how the equation $\int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_2} F \cdot d\gamma_2$ follows from Stokes' theorem.
4. Does Stokes' theorem imply that $\int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_3} F \cdot d\gamma_3$ holds?

Exercise 26.e.32 (*12.4 — True or False* (important)): Let $U := \mathbb{R}^3 \setminus \{0\}$ and $F \in C^1(U, \mathbb{R}^3)$. For $r > 0$ set

$$I_r := \int_{\partial B_r} \langle F, n\rangle \, dS$$

where n denotes the outward normal. Which of the following statements are always true?

(a) If F is divergence-free, then $I_r = 0$ for all $r > 0$.
(b) If F is divergence-free, then the value of I_r does not depend on r.
(c) If F is curl-free, then the value of I_r does not depend on r.
(d) If $F = \mathrm{curl}(G)$ for a vector field $G \in C^1(U, \mathbb{R}^3)$ and $|F(x)| \le C/|x|$, then $I_r = 0$ for all $r > 0$.

Exercise 26.e.33 (*12.5 — Conservative vector fields* (important)): For which values of $\lambda \in \mathbb{R}$ is the vector field $F : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$F(x,y) = \begin{pmatrix} \lambda x e^y \\ (y + 1 + x^2)e^y \end{pmatrix}$$

conservative? Determine a potential for F for these values.

Exercise 26.e.34 (*12.8 — Multiple Choice* (important)): Let $Q = [a,b] \times [c,d] \subseteq \mathbb{R}^2$ and $f \in C(Q, \mathbb{R}^2)$. Let n be an outward normal field on Q and

$$I_b = \int_{[a,b]\times\{c\}} \langle f, n\rangle, \qquad I_t = \int_{[a,b]\times\{d\}} \langle f, n\rangle, \qquad I_l = \int_{\{a\}\times[c,d]} \langle f, n\rangle, \qquad I_r = \int_{\{b\}\times[c,d]} \langle f, n\rangle.$$

Which of the following expressions corresponds to the outer flux of f through ∂Q?

(a) $I_b - I_t + I_r - I_l$
(b) $I_b + I_t + I_r + I_l$
(c) $I_b + I_t - I_r - I_l$
(d) $-I_b + I_t - I_r + I_l$

Solutions (Section 26.f)

Solution of Exercise 26.e.31:

1. The domain is $\{(x,y,z) \in \mathbb{R}^3 : (x,y) \neq (0,0)\}$, i.e. $\mathbb{R}^3$ with the z-axis removed. It is connected (path-connected: any two points can be joined by a path avoiding the axis), but not simply connected: γ_1, for instance, which winds once around the axis, cannot be contracted to a point without crossing it. (It is also not convex: the segment from $(-1,0,0)$ to $(1,0,0)$ leaves the domain.)

2. Writing $F = (F^1, F^2, F^3)$ with $F^1 = 2(xz+y)/(x^2+y^2)$, $F^2 = 2(yz-x)/(x^2+y^2)$, $F^3 = \log(x^2+y^2)$, a direct computation of each pair of mixed partials gives

$$\partial_y F^3 = \partial_z F^2 = \frac{2y}{x^2+y^2}, \qquad \partial_z F^1 = \partial_x F^3 = \frac{2x}{x^2+y^2}, \qquad \partial_x F^2 = \partial_y F^1 = \frac{2(x^2-y^2-2xyz)}{(x^2+y^2)^2},$$

so $\operatorname{curl} F = (\partial_y F^3 - \partial_z F^2,\ \partial_z F^1 - \partial_x F^3,\ \partial_x F^2 - \partial_y F^1) = (0,0,0)$: F is curl-free. On γ_1, $x = \cos t$, $y = \sin t$, $z = 0$, $x^2+y^2 = 1$, so $F(\gamma_1(t)) = (2\sin t,\ -2\cos t,\ 0)$ and $\gamma_1'(t) = (-\sin t, \cos t, 0)$, giving

$$\int_{\gamma_1} F \cdot d\gamma_1 = \int_0^{2\pi} \big(2\sin t \cdot (-\sin t) + (-2\cos t)\cdot \cos t\big)\, dt = \int_0^{2\pi} -2\, dt = -4\pi.$$

Since this is nonzero on a closed loop, F is **not** conservative. It is curl-free, but its domain is not simply connected (part 1), so curl-free does not force conservative here.

3. Let $S := \{(x,y,z) : \sqrt{x^2+y^2} = 1 + z/2,\ 0 \le z \le 2\}$, the cone swept out by linearly interpolating the radius from γ_1 ($z=0$, radius 1) to γ_2 ($z = 2$, radius 2), parametrized by $H(s,t) := \big((1+s)\cos t, (1+s)\sin t, 2s\big)$ for $(s,t) \in [0,1]\times[0,2\pi]$. Since the radius $1+s \ge 1 > 0$ throughout, S never meets the z-axis and lies entirely in the domain of F; its oriented boundary (outward normal) is γ_2 minus γ_1. Since $\operatorname{curl} F = 0$ on S by part 2, Theorem 25.c.12 gives

$$\int_{\gamma_2} F \cdot d\gamma_2 - \int_{\gamma_1} F\cdot d\gamma_1 = \int_{\partial S} F \cdot ds = \int_S \operatorname{curl} F \cdot d\vec{n} = 0,$$

i.e. $\int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_2} F \cdot d\gamma_2$.

4. No. The disk $D := \{(x,y,z) : (x-6)^2 + y^2 \le 9,\ z = 3\}$, oriented upward, has oriented boundary exactly γ_3, and D lies entirely in the domain of F (it does not meet the z-axis, since its center $(6,0,3)$ is at distance 6 from the axis and its radius is only 3). By Theorem 25.c.12 with $\operatorname{curl} F = 0$,

$$\int_{\gamma_3} F \cdot d\gamma_3 = \int_{\partial D} F \cdot ds = \int_D \operatorname{curl} F \cdot d\vec{n} = 0.$$

Since $\int_{\gamma_1} F\cdot d\gamma_1 = -4\pi \neq 0 = \int_{\gamma_3} F \cdot d\gamma_3$, the two do **not** agree, and Stokes' theorem gives no reason to expect they would: unlike the pair (γ_1, γ_2), there is no surface joining γ_1 and γ_3 that stays inside the domain, since γ_1 winds around the excluded axis and γ_3 does not. **Q.E.D.**

Solution of Exercise 26.e.32:

(a) False. The Newton potential's gradient $F(x) = \nabla(1/|x|) = -x|x|^{-3}$ is divergence-free on U (it is V_{-3} up to sign, from Exercise 23.b.9, with $n=3$), yet $I_r = \int_{\partial B_r} F \cdot n\, dS = -\operatorname{vol}_2(\partial B_r) \neq 0$.

(b) True. Applying Theorem 23.b.4 on the shell $B_{r_2} \setminus B_{r_1} \Subset U$ for $0 < r_1 < r_2$, whose boundary is ∂B_{r_2} (outward) together with ∂B_{r_1} with reversed orientation,

$$0 = \int_{B_{r_2}\setminus B_{r_1}} \operatorname{div} F\, dx = I_{r_2} - I_{r_1},$$

so I_r is independent of r.

(c) False. Take $F(x) = x = \frac{1}{2}\nabla|x|^2$, which is curl-free (being a gradient). Then $\operatorname{div} F \equiv 3$, so by Theorem 23.b.4, $I_r = \int_{B_r} 3\, dx = 3\operatorname{vol}_3(B_r) = 4\pi r^3$, which depends on r.

(d) True. If $F = \operatorname{curl} G$ then $\operatorname{div} F = \operatorname{div}(\operatorname{curl} G) = 0$ (a standard vector identity), so part (b) applies and I_r is independent of r. On the other hand,

$$|I_r| \le \int_{\partial B_r} |\langle F, n\rangle|\, dS \le \frac{C}{r}\operatorname{vol}_2(\partial B_r) = \frac{C}{r}\cdot 4\pi r^2 = 4\pi C r \longrightarrow 0 \quad \text{as } r \to 0^+.$$

Since I_r is constant in r and this constant is a limit of values tending to 0, $I_r = 0$ for all $r > 0$. **Q.E.D.**

Solution of Exercise 26.e.33:

Writing $F = (F^1, F^2)$ with $F^1 = \lambda x e^y$, $F^2 = (y+1+x^2)e^y$, the integrability condition is $\partial_y F^1 = \partial_x F^2$:

$$\partial_y F^1 = \lambda x e^y, \qquad \partial_x F^2 = 2xe^y.$$

Since $\mathbb{R}^2$ is convex (so integrability is equivalent to conservativity here), F is conservative if and only if $\lambda = 2$.

For $\lambda = 2$, seek f with $\partial_x f = F^1 = 2xe^y$: integrating in x, $f(x,y) = x^2e^y + g(y)$ for some function g. Then $\partial_y f = x^2e^y + g'(y)$ must equal $F^2 = (y+1+x^2)e^y$, so $g'(y) = (y+1)e^y$; integrating by parts, $\int (y+1)e^y\,dy = ye^y - e^y + e^y = ye^y$ (up to an additive constant). Hence

$$f(x,y) = \boxed{(x^2+y)e^y} \qquad \text{(unique up to an additive constant)},$$

which one checks directly satisfies $\nabla f = F$. **Q.E.D.**

Solution of Exercise 26.e.34:

The four quantities I_b, I_t, I_l, I_r are already defined using the same outward normal field n on each of the four edges of ∂Q, so the total outer flux is simply their sum, with no relative sign corrections needed: $\boxed{I_b + I_t + I_r + I_l}$, answer **(b)**. **Q.E.D.**

Ordinary Differential Equations (Appendix A)

AI-Note A.1: This appendix compiles the material on Ordinary Differential Equations (ODEs). During the semester, this topic was covered in supplementary sessions.

Turning an n-th order ODE into an $n \times n$ system (Section A.a)

Given an n-th order linear ODE, we can rewrite it as a first-order problem in higher dimensions. Consider the example equation:

$$y(t) + y'(t) + 2y''(t) = f(t).$$

We define $u_1 := y$, which gives:

$$u_1 + u_1' + 2u_1'' = f.$$

Letting $u_2 := u_1'$, we can rewrite the second derivative as $u_1'' = u_2'$. The system becomes:

$$\begin{aligned} u_1 + u_2 + 2u_2' &= f \\ u_2 &= u_1'. \end{aligned}$$

Solving for the derivatives u_1' and u_2', we get:

$$\begin{aligned} u_1' &= u_2 \\ u_2' &= -\frac{1}{2}u_1 - \frac{1}{2}u_2 + \frac{1}{2}f. \end{aligned}$$

In matrix notation, this is:

$$\begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1/2 & -1/2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} 0 \\ f/2 \end{pmatrix},$$

or more compactly,

$$u'(t) = Au(t) + F(t).$$

This reduction to a first-order system can be done with any n-th order ODE. We therefore focus on solving systems of the form $u'(t) = Au(t) + F(t)$.

Solving linear systems of ODEs (Section A.b)

Diagonalizable matrices (Subsection A.b.1)

We first look at the homogeneous case where $F \equiv 0$, so $u'(t) = Au(t)$. Suppose the matrix A is diagonalizable, meaning we can write $A = PDP^{-1}$ where D is a diagonal matrix and P is invertible:

$$D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

Then we have:

$$\begin{aligned} & u'(t) = Au(t) \\ \iff\ & u'(t) = PDP^{-1}u(t) \\ \iff\ & P^{-1}u'(t) = DP^{-1}u(t) \\ \iff\ & (P^{-1}u(t))' = D(P^{-1}u(t)). \end{aligned}$$

By defining a new variable $v(t) := P^{-1}u(t)$, the system uncouples:

$$v'(t) = Dv(t) \iff \begin{pmatrix} v_1'(t) \\ \vdots \\ v_n'(t) \end{pmatrix} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} v_1(t) \\ \vdots \\ v_n(t) \end{pmatrix}.$$

This yields n independent 1-dimensional equations $v_i'(t) = \lambda_i v_i(t)$, which have the solutions:

$$v_i(t) = C_i e^{\lambda_i t} \quad \text{for } i \in \{1, \ldots, n\}.$$

The solution to the original system is then recovered by $u(t) = Pv(t)$. By diagonalizing, we were able to uncouple the $n \times n$ system and solve the scalar equations without a problem.

Non-diagonalizable matrices (Subsection A.b.2)

What if A is not diagonalizable? Then we have to use the Jordan Normal Form: $A = PJP^{-1}$. Following the same substitution $v(t) = P^{-1}u(t)$, we obtain $v'(t) = Jv(t)$. We now focus on each Jordan block in J. For a block with eigenvalue λ_i, the system looks like:

$$\begin{pmatrix} v_j'(t) \\ \vdots \\ v_{j+d}'(t) \end{pmatrix} = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix} \begin{pmatrix} v_j(t) \\ \vdots \\ v_{j+d}(t) \end{pmatrix}.$$

This translates to the equations:

$$\begin{aligned} v_j'(t) &= \lambda_i v_j(t) + v_{j+1}(t) \\ &\vdots \\ v_{j+d}'(t) &= \lambda_i v_{j+d}(t). \end{aligned}$$

We first solve for $v_{j+d}(t)$ from the last equation. Then we plug it into the preceding equation to find $v_{j+d-1}(t)$, and continue this process upwards.

Since this back-substitution can be time-consuming, it is useful to use matrix exponentials. For a system $u'(t) = Au(t)$, the formal solution is:

$$u(t) = e^{tA}u_0,$$

where u_0 is given by the initial conditions. Evaluating the matrix exponential e^{tA} provides a direct way to solve the system.

Remark A.b.1 (*Calculating the matrix exponential in practice*)**:** When computing $\exp(A) = I + \sum_{m=1}^{\infty} \frac{A^m}{m!}$, the most common strategies are:

- **(a)** **Look for a pattern:** Sometimes $A^2 = A$, or $A^3 = 2A$, or A is nilpotent ($A^n = 0$ for $n \geq k$). In these cases, the infinite series can be simplified directly.
- **(b)** **Diagonalization:** If A is diagonalizable ($A = BDB^{-1}$), then $\exp(A) = B\exp(D)B^{-1}$, where $\exp(D)$ is the diagonal matrix of exponentials.
- **(c)** **Jordan Normal Form (JNF):** If A is not diagonalizable, one can use the JNF, though this is often more tedious and less frequently required.

Example A.b.2 (*Solving a 2×2 system using matrix exponentials*)**:** Consider the system given by

$$\begin{aligned} u_1(t) + 2u_2(t) &= u_1'(t) \\ 2u_1(t) + u_2(t) &= u_2'(t). \end{aligned}$$

This can be written as $u'(t) = Au(t)$ with $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. The eigenvalues of A are found via $\det(A - \lambda I) = (1-\lambda)^2 - 4 = 0$, giving $\lambda_1 = 3$ and $\lambda_2 = -1$. The corresponding eigenvectors are $w_1 = (1,1)^\mathsf{T}$ and $w_2 = (1,-1)^\mathsf{T}$. Since A is diagonalizable, we have $A = PDP^{-1}$ with $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$. The general solution is then given by:

$$u(t) = \sum_{i=1}^{2} C_i e^{\lambda_i t} w_i = C_1 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Now consider a non-diagonalizable example where the system is already in Jordan normal form, $u'(t) = Ju(t)$ with $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Using the matrix exponential, the solution is $u(t) = e^{tJ}u(0)$. We can split $J = D + N$ where $D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is diagonal and $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is nilpotent ($N^2 = 0$). Since D and N commute, $e^{tJ} = e^{tD}e^{tN}$. We calculate:

$$e^{tD} = \begin{pmatrix} e^t & 0 \\ 0 & e^t \end{pmatrix}, \qquad e^{tN} = I + tN = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Multiplying these gives:

$$e^{tJ} = \begin{pmatrix} e^t & te^t \\ 0 & e^t \end{pmatrix}.$$

Thus, the solution is $u_1(t) = e^t u_1(0) + te^t u_2(0)$ and $u_2(t) = e^t u_2(0)$.

Example A.b.3 (*Solving an initial value problem*): Consider the initial value problem (Michael's Bsp. 23.1.4):

$$\begin{aligned} y_1' &= 3y_1 - 2y_2, \\ y_2' &= -y_1 + 2y_2, \end{aligned}$$

with initial conditions $y_1(0) = 1$ and $y_2(0) = 0$. The system is $y' = Ay$ with $A = \begin{pmatrix} 3 & -2 \\ -1 & 2 \end{pmatrix}$. We find the eigenvalues:

$$\det(A - \lambda I) = (3-\lambda)(2-\lambda) - 2 = \lambda^2 - 5\lambda + 4 = (\lambda - 4)(\lambda - 1) = 0.$$

The eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 1$. The corresponding eigenvectors are: For $\lambda_1 = 4$: $\begin{pmatrix} -1 & -2 \\ -1 & -2 \end{pmatrix} v = 0 \implies v_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$. For $\lambda_2 = 1$: $\begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix} v = 0 \implies v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. The general solution is $y(t) = C_1 e^{4t} \begin{pmatrix} -2 \\ 1 \end{pmatrix} + C_2 e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Using the initial conditions $y(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$\begin{aligned} -2C_1 + C_2 &= 1 \\ C_1 + C_2 &= 0 \implies C_2 = -C_1. \end{aligned}$$

Substituting this gives $-3C_1 = 1$, so $C_1 = -1/3$ and $C_2 = 1/3$. The particular solution is:

$$y(t) = -\frac{1}{3} e^{4t} \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \frac{1}{3} e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2e^{4t} + e^t \\ -e^{4t} + e^t \end{pmatrix}.$$

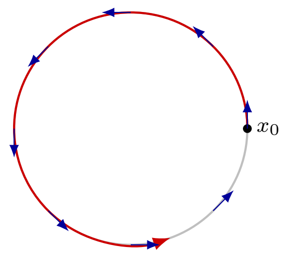
Autonomous systems and phase portraits (Subsection A.b.3)

A system $u'(t) = F(u(t))$ in which F does not depend on t explicitly is called "*autonomous*" (German: „*autonom*“). Its flow can then be pictured directly as a vector field on the state space, independent of time: at each point x, the arrow $F(x)$ shows the instantaneous velocity of the trajectory passing through x. The three examples below (two linear, one not) are the standard pictures.

Example A.b.4 (*A pure rotation*)**:** The system $u_1' = -u_2$, $u_2' = u_1$ is $u' = Au$ with $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The unique solution with $u(0) = x_0$ is

$$u(t) = e^{tA} x_0 = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} x_0,$$

a rigid rotation of x_0 around the origin at unit angular speed: F is the velocity field of a fluid rotating rigidly counterclockwise, and every trajectory is a circle centred at the origin.

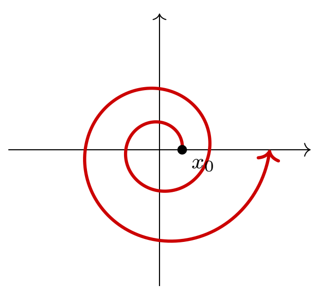


$u(t) = e^{tA} x_0$, circular trajectories

Example A.b.5 (*A rotating spiral*)**:** The system $u_1' = u_1 - u_2$, $u_2' = u_1 + u_2$ is $u' = Au$ with $A = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) = \mathrm{id} + B$, where $B := \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is the generator of Example A.b.4. Since id and B commute, $e^{tA} = e^{t\,\mathrm{id}} e^{tB}$, so

$$u(t) = e^{tA} x_0 = e^t \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} x_0 :$$

the same rotation as before, now with an exponentially growing radius. The trajectory is an outward spiral.



$u(t) = e^{tA} x_0$: rotation combined with e^t growth

Example A.b.6 (*A nonlinear system with a limit cycle*)**:** Consider the nonlinear autonomous system

$$u' = u - v - (u^2 + v^2)u, \qquad v' = u + v - (u^2 + v^2)v,$$

with initial value $(u(0), v(0)) = (x_0, y_0)$. If $(x_0, y_0) = (0, 0)$, the constant function $(u, v) \equiv (0, 0)$ is a solution, and Theorem A.d.11 (the right-hand side is C^1, Exercise A.d.14) shows it is the **only** one.

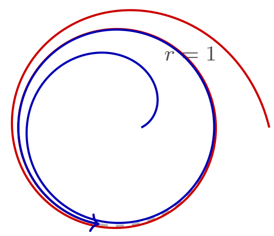
For $(x_0, y_0) \neq (0, 0)$, switch to polar coordinates $u = r\cos\varphi$, $v = r\sin\varphi$ with $r^2 = u^2 + v^2$. Differentiating $r^2 = u^2 + v^2$ and substituting the system,

$$(r^2)' = 2uu' + 2vv' = 2\big(u^2 - uv - (u^2 + v^2)u^2\big) + 2\big(uv + v^2 - (u^2 + v^2)v^2\big) = 2(r^2 - r^4),$$

so, since $(r^2)' = 2rr'$, the radius satisfies the autonomous scalar equation

$$r' = r - r^3, \qquad r(0) = \sqrt{x_0^2 + y_0^2}.$$

Separating variables and solving gives $r(t) = \exp(t)/\sqrt{A + \exp(2t)}$ for $A := 1/(x_0^2 + y_0^2) - 1 \in (-1, \infty)$, and $r(t) \to 1$ as $t \to \infty$ **regardless of the initial value**. Therefore the circle $r = 1$ is a "*limit cycle*" (German: „*Grenzzykel*"), attracting every trajectory except the equilibrium at the origin. A short computation with the angle ($u\varphi' = u - r^2 u$ from the system, matched against u', and symmetrically for v) gives $\varphi' \equiv 1$ away from the origin, so the motion is a steady rotation combined with the radial approach to $r = 1$: trajectories starting outside the unit circle spiral in, and those starting inside spiral out, both converging to the same circular orbit.



qualitative sketch: outside (red) spirals in, inside (blue) spirals out, both approaching the limit cycle $r = 1$

Inhomogeneous linear ODEs (Section A.c)

Given the n-th order linear inhomogeneous ODE

$$Ly(x) = b(x), \qquad \text{where } L = \sum_{k=0}^{n} a_k \frac{d^k}{dx^k},$$

the general solution is always of the form

$$y(x) = y_{\text{hom}}(x) + y_{\text{part}}(x).$$

The homogeneous solution $y_{\text{hom}}(x)$ fulfills $Ly_{\text{hom}}(x) = 0$, or $\sum_{k=0}^{n} a_k y_{\text{hom}}^{(k)}(x) = 0$. This is solved using the general Ansatz $y_{\text{hom}}(x) = Ce^{rx}$, which leads to the characteristic equation:

$$\sum_{k=0}^{n} a_k C r^k e^{rx} = 0 \implies \sum_{k=0}^{n} a_k r^k = 0,$$

yielding roots $r_1, \ldots, r_n$ (assuming they are all distinct).

The particular solution $y_{\text{part}}(x)$ solves $Ly_{\text{part}}(x) = b(x)$ and is found using a fitting Ansatz, often an *"educated guess"* based on the form of the inhomogeneous term $b(x)$.

Inhomogeneous Term $b(x)$	**Ansatz for** $y_{\text{part}}(x)$
$\sum_{i=0}^{m} b_i x^i$	$\sum_{i=0}^{m} A_i x^i$
$e^{\alpha x} \sum_{i=0}^{m} b_i x^i$	$e^{\alpha x} \sum_{i=0}^{m} A_i x^i$
$\sin(\omega x) \sum_{i=0}^{m} b_i x^i + \cos(\omega x) \sum_{i=0}^{m} c_i x^i$	$\sin(\omega x) \sum_{i=0}^{m} A_i x^i + \cos(\omega x) \sum_{i=0}^{m} B_i x^i$
$\sinh(\omega x) \sum_{i=0}^{m} b_i x^i + \cosh(\omega x) \sum_{i=0}^{m} c_i x^i$	$\sinh(\omega x) \sum_{i=0}^{m} A_i x^i + \cosh(\omega x) \sum_{i=0}^{m} B_i x^i$
$e^{\alpha x} \sin(\omega x) \sum_{i=0}^{m} b_i x^i + e^{\alpha x} \cos(\omega x) \sum_{i=0}^{m} c_i x^i$	$e^{\alpha x} \sin(\omega x) \sum_{i=0}^{m} A_i x^i + e^{\alpha x} \cos(\omega x) \sum_{i=0}^{m} B_i x^i$

Table A.1: Standard Ansätze for solving inhomogeneous linear ODEs. **Special case:** If a part of the chosen function for $y_{\text{part}}(x)$ is already present in the solution of the homogeneous problem, the Ansatz is additionally multiplied by x.

Example A.c.7 (*Solving an inhomogeneous linear ODE*): Consider the second-order ODE (Michael's Bsp. 24.2.1):

$$y'' - 4y' + 13y = xe^x.$$

First, we solve the homogeneous problem $y'' - 4y' + 13y = 0$. The characteristic polynomial is $\lambda^2 - 4\lambda + 13 = 0$, with roots:

$$\lambda_{1,2} = \frac{4 \pm \sqrt{16-52}}{2} = 2 \pm 3i.$$

The homogeneous solution is therefore:

$$y_{\text{hom}}(x) = e^{2x}\big(C_1\cos(3x) + C_2\sin(3x)\big).$$

Next, we find a particular solution for the inhomogeneous term $b(x) = xe^x$. According to Table A.1, we use the Ansatz $y_{\text{part}}(x) = (Ax+B)e^x$. Since $\lambda = 1$ is not a root of the characteristic polynomial, we do not need to multiply by an extra factor of x. We compute the derivatives:

$$\begin{aligned} y'_{\text{part}}(x) &= Ae^x + (Ax+B)e^x = (Ax+A+B)e^x, \\ y''_{\text{part}}(x) &= Ae^x + (Ax+A+B)e^x = (Ax+2A+B)e^x. \end{aligned}$$

Plugging these into the original ODE yields:

$$\begin{aligned} (Ax+2A+B)e^x - 4(Ax+A+B)e^x + 13(Ax+B)e^x &= xe^x \\ \big((A-4A+13A)x + (2A+B-4A-4B+13B)\big)e^x &= xe^x \\ (10Ax - 2A + 10B)e^x &= xe^x. \end{aligned}$$

Comparing coefficients, we get $10A = 1 \implies A = \frac{1}{10}$, and $-2A + 10B = 0 \implies 10B = \frac{2}{10} \implies B = \frac{1}{50}$. The particular solution is $y_{\text{part}}(x) = \left(\frac{1}{10}x + \frac{1}{50}\right)e^x$. The general solution is then $y(x) = y_{\text{hom}}(x) + y_{\text{part}}(x)$.

Exercise A.c.8 (*Three linear ODEs*):

1. Determine the general solution on $I = \mathbb{R}$ of $y'' - 4y' + 4y = \sin t$.
2. Determine the solution on $I = \mathbb{R}$ of the initial value problem $y'' - y = t$, $y(0) = 1$, $y'(0) = 3$.
3. Find a solution on $I = (0, \infty)$ of the initial value problem $y' - \left(\frac{4}{t} + 1\right)y = t^4$, $y(1) = 1$.

Proof of Exercise A.c.8:

1. The characteristic polynomial $r^2 - 4r + 4 = (r-2)^2$ has the double root $r = 2$, so $y_{\text{hom}}(t) = (C_1 + C_2 t)e^{2t}$. For the particular solution, try $y_{\text{part}}(t) := A\sin t + B\cos t$ (Table A.1, with $\omega = 1$ not a root of the characteristic polynomial, so no extra factor of t is needed). Substituting,

$$y''_{\text{part}} - 4y'_{\text{part}} + 4y_{\text{part}} = (3A + 4B)\sin t + (3B - 4A)\cos t \stackrel{!}{=} \sin t,$$

so $3A + 4B = 1$ and $-4A + 3B = 0$, giving $B = \frac{4}{3}A$ and then $A = \frac{3}{25}$, $B = \frac{4}{25}$. The general solution is

$$y(t) = (C_1 + C_2 t)e^{2t} + \tfrac{3}{25}\sin t + \tfrac{4}{25}\cos t.$$

2. The characteristic polynomial $r^2 - 1$ has roots $r = \pm 1$, so $y_{\text{hom}}(t) = C_1 e^t + C_2 e^{-t}$. Since $r = 0$ is not a root, try $y_{\text{part}}(t) := At + B$; then $y''_{\text{part}} - y_{\text{part}} = -At - B \stackrel{!}{=} t$ gives $A = -1$, $B = 0$, so $y_{\text{part}}(t) = -t$. The general solution is $y(t) = C_1 e^t + C_2 e^{-t} - t$, with $y'(t) = C_1 e^t - C_2 e^{-t} - 1$. The initial conditions give $C_1 + C_2 = 1$ and $C_1 - C_2 = 4$, so $C_1 = \frac{5}{2}$, $C_2 = -\frac{3}{2}$, and

$$y(t) = \tfrac{5}{2}e^t - \tfrac{3}{2}e^{-t} - t.$$

3. This is a first-order linear ODE, $y' + p(t)y = t^4$ with $p(t) := -\left(\frac{4}{t} + 1\right)$, so the standard integrating factor is $\mu(t) := \exp\left(\int p(t)\,dt\right) = \exp\left(-4\ln t - t\right) = t^{-4}e^{-t}$.

AI-Note A.2: The script's own hint suggests $u(t) := e^{H(t)}y(t)$ with $H'(t) = \frac{4}{t} + 1$, giving a **positive** exponent. Checking it directly, $u' = e^H(H'y + y')$, and substituting $y' = H'y + t^4$ (from the ODE) gives $u' = e^H(2H'y + t^4)$, which still contains y. The substitution therefore does not decouple. The sign must be negative, $u(t) := e^{-H(t)}y(t) = t^{-4}e^{-t}y(t)$: then $u' = e^{-H}(y' - H'y) = e^{-H}\big(y' - (\frac{4}{t} + 1)y\big) = e^{-H}t^4$ directly from the ODE, which **does** decouple. Used below.

So with $u(t) := t^{-4}e^{-t}y(t)$,

$$u'(t) = t^{-4}e^{-t}t^4 = e^{-t} \qquad \Longrightarrow \qquad u(t) = -e^{-t} + C$$

for a constant C. Hence $y(t) = t^4e^t\big(C - e^{-t}\big) = Ct^4e^t - t^4$. The initial condition $y(1) = 1$ gives $Ce - 1 = 1$, i.e. $C = 2/e$, so

$$y(t) = 2t^4e^{t-1} - t^4, \qquad t \in (0, \infty).$$

Q.E.D.

Example A.c.9 (*Pure resonance of a pumped harmonic oscillator*): Consider the forced harmonic oscillator $y : [0, \infty) \to \mathbb{R}$,

$$y''(t) + ky(t) = \sin(\omega t) \quad \text{for } t > 0, \qquad y(0) = y'(0) = 0,$$

with $k > 0$. The homogeneous equation $z'' + kz = 0$ has general solution $z(t) = C_1\cos(\sqrt{k}t) + C_2\sin(\sqrt{k}t)$. For $\omega \neq \sqrt{k}$, the Ansatz $y_{\text{part}}(t) := A\sin(\omega t) + B\cos(\omega t)$ gives, on substitution, $A(k-\omega^2) = 1$ and $B(k-\omega^2) = 0$, so $B = 0$ and $A = 1/(k-\omega^2)$, giving $y_{\text{part}}(t) = \sin(\omega t)/(k-\omega^2)$. Imposing $y(0) = y'(0) = 0$ on the general solution $y = z + y_{\text{part}}$ fixes $C_1 = 0$ and $C_2 = -\omega/\big(\sqrt{k}(k-\omega^2)\big)$, giving the complete solution, for $\omega \neq \sqrt{k}$,

$$y(t) = \frac{1}{\sqrt{k}+\omega}\left(\frac{\sin(\sqrt{k}t)}{\sqrt{k}} - \frac{\sin(\sqrt{k}t) - \sin(\omega t)}{\sqrt{k}-\omega}\right).$$

Taking the limit $\omega \to \sqrt{k}$ (“*resonance*”: the forcing frequency matches the system’s natural frequency) gives

$$y(t) = \frac{1}{2\sqrt{k}}\left(\frac{\sin(\sqrt{k}t)}{\sqrt{k}} - t\cos(\sqrt{k}t)\right).$$

The term $t\cos(\sqrt{k}t)$ shows the amplitude of the oscillation now grows **linearly** in t, without bound. This is the phenomenon behind bridges failing under marching troops and, per popular-science lore not verified here, opera singers shattering glasses.

Remark A.c.10: This is the concrete instance of the *“special case”* noted in the caption of Table A.1: the naive Ansatz $A\sin(\omega t) + B\cos(\omega t)$ fails exactly when $\omega = \sqrt{k}$, because $\sin(\sqrt{k}t)$ and $\cos(\sqrt{k}t)$ already solve the homogeneous equation. The fix is the same one the table prescribes. Multiplying the Ansatz by t is visibly what appears in the limiting solution above.

Duhamel’s Principle (Subsection A.c.1)

For a system of inhomogeneous linear ODEs $u'(t) = Au(t) + f(t)$, the solution can be constructed by combining the homogeneous solution and a particular solution obtained via Duhamel’s principle:

$$u(t) = \underbrace{e^{At}u(0)}_{\text{homogeneous solution}} + \underbrace{\int_0^t e^{(t-s)A}f(s)\,ds}_{\text{inhomogeneous part}}.$$

This elegantly generalizes the 1-dimensional case where the solution to $u'(t) = au(t) + f(t)$ is given by $u(t) = e^{at}u(0) + \int_0^t e^{a(t-s)}f(s)\,ds$.

The Cauchy–Lipschitz / Picard–Lindelöf theorem (Section A.d)

The Cauchy–Lipschitz theorem (also known as the Picard–Lindelöf theorem) guarantees the existence and uniqueness of solutions to first-order initial value problems, provided the right-hand side is Lipschitz continuous with respect to the state variable. For an equation of the form $y'(x) = F(x, y(x))$, one first puts the differential equation in normal form to determine if the theorem applies, which requires checking the properties of F.

Theorem A.d.11 (*Picard–Lindelöf, local existence and uniqueness*): Let $U \subseteq \mathbb{R}\times\mathbb{R}^n$ be an open set, $(t_0, x_0) \in U$, and $F : U \to \mathbb{R}^n$ be continuous. If F is locally Lipschitz continuous in x, i.e.,

$$\forall (t_1, x_1) \in U \ \ \exists \varepsilon > 0, L \geq 0 \ \text{ such that}$$
$$\forall (t, x_2), (t, x_3) \in B_\varepsilon(t_1, x_1) \cap U, \ \ \|F(t, x_2) - F(t, x_3)\| \leq L\|x_2 - x_3\|,$$

then there exists an interval $(a, b) \subseteq \mathbb{R}$ with $t_0 \in (a, b)$ such that a differentiable function $u : (a, b) \to \mathbb{R}^n$ exists which satisfies $(t, u(t)) \in U$ for all $t \in (a, b)$, and u is a solution to the initial value problem:

$$\begin{cases} u'(t) = F(t, u(t)) \\ u(t_0) = x_0. \end{cases}$$

Furthermore, for any other solution $v : (\tilde{a}, \tilde{b}) \to \mathbb{R}^n$ of the same initial value problem, we have $(\tilde{a}, \tilde{b}) \subseteq (a, b)$ and $u|_{(\tilde{a},\tilde{b})} = v$. As $t \to a$ or $t \to b$, the limit $\lim_{t\to a}(t, u(t))$ (respectively $\lim_{t\to b}(t, u(t))$) either does not exist or does not lie in U.

A simplified, highly practical formulation is the following:

Corollary A.d.12 (*Picard–Lindelöf, simplified version*): Let $I \subseteq \mathbb{R}$ be a compact interval, $\Omega \subseteq \mathbb{R}^n$ an open interval (or open set), $(t_0, y_0) \in I \times \Omega$, and $f : I \times \Omega \to \mathbb{R}^n$ continuous. Suppose f satisfies the Lipschitz condition

$$\|f(t, y_1) - f(t, y_2)\| \leq L\|y_1 - y_2\| \qquad \forall t \in I \text{ and } \forall y_1, y_2 \in \Omega.$$

Then there exists a $\delta > 0$ such that the initial value problem $u' = f(t, u)$ with $u(t_0) = y_0$ has a unique solution $u : I \cap [t_0 - \delta, t_0 + \delta] \to \mathbb{R}^n$. If $\Omega = \mathbb{R}^n$, one can choose the domain to be the entire interval I.

Example A.d.13 (*Applying Picard–Lindelöf*): Let $I = [0, \infty)$ and $\Omega = [0, \infty)$. Consider the function $f : I\times\Omega \to \mathbb{R}$ given by $f(t, y) = \frac{y}{1+t^2}$. We want to prove that the differential equation $u'(t) - \frac{u(t)}{1+t^2} = 0$ has a unique solution for $u : (0, \infty) \to \mathbb{R}$.

We rewrite the ODE as $u'(t) = f(t, u(t))$. To apply the Picard–Lindelöf theorem, we check if f satisfies the Lipschitz condition in y:

$$|f(t, y_1) - f(t, y_2)| = \left|\frac{y_1}{1+t^2} - \frac{y_2}{1+t^2}\right| = \frac{1}{1+t^2}|y_1 - y_2| \leq 1 \cdot |y_1 - y_2|.$$

Since $\frac{1}{1+t^2} \leq 1$ for all $t \in [0, \infty)$, the function f is Lipschitz continuous in y with Lipschitz constant $L = 1$. Thus, by Picard–Lindelöf, the ODE possesses a unique solution.

Exercise A.d.14 (*C^1 in space implies locally Lipschitz*): Let $U \subseteq \mathbb{R}\times\mathbb{R}^n$ be open and $F : U \to \mathbb{R}^n$ be continuous, such that the partial derivatives $\partial_{x_k}F$ for $k \in \{1, \ldots, n\}$ exist and are continuous on U. Show that F is locally Lipschitz in x and thus satisfies the hypotheses of Theorem A.d.11.

Proof of Exercise A.d.14:

Fix $(t_1, x_1) \in U$. Since U is open, choose $\varepsilon > 0$ with the closed ball $\overline{B_\varepsilon(t_1, x_1)} \subseteq U$; this closed ball is compact. Each $\partial_{x_k}F$ is continuous, hence bounded on this compact set, say by M_k; set $L := \sum_{k=1}^n M_k$.

For $(t, x_2), (t, x_3) \in B_\varepsilon(t_1, x_1)$, the straight segment $s \mapsto x_3 + s(x_2 - x_3)$, $s \in [0, 1]$, stays inside the ball (which is convex), so $g(s) := F(t,\, x_3 + s(x_2 - x_3))$ is differentiable on $[0, 1]$ with $g'(s) = \sum_{k=1}^n \partial_{x_k}F(t, x_3 + s(x_2 - x_3))\,(x_2 - x_3)_k$ by the chain rule. Hence

$$\|F(t, x_2) - F(t, x_3)\| = \|g(1) - g(0)\| = \left\|\int_0^1 g'(s)\,ds\right\| \leq \int_0^1 \sum_{k=1}^n |\partial_{x_k}F(t, x_3 + s(x_2 - x_3))|\,|(x_2 - x_3)_k|\,ds$$
$$\leq \sum_{k=1}^n M_k\,|(x_2 - x_3)_k| \leq \Big(\sum_{k=1}^n M_k\Big)\|x_2 - x_3\| = L\|x_2 - x_3\|,$$

using $|(x_2 - x_3)_k| \le \|x_2 - x_3\|$ in the last step. So F is L-Lipschitz in x on $B_\varepsilon(t_1, x_1) \cap U$, and since $(t_1, x_1) \in U$ was arbitrary, F is locally Lipschitz in x. **Q.E.D.**

Remark A.d.15: This is the standard shortcut for checking the hypothesis of Theorem A.d.11 in practice: one essentially never verifies the Lipschitz estimate by hand, but instead checks that F is C^1 in the spatial variables. That is exactly the regularity most vector fields one writes down already have.

Lemma A.d.16 (*Grönwall's inequality*): Let $u \in C^1([a,b])$ satisfy $u'(t) \le \beta(t)u(t)$ for all $t \in [a,b]$, where $\beta \in C^0([a,b])$. Then

$$u(t) \;\le\; u(a)\exp\biggl(\int_a^t \beta(s)\,ds\biggr) \qquad \text{for all } t \in [a,b].$$

Proof:

Let $v(t) := \exp\bigl(\int_a^t \beta(s)\,ds\bigr) > 0$, so $v'(t) = \beta(t)v(t)$, and set $f(t) := u(t)/v(t) \in C^1([a,b])$. Then

$$f'(t) = \frac{u'(t)v(t) - u(t)v'(t)}{v(t)^2} \le \frac{\beta(t)u(t)v(t) - u(t)\beta(t)v(t)}{v(t)^2} = 0,$$

using the hypothesis $u'(t) \le \beta(t)u(t)$ in the numerator. So f is decreasing on $[a,b]$, giving $u(t)/v(t) = f(t) \le f(a) = u(a)/v(a) = u(a)$, i.e. $u(t) \le u(a)v(t)$, which is the claim. **Q.E.D.**

Remark A.d.17 (*What Grönwall is for*): Grönwall's inequality is the standard tool for turning a **differential** bound into an **integral** one. The official script uses it to give a clean uniqueness proof for linear ODEs: if z solves an order-m linear equation $Lz = 0$, the quantity $u(t) := z(t)^2 + z'(t)^2 + \cdots + z^{(m-1)}(t)^2$ satisfies a bound $u'(t) \le Cu(t)$ for a constant C depending only on the coefficients of L, and Grönwall with $u(t_0) = 0$ (zero initial data) forces $u \equiv 0$, hence $z \equiv 0$. This is the *"zero in, zero out"* principle behind uniqueness for linear ODEs. The same device, run on $u(t) := \|v_1(t) - v_2(t)\|^2$ for two solutions v_1, v_2 of a Lipschitz system agreeing at one time, gives an alternative route to the uniqueness half of Theorem A.d.11.

Picard iteration (Subsection A.d.1)

The proof of Theorem A.d.11 is a Banach fixed-point argument (Theorem 6.c.17), and fixed-point arguments are **constructive**: the sequence of approximations built along the way converges to the solution. So Picard–Lindelöf does not merely assert that a solution exists. It hands over a procedure for computing one.

The starting point is that an initial value problem can be rewritten as an integral equation. Integrating $y'(s) = F(s, y(s))$ from t_0 to t and using $y(t_0) = u_0$,

$$y'(t) = F(t, y(t)), \quad y(t_0) = u_0 \qquad \Longleftrightarrow \qquad y(t) = u_0 + \int_{t_0}^t F\bigl(s, y(s)\bigr)\,ds.$$

The right-hand side exhibits y as a fixed point of the operator sending a function to $u_0 + \int_{t_0}^{\cdot} F(s,\cdot)\,ds$, and the Lipschitz hypothesis is exactly what makes that operator a contraction on a small enough time interval.

Definition A.d.18 (*Picard iteration*): Starting from the constant function $y_0(t) :\equiv u_0$, the "*Picard iteration*" (German: „*Picard-Iteration*“) is the sequence

$$y_n(t) := u_0 + \int_{t_0}^t F\bigl(s,\, y_{n-1}(s)\bigr)\,ds, \qquad n \ge 1.$$

Under the hypotheses of Theorem A.d.11, $(y_n)_{n\in\mathbb{N}}$ converges uniformly to the exact solution on every compact subinterval of the maximal interval of existence.

AI-Example A.d.19 (*Picard iteration recovers a known solution*): Take $y' = y$ with $y(0) = 1$, so $F(t,y) = y$, $t_0 = 0$ and $u_0 = 1$. Then $y_0 \equiv 1$ and

$$\begin{aligned} y_1(t) &= 1 + \int_0^t 1\,ds = 1 + t, \\ y_2(t) &= 1 + \int_0^t (1+s)\,ds = 1 + t + \tfrac{t^2}{2}, \\ y_3(t) &= 1 + \int_0^t \Big(1 + s + \tfrac{s^2}{2}\Big) ds = 1 + t + \tfrac{t^2}{2} + \tfrac{t^3}{6}. \end{aligned}$$

An induction gives $y_n(t) = \sum_{k=0}^n t^k/k!$, the n-th partial sum of the exponential series. The iteration converges to e^t, which is indeed the solution. Pleasingly, it reconstructs the Taylor series of e^t one term per step, since each integration raises the degree by exactly one.

Theorem A.d.20 (*Peano existence theorem*): Let $I := [t_0 - a,\ t_0 + a] \subseteq \mathbb{R}$ and $B := \overline{B_b(x_0)} \subseteq \mathbb{R}^n$, and let $f : I \times B \to \mathbb{R}^n$ be **continuous**, with no Lipschitz condition in x assumed. Put $M := \max_{I\times B}\|f\|$, which is finite by compactness, and $h := \min\{a,\ b/M\}$. Then the initial value problem

$$\dot{x}(t) = f\big(t, x(t)\big), \qquad x(t_0) = x_0$$

has at least one C^1 solution $x : [t_0 - h,\ t_0 + h] \to B$.

Remark A.d.21 (*Peano versus Picard–Lindelöf*): Dropping the Lipschitz hypothesis costs exactly uniqueness and nothing else. The standard witness is $\dot{x} = \sqrt{|x|}$ with $x(0) = 0$: the right-hand side is continuous everywhere but not Lipschitz at $x = 0$, since $\sqrt{s}/s \to \infty$ as $s \downarrow 0$. Both

$$x(t) \equiv 0 \qquad \text{and} \qquad x(t) = \tfrac{1}{4}t^2 \quad (t \geq 0)$$

solve it, as do the infinitely many solutions that wait at 0 until an arbitrary time $\tau > 0$ and only then take off along $\frac{1}{4}(t-\tau)^2$. Peano's theorem still applies and still delivers a solution; it simply cannot say which one.

Note also where the two theorems get their machinery. Picard–Lindelöf is a fixed-point argument. It runs the Banach fixed-point theorem (Theorem 6.c.17) on the integral operator, and the Lipschitz constant is precisely what makes that operator a contraction. Peano has no contraction to work with and instead argues by compactness, extracting a convergent subsequence from a family of approximate solutions. The theorem licensing that extraction is Ascoli–Arzelà (Theorem 7.d.43), whose equicontinuity hypothesis is supplied by the bound $|F| \leq C$: it makes all the approximate solutions Lipschitz with the same constant C. That is also why it gives no uniqueness: a compactness argument produces **a** limit, never **the** limit.

Dependence on the initial condition (Subsection A.d.2)

Picard–Lindelöf produces, for each starting point, one solution. In applications one rarely has the starting point exactly: it is measured, and therefore wrong by a little. So the question is not only whether a solution exists but whether it **moves continuously** when the initial condition is perturbed, and more sharply whether it is differentiable in that data.

Write $u(t;x_0)$ for the value at time t of the solution of

$$u' = F(t,u), \qquad u(0) = x_0,$$

so that $u(\cdot\,;x_0)$ is the trajectory through x_0 and $u(t;\cdot)$ records where each starting point has been carried by time t.

Theorem A.d.22 (*Differentiability in the initial condition*): Let $\Omega \subseteq \mathbb{R}\times\mathbb{R}^n$ be open and convex and let $F : \Omega \to \mathbb{R}^n$ be continuous and C^1 in its spatial variable. Then $x_0 \mapsto u(t;x_0)$ is continuously differentiable, on the set of x_0 for which the solution exists up to time t.

Remark A.d.23 (*Why this is more than a technicality*): The hypothesis is barely stronger than the one Picard–Lindelöf already needs (C^1 in space rather than locally Lipschitz in space), and the conclusion is what makes the whole subject usable. It says the **flow** $x_0 \mapsto u(t;x_0)$ is a C^1 map, so the trajectories of an ODE do not merely exist individually but fit together smoothly into a family. Two consequences worth naming:

- **Errors stay controlled.** A measurement error ε in x_0 produces an error in $u(t;x_0)$ that is $O(\varepsilon)$ for fixed t, rather than something uncontrolled. The constant does typically grow with t. Differentiability at each fixed time is not a statement about long-run stability, and chaotic systems are exactly the ones where that constant grows fast.
- **The derivative itself solves an ODE.** Differentiating $u' = F(t,u)$ in x_0 gives that $J(t) := \partial u(t;x_0)/\partial x_0$ satisfies the linear system $J' = \mathrm{J}_u\, F(t, u(t;x_0))\, J$ with $J(0) = \mathrm{id}_n$, the "*variational equation*". So the sensitivity is computable without ever solving the original equation twice.

Glossary — English and German terms (Appendix B)

The lecture and the exercise sheets for this course exist in both English and German, and the two vocabularies do not always line up in the obvious way. Note that *„Abschluss“* is the closure and not a *“conclusion”*, *„Rand“* is the boundary and not a *“rim”*, and *„Satz von Schwarz“* is what English-language texts usually call Clairaut's theorem. Each term below is mirrored once in the main text, at the point where it is first introduced; this appendix collects those pairs in one place so the document can be used alongside German material.

Canonical German wording follows Jérôme Paschoud's topic-named files. The **Ch.** column gives the chapter in which the term is first introduced; a letter there is an appendix.

A — E

English	German	Ch.
accumulation point	Häufungspunkt	4
antisymmetric k-linear form	antisymmetrische k-lineare Form	24
boundary	Rand	3
bounded	beschränkt	7
chain rule	Kettenregel	10
closed	abgeschlossen	3
closure	Abschluss	3
compact	kompakt	7
complete metric space	vollständiger metrischer Raum	6
connected	zusammenhängend	8
continuity equation	Kontinuitätsgleichung	23
continuous	stetig	5
contraction	Kontraktion	6
convex	konvex	14
critical point	kritischer Punkt	12
cuspidal cubic	Neilsche Parabel	13
degenerate (critical point)	entartet	12
diffeomorphism	Diffeomorphismus	15
differentiable	differenzierbar	9
differential (the)	das Differential	10
differential p-form	Differentialform	24
directional derivative	Richtungsableitung	10
disconnected	unzusammenhängend	8

F — L

English	German	Ch.
figure eight	Lemniskate	C
fixed point	Fixpunkt	6
flux	Fluss	23
Frobenius norm	Frobeniusnorm	2
Gabriel's horn	Torricellis Trompete	21
Gram determinant	Gram-Determinante	21
gradient	Gradient	9
Hessian matrix	Hesse-Matrix	11
Hölder's inequality	Höldersche Ungleichung	1
homeomorphism	Homöomorphismus	15
Hurwitz criterion	Hurwitz-Kriterium	12
isometry	Isometrie	21
Jacobi matrix	Jacobi-Matrix	9
Jordan measurable	Jordan-messbar	20
Lagrangian	Lagrange-Funktion	13
Lebesgue number	Lebesgue-Zahl	7
Lebesgue's covering lemma	Lebesguesches Überdeckungslemma	7
length	Länge	21
level set	Niveaumenge	13
Lipschitz continuous	Lipschitz-stetig	5
local minimum	lokales Minimum	12

M — P

English	German	Ch.
metric	Metrik	2
metric space	metrischer Raum	2
multi-index	Multiindex	11
negative definite	negativ definit	12
negative semidefinite	negativ semidefinit	14
norm	Norm	2
normal space	Normalraum	17
open	offen	3
open cover	offene Überdeckung	7
orientable	orientierbar	25
partial derivative	partielle Ableitung	9
path	Weg	8
path-connected	wegzusammenhängend	8
positive definite	positiv definit	12
positive semidefinite	positiv semidefinit	14
pseudo-metric	Pseudometrik	2
pullback	Zurückziehung	25

Q — Z

English	German	Ch.
saddle point	Sattelpunkt	12
scalar field	Skalarfeld	10
inner product / scalar product	Skalarprodukt	2
Schwarz's theorem	Satz von Schwarz	12
sequence	Folge	1
shadow price	Schattenpreis	13
space	Raum	2
submanifold	Untermannigfaltigkeit	17
submultiplicative	submultiplikativ	2
support (of a partition function)	Träger	25
tangent space	Tangentialraum	17
totally bounded	total beschränkt	7
trapezoidal rule (Gauss)	Gaußsche Trapezformel	26
uniform convergence	gleichmässige Konvergenz	4
uniformly continuous	gleichmässig stetig	5
velocity (of a curve)	Geschwindigkeit	9
wedge product	Dachprodukt	24
interior	Inneres	3

AI-Note B.1: Three conventions in this table are worth flagging, because they are places where the German and English literature genuinely diverge rather than merely translate.

- ***„Satz von Schwarz“*** is the symmetry of mixed partial derivatives (Theorem 11.b.8). English texts more often call it **Clairaut’s theorem**; both names are used in this document, with Schwarz first because that is what the lecture uses.
- ***„Neilsche Parabel“*** names the curve $y^2 = x^3$ after William Neile. The English **cuspidal cubic** is descriptive rather than eponymous, so searching for one name will not find the other.
- ***„gleichmässig“*** carries both English senses. It is *“uniform”* in **uniform convergence** and in **uniformly continuous**, which are different notions (Exercise 4.a.14, Definition 5.a.4). The German gives no hint that these are distinct, and neither does the English.

Mid-Semester Repetition Quiz (Appendix C)

Corsin devoted the first half of one exercise class to a quiz revisiting everything up to this point: differentiability, the chain rule, extrema, the implicit function theorem, submanifolds. It ranges across Chapters 1 to 17 and belongs to no single chapter, so it is collected here, where it also serves as a self-test before the integration half of the course.

The quiz (Section C.a)

AI-Note C.1: Corsin credits the questions to Prof. Serra and Prof. Lang, and ran them from a shared Kahoot at the mid-semester repetition session. On his pages, green marks the **correct** statement and red an **incorrect** one; that colouring is rendered below as explicit **True**/**False** verdicts.

Question C.a.1: Let $U := \{(x, y) \in \mathbb{R}^2 : 1 \leq x^2 + y^2 \leq 4\}$ and $f(x, y) := \sin(xy) - y^4$. Which of the following holds?

- **(a)** $f(U) = [a, b]$ for some $a < b$.
- **(b)** $f(U)$ is disconnected and closed with two connected components.

Answer C.a.2: (a) is true, (b) is false. U is compact and connected and f is continuous, so $f(U) \subseteq \mathbb{R}$ is compact (Item **(a)**) and connected (Theorem 8.a.6). The connected subsets of $\mathbb{R}$ are exactly the intervals, and a compact one is closed and bounded, so $f(U) = [a, b]$.

AI-Note C.2: Worth noting how little computation this needs: f is never evaluated. The annulus U is connected (it is **not** simply connected, but that is a different property and irrelevant here), and the two preservation theorems do the rest. Recognising that a question is asking about preserved properties rather than about values is most of the work.

Question C.a.3: Let $X \subseteq \mathbb{R}^2$ with $X \neq \mathbb{R}^2$ and $X \neq \emptyset$. If X is complete, is it closed?

Answer C.a.4: True. Let $(a_n)_{n=0}^{\infty}$ be a convergent sequence with elements in X. Then (a_n) is Cauchy, and by completeness it has a limit point in X. So X is sequentially closed, hence closed (Proposition 4.a.7).

AI-Note C.3: The limit is taken in $\mathbb{R}^2$, and completeness of X is what forces it back into X. The converse also holds here, but only because the ambient space $\mathbb{R}^2$ is itself complete: a closed subset of a complete space is complete. In a non-complete ambient space closedness gives nothing. Note that $\mathbb{Q}$ is closed in $\mathbb{Q}$ and not complete.

Question C.a.5: Let $X \subseteq \mathbb{R}^2$ as above. If X is not open, is it closed?

Answer C.a.6: False.

AI-Note C.4: Corsin marks this false but gives no counterexample; the half-open square $X := [0, 1] \times (0, 1)$ from Section 3.a is one, being neither open nor closed. *"Open"* and *"closed"* are not opposites and not exhaustive: $\emptyset$ and $\mathbb{R}^2$ are both, most sets are neither, and the excluded values $X \neq \emptyset$, $X \neq \mathbb{R}^2$ in the hypothesis are there precisely to rule out the two clopen cases.

Question C.a.7: Let

$$f(x, y, z) := \frac{xy^2}{x^2 + y^2 + z^2} \quad \text{for } (x, y, z) \neq 0, \qquad f(0, 0, 0) := 0.$$

Is $f \in C^1(\mathbb{R}^3)$?

Answer C.a.8: False. We compute

$$\partial_x f = \frac{y^2}{x^2+y^2+z^2} - 2\,\frac{x^2y^2}{(x^2+y^2+z^2)^2}.$$

In spherical coordinates $x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, $z = r\cos\theta$, every power of r cancels:

$$\partial_x f = \sin^2\theta\sin^2\varphi - 2\sin^4\theta\cos^2\varphi\sin^2\varphi.$$

The limit $\lim_{r\to 0}\partial_x f(r,\theta,\varphi)$ is therefore ill-defined, since it depends on the direction (θ,φ) along which the origin is approached. Consequently $\partial_x f$ is not sequentially continuous at 0.

AI-Note C.5: The cancellation of r is the whole argument, and it is worth naming the reason: $\partial_x f$ is "*homogeneous of degree* 0", so it is constant along each ray from the origin and takes a different constant on different rays. Compare AI-Example 10.b.8, where the same trick (restrict to a ray, watch the parameter cancel) exposed a function whose directional derivatives all existed without it being differentiable.

Question C.a.9: Let $f \in C^\infty(\mathbb{R}^3)$ and $\alpha \in \mathbb{R}$ be such that

$$\nabla f(0) = (0,0,0), \qquad \operatorname{Hess} f(0) = \begin{pmatrix} 1 & \alpha & 0 \\ \alpha & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Which of the following is **false**?

(a) For all $\alpha \in \mathbb{R}$, 0 is not a local maximum point of f.
(b) For $\alpha > \sqrt{2}$, 0 is a saddle point.
(c) For $\alpha \neq \pm\sqrt{2}$, $\nabla f : \mathbb{R}^3 \to \mathbb{R}^3$ is a local diffeomorphism around 0.
(d) For $\alpha < \sqrt{2}$, 0 is a local minimum point of f.

Answer C.a.10: (d) is the false one.

(a) is true, since $1 > 0$ is clearly an eigenvalue of $\operatorname{Hess} f(0)$ for any choice of α (the third standard basis vector is an eigenvector), so $\operatorname{Hess} f(0)$ is never negative semidefinite and 0 can never be a local maximum.

(b) is true. Let $\lambda_1, \lambda_2, \lambda_3 = 1$ be the eigenvalues of $\operatorname{Hess} f(0)$. Then

$$\lambda_1\lambda_2 = \det\big(\operatorname{Hess} f(0)\big) = 2 - \alpha^2 < 0 \quad \text{for } \alpha > \sqrt{2},$$

so $\operatorname{Hess} f(0)$ has both positive and negative eigenvalues, making it indefinite, and 0 is a saddle point by Theorem 12.b.10.

(c) is true. For these values $\det(\operatorname{Hess} f(0)) \neq 0$, and therefore the inverse function theorem (Theorem 15.a.1) applies. Note that $\operatorname{Hess} f(0) = D(\nabla f)_0$: the Hessian of f **is** the differential of the gradient map.

(d) is false, because the same reasoning as in **(b)** gives that for $\alpha < -\sqrt{2} < \sqrt{2}$ we again have $2 - \alpha^2 < 0$, so 0 is a saddle point of f, not a local minimum. The statement is true only on $-\sqrt{2} < \alpha < \sqrt{2}$.

AI-Note C.6: Corsin's third bullet reads "*for* $\alpha \neq \sqrt{2}$", but $\det(\operatorname{Hess} f(0)) = 2 - \alpha^2$ vanishes at $\alpha = -\sqrt{2}$ as well, so it must read $\alpha \neq \pm\sqrt{2}$; typeset with $\pm$ above. The oversight is a conspicuous one here, since the **next** bullet is false for exactly the reason the third one omits. The quiz is built around the missing negative root.

Question C.a.11: Let $V := \{(x,y) \in \mathbb{R}^2 : y^4 + y^2 = x^3 + x\}$. Decide:

(a) V is a graph over y in a neighbourhood of $(0,0)$.
(b) V is a graph over x in a neighbourhood of $(0,0)$.

Answer C.a.12: (a) True. Notice that $V = f^{-1}\{0\}$ for $f(x,y) := y^4 + y^2 - x^3 - x$. Then

$$\mathrm{J}\, f(x,y) = (\partial_x f,\ \partial_y f) = (-3x^2 - 1,\ 4y^3 + 2y), \qquad \mathrm{J}\, f(0,0) = (-1,\ 0).$$

Since $\partial_x f(0,0) = -1 \neq 0$, the implicit function theorem (Theorem 16.a.1) applies with x as the variable to solve for: in a neighbourhood of $(0,0)$,

$$(x,y) \in V \iff f(x,y) = 0 \iff x = x(y).$$

The theorem does **not** apply for y, since $\partial_y f(0,0) = 0$.

(b) False. Solving the defining equation for y^2 by the quadratic formula gives

$$y^2 = -\tfrac{1}{2} + \sqrt{\tfrac{1}{4} + x^3 + x},$$

which has no solution for $x \in (-\varepsilon, 0)$: there $x^3 + x < 0$, so the square root is smaller than $\frac{1}{2}$ and the right-hand side is negative, while $y^2 \geq 0$. So V contains no points at all just to the left of the origin, and cannot be a graph over x there.

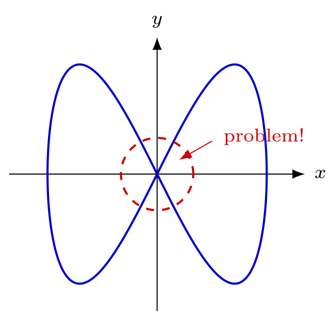
AI-Note C.7: The two parts are settled by different means, and the asymmetry is the point. Part **(a)** is decided by **checking a derivative**; part **(b)** needs an **explicit computation** showing the set is empty on one side. That is because the implicit function theorem yields no negative results: failure of its hypothesis proves nothing on its own, exactly as discussed after Theorem 16.a.1.

Question C.a.13: Let $\gamma : I \to \mathbb{R}^2$ be injective and smooth, with I an open interval. Is $\gamma(I)$ then a smooth submanifold of $\mathbb{R}^2$?

Answer C.a.14: False. Consider

$$\gamma : (0, 2\pi) \to \mathbb{R}^2, \qquad t \mapsto \big(\sin t,\ \sin 2t\big),$$

whose image is a "*figure eight*" (German: „*Lemniskate*“). This γ is injective and smooth, but there is no graphical representation of $\operatorname{im}(\gamma)$ around $(0,0)$.



AI-Note C.8: Two things are worth extracting. First, how injectivity survives a self-crossing at all: the two parameter ends $t \to 0^+$ and $t \to 2\pi^-$ both send $\gamma(t) \to (0,0)$, but neither **attains** it, so the image passes through the origin only once (via $t = \pi$) while being approached from two further directions.

Second, this locates precisely which hypothesis of Theorem 17.a.9 fails. It is not injectivity, and not injectivity of $D\gamma_t$; it is the requirement that the parametrization be a **homeomorphism** onto its image. Here γ^{-1} is discontinuous at $(0,0)$: points of the image arbitrarily close to the origin have parameters near 0 and near 2π, far apart in I. That the resulting set is not a submanifold can then be confirmed by the component count of AI-Example 17.a.6. Deleting the crossing point leaves four pieces, while deleting a point from an interval leaves two.

The quiz ends here, and with it the differential-topology half of the course. The rest of the week starts integration in several variables.

Problem Sheet Index (Appendix D)

The problem sheets are not reproduced as blocks anywhere in this document. Every problem is quoted verbatim from the official sheet and placed next to the material it tests, with its solution in the **Solutions** section of the chapter that hosts it. That is the right arrangement for reading the notes and the wrong one for answering *"what is on sheet 7?"*, so this appendix restores the sheet view.

All statements are quoted from `exercises/ExN_Analysis2_eng.pdf`; attribution: Prof. Joaquim Serra, D-MATH, ETH Zürich.

Notation D.1 (*Corsin's colour key*): From sheet 4 onwards Corsin opens each file with a colour-coded *Recommended exercises* page. The key, as given there: blue = **important**, orange = **semi-important**, red = **optional**. The official sheets additionally mark harder problems with **(*)**. Each problem carries its tag in its own title throughout this document; the tables below collect them, together with Corsin's own comments.

AI-Note D.1: Sheets 1–3 have no priority page (Corsin's *Recommended exercises* pages start with sheet 4), so no problem on them is singled out, and all of them are reproduced and solved.

Sheets 1–3 (Section D.a)

Problem	Topic	Now in
1.1	Recap questions	Section 1.a
1.2	From equations to drawings	Section 1.c
1.3	From drawings to equations	Section 1.c
1.4	Young's inequality	Section 1.b
2.1	Metric spaces	Section 2.b
2.2	Multiple choice (closure)	Section 3.c
2.3	Multiple choice (distance from a set)	Section 3.c
2.4	Boundary, interior	Section 3.c
2.5	Product of metric spaces	Section 2.b
2.6	Continuity of the distance	Section 5.a
2.7	Continuity of the composition	Section 5.a
3.1	Lipschitz or not	Section 5.b
3.2	Problems at the origin	Section 5.a
3.3	Open, closed, complete and compact	Section 7.c
3.4	Multiple choice (open, complete, compact)	Section 7.c
3.5	Multiple choice (fixed points)	Section 6.c
3.6	Multiple choice (uniform continuity)	Section 5.b
3.7	Cauchy–Schwarz	Section 2.c
3.8	Inner product and norm	Section 2.c
3.9	All norms are equivalent	Section 2.c
3.10	Hilbert–Schmidt norm	Section 2.c
3.11	(*) Uniform continuity and moduli	Section 5.b

Sheet 4 (Section D.b)

Problem	Priority	Corsin's note	Now in
4.1	important	"Use the chain rule"	Section 9.b
4.2	important	"Recall that path connected implies connected"	Section 8.b
4.3	semi-important (parts 1, 2, 3, 5); **parts 4 and 6 important**	"For 3., use Cauchy–Schwarz in $\mathbb{R}^n$ applied to convenient vectors."	Section 2.c
4.4	optional	—	Section 2.c
4.5	optional	—	Section 11.a
4.6	important (parts 1, 2, 3)	—	Section 10.d

AI-Note D.2: Problems **4.4** and **4.5** appear in this document for the first time. They had been transcribed into `content/exercise-sheets/week-04.tex`, but that file was never `\input` by anything, so neither problem had ever reached the built PDF and neither carried a solution. Both are now placed and solved.

Sheet 5 (Section D.c)

Problem	Priority	Corsin's note	Now in
5.1	important	"Recall the linearity of the differential and that $C^1 \Rightarrow$ differentiable"	Section 10.b
5.2	important	"Compare with the example from the script and recall that continuous functions on compact sets attain their extrema."	Section 13.a
5.3	important	"There is no difference between $x \to 0$ and $\lvert x\rvert \to 0$."	Section 11.b
5.4	semi-important	"Use the definition of directional derivative, $\partial_v f(x) = \frac{\partial}{\partial t}\big\rvert_{t=0} f(x+tv)$. Solve a few of these, but not all."	Section 10.b
5.5	optional	—	Section 11.b
5.6	optional	"Interesting, but irrelevant for the course"	Section 11.b
5.7	optional	"Interesting for physicists, will be covered in classical mechanics."	Section 10.d

AI-Note D.3: Problems **5.4–5.7** appear here for the first time, for the same reason as 4.4 and 4.5: they lived in `content/exercise-sheets/week-05.tex`, which nothing ever `\input`. All four are now placed and solved.

Sheet 6 (Section D.d)

Problem	Priority	Corsin's note	Now in
6.1	unmarked	—	not transcribed
6.2	important	"This is what we covered on Friday. Find a Lemma in your linear algebra notes which relates the determinant and trace to the eigenvalues."	Section 12.b
6.3	semi-important	"By using $\alpha+\beta+\gamma = \pi$ for the angles, you can express this as an optimization problem in two variables."	not transcribed
6.4	unmarked	—	not transcribed
6.5	optional	"See 'data analysis'."	not transcribed
6.6	optional	"Do two or three, not all of them."	not transcribed
6.7	important	—	Section 14.a
6.8	important	—	Section 12.b
6.9	important	—	Section 12.b
6.10–6.12	important	"These are good exercises, but how many optimization problems do you really need?"	Section 13.a, Section 12.b

AI-Note D.4: Sheet 6 is incomplete. Only the problems Corsin tagged **important** (6.2 and 6.7–6.12) were ever transcribed, and 6.1 and 6.3–6.6 have not been. Everything that **is** transcribed is solved. This is a gap in the source material, not a casualty of the restructure.

Sheet 7 (Section D.e)

Problem	Priority	Corsin's note	Now in
7.1	important	"First think of a diffeomorphism $(a, b) \leftrightarrow \mathbb{R}$ and adjust it such that $(a, b) = (0, 1)$."	Section 15.a
7.2–7.4	unmarked	—	not transcribed
7.5	important	"This is a great exercise to help with understanding of IFT."	Section 16.a
7.6	important	"You will need these coordinates very often over the next weeks."	Section 17.c
7.7	semi-important	"In **1)**, only the Möbius strip should be enough."	not transcribed
7.8	semi-important	"Do **1)** and **2)**. The others will be in the lecture."	not transcribed
7.9	unmarked	—	not transcribed

AI-Note D.5: Corsin's handwritten notes read "In 7)," and "Do 7) and 2)", but neither exercise 7.7 nor 7.8 has a sub-item numbered **7**). Both are numbered **1.**–**3.** (7.7) and **1.**–**4.** (7.8). Read as **1)** above (a plausible misreading of Corsin's handwritten "1", which this week is slightly closed at the top); this also makes the notes self-consistent, since the referenced sub-items (the Möbius strip inside 7.7's part 1, and the parametrization/tangent-space parts of 7.8) exist under number 1, not 7.

AI-Note D.6: Like sheet 6, sheet 7 is only partly transcribed. The three Corsin tagged **important**, namely 7.1, 7.5 and 7.6, are in the document and solved. Problems 7.2–7.4 and 7.7–7.9 were never transcribed.

Sheet 8 (Section D.f)

Problem	Priority	Now in
8.1	—	not transcribed
8.2	important	Section 19.a
8.3	—	not transcribed
8.4	important	Section 19.a
8.5	important	Section 20.a
8.6–8.8	—	not transcribed

AI-Note D.7: Sheet 8's three transcribed problems had **no** solutions at all until this restructure. The old chapter's **Solutions** section held nothing but a placeholder note, which had already gone stale, since the statements were in fact transcribed. All three are now solved.

Sheet 8 is the one place where a second tutor's hint file was mined, namely Sascha Brack's annotated copy of the sheet. That material stays, but per a standing decision no more of it is added: those files are annotated copies of the official sheet rather than independent notes, so the yield is cross-tutor priority agreement plus short margin hints, which is not worth the reading cost next to Corsin's own priority page.

Sheet 9 (Section D.g)

Only **9.1** and **9.6** carry an explicit colour marker on Corsin's page; **9.2** is annotated with a hint but no clearly distinguishable colour, so it is listed as unmarked below even though a hint is quoted for it. **9.3–9.5** and **9.7** carry neither colour nor hint.

Problem	Priority	Corsin's note	Now in
9.1	important	(hint on coordinates, quoted with the problem)	Section 21.b
9.2	unmarked	(hint on the domain of part 2, quoted with the problem)	Section 21.c
9.3–9.5	unmarked	—	not transcribed
9.6	semi-important	"This may become important in later courses. $\langle \cdot, \cdot \rangle$ is the standard inner prod."	not transcribed
9.7	unmarked	—	not transcribed

AI-Note D.8: Only 9.1 is tagged important, but its companion 9.2 carries an explicit worked hint from Corsin (unlike 9.3–9.5 and 9.7, which carry nothing at all), so both are reproduced and solved. 9.3–9.7 were never transcribed.

Sheet 10 (Section D.h)

Problem	Priority	Corsin's note	Now in
10.1	important	"Use spherical coordinates with $r \equiv 1$."	Section 21.b
10.2	important	(worked parametrization of the torus, quoted with the problem)	Section 21.b
10.3	important	"1.)–4.), 5.)"	Section 21.c
10.4–10.6	semi-important	—	not transcribed

AI-Note D.9: For **10.3**, Corsin's note *"1.)–4.), 5.)"* appears to flag that the starred sub-part **5.** (the harder *"if and only if"* characterization of a bounded C^1 domain, marked (*) on the official sheet) is worth attempting alongside the more routine parts 1–4, not skipped as the (*) marker might suggest. 10.4–10.6 (tractrix, logarithmic spiral, moment of inertia of an ellipse) were never transcribed.

Sheet 11 (Section D.i)

Problem	Priority	Now in
11.1	important	Section 22.b
11.2–11.4	semi-important	not transcribed
11.5	important	Section 23.b
11.6–11.9	optional	not transcribed

AI-Note D.10: Problems 11.2–11.4 and 11.6–11.9 were never transcribed, consistent with the standing decision to follow Corsin's priority marking. Between them they cover improper-integral classification, a standard flux computation, a cylinder's fluxes and volumes, the heat kernel, a divergence-theorem reality check, a multiple-choice question on Lipschitz continuity, and differential-form computations.

Sheet 12 (Section D.j)

Problem	Priority	Now in
12.1	semi-important	not transcribed
12.2	optional	not transcribed
12.3	important	Section 26.e
12.4	important	Section 26.e
12.5	important	Section 26.e
12.6–12.7	semi-important	not transcribed
12.8	important	Section 26.e
12.9	important	Section 25.b

AI-Note D.11: 12.1 (rigorous concatenation of paths), 12.2 (existence of a potential via a path integral, building the tool used implicitly in Exercise 26.e.33), 12.6 (a Stokes reality check structurally identical to Exercise 26.e.31 part 3) and 12.7 (four standard curl/div product-rule identities) were never transcribed.

Sheet 13 (Section D.k)

AI-Note D.12: There is no sheet 13 anywhere in this document, and none was ever transcribed. The old week-13 chapter carried no `\exercisesheet` heading at all. Whether a thirteenth sheet was issued is not recorded in any source available here. The two exercises in Chapter 26 that are not from an official sheet (Exercise 26.e.29 and Example 26.e.30) come from Corsin's own notes.